PRACTICE BOOK

2255 unit questions · 8 subjects · 10 full-length mocks

Layout modelled on standard GATE / Made Easy practice books:
unit questions, then compact solutions — not a theory text.

Stems follow official DA 2024 / 2025 / 2026 habits:
Bayes and named distributions · small matrices · Python defaults
quicksort recurrences · BCNF / SQL NULL · PCA orthogonality
LOOCV · uninformed vs A* · FOL implication vs conjunction

Official 2027 DA syllabus (IIT Madras) is unchanged from 2026.
Mocks use the latest confirmed paper: 65 Q, 100 marks, 3 hours.

No official PYQ is copied. Similar patterns only.

How to use this book

This is a question book, not a textbook. Each unit is a dense drill in the style of standard GATE practice volumes: numbered stems, four options for MCQ/MSQ, a blank for NAT, then a compact answer key. Work a column with a timer. Open solutions only after you finish.

Official GATE DA paper (confirmed 2026, unchanged for 2027)

65 questions, 100 marks, 3 hours. General Aptitude 10 questions / 15 marks (Q.1-5 one mark, Q.6-10 two marks). DA 55 questions / 85 marks (Q.11-35 one mark, Q.36-65 two marks). MCQ negative marking is -1/3 (1-mark) and -2/3 (2-mark). MSQ and NAT have no negative marking. The ten mock papers in this book use that exact skeleton.

What was analysed

Official DA 2024 (IISc Bengaluru), DA 2025 (IIT Roorkee) and DA 2026 (IIT Guwahati) were read for habit, not copied: 2x2 eigenproblems, a named distribution with a number, a 6-8 line Python snippet with a mutable default, SQL three-valued logic, BCNF vs 3NF on a tiny FD set, PCA axes at 90 degrees, LOOCV counting, A* vs uninformed search, and FOL implication versus conjunction. New numbers and wording throughout. Predictions in older drafts were preparation priorities, not guarantees — this edition does not claim to forecast 2027.

Marking while you practise

MCQ: choose one option. A wrong MCQ should be scored with the official negative. MSQ: one or more options; no partial credit in this key (all-correct or zero). NAT: a number; GATE accepts a range, so treat a clearly equivalent form as correct.

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Unit questions 2255 · mock questions 650 · unofficial · not affiliated with GATE / IIT / IISc.

UNIT 1 Probability and Statistics · 264 questions

MCQ negative -1/3 or -2/3. MSQ / NAT no negative. Work first, then the solutions section.

Q.1 MCQ · 2 marks · P&S; / counting

A lab has 8 data scientists and 6 engineers. A review panel of 5 is formed with at least 2 engineers. The number of possible panels is

(A) 1526
(B) 560
(C) 56
(D) 2002

Q.2 MCQ · 1 mark · P&S; / axioms

A finite sample space Omega = {omega1, omega2, omega3} has P({omega1})=0.2 and P({omega2})=0.5. Which assignment is the only one compatible with Kolmogorov axioms?

(A) P({omega3})=0.3
(B) P(Omega)=0.7
(C) P({omega3})=-0.3
(D) P({omega3})=0.7

Q.3 MCQ · 1 mark · P&S; / independence

Events A,B satisfy P(A)=0.4, P(B)=0.5. They CANNOT be both mutually exclusive and independent because

(A) 0.4+0.5 > 1
(B) P(Omega) would fail
(C) independence needs P(A n B)=0.2 while exclusivity needs P(A n B)=0
(D) they would have to be the same event

Q.4 MCQ · 2 marks · P&S; / joint

The joint PMF of (X,Y) on {0,1}x{0,1} is p(0,0)=0.1, p(0,1)=0.2, p(1,0)=0.3, p(1,1)=0.4. Then P(X=1|Y=1) equals

(A) 0.4/0.7
(B) 0.7
(C) 0.4
(D) 0.4/0.6

Q.5 MCQ · 2 marks · P&S; / Bayes

Two factories ship phones. Factory F1 ships 2000 phones with 2% defective; F2 ships 3000 phones with 6% defective. A phone is drawn uniformly from the 5000 and is defective. P(it came from F1) is closest to

(A) 0.18
(B) 0.50
(C) 0.40
(D) 0.02

Q.6 MCQ · 1 mark · P&S; / cond-exp

A two-stage experiment draws Y then X. The law of total expectation identifies E[X] with

(A) E[X^2]
(B) E[X|Y]
(C) Var(E[X|Y])
(D) E[E[X|Y]]

Q.7 MCQ · 1 mark · P&S; / descriptive

The sample average of 40 observations is 25. After including a new observation 65, the updated sample average is

(A) 30
(B) 25.5
(C) 45
(D) 26

Q.8 MCQ · 1 mark · P&S; / covariance

Which statement is TRUE for finite-variance random variables?

(A) Independent = > Cov=0, but Cov=0 need not imply independent
(B) Corr is defined even if Var(X)=0
(C) Cov=0 = > independent
(D) Var(X+Y)=Var(X)+Var(Y) always

Q.9 MCQ · 1 mark · P&S; / discrete

A PMF on {1,2,3} is p(1)=c, p(2)=2c, p(3)=3c. Then c equals

(A) 1
(B) 1/2
(C) 1/6
(D) 1/3

Q.10 MCQ · 1 mark · P&S; / binomial

X~Binomial(n=4, p=1/2). Then P(X=2)/P(X=1) equals

(A) 3/2
(B) 2
(C) 1
(D) C(4,2)

Q.11 MCQ · 1 mark · P&S; / Bernoulli

A 3x3 random matrix has i.i.d. Bernoulli(1/2) entries. The probability that the first row sums to 3 (all ones) is

(A) C(9,3)/2^9
(B) 1/27
(C) 1/2
(D) 1/8

Q.12 MCQ · 1 mark · P&S; / Poisson

Consider (i) a Poisson random variable has equal mean and variance; (ii) a Geometric(p) random variable on {1,2,...} has equal mean and variance. Which is correct?

(A) both true
(B) (i) false, (ii) true
(C) (i) true, (ii) false
(D) both false

Q.13 MCQ · 1 mark · P&S; / continuous

A waiting time X is modelled with a density f (no atoms). For any fixed a, P(X=a) equals

(A) f(a)
(B) 0
(C) F(a)
(D) 1

Q.14 MCQ · 1 mark · P&S; / uniform

X~Uniform(0,1), Y=1-X. Then (X,Y) is

(A) independent with Y~Uniform(0,1) but a different copula
(B) jointly standard normal
(C) Uniform on the segment x+y=1, 0 < x < 1 (hence dependent)
(D) i.i.d. Uniform(0,1)

Q.15 MCQ · 1 mark · P&S; / exponential

A waiting-time model is called memoryless if P(X > s+t | X > s)=P(X > t). Which option lists memoryless laws?

(A) Binomial and Exponential
(B) Uniform(0,1) and Exponential
(C) Normal and Geometric
(D) Exponential on (0,inf) and Geometric on {1,2,...}

Q.16 MCQ · 1 mark · P&S; / normal

Z~N(0,1) and X=3Z+4. Then X is distributed as

(A) N(3, 4)
(B) N(4, 3)
(C) N(4, 9)
(D) N(4, 1)

Q.17 MCQ · 1 mark · P&S; / chi-squared

Noise samples Z1,...,Zk are i.i.d. N(0,1). The residual sum of squares Z1^2+...+Zk^2 follows

(A) chi^2 with k degrees of freedom
(B) Uniform(0,k)
(C) N(0,k)
(D) t_k

Q.18 MCQ · 1 mark · P&S; / t-dist

In the usual t construction, Z~N(0,1) is independent of V~chi^2_nu. The ratio Z/sqrt(V/nu) follows

(A) N(0,1)
(B) F(nu,1)
(C) chi^2_nu
(D) t with nu degrees of freedom

Q.19 MCQ · 2 marks · P&S; / cond-pdf

X~Uniform(0,1). Given X=x, Y~Uniform(0,x). Then the joint density on 0 < y < x < 1 is

(A) f(x,y)=1
(B) f(x,y)=x
(C) f(x,y)=1/x
(D) f(x,y)=2

Q.20 MCQ · 1 mark · P&S; / CLT

i.i.d. observations with mean mu and variance sigma^2 in (0,inf). For large n the sample mean Xbar is approximately

(A) exactly t_{n-1}
(B) N(mu, sigma^2)
(C) chi^2_n
(D) N(mu, sigma^2/n)

Q.21 MCQ · 1 mark · P&S; / CI

n i.i.d. observations with known sigma. The usual 95% z-confidence interval for the mean mu is

(A) mu +/- sigma
(B) Xbar +/- t_{n-1, 0.025} * sigma/sqrt(n)
(C) Xbar +/- z_{0.025} * sigma
(D) Xbar +/- z_{0.025} * sigma/sqrt(n)

Q.22 MCQ · 1 mark · P&S; / z-test

A two-sided z-test of H0: mu=mu0 at level alpha (known sigma) rejects when

(A) p-value > alpha
(B) Z > z_alpha only, ignoring the lower tail
(C) |Z| > z_{alpha/2}
(D) Z=0

Q.23 MCQ · 1 mark · P&S; / t-test

A one-sample t-test (unknown sigma, Normal i.i.d. data) of H0: mu=mu0 uses the statistic

(A) z = (Xbar-mu0)/(sigma/sqrt(n)) with sigma replaced by 0
(B) F = Xbar / S
(C) t = (Xbar-mu0)/(S/sqrt(n)) ~ t_{n-1} under H0
(D) chi^2 = n S^2

Q.24 MCQ · 1 mark · P&S; / chi2-test

A chi-squared goodness-of-fit test with k cells, probabilities fully specified (no parameter estimated), has under H0 the statistic sum (O-E)^2/E approximately

(A) chi^2_k
(B) t_{k-1}
(C) chi^2_{k-1}
(D) N(0,1)

Q.25 MCQ · 1 mark · P&S; / mixed

Consider (i) for X~Poisson(lambda), mean equals variance; (ii) for Z~N(0,4), mean is 0 and variance is 1. Which is correct?

(A) both false
(B) both true
(C) (i) false, (ii) true
(D) (i) true, (ii) false

Q.26 MCQ · 2 marks · P&S; / multinomial

A fair three-sided spinner (labels A,B,C) is spun 5 times independently. P(exactly 2 A's, 2 B's, 1 C) equals

(A) 1/3^5
(B) 30 / 3^5
(C) 5/3
(D) C(5,2)/3^5

Q.27 MCQ · 1 mark · P&S; / tests

In hypothesis testing, a Type I error is

(A) rejecting H0 when H0 is false
(B) computing a CI instead of a test
(C) failing to reject H0 when H0 is false
(D) rejecting H0 when H0 is true

Q.28 MCQ · 1 mark · P&S; / LLN

i.i.d. Xi with finite mean mu. The weak law of large numbers states that Xbar_n converges to mu

(A) only when the Xi are Normal
(B) almost surely only if the Xi are bounded, and never in probability otherwise
(C) in distribution to Uniform(0,1)
(D) in probability

Q.29 MCQ · 2 marks · P&S; / counting

Seven distinct faculty sit around a circular conference table. Two particular faculty refuse to sit next to each other. The number of distinct seating arrangements, treating rotations as identical, is

(A) 720
(B) 480
(C) 3600
(D) 240

Q.30 MCQ · 1 mark · P&S; / axioms

For events A,B with P(A)=0.6, P(B)=0.7, the smallest possible value of P(A n B) is

(A) 0.3
(B) 0.42
(C) 0
(D) 0.6

Q.31 MCQ · 2 marks · P&S; / independence

Two fair coins are tossed independently. Let A = first coin is heads, B = second coin is heads, C = the two coins show different faces. Then

(A) A,B,C are pairwise independent but not mutually independent
(B) A,B,C are mutually independent
(C) P(A n B n C)=1/4
(D) A and C are disjoint

Q.32 MCQ · 1 mark · P&S; / joint

A joint table of (Job type J, Delay D) is: intern: on-time 20, delayed 10; full-time: on-time 40, delayed 30 (counts). Assuming uniform over the 100 jobs, P(delayed | intern) equals

(A) 10/100
(B) 1/3
(C) 1/2
(D) 40/70

Q.33 MSQ · 2 marks · P&S; / Bayes

`A binary classifier for a rare class with prevalence 0.01 has sensitivity 0.99 and specificity 0.99. Which claims are correct?`

(A) Sensitivity equals P(predicted 1 | class=1)
(B) P(predicted 1 | class=0) = 0.01
(C) Specificity equals P(predicted 1 | class=0)
(D) P(class=1 | predicted 1) is far below 0.99 because of the low prevalence

Q.34 MCQ · 1 mark · P&S; / cond-exp

For a square-integrable response X and a discrete grouping variable Y, the law of total variance expands Var(X) as

(A) E[Var(X|Y)] only
(B) Var(E[X|Y]) only
(C) E[Var(X|Y)] + Var(E[X|Y])
(D) E[X^2] - E[X]

Q.35 MCQ · 1 mark · P&S; / descriptive

Incomes have min 40, max 160, mean 88, sd 16 (all in thousands). The z-score of income 104 is

(A) 0.1
(B) (104-88)/160
(C) (104-40)/(160-40)
(D) 1

Q.36 MCQ · 1 mark · P&S; / covariance

Let X~Bernoulli(p) and Y=1-X. Then Corr(X,Y) equals

(A) 1
(B) 0
(C) -1
(D) p(1-p)

Q.37 MCQ · 1 mark · P&S; / discrete

Let F be the CDF of an integer-valued exam score. Which property must F satisfy?

(A) right-continuous and non-decreasing, with jumps at atoms of the law
(B) left-continuous
(C) a density
(D) strictly increasing

Q.38 MSQ · 2 marks · P&S; / binomial

X~Binomial(n=6, p=1/3). Which identities hold?

(A) P(X=0)=(2/3)^6
(B) X is approximately N(2, 4/3) with no error even for n=6 (exact, not approximate)
(C) Var(X)=6*(1/3)*(2/3)=4/3
(D) E[X]=2

Q.39 NAT · 2 marks · P&S; / Bernoulli

A 4x4 matrix of i.i.d. Bernoulli(1/2) entries. P(the first row sums to 2 AND the first column sums to 2). The overlapping (1,1) entry is shared. This probability, rounded to three decimals, is _____.

NAT · enter the numerical value

Q.40 MCQ · 2 marks · P&S; / Poisson

If X~Poisson(lambda) and Y~Poisson(mu) are independent, then the conditional law of X given X+Y=n is

(A) Binomial(n, 1/2) always
(B) Uniform{0,...,n}
(C) Binomial(n, lambda/(lambda+mu))
(D) Poisson(lambda/(lambda+mu))

Q.41 MCQ · 1 mark · P&S; / continuous

Which listed property is required of every CDF F: R - > [0,1] of a real random variable?

(A) a density
(B) symmetric about 0
(C) non-decreasing and right-continuous with limits 0 at -inf and 1 at +inf
(D) decreasing

Q.42 MCQ · 2 marks · P&S; / uniform

The spacings of n i.i.d. Uniform(0,1) order statistics are distributed as

(A) i.i.d. Uniform(0,1)
(B) i.i.d. Normal
(C) degenerate at 1/n
(D) normalised i.i.d. exponentials (Dirichlet(1,..,1) on the simplex)

Q.43 MCQ · 2 marks · P&S; / exponential

`P(X > =3)=0.2 for an Exponential random variable X with mean 1/lambda. Then lambda equals`

(A) (1/3) ln 5
(B) ln 0.2
(C) 3 ln 5
(D) ln 5

Q.44 MCQ · 1 mark · P&S; / normal

A z-test pivot is claimed to be standard normal. Which option correctly states the mean and variance of N(0,1)?

(A) mean 0 and variance 0
(B) mean 0 and variance 1
(C) mean 1 and variance 0
(D) mean 1 and variance 1

Q.45 MCQ · 1 mark · P&S; / chi-squared

A one-sample variance CI for Normal data uses which pivotal quantity?

(A) S^2 ~ N(0,1)
(B) (n-1) S^2 / sigma^2 ~ chi^2_{n-1}
(C) n S^2 / mu ~ t_{n-1}
(D) (Xbar-mu)/S ~ chi^2

Q.46 MCQ · 1 mark · P&S; / t-dist

Compared with N(0,1), a t_nu density (nu finite) has

(A) heavier tails and a peak a little lower at 0
(B) support (0,inf)
(C) mean equal to nu
(D) lighter tails

Q.47 MCQ · 1 mark · P&S; / cond-pdf

If f_{X|Y}(x|y)=f_X(x) for (almost) all x,y in the support, then

(A) the joint density cannot exist
(B) X and Y are independent
(C) Y is Normal
(D) X is constant

Q.48 MCQ · 2 marks · P&S; / CLT

Y~Binomial(200, 0.4). Using the CLT without continuity correction, P(70 < = Y < = 90) is approximately

(A) Phi(1)-Phi(-1)
(B) Phi(90)-Phi(70)
(C) Phi(1.44)-Phi(-1.44) because mean=80 and sd=sqrt(48)~6.93, (90-80)/6.93~1.44
(D) exactly 0.5

Q.49 MCQ · 1 mark · P&S; / CI

A 99% interval with the same data is, compared with a 95% z-interval (known sigma),

(A) wider, because z_{0.005} > z_{0.025}
(B) narrower
(C) identical
(D) shifted but of equal length

Q.50 MCQ · 1 mark · P&S; / z-test

For the z=1.5 two-sided test of H0: mu=50, using z_{0.025}=1.96, the decision at 5% is

(A) reject H0
(B) do not reject H0
(C) reject only the lower tail
(D) the test is undefined

Q.51 MCQ · 1 mark · P&S; / t-test

For t=2 on 8 df, using the critical value t_{8, 0.025}~2.306, a two-sided 5% test

(A) rejects H0
(B) has p-value 0
(C) requires known sigma
(D) does not reject H0

Q.52 MCQ · 1 mark · P&S; / chi2-test

A 2x2 contingency table independence test (no cell fusion) uses chi^2 with degrees of freedom

(A) 2
(B) 3
(C) 1
(D) 4

Q.53 MCQ · 1 mark · P&S; / mixed

Four fair coins are tossed independently. Let A be the event of at least three heads and B the event of at least three tails. Then P(A n B) equals

(A) 1/16
(B) 1/8
(C) 0
(D) 1/2

Q.54 NAT · 1 mark · P&S; / multinomial

Dice: a fair six-sided die is rolled 4 times. The number of outcome tuples with each of four distinct faces (a permutation of four different numbers) is P(6,4)= _____.

NAT · enter the numerical value

Q.55 MCQ · 1 mark · P&S; / tests

Power of a level-alpha test against a particular alternative is

(A) always equal to alpha
(B) the p-value
(C) P(Type I error)
(D) 1 - P(Type II error | that alternative)

Q.56 MCQ · 2 marks · P&S; / counting

From a standard 52-card deck, a 5-card hand is dealt. The number of hands that are flushes (all five cards of one suit, including straight flushes) is

(A) 5148
(B) 649740.0
(C) 0
(D) 5108

Q.57 MCQ · 1 mark · P&S; / axioms

For events A,B with P(A)=0.6, P(B)=0.7, the largest possible value of P(A n B) is

(A) 0.6
(B) 0.42
(C) 1.3
(D) 0.7

Q.58 MCQ · 1 mark · P&S; / independence

If A and B are mutually exclusive and P(A) > 0, P(B) > 0, then P(A|B) equals

(A) 0
(B) P(A)/P(B)
(C) 1
(D) P(A)

Q.59 MCQ · 2 marks · P&S; / joint

Random variables X,Y have joint PMF p(x,y)=c*(x+y) for x=1,2 and y=1,2,3. The normalising constant c equals

(A) 1/18
(B) 1/21
(C) 1/12
(D) 1/9

Q.60 NAT · 2 marks · P&S; / Bayes

A rare plant disease affects 2% of a greenhouse. A leaf test detects the disease with probability 0.95 if present and falsely flags 5% of healthy plants. Given a positive test, P(diseased) equals _____ (rounded to two decimals).

NAT · enter the numerical value

Q.61 MCQ · 1 mark · P&S; / cond-exp

X is Poisson(lambda). Then E[X(X-1)] equals

(A) lambda^2 - lambda
(B) lambda^2
(C) lambda^2 + lambda
(D) lambda

Q.62 MCQ · 1 mark · P&S; / descriptive

For a unimodal right-skewed histogram of positive waiting times, which ordering is typical?

(A) mode < median < mean
(B) mean < median < mode
(C) mode < mean < median always fails
(D) mean = median = mode

Q.63 MCQ · 1 mark · P&S; / covariance

X~Uniform(-1,1) and Y=X^2. Then Corr(X,Y) equals

(A) 0
(B) 1/2
(C) 1
(D) -1

Q.64 MCQ · 1 mark · P&S; / discrete

A random variable X with PMF p(x)= (1-q) q^x for x=0,1,2,... (0 < q < 1) is

(A) Binomial with n=inf
(B) Poisson(q)
(C) Uniform
(D) Geometric starting at 0 with success probability 1-q

Q.65 NAT · 2 marks · P&S; / binomial

A packet of 8 seeds germinates independently with probability 0.5. P(exactly 3 germinate) equals C(8,3)/256. The probability as a simplified fraction C(8,3)/256 equals, in decimal, _____ (rounded to three decimals).

NAT · enter the numerical value

Q.66 NAT · 1 mark · P&S; / Bernoulli

A bit-flip channel writes a bit correctly with probability 0.9 (Bernoulli success). The variance of a single written bit-error indicator (1 if error) is _____ (two decimals).

NAT · enter the numerical value

Q.67 MCQ · 1 mark · P&S; / Poisson

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Rare independent events in 50 trials with success
probability 0.02 are approximated by Poisson(lambda)
with lambda equal to
```

(A) 50
(B) 0.02
(C) 1
(D) 2.5

Q.68 MCQ · 2 marks · P&S; / continuous

The CDF of X is 0 for x < 0, x^2 for 0 < =x < =1, and 1 for x > 1. Then P(0.25 < X < = 0.75) equals

(A) 0.5625
(B) 0.5
(C) 0.75
(D) 0.25

Q.69 MSQ · 2 marks · P&S; / uniform

X~Uniform(0,2). Which are true?

(A) P(X=1)=1/2
(B) E[X]=1
(C) f_X(x)=1/2 for x in (0,2)
(D) Var(X)=(2-0)^2/12=1/3

Q.70 MCQ · 2 marks · P&S; / exponential

X~Exponential with mean 1/ln 2. Let Y=floor(X). Then P(Y=ell) = (1-q) q^ell for ell=0,1,2,... The value of q is

(A) 2
(B) 1/2
(C) 1/4
(D) ln 2

Q.71 MCQ · 1 mark · P&S; / normal

X~N(mu, sigma^2), Z=(X-mu)/sigma. Then P(|X-mu| < sigma) equals

(A) 1/2
(B) 2Phi(sigma)-1
(C) Phi(1)
(D) P(|Z| < 1) = 2Phi(1)-1

Q.72 MSQ · 2 marks · P&S; / chi-squared

Let X1,...,Xn (n > =2) be i.i.d. N(0,1) and Xbar the sample mean. Which statements are correct?

(A) sum_{i=1}^n X_i^2 ~ chi^2_n
(B) X1^2 + X2^2 ~ Exponential with mean 2
(C) (sqrt(n) Xbar)^2 ~ chi^2_2
(D) sum_{i=1}^n (X_i - Xbar)^2 ~ chi^2_{n-1}

Q.73 MCQ · 1 mark · P&S; / t-dist

Student's t with one degree of freedom is equal in law to which named distribution?

(A) Uniform(-1,1)
(B) Exponential(1)
(C) standard Cauchy
(D) chi^2_1

Q.74 MCQ · 2 marks · P&S; / cond-pdf

Given Y=y, X is N(y, 1), and Y is N(0,1). Then the joint of (X,Y) is

(A) degenerate on X=Y
(B) independent N(0,1)
(C) X~N(0,1) independent of Y
(D) bivariate normal with Cov(X,Y)=1

Q.75 MCQ · 1 mark · P&S; / CLT

The Lindeberg-Lévy CLT requires which of the following as a sufficient set of hypotheses?

(A) i.i.d. Cauchy
(B) only uncorrelated, no moment assumption
(C) finite sample size n=2
(D) i.i.d. with finite positive variance

Q.76 MCQ · 2 marks · P&S; / CI

An approximate large-sample 95% CI for a proportion p-hat=0.4 based on n=100 uses se=sqrt(0.4*0.6/100)=0.049. Using 1.96 the interval is about

(A) (0.35, 0.45)
(B) (0.40, 0.50)
(C) (0.30, 0.50)
(D) (0,1)

Q.77 MCQ · 2 marks · P&S; / z-test

A z-test of H0: p=0.5 vs H1: p > 0.5 with n=100, p-hat=0.58 has z=(0.58-0.5)/sqrt(0.5*0.5/100)=1.6. Using z_{0.05}=1.645, at 5% one-sided we

(A) reject because p-hat > 0.5
(B) reject H0
(C) do not reject H0 (z=1.6 < 1.645)
(D) cannot compute se

Q.78 MCQ · 1 mark · P&S; / t-test

A two-sample unpaired t-test with equal-variance assumption uses a pooled variance estimator and df equal to

(A) min(n1,n2)
(B) n1+n2-2
(C) n1+n2
(D) n1-1 only

Q.79 MCQ · 1 mark · P&S; / chi2-test

Pearson chi^2 for a 2x2 table can be written n(ad-bc)^2 / ((a+b)(c+d)(a+c)(b+d)). If a=b=c=d=10, the statistic equals

(A) 1
(B) n
(C) 0
(D) 40

Q.80 MCQ · 1 mark · P&S; / mixed

A fair six-sided die is rolled 4 times independently. P(at least one six) equals

(A) (5/6)^4
(B) 4/6
(C) 1
(D) 1-(5/6)^4

Q.81 MSQ · 2 marks · P&S; / tests

Which statements are correct about GATE-style hypothesis tests?

(A) z-tests require a known sigma (or a large-sample plug-in)
(B) A chi^2 GoF statistic is always exactly chi^2 even for n=5 sparse cells
(C) Failing to reject H0 does not prove H0 is true
(D) A small p-value is evidence against H0 (under the modelling assumptions)

Q.82 MCQ · 2 marks · P&S; / counting

A reading list is formed by choosing 3 books from 5 distinct ML texts, 4 distinct systems texts, and 3 distinct theory texts, with at least one ML text. The number of lists (order does not matter) is

(A) 185
(B) 105
(C) 220
(D) 10

Q.83 MCQ · 1 mark · P&S; / axioms

Consider statements about a probability measure P on a sample space Omega. (i) P(empty set)=0. (ii) If A subset B then P(A) < = P(B). Which option is correct?

(A) Both (i) and (ii) are true
(B) (i) true, (ii) false
(C) Both false
(D) (i) false, (ii) true

Q.84 MCQ · 1 mark · P&S; / independence

Let A,B be independent. Which one is NOT necessarily true?

(A) A^c and B are independent
(B) A and B^c are independent
(C) P(A u B)=P(A)+P(B)
(D) A^c and B^c are independent

Q.85 MCQ · 1 mark · P&S; / joint

X and Y are independent. X is uniform on {1,2} and Y is uniform on {0,1}. Then P(X+Y=2) equals

(A) 1/4
(B) 1/8
(C) 3/8
(D) 1/2

Q.86 NAT · 2 marks · P&S; / Bayes

In a mailbox, 30% of messages are spam. A filter tags spam with probability 0.90 and tags ham as spam with probability 0.05. Given a tagged message, P(spam) equals _____ (two decimals).

NAT · enter the numerical value

Q.87 MCQ · 2 marks · P&S; / cond-exp

Let Z = a + bX + error with E[error]=0 and E[error(X-E[X])]=0. Given E[X]=1, Var(X)=4, Cov(Z,X)=-2 and E[Z]=3, the pair (a,b) is

(A) a=2, b=-1
(B) a=3, b=-2
(C) a=3.5, b=-0.5
(D) a=-2, b=1

Q.88 MCQ · 2 marks · P&S; / descriptive

Five integers from 0 to 20 (repetitions allowed) have mean 10, median 12, and a unique mode 14. Which statement is TRUE?

(A) All five integers must equal 10
(B) The unique mode must be the median
(C) After sorting x1 < =x2 < =x3 < =x4 < =x5 we must have x3=12, the unique mode 14 appears at least twice, and the mean constraint is x1+...+x5=50
(D) Such a sample is impossible because mode 14 > median 12

Q.89 MCQ · 2 marks · P&S; / covariance

X~Uniform(-1,1). Given X=x, Y is Uniform(x^2-0.2, x^2+0.2). Then Corr(X,Y) equals

(A) -1
(B) 1
(C) 0
(D) 0.2

Q.90 MCQ · 1 mark · P&S; / discrete
Which of the following is a valid PMF on {0,1,2}?
(A) p=(0.2, 0.3, 0.6)
(B) p=(0.2, 0.3, 0.5)
(C) p=(-0.1, 0.5, 0.6)
(D) p=(0.2, 0.2, 0.2)

Q.91 NAT · 2 marks · P&S; / binomial
A basketball player hits a free throw with probability 0.6 independently. In 5 attempts, P(exactly 4 hits) equals _____ (rounded to three decimals).
NAT · enter the numerical value

Q.92 MSQ · 2 marks · P&S; / Poisson
Let N~Poisson(lambda) with lambda > 0. Which are true?
(A) E[N(N-1)]=lambda^2
(B) The MGF is exp(lambda(e^t - 1))
(C) N takes values in (0, inf) with a density
(D) P(N=0)=e^{-lambda}

Q.93 MCQ · 1 mark · P&S; / continuous
If F is the CDF of a continuous r.v. with density f, then almost everywhere
(A) F'(x)=f(x)
(B) F(x)=f(x)
(C) f'(x)=F(x)
(D) F(x)=1-f(x)

Q.94 NAT · 1 mark · P&S; / uniform
Two independent Uniform(0,1) r.v.s U,V. P(U+V < =1) equals _____ (one decimal).
NAT · enter the numerical value

Q.95 MSQ · 2 marks · P&S; / exponential
Let X~Exponential(rate=lambda). Which are true?
(A) The MGF is lambda/(lambda-t) for t < lambda
(B) P(X=0)=lambda
(C) E[X^2]=2/lambda^2
(D) P(X > s+t | X > s)=P(X > t) for s,t > 0

Q.96 MCQ · 1 mark · P&S; / normal
Let Z1,Z2,Z3 be i.i.d. N(0,1) and W=Z1^2+Z2^2+Z3^2. Then Var(W) equals
(A) 6
(B) 9
(C) 2
(D) 3

Q.97 NAT · 1 mark · P&S; / chi-squared
If Z~N(0,1) then P(Z^2 < = 3.84) with 3.84 ~ chi^2_{1, 0.95} quantile is approximately _____ (two decimals).
NAT · enter the numerical value

Q.98 MCQ · 1 mark · P&S; / t-dist
A one-sample t-interval for mu (unknown sigma, Normal data) uses critical values from
(A) F_{1,1}
(B) t_{n-1}
(C) N(0,1) even for n=3
(D) chi^2_n

Q.99 NAT · 2 marks · P&S; / cond-pdf
Joint density f(x,y)=e^{-x-y} for x > 0,y > 0. Then f_{X|Y}(x|y) for x > 0 equals e^{-x} (does not depend on y). The value of f_{X|Y}(1|5) rounded to two decimals is _____.
NAT · enter the numerical value

Q.100 MCQ · 2 marks · P&S; / CLT
A de Moivre-Laplace application: n=16 independent Bernoulli(1/2). Approximate P(S=8) using a continuity correction as P(7.5 < S* < 8.5) where S*~N(8,4). This equals Phi((8.5-8)/2)-Phi((7.5-8)/2) = Phi(0.25)-Phi(-0.25). Using Phi(0.25)~0.5987 the approximation rounded to two decimals is
(A) 0.50
(B) 0.08
(C) 0.20
(D) 0.40

Q.101 MCQ · 1 mark · P&S; / CI
Which statement about a 95% CI for mu is correct (repeated sampling interpretation)?
(A) About 95% of such random intervals contain the true mu
(B) Width does not depend on n
(C) The interval always contains Xbar at an endpoint
(D) P(mu in this already-computed interval)=0.95 as a posterior

Q.102 MSQ · 2 marks · P&S; / z-test
n=100, sigma=10, Xbar=103, H0: mu=100 vs H1: mu != 100, alpha=0.05, z_crit=1.96. Which are true?
(A) An approximate p-value is 2(1-Phi(3)) ~ 0.0027
(B) The 95% z-interval for mu is centred at 100
(C) z = (103-100)/(10/10) = 3
(D) The two-sided test rejects H0

Q.103 NAT · 1 mark · P&S; / t-test
n=9, Xbar=4, s=3, H0: mu=2. The t statistic equals _____ (integer).
NAT · enter the numerical value

Q.104 MSQ · 2 marks · P&S; / chi2-test
A Pearson chi-squared test of independence is run on an r x c table with large n. Which are true?
(A) The test requires the two classification variables to be Normal
(B) Expected cell counts use the product of fitted marginals
(C) Degrees of freedom equal (r-1)(c-1)
(D) A large statistic (with small p) is evidence against independence

Q.105 MCQ · 2 marks · P&S; / mixed
Matching: Column P = memoryless continuous; Q = mean=variance; R = limits of t_n. Column 2: 1 Exponential, 2 Poisson, 3 N(0,1). The correct match is
(A) P-1, Q-2, R-3
(B) P-2, Q-1, R-3
(C) P-3, Q-2, R-1
(D) P-1, Q-3, R-2

Q.106 MSQ · 2 marks · P&S; / tests

In a level-alpha two-sided z-test of H0: mu=mu0 (sigma known), which are true?

(A) Failing to reject H0 proves mu=mu0
(B) Power is a function of the true mu, n, sigma and alpha
(C) Type I error probability is alpha when H0 is true (under the Normal/CLT model)
(D) p-value < alpha is the same decision as rejecting at level alpha

Q.107 MCQ · 2 marks · P&S; / counting

Let S be a 4-element set. The number of functions f: S - > S that are bijections AND satisfy f(f(x))=x for every x in S (involutions that are permutations) is

(A) 24
(B) 16
(C) 10
(D) 4

Q.108 MCQ · 1 mark · P&S; / axioms

Two events A and B satisfy P(A u B)=P(A)+P(B). This equality holds if and only if

(A) A subset B
(B) P(A n B)=0
(C) P(A)=P(B)
(D) A and B are independent

Q.109 MCQ · 2 marks · P&S; / independence

A fair six-sided die is rolled twice independently. Let D be the event that the sum is 7 and E the event that the first roll is 3. Then P(D n E) equals

(A) 1/12
(B) 1/36
(C) 1/6
(D) 0

Q.110 MCQ · 1 mark · P&S; / joint

For a joint density f_{X,Y}(x,y), the conditional density f_{X|Y}(x|y) equals (when f_Y(y) > 0)

(A) f_{X,Y}(x,y)/f_Y(y)
(B) F_X(x)
(C) f_X(x)
(D) f_Y(y)

Q.111 NAT · 2 marks · P&S; / Bayes

Plant A makes 70% of chips (defect rate 1%); plant B makes the rest (defect rate 4%). A randomly chosen chip is defective. P(it came from plant B) equals _____ (two decimals).

NAT · enter the numerical value

Q.112 MCQ · 2 marks · P&S; / cond-exp

If E[X|Y]=c almost surely for a constant c, then

(A) E[X]=c and Cov(X,Y)=0
(B) X=c almost surely
(C) Var(X)=0
(D) X is independent of Y necessarily

Q.113 NAT · 2 marks · P&S; / descriptive

A data set of n=5 values has sum 20 and sum of squares 100. The population variance (divide by n) is _____.

NAT · enter the numerical value

Q.114 MCQ · 1 mark · P&S; / covariance

The correlation matrix of a non-degenerate random vector must be

(A) symmetric positive semidefinite with ones on the diagonal
(B) orthogonal
(C) skew-symmetric
(D) diagonal with possibly negative entries

Q.115 MSQ · 2 marks · P&S; / discrete

Let X be integer-valued with CDF F. Which statements are correct?

(A) P(X=k)=F(k)-F(k-1) for integer k
(B) F may have jump discontinuities
(C) F is left-continuous
(D) F is non-decreasing

Q.116 NAT · 2 marks · P&S; / binomial

A 10-question MCQ is guessed independently with 4 options each. P(at least one correct) = 1-(3/4)^10. This value rounded to two decimals is _____.

NAT · enter the numerical value

Q.117 NAT · 1 mark · P&S; / Poisson

Calls arrive as Poisson with mean 3 per minute. P(no call in a given minute) equals e^{-3}. Rounded to three decimals this is _____.

NAT · enter the numerical value

Q.118 MSQ · 2 marks · P&S; / continuous

Let f be a PDF on R. Which properties MUST hold?

(A) int_{-inf}^{inf} f(x) dx = 1
(B) f is continuous
(C) f(x) > = 0 for (almost) all x
(D) f(x) < = 1 for all x

Q.119 MSQ · 2 marks · P&S; / exponential

Waiting times W1, W2 are i.i.d. Exponential(rate=1). Which are true?

(A) max(W1,W2) ~ Exponential(rate=1)
(B) W1+W2 ~ Gamma(shape=2, rate=1) (Erlang-2)
(C) min(W1,W2) ~ Exponential(rate=2)
(D) P(W1 < W2)=1/2

Q.120 MCQ · 1 mark · P&S; / normal

Independent X~N(1,4) and Y~N(2,9). Then X+Y is

(A) N(3, 5)
(B) not normal
(C) N(3, 13)
(D) N(3, 36)

Q.121 MSQ · 2 marks · P&S; / t-dist

Which are true for Student's t_nu as nu - > infinity?

(A) t_nu becomes Uniform(-1,1)
(B) t_nu converges in distribution to N(0,1)
(C) the variance of t_nu (for nu > 2) tends to 1
(D) P(|t_nu| > 2) tends to P(|Z| > 2) for Z~N(0,1)

Q.122 NAT · 2 marks · P&S; / cond-pdf

Joint density f(x,y)=8xy on 0 < y < x < 1. The conditional f_{Y|X}(y|x) for 0 < y < x is 2y/x^2. Hence f_{Y|X}(0.2|0.5) equals _____.

NAT · enter the numerical value

Q.123 MCQ · 1 mark · P&S; / CLT
Which quantity is asymptotically N(0,1) under i.i.d. finite-variance sampling, with mu, sigma known?
(A) S^2 / sigma^2
(B) sqrt(n) (Xbar - mu)/sigma
(C) Xbar - mu
(D) n (Xbar - mu)

Q.124 MCQ · 1 mark · P&S; / CI
If a 95% CI for mu is (2.1, 3.7), then a two-sided t-test of H0: mu=3 at level 5%
(A) does not reject H0
(B) is inapplicable because 3 is an integer
(C) rejects H0
(D) has p-value exactly 0.05

Q.125 NAT · 1 mark · P&S; / z-test
A one-sided z-test has z=1.64. Using Phi(1.64)~0.9495, the p-value P(Z > 1.64) rounded to two decimals is _____.
NAT · enter the numerical value

Q.126 NAT · 2 marks · P&S; / chi2-test
A die is rolled 60 times. Under fairness E_i=10 for each face. Observed counts 8,12,9,11,7,13. The Pearson chi^2 statistic sum (O-E)^2/E equals _____ (one decimal).
NAT · enter the numerical value

Q.127 MCQ · 1 mark · P&S; / mixed
Let phi be convex and X integrable so that E[phi(X)] exists. Jensen's inequality states that
(A) phi(E[X]) > = E[phi(X)] always
(B) Var(X) < =0
(C) E[X^2] < = (E[X])^2
(D) phi(E[X]) < = E[phi(X)]

Q.128 MCQ · 2 marks · P&S; / counting
There are 6 distinct courses. The number of ways to assign each course a slot in {morning, afternoon, evening} so that every slot is used at least once is
(A) 540
(B) 20
(C) 729
(D) 216

Q.129 MCQ · 2 marks · P&S; / axioms
A die with faces {1,2,3,4,5,6} is loaded so that P(face k) is proportional to k. Then P(even face) equals
(A) 9/21
(B) 1/2
(C) 6/21
(D) 12/21

Q.130 MCQ · 1 mark · P&S; / independence
Consider (i) mutually exclusive events with positive probability are dependent; (ii) Cov(X,Y)=0 always implies X,Y independent. Which is correct?
(A) both false
(B) both true
(C) (i) true, (ii) false
(D) (i) false, (ii) true

Q.131 MCQ · 1 mark · P&S; / joint
X,Y have joint density f(x,y)=1 on the unit square (0,1)^2. Then X and Y are
(A) uniform but dependent
(B) neither uniform nor independent
(C) independent but not uniform
(D) independent Uniform(0,1)

Q.132 NAT · 2 marks · P&S; / Bayes
Tomorrow rains with probability 0.2. A forecast says 'rain' with probability 0.8 if it will rain and with probability 0.1 if it will not. Given a 'rain' forecast, P(rain) equals _____ (two decimals).
NAT · enter the numerical value

Q.133 MCQ · 2 marks · P&S; / cond-exp
A biased coin with P(H)=1/3 is tossed independently until two consecutive heads appear. The expected number of tosses is
(A) 6
(B) 12
(C) 3
(D) 9

Q.134 NAT · 1 mark · P&S; / descriptive
The interquartile range of the ordered sample 2,4,6,8,10,12,14 is Q3-Q1. With Q1=4 and Q3=12 (seven-point convention using medians of halves excluding the overall median), IQR equals _____.
NAT · enter the numerical value

Q.135 MCQ · 2 marks · P&S; / covariance
Let X,Y be independent, X~Bernoulli(0.4), Y~Normal(0, 25). Then Var((2X-1)Y) equals
(A) 9
(B) 25
(C) 16
(D) 10

Q.136 MSQ · 2 marks · P&S; / discrete
X takes values in {0,1,2} with p=(0.2,0.5,0.3). Which are true?
(A) P(X=1.7)=0.7
(B) F(1.7)=P(X < =1.7)=0.7
(C) E[X]=1.1
(D) The mode is 1

Q.137 NAT · 2 marks · P&S; / binomial
Four independent servers fail with probability 0.1 each. P(exactly two fail) equals _____ (rounded to four decimals).
NAT · enter the numerical value

Q.138 NAT · 1 mark · P&S; / Poisson
A Poisson process has rate 5 per hour. The expected waiting time (in hours) until the first event is _____ (one decimal).
NAT · enter the numerical value

Q.139 NAT · 2 marks · P&S; / continuous
f(x)=3x^2 on (0,1). Then Var(X) equals _____ (four decimals).
NAT · enter the numerical value

Q.140 NAT · 2 marks · P&S; / exponential

X is Exponential with mean 5. If P(X > 5)=e^{-1}, then P(X > 10 | X > 5) equals _____ (rounded to two decimals).

NAT · enter the numerical value

Q.141 MCQ · 1 mark · P&S; / normal

The 68-95-99.7 rule says that about 95% of N(mu,sigma^2) mass lies in

(A) [mu-sigma, mu+sigma]
(B) [mu-3sigma, mu+sigma]
(C) [mu-2sigma, mu+2sigma]
(D) [0,1]

Q.142 MSQ · 2 marks · P&S; / t-dist

A one-sample t-test is applied to n=9 allegedly i.i.d. Normal observations. Which are true?

(A) Under H0 the t statistic has 8 degrees of freedom
(B) The t critical value is smaller than the z critical value for the same tail probability
(C) If sigma were known one would use a z-test instead
(D) Heavy-tailed true data (not Normal, small n) make the nominal 5% level untrustworthy

Q.143 NAT · 2 marks · P&S; / cond-pdf

X~Uniform(0,1), Y|X=x ~ Uniform(0,x). Then E[Y] equals _____ (two decimals).

NAT · enter the numerical value

Q.144 MSQ · 2 marks · P&S; / CLT

Let X1,...,Xn be i.i.d. with mean mu and finite variance sigma^2 > 0. Which are true for large n?

(A) Xbar is approximately N(mu, sigma^2/n)
(B) sqrt(n)(Xbar-mu)/sigma is approximately N(0,1)
(C) Xbar converges in probability to mu (WLLN)
(D) Xbar is exactly t_{n-1} for every n, even if the Xi are Uniform

Q.145 MSQ · 2 marks · P&S; / CI

n i.i.d. N(mu, sigma^2) with sigma known. A 95% z-interval I_n = Xbar +/- 1.96 sigma/sqrt(n). Which are true?

(A) P_mu(I_n contains mu)=0.95 for every n > =1
(B) Replacing 1.96 by 2.58 produces a 99% interval of the same centre
(C) The interval is a posterior 95% credible set under every prior
(D) The length of I_n tends to 0 as n tends to inf

Q.146 NAT · 1 mark · P&S; / chi2-test

A 2x3 contingency table (independence, no estimated extra parameters beyond margins) has chi^2 degrees of freedom equal to _____.

NAT · enter the numerical value

Q.147 MCQ · 1 mark · P&S; / mixed

Assume the MGF M(t)=E[e^{tX}] exists in a neighbourhood of 0. Then M'(0) equals

(A) Var(X)
(B) E[X]
(C) 1
(D) E[X^2]

Q.148 MCQ · 1 mark · P&S; / counting

A fair coin is tossed until the first head appears, but at most 4 tosses are made. The size of the sample space if an outcome is recorded as a finite string over {H,T} (stopping at H or at 4 tosses) is

(A) 8
(B) 16
(C) 5
(D) 4

Q.149 MSQ · 2 marks · P&S; / axioms

Let A,B be events in a probability space. Which of the following are ALWAYS true?

(A) P(A n B) = P(A)P(B)
(B) P(A u B) = P(A)+P(B)-P(A n B)
(C) P(A^c) = 1-P(A)
(D) P(A u B) < = P(A)+P(B)

Q.150 MSQ · 2 marks · P&S; / independence

A and B are independent with 0 < P(A),P(B) < 1. Which statements MUST be true?

(A) P(A n B)=P(A)P(B)
(B) P(A u B)=P(A)+P(B)-P(A)P(B)
(C) P(A|B)=P(A)
(D) A and B are disjoint

Q.151 MCQ · 2 marks · P&S; / joint

A joint PMF is given by the table X\Y | 0 1 2 and rows X=0: 0.05, 0.10, 0.15; X=1: 0.10, 0.20, 0.40. Which statement is TRUE?

(A) P(Y=2 | X=1) = 0.40/0.70
(B) The table already fails to be a PMF because the entries sum to more than 1
(C) X and Y are independent because no cell is zero
(D) P(X=0)=0.40

Q.152 NAT · 2 marks · P&S; / Bayes

Three drawers: D1 has 3 gold 1 silver coins; D2 has 1 gold 3 silver; D3 has 2 gold 2 silver. A drawer is chosen uniformly. A gold coin is drawn. P(drawer was D1 | gold) equals _____ (two decimals).

NAT · enter the numerical value

Q.153 MSQ · 2 marks · P&S; / cond-exp

X,Y square-integrable. Which identities always hold?

(A) Var(X)=E[Var(X|Y)]+Var(E[X|Y])
(B) E[X|Y]=E[X] almost surely even when X and Y are dependent
(C) E[E[X|Y]]=E[X]
(D) E[X g(Y)]=E[E[X|Y] g(Y)] for bounded g

Q.154 NAT · 1 mark · P&S; / descriptive

Five exam scores have mean 12. After a sixth score 18 is included, the new mean is _____ (one decimal).

NAT · enter the numerical value

Q.155 MSQ · 2 marks · P&S; / covariance

Which claims are correct?

(A) Cov(X,X)=0
(B) |Corr(X,Y)| < = 1 whenever the correlation is defined
(C) Var(X+Y) < = (sd X + sd Y)^2 always (finite variances)
(D) Corr(aX+b, cY+d) = sign(ac) Corr(X,Y) for a,c != 0

Q.156 NAT · 2 marks · P&S; / discrete
X has PMF P(X=-1)=0.2, P(X=0)=0.3, P(X=2)=0.5. Then Var(X) equals _____ (two decimals).
NAT · enter the numerical value

Q.157 NAT · 2 marks · P&S; / binomial
A fair die is rolled 5 times. P(exactly two sixes) equals _____ (rounded to three decimals).
NAT · enter the numerical value

Q.158 NAT · 1 mark · P&S; / continuous
Density f(x)=2x on (0,1). The median m satisfies F(m)=1/2, so m equals _____ (rounded to two decimals).
NAT · enter the numerical value

Q.159 NAT · 1 mark · P&S; / exponential
Two independent Exponential(rate=1) clocks. The expected time until the first rings is _____ (one decimal).
NAT · enter the numerical value

Q.160 MCQ · 2 marks · P&S; / normal
Z~N(0,1) with pdf g and CDF G; Y~t_1 with pdf h and CDF H. Let c > 0 satisfy g(c)=h(c). Which statement is TRUE?
(A) G(0)=H(0)=1/2 because both densities are even
(B) h is the N(0,1) density
(C) t_1 has lighter tails than N(0,1)
(D) G is the CDF of a Uniform(0,1)

Q.161 NAT · 1 mark · P&S; / t-dist
Var of t_nu for nu=4 equals nu/(nu-2) = _____.
NAT · enter the numerical value

Q.162 NAT · 2 marks · P&S; / cond-pdf
Bivariate density f(x,y)= (x+y)/8 on 0 < x < 2, 0 < y < 2. The marginal f_X(1) equals int_0^2 (1+y)/8 dy = _____ (three decimals).
NAT · enter the numerical value

Q.163 NAT · 2 marks · P&S; / CLT
n=25 i.i.d. Uniform(0,12), which have variance 12. Then sd of Xbar is _____ (two decimals).
NAT · enter the numerical value

Q.164 NAT · 2 marks · P&S; / CI
n=16, sigma=8, Xbar=50, z_{0.025}=1.96. Half-width of the 95% z-interval is 1.96*8/4 = _____ (two decimals).
NAT · enter the numerical value

Q.165 MCQ · 2 marks · P&S; / mixed
Two independent Exp(1) r.v.s X,Y. Then X/(X+Y) is distributed as
(A) Uniform(0,1)
(B) N(0,1)
(C) Exp(1)
(D) Beta(2,1)

Q.166 MSQ · 2 marks · P&S; / counting
Let S be a finite n-set. Which counting formulae are correct?
(A) The number of 2-element sequences with repetition from S is C(n,2)
(B) The number of k-subsets is C(n,k)
(C) The number of permutations of S is n!
(D) The number of injections from a k-set to S is P(n,k)=n!/(n-k)! for k < =n

Q.167 MSQ · 2 marks · P&S; / axioms
Let A,B be events with 0 < P(A) < 1 and 0 < P(B) < 1. Which statements MUST be true?
(A) P(A n B) < = min(P(A), P(B))
(B) P(A|B) = P(B|A)
(C) P(A|B) + P(A^c|B) = 1
(D) P(A u B) > = max(P(A), P(B))

Q.168 MSQ · 2 marks · P&S; / independence
Which constructions yield independent events (assuming fair independent tosses of two coins)?
(A) A = first heads, B = coins show equal faces
(B) A = empty event, B = any event
(C) A = first heads, B = second heads
(D) A = first heads, B = at least one head (these are dependent)

Q.169 MCQ · 1 mark · P&S; / joint
Let f_{X,Y}(x,y)=e^{-x-y} for x > 0, y > 0. Then X and Y are
(A) dependent exponentials
(B) i.i.d. Exponential(1)
(C) jointly uniform
(D) i.i.d. Normal

Q.170 NAT · 2 marks · P&S; / Bayes
A box contains two coins: one fair and one two-headed. A coin is picked at random and flipped; it shows heads. P(the coin is two-headed | heads) equals _____ (two decimals).
NAT · enter the numerical value

Q.171 NAT · 2 marks · P&S; / cond-exp
Y is Uniform on {1,2,3}. Given Y=y, X is Uniform on {1,...,y} (discrete). Then E[X] equals _____ (rounded to two decimals).
NAT · enter the numerical value

Q.172 NAT · 1 mark · P&S; / descriptive
For {4, 8, 10, 12, 16}, the population standard deviation (divide by n, not n-1) equals _____ (two decimals).
NAT · enter the numerical value

Q.173 MSQ · 2 marks · P&S; / covariance
X,Y have finite positive variances. Which are true?
(A) Corr(X,Y)=0 iff X and Y are independent
(B) Corr(X,Y)=1 iff Y=a+bX a.s. for some b > 0
(C) Corr(X,Y)=Cov(X,Y)/(sdX sdY)
(D) |Corr(X,Y)| < =1

Q.174 NAT · 2 marks · P&S; / discrete
X ~ Geometric(p=1/3) on {1,2,...}. Then P(X > 4 | X > 2) equals P(X > 2) by memorylessness, and P(X > 2)=(2/3)^2. The conditional probability equals _____ (rounded to two decimals).
NAT · enter the numerical value

Q.175 NAT · 2 marks · P&S; / binomial
Eight emails, each independently spam with probability 1/4. P(exactly 2 spam) equals _____ (rounded to four decimals).
NAT · enter the numerical value

Q.176 MSQ · 2 marks · P&S; / normal
Let X~N(0,1). Which are true?
(A) P(X > 0)=0
(B) P(X > a)=P(X < -a) for all a
(C) The density is even: f(x)=f(-x)
(D) E[X^3]=0

Q.177 NAT · 1 mark · P&S; / cond-pdf
Let Y ~ N(0,1) and, given Y=y, let X ~ N(y, 1). Then Var(X) equals _____.
NAT · enter the numerical value

Q.178 NAT · 1 mark · P&S; / CLT
Sum S_n of n i.i.d. with mean 0 variance 1. Approximate P(S_n / sqrt(n) < = 0) for large n is _____ (one decimal).
NAT · enter the numerical value

Q.179 NAT · 2 marks · P&S; / CI
To have a 95% z-interval half-width at most 0.5 when sigma=2 and z=1.96, the smallest integer n satisfying n > = (1.96*2/0.5)^2 is _____.
NAT · enter the numerical value

Q.180 MCQ · 1 mark · P&S; / mixed
A Poisson process of rate 2 is observed for time 3. The count N(3) is distributed as
(A) Poisson(6)
(B) Exp(2)
(C) Binomial(3,2)
(D) Poisson(2)

Q.181 NAT · 2 marks · P&S; / counting
The number of onto (surjective) functions from a 4-element feature-index set to a 3-element label set {red, green, blue} is _____.
NAT · enter the numerical value

Q.182 NAT · 1 mark · P&S; / axioms
Three events A,B,C are pairwise disjoint with P(A)=0.1, P(B)=0.25, P(C)=0.15. Then P(A u B u C) equals _____ (two decimals).
NAT · enter the numerical value

Q.183 MSQ · 2 marks · P&S; / independence
A fair coin is tossed twice independently. Let H1 = first toss heads, Eq = both tosses equal, Xor = tosses differ. Which of the following pairs are independent?
(A) H1 and (H1 n Eq)
(B) Eq and Xor
(C) H1 and Xor
(D) H1 and Eq

Q.184 MCQ · 1 mark · P&S; / joint
A survey of 80 students yields the counts (laptop, yes-coffee)=30, (laptop, no-coffee)=10, (no-laptop, yes-coffee)=20, (no-laptop, no-coffee)=20. P(coffee | laptop) equals
(A) 50/80
(B) 30/40
(C) 30/80
(D) 10/40

Q.185 NAT · 2 marks · P&S; / Bayes
A server crash is due to hardware with prior 0.3, otherwise software. A diagnostic flags hardware with probability 0.9 if hardware and 0.2 if software. Given a hardware flag, P(true hardware fault) equals _____ (two decimals).
NAT · enter the numerical value

Q.186 NAT · 2 marks · P&S; / cond-exp
A fair die is rolled. If the face is even, a fair coin is tossed once and X=number of heads; if odd, the coin is tossed twice and X=number of heads. E[X] equals _____ (two decimals).
NAT · enter the numerical value

Q.187 NAT · 1 mark · P&S; / descriptive
z-score of a value 82 when the attribute has mean 70 and sd 8 equals _____ (one decimal).
NAT · enter the numerical value

Q.188 NAT · 2 marks · P&S; / covariance
Var(X)=4, Var(Y)=9, Corr(X,Y)=0.5. Then Var(2X-Y) equals _____.
NAT · enter the numerical value

Q.189 NAT · 1 mark · P&S; / discrete
X takes values 1,2,3,4 each with probability 1/4. Then P(X < = 2.5) equals _____ (two decimals).
NAT · enter the numerical value

Q.190 NAT · 2 marks · P&S; / binomial
A batter gets a hit with probability 0.3 independently in 8 at-bats. P(at least one hit)=1-(0.7)^8. Rounded to two decimals this is _____.
NAT · enter the numerical value

Q.191 MSQ · 2 marks · P&S; / normal
X~N(2, 16) and Z=(X-2)/4. Which are true?
(A) Z ~ N(0,1)
(B) Var(X+Z)=16+1 because X and Z are independent
(C) P(|X-2| < 4)=P(|Z| < 1)
(D) P(X > 2)=1/2

Q.192 NAT · 2 marks · P&S; / CI
n=9 Normal observations, s=3, t_{8, 0.025}=2.306. Half-width of the 95% t-interval is 2.306*s/sqrt(n) = _____ (two decimals).
NAT · enter the numerical value

Q.193 MCQ · 1 mark · P&S; / mixed
Let I_A be the 0-1 indicator of an event A on a probability space. Then E[I_A] equals
(A) P(A)
(B) 1-P(A) only
(C) Var(A)
(D) P(A)^2

Q.194 NAT · 2 marks · P&S; / counting
Five boys and four girls sit in a row so that no two girls are adjacent. The number of seating arrangements of these 9 distinct people is _____.
NAT · enter the numerical value

Q.195 NAT · 1 mark · P&S; / axioms
A sample space has 8 equally likely outcomes. Event A contains 3 of them. P(A^c) equals _____ (three decimals).
NAT · enter the numerical value

Q.196 NAT · 1 mark · P&S; / independence
Three independent alarms each fail with probability 0.2. The probability that at least one works is _____ (three decimals).
NAT · enter the numerical value

Q.197 MSQ · 2 marks · P&S; / joint
A 2x2 joint PMF on {0,1}x{0,1} has all four cells equal to 1/4. Which are true?
(A) P(X=Y)=1/2
(B) Corr(X,Y)=0
(C) P(X=1,Y=1)=1/2
(D) X and Y are independent Bernoulli(1/2)

Q.198 NAT · 2 marks · P&S; / Bayes
An oil well is wet with prior 0.4. A seismic test is positive with probability 0.85 if wet and 0.25 if dry. Given a positive test, P(wet) equals _____ (two decimals).
NAT · enter the numerical value

Q.199 NAT · 2 marks · P&S; / cond-exp
N~Poisson(3) and given N=n, X is Binomial(n, 0.5) (X=0 if n=0). Then E[X] equals _____ (one decimal).
NAT · enter the numerical value

Q.200 NAT · 2 marks · P&S; / covariance
X is Uniform on {1,2,3} and Y=X^2. Then Cov(X,Y)=E[X^3]-E[X]E[X^2]. This covariance equals _____ (rounded to two decimals).
NAT · enter the numerical value

Q.201 NAT · 2 marks · P&S; / binomial
A login succeeds with probability 0.4 independently. In 5 attempts, P(first success occurs on the 5th attempt) equals _____ (rounded to four decimals). (This is a geometric probability, not a binomial count.)
NAT · enter the numerical value

Q.202 NAT · 1 mark · P&S; / normal
If X~N(0, sigma^2), the MGF E[e^{tX}] equals exp(sigma^2 t^2 / 2). For sigma=2 and t=1 the MGF equals e^k with k=_____.
NAT · enter the numerical value

Q.203 NAT · 1 mark · P&S; / CI
A 95% z-interval is built from n=16, sigma=8, Xbar=50 and z_{0.025}=1.96. Its lower endpoint is _____ (two decimals).
NAT · enter the numerical value

Q.204 MCQ · 2 marks · P&S; / mixed
Five i.i.d. N(0,1) r.v.s. Let S = X1^2+...+X5^2 and T = sqrt(5) Xbar. Which is TRUE?
(A) S ~ N(0,5)
(B) T ~ chi^2_5
(C) S and T are equal
(D) S ~ chi^2_5 and T^2 ~ chi^2_1, and S-T^2 ~ chi^2_4 independent of T

Q.205 NAT · 2 marks · P&S; / counting
The number of onto functions from a 5-element domain to a 3-element codomain is _____.
NAT · enter the numerical value

Q.206 NAT · 1 mark · P&S; / independence
Three independent alarms each fail with probability 0.1. The probability that all three fail is _____ (three decimals).
NAT · enter the numerical value

Q.207 NAT · 2 marks · P&S; / joint
Joint PMF p(x,y)=(x+y)/21 for x=1,2 and y=1,2,3. Then P(X=2) equals _____ (rounded to two decimals).
NAT · enter the numerical value

Q.208 NAT · 2 marks · P&S; / Bayes
City buses are 80% blue and 20% green. A witness identifies a bus as green; the witness is correct 75% of the time (and otherwise names the other colour). P(bus is green | identified green) equals _____ (two decimals).
NAT · enter the numerical value

Q.209 NAT · 2 marks · P&S; / cond-exp
N~Poisson(4) and given N=n, S is the sum of n i.i.d. Uniform{1,2} (S=0 if n=0). E[S] equals _____ (integer).
NAT · enter the numerical value

Q.210 NAT · 1 mark · P&S; / covariance
A 2D sample of 4 points: (1,2), (2,4), (3,6), (4,8). The (Pearson) sample correlation is _____.
NAT · enter the numerical value

Q.211 NAT · 1 mark · P&S; / binomial
A shopper buys a warranty on a visit with probability 0.3, independently across 7 visits. The variance of the number of warranties sold is _____ (two decimals).
NAT · enter the numerical value

Q.212 MCQ · 2 marks · P&S; / mixed
The matrix [[4, 3], [3, 1]] is claimed as a covariance matrix. This claim is
(A) true, because the diagonal is positive
(B) false, because a 2x2 covariance must be PSD and 4*1-9=-5 < 0
(C) false only because 3 > 1
(D) true, because it is symmetric

Q.213 NAT · 2 marks · P&S; / counting
How many 5-digit integers (no leading zero) can be formed using digits 0-9 if repetition is not allowed?
NAT · enter the numerical value

Q.214 NAT · 2 marks · P&S; / joint
Joint PMF: P(X=0,Y=0)=0.15, P(X=0,Y=1)=0.25, P(X=1,Y=0)=0.20, P(X=1,Y=1)=0.40. Then Cov(X,Y) equals _____ (two decimals).
NAT · enter the numerical value

Q.215 NAT · 2 marks · P&S; / Bayes
A malware scanner flags 98% of infected files and 4% of clean files. Prevalence of infection is 1%. Given a flag, P(infected) equals _____ (two decimals).
NAT · enter the numerical value

Q.216 NAT · 2 marks · P&S; / cond-exp
A gambler starts with 0. Each step a fair coin adds +1 or -1. Let T be the first time the fortune is +2 or -2. E[T] equals _____ (integer).
NAT · enter the numerical value

Q.217 NAT · 1 mark · P&S; / binomial
A byte is 8 independent bits, each equal to 1 with probability 0.5. P(the byte has even parity), i.e. even number of 1s, equals _____ (two decimals).
NAT · enter the numerical value

Q.218 MCQ · 1 mark · P&S; / mixed
Negative-binomial flavour: independent Bernoulli(p=1/2) trials. The number of trials T until (and including) the 3rd success has E[T] equal to
(A) 1.5
(B) 6
(C) 3
(D) 8

Q.219 NAT · 2 marks · P&S; / counting
Four married couples sit around a circular table so that spouses sit together (each couple occupies two adjacent seats). Treating rotations as the same, the number of seating arrangements is _____.
NAT · enter the numerical value

Q.220 NAT · 2 marks · P&S; / joint
Joint density f(x,y)=8xy on the triangle 0 < y < x < 1. Then P(Y < 1/2) equals _____ (four decimals).
NAT · enter the numerical value

Q.221 NAT · 2 marks · P&S; / Bayes
Among donors, 40% are type A, 10% type B, 50% type O. A mistyped record lists a donor as type A with probability 0.9 if truly A, 0.05 if B, 0.05 if O. Given a listed-A record, P(true type A) equals _____ (two decimals).
NAT · enter the numerical value

Q.222 NAT · 1 mark · P&S; / cond-exp
A mixed distribution: with probability 0.4, X~Bernoulli(0.5); with probability 0.6, X is identically 1. Then E[X] equals _____ (two decimals).
NAT · enter the numerical value

Q.223 NAT · 1 mark · P&S; / binomial
Among 6 independent TCP packets, each is lost with probability 0.2. The expected number of lost packets is _____ (one decimal).
NAT · enter the numerical value

Q.224 MCQ · 2 marks · P&S; / mixed
A random variable X has MGF M(t)=exp(5t + 8 t^2). Then X is
(A) N(5, 8)
(B) N(8, 5)
(C) N(5, 16)
(D) Poisson(5)

Q.225 NAT · 2 marks · P&S; / counting
Using inclusion-exclusion, the number of integers from 1 to 100 that are divisible by 2 or 3 or 5 is _____.
NAT · enter the numerical value

Q.226 NAT · 2 marks · P&S; / joint
On the unit square with uniform joint density, P(Y > X+0.5) equals _____ (three decimals).
NAT · enter the numerical value

Q.227 NAT · 2 marks · P&S; / Bayes
Radar: an object is an aircraft with prior 0.05, else a bird. The radar alarm sounds with probability 0.99 on aircraft and 0.10 on birds. Given an alarm, P(aircraft) equals _____ (two decimals).
NAT · enter the numerical value

Q.228 NAT · 1 mark · P&S; / cond-exp
X given Y=y is Exponential with mean y, and Y is Exponential with mean 2. Then E[X] equals _____ (integer).
NAT · enter the numerical value

Q.229 NAT · 1 mark · P&S; / binomial
Each day of a 7-day week is rainy independently with probability 0.3. P(no rainy day) equals _____ (rounded to three decimals).
NAT · enter the numerical value

Q.230 MCQ · 2 marks · P&S; / mixed
Quantile: X~Exp(rate=ln 2) so median m satisfies P(X <=m)=1/2. Then m equals
(A) ln 2
(B) 2
(C) 1
(D) 1/2

Q.231 NAT · 2 marks · P&S; / counting
The number of integer solutions of w+x+y+z=8 with each variable > =0 and w > =2 is _____.
NAT · enter the numerical value

Q.232 NAT · 2 marks · P&S; / joint
If f(x,y)=e^{-x-y} for x > 0,y > 0, then P(X+Y < = 1) equals 1-2/e. Numerically this is _____ (rounded to two decimals).
NAT · enter the numerical value

Q.233 NAT · 2 marks · P&S; / Bayes
A student knows the answer with probability 0.7, otherwise guesses uniformly among 4 options. Given a correct response, P(the student knew it) equals _____ (two decimals).
NAT · enter the numerical value

Q.234 NAT · 1 mark · P&S; / binomial
Each of 12 students passes a qualifier independently with probability 0.8. The expected number who pass is _____ (one decimal).
NAT · enter the numerical value

Q.235 MCQ · 2 marks · P&S; / mixed
Paired t-test vs two-sample t-test: 12 students measured before and after a workshop. The appropriate 1-sample analysis is a t-test on
(A) the 12 differences, with 11 degrees of freedom
(B) a z-test with sigma=0
(C) 24 independent numbers, df=22
(D) the 12 before scores only

Q.236 NAT · 1 mark · P&S; / counting
A 4-person working group is chosen from 9 researchers and then one of the 4 is designated chair. The number of possible (group, chair) outcomes is _____.
NAT · enter the numerical value

Q.237 NAT · 1 mark · P&S; / joint
Joint density f(x,y)=8xy on 0 < y < x < 1. The marginal f_X(x) for x in (0,1) is 4x^3. Hence f_X(1/2) equals _____ (one decimal).
NAT · enter the numerical value

Q.238 NAT · 2 marks · P&S; / Bayes
Packets drop due to congestion (prior 0.6) or due to a link fault. A monitor reports 'congestion-like' with probability 0.8 under congestion and 0.3 under fault. Given that report, P(congestion) equals _____ (two decimals).
NAT · enter the numerical value

Q.239 MCQ · 1 mark · P&S; / mixed
A sample of size 5 has values 1,2,3,4,100. Compared with the mean, the median is
(A) 3, and it is far less affected by the 100
(B) 22, the same as the mean
(C) 1
(D) 100

Q.240 NAT · 1 mark · P&S; / counting
From 10 distinct papers, 3 are chosen for an oral exam and then ordered (first, second, third). The number of possible ordered triples is _____.
NAT · enter the numerical value

Q.241 NAT · 1 mark · P&S; / joint
Using the same joint table P(X=1,Y=1)=0.10, P(X=1,Y=2)=0.20, P(X=2,Y=1)=0.30, P(X=2,Y=2)=0.40, the value of P(Y=1|X=1) is _____ (rounded to two decimals).
NAT · enter the numerical value

Q.242 NAT · 2 marks · P&S; / Bayes
At customs, 5% of travellers carry undeclared goods. A scanner alarms with probability 0.92 if undeclared goods are present and 0.08 otherwise. Given an alarm, P(undeclared goods) equals _____ (two decimals).
NAT · enter the numerical value

Q.243 MCQ · 1 mark · P&S; / mixed
An ANOVA-free chi-squared: a spinner with 4 labelled faces is spun 40 times. Under fairness each expected count is 10. If all observed counts equal 10, the Pearson statistic is
(A) 40
(B) 10
(C) 0, and one does not reject fairness
(D) 3

Q.244 NAT · 1 mark · P&S; / counting
The multinomial coefficient 6!/(2!2!2!) counts the number of words of length 6 with two A's, two B's and two C's. Its value is _____.
NAT · enter the numerical value

Q.245 NAT · 2 marks · P&S; / Bayes
A tumour is present with prior 0.01. An MRI is positive with probability 0.95 if a tumour is present and 0.02 otherwise. Given a positive MRI, P(tumour) equals _____ (two decimals).
NAT · enter the numerical value

Q.246 MSQ · 2 marks · P&S; / mixed
Let X~Binomial(n,p) and Y~Poisson(np) with the same mean. Which comparisons are correct for 0 < p < 1, n >=2?
(A) X and Y are equal almost surely
(B) E[X]=E[Y]
(C) P(X=0)=(1-p)^n which is not equal to e^{-np} in general
(D) Var(X)=np(1-p) < np = Var(Y)

Q.247 NAT · 1 mark · P&S; / counting
A robot on a grid may move only right or up. The number of shortest paths from (0,0) to (4,3) is _____.
NAT · enter the numerical value

Q.248 NAT · 2 marks · P&S; / Bayes
Lots: 90% of lots are good (defect rate 1%) and 10% are bad (defect rate 20%). One item drawn from a random lot is defective. P(the lot is bad) equals _____ (two decimals).
NAT · enter the numerical value

Q.249 MSQ · 2 marks · P&S; / mixed
A data set of 7 distinct numbers is summarised. Which are true?
(A) The sample mean is sensitive to that largest value
(B) Changing the largest value does not change the median
(C) The unique mode of 7 distinct numbers is the largest
(D) The sample median is the 4th order statistic

Q.250 NAT · 1 mark · P&S; / counting
A password is a length-4 string over {A,B,C,D,E} with no two consecutive characters equal. The number of such passwords is _____.
NAT · enter the numerical value

Q.251 NAT · 2 marks · P&S; / Bayes
A login failure is a brute-force attack with prior 0.15, else a user typo. An anomaly score is high with probability 0.88 on attacks and 0.12 on typos. Given a high score, P(attack) equals _____ (two decimals).
NAT · enter the numerical value

Q.252 MSQ · 2 marks · P&S; / mixed
Let F be the CDF of Uniform(0,1). Which are true?
(A) F is differentiable at 0.3 with derivative 1
(B) F has a jump at 0
(C) F(x)=0 for x < 0, F(x)=x for x in [0,1], F(x)=1 for x > 1
(D) F is continuous

Q.253 NAT · 1 mark · P&S; / counting
A 9-match season is split into 4 wins, 3 draws, 2 losses. The number of result sequences is 9!/(4!3!2!) = _____.
NAT · enter the numerical value

Q.254 NAT · 2 marks · P&S; / mixed
Three fair coins. Let T = at least two heads and W = at least one tail. P(T n W) equals _____ (three decimals).
NAT · enter the numerical value

Q.255 NAT · 1 mark · P&S; / counting
The number of derangements of 4 distinct recommendation letters into 4 envelopes (no letter in the matching envelope) is _____.
NAT · enter the numerical value

Q.256 NAT · 2 marks · P&S; / mixed
Order statistics: 3 i.i.d. Uniform(0,1). E[max] equals 3/4 = _____ (two decimals).
NAT · enter the numerical value

Q.257 NAT · 1 mark · P&S; / counting
The number of ways to distribute 7 identical USB sticks to 3 distinct interns (each may get zero) is _____.
NAT · enter the numerical value

Q.258 NAT · 2 marks · P&S; / mixed
3 i.i.d. Uniform(0,1). E[min] equals 1/4 = _____ (two decimals).
NAT · enter the numerical value

Q.259 NAT · 1 mark · P&S; / counting
A 6-character OTP is formed from digits 0-9 with all characters distinct. The number of possible OTPs is _____.
NAT · enter the numerical value

Q.260 NAT · 2 marks · P&S; / mixed
A hypergeometric flavour without the name: an urn has 5 red and 7 blue tokens. Three tokens are drawn without replacement. P(all red) = C(5,3)/C(12,3) equals _____ (rounded to three decimals).
NAT · enter the numerical value

Q.261 NAT · 1 mark · P&S; / counting
In a workshop of 12 people, every pair shakes hands exactly once. The number of handshakes is _____.
NAT · enter the numerical value

Q.262 NAT · 2 marks · P&S; / mixed
A coupon-collector with 4 coupon types, uniform independent draws. Expected draws to complete the set is 4(1+1/2+1/3+1/4) = _____ (one decimal).
NAT · enter the numerical value

Q.263 NAT · 1 mark · P&S; / mixed
Skewness of N(mu, sigma^2) is 0. For Exp(rate=1) the skewness E[(X-1)^3]/1^3 equals 2. The difference (Exp skewness minus Normal skewness) is _____.
NAT · enter the numerical value

Q.264 NAT · 1 mark · P&S; / mixed
A 2x2 covariance matrix [[4, 1], [1, 9]] has Corr(X,Y) equal to 1/(2*3) = _____ (rounded to two decimals).
NAT · enter the numerical value

UNIT 1 SOLUTIONS

Q.1 [MCQ] A — Count panels by number of engineers e=2,3,4,5: C(8,5-e)C(6,e). Sum equals 1526.

Q.2 [MCQ] A — Non-negativity and P(Omega)=1 force P({omega3})=1-0.2-0.5=0.3.

Q.3 [MCQ] C — Independent nonempty-probability events have positive intersection probability P(A)P(B) > 0, hence cannot be disjoint.

Q.4 [MCQ] D — P(Y=1)=0.2+0.4=0.6, so P(X=1|Y=1)=0.4/0.6=2/3.

Q.5 [MCQ] A — Defectives: 40 from F1, 180 from F2. P(F1|D)=40/220 ~ 0.182.

Q.6 [MCQ] D — Tower property: E[E[X|Y]]=E[X].

Q.7 [MCQ] D — (40*25+65)/41=1065/41=26.

Q.8 [MCQ] A — Independence implies uncorrelatedness; the converse fails. Additivity of variance needs uncorrelatedness. Correlation needs positive variances.

Q.9 [MCQ] C — c+2c+3c=1 = > 6c=1.

Q.10 [MCQ] A — C(4,2)/C(4,1)=6/4=3/2, and the p-powers match.

Q.11 [MCQ] D — Three independent fair bits: 2^{-3}=1/8.

Q.12 [MCQ] C — Poisson: mean=var=lambda. Geometric mean 1/p, variance (1-p)/p^2, equal only for special p (p= (sqrt(5)-1)/2 not generic). DA 2024 used Poisson+N(0,1); this pairs Poisson with Geometric instead.

Q.13 [MCQ] B — Atoms have probability 0 under a density.

Q.14 [MCQ] C — Y is a deterministic function of X, supported on the line x+y=1.

Q.15 [MCQ] D — The memoryless laws on R+ and on N are Exponential and Geometric.

Q.16 [MCQ] C — Affine image of a Gaussian is Gaussian with mean 4 and variance 9.

Q.17 [MCQ] A — Definition of chi^2_k.

Q.18 [MCQ] D — Definition of Student's t.

Q.19 [MCQ] C — f_X(x)=1 on (0,1) and f_{Y|X}(y|x)=1/x on (0,x), product 1/x.

Q.20 [MCQ] D — CLT scaling: variance of the mean is sigma^2/n.

Q.21 [MCQ] D — SE is sigma/sqrt(n); z not t when sigma is known.

Q.22 [MCQ] C — Two-sided rejection region is both tails.

Q.23 [MCQ] C — Definition of the one-sample t statistic.

Q.24 [MCQ] C — One linear constraint sum O=n, hence k-1 df.

Q.25 [MCQ] D — Poisson mean=var. N(0,4) has variance 4, not 1..)

Q.26 [MCQ] B — Multinomial: 5!/(2!2!1!) * (1/3)^5 = 30/243.

Q.27 [MCQ] D — Type I = false rejection; Type II = false non-rejection.

Q.28 [MCQ] D — WLLN is convergence in probability. SLLN is a.s. (under slightly stronger hypotheses).

Q.29 [MCQ] B — Circular arrangements of 7 are 6!. Treat the pair as glued: 2*5!. Subtract to get 6!-2*5! = 480.

Q.30 [MCQ] A — Frechet bound: P(A n B) > = P(A)+P(B)-1 = 0.3.

Q.31 [MCQ] A — Pairwise: each pair has P=1/4 = (1/2)(1/2). But A n B n C is empty (two heads cannot differ), so P(A n B n C)=0 != 1/8.

Q.32 [MCQ] B — Among 30 intern jobs, 10 delayed: 10/30=1/3.

Q.33 [MSQ] A;B;D — PPV is pulled down by the 99:1 prior odds. FPR=1-specificity=0.01. Specificity is P(predicted 0 | class=0), not the FPR.

Q.34 [MCQ] C — Var(X)=E[Var(X|Y)]+Var(E[X|Y]).

Q.35 [MCQ] D — z=(104-88)/16=1. Min/max are distractors (min-max scaling).

Q.36 [MCQ] C — Y is an affine decreasing function of X with slope -1, so correlation is -1 (for 0 < p < 1).

Q.37 [MCQ] A — CDFs are always right-continuous and non-decreasing; discrete laws add jumps. They need not be left-continuous or strictly increasing.

Q.38 [MSQ] A;C;D — Mean, variance and empty-success probability are exact. CLT is an approximation, not an identity for n=6.

Q.39 [NAT] 0.141 — Condition on a11. Each case multiplies two independent Binomial(3,1/2) probabilities: 0.5*(3/8)^2 + 0.5*(3/8)^2 = (3/8)^2 = 0.140625 ~ 0.141.

Q.40 [MCQ] C — Classic Poisson-binomial relationship.

Q.41 [MCQ] C — The characterisation of CDFs.

Q.42 [MCQ] D — Classic fact: n+1 spacings including ends are like n+1 i.i.d. exp / sum.

Q.43 [MCQ] A — e^{-3 lambda}=0.2 = > 3 lambda = ln 5, lambda=(ln 5)/3.

Q.44 [MCQ] B — Definition of N(0,1).

Q.45 [MCQ] B — Gaussian sample variance pivot.

Q.46 [MCQ] A — t is symmetric about 0 with polynomial tails; as nu- > inf it approaches N(0,1).

Q.47 [MCQ] B — Conditional equals marginal iff independence.

Q.48 [MCQ] C — np=80, npq=48, sd~6.928. z=(90-80)/6.928~1.44 and (70-80)/6.928~-1.44.

Q.49 [MCQ] A — Larger confidence needs a larger critical value.

Q.50 [MCQ] B — |1.5| < 1.96 so insufficient evidence against H0 at 5%.

Q.51 [MCQ] D — |2| < 2.306, so do not reject at 5%.

Q.52 [MCQ] C — (2-1)(2-1)=1.

Q.53 [MCQ] C — Cannot have > =3 heads and > =3 tails in 4 tosses.

Q.54 [NAT] 360 — P(6,4)=360.

Q.55 [MCQ] D — Power = 1 - beta(alternative).

Q.56 [MCQ] A — Choose suit in 4 ways then 5 ranks from 13: 4*C(13,5)=5148. (Straight-flush exclusion is not asked.)

Q.57 [MCQ] A — P(A n B) < = min(P(A),P(B))=0.6.

Q.58 [MCQ] A — A n B is empty so the conditional probability is 0. They are dependent.

Q.59 [MCQ] B — Sum of (x+y): (2+3+4)+(3+4+5)=21, so c=1/21.

Q.60 [NAT] 0.28 — Bayes: 0.019/(0.019+0.049)=0.019/0.068 ~ 0.28.

Q.61 [MCQ] B — Falling factorial: E[X(X-1)]=lambda^2 for Poisson.

Q.62 [MCQ] A — Right skew pulls the mean toward the long right tail.

Q.63 [MCQ] A — X is symmetric about 0 and Y is even, so E[XY]=E[X^3]=0=E[X]E[Y]. Uncorrelated but dependent.

Q.64 [MCQ] D — This is the Geometric(number of failures) law.

Q.65 [NAT] 0.219 — C(8,3)=56, 56/256=0.219.

Q.66 [NAT] 0.09 — Error ~ Bernoulli(0.1), variance 0.09.

Q.67 [MCQ] C — np=1, the Poisson limit mean.

Q.68 [MCQ] B — F(0.75)-F(0.25)=0.5625-0.0625=0.5. Density 2x on (0,1).

Q.69 [MSQ] B;C;D — Continuous uniform has P(X=1)=0. Mean is the midpoint, variance the interval-length formula.

Q.70 [MCQ] B — P(Y > =ell)=P(X > =ell)=e^{-ell ln 2}=2^{-ell}, a geometric ratio q=1/2.

Q.71 [MCQ] D — Standardise to a N(0,1) probability.

Q.72 [MSQ] A;B;D — The last claim would require 2 degrees of freedom but sqrt(n) Xbar is N(0,1), so its square is chi^2_1. chi^2_2 is Exp with mean 2.

Q.73 [MCQ] C — Student t with 1 df is Cauchy.

Q.74 [MCQ] D — E[X|Y]=Y so Cov(X,Y)=Var(Y)=1, Var(X)=E[Var(X|Y)]+Var(E[X|Y])=1+1=2. Bivariate Gaussian.

Q.75 [MCQ] D — Classical CLT: i.i.d. and finite variance. Cauchy has no mean.

Q.76 [MCQ] C — 0.4 +/- 1.96*0.049 ~ 0.4 +/- 0.096, i.e. about (0.30, 0.50).

Q.77 [MCQ] C — The statistic falls just short of the 5% critical value.

Q.78 [MCQ] B — Two estimated means and one pooled variance: n1+n2-2.

Q.79 [MCQ] C — ad-bc=0, so the statistic is 0 (perfect independence in the table).

Q.80 [MCQ] D — Complement of no sixes.

Q.81 [MSQ] A;C;D — p-values are not proof of H0. Chi^2 is an approximation requiring adequate expected counts.

Q.82 [MCQ] A — Total C(12,3) minus lists with no ML book C(7,3) = 220-35=185.

Q.83 [MCQ] A — Both are standard consequences of Kolmogorov axioms (monotonicity and empty-set probability).

Q.84 [MCQ] C — Independence is preserved by complements, but the union formula without subtracting the product is exclusivity, not independence.

Q.85 [MCQ] D — Favourable independent pairs: (1,1) and (2,0), each with probability 1/4, total 1/2.

Q.86 [NAT] 0.89 — 0.27/(0.27+0.035)=0.27/0.305 ~ 0.89.

Q.87 [MCQ] C — b=Cov(Z,X)/Var(X)=-2/4=-1/2. Then E[Z]=a+b E[X] = > 3=a-0.5 = > a=3.5.

Q.88 [MCQ] C — Median is the middle of five. A unique mode 14 must occur more often than others and at least twice, which is compatible with median 12 if 14 occupies the upper end. Mean 10 is a sum constraint, not impossibility.

Q.89 [MCQ] C — E[Y|X]=X^2, so E[XY]=E[X^3]=0 and E[X]=0, hence Cov=0.

Q.90 [MCQ] B — Must be non-negative and sum to 1. Only (0.2,0.3,0.5) works.

Q.91 [NAT] 0.259 — C(5,4)(0.6)^4(0.4)=5*0.1296*0.4=0.259.

Q.92 [MSQ] A;B;D — Poisson is a discrete law on {0,1,2,...}; it has no density on the line. The factorial-moment and MGF formulae hold.

Q.93 [MCQ] A — Fundamental theorem: the density is the derivative of the CDF.

Q.94 [NAT] 0.5 — Lower triangle of the unit square, area 1/2.

Q.95 [MSQ] A;C;D — Memoryless, second moment 2/l^2, MGF formula. A continuous law has P(X=0)=0.

Q.96 [MCQ] A — W~chi^2_3, mean 3, variance 2*3=6.

Q.97 [NAT] 0.95 — chi^2_1 quantile 0.95 is 1.96^2~3.84.

Q.98 [MCQ] B — The t pivot has n-1 degrees of freedom.

Q.99 [NAT] 0.37 — Independence: conditional density is e^{-x}, so e^{-1}~0.37.

Q.100 [MCQ] C — 2*0.5987-1=0.1974 ~ 0.20.

Q.101 [MCQ] A — Frequentist coverage is a property of the procedure, not of one realised interval as a posterior probability.

Q.102 [MSQ] A;C;D — SE=1, z=3 > 1.96 so reject. The CI is centred at Xbar=103, not at the H0 value.

Q.103 [NAT] 2 — SE=3/3=1, t=(4-2)/1=2.

Q.104 [MSQ] B;C;D — It is a categorical test; Normality of the labels is not assumed. Expected counts are n * p_row * p_col under independence.

Q.105 [MCQ] A — Exponential is memoryless, Poisson has mean=var, t_n - > N(0,1).

Q.106 [MSQ] B;C;D — Standard Neyman-Pearson facts. Non-rejection is not a proof of H0.

Q.107 [MCQ] C — Count involutions in S_4: the identity, 6 transpositions, and 3 products of two disjoint transpositions, total 10.

Q.108 [MCQ] B — Additivity of P on a union requires the intersection to be null (almost surely disjoint).

Q.109 [MCQ] B — D n E = {(3,4)} only, probability 1/36. Also P(D)=6/36, P(E)=1/6, product 1/36, so D and E are independent.

Q.110 [MCQ] A — Definition of conditional density: joint over marginal.

Q.111 [NAT] 0.63 — P(B|D)=0.012/(0.007+0.012)=0.012/0.019 ~ 0.63.

Q.112 [MCQ] A — Tower gives E[X]=c. Also E[XY]=E[E[X|Y] Y]=c E[Y] so Cov=0. Independence and degeneracy need not hold.

Q.113 [NAT] 4 — E[X^2]-(E[X])^2=20-16=4.

Q.114 [MCQ] A — Corr = D^{-1/2} Cov D^{-1/2} is a Gram matrix, hence SPD/PSD, with unit diagonal.

Q.115 [MSQ] A;B;D — Standard discrete CDF facts. F is right-continuous, not left-continuous.

Q.116 [NAT] 0.94 — 1-0.75^10 ~ 1-0.0563=0.94.

Q.117 [NAT] 0.05 — e^{-3}~0.050.

Q.118 [MSQ] A;C — Non-negativity and unit integral. Densities may exceed 1 (e.g. Uniform(0,0.1)) and need not be continuous.

Q.119 [MSQ] B;C;D — Min of independent exponentials adds rates. Sum is Erlang. Symmetry gives P(W1 < W2)=1/2.

Q.120 [MCQ] C — Sum of independent Gaussians is Gaussian with added means and variances.

Q.121 [MSQ] B;C;D — Standard limit of t. Variance is nu/(nu-2) - > 1.

Q.122 [NAT] 1.6 — 2*0.2/0.25=1.6.

Q.123 [MCQ] B — The classical CLT statement.

Q.124 [MCQ] A — 3 lies inside the interval, so H0 is not rejected at 5%.

Q.125 [NAT] 0.05 — 1-0.9495=0.0505 ~ 0.05.

Q.126 [NAT] 2.8 — (4+4+1+1+9+9)/10=2.8.

Q.127 [MCQ] D — Convex: phi of mean < = mean of phi. Variance identity is the square case.

Q.128 [MCQ] A — Onto functions from 6 courses to 3 slots: 3^6 - C(3,1)2^6 + C(3,2)= 729-192+3=540.

Q.129 [MCQ] D — Weights 1+...+6=21; even faces 2+4+6=12, so 12/21.

Q.130 [MCQ] C — (i) holds because P(A n B)=0 != P(A)P(B). Uncorrelated does not imply independent (classic counterexamples).

Q.131 [MCQ] D — f factors as 1*1 and each marginal is Uniform(0,1).

Q.132 [NAT] 0.67 — 0.16/(0.16+0.08)=0.16/0.24=2/3 ~ 0.67.

Q.133 [MCQ] B — States 0,1,2=done. e0=1+(2/3)e0+(1/3)e1, e1=1+(2/3)e0+(1/3)*0. Solve: e1=1+(2/3)e0, e0=1+(2/3)e0+(1/3)+(2/9)e0 = > (1/9)e0=4/3 = > e0=12.

Q.134 [NAT] 8 — Q1=4, Q3=12, IQR=8.

Q.135 [MCQ] B — (2X-1)= +/-1 with probabilities 0.4 and 0. (2X-1)^2=1, E[Y]=0 so second term 0, E[Y^2]=25. Twist on DA 2026 Q44 (which used p=0.3, var 100).

Q.136 [MSQ] B;C;D — E[X]=0.5+0.6=1.1. CDF at 1.7 includes 0 and 1. A discrete law has P(X=1.7)=0.

Q.137 [NAT] 0.0486 — C(4,2)*0.01*0.81=0.0486.

Q.138 [NAT] 0.2 — Waiting time is Exponential(5), mean 1/5=0.2.

Q.139 [NAT] 0.0375 — E[X^2]=int 3x^4=3/5=0.6, Var=0.6-0.5625=0.0375.

Q.140 [NAT] 0.37 — Memoryless: P(X > 5)=e^{-1}~0.37.)

Q.141 [MCQ] C — Two-sigma interval is the 95% heuristic (more precisely 1.96).

Q.142 [MSQ] A;C;D — t_{n-1} is heavier-tailed than N(0,1), so |t|_{crit} > z_crit. Normality matters for small n.

Q.143 [NAT] 0.25 — E[Y|X]=X/2 so E[Y]=1/4.

Q.144 [MSQ] A;B;C — CLT and WLLN apply. The exact t law needs Gaussian data and an independent sample sd in the denominator, not the raw mean.

Q.145 [MSQ] A;B;D — Exact Gaussian z-interval has exact coverage. Length shrinks. Larger z widens a still-centred interval.

Q.146 [NAT] 2 — (2-1)(3-1)=2.

Q.147 [MCQ] B — The first derivative of the MGF at 0 is the mean.

Q.148 [MCQ] C — Outcomes: H, TH, TTH, TTTH, TTTT. Five atoms.

Q.149 [MSQ] B;C;D — Inclusion-exclusion and complement are axioms/theorems; Boole's inequality always holds. Product formula needs independence.

Q.150 [MSQ] A;B;C — Independence is the product formula, which implies P(A|B)=P(A) and the inclusion-exclusion form. Disjointness would force a zero product, a contradiction.

Q.151 [MCQ] A — Row X=1 sums to 0.70, so the conditional is 0.40/0.70. Entries sum to 1.00 so it is a PMF; independence would need a product structure (fails: 0.05 vs 0.15*0.30? P(X=0)=0.30, P(Y=0)=0.15, product 0.045 != 0.05).

Q.152 [NAT] 0.5 — P(G)=(1/3)(3/4+1/4+1/2)= (1/3)(1.5)=0.5. P(D1|G)=(0.25)/0.5=0.50.

Q.153 [MSQ] A;C;D — Tower property, law of total variance, and the defining projection property. E[X|Y]=E[X] is independence-like (mean independence), not automatic.

Q.154 [NAT] 13 — Total 60+18=78, mean 13.0.

Q.155 [MSQ] B;C;D — Cauchy-Schwarz gives |rho| < =1 and the triangle inequality for sd. Affine transforms pick up the sign of the slopes. Cov(X,X)=Var(X), not 0.

Q.156 [NAT] 1.56 — 2.2-0.64=1.56.

Q.157 [NAT] 0.161 — C(5,2)*(1/6)^2*(5/6)^3=10*(1/36)*(125/216)=1250/7776 ~ 0.161.

Q.158 [NAT] 0.71 — m^2=1/2, m=sqrt(1/2)~0.71.

Q.159 [NAT] 0.5 — Minimum of two i.i.d. Exp(1) is Exp(2), mean 1/2.

Q.160 [MCQ] A — Both laws are symmetric about 0 so both CDFs equal 1/2 at 0. t_1 (Cauchy) has heavier tails.

Q.161 [NAT] 2 — 4/(4-2)=2.

Q.162 [NAT] 0.5 — int_0^2 (1+y)/8 dy = (y + y^2/2)/8 |_0^2 = (2+2)/8=0.5.

Q.163 [NAT] 0.69 — Var Uniform(0,12)=144/12=12, sd of mean = sqrt(12)/5 ~ 0.693. Report 0.69.

Q.164 [NAT] 3.92 — SE=2, half-width 3.92.

Q.165 [MCQ] A — For i.i.d. Exp, the normalised first is Uniform (Dirichlet/Beta(1,1)).

Q.166 [MSQ] B;C;D — Sequences with repetition: n^2, not C(n,2). C(n,2) counts unordered pairs.

Q.167 [MSQ] A;C;D — Monotonicity gives the two inequalities. Conditioning on B, A and A^c partition the space. Bayes does not force P(A|B)=P(B|A).

Q.168 [MSQ] A;B;C — First and second heads are independent; first heads and equality: P both = 1/4 = (1/2)(1/2). Empty event is independent of everything. 'At least one head' is dependent on first heads.

Q.169 [MCQ] B — The joint factors as e^{-x} e^{-y} on the positive quadrant.

Q.170 [NAT] 0.67 — P(H)=0.25+0.5=0.75, P(two-headed|H)=0.5/0.75=2/3 ~ 0.67.

Q.171 [NAT] 1.5 — E[X|Y=1]=1, E[X|Y=2]=1.5, E[X|Y=3]=2, average 1.5.

Q.172 [NAT] 4 — Variance (36+4+0+4+36)/5=16, sd=4.00? 80/5=16, sd=4. Let me use 4.

Q.173 [MSQ] B;C;D — Cauchy-Schwarz: |rho|=1 iff linear relation a.s. with the sign of b. Zero correlation is weaker than independence.

Q.174 [NAT] 0.44 — Memoryless: P(X > 2+2 | X > 2)=P(X > 2)=4/9 ~ 0.44.

Q.175 [NAT] 0.3115 — C(8,2)*(1/16)*(0.75)^6=28*0.0625*0.177978515625 ~ 0.3115.

Q.176 [MSQ] B;C;D — Standard normal is symmetric about 0, odd moments vanish, P(X > 0)=1/2.

Q.177 [NAT] 2 — 1 + Var(Y)=2.

Q.178 [NAT] 0.5 — The standardised sum is approx N(0,1).

Q.179 [NAT] 62 — (3.92/0.5)^2=7.84^2=61.4656, so n=62.

Q.180 [MCQ] A — Poisson process counts are Poisson(rate * time).

Q.181 [NAT] 36 — Inclusion-exclusion: 3^4 - C(3,1)2^4 + C(3,2)1^4 = 81-48+3 = 36. Equivalently 3! S(4,3) = 6*6 = 36.

Q.182 [NAT] 0.5 — Disjoint union: 0.1+0.25+0.15=0.50.

Q.183 [MSQ] C;D — P(H1 n Eq)=P(HH)=1/4=P(H1)P(Eq). Same for Xor. But Eq and Xor are complementary, hence disjoint with positive probability, so dependent.

Q.184 [MCQ] B — Among 40 laptop students, 30 drink coffee: 3/4.

Q.185 [NAT] 0.66 — 0.27/(0.27+0.14)=0.27/0.41 ~ 0.66.

Q.186 [NAT] 0.75 — P(even)=1/2, E[X|even]=1/2; E[X|odd]=1. Total 0.75.

Q.187 [NAT] 1.5 — (82-70)/8=1.5.

Q.188 [NAT] 13 — 4VarX+VarY-4 Cov=16+9-12=13.

Q.189 [NAT] 0.5 — X=1 or 2, probability 1/2.

Q.190 [NAT] 0.94 — 0.7^8 ~ 0.0576, so 0.94.

Q.191 [MSQ] A;C;D — Z is an affine function of X, hence perfectly dependent with X; Var(X+Z)=Var(1.25 X - 0.5) is not 17.

Q.192 [NAT] 2.31 — SE=1, half-width 2.31.

Q.193 [MCQ] A — Fundamental bridge: expectation of an indicator is the probability.

Q.194 [NAT] 43200 — Seat boys first: 5! ways, creating 6 gaps; choose 4 gaps for girls: C(6,4)*4!. Product is 43200.

Q.195 [NAT] 0.625 — P(A)=3/8 so P(A^c)=5/8=0.625.

Q.196 [NAT] 0.992 — Complement of all fail: 1-0.008=0.992.

Q.197 [MSQ] A;B;D — Product structure 1/4=(1/2)(1/2). P(X=Y)=P(00)+P(11)=1/2. P(11)=1/4 not 1/2.

Q.198 [NAT] 0.69 — 0.34/(0.34+0.15)=0.34/0.49 ~ 0.69.

Q.199 [NAT] 1.5 — E[X]=E[E[X|N]]=E[0.5 N]=1.5.

Q.200 [NAT] 2.67 — E[X]=2, E[X^2]=14/3, E[X^3]=12, Cov=12-2*(14/3)=12-28/3=8/3~2.67.

Q.201 [NAT] 0.0518 — Four failures then one success: (0.6)^4 * 0.4 = 0.0518.

Q.202 [NAT] 2 — 4*1/2=2.

Q.203 [NAT] 46.08 — 50-3.92=46.08.

Q.204 [MCQ] D — Cochran: sample ss splits into n(Xbar)^2 ~ chi^2_1 and residual chi^2_{n-1}, independent.

Q.205 [NAT] 150 — 3^5 - C(3,1)2^5 + C(3,2)1^5 = 243-96+3 = 150. Also 3! S(5,3)=6*25=150.

Q.206 [NAT] 0.001 — 0.1^3=0.001.

Q.207 [NAT] 0.57 — P(X=2)=((2+1)+(2+2)+(2+3))/21=12/21=0.571..., report 0.57.

Q.208 [NAT] 0.43 — 0.15/(0.15+0.20)=0.15/0.35 ~ 0.43.

Q.209 [NAT] 6 — E[S|N]=N*1.5 so E[S]=6.

Q.210 [NAT] 1 — Y=2X exactly on the sample, so r=1.

Q.211 [NAT] 1.47 — np(1-p)=7*0.21=1.47.

Q.212 [MCQ] B — Symmetric is necessary but not sufficient: eigenvalues must be > =0, equivalently det > =0 here.

Q.213 [NAT] 27216 — First digit: 9 choices (1-9). Then 9 remaining (including 0) for the second, then 8,7,6. Product 27216.

Q.214 [NAT] 0.01 — E[XY]=P(1,1)=0.40, E[X]=0.60, E[Y]=0.65, so 0.40-0.39=0.01.

Q.215 [NAT] 0.2 — 0.0098/(0.0098+0.0396)=0.0098/0.0494 ~ 0.20.

Q.216 [NAT] 4 — Simple symmetric random walk hitting time of +/-a from 0 has mean a^2=4.

Q.217 [NAT] 0.5 — For a fair independent bits, even and odd parity are equally likely: 0.50.

Q.218 [MCQ] B — Sum of 3 geometric(1/2) waiting times: 3*(1/p)=6.

Q.219 [NAT] 96 — Treat each couple as a block: 3! circular block arrangements, and each block can be ordered 2 ways: 6*16=96.

Q.220 [NAT] 0.4375 — Integrate 8xy over y < 1/2 inside the triangle: x in (0,1/2] with y < x plus x in (1/2,1] with y < 1/2, giving 1/16+3/8=0.4375.

Q.221 [NAT] 0.92 — 0.36/(0.36+0.005+0.025)=0.36/0.39 ~ 0.92.

Q.222 [NAT] 0.8 — E[E[X|regime]]=0.4*0.5+0.6*1=0.80.

Q.223 [NAT] 1.2 — Binomial mean np=6*0.2=1.2.

Q.224 [MCQ] C — Normal MGF exp(mu t + sigma^2 t^2 / 2) so sigma^2/2=8, sigma^2=16, mu=5.

Q.225 [NAT] 74 — |A u B u C| = 50+33+20 -16 -10 -6 +3 = 74.

Q.226 [NAT] 0.125 — The region is a right triangle of legs 0.5, area 0.125.

Q.227 [NAT] 0.34 — 0.0495/(0.0495+0.095)=0.0495/0.1445 ~ 0.34.

Q.228 [NAT] 2 — E[X]=E[E[X|Y]]=E[Y]=2.

Q.229 [NAT] 0.082 — 0.7^7 ~ 0.082.

Q.230 [MCQ] C — Survival e^{-(ln 2) m}=1/2 = > (ln 2) m = ln 2 = > m=1.

Q.231 [NAT] 84 — Set w'=w-2 > =0 so w'+x+y+z=6: C(9,3)=84.

Q.232 [NAT] 0.26 — Integrate e^{-x-y} over x+y < =1, x,y > 0: 1-2e^{-1} ~ 0.264.

Q.233 [NAT] 0.9 — 0.70/(0.70+0.075)=0.70/0.775 ~ 0.90.

Q.234 [NAT] 9.6 — 12*0.8=9.6.

Q.235 [MCQ] A — Pairing reduces to a one-sample t on differences; df=n-1=11.

Q.236 [NAT] 504 — C(9,4)*4 = 126*4 = 504. Equivalently P(9,1)*C(8,3)=9*56=504.

Q.237 [NAT] 0.5 — 4*(1/2)^3=0.5.

Q.238 [NAT] 0.8 — 0.48/(0.48+0.12)=0.48/0.60=0.80.

Q.239 [MCQ] A — Median is 3. Mean is 22. The outlier inflates the mean.

Q.240 [NAT] 720 — P(10,3)=720.

Q.241 [NAT] 0.33 — P(X=1)=0.30 so 0.10/0.30=1/3 ~ 0.33.

Q.242 [NAT] 0.38 — 0.046/(0.046+0.076)=0.046/0.122 ~ 0.38.

Q.243 [MCQ] C — O=E in every cell = > chi^2=0.

Q.244 [NAT] 90 — 720/8=90.

Q.245 [NAT] 0.32 — 0.0095/(0.0095+0.0198)=0.0095/0.0293 ~ 0.32.

Q.246 [MSQ] B;C;D — Poisson has the larger variance np vs np(1-p). They are different laws; Poisson is a limit, not an a.s. identity.

Q.247 [NAT] 35 — Need 4 right and 3 up, in some order: C(7,3)=35.

Q.248 [NAT] 0.69 — 0.020/(0.009+0.020)=0.020/0.029 ~ 0.69.

Q.249 [MSQ] A;B;D — For odd n the median is the middle order statistic, robust to extremes. With all values distinct there are 7 modes (or none unique), not 'the largest'.

Q.250 [NAT] 320 — First char 5 choices; each later char has 4 choices (not equal to previous): 5*4^3=320.

Q.251 [NAT] 0.56 — 0.132/(0.132+0.102)=0.132/0.234 ~ 0.56.

Q.252 [MSQ] A;C;D — Continuous uniform CDF is continuous (no jumps) and has density 1 on (0,1).

Q.253 [NAT] 1260 — 362880/(24*6*2)=1260.

Q.254 [NAT] 0.375 — T n W is exactly two heads (three heads has no tail). Probability 3/8=0.375.

Q.255 [NAT] 9 — !4 = 4!(1-1+1/2-1/6+1/24) = 9.

Q.256 [NAT] 0.75 — F_max(t)=t^3, density 3t^2, mean 3/4.

Q.257 [NAT] 36 — Stars and bars C(9,2)=36.

Q.258 [NAT] 0.25 — Density of min is 3(1-t)^2, mean 1/4.

Q.259 [NAT] 151200 — P(10,6) = 10!/4! = 151200.

Q.260 [NAT] 0.045 — 10/220=1/22~0.045.

Q.261 [NAT] 66 — C(12,2)=66.

Q.262 [NAT] 8.3 — 4*(1+0.5+0.333+0.25)=8.3.

Q.263 [NAT] 2 — Exponential(1) has skewness 2, Gaussian has 0.

Q.264 [NAT] 0.17 — rho=1/(sd 2 * sd 3)=1/6~0.17.

UNIT 2 Linear Algebra · 276 questions

MCQ negative -1/3 or -2/3. MSQ / NAT no negative. Work first, then the solutions section.

Q.1 MSQ · 2 marks · Subspace
Which of the following are subspaces of R^3?
(A) {x : x1+x2+x3=0}
(B) {x : x1+x2+x3=1}
(C) {0}
(D) {x : x1 x2 = 0}

Q.2 MCQ · 1 mark · Span
span{(1,0),(0,1),(1,1)} in R^2 equals
(A) a line
(B) R^2
(C) {0}
(D) R^3

Q.3 MSQ · 2 marks · Independence
Which sets in R^3 are linearly independent?
(A) {(1,0,0),(0,1,0)}
(B) {(1,1,1),(2,2,2)}
(C) {(1,0,0),(0,1,0),(0,0,1)}
(D) {(0,0,0),(1,0,0)}

Q.4 MCQ · 1 mark · Basis
Any two bases of a finite-dimensional vector space
(A) can have different cardinalities
(B) have equally many vectors
(C) must be orthogonal
(D) must contain e1

Q.5 NAT · 1 mark · Dim
dim of the real vector space of 2x3 matrices is _____.
NAT · enter the numerical value

Q.6 NAT · 1 mark · Sum
dim U=3, dim W=3, dim(U intersect W)=1 inside R^5. Then dim(U+W) is _____.
NAT · enter the numerical value

Q.7 MCQ · 1 mark · Direct
The sum U+W is direct if and only if
(A) U intersect W = {0}
(B) U=W
(C) dim U=0
(D) U+W=R^n always

Q.8 NAT · 1 mark · Rank
Rank of [[1,2,3],[2,4,6],[0,0,0]] is _____.
NAT · enter the numerical value

Q.9 NAT · 1 mark · Nullity
A 3x5 matrix of rank 2 has nullity _____.
NAT · enter the numerical value

Q.10 MCQ · 1 mark · Rank-nullity
For a linear map A: R^n - > R^m, rank(A)+nullity(A) equals
(A) m
(B) n
(C) m+n
(D) rank(A^T) always plus n

Q.11 MCQ · 1 mark · Rank transpose
rank(A) always equals
(A) rank(A^T)
(B) nullity(A)
(C) n
(D) 0

Q.12 MCQ · 1 mark · Consistency
The system [[1,2],[2,4]] [x,y]^T = [3,6]^T is
(A) inconsistent
(B) consistent with infinitely many solutions
(C) a unique solution
(D) empty of solutions and of variables

Q.13 MCQ · 1 mark · Consistency2
[[1,2],[2,4]] [x,y]^T = [3,5]^T is
(A) consistent
(B) inconsistent
(C) underdetermined with a unique solution
(D) homogeneous

Q.14 NAT · 1 mark · Free
RREF of a 3x4 matrix has two pivot columns. Number of free variables is _____.
NAT · enter the numerical value

Q.15 NAT · 1 mark · Det2
det([[4,3],[2,1]]) is _____.
NAT · enter the numerical value

Q.16 NAT · 1 mark · Det2b
det([[0,-1],[1,0]]) is _____.
NAT · enter the numerical value

Q.17 NAT · 1 mark · Det3
det of the upper-triangular matrix diag(2,-1,5) is _____.
NAT · enter the numerical value

Q.18 MCQ · 1 mark · Det product
det(AB) equals
(A) det A + det B
(B) det(A) det(B)
(C) det(A+B)
(D) trace(AB)

Q.19 MCQ · 1 mark · Det inv
If det A=2 then det(A^{-1}) is
(A) 2
(B) 1/2
(C) -2
(D) 0

Q.20 NAT · 1 mark · Det scale
det(3I_2) is _____.
NAT · enter the numerical value

Q.21 MCQ · 1 mark · Eig 2x2
Let M=[[2,-1],[3,1]]. Which statement is TRUE?
(A) Both eigenvalues are non-negative reals
(B) The eigenvalues form a complex conjugate pair
(C) One eigenvalue is zero and the other is 3
(D) One eigenvalue is positive real and the other is negative real

Q.22 MCQ · 1 mark · Eig 2x2b
Let A=[[0,-1],[1,0]]. The eigenvalues are
(A) 1 and -1
(B) i and -i
(C) 0 and 0
(D) 2 and 0

Q.23 NAT · 1 mark · Trace eig
Sum of eigenvalues of [[4,1],[2,3]] (counted with multiplicity) is _____.
NAT · enter the numerical value

Q.24 NAT · 1 mark · Prod eig
Product of eigenvalues of [[4,1],[2,3]] is _____.
NAT · enter the numerical value

Q.25 MCQ · 1 mark · Eig vec
A=[[2,0],[0,3]]. A corresponding eigenvector for lambda=3 is
(A) [1,0]^T
(B) [0,1]^T
(C) [1,1]^T
(D) [0,0]^T

Q.26 MCQ · 1 mark · Diag
A 2x2 real matrix is diagonalizable over R if it has
(A) a repeated eigenvalue always
(B) two independent eigenvectors
(C) det=0
(D) trace=0

Q.27 MCQ · 1 mark · Not diag
[[0,1],[0,0]] is
(A) diagonalizable
(B) not diagonalizable (Jordan block)
(C) orthogonal
(D) full rank

Q.28 NAT · 1 mark · Char
Characteristic polynomial of [[0,1],[-2,3]] is t^2-3t+2. The roots (eigenvalues) sum to _____.
NAT · enter the numerical value

Q.29 MCQ · 1 mark · Cayley
Cayley-Hamilton says a square matrix
(A) has distinct eigenvalues
(B) satisfies its own characteristic equation
(C) is invertible
(D) has det 1

Q.30 NAT · 1 mark · Idempotent
If P^2=P, the possible eigenvalues of P are only 0 and _____.
NAT · enter the numerical value

Q.31 MCQ · 1 mark · Projection
An orthogonal projection matrix P satisfies
(A) P^T=P and P^2=P
(B) P^T=-P
(C) P^2=I always
(D) det P=1 always

Q.32 MCQ · 1 mark · Orthogonal mtx
A real matrix Q is orthogonal iff
(A) Q^T Q = I
(B) Q^2=Q
(C) det Q=0
(D) trace Q=0

Q.33 NAT · 1 mark · Orth det
If Q is orthogonal, |det Q| equals _____.
NAT · enter the numerical value

Q.34 MCQ · 1 mark · Rotation
M=[[cos theta, -sin theta],[sin theta, cos theta]] with theta=2*pi/5. Then M^5 equals
(A) M
(B) M^2
(C) I_2
(D) 0

Q.35 MCQ · 1 mark · Rotation 2026
Same rotation by 2*pi/5. M^{2026} equals which of the following?
(A) M
(B) M^2
(C) M^{-1}
(D) I_2

Q.36 NAT · 1 mark · Proj 1d
The orthogonal projection of (3,4) onto span{(1,0)} is the vector (a,0). a equals _____.
NAT · enter the numerical value

Q.37 NAT · 1 mark · Proj formula
P = uu^T for unit u=(3/5,4/5). P_{11} equals _____ (two decimals).
NAT · enter the numerical value

Q.38 MCQ · 1 mark · Quad
The quadratic form [x,y] [[2,0],[0,-3]] [x,y]^T is
(A) positive definite
(B) negative definite
(C) indefinite
(D) zero

Q.39 MCQ · 1 mark · PD
A symmetric matrix is positive definite iff
(A) all eigenvalues are positive
(B) det > 0 only
(C) trace > 0 only
(D) it is orthogonal

Q.40 NAT · 1 mark · Quad eval
q(x,y)=x^2+2xy+y^2 at (1,-1) equals _____.
NAT · enter the numerical value

Q.41 MCQ · 1 mark · LU
An LU factorisation without pivoting exists for a square A if
(A) all leading principal minors are nonzero
(B) A is singular
(C) A is orthogonal
(D) trace A=0

Q.42 MCQ · 1 mark · SVD
For any real m x n matrix, SVD writes A as
(A) U Sigma V^T with U,V orthogonal and Sigma rectangular diagonal
(B) QR only
(C) P^2=P
(D) LDL with no signs

Q.43 NAT · 1 mark · SVD vals
Singular values of diag(3,-2) are 3 and _____.
NAT · enter the numerical value

Q.44 NAT · 1 mark · Fro
Frobenius norm of [[1,0],[0,2]] is _____ (one decimal, sqrt(5)~2.2).
NAT · enter the numerical value

Q.45 NAT · 1 mark · Op2
Spectral norm of diag(1,4) is _____.
NAT · enter the numerical value

Q.46 MCQ · 1 mark · Left inv
A 3x2 matrix of rank 2 has a
(A) right inverse (2x3)
(B) left inverse (2x3)
(C) two-sided inverse 3x3
(D) zero nullity in R^3

Q.47 NAT · 1 mark · Kernel dim
A 2x3 matrix of rank 2: dim ker A is _____.
NAT · enter the numerical value

Q.48 MCQ · 1 mark · Row eq
Row-equivalent matrices have the same
(A) eigenvalues always
(B) row space (hence same rank)
(C) determinants always
(D) singular values always

Q.49 NAT · 1 mark · Trace cyclic
tr(AB) for A=[[1,2],[0,1]], B=[[0,1],[1,0]] equals tr(BA)=_____.
NAT · enter the numerical value

Q.50 MCQ · 1 mark · Sim
Similar matrices A and P^{-1}AP share
(A) the characteristic polynomial
(B) the (1,1) entry
(C) the Frobenius inner product with I only if P=I
(D) nothing in general

Q.51 NAT · 1 mark · Null space vec
A=[[1,1,1],[2,2,2]]. A basis vector of ker A can be (1,-1,0). Another independent one is (1,0,c) with c=_____.
NAT · enter the numerical value

Q.52 MCQ · 1 mark · Partition
If A is block-diagonal diag(B,C) with B,C square, then det A equals
(A) det B + det C
(B) det B * det C
(C) 0
(D) tr B * tr C

Q.53 NAT · 1 mark · Block det
det diag([[1,0],[0,1]], [2]) is _____.
NAT · enter the numerical value

Q.54 MCQ · 1 mark · Idempotent proj
For a projection (not necessarily orthogonal) P, I-P is
(A) also a projection onto ker P along the complementary image
(B) orthogonal always
(C) invertible always
(D) nilpotent

Q.55 NAT · 1 mark · Rank proj
If P is a projection R^n- > R^n of rank k, then tr(P) equals _____.
NAT · enter the numerical value

Q.56 MCQ · 1 mark · Normal
A real symmetric matrix is
(A) always orthogonally diagonalizable
(B) never diagonalizable
(C) always singular
(D) always orthogonal

Q.57 MCQ · 1 mark · Skew
A real skew-symmetric n x n matrix with n odd
(A) is invertible
(B) has det 0
(C) has all eigenvalues real and positive
(D) equals I

Q.58 NAT · 1 mark · Skew 2
det of [[0,2],[-2,0]] is _____.
NAT · enter the numerical value

Q.59 MCQ · 1 mark · Gauss
After one elimination step on [[2,1,1],[2,2,3]] the (2,1) entry becomes
(A) 2
(B) 0
(C) 1
(D) 3

Q.60 NAT · 1 mark · Pivot
The first pivot of [[0,1],[2,3]] after a row swap is _____.
NAT · enter the numerical value

Q.61 MCQ · 1 mark · Unique sol
Ax=b has a unique solution for every b in R^n iff A is
(A) square and invertible
(B) any n x (n+1)
(C) singular
(D) a projection

Q.62 NAT · 1 mark · Homog
A nontrivial solution of Ax=0 exists iff rank A is strictly less than the number of columns. For 3x3 rank 2, number of independent nontrivial solutions (dim ker) is _____.
NAT · enter the numerical value

Q.63 MCQ · 1 mark · Col space
Col(A) is
(A) the span of the columns of A
(B) the left nullspace
(C) always R^n
(D) ker(A^T) always

Q.64 MCQ · 1 mark · Four fund
ker(A) and row(A) in R^n are
(A) the same space
(B) orthogonal complements
(C) both {0}
(D) equal to Col(A)

Q.65 NAT · 1 mark · Rank product
rank(AB) is at most min(rank A, rank B). If A is 2x3 rank 2 and B is 3x2 rank 2, rank(AB) is at most _____.
NAT · enter the numerical value

Q.66 MCQ · 1 mark · Outer
uv^T for nonzero u in R^m, v in R^n has rank
(A) 0
(B) 1
(C) min(m,n) always
(D) m+n

Q.67 NAT · 1 mark · All ones eig
The all-ones 3x3 matrix J has eigenvalues 3,0,0. Rank(J) is _____.
NAT · enter the numerical value

Q.68 MCQ · 1 mark · Stochastic
A row-stochastic matrix has
(A) each row summing to 1
(B) each column summing to 0
(C) trace 1
(D) det 1

Q.69 NAT · 1 mark · Stoch eig
[[0.5,0.5],[0.2,0.8]] has 1 as an eigenvalue. The other eigenvalue is trace-1=_____. (one decimal)
NAT · enter the numerical value

Q.70 MCQ · 1 mark · Least sq
The normal equation for min ||Ax-b||_2 (A full column rank) is
(A) A^T A x = A^T b
(B) A x = b
(C) x = A b
(D) A A^T x = b

Q.71 NAT · 1 mark · LS 1d
Best least-squares slope through origin for points (1,2),(2,2): xhat = (A^T b)/(A^T A) with A=(1,2)^T, b=(2,2)^T equals _____ (one decimal).
NAT · enter the numerical value

Q.72 MCQ · 1 mark · Pseudo
The Moore-Penrose pseudoinverse of a full-column-rank A is
(A) (A^T A)^{-1} A^T
(B) A^T (A A^T)^{-1}
(C) A^{-1}
(D) A^T

Q.73 NAT · 1 mark · Cond
For diag(10, 0.1), the 2-norm condition number is _____.
NAT · enter the numerical value

Q.74 MCQ · 1 mark · Ill
A large condition number means
(A) A is close to singular in relative perturbation sense
(B) A is orthogonal
(C) det A=1
(D) A=0

Q.75 NAT · 1 mark · Inv 2x2
Inverse of [[1,1],[0,1]] is [[1,-1],[0,1]]. The (1,2) entry of the inverse is _____.
NAT · enter the numerical value

Q.76 MCQ · 1 mark · Triang eig
Eigenvalues of a triangular matrix are
(A) the diagonal entries
(B) always 0
(C) the row sums
(D) singular values

Q.77 NAT · 1 mark · Nilp
N=[[0,1],[0,0]]. N^2 equals the zero matrix. Rank(N) is _____.
NAT · enter the numerical value

Q.78 MCQ · 1 mark · Min poly
The minimal polynomial divides the characteristic polynomial and has the same
(A) roots (the eigenvalues)
(B) degree always n
(C) constant term 1
(D) leading term n

Q.79 NAT · 1 mark · Geom mult
For a Jordan block of size 3 with lambda=2, geometric multiplicity of 2 is _____.
NAT · enter the numerical value

Q.80 MCQ · 1 mark · Alg vs geo
Always
(A) geo multiplicity < = algebraic multiplicity
(B) geo > alg
(C) they are equal for every matrix
(D) geo=0

Q.81 NAT · 1 mark · Dim hom
Solution space of x+y+z=0 in R^3 has dimension _____.
NAT · enter the numerical value

Q.82 MCQ · 1 mark · Affine
The set {x: Ax=b} for consistent Ax=b is
(A) a subspace if b!=0
(B) an affine translate of ker A
(C) always {0}
(D) a sphere

Q.83 NAT · 1 mark · Ones ker
Number of independent solutions of x1+x2+x3+x4=0 is _____.
NAT · enter the numerical value

Q.84 MCQ · 1 mark · Onb
An orthonormal basis of R^2 is
(A) {(1,0),(0,1)}
(B) {(1,1),(1,-1)} without scaling
(C) {(1,0),(1,0)}
(D) {(0,0),(1,0)}

Q.85 NAT · 1 mark · GS
Gram-Schmidt on (1,1) then (0,1): the second orthonormal vector's first component is _____ (two decimals, -1/sqrt(2)~-0.71).
NAT · enter the numerical value

Q.86 MCQ · 1 mark · QR
QR factorisation writes a full-column-rank A as
(A) Q orthogonal times R upper triangular
(B) LU
(C) SVD only
(D) P symmetric

Q.87 NAT · 1 mark · Hadamard
A 2x2 Hadamard matrix [[1,1],[1,-1]] has det _____.
NAT · enter the numerical value

Q.88 MCQ · 1 mark · Perm
A permutation matrix is
(A) orthogonal (hence invertible, P^{-1}=P^T)
(B) singular
(C) a projection of rank 0
(D) symmetric always

Q.89 NAT · 1 mark · Perm det
det of the permutation matrix for the transposition (12) in R^2, i.e. [[0,1],[1,0]], is _____.
NAT · enter the numerical value

Q.90 MCQ · 1 mark · Block mul
[[A,B],[0,C]] [[X,Y],[0,Z]] equals [[AX, AY+BZ],[0, CZ]] when shapes allow. This is
(A) false in general
(B) the rule for conforming block-upper-triangular products
(C) only for scalars
(D) LU

Q.91 NAT · 1 mark · Trace I
tr(I_5) is _____.
NAT · enter the numerical value

Q.92 NAT · 1 mark · Rank I
rank(I_5) is _____.
NAT · enter the numerical value

Q.93 MCQ · 1 mark · Scalar
The matrix aI_n has eigenvalues
(A) all equal to a
(B) 0 and a
(C) 1 repeated
(D) a^n

Q.94 NAT · 1 mark · Det 2I
det(2 I_3) is _____.
NAT · enter the numerical value

Q.95 MCQ · 1 mark · Commute
If AB=BA and A is diagonalizable with distinct eigenvalues, then B
(A) is a polynomial in A (in particular shares the eigenbasis)
(B) must be 0
(C) must be I
(D) cannot exist

Q.96 NAT · 1 mark · Poly eig
If Av=2v and p(t)=t^2-t, then p(A)v equals _____ v, i.e. the scalar is _____.
NAT · enter the numerical value

Q.97 MCQ · 1 mark · Invar
A subspace W is A-invariant if
(A) A(W) subset W
(B) W subset ker A
(C) A is invertible
(D) W=R^n only

Q.98 NAT · 1 mark · Image dim
dim im T for T: R^4- > R^3 of rank 2 is _____.
NAT · enter the numerical value

Q.99 MCQ · 1 mark · Iso
A linear map T: R^n- > R^n is an isomorphism iff
(A) ker T={0}
(B) T=0
(C) rank T=0
(D) T is a projection

Q.100 NAT · 1 mark · Change basis
If P has columns the new basis, [v]_new = P^{-1} v_old. For P=I this map is the identity, so the (1,1) entry of P^{-1} is _____.
NAT · enter the numerical value

Q.101 MCQ · 1 mark · Coord
The matrix of a linear map T with respect to bases changes by
(A) similarity (or the rectangular analogue P^{-1} A Q)
(B) addition of I
(C) transposition only
(D) nothing

Q.102 NAT · 1 mark · Dot
The standard inner product of (1,2,3) and (1,-1,1) is _____.
NAT · enter the numerical value

Q.103 MCQ · 1 mark · Cauchy
|(u,v)| < = ||u|| ||v|| is
(A) Cauchy-Schwarz
(B) triangle only
(C) Pythagoras
(D) Parseval only on C^n

Q.104 NAT · 1 mark · Norm
Euclidean norm of (3,4) is _____.
NAT · enter the numerical value

Q.105 MCQ · 1 mark · Orth set
A set of nonzero orthogonal vectors is
(A) linearly dependent
(B) linearly independent
(C) a basis of R^n always
(D) empty of inner products

Q.106 NAT · 1 mark · Onb size
An orthonormal set in R^n has size at most _____.
NAT · enter the numerical value

Q.107 MCQ · 1 mark · Compl
The orthogonal complement of a k-dimensional subspace of R^n has dimension
(A) k
(B) n-k
(C) n
(D) 0

Q.108 NAT · 1 mark · Perp
In R^3, the orthogonal complement of span{(1,0,0),(0,1,0)} has dimension _____.
NAT · enter the numerical value

Q.109 MCQ · 1 mark · Proj unique
The orthogonal projection of v onto a subspace W is the unique w in W such that
(A) v-w is orthogonal to W
(B) w=0
(C) ||w|| is maximal among all of R^n
(D) w=v always

Q.110 NAT · 1 mark · Refl
Householder reflection I-2uu^T for unit u is orthogonal. det equals _____.
NAT · enter the numerical value

Q.111 MCQ · 1 mark · PSD
A Gram matrix A^T A is always
(A) positive semidefinite
(B) indefinite
(C) orthogonal
(D) nilpotent

Q.112 NAT · 1 mark · Gram rank
rank(A^T A) equals rank(A). If A is 4x3 of rank 2, rank(A^T A) is _____.
NAT · enter the numerical value

Q.113 MCQ · 1 mark · Cholesky
A symmetric positive definite matrix admits
(A) a unique Cholesky factor R^T R with R upper triangular, positive diagonal
(B) no factorisation
(C) only SVD with zero values
(D) P^2=P

Q.114 NAT · 1 mark · SPD check
Leading minors of [[2,1],[1,2]] are 2 and 3, both positive, so the matrix is SPD. det equals _____.
NAT · enter the numerical value

Q.115 MCQ · 1 mark · Quad classify
q=x^2+y^2 is
(A) positive definite on R^2
(B) indefinite
(C) negative definite
(D) zero

Q.116 MCQ · 1 mark · Sig
The inertia of a real symmetric matrix (numbers of +, -, 0 eigenvalues) is
(A) invariant under congruence
(B) changed by any invertible congruence
(C) the same as the Jordan form always
(D) always (n,0,0)

Q.117 NAT · 1 mark · Inertia
diag(1,-2,0) has how many negative eigenvalues? _____.
NAT · enter the numerical value

Q.118 MCQ · 1 mark · Row op det
Swapping two rows of A multiplies det by
(A) +1
(B) -1
(C) 0
(D) 2

Q.119 NAT · 1 mark · Scale row
Multiply one row of I_3 by 5. The determinant becomes _____.
NAT · enter the numerical value

Q.120 MCQ · 1 mark · Cramer
Cramer's rule expresses each x_i as
(A) det(A_i)/det(A) when det A !=0
(B) tr A
(C) rank A
(D) ||A||

Q.121 NAT · 1 mark · Cramer x
[[1,1],[0,1]] [x,y]^T=[1,1]^T. Then y=_____.
NAT · enter the numerical value

Q.122 NAT · 1 mark · Cramer x2
Same system: x equals _____.
NAT · enter the numerical value

Q.123 MCQ · 1 mark · Adjugate
A^{-1} = (1/det A) adj(A) when det A !=0. adj(A) is
(A) the transpose of the cofactor matrix
(B) A^T
(C) A^2
(D) 0

Q.124 NAT · 1 mark · Cofactor
Cofactor C_{11} of [[1,2],[3,4]] is det[4]=_____.
NAT · enter the numerical value

Q.125 NAT · 1 mark · Adj 2
adj([[1,2],[3,4]])=[[4,-2],[-3,1]]. det A=-2 so A^{-1} has (1,1) entry 4/(-2)=_____.
NAT · enter the numerical value

Q.126 MCQ · 1 mark · Vandermonde
The Vandermonde matrix on distinct nodes is
(A) invertible
(B) always singular
(C) orthogonal
(D) nilpotent

Q.127 NAT · 1 mark · Van 2
det [[1,1],[x,y]] = y-x. For x=2,y=5 this det is _____.
NAT · enter the numerical value

Q.128 MCQ · 1 mark · Companion
The companion matrix of t^2-3t+2 has characteristic polynomial
(A) t^2-3t+2
(B) t^2+1
(C) t-1
(D) 0

Q.129 NAT · 1 mark · Comp eig
Roots of t^2-3t+2=0 are 1 and _____.
NAT · enter the numerical value

Q.130 MCQ · 1 mark · Markov
A column-stochastic matrix has 1 as eigenvalue because
(A) columns sum to 1 so A^T 1 = 1 (or A1=1 depending on convention)
(B) trace=1
(C) det=1
(D) rank=1 always

Q.131 NAT · 1 mark · Perron
The Perron root of a positive matrix is simple and positive. For [[1,1],[1,1]] it is _____.
NAT · enter the numerical value

Q.132 MCQ · 1 mark · Graph lap
The Laplacian of an undirected graph is PSD with kernel dimension equal to
(A) the number of connected components
(B) the number of edges
(C) n always
(D) 0 always

Q.133 NAT · 1 mark · Path3
Path on 3 vertices: Laplacian eigenvalues include 0. Rank of L is _____.
NAT · enter the numerical value

Q.134 MCQ · 1 mark · Adj spec
The adjacency matrix of an undirected simple graph is
(A) real symmetric (hence orthogonally diagonalizable)
(B) skew and invertible
(C) a projection
(D) nilpotent of index 2 always

Q.135 NAT · 1 mark · K2
Complete graph K2 has adjacency [[0,1],[1,0]]. Eigenvalues 1 and _____.
NAT · enter the numerical value

Q.136 MCQ · 1 mark · Incidence
For an oriented graph, B B^T is
(A) the Laplacian (unnormalized)
(B) I
(C) A^2
(D) 0

Q.137 NAT · 1 mark · Cycle even
A 4-cycle is bipartite so -2 is an adjacency eigenvalue of C4 (regular of degree 2). The largest adjacency eigenvalue is _____.
NAT · enter the numerical value

Q.138 MCQ · 1 mark · Shift
`A-lambda I is singular iff`
(A) lambda is an eigenvalue
(B) A=0
(C) lambda=0 always
(D) A is orthogonal

Q.139 NAT · 1 mark · Eig 0
A is singular iff 0 is an eigenvalue. det([[1,1],[2,2]]) is _____.
NAT · enter the numerical value

Q.140 MCQ · 1 mark · Left eig
`A left eigenvector satisfies w^T A = lambda w^T. For real symmetric A, left and right eigenspaces`
(A) coincide
(B) are always orthogonal to each other in a different sense than usual
(C) never exist
(D) are {0}

Q.141 NAT · 1 mark · Rayleigh
For symmetric A, the Rayleigh quotient v^T A v / v^T v at an eigenvector equals the eigenvalue. For A=diag(2,5) and v=e2 it is _____.
NAT · enter the numerical value

Q.142 MCQ · 1 mark · Min max
The smallest eigenvalue of a real symmetric A is the minimum of the Rayleigh quotient. For diag(2,5) that minimum is _____.
(A) 2
(B) 5
(C) 7
(D) 0

Q.143 NAT · 1 mark · Weyl
Eigenvalues of A and A+E differ by at most ||E|| (in a suitable norm). If ||E||=0.1 and gap is 5, a crude sin Theta bound 0.1/5 equals _____ (two decimals).
NAT · enter the numerical value

Q.144 MCQ · 1 mark · Exp
exp(tA) for a square A is the fundamental matrix of u'=Au. exp(0)=
(A) I
(B) 0
(C) A
(D) tI

Q.145 NAT · 1 mark · Exp diag
exp(diag(0, ln 2)) = diag(1,2). The (2,2) entry is _____.
NAT · enter the numerical value

Q.146 MCQ · 1 mark · Nilp exp
For N^2=0, exp(N)=
(A) I+N
(B) I
(C) 0
(D) N

Q.147 NAT · 1 mark · Jordan
Number of Jordan blocks for lambda equals geo multiplicity. For a single 2x2 block, geo multiplicity is _____.
NAT · enter the numerical value

Q.148 MCQ · 1 mark · Defective
A defective eigenvalue has
(A) geo multiplicity < algebraic multiplicity
(B) geo > alg
(C) no eigenvector
(D) det=1

Q.149 NAT · 1 mark · Index
Index of eigenvalue 0 for [[0,1],[0,0]] (size of largest block) is _____.
NAT · enter the numerical value

Q.150 MCQ · 1 mark · Schur
Over C every square matrix is unitarily similar to
(A) upper triangular (Schur form)
(B) a projection
(C) I
(D) a permutation

Q.151 MCQ · 1 mark · Real schur
A real matrix with a complex conjugate eigenvalue pair has a real Schur form containing
(A) a 2x2 block
(B) only 1x1 blocks always
(C) a zero row
(D) P^2=P

Q.152 NAT · 1 mark · Rot det
det of a 2x2 rotation is _____.
NAT · enter the numerical value

Q.153 MCQ · 1 mark · Reflect det
det of a Householder reflector is
(A) -1
(B) 1
(C) 0
(D) 2

Q.154 NAT · 1 mark · Pauli
sigma_z = [[1,0],[0,-1]] has eigenvalues 1 and _____.
NAT · enter the numerical value

Q.155 MCQ · 1 mark · Hermitian
A complex matrix with A^*=A has
(A) real eigenvalues
(B) pure imaginary eigenvalues
(C) det 0
(D) trace 0 always

Q.156 NAT · 1 mark · Hermit eig
[[0,-i],[i,0]] is Hermitian. Eigenvalues are 1 and _____.
NAT · enter the numerical value

Q.157 MCQ · 1 mark · Unitary
A unitary matrix has
(A) eigenvalues on the unit circle
(B) eigenvalues 0,1
(C) det 0
(D) trace n

Q.158 NAT · 1 mark · Fourier 2
The 2-point DFT matrix F/sqrt(2) with F=[[1,1],[1,-1]] is unitary up to the scale already. F^2 equals 2I. det F is _____.
NAT · enter the numerical value

Q.159 MCQ · 1 mark · Norm equiv
On finite-dimensional R^n all norms are equivalent. In particular ||x||_1 / ||x||_2 is bounded by
(A) sqrt(n)
(B) 1/n
(C) 0
(D) n^2

Q.160 NAT · 1 mark · L1
||(3,-4)||_1 is _____.
NAT · enter the numerical value

Q.161 NAT · 1 mark · Linf
||(3,-4)||_inf is _____.
NAT · enter the numerical value

Q.162 MCQ · 1 mark · Op norm
The operator 2-norm of A is the largest
(A) singular value
(B) eigenvalue always (even if complex)
(C) diagonal entry
(D) Frobenius entry

Q.163 NAT · 1 mark · Sigma min
For an invertible A, ||A^{-1}||_2 = 1/sigma_min. If sigma = 4 and 0.5, ||A^{-1}||_2 is _____.
NAT · enter the numerical value

Q.164 MCQ · 1 mark · Low rank
Best rank-k approximation in Frobenius (or 2) norm is truncated SVD (Eckart-Young). For a rank-1 matrix the best rank-0 approx error is
(A) the largest singular value (i.e. ||A||)
(B) 0
(C) det A
(D) n

Q.165 NAT · 1 mark · EY
A=diag(5,1). Best rank-1 Frobenius error is the discarded sigma=_____.
NAT · enter the numerical value

Q.166 MCQ · 1 mark · PCA link
PCA on centered data uses the leading eigenvectors of the covariance (a PSD matrix). The first PC maximises
(A) variance of the projection
(B) the mean
(C) k in k-means
(D) Gini

Q.167 NAT · 1 mark · PCA frac
Eigenvalues of covariance 4 and 1. Fraction of variance on PC1 is _____ (one decimal).
NAT · enter the numerical value

Q.168 MCQ · 1 mark · Whitening
If Sigma = U D U^T with D > 0, the whitening map D^{-1/2} U^T sends the Gaussian to
(A) isotropic (covariance I)
(B) a Poisson
(C) a Bernoulli
(D) k-means

Q.169 NAT · 1 mark · Mahalanobis
For Sigma=I, Mahalanobis distance of (3,4) from 0 is _____.
NAT · enter the numerical value

Q.170 MCQ · 1 mark · Partition inv
Block inversion of [[A,B],[C,D]] uses the Schur complement D-C A^{-1} B when A is invertible. If B=0,C=0 the inverse is
(A) diag(A^{-1}, D^{-1})
(B) diag(A,D)
(C) 0
(D) I

Q.171 NAT · 1 mark · Schur scalar
Schur complement of 2 in [[2,1],[1,2]] is 2-1*(1/2)*1=_____. (one decimal)
NAT · enter the numerical value

Q.172 MCQ · 1 mark · Kronecker
Eigenvalues of A tensor I_k are those of A, each repeated k times. For A=diag(2,5) and k=2, the largest eigenvalue is
(A) 2
(B) 5
(C) 10
(D) 7

Q.173 NAT · 1 mark · Kron dim
A 2x2 tensored with I_3 is 6x6. Size n equals _____.
NAT · enter the numerical value

Q.174 MCQ · 1 mark · Vec
vec(AXB)=(B^T tensor A) vec(X). This identity is used in
(A) Lyapunov / Kronecker linear systems
(B) k-means only
(C) SQL
(D) A* search

Q.175 NAT · 1 mark · Lyap
For 1x1, AX+XA^T=Q with A=2, Q=8, X is _____.
NAT · enter the numerical value

Q.176 MCQ · 1 mark · Gershgorin
Every eigenvalue lies in at least one Gershgorin disc. For [[3,1],[0,2]] the discs are centered at 3 (radius 1) and 2 (radius 0). 2 is
(A) an eigenvalue (and the radius-0 disc is a point)
(B) not an eigenvalue
(C) the determinant
(D) the nullity

Q.177 NAT · 1 mark · Gersh r
For [[10,2],[1,4]] the disc at 10 has radius _____.
NAT · enter the numerical value

Q.178 MCQ · 1 mark · Strict diag dom
A strictly diagonally dominant matrix is
(A) nonsingular
(B) always orthogonal
(C) a projection
(D) nilpotent

Q.179 NAT · 1 mark · DD
[[3,-1],[-1,3]] is SPD (diagonally dominant with positive diag). det is _____.
NAT · enter the numerical value

Q.180 MCQ · 1 mark · M-matrix
A nonsingular M-matrix has inverse
(A) entrywise nonnegative
(B) skew
(C) orthogonal
(D) nilpotent

Q.181 NAT · 1 mark · Inv pos
[[2,-1],[-1,2]]^{-1} = (1/3)[[2,1],[1,2]]. The (1,1) entry as a simplified fraction 2/3 in decimal (two places) is _____.
NAT · enter the numerical value

Q.182 MCQ · 1 mark · Toeplitz
A Toeplitz matrix is constant along
(A) diagonals
(B) rows
(C) columns
(D) the kernel

Q.183 NAT · 1 mark · Toep 2
The Toeplitz [[2,1],[1,2]] has det _____.
NAT · enter the numerical value

Q.184 MCQ · 1 mark · Hankel
A Hankel matrix is constant along
(A) anti-diagonals
(B) the main diagonal only
(C) the border
(D) eigenvectors

Q.185 NAT · 1 mark · Circ
The circulant generated by (1,0,0) is I_3. det is _____.
NAT · enter the numerical value

Q.186 MCQ · 1 mark · DFT circ
Circulants are diagonalised by the DFT matrix. Eigenvalues are
(A) the DFT of the first row/column
(B) all 1
(C) all 0
(D) the pivots of LU always

Q.187 NAT · 1 mark · Circ eig
Circulant of (1,1) in 2D is [[1,1],[1,1]]. Eigenvalues 2 and _____.
NAT · enter the numerical value

Q.188 NAT · 1 mark · Sherman0
Sherman-Morrison for 1x1: (1+3)^{-1} as a decimal (two places) is _____.
NAT · enter the numerical value

Q.189 MCQ · 1 mark · Woodbury
The Woodbury identity generalises Sherman-Morrison to
(A) A+U C V
(B) A^2
(C) QR
(D) P^2=P

Q.190 NAT · 1 mark · Update det
det(I+uv^T)=1+v^T u. For u=v=(1,0)^T this det is _____.
NAT · enter the numerical value

Q.191 MCQ · 1 mark · Matrix det lemma
Same identity. For orthogonal projector uv^T of rank 1, det(I-uv^T) with ||u||=1,v=u is
(A) 0
(B) 1
(C) 2
(D) -1

Q.192 NAT · 1 mark · I-P rank
If P is rank-1 projection on R^3, rank(I-P) is _____.
NAT · enter the numerical value

Q.193 MCQ · 1 mark · Trace cyclic 2
tr(ABC)=tr(BCA)=tr(CAB) when products are defined. tr(AB)-tr(BA) is
(A) 0
(B) 1
(C) n
(D) det(AB)

Q.194 NAT · 1 mark · Commutator tr
tr([A,B]) is always _____.
NAT · enter the numerical value

Q.195 MCQ · 1 mark · sl n
Matrices of trace 0 form a subspace of M_n(R) of dimension
(A) n^2-1
(B) n^2
(C) n
(D) 1

Q.196 NAT · 1 mark · sl2
dim of 2x2 trace-zero matrices is _____.
NAT · enter the numerical value

Q.197 MCQ · 1 mark · so n
Real skew-symmetric n x n matrices have dimension n(n-1)/2. For n=3 that is _____.
(A) 3
(B) 6
(C) 9
(D) 1

Q.198 NAT · 1 mark · so3
dim so(3) is _____.
NAT · enter the numerical value

Q.199 MCQ · 1 mark · GL
GL_n(R) is the set of
(A) invertible n x n real matrices
(B) all matrices
(C) projections
(D) symmetric matrices

Q.200 NAT · 1 mark · GL det
The matrix [[1,0],[0,-1]] is in GL_2(R) because det equals _____.
NAT · enter the numerical value

Q.201 MCQ · 1 mark · O n
O(n) is {Q: Q^T Q=I}. It has two components according to
(A) det = +-1
(B) trace
(C) rank
(D) nilpotency

Q.202 NAT · 1 mark · SO2
A rotation by 0 is I_2, det _____.
NAT · enter the numerical value

Q.203 MCQ · 1 mark · SPD cone
The set of SPD matrices is
(A) an open convex cone
(B) a subspace
(C) a finite set
(D) O(n)

Q.204 NAT · 1 mark · Mid SPD
The average of I and 2I is 1.5 I, which is SPD. Smallest eigenvalue is _____ (one decimal).
NAT · enter the numerical value

Q.205 MCQ · 1 mark · Log SPD
The principal log of an SPD matrix is the symmetric log via the spectral calculus. log I =
(A) 0
(B) I
(C) -I
(D) undefined

Q.206 NAT · 1 mark · Sqrt SPD
The unique SPD square root of diag(4,9) is diag(2,3). Trace is _____.
NAT · enter the numerical value

Q.207 MCQ · 1 mark · Polar
The polar factorisation A=U P with P=SPD and U orthogonal (for invertible A). P equals
(A) sqrt(A^T A)
(B) A
(C) A^{-1}
(D) 0

Q.208 NAT · 1 mark · Polar 1d
For A=[-3] in R^{1x1}, the polar 'U' is sign=-1 and P=_____.
NAT · enter the numerical value

Q.209 MCQ · 1 mark · Thin SVD
Economy SVD of a tall full-column-rank A has U of size
(A) m x n (thin)
(B) m x m always only
(C) n x n only
(D) 1 x 1

Q.210 NAT · 1 mark · Full SVD U
Full SVD of a 3x2 matrix has U of size 3x3. Number of rows of U is _____.
NAT · enter the numerical value

Q.211 MCQ · 1 mark · Left null
The left nullspace is ker(A^T). Its dimension is m-rank(A). For 3x2 rank 2 this is _____.
(A) 1
(B) 2
(C) 3
(D) 0

Q.212 NAT · 1 mark · Left null dim
m=5, rank=3, dim left nullspace is _____.
NAT · enter the numerical value

Q.213 MCQ · 1 mark · RREF unique
The reduced row echelon form of a matrix is
(A) unique
(B) highly non-unique
(C) always I
(D) always 0

Q.214 NAT · 1 mark · Pivots count
Number of pivots in RREF equals the rank. For a rank-3 matrix that number is _____.
NAT · enter the numerical value

Q.215 MCQ · 1 mark · Full rank sq
A square matrix has full rank iff
(A) it is invertible
(B) trace !=0
(C) all entries nonzero
(D) it is symmetric

Q.216 NAT · 1 mark · Inv check
det([[1,2],[3,5]])=5-6=_____.
NAT · enter the numerical value

Q.217 MCQ · 1 mark · 2x2 inv formula
[[a,b],[c,d]]^{-1} = (1/(ad-bc)) [[d,-b],[-c,a]]. For ad-bc=1, the (1,1) entry is
(A) d
(B) a
(C) 0
(D) 1 always

Q.218 NAT · 1 mark · Formula
Inverse of [[2,0],[0,4]] is diag(1/2, 1/4). The (2,2) entry as a decimal is _____.
NAT · enter the numerical value

Q.219 MCQ · 1 mark · PA=LU
Partial pivoting writes PA=LU. P is a
(A) permutation matrix
(B) projection
(C) diagonal of eigenvalues
(D) SVD factor

Q.220 NAT · 1 mark · Pivots LU
Without pivoting, LU of [[1e-20,1],[1,1]] is numerically unsafe because the first pivot is ~0. The exact (1,1) pivot is 10^{-20}, approximately _____. (enter 0)
NAT · enter the numerical value

Q.221 MCQ · 1 mark · Growth
The growth factor in GE measures
(A) how large entries become relative to ||A||
(B) the rank
(C) the trace
(D) k-means k

Q.222 NAT · 1 mark · Need 260 pad unique
Number of leading principal minors of an n x n matrix is n. For n=4 that is _____.
NAT · enter the numerical value

Q.223 MCQ · 1 mark · Sylvester PD
Sylvester's criterion: SPD iff all leading principal minors are positive. For [[1,0],[0,2]] they are 1 and 2, so the matrix is
(A) SPD
(B) indefinite
(C) NSD
(D) singular

Q.224 NAT · 1 mark · Minor 2
Leading 2x2 minor of diag(1,2,3) is _____.
NAT · enter the numerical value

Q.225 MCQ · 1 mark · Bordered skip
A and B are congruent if B=P^T A P for invertible P. Congruence of symmetric matrices preserves
(A) inertia
(B) eigenvalues exactly
(C) the (1,2) entry
(D) nilpotency index always

Q.226 NAT · 1 mark · Cong I
P^T I P = P^T P, which is SPD if P is invertible. For P=2I_2, P^T P = 4 I, det _____.
NAT · enter the numerical value

Q.227 MCQ · 1 mark · Isometry
A linear isometry of R^n Euclidean is an
(A) orthogonal matrix (possibly det -1)
(B) any invertible matrix
(C) any symmetric matrix
(D) a projection

Q.228 NAT · 1 mark · Preserve
||Qx|| for orthogonal Q and x=(3,4) is _____.
NAT · enter the numerical value

Q.229 MCQ · 1 mark · Volume
|det A| is the volume scaling of the unit cube under A. For a 90-degree rotation this factor is
(A) 1
(B) 0
(C) 2
(D) pi

Q.230 NAT · 1 mark · Shear vol
Shear [[1,1],[0,1]] has det _____.
NAT · enter the numerical value

Q.231 MCQ · 1 mark · Orient
det A > 0 means A is
(A) orientation-preserving
(B) singular
(C) a reflection
(D) PSD

Q.232 NAT · 1 mark · Refl orient
diag(1,-1) has det _____.
NAT · enter the numerical value

Q.233 NAT · 1 mark · Skew 90
[[0,-1],[1,0]] represents rotation by 90 degrees. Its squared Frobenius norm is _____.
NAT · enter the numerical value

Q.234 MCQ · 1 mark · Area para
Area of the parallelogram spanned by (1,0) and (1,1) is |det| = _____.
(A) 1
(B) 0
(C) 2
(D) sqrt(2)

Q.235 NAT · 1 mark · Volume para
Volume of the parallelepiped of e1,e2,e3 is _____.
NAT · enter the numerical value

Q.236 MCQ · 1 mark · Lin dep test
Vectors are dependent iff the matrix with those columns has
(A) det 0 (when square) / rank < number of vectors
(B) trace 0
(C) positive entries
(D) Q^T Q=I

Q.237 NAT · 1 mark · Three vec
Columns (1,0,0),(0,1,0),(1,1,0) in R^3: rank is _____.
NAT · enter the numerical value

Q.238 MCQ · 1 mark · Hyperplane
A hyperplane through the origin in R^n is a subspace of dimension
(A) n-1
(B) n
(C) 1
(D) 0

Q.239 NAT · 1 mark · Codim
Codimension of a hyperplane in R^7 is _____.
NAT · enter the numerical value

Q.240 MCQ · 1 mark · Direct sum R
R^2 = span{e1} oplus span{e2} is a
(A) direct sum (internal)
(B) not a sum
(C) tensor product only
(D) quotient only

Q.241 NAT · 1 mark · Sum dim 2
dim(span{e1}+span{e2}) in R^2 is _____.
NAT · enter the numerical value

Q.242 MCQ · 1 mark · Quotient
dim(V/W) = dim V - dim W. For V=R^4, W a line, this is _____.
(A) 3
(B) 4
(C) 1
(D) 0

Q.243 NAT · 1 mark · Quot
dim(R^5 / R^2) if R^2 is a subspace is _____.
NAT · enter the numerical value

Q.244 MCQ · 1 mark · Dual
The dual space V* has the same dimension as V when dim V is finite. dim (R^n)* is _____.
(A) n
(B) n^2
(C) 1
(D) 0

Q.245 NAT · 1 mark · Dual n
dim (R^6)* is _____.
NAT · enter the numerical value

Q.246 MCQ · 1 mark · Annihilator
dim W^0 = dim V - dim W. For a plane in R^4, dim annihilator is _____.
(A) 2
(B) 4
(C) 1
(D) 0

Q.247 NAT · 1 mark · Ann
Annihilator of {0} in R^n has dimension _____.
NAT · enter the numerical value

Q.248 MCQ · 1 mark · Rank A A.T
rank(A A^T)=rank(A). For A 2x5 rank 2, rank(A A^T) is _____.
(A) 2
(B) 5
(C) 0
(D) 7

Q.249 NAT · 1 mark · Gram rows
A A^T is 2x2 when A is 2 x anything. If A has orthonormal rows, A A^T = I_2, det _____.
NAT · enter the numerical value

Q.250 MCQ · 1 mark · Frame
Rows orthonormal means A A^T=I (a tight frame / Parseval in that case). Then A is a
(A) right inverse of A^T, i.e. A is a co-isometry
(B) singular always
(C) projection of rank 0
(D) nilpotent

Q.251 NAT · 1 mark · Coiso
If A is 2x3 with orthonormal rows, A A^T is I_2. Number of rows is _____.
NAT · enter the numerical value

Q.252 MCQ · 1 mark · Isometry tall
A tall matrix with orthonormal columns satisfies A^T A=I. Then ||Ax||=||x||. For A 3x2 such, rank is _____.
(A) 2
(B) 3
(C) 1
(D) 0

Q.253 MCQ · 1 mark · Trace inner
< A,B > _F = tr(A^T B) is an inner product on matrices. ||I_2||_F is _____.
(A) sqrt(2)
(B) 2
(C) 1
(D) 0

Q.254 NAT · 1 mark · Frob I
||I_3||_F is _____ (one decimal, sqrt(3)~1.7).
NAT · enter the numerical value

Q.255 MCQ · 1 mark · Onb matrices
E_{ij} the matrix units form an orthonormal basis of M_{m n} for the Frobenius inner product. There are how many? mn. For 2x2: _____.
(A) 4
(B) 2
(C) 8
(D) 1

Q.256 NAT · 1 mark · Units
Number of matrix units in M_{3,2} is _____.
NAT · enter the numerical value

Q.257 MCQ · 1 mark · Vec inner
< A,B > _F = vec(A)·vec(B). For A=B=I_2 this inner product is _____.
(A) 2
(B) 4
(C) 1
(D) 0

Q.258 NAT · 1 mark · Vec len
vec(I_2) has Euclidean length sqrt(2) ~ _____ (one decimal).
NAT · enter the numerical value

Q.259 MCQ · 1 mark · Hadamard prod
The entrywise (Hadamard) product of two PSD matrices is PSD (Schur product theorem). I odot I equals
(A) I
(B) 0
(C) all-ones
(D) 2I

Q.260 NAT · 1 mark · Hadamard ones
(J odot I) for 2x2 is I. det is _____.
NAT · enter the numerical value

Q.261 MCQ · 1 mark · All-ones eig more
J_n /n is the projection onto span{1}. For n=4 its rank is _____.
(A) 1
(B) 4
(C) 0
(D) 2

Q.262 NAT · 1 mark · Centering
The centering matrix I-J/n for n=3 has rank _____.
NAT · enter the numerical value

Q.263 MCQ · 1 mark · Centering PSD
I-J/n is
(A) a projection (hence PSD with eigs 0,1)
(B) indefinite
(C) orthogonal with det -1 always
(D) nilpotent

Q.264 NAT · 1 mark · Cent eig
I-J/n has eigenvalues 1 (n-1 times) and one 0. For n=5 number of 1s is _____.
NAT · enter the numerical value

Q.265 MCQ · 1 mark · Covariance as Gram
A sample covariance (up to scale) is a Gram matrix of centered data, hence
(A) PSD
(B) orthogonal
(C) a permutation
(D) nilpotent

Q.266 NAT · 1 mark · Sample cov rank
n=3 points in R^2, centered, the 2x2 covariance has rank at most _____.
NAT · enter the numerical value

Q.267 MCQ · 1 mark · n-1
Centered n points span at most n-1 dimensions. For n=2 that max is _____.
(A) 1
(B) 2
(C) 0
(D) 3

Q.268 NAT · 1 mark · Two pts
Two centered points in R^100 lie in a subspace of dimension at most _____.
NAT · enter the numerical value

Q.269 MCQ · 1 mark · Subspace
The vector space of 2x2 real symmetric matrices has dimension
(A) 2
(B) 3
(C) 4
(D) 1

Q.270 NAT · 1 mark · Independence
Maximum size of a linearly independent set in R^5 is _____.
NAT · enter the numerical value

Q.271 NAT · 1 mark · Rank
The 5x5 all-ones matrix has rank _____.
NAT · enter the numerical value

Q.272 NAT · 1 mark · Nullity
An invertible 4x4 matrix has nullity _____.
NAT · enter the numerical value

Q.273 NAT · 1 mark · LU
For A=[[2,1],[4,3]], L has 1s on the diagonal and L_{21} equals _____.
NAT · enter the numerical value

Q.274 MCQ · 1 mark · Subspace
Polynomials of degree exactly 3 (with the zero polynomial excluded) form
(A) a subspace of P_3
(B) not a subspace, because 0 is not of exact degree 3
(C) P_2
(D) R^3

Q.275 NAT · 1 mark · Rank
Rank of diag(2,0,3,0) is _____.
NAT · enter the numerical value

Q.276 NAT · 1 mark · Subspace
Dimension of {x in R^4 : x1=x2 and x3=0} is _____.
NAT · enter the numerical value

UNIT 2 SOLUTIONS

Q.1 [MSQ] A;C — Must contain 0 and be closed under addition. The product constraint is not a subspace; the affine plane misses 0.

Q.2 [MCQ] B — The first two vectors already form a basis.

Q.3 [MSQ] A;C — A scalar multiple pair is dependent; the zero vector kills independence.

Q.4 [MCQ] B — Dimension is well-defined.

Q.5 [NAT] 6 — Six entries.

Q.6 [NAT] 5 — 3+3-1=5.

Q.7 [MCQ] A — Definition of internal direct sum.

Q.8 [NAT] 1 — Row2=2*Row1; Row3=0.

Q.9 [NAT] 3 — 5-2 by rank-nullity.

Q.10 [MCQ] B — Domain dimension.

Q.11 [MCQ] A — Row rank = column rank.

Q.12 [MCQ] B — Second equation is twice the first; one free variable.

Q.13 [MCQ] B — 3*2=6 != 5.

Q.14 [NAT] 2 — 4-2.

Q.15 [NAT] -2 — 4-6=-2.

Q.16 [NAT] 1 — Rotation by 90 degrees, det=1.

Q.17 [NAT] -10 — Product of diagonals.

Q.18 [MCQ] B — Multiplicativity.

Q.19 [MCQ] B — det A * det A^{-1}=1.

Q.20 [NAT] 9 — 3^2.

Q.21 [MCQ] B — Trace 3, det 5, discriminant 9-20 < 0.

Q.22 [MCQ] B — Rotation by 90 degrees: characteristic t^2+1=0.

Q.23 [NAT] 7 — Equals trace.

Q.24 [NAT] 10 — Equals det=12-2=10.

Q.25 [MCQ] B — Ae2=3 e2.

Q.26 [MCQ] B — A basis of eigenvectors.

Q.27 [MCQ] B — Nilpotent index 2, single eigenvector line.

Q.28 [NAT] 3 — Trace.

Q.29 [MCQ] B — p(A)=0.

Q.30 [NAT] 1 — lambda^2=lambda.

Q.31 [MCQ] A — Symmetric idempotent.

Q.32 [MCQ] A — Columns orthonormal.

Q.33 [NAT] 1 — det(Q^T Q)=1.

Q.34 [MCQ] C — Rotation by 2pi.

Q.35 [MCQ] A — 2026 = 5*405 + 1, so M^{2026}=M.

Q.36 [NAT] 3 — First coordinate.

Q.37 [NAT] 0.36 — (3/5)^2=0.36.

Q.38 [MCQ] C — Eigenvalues 2 and -3.

Q.39 [MCQ] A — Sylvester / spectral.

Q.40 [NAT] 0 — (x+y)^2.

Q.41 [MCQ] A — Classical criterion.

Q.42 [MCQ] A — Existence of SVD.

Q.43 [NAT] 2 — Absolute eigenvalues for diagonal.

Q.44 [NAT] 2.2 — sqrt(1+4)=sqrt5~2.236.

Q.45 [NAT] 4 — Largest |eigenvalue| here.

Q.46 [MCQ] B — Full column rank = > left inverse.

Q.47 [NAT] 1 — 3-2.

Q.48 [MCQ] B — Elementary row ops.

Q.49 [NAT] 2 — AB=[[2,1],[1,0]], tr=2.

Q.50 [MCQ] A — Same eigenvalues and algebraic multiplicities.

Q.51 [NAT] -1 — x+y+z=0.

Q.52 [MCQ] B — Block triangular/diagonal.

Q.53 [NAT] 2 — 1*2.

Q.54 [MCQ] A — (I-P)^2=I-P.

Q.55 [NAT] k — Trace equals rank for projections.

Q.56 [MCQ] A — Spectral theorem.

Q.57 [MCQ] B — det A = det(A^T)=det(-A)=(-1)^n det A, n odd = > det=0.

Q.58 [NAT] 4 — 4.

Q.59 [MCQ] B — R2 < - R2-R1.

Q.60 [NAT] 2 — Swap to put 2 in (1,1).

Q.61 [MCQ] A — Full rank square.

Q.62 [NAT] 1 — 1.

Q.63 [MCQ] A — Definition. Also equals ker(A^T)^perp.

Q.64 [MCQ] B — Fundamental theorem.

Q.65 [NAT] 2 — min=2.

Q.66 [MCQ] B — All columns are multiples of u.

Q.67 [NAT] 1 — One nonzero eigenvalue.

Q.68 [MCQ] A — 1 is always an eigenvalue.

Q.69 [NAT] 0.3 — trace=1.3.

Q.70 [MCQ] A — Normal equations.

Q.71 [NAT] 1.2 — (2+4)/(1+4)=6/5=1.2.

Q.72 [MCQ] A — Left inverse.

Q.73 [NAT] 100 — 10/0.1.

Q.74 [MCQ] A — Sensitivity of Ax=b.

Q.75 [NAT] -1 — Shear inverse.

Q.76 [MCQ] A — char poly product (t-a_ii).

Q.77 [NAT] 1 — One independent column.

Q.78 [MCQ] A — Same linear factors, possibly lower multiplicity.

Q.79 [NAT] 1 — dim ker(A-2I)=1.

Q.80 [MCQ] A — Standard.

Q.81 [NAT] 2 — One independent constraint.

Q.82 [MCQ] B — Particular + homogeneous.

Q.83 [NAT] 3 — Hyperplane.

Q.84 [MCQ] A — Standard. The second needs 1/sqrt(2).

Q.85 [NAT] -0.71 — v2=(0,1)-proj= (-1/2,1/2), normalize.

Q.86 [MCQ] A — Thin or full QR.

Q.87 [NAT] -2 — -1-1=-2.

Q.88 [MCQ] A — Rows of I permuted.

Q.89 [NAT] -1 — Odd permutation.

Q.90 [MCQ] B — Block multiplication.

Q.91 [NAT] 5 — 5.

Q.92 [NAT] 5 — 5.

Q.93 [MCQ] A — a times I.

Q.94 [NAT] 8 — 2^3.

Q.95 [MCQ] A — Classical fact for distinct eigs.

Q.96 [NAT] 2 — p(2)=4-2=2.

Q.97 [MCQ] A — Definition.

Q.98 [NAT] 2 — Rank.

Q.99 [MCQ] A — Equivalent: invertible, full rank.

Q.100 [NAT] 1 — I.

Q.101 [MCQ] A — Change of basis.

Q.102 [NAT] 2 — 1-2+3=2.

Q.103 [MCQ] A — CS.

Q.104 [NAT] 5 — 5.

Q.105 [MCQ] B — Standard lemma.

Q.106 [NAT] n — Independence.

Q.107 [MCQ] B — dim W + dim W^perp = n.

Q.108 [NAT] 1 — The z-axis.

Q.109 [MCQ] A — Normal equations.

Q.110 [NAT] -1 — One eigenvalue -1.

Q.111 [MCQ] A — x^T A^T A x = ||Ax||^2 > =0.

Q.112 [NAT] 2 — Same rank.

Q.113 [MCQ] A — Cholesky.

Q.114 [NAT] 3 — 4-1=3.

Q.115 [MCQ] A — ||(x,y)||^2.
Q.116 [MCQ] A — Sylvester's law of inertia.
Q.117 [NAT] 1 — One.
Q.118 [MCQ] B — Alternating.
Q.119 [NAT] 5 — Multilinear.
Q.120 [MCQ] A — Replace column i by b.
Q.121 [NAT] 1 — Back substitution.
Q.122 [NAT] 0 — x+y=1, y=1 = > x=0.
Q.123 [MCQ] A — Classical adjoint.
Q.124 [NAT] 4 — Minor.
Q.125 [NAT] -2 — -2.
Q.126 [MCQ] A — Distinct x_i.
Q.127 [NAT] 3 — 3.
Q.128 [MCQ] A — By construction.
Q.129 [NAT] 2 — Factors (t-1)(t-2).
Q.130 [MCQ] A — All-ones eigenvector.
Q.131 [NAT] 2 — Eigenvalues 2,0.
Q.132 [MCQ] A — Standard.
Q.133 [NAT] 2 — Connected = > rank n-1.
Q.134 [MCQ] A — A_{ij}=A_{ji}.
Q.135 [NAT] -1 — det(t^2-1).
Q.136 [MCQ] A — B incidence.
Q.137 [NAT] 2 — Degree of regular graph.
Q.138 [MCQ] A — Definition.
Q.139 [NAT] 0 — Rows dependent.
Q.140 [MCQ] A — A=A^T.
Q.141 [NAT] 5 — 5.
Q.142 [MCQ] A — Courant-Fischer.
Q.143 [NAT] 0.02 — Davis-Kahan flavour.
Q.144 [MCQ] A — Series.
Q.145 [NAT] 2 — e^{ln 2}.
Q.146 [MCQ] A — Series truncates.
Q.147 [NAT] 1 — 1.
Q.148 [MCQ] A — Definition.
Q.149 [NAT] 2 — N^2=0, N!=0.
Q.150 [MCQ] A — Schur.
Q.151 [MCQ] A — 2x2 rotation-scale blocks.
Q.152 [NAT] 1 — 1.
Q.153 [MCQ] A — One -1 eigenvalue.
Q.154 [NAT] -1 — diag.
Q.155 [MCQ] A — Hermitian spectral theorem.
Q.156 [NAT] -1 — Squares to I, det=-1.
Q.157 [MCQ] A — U^* U=I.
Q.158 [NAT] -2 — Same as Hadamard.
Q.159 [MCQ] A — CS or equivalence constants.
Q.160 [NAT] 7 — 7.
Q.161 [NAT] 4 — max abs.
Q.162 [MCQ] A — ||A||_2 = sigma_max.
Q.163 [NAT] 2 — 1/0.5.
Q.164 [MCQ] A — ||A - 0|| = sigma_1.
Q.165 [NAT] 1 — 1.
Q.166 [MCQ] A — Same as largest Rayleigh quotient.
Q.167 [NAT] 0.8 — 4/5.
Q.168 [MCQ] A — Affine.
Q.169 [NAT] 5 — Euclidean.
Q.170 [MCQ] A — Block diagonal.
Q.171 [NAT] 1.5 — 2-0.5.
Q.172 [MCQ] B — 5 repeated twice.
Q.173 [NAT] 6 — 2*3.
Q.174 [MCQ] A — Standard.
Q.175 [NAT] 2 — 2X+2X=8, X=2.
Q.176 [MCQ] A — Triangular, eigs 3 and 2.
Q.177 [NAT] 2 — Off-diagonal abs sum.
Q.178 [MCQ] A — Levy-Desplanques.
Q.179 [NAT] 8 — 9-1=8.
Q.180 [MCQ] A — Classic.
Q.181 [NAT] 0.67 — 2/3.
Q.182 [MCQ] A — T_{ij}=t_{i-j}.
Q.183 [NAT] 3 — 4-1.
Q.184 [MCQ] A — H_{ij}=h_{i+j}.
Q.185 [NAT] 1 — 1.
Q.186 [MCQ] A — Classic.
Q.187 [NAT] 0 — J.
Q.188 [NAT] 0.25 — 1/4.
Q.189 [MCQ] A — Low-rank updates.
Q.190 [NAT] 2 — 1+1=2. Matrix [[2,0],[0,1]].
Q.191 [MCQ] A — 1-1=0, I-P is singular along u.
Q.192 [NAT] 2 — Complement.
Q.193 [MCQ] A — Cyclic.
Q.194 [NAT] 0 — tr AB - tr BA=0.
Q.195 [MCQ] A — One linear constraint.
Q.196 [NAT] 3 — 4-1.
Q.197 [MCQ] A — 3.
Q.198 [NAT] 3 — 3.
Q.199 [MCQ] A — Group under multiplication.
Q.200 [NAT] -1 — nonzero.
Q.201 [MCQ] A — SO(n) is det 1.
Q.202 [NAT] 1 — 1.
Q.203 [MCQ] A — Not a subspace (I and -I).
Q.204 [NAT] 1.5 — 1.5.
Q.205 [MCQ] A — log 1 = 0.
Q.206 [NAT] 5 — 5.
Q.207 [MCQ] A — Analog of z=e^{i theta} |z|.
Q.208 [NAT] 3 — abs.
Q.209 [MCQ] A — Thin U.
Q.210 [NAT] 3 — m.
Q.211 [MCQ] A — 1.
Q.212 [NAT] 2 — 5-3.
Q.213 [MCQ] A — Uniqueness of RREF.
Q.214 [NAT] 3 — 3.
Q.215 [MCQ] A — Equivalent.
Q.216 [NAT] -1 — nonzero = > invertible.
Q.217 [MCQ] A — d.
Q.218 [NAT] 0.25 — 0.25.
Q.219 [MCQ] A — Row swaps.
Q.220 [NAT] 0 — Tiny pivot.
Q.221 [MCQ] A — Stability.
Q.222 [NAT] 4 — 1x1 up to 4x4.
Q.223 [MCQ] A — Both positive.
Q.224 [NAT] 2 — 1*2.
Q.225 [MCQ] A — Sylvester inertia.
Q.226 [NAT] 16 — 16.
Q.227 [MCQ] A — Preserves norms.
Q.228 [NAT] 5 — Same as ||x||.
Q.229 [MCQ] A — |det|=1.
Q.230 [NAT] 1 — Volume preserved.
Q.231 [MCQ] A — GL+.
Q.232 [NAT] -1 — Orientation reversing.
Q.233 [NAT] 2 — 1+1.
Q.234 [MCQ] A — 1.
Q.235 [NAT] 1 — det I=1.
Q.236 [MCQ] A — Definition via rank.

Q.237 [NAT] 2 — Third = sum of first two.
Q.238 [MCQ] A — One linear constraint.
Q.239 [NAT] 1 — 1.
Q.240 [MCQ] A — Intersection 0, sum whole space.
Q.241 [NAT] 2 — 2.
Q.242 [MCQ] A — 3.
Q.243 [NAT] 3 — 5-2.
Q.244 [MCQ] A — Coordinate functionals.
Q.245 [NAT] 6 — 6.
Q.246 [MCQ] A — 2.
Q.247 [NAT] n — Whole dual.
Q.248 [MCQ] A — 2x2 Gram of rows, rank 2.
Q.249 [NAT] 1 — 1.
Q.250 [MCQ] A — A A^T=I.
Q.251 [NAT] 2 — 2.
Q.252 [MCQ] A — Full column rank.
Q.253 [MCQ] A — sqrt(tr I)=sqrt(2).
Q.254 [NAT] 1.7 — sqrt(3).
Q.255 [MCQ] A — 4.
Q.256 [NAT] 6 — 6.
Q.257 [MCQ] A — tr I=2.
Q.258 [NAT] 1.4 — 1.414.
Q.259 [MCQ] A — Entrywise 1 on the diagonal.
Q.260 [NAT] 1 — 1.
Q.261 [MCQ] A — Rank-1 projection.
Q.262 [NAT] 2 — Projects orthogonal to 1.
Q.263 [MCQ] A — Symmetric idempotent.
Q.264 [NAT] 4 — 4.
Q.265 [MCQ] A — Always.
Q.266 [NAT] 2 — At most min(2, n-1)=2.
Q.267 [MCQ] A — They lie on a line through 0 after centering.
Q.268 [NAT] 1 — n-1=1.
Q.269 [MCQ] B — Independent entries a,b,c in [[a,b],[b,c]].
Q.270 [NAT] 5 — Equals dim.
Q.271 [NAT] 1 — Every column equals the first.
Q.272 [NAT] 0 — Trivial kernel.
Q.273 [NAT] 2 — 4/2=2.
Q.274 [MCQ] B — Missing the zero vector.
Q.275 [NAT] 2 — Two nonzero pivots.
Q.276 [NAT] 2 — Free variables x1 and x4.

UNIT 3 Calculus and Optimization · 267 questions

MCQ negative -1/3 or -2/3. MSQ / NAT no negative. Work first, then the solutions section.

Q.1 NAT · 1 mark · Limit
lim_{x- > 0} (sin 5x)/x equals _____.
NAT · enter the numerical value

Q.2 NAT · 1 mark · Limit inf
lim_{n- > infty} n(a^{1/n}-1) for a=e equals _____.
NAT · enter the numerical value

Q.3 MCQ · 1 mark · Squeeze
If -x^2 < = f(x) < = x^2 for all x, then lim_{x- > 0} f(x) is
(A) 0
(B) 1
(C) does not exist
(D) infty

Q.4 NAT · 1 mark · One sided
lim_{x- > 0+} |x|/x equals _____.
NAT · enter the numerical value

Q.5 MCQ · 1 mark · One sided 2
lim_{x- > 0} |x|/x
(A) is 1
(B) is -1
(C) does not exist
(D) is 0

Q.6 NAT · 1 mark · Inf
lim_{x- > 0+} 1/x is +infty. lim_{x- > 0-} 1/x is _____. (enter -1 as a flag for -infty)
NAT · enter the numerical value

Q.7 MCQ · 1 mark · Cont
f(x)=sin x / x for x!=0, f(0)=1 is
(A) discontinuous at 0
(B) continuous at 0
(C) undefined as a function on R
(D) differentiable nowhere

Q.8 MCQ · 1 mark · Cont 2
f(x)=1 for x rational, 0 for x irrational is continuous at
(A) every x
(B) no x
(C) 0 only
(D) integers only

Q.9 MCQ · 1 mark · IVT
f continuous on [0,1], f(0)=-1, f(1)=2. Then f has a root in (0,1)
(A) always
(B) never
(C) only if f is polynomial
(D) only if f is monotone

Q.10 NAT · 1 mark · IVT count
f(x)=x^3-x-1 on [1,2]: f(1)=-1 < 0 < f(2)=5. Number of roots guaranteed by IVT in [1,2] is at least _____.
NAT · enter the numerical value

Q.11 MCQ · 1 mark · Diff = > cont
If f is differentiable at a then f is
(A) continuous at a
(B) analytic at a
(C) monotone near a
(D) polynomial

Q.12 MCQ · 1 mark · Cont not diff
f(x)=|x| at 0 is
(A) differentiable
(B) continuous but not differentiable
(C) discontinuous
(D) twice differentiable

Q.13 NAT · 1 mark · Deriv
d/dx (x^3) at x=2 equals _____.
NAT · enter the numerical value

Q.14 NAT · 1 mark · Deriv 2
d/dx (e^{2x}) at x=0 equals _____.
NAT · enter the numerical value

Q.15 NAT · 1 mark · Product
d/dx (x sin x) at x=0 equals _____.
NAT · enter the numerical value

Q.16 NAT · 1 mark · Quotient
d/dx (x/(1+x)) at x=0 equals _____.
NAT · enter the numerical value

Q.17 NAT · 1 mark · Chain
d/dx sin(x^2) at x=0 equals _____.
NAT · enter the numerical value

Q.18 NAT · 1 mark · Chain 2
d/dx ln(1+x^2) at x=1 equals _____.
NAT · enter the numerical value

Q.19 MCQ · 1 mark · Not diff
f(x)=x^{1/3} at 0 has
(A) infinite derivative (vertical tangent) / not finite f'
(B) f'=0
(C) f'=1
(D) a jump

Q.20 NAT · 1 mark · Higher
f(x)=x^4. f^{(4)}(0) equals _____.
NAT · enter the numerical value

Q.21 NAT · 1 mark · Higher 2
f(x)=e^x. f^{(n)}(0) equals _____.
NAT · enter the numerical value

Q.22 MCQ · 1 mark · Rolle
Rolle's theorem requires f(a)=f(b) and f continuous on [a,b], differentiable on (a,b). Then there is c in (a,b) with
(A) f'(c)=0
(B) f(c)=0
(C) f''(c)=0
(D) f'(c)=1

Q.23 MCQ · 1 mark · MVT
MVT: there is c in (a,b) with f'(c)=(f(b)-f(a))/(b-a), assuming
(A) f continuous on [a,b] and differentiable on (a,b)
(B) f only integrable
(C) f polynomial of degree > =3
(D) f never zero

Q.24 NAT · 1 mark · MVT val
f(x)=x^2 on [1,3]. The MVT c satisfies 2c=(9-1)/2=4, so c=_____.
NAT · enter the numerical value

Q.25 MCQ · 1 mark · Increasing
If f'(x) > 0 on (a,b) then f is
(A) strictly increasing there
(B) constant
(C) strictly decreasing
(D) unbounded

Q.26 NAT · 1 mark · Crit
f(x)=x^3-3x. Critical points solve 3x^2-3=0, so x=+-1. The local maximum value is f(-1)=_____.
NAT · enter the numerical value

Q.27 NAT · 1 mark · Crit min
Same f. Local minimum value f(1) is _____.
NAT · enter the numerical value

Q.28 MCQ · 1 mark · 2nd test
f'(c)=0, f''(c) > 0 implies
(A) local min
(B) local max
(C) global min always
(D) inflection necessarily

Q.29 MCQ · 1 mark · Inflexion
f(x)=x^3 at 0 has f'(0)=f''(0)=0. It is
(A) a local min
(B) a local max
(C) an inflection (not a local extremum)
(D) discontinuous

Q.30 NAT · 1 mark · Closed
f(x)=x^3-3x on [-2,2]. Evaluate at critical points and endpoints: f(-2)=-2, f(-1)=2, f(1)=-2, f(2)=2. The global maximum on [-2,2] is _____.
NAT · enter the numerical value

Q.31 NAT · 1 mark · Closed min
Same. Global minimum on [-2,2] is _____.
NAT · enter the numerical value

Q.32 MCQ · 1 mark · Endpoint
On a closed bounded interval a continuous function
(A) attains min and max
(B) may miss its max
(C) is differentiable
(D) is polynomial

Q.33 NAT · 1 mark · Taylor e
Taylor of e^x at 0 up to x^2 is 1+x+x^2/2. At x=0.2 this approximation is _____ (two decimals).
NAT · enter the numerical value

Q.34 NAT · 1 mark · Taylor sin
sin x = x - x^3/6 + O(x^5). The cubic approximation of sin(0.3) uses 0.3 - 0.027/6. 0.027/6=0.0045 so the value is _____ (three decimals).
NAT · enter the numerical value

Q.35 MCQ · 1 mark · Taylor rem
Lagrange remainder after n terms involves
(A) f^{(n+1)}(xi) (x-a)^{n+1}/(n+1)!
(B) only f(a)
(C) the integral of f
(D) f'(a) only

Q.36 NAT · 1 mark · Taylor ln
ln(1+x)=x-x^2/2+x^3/3-... for |x| < 1. Truncating after x for x=0.2 gives error about x^2/2=_____. (two decimals)
NAT · enter the numerical value

Q.37 MCQ · 1 mark · Analytic
e^{-1/x^2} (with value 0 at 0) is C^infty with all derivatives 0 at 0, hence its Taylor series at 0
(A) equals the function
(B) is 0, not equal to the function for x!=0
(C) diverges
(D) is 1

Q.38 NAT · 1 mark · LHopital
lim_{x- > 0} (sin x - x)/x^3 = -1/6. Numerically -1/6 as three decimals is _____.
NAT · enter the numerical value

Q.39 MCQ · 1 mark · LHopital hyp
0/0 form of lim f/g with f,g differentiable, g' !=0 near a (except possibly a), then the limit of f'/g' if it exists equals the original. This is
(A) L'Hopital
(B) IVT
(C) MVT only
(D) Taylor with remainder 0

Q.40 NAT · 1 mark · Exp limit
lim_{x- > infty} (1+2/x)^x = e^2 ~ _____ (one decimal).
NAT · enter the numerical value

Q.41 NAT · 1 mark · Log deriv
d/dx log_10 x = 1/(x ln 10). At x=1, 1/ln 10 ~ _____ (two decimals).
NAT · enter the numerical value

Q.42 NAT · 1 mark · Inv deriv
d/dx arctan x at x=1 equals _____ (one decimal).
NAT · enter the numerical value

Q.43 NAT · 1 mark · Inv sin
d/dx arcsin x at x=0 equals _____.
NAT · enter the numerical value

Q.44 MCQ · 1 mark · Implicit
x^2+y^2=1, dy/dx at (3/5,4/5) is
(A) -3/4
(B) 3/4
(C) -4/3
(D) 0

Q.45 NAT · 1 mark · Related
A circle A=pi r^2. If dr/dt=2 at r=3, dA/dt = 2 pi r dr/dt = 12 pi. Enter the integer coefficient of pi: _____.
NAT · enter the numerical value

Q.46 MCQ · 1 mark · Convex
f'' > =0 on an interval means f is
(A) convex (concave up)
(B) concave down
(C) linear only
(D) bounded

Q.47 NAT · 1 mark · Jensen
For convex f, (f(a)+f(b))/2 > = f((a+b)/2). For f(x)=x^2, a=0,b=2: left 2, right f(1)=_____.
NAT · enter the numerical value

Q.48 NAT · 1 mark · Opt wire
A wire of length 12 is bent into a rectangle with width w and length 6-w (perimeter 12). Area w(6-w) is max at w=_____.
NAT · enter the numerical value

Q.49 NAT · 1 mark · Opt area
Max area for that rectangle is _____.
NAT · enter the numerical value

Q.50 NAT · 1 mark · Opt box
An open box from a 12 by 12 square by cutting squares of side x: V=x(12-2x)^2. V'(x)=0 at x=2 (the physical critical point in (0,6)). V(2) is _____.
NAT · enter the numerical value

Q.51 MCQ · 1 mark · Opt 1st
For a differentiable f on an open interval, an interior local extremum occurs only where
(A) f'=0 or f' DNE
(B) f=0
(C) f''=0
(D) f is linear

Q.52 NAT · 1 mark · Profit
R(q)=10q-q^2, C(q)=q. Profit P=9q-q^2. Max at q=_____.
NAT · enter the numerical value

Q.53 NAT · 1 mark · Profit val
That max profit is _____ (one decimal).
NAT · enter the numerical value

Q.54 NAT · 1 mark · MLE bern
Bernoulli likelihood p^3 (1-p)^1. log L' = 3/p - 1/(1-p)=0 = > p=_____. (two decimals)
NAT · enter the numerical value

Q.55 NAT · 1 mark · LS 1d
Min_b sum_{i=1}^2 (y_i - b)^2 for y=(1,3) is at b= the mean _____.
NAT · enter the numerical value

Q.56 NAT · 1 mark · LS slope
Min_m sum (y_i - m x_i)^2, x=(1,2), y=(2,2): m=(2+4)/(1+4)=_____. (one decimal)
NAT · enter the numerical value

Q.57 MCQ · 1 mark · Convex opt
A strictly convex differentiable f on R has at most one minimiser. f(x)=x^2 has minimiser
(A) 0
(B) 1
(C) none
(D) all x

Q.58 NAT · 1 mark · Exp decay
d/dx e^{-5x} at 0 equals _____.
NAT · enter the numerical value

Q.59 NAT · 1 mark · Sigmoid
f(x)=1/(1+e^{-x}). f'(x)=f(1-f). At the point where f=0.4, f' equals _____ (two decimals).
NAT · enter the numerical value

Q.60 NAT · 1 mark · Sigmoid 2
Same f. f(0) equals _____ (one decimal).
NAT · enter the numerical value

Q.61 MCQ · 1 mark · Logistic deriv
The unique maximiser of f(1-f) for f in (0,1) is at f=
(A) 0.5
(B) 0
(C) 1
(D) 0.25

Q.62 NAT · 1 mark · Newton
Newton for g(x)=x^2-2 starting at x0=1: x1=1- (1-2)/2 = 1.5. x1 equals _____ (one decimal).
NAT · enter the numerical value

Q.63 NAT · 1 mark · Newton 2
Next Newton step from 1.5: 1.5 - (2.25-2)/(3) = 1.5-0.25/3. Value _____ (two decimals).
NAT · enter the numerical value

Q.64 MCQ · 1 mark · GD
Gradient descent on f(x)=x^2 with step 0.1 from x=1: next x is
(A) 0.8
(B) 1.2
(C) 0
(D) 2

Q.65 NAT · 1 mark · GD 2
f(x)=(x-3)^2, step 0.5 from x=5. f'=2(x-3)=4, next x=5-0.5*4=_____.
NAT · enter the numerical value

Q.66 MCQ · 1 mark · Unimodal
A unique critical point that is a local min for a coercive C^2 function on R is
(A) the global min
(B) only local
(C) a max
(D) an inflection

Q.67 NAT · 1 mark · Poly min
f(x)=x^4-2x^2. f'=4x^3-4x=4x(x^2-1). Local minima at +-1, value f(1)=_____.
NAT · enter the numerical value

Q.68 NAT · 1 mark · Asymp
lim_{x- > infty} (sqrt(x^2+x)-x) = 1/2. Enter the limit as a decimal (one decimal).
NAT · enter the numerical value

Q.69 MCQ · 1 mark · Oscillate
lim_{x- > infty} sin x
(A) is 0
(B) is 1
(C) does not exist
(D) is infty

Q.70 NAT · 1 mark · Avg value
Average of f(x)=x on [0,2] is (1/2) int_0^2 x dx = _____.
NAT · enter the numerical value

Q.71 MCQ · 1 mark · FTC
d/dx int_0^x t^2 dt equals
(A) x^2
(B) x^3/3
(C) 2x
(D) 0

Q.72 NAT · 1 mark · FTC 2
d/dx int_0^{x^2} e^t dt = e^{x^2} * 2x. At x=0 this is _____.
NAT · enter the numerical value

Q.73 MCQ · 1 mark · Not needed
GATE DA calculus is single-variable: no partials, no Lagrange. The number of variables in Section 3 optimisation is
(A) 1
(B) 2
(C) n
(D) unspecified

Q.74 NAT · 1 mark · Soft
The function |x| is minimised at x=_____.
NAT · enter the numerical value

Q.75 NAT · 1 mark · Abs crit
d/dx |x-3| does not exist at x=_____.
NAT · enter the numerical value

Q.76 MCQ · 1 mark · Piecewise
f(x)=x^2 for x < 0, f(x)=x for x > =0. At 0, f is
(A) C^1
(B) continuous but f' has a jump (left 0, right 1)
(C) discontinuous
(D) analytic

Q.77 NAT · 1 mark · Left deriv
Left derivative of that f at 0 is _____.
NAT · enter the numerical value

Q.78 NAT · 1 mark · Right deriv
Right derivative of that f at 0 is _____.
NAT · enter the numerical value

Q.79 MCQ · 1 mark · Lipschitz
f(x)=sqrt(|x|) near 0 is
(A) Lipschitz
(B) not Lipschitz (unbounded slope)
(C) C^1
(D) analytic

Q.80 NAT · 1 mark · Holder
sqrt(x) on [0,1] is Holder-1/2. The derivative at 0+ is _____. (enter 1 to mean +infty as a flag)
NAT · enter the numerical value

Q.81 NAT · 1 mark · Poly crit count
A degree-4 polynomial has at most how many critical points? _____.
NAT · enter the numerical value

Q.82 MCQ · 1 mark · Fund alg
A nonconstant polynomial of odd degree R- > R
(A) has at least one real root
(B) has no real root
(C) is always positive
(D) is bounded

Q.83 NAT · 1 mark · Cub root
x^3-x-1=0 has a unique real root (since f' =3x^2-1 has two zeros but f increases overall from -infty to infty with one sign change). f(1)=_____.
NAT · enter the numerical value

Q.84 NAT · 1 mark · Bisection
Bisection on [1,2] for that cubic: midpoint 1.5, f(1.5)=3.375-1.5-1=0.875 > 0 so the root is in [1, 1.5]. New interval length is _____ (one decimal).
NAT · enter the numerical value

Q.85 MCQ · 1 mark · Conv Newton
Newton on a simple root of a C^2 function is typically
(A) quadratically convergent
(B) linear with rate 1/2 always
(C) divergent always
(D) cubic always

Q.86 NAT · 1 mark · Harmonic
d/dx (1/x) at x=2 equals _____ (two decimals).
NAT · enter the numerical value

Q.87 NAT · 1 mark · Power
d/dx x^{1/2} at x=4 equals _____ (two decimals).
NAT · enter the numerical value

Q.88 MCQ · 1 mark · Log log
d/dx ln(ln x) for x > e is 1/(x ln x). At x=e this derivative is
(A) 1/e
(B) 0
(C) undefined / infinite because ln e=1 wait: 1/(e*1)=1/e
(D) e

Q.89 NAT · 1 mark · Exp base
d/dx 2^x = 2^x ln 2. At x=0 the derivative is ln 2 ~ _____ (two decimals).
NAT · enter the numerical value

Q.90 NAT · 1 mark · Cosh
d/dx cosh x = sinh x. sinh(0) is _____.
NAT · enter the numerical value

Q.91 NAT · 1 mark · Tanh
d/dx tanh x = 1-tanh^2 x. At 0 this is _____.
NAT · enter the numerical value

Q.92 MCQ · 1 mark · Even deriv
The derivative of an even C^1 function is
(A) odd
(B) even
(C) zero
(D) positive

Q.93 NAT · 1 mark · Odd f
sin is odd. sin'(0)=_____.
NAT · enter the numerical value

Q.94 MCQ · 1 mark · Periodic deriv
The derivative of a C^1 periodic function is
(A) periodic with the same period
(B) never periodic
(C) constant
(D) always zero mean only if we integrate

Q.95 NAT · 1 mark · Sin max
Max of sin x + cos x = sqrt(2) sin(x+pi/4). The maximum value is _____ (three decimals, sqrt2~1.414).
NAT · enter the numerical value

Q.96 NAT · 1 mark · Amp
R in a sin x + b cos x is sqrt(a^2+b^2). For 3 sin x + 4 cos x, R=_____.
NAT · enter the numerical value

Q.97 MCQ · 1 mark · Bound
|f'(x)| < =M implies f is
(A) M-Lipschitz
(B) constant
(C) unbounded
(D) polynomial

Q.98 NAT · 1 mark · Lip const
sin is 1-Lipschitz. |sin 0.3 - sin 0| < = _____. (one decimal)
NAT · enter the numerical value

Q.99 MCQ · 1 mark · Mono test
f'=x^2 > =0 so f(x)=x^3/3 is
(A) nondecreasing on R
(B) decreasing
(C) constant
(D) unbounded below only

Q.100 NAT · 1 mark · Inflect count
f=x^4 has f''=12 x^2 > =0, inflection points: _____. (number of them)
NAT · enter the numerical value

Q.101 MCQ · 1 mark · Change sign 2nd
Inflection requires f'' changing sign (under extra regularity). For x^3, f''=6x changes sign at
(A) 0
(B) 1
(C) nowhere
(D) infty

Q.102 NAT · 1 mark · Taylor poly
The nth Taylor polynomial of 1/(1-x) at 0 is 1+x+...+x^n. At x=1/2, n=1: 1+1/2=_____. (one decimal)
NAT · enter the numerical value

Q.103 NAT · 1 mark · Geom sum
True 1/(1-1/2)=2. The n=1 error is 0.5. Enter the true value _____.
NAT · enter the numerical value

Q.104 MCQ · 1 mark · Radius
Taylor of 1/(1-x) at 0 has radius of convergence
(A) 1
(B) 0
(C) infty
(D) 2

Q.105 MCQ · 1 mark · Exp radius
The Taylor series of e^x at 0 has radius
(A) 0
(B) 1
(C) infty
(D) e

Q.106 MCQ · 1 mark · Termwise
Inside the radius one may differentiate a power series termwise. For sum x^n, the derivative is
(A) sum n x^{n-1}
(B) sum x^n
(C) 0
(D) 1/(1-x)

Q.107 NAT · 1 mark · Arctan ser
arctan x = x-x^3/3+... At x=1 the series is the Leibniz series for pi/4. Truncating after one term gives _____.
NAT · enter the numerical value

Q.108 NAT · 1 mark · Pi4
pi/4 ~ _____ (three decimals).
NAT · enter the numerical value

Q.109 NAT · 1 mark · Leibniz err
Error bound after first term of Leibniz pi/4 series is 1/3 ~ _____ (two decimals).
NAT · enter the numerical value

Q.110 NAT · 1 mark · Opt fence
Three sides of a rectangular pen against a wall, total fence 20: 2w+L=20, A=w(20-2w). Max at w=_____.
NAT · enter the numerical value

Q.111 NAT · 1 mark · Fence A
Max area is _____.
NAT · enter the numerical value

Q.112 NAT · 1 mark · Can
Cylinder volume 16 pi, min surface: r^3=16/pi * pi wait V=pi r^2 h=16 pi = > r^2 h=16. S=2pi r^2+2pi r h. Standard r=2, h=4. h/r=_____.
NAT · enter the numerical value

Q.113 NAT · 1 mark · Can r
That r is _____.
NAT · enter the numerical value

Q.114 NAT · 1 mark · AMGM 1var
Max of x(1-x) on [0,1] is at x=1/2, value _____ (two decimals).
NAT · enter the numerical value

Q.115 NAT · 1 mark · Beam
Strength of a beam ~ x (a-x)^2. For a=3, f=x(3-x)^2, f'= (3-x)^2 - 2x(3-x)= (3-x)(3-3x). Crit in (0,3) at x=_____.
NAT · enter the numerical value

Q.116 NAT · 1 mark · Snell skip
Light time x/v1 + (L-x)/v2. This is 1-var calculus. If v1=v2 the min is the straight path, x= L/2 when geometry is symmetric. For L=10, x=_____.
NAT · enter the numerical value

Q.117 MCQ · 1 mark · Coercive
f(x)=x^4-x is coercive (f- > +infty as |x|- > infty) so a global min exists. f'(x)=4x^3-1=0 at x=4^{-1/3}. This is the unique crit pt, hence the global min. f''(that point) is
(A) positive
(B) negative
(C) zero
(D) undefined

Q.118 NAT · 1 mark · Lin opt
Min of 3x+4 on [0,1] is at x=0, value _____.
NAT · enter the numerical value

Q.119 NAT · 1 mark · Lin max
Max of 3x+4 on [0,1] is _____.
NAT · enter the numerical value

Q.120 MCQ · 1 mark · KKT 1d
Min f on [a,b]: candidates are interior crit pts and endpoints. This is the 1-var KKT. For |x| on [-1,2] the min is at
(A) 0
(B) -1
(C) 2
(D) 1

Q.121 NAT · 1 mark · KKT max
Max of |x| on [-1,2] is _____.
NAT · enter the numerical value

Q.122 NAT · 1 mark · Quad min
f(x)=x^2-4x+7 = (x-2)^2+3. Min value is _____.
NAT · enter the numerical value

Q.123 NAT · 1 mark · Quad argmin
Argmin is _____.
NAT · enter the numerical value

Q.124 MCQ · 1 mark · No min open
f(x)=x on (0,1) has
(A) no min and no max
(B) min 0
(C) max 1
(D) min and max

Q.125 MCQ · 1 mark · Attained
A continuous function on a compact set
(A) attains its inf and sup
(B) need not attain inf
(C) is unbounded
(D) is polynomial

Q.126 NAT · 1 mark · Comp
[0,1] is compact. Max of e^{-x} on [0,1] is _____.
NAT · enter the numerical value

Q.127 NAT · 1 mark · Comp min
Min of e^{-x} on [0,1] is e^{-1} ~ _____ (two decimals).
NAT · enter the numerical value

Q.128 MCQ · 1 mark · Diff quot
f'(a)=lim h- > 0 (f(a+h)-f(a))/h. For f(x)=x^2, a=3, the difference quotient at h=0.1 is (9.61-9)/0.1=_____.
(A) 6.1
(B) 6
(C) 9
(D) 0.1

Q.129 NAT · 1 mark · True deriv
That true derivative is _____.
NAT · enter the numerical value

Q.130 MCQ · 1 mark · Caratheodory
A function is differentiable at a iff f(x)=f(a)+phi(x)(x-a) with phi continuous at a, phi(a)=f'(a). For x^2 at 0, phi(x)=x, phi(0)=_____.
(A) 0
(B) 1
(C) 2
(D) undefined

Q.131 NAT · 1 mark · Phi
For x^2 at 3, phi(x)=x+3, phi(3)=_____.
NAT · enter the numerical value

Q.132 MCQ · 1 mark · C1 not C2
f(x)=x^{4/3} is C^1 but f'' blows at 0. f'(x)=(4/3) x^{1/3}, f'(0)=_____.
(A) 0
(B) infty
(C) 1
(D) 4/3

Q.133 NAT · 1 mark · Blow
f''(x)=(4/9) x^{-2/3} at x=0 is infinite. Enter 1 as a flag.
NAT · enter the numerical value

Q.134 MCQ · 1 mark · Bump
A C^infty bump supported in [-1,1] exists. Its Taylor series at 0 may be
(A) nonzero
(B) identically 0 if the bump is flat enough at 0, or not; a standard bump is not analytic
(C) a polynomial of degree 1
(D) e^x

Q.135 NAT · 1 mark · Poly id
The only C^infty functions equal to their Taylor series everywhere are (real) analytic ones. e^x is analytic. e^x at 0, 0th coefficient is _____.
NAT · enter the numerical value

Q.136 MCQ · 1 mark · Maclaurin
Maclaurin series is Taylor at
(A) 0
(B) 1
(C) infty
(D) a generic a

Q.137 NAT · 1 mark · Cos mac
cos x = 1 - x^2/2 + ... Coefficient of x^2 is _____ (one decimal).
NAT · enter the numerical value

Q.138 NAT · 1 mark · Binom
Newton binomial (1+x)^a = 1 + a x + a(a-1)/2 x^2+... For a=1/2, coeff of x is _____ (one decimal).
NAT · enter the numerical value

Q.139 NAT · 1 mark · Sqrt
(1+x)^{1/2} at x=0.2, linear approx 1+0.1=_____. (one decimal)
NAT · enter the numerical value

Q.140 NAT · 1 mark · True sqrt
sqrt(1.2)~ _____ (two decimals).
NAT · enter the numerical value

Q.141 MCQ · 1 mark · Error term
The remainder after linear Taylor of sqrt(1+x) is negative because the second derivative of sqrt is negative (concave). So the linear approx
(A) overestimates
(B) underestimates
(C) is exact
(D) is random

Q.142 NAT · 1 mark · 2nd deriv sqrt
d^2/dx^2 (1+x)^{1/2} at 0 is (1/2)(-1/2)=_____. (two decimals)
NAT · enter the numerical value

Q.143 MCQ · 1 mark · Opt 2nd
A critical point with f''=0 is inconclusive (x^3, x^4, -x^4). For x^4, f''(0)=0 but it is a
(A) min
(B) max
(C) inflection
(D) none of these / saddle in 1d doesn't apply; it is a min

Q.144 NAT · 1 mark · x4
min of x^4 is _____.
NAT · enter the numerical value

Q.145 MCQ · 1 mark · Unbdd below
-x^4 has
(A) a global max at 0 and no global min
(B) a global min
(C) no extremum
(D) only inflections

Q.146 NAT · 1 mark · Local vs global
f(x)=x^3-3x has a local max that is not global because f- > infty as x- > infty. The local max value is _____.
NAT · enter the numerical value

Q.147 MCQ · 1 mark · Asymp line
f(x)=x+1/x for x > 0 has min at x=1, f=2. As x- > infty, f(x)-x tends to
(A) 0
(B) 1
(C) infty
(D) -infty

Q.148 NAT · 1 mark · Min x+1/x
Min of x+1/x for x > 0 is _____.
NAT · enter the numerical value

Q.149 NAT · 1 mark · Argmin x+1/x
Attained at x=_____.
NAT · enter the numerical value

Q.150 MCQ · 1 mark · x+1/x x < 0
For x < 0, x+1/x has a local max at x=-1, value
(A) -2
(B) 2
(C) 0
(D) -1

Q.151 NAT · 1 mark · Neg
That local max value is _____.
NAT · enter the numerical value

Q.152 MCQ · 1 mark · No global x+1/x
On R\{0}, x+1/x is unbounded above and below, so
(A) no global min or max
(B) global min 2
(C) global max -2
(D) global min -2

Q.153 NAT · 1 mark · Gate 1var
Official DA syllabus optimisation involves a single variable. Number of variables: _____.
NAT · enter the numerical value

Q.154 MCQ · 1 mark · Partial off
Partial derivatives of F(x,y) are
(A) in the DA 2027 calculus syllabus
(B) NOT in the official DA calculus syllabus (single variable only)
(C) the same as d/dx
(D) Lagrange multipliers

Q.155 NAT · 1 mark · Hess off
Hessian matrices are multivariable. Enter 0 if they are outside DA calculus syllabus.
NAT · enter the numerical value

Q.156 MCQ · 1 mark · Taylor 1d
The 2nd-order Taylor of f at a is f(a)+f'(a)h+(1/2)f''(a)h^2. For f=e^x, a=0, h=1 this is 1+1+1/2=____.
(A) 2.5
(B) e
(C) 1
(D) 3

Q.157 NAT · 1 mark · e approx
e ~ _____ (two decimals).
NAT · enter the numerical value

Q.158 NAT · 1 mark · Better
3rd order 1+1+1/2+1/6=_____ (three decimals).
NAT · enter the numerical value

Q.159 MCQ · 1 mark · Remainder e
Lagrange remainder after n=2 for e^x at 0, x=1: e^xi / 6 for some xi in (0,1), hence in (1/6, e/6). It is less than e/6~0.45. The actual error e-2.5~0.218. This confirms
(A) the bound
(B) that Taylor is useless
(C) divergence
(D) that f''=0

Q.160 NAT · 1 mark · Err e
e-2.5 ~ _____ (two decimals).
NAT · enter the numerical value

Q.161 MCQ · 1 mark · Opt word
A farmer maximises area. This is 1-var calculus once the constraint is used to eliminate a variable. The constraint 2x+y=P is used to write A=x(P-2x). This is
(A) substitution
(B) Lagrange on 2 vars (off syllabus as a method, though equivalent)
(C) linear programming in many vars
(D) k-means

Q.162 NAT · 1 mark · P=20 A
P=20, A=x(20-2x), A'=20-4x=0, x=5, A=_____.
NAT · enter the numerical value

Q.163 MCQ · 1 mark · Second enough
On R, f'' > 0 everywhere means f is strictly convex, so a critical point is a
(A) unique global min
(B) max
(C) saddle
(D) inflection

Q.164 NAT · 1 mark · Strict convex
x^2 is strictly convex. Number of minimisers is _____.
NAT · enter the numerical value

Q.165 MCQ · 1 mark · Not strict
x^4 is strictly convex? f''=12 x^2 > =0 but 0 at a point: still strictly convex actually. f(tx+(1-t)y) < t f + (1-t)f for x!=y. Yes. Number of mins of x^4 is
(A) 1
(B) 2
(C) infty
(D) 0

Q.166 NAT · 1 mark · Flat min
x^4 min value _____.
NAT · enter the numerical value

Q.167 MCQ · 1 mark · GD step too big
GD on x^2, step size eta=2: x < - x-2*(2x)= -3x, which diverges if x0!=0. So eta must be < 1/L with L=2, i.e. eta < 0.5 for contraction. eta=2 is
(A) too large
(B) optimal
(C) too small always
(D) Newton

Q.168 NAT · 1 mark · L smooth
x^2 has Lipschitz gradient with L=_____.
NAT · enter the numerical value

Q.169 MCQ · 1 mark · Newton 1d
Newton is GD with step 1/f''(x). For x^2, f''=2, so Newton is x < - x - (2x)/2 =0 in one step. From any x0!=0, Newton lands on
(A) 0
(B) x0
(C) 1
(D) infty

Q.170 NAT · 1 mark · Newton quad
Number of Newton steps to minimise a positive-definite quadratic in 1D (exactly) is _____.
NAT · enter the numerical value

Q.171 MCQ · 1 mark · Secant
The secant method replaces f'' by a finite difference. It typically converges with order
(A) golden ratio ~1.618
(B) 2 exactly always
(C) 1 always
(D) 3

Q.172 MCQ · 1 mark · Bisection opt
Ternary / golden-section search minimises a unimodal f without derivatives. Each step shrinks the interval by a constant factor. It is
(A) derivative-free 1-var optimisation
(B) Newton
(C) simplex in 10D
(D) SGD

Q.173 NAT · 1 mark · Unimodal box
V(x)=x(12-2x)^2 on (0,6) is unimodal with a unique max. That max is at x=_____.
NAT · enter the numerical value

Q.174 MCQ · 1 mark · Log concave
A Gaussian density is log-concave, so 1-var MLE for the mean of N(mu,1) is the sample mean, unique. For one sample x=4, MLE is _____.
(A) 4
(B) 0
(C) 1
(D) infty

Q.175 NAT · 1 mark · Nll
NLL of N(mu,1) at x=4 is (1/2)(4-mu)^2 + const. Min at mu=_____.
NAT · enter the numerical value

Q.176 MCQ · 1 mark · Poisson MLE
i.i.d. Poisson, MLE of lambda is the sample mean. For data 1,2,3, MLE is _____.
(A) 2
(B) 1
(C) 3
(D) 6

Q.177 NAT · 1 mark · Pois mean
Mean of 1,2,3 is _____.
NAT · enter the numerical value

Q.178 MCQ · 1 mark · Constrained 1d
Min x^2 on [1,2] is at the boundary x=1, value 1. The unconstrained min 0 is infeasible. So constraints matter. The constrained min value is
(A) 1
(B) 0
(C) 4
(D) 2

Q.179 NAT · 1 mark · Bd
Min x^2 on [1,2] is _____.
NAT · enter the numerical value

Q.180 NAT · 1 mark · Bd max
Max x^2 on [1,2] is _____.
NAT · enter the numerical value

Q.181 MCQ · 1 mark · Ineq
For min f on x > =0, if the unconstrained min is at x=-1 < 0, the constrained min is at 0 (if f' (0) > 0 wait: if f increases away from -1, f'(0) > 0 so 0 is min on [0,infty) if f' > =0 on [0,infty). Example f=(x+1)^2, f'(0)=2 > 0, min on [0,infty) at
(A) 0
(B) -1
(C) infty
(D) 1

Q.182 NAT · 1 mark · Ineq val
f=(x+1)^2 at 0 equals _____.
NAT · enter the numerical value

Q.183 NAT · 1 mark · More limits 1
lim_{x- > 0} (1-cos x)/x equals _____.
NAT · enter the numerical value

Q.184 NAT · 1 mark · More limits 2
lim_{x- > infty} x / e^x equals _____.
NAT · enter the numerical value

Q.185 NAT · 1 mark · More limits 3
lim_{x- > infty} ln x / x equals _____.
NAT · enter the numerical value

Q.186 NAT · 1 mark · More limits 4
lim_{x- > 0+} x^x equals _____.
NAT · enter the numerical value

Q.187 MCQ · 1 mark · e def
lim (1+1/n)^n equals
(A) e
(B) 1
(C) 0
(D) pi

Q.188 NAT · 1 mark · More der 1
d/dx (x ln x) at x=1 equals _____.
NAT · enter the numerical value

Q.189 NAT · 1 mark · More der 2
d/dx (sin x / x) at 0, after filling f(0)=1, equals _____.
NAT · enter the numerical value

Q.190 NAT · 1 mark · More der 3
d/dx arctan(x^2) at x=1 equals _____.
NAT · enter the numerical value

Q.191 NAT · 1 mark · More der 4
d/dx (ln(sin x)) at x=pi/2 equals _____.
NAT · enter the numerical value

Q.192 NAT · 1 mark · More der 5
d/dx (x^x) = x^x (ln x +1). At x=1 this is _____.
NAT · enter the numerical value

Q.193 NAT · 1 mark · More opt 1
Min of e^x + e^{-x} is at x=0, value _____.
NAT · enter the numerical value

Q.194 NAT · 1 mark · More opt 2
Min of x ln x on (0,1]: wait x ln x - > 0 at 0+, min is at 1/e, value -1/e. -1/e ~ _____ (two decimals).
NAT · enter the numerical value

Q.195 NAT · 1 mark · More opt 3
Argmin of x ln x on (0,infty) is 1/e ~ _____ (two decimals).
NAT · enter the numerical value

Q.196 NAT · 1 mark · More opt 4
A rectangle inscribed in the semicircle y=sqrt(1-x^2), x in [-1,1], height y, width 2x, A=2x sqrt(1-x^2). Max: let u=x^2, A^2=4 u (1-u), max at u=_____. (one decimal)
NAT · enter the numerical value

Q.197 NAT · 1 mark · More opt 5
Then x=1/sqrt(2), A=1. Max area equals _____.
NAT · enter the numerical value

Q.198 NAT · 1 mark · More tay 1
Taylor of ln(1-x) at 0 starts -x -x^2/2 -x^3/3-... Coefficient of x is _____.
NAT · enter the numerical value

Q.199 NAT · 1 mark · More tay 2
Taylor of 1/(1+x) at 0 starts 1-x+x^2-... At x=0.1 the linear approx is _____. (one decimal)
NAT · enter the numerical value

Q.200 NAT · 1 mark · More tay 3
True 1/1.1~ _____ (two decimals).
NAT · enter the numerical value

Q.201 NAT · 1 mark · More tay 4
cos(0.2)~ 1-0.02=0.98 from quadratic. 0.02=0.04/2. Enter the quadratic approx _____ (two decimals).
NAT · enter the numerical value

Q.202 NAT · 1 mark · More tay 5
Actual cos(0.2)~0.9801. Error after quadratic is about _____ (four decimals).
NAT · enter the numerical value

Q.203 MCQ · 1 mark · Diff disc
A derivative, if it exists on an interval, has the intermediate value property (Darboux) even if f' is discontinuous. So f' cannot have a simple jump. f(x)=x^2 sin(1/x) has f' existing everywhere, f' discontinuous at 0. f'(0)=_____.
(A) 0
(B) 1
(C) does not exist
(D) infty

Q.204 NAT · 1 mark · Oscill deriv
f'(0) for x^2 sin(1/x) is _____.
NAT · enter the numerical value

Q.205 MCQ · 1 mark · Nowhere deriv
Weierstrass function is continuous everywhere, differentiable nowhere. It shows continuity
(A) does not imply differentiability
(B) implies differentiability
(C) equals analyticity
(D) is impossible

Q.206 NAT · 1 mark · Poly always
Polynomials are C^infty. Number of times a degree-5 polynomial can be differentiated before becoming identically 0 is _____.
NAT · enter the numerical value

Q.207 MCQ · 1 mark · Degree drop
Each differentiation of a nonconstant polynomial drops degree by
(A) 1
(B) 0
(C) 2
(D) the leading coefficient

Q.208 NAT · 1 mark · Lead
Leading coefficient of d/dx (3x^5) is _____.
NAT · enter the numerical value

Q.209 MCQ · 1 mark · Const deriv
f'=0 on an interval implies f is
(A) constant there
(B) linear
(C) zero
(D) unbounded

Q.210 NAT · 1 mark · Antideriv val
int_0^2 2x dx equals _____.
NAT · enter the numerical value

Q.211 NAT · 1 mark · Area
Area under y=x on [0,2] is _____.
NAT · enter the numerical value

Q.212 MCQ · 1 mark · FTC2
int_a^b f' = f(b)-f(a) if f' is integrable. For f(x)=x^3, a=0,b=1, the integral of 3x^2 is
(A) 1
(B) 0
(C) 3
(D) 1/4

Q.213 NAT · 1 mark · Net change
If v(t)=t is velocity, displacement from 0 to 2 is int v = _____.
NAT · enter the numerical value

Q.214 MCQ · 1 mark · Mean value int
There is c in (a,b) with f(c) equal to the average of f, if f is continuous. For f(x)=x on [0,2], c=_____.
(A) 1
(B) 0
(C) 2
(D) 4

Q.215 NAT · 1 mark · Avg val 2
Average of x^2 on [0,1] is 1/3 ~ _____ (two decimals).
NAT · enter the numerical value

Q.216 NAT · 1 mark · Final 1
lim_{x- > 0} (e^x -1 -x)/x^2 = 1/2. Enter _____ (one decimal).
NAT · enter the numerical value

Q.217 NAT · 1 mark · Final 2
lim_{x- > infty} (ln(x+1)-ln x)=0. Enter _____.
NAT · enter the numerical value

Q.218 NAT · 1 mark · Final 3
d/dx (x^3 e^{-x}) at 0 equals _____.
NAT · enter the numerical value

Q.219 NAT · 1 mark · Final 4
Critical points of x e^{-x} : f'=e^{-x}(1-x)=0 = > x=_____.
NAT · enter the numerical value

Q.220 NAT · 1 mark · Final 5
Max of x e^{-x} for x > 0 is 1/e ~ _____ (two decimals).
NAT · enter the numerical value

Q.221 NAT · 1 mark · Final 6
f(x)=ln x / x on (1,infty) max at x=e, value 1/e ~ _____ (two decimals).
NAT · enter the numerical value

Q.222 NAT · 1 mark · Final 7
Argmax of ln x / x is e ~ _____ (two decimals).
NAT · enter the numerical value

Q.223 NAT · 1 mark · Final 8
lim_{x- > 0} x / sin(3x) = 1/3 ~ _____ (three decimals).
NAT · enter the numerical value

Q.224 NAT · 1 mark · Final 9
d/dx (tan x) at 0 equals _____.
NAT · enter the numerical value

Q.225 NAT · 1 mark · Final 10
The unique solution of f'=f, f(0)=1 is e^x. f(1)~ _____ (two decimals).
NAT · enter the numerical value

Q.226 MSQ · 2 marks · Limit facts
Which limits equal 0?
(A) lim_{x- > infty} (ln x)/x
(B) lim_{x- > infty} x/e^x
(C) lim_{x- > 0+} x ln x
(D) lim_{x- > 0} (sin x)/x

Q.227 MSQ · 2 marks · Diff facts
Which are differentiable at 0?
(A) x^2 sin(1/x) (with 0 at 0)
(B) |x|
(C) x^{1/3}
(D) e^x

Q.228 MSQ · 2 marks · Theorems
Which apply to f(x)=x^2 on [0,1]?
(A) EVT (attains min/max)
(B) MVT
(C) Rolle if we shift to match endpoints? f(0)!=f(1) so Rolle fails
(D) IVT on the range [0,1]

Q.229 NAT · 1 mark · Add1
lim_{x- > 0} (tan 3x)/x equals _____.
NAT · enter the numerical value

Q.230 NAT · 1 mark · Add2
d/dx (x^x) at x=e: e^e (1+1)=2 e^e. Too large. At x=1 it was 1. d/dx (ln x)^2 at x=e equals 2 ln e * (1/e)= _____. (two decimals)
NAT · enter the numerical value

Q.231 NAT · 1 mark · Add3
Min of (x-1)^2+(x-4)^2 is at the mean of 1 and 4, x=_____ (one decimal).
NAT · enter the numerical value

Q.232 NAT · 1 mark · Add4
That min value is _____ (one decimal).
NAT · enter the numerical value

Q.233 NAT · 1 mark · Add5
lim_{h- > 0} (cos h -1)/h^2 = -1/2. Enter _____ (one decimal).
NAT · enter the numerical value

Q.234 NAT · 1 mark · Add6
f(x)=1/x, f'(2)=_____ (two decimals).
NAT · enter the numerical value

Q.235 NAT · 1 mark · Add7
The tangent to y=x^2 at x=1 has slope 2 and equation y=2x-1. y-intercept is _____.
NAT · enter the numerical value

Q.236 NAT · 1 mark · Add8
Normal line at that point has slope _____ (one decimal).
NAT · enter the numerical value

Q.237 NAT · 1 mark · Add9
Linearisation of sqrt(x) at x=4: 2+(x-4)/4. At x=4.4 this is 2.1. Enter _____ (one decimal).
NAT · enter the numerical value

Q.238 NAT · 1 mark · Add10
True sqrt(4.4)~2.098. The linearisation overestimated, consistent with concavity. Enter 1 if true.
NAT · enter the numerical value

Q.239 MCQ · 1 mark · Add11
A function with f'=0 at an interior point of a closed interval
(A) may fail to be an extremum (e.g. x^3 on [-1,1] at 0)
(B) must be a global max
(C) must be a global min
(D) cannot exist

Q.240 NAT · 1 mark · Add12
On [-1,1], x^3 has max _____.
NAT · enter the numerical value

Q.241 NAT · 1 mark · Add13
On [-1,1], x^3 has min _____.
NAT · enter the numerical value

Q.242 MCQ · 1 mark · Add14
If f'' changes from + to - at c with f'(c)=0, then c is a
(A) local max
(B) local min
(C) neither
(D) discontinuity

Q.243 NAT · 1 mark · Add15
f= -x^2, f'(0)=0, f''(0)=-2 < 0, local max value _____.
NAT · enter the numerical value

Q.244 NAT · 1 mark · Add16
lim_{x- > infty} (x^2+3x)/(2x^2-1) equals _____ (one decimal).
NAT · enter the numerical value

Q.245 NAT · 1 mark · Add17
lim_{x- > 0} ((1+x)^5 -1)/x equals _____.
NAT · enter the numerical value

Q.246 NAT · 1 mark · Add19
The critical point of f(x)=x-ln x (x > 0) solves 1-1/x=0, x=_____.
NAT · enter the numerical value

Q.247 NAT · 1 mark · Add20
f(1)=1-0=_____. This is the global min of x-ln x.
NAT · enter the numerical value

Q.248 MCQ · 1 mark · Add21
x-ln x - > infty as x- > 0+ and as x- > infty, unique crit pt, so the crit pt is a
(A) global min
(B) global max
(C) neither
(D) saddle

Q.249 NAT · 1 mark · Add22
lim_{x- > 0} sinh x / x equals _____.
NAT · enter the numerical value

Q.250 NAT · 1 mark · Add23
d/dx (arcsin x + arccos x) equals _____.
NAT · enter the numerical value

Q.251 NAT · 1 mark · Add24
arcsin x + arccos x = _____ * pi/2, i.e. the constant is pi/2 so enter 1 as the multiplier of pi/2.
NAT · enter the numerical value

Q.252 NAT · 1 mark · Add25
The most negative slope of sin x is _____.
NAT · enter the numerical value

Q.253 MSQ · 2 marks · Add26
Which functions are convex on R?
(A) e^x
(B) x^2
(C) -ln(1+x^2) is NOT convex everywhere? Wait - skip
(D) cosh x

Q.254 NAT · 1 mark · Add27
The unique min of cosh x is _____.
NAT · enter the numerical value

Q.255 NAT · 1 mark · Add28
lim_{x- > infty} tanh x equals _____.
NAT · enter the numerical value

Q.256 NAT · 1 mark · Add29
d/dx (x^2 ln x) at x=1 equals _____.
NAT · enter the numerical value

Q.257 NAT · 1 mark · Add30
The third derivative of e^{2x} at 0 is 8. Enter _____.
NAT · enter the numerical value

Q.258 NAT · 1 mark · Limit
lim_{x- > 0} (1-cos 2x)/x^2 equals _____.
NAT · enter the numerical value

Q.259 NAT · 1 mark · Inf
inf of x on (0,1) is _____.
NAT · enter the numerical value

Q.260 NAT · 1 mark · Rolle
f(x)=x^2-1 on [-1,1]. A point given by Rolle is x=_____.
NAT · enter the numerical value

Q.261 NAT · 1 mark · Phi
Golden ratio (1+sqrt5)/2 ~ _____ (two decimals).
NAT · enter the numerical value

Q.262 NAT · 1 mark · Limit
lim_{x- > 0} (e^{3x}-1)/x equals _____.
NAT · enter the numerical value

Q.263 NAT · 1 mark · Limit
lim_{x- > infty} (3x^2+1)/(x^2-7) equals _____.
NAT · enter the numerical value

Q.264 NAT · 1 mark · Limit
lim_{x- > infty} (2x+sin x)/x equals _____.
NAT · enter the numerical value

Q.265 MCQ · 1 mark · Limit
lim_{x- > 0+} x ln x equals
(A) 0
(B) +infty
(C) -infty
(D) 1

Q.266 NAT · 1 mark · Limit
lim_{x- > 0} ln(1+4x)/x equals _____.
NAT · enter the numerical value

Q.267 MCQ · 1 mark · Limit
lim_{x- > 0} (tan x - x)/x^3 equals
(A) 0
(B) 1/3
(C) 1/2
(D) infty

UNIT 3 SOLUTIONS

Q.1 [NAT] 5 — sin(5x)/(5x)*5 - > 5.
Q.2 [NAT] 1 — a^{1/n}=e^{(ln a)/n}=1+(ln a)/n+..., factor ln a; ln e=1.
Q.3 [MCQ] A — Squeeze.
Q.4 [NAT] 1 — Positive side.
Q.5 [MCQ] C — Left -1, right +1.
Q.6 [NAT] -1 — Diverges to -infty.
Q.7 [MCQ] B — Removable, filled by the limit.
Q.8 [MCQ] B — Thomae's opposite: Dirichlet function.
Q.9 [MCQ] A — IVT.
Q.10 [NAT] 1 — At least one.
Q.11 [MCQ] A — Standard.
Q.12 [MCQ] B — Corner.
Q.13 [NAT] 12 — 3x^2=12.
Q.14 [NAT] 2 — 2e^{2x}.
Q.15 [NAT] 0 — sin+x cos, at 0 is 0.
Q.16 [NAT] 1 — 1/(1+x)^2.
Q.17 [NAT] 0 — 2x cos(x^2).
Q.18 [NAT] 1 — 2x/(1+x^2)=2/2=1.
Q.19 [MCQ] A — Difference quotient - > infty.
Q.20 [NAT] 24 — 4!=24.
Q.21 [NAT] 1 — All 1.
Q.22 [MCQ] A — Rolle.
Q.23 [MCQ] A — MVT.
Q.24 [NAT] 2 — 2.
Q.25 [MCQ] A — MVT.
Q.26 [NAT] 2 — f(-1)=-1+3=2.
Q.27 [NAT] -2 — -1-3 wait: 1-3=-2.
Q.28 [MCQ] A — Second derivative test.
Q.29 [MCQ] C — Odd cubic.
Q.30 [NAT] 2 — 2.
Q.31 [NAT] -2 — -2.
Q.32 [MCQ] A — Extreme value theorem.
Q.33 [NAT] 1.22 — 1+0.2+0.02=1.22.
Q.34 [NAT] 0.296 — 0.3-0.0045=0.2955~0.296.
Q.35 [MCQ] A — Standard.
Q.36 [NAT] 0.02 — 0.02.
Q.37 [MCQ] B — Classic smooth non-analytic.
Q.38 [NAT] -0.167 — -0.1666...
Q.39 [MCQ] A — L'Hopital.
Q.40 [NAT] 7.4 — e^2~7.389.
Q.41 [NAT] 0.43 — 1/2.302~0.43.
Q.42 [NAT] 0.5 — 1/(1+x^2)=1/2.
Q.43 [NAT] 1 — 1/sqrt(1-x^2).
Q.44 [MCQ] A — 2x+2y y'=0 = > y'=-x/y.
Q.45 [NAT] 12 — 12.
Q.46 [MCQ] A — Convex.
Q.47 [NAT] 1 — 1 < =2.
Q.48 [NAT] 3 — Square.
Q.49 [NAT] 9 — 3*3.
Q.50 [NAT] 128 — 2*8^2=128.
Q.51 [MCQ] A — Fermat.
Q.52 [NAT] 4.5 — P'=9-2q=0.
Q.53 [NAT] 20.3 — P(4.5)=9*4.5-20.25=40.5-20.25=20.25.
Q.54 [NAT] 0.75 — 3/4.
Q.55 [NAT] 2 — 2.
Q.56 [NAT] 1.2 — 6/5.
Q.57 [MCQ] A — f'=2x.
Q.58 [NAT] -5 — -5.
Q.59 [NAT] 0.24 — 0.4*0.6=0.24. GATE 2024 Q.33 pattern.
Q.60 [NAT] 0.5 — 1/2.
Q.61 [MCQ] A — Vertex of parabola.
Q.62 [NAT] 1.5 — 1.5.
Q.63 [NAT] 1.42 — 1.5-0.0833=1.4167.
Q.64 [MCQ] A — x-0.1*2x=0.8.
Q.65 [NAT] 3 — Lands on the minimiser.
Q.66 [MCQ] A — Coercive + unique crit pt.
Q.67 [NAT] -1 — 1-2=-1.
Q.68 [NAT] 0.5 — Rationalise.
Q.69 [MCQ] C — Oscillates.
Q.70 [NAT] 1 — 1.
Q.71 [MCQ] A — FTC.
Q.72 [NAT] 0 — 0.
Q.73 [MCQ] A — Syllabus.
Q.74 [NAT] 0 — 0.
Q.75 [NAT] 3 — Corner.
Q.76 [MCQ] B — Not C^1.
Q.77 [NAT] 0 — 2x- > 0.
Q.78 [NAT] 1 — 1.
Q.79 [MCQ] B — Derivative blows up.
Q.80 [NAT] 1 — Infinite.
Q.81 [NAT] 3 — Degree of derivative is 3.
Q.82 [MCQ] A — Limits +-infty opposite.
Q.83 [NAT] -1 — -1.
Q.84 [NAT] 0.5 — 0.5.
Q.85 [MCQ] A — Standard.
Q.86 [NAT] -0.25 — -1/4.
Q.87 [NAT] 0.25 — 1/(2sqrt x)=1/4.
Q.88 [MCQ] A — 1/e.
Q.89 [NAT] 0.69 — 0.693.
Q.90 [NAT] 0 — 0.
Q.91 [NAT] 1 — 1.
Q.92 [MCQ] A — f(-x)=f(x) = > -f'(-x)=f'(x).
Q.93 [NAT] 1 — cos 0=1.
Q.94 [MCQ] A — Chain of translation.
Q.95 [NAT] 1.414 — sqrt(2).
Q.96 [NAT] 5 — 5.
Q.97 [MCQ] A — MVT.
Q.98 [NAT] 0.3 — 0.3.
Q.99 [MCQ] A — f' > =0.
Q.100 [NAT] 0 — f'' vanishes at 0 but does not change sign.
Q.101 [MCQ] A — 0.
Q.102 [NAT] 1.5 — 1.5.
Q.103 [NAT] 2 — 2.
Q.104 [MCQ] A — Sings at x=1.
Q.105 [MCQ] C — Entire.
Q.106 [MCQ] A — And equals 1/(1-x)^2.
Q.107 [NAT] 1 — 1, while pi/4~0.785.
Q.108 [NAT] 0.785 — 0.785.
Q.109 [NAT] 0.33 — 0.33.
Q.110 [NAT] 5 — A'=20-4w=0.
Q.111 [NAT] 50 — 5*10=50.
Q.112 [NAT] 2 — h=2r.
Q.113 [NAT] 2 — 2.
Q.114 [NAT] 0.25 — 0.25.
Q.115 [NAT] 1 — 3-3x=0 = > x=1.
Q.116 [NAT] 5 — 5.
Q.117 [MCQ] A — 12 x^2 > 0.
Q.118 [NAT] 4 — Increasing.
Q.119 [NAT] 7 — At x=1.

Q.120 [MCQ] A — 0.
Q.121 [NAT] 2 — Endpoint 2.
Q.122 [NAT] 3 — 3.
Q.123 [NAT] 2 — 2.
Q.124 [MCQ] A — Open interval.
Q.125 [MCQ] A — Weierstrass.
Q.126 [NAT] 1 — At 0.
Q.127 [NAT] 0.37 — 0.368.
Q.128 [MCQ] A — 6.1, while f'=6.
Q.129 [NAT] 6 — 2*3.
Q.130 [MCQ] A — 0.
Q.131 [NAT] 6 — 6.
Q.132 [MCQ] A — 0.
Q.133 [NAT] 1 — Unbounded.
Q.134 [MCQ] B — Non-analytic smooth.
Q.135 [NAT] 1 — 1.
Q.136 [MCQ] A — a=0.
Q.137 [NAT] -0.5 — -1/2.
Q.138 [NAT] 0.5 — 1/2.
Q.139 [NAT] 1.1 — 1.1.
Q.140 [NAT] 1.10 — 1.095~1.10.
Q.141 [MCQ] A — Concave down.
Q.142 [NAT] -0.25 — -1/4.
Q.143 [MCQ] A — Min.
Q.144 [NAT] 0 — 0.
Q.145 [MCQ] A — Goes to -infty.
Q.146 [NAT] 2 — 2.
Q.147 [MCQ] A — f(x)-x=1/x - > 0.
Q.148 [NAT] 2 — AM-GM.
Q.149 [NAT] 1 — 1.
Q.150 [MCQ] A — -2.
Q.151 [NAT] -2 — -2.
Q.152 [MCQ] A — Both sides.
Q.153 [NAT] 1 — 1.
Q.154 [MCQ] B — Stay 1D.
Q.155 [NAT] 0 — Outside.
Q.156 [MCQ] A — 2.5 vs e~2.718.
Q.157 [NAT] 2.72 — 2.718.
Q.158 [NAT] 2.667 — 2.666...
Q.159 [MCQ] A — Bound holds.
Q.160 [NAT] 0.22 — 0.218.
Q.161 [MCQ] A — Eliminate.
Q.162 [NAT] 50 — Same as fence.
Q.163 [MCQ] A — Convex.
Q.164 [NAT] 1 — One.
Q.165 [MCQ] A — 0 only.
Q.166 [NAT] 0 — 0.
Q.167 [MCQ] A — Unstable.
Q.168 [NAT] 2 — f''=2.
Q.169 [MCQ] A — Quadratic solved exactly.
Q.170 [NAT] 1 — One.
Q.171 [MCQ] A — Superlinear.
Q.172 [MCQ] A — 1D unimodal.
Q.173 [NAT] 2 — 2.
Q.174 [MCQ] A — 4.
Q.175 [NAT] 4 — 4.
Q.176 [MCQ] A — 2.
Q.177 [NAT] 2 — 2.
Q.178 [MCQ] A — 1.
Q.179 [NAT] 1 — 1.
Q.180 [NAT] 4 — 4.
Q.181 [MCQ] A — 0.
Q.182 [NAT] 1 — 1.
Q.183 [NAT] 0 — x/2 * (1-cos)/ (x^2/2) - > 0.
Q.184 [NAT] 0 — Exp dominates.
Q.185 [NAT] 0 — Log vs linear.
Q.186 [NAT] 1 — e^{x ln x}- > e^0=1.
Q.187 [MCQ] A — Definition of e.
Q.188 [NAT] 1 — ln x +1 at 1 is 1.
Q.189 [NAT] 0 — Even function, odd derivative.
Q.190 [NAT] 1 — 2x/(1+x^4)=2/2=1.
Q.191 [NAT] 0 — cot(pi/2)=0.
Q.192 [NAT] 1 — 1*(0+1).
Q.193 [NAT] 2 — 2 cosh(0)=2.
Q.194 [NAT] -0.37 — -0.368.
Q.195 [NAT] 0.37 — 0.368.
Q.196 [NAT] 0.5 — 1/2.
Q.197 [NAT] 1 — 2*(1/sqrt2)*1/sqrt2=1.
Q.198 [NAT] -1 — -1.
Q.199 [NAT] 0.9 — 0.9.
Q.200 [NAT] 0.91 — 0.909.
Q.201 [NAT] 0.98 — 0.98.
Q.202 [NAT] 0.0001 — Very small, next term x^4/24.
Q.203 [MCQ] A — Squeeze x sin.
Q.204 [NAT] 0 — 0.
Q.205 [MCQ] A — Classic.
Q.206 [NAT] 6 — 6th derivative vanishes. After 5 nonzero in general, 6th is 0 so 6.
Q.207 [MCQ] A — 1.
Q.208 [NAT] 15 — 15 x^4.
Q.209 [MCQ] A — MVT.
Q.210 [NAT] 4 — x^2 from 0 to 2.
Q.211 [NAT] 2 — 2.
Q.212 [MCQ] A — 1.
Q.213 [NAT] 2 — 2.
Q.214 [MCQ] A — Average 1.
Q.215 [NAT] 0.33 — 0.33.
Q.216 [NAT] 0.5 — Taylor.
Q.217 [NAT] 0 — ln(1+1/x).
Q.218 [NAT] 0 — Product, 0.
Q.219 [NAT] 1 — 1.
Q.220 [NAT] 0.37 — 0.368.
Q.221 [NAT] 0.37 — Same number different function.
Q.222 [NAT] 2.72 — e.
Q.223 [NAT] 0.333 — 1/3.
Q.224 [NAT] 1 — sec^2 0=1.
Q.225 [NAT] 2.72 — e.
Q.226 [MSQ] A;B;C — (sin x)/x - > 1, the others vanish.
Q.227 [MSQ] A;D — |x| has a corner; cube root has vertical tangent.
Q.228 [MSQ] A;B;D — Rolle needs equal endpoint values.
Q.229 [NAT] 3 — 3 sec^2(0).
Q.230 [NAT] 0.74 — 2/e~0.736.
Q.231 [NAT] 2.5 — 2.5.
Q.232 [NAT] 4.5 — 2*(1.5)^2=4.5.
Q.233 [NAT] -0.5 — -0.5.
Q.234 [NAT] -0.25 — -1/4.
Q.235 [NAT] -1 — -1.
Q.236 [NAT] -0.5 — -1/2.
Q.237 [NAT] 2.1 — 2.1.
Q.238 [NAT] 1 — Concave down.
Q.239 [MCQ] A — Inflection.
Q.240 [NAT] 1 — Endpoint.

Q.241 [NAT] -1 — Endpoint.
Q.242 [MCQ] A — First-derivative test / sign of f''.
Q.243 [NAT] 0 — 0.
Q.244 [NAT] 0.5 — 1/2.
Q.245 [NAT] 5 — Derivative of (1+x)^5 at 0.
Q.246 [NAT] 1 — 1.
Q.247 [NAT] 1 — 1.
Q.248 [MCQ] A — Coercive on (0,infty).
Q.249 [NAT] 1 — Same as sin for the odd analytic function with f'(0)=1.
Q.250 [NAT] 0 — They sum to pi/2.
Q.251 [NAT] 1 — pi/2.
Q.252 [NAT] -1 — cos=-1.
Q.253 [MSQ] A;B;D — e^x, x^2, cosh are convex.
Q.254 [NAT] 1 — cosh 0=1.
Q.255 [NAT] 1 — 1.
Q.256 [NAT] 1 — 2x ln x + x, at 1 is 1.
Q.257 [NAT] 8 — 8 e^{2x}.
Q.258 [NAT] 2 — 1-cos u ~ u^2/2 with u=2x gives 2.
Q.259 [NAT] 0 — 0 not attained.
Q.260 [NAT] 0 — f'=2x.
Q.261 [NAT] 1.62 — 1.618.
Q.262 [NAT] 3 — Derivative of e^{3x} at 0.
Q.263 [NAT] 3 — Leading coefficients.
Q.264 [NAT] 2 — sin/x- > 0.
Q.265 [MCQ] A — Standard: slower than any positive power.
Q.266 [NAT] 4 — ln(1+u)/u - > 1 with u=4x.
Q.267 [MCQ] B — Taylor: x+x^3/3 -x over x^3.

UNIT 4 Programming, Data Structures and Algorithms · 286 questions

MCQ negative -1/3 or -2/3. MSQ / NAT no negative. Work first, then the solutions section.

Q.1 MCQ · 2 marks · Python

```
Consider the Python program
def push_tag(tag, bag=[]):
bag.append(tag)
return bag
print(push_tag('a'))
print(push_tag('b'))
print(push_tag('c', []))
Which option is the exact printed output?
```

(A) ['a']\n['b']\n['c']
(B) ['a']\n['a', 'b']\n['c']
(C) ['a']\n['a', 'b']\n['a', 'b', 'c']
(D) ['a']\n['b']\n['a', 'b', 'c']

Q.2 NAT · 1 mark · Stacks and queues

An empty stack S is processed by: push 3, push 7, push 1, pop, push 5, pop, pop. The integer that remains in S is

NAT · enter the numerical value

Q.3 MCQ · 2 marks · Trees

A binary tree T has preorder R A C D B E and inorder C A D R E B. The postorder of T is

(A) C D A E B R
(B) C D A R E B
(C) A C D B E R
(D) C A D E B R

Q.4 MCQ · 2 marks · Hash tables

A hash table of size 10 (indices 0..9) uses h(x)=3x mod 10 and linear probing. Keys 1,4,5,6,14,15 are inserted into an empty table. The indices of 14 and 15 are, respectively (this is a NEW table and NEW hash; do not reuse an official paper)

(A) 4 and 6
(B) 5 and 6
(C) 4 and 7
(D) 6 and 6

Q.5 NAT · 2 marks · Searching

Sorted array HostelWaitlist = [2, 5, 8, 11, 14, 17, 20, 23, 26, 29]. Recursive/iterative binary search uses a three-way comparison at mid = floor((lo+hi)/2) and continues while lo < =hi. Number of mid-key comparisons until the search for 17 terminates is

NAT · enter the numerical value

Q.6 NAT · 2 marks · Sorting

Bubble sort (adjacent swaps, ascending, full n-1 passes, no early-stop) on QuizMarks = [4, 3, 2, 1, 5]. The number of swaps is

NAT · enter the numerical value

Q.7 MCQ · 2 marks · Graphs

Undirected unweighted graph Campus has edges Admin-Lab, Lab-Canteen, Canteen-Hostel, Admin-Library, Library-Hostel. BFS from Admin scans neighbours in alphabetical order. The dequeue order is

(A) Admin Lab Library Canteen Hostel
(B) Admin Lab Canteen Hostel Library
(C) Hostel Canteen Library Lab Admin
(D) Admin Canteen Hostel Lab Library

Q.8 MCQ · 1 mark · Python

```
Consider the Python program
def tally(k, box=None):
if box is None:
box = []
box.append(k)
return box
print(tally(4))
print(tally(5))
The printed output is
```

(A) [4]\n[4, 5]
(B) [4]\n[5]
(C) [4, 5]\n[4, 5]
(D) [4]\n[4]

Q.9 MCQ · 1 mark · Stacks and queues

An empty queue (FIFO) is processed by: enqueue 4, enqueue 9, enqueue 2, dequeue, enqueue 6, dequeue. The front-to-rear contents are

(A) 2, 6
(B) 4, 9
(C) 6, 2
(D) 9, 2, 6

Q.10 NAT · 1 mark · Trees

For the same tree with preorder R A C D B E and inorder C A D R E B, the height of T (root height 0) is

NAT · enter the numerical value

Q.11 NAT · 1 mark · Hash tables

Hash table size 11, h(x)=x mod 11, linear probing. Insert 12,22,32,42 into an empty table. The index where 42 is stored is

NAT · enter the numerical value

Q.12 NAT · 2 marks · Searching

Sorted array RollSheet = [101, 205, 308, 412, 519, 623, 730]. Recursive/iterative binary search uses a three-way comparison at mid = floor((lo+hi)/2) and continues while lo < =hi. Number of mid-key comparisons until the search for 412 terminates is

NAT · enter the numerical value

Q.13 NAT · 2 marks · Sorting

Bubble sort (adjacent swaps, ascending, full n-1 passes, no early-stop) on LabQueue = [5, 1, 4, 2, 8]. The number of swaps is

NAT · enter the numerical value

Q.14 MCQ · 2 marks · Graphs

Undirected unweighted graph Campus has edges Admin-Lab, Lab-Canteen, Canteen-Hostel, Admin-Library, Library-Hostel. Recursive DFS from Admin with neighbours tried in alphabetical order. The discovery order is

(A) Admin Lab Canteen Hostel Library
(B) Admin Lab Library Canteen Hostel
(C) Library Hostel Canteen Lab Admin
(D) Library Lab Hostel Canteen Admin

Q.15 MCQ · 1 mark · Python

```
After executing
shelf = [10, 20, 30]
rack = shelf
rack.append(40)
print(shelf)
the value printed for shelf is
```

(A) [10, 20, 30]
(B) [10, 20, 30, 40]
(C) [40]
(D) [10, 20, 40]

Q.16 MCQ · 2 marks · Stacks and queues

A double-ended queue D is empty. Operations: insertRear(1), insertFront(2), insertRear(3), deleteFront, insertFront(4), deleteRear. Front-to-rear contents of D are

(A) 4, 1
(B) 1, 3
(C) 4, 1, 3
(D) 2, 1

Q.17 NAT · 1 mark · Trees

For the binary tree with preorder R A C D B E and inorder C A D R E B, the number of leaf nodes is

NAT · enter the numerical value

Q.18 NAT · 2 marks · Hash tables

Table size 11, h(x)=x mod 11, linear probing. Insert 10,22,31,4,15,28,17. Number of probes used to insert 17 (counting the successful slot) is

NAT · enter the numerical value

Q.19 NAT · 2 marks · Searching

Let A be a sorted array of 1000 distinct integers. Recursive binary search for a missing key y uses a three-way comparison at mid and loops while lo < =hi. An upper bound on the number of comparisons is the number of times n=1000 can be halved, about ceil(log2(1001)). The exact worst-case comparison count of this implementation (floor mid, lo=mid+1 or hi=mid-1) on n=1000 is

NAT · enter the numerical value

Q.20 NAT · 2 marks · Sorting

Insertion sort (adjacent transpositions to place the key) on QuizMarks = [4, 3, 2, 1, 5]. Number of swaps is

NAT · enter the numerical value

Q.21 NAT · 1 mark · Graphs

In graph Campus (edges Admin-Lab, Lab-Canteen, Canteen-Hostel, Admin-Library, Library-Hostel), BFS from Admin with alphabetical neighbour order visits how many vertices?

NAT · enter the numerical value

Q.22 MCQ · 1 mark · Python

```
After executing
p = [1, 2, 3]
q = p
p = p + [4]
print(p)
print(q)
the pair (p, q) is
```

(A) ([1, 2, 3, 4], [1, 2, 3, 4])
(B) ([1, 2, 3, 4], [1, 2, 3])
(C) ([1, 2, 3], [1, 2, 3, 4])
(D) ([1, 2, 3, 4], [1, 2, 3, 4, 4])

Q.23 NAT · 1 mark · Stacks and queues

Characters of GATE are pushed on an empty stack in left-to-right order, then all are popped. If A=1, G=2, T=3, E=4, the first popped value (the code of that character) is

NAT · enter the numerical value

Q.24 MSQ · 2 marks · Trees

Binary tree U has preorder P Q S T R and inorder S Q T P R. Which statements are true?

(A) R is the right child of P and is a leaf
(B) Q is the left child of P
(C) S is the right child of Q
(D) The postorder is S T Q R P

Q.25 MCQ · 2 marks · Hash tables

Open-address table of size 7 with h(x)=x mod 7 and linear probing. Keys 50, 15, 36, 8 are inserted (empty start). Key 8 occupies index

(A) 1
(B) 2
(C) 3
(D) 4

Q.26 MCQ · 1 mark · Searching

Binary search runs in O(log n) worst case on which input?

(A) a linked list of n integers in increasing order
(B) an array of n integers in increasing order
(C) an array of n integers in arbitrary order
(D) a linked list of n integers in arbitrary order

Q.27 NAT · 2 marks · Sorting

Insertion sort (adjacent transpositions to place the key) on LabQueue = [5, 1, 4, 2, 8]. Number of swaps is

NAT · enter the numerical value

Q.28 MCQ · 2 marks · Graphs

Undirected unweighted graph Harbor has edges Pier-Dock, Dock-Yard, Yard-Gate, Pier-Shed, Shed-Gate, Dock-Shed. BFS from Pier scans neighbours in alphabetical order. The dequeue order is

(A) Pier Dock Shed Yard Gate
(B) Pier Dock Shed Gate Yard
(C) Gate Yard Shed Dock Pier
(D) Dock Gate Pier Shed Yard

Q.29 MCQ · 2 marks · Python

```
After executing
p = [1, 2, 3]
q = p
p += [4]
print(p is q, p, q)
the printed line is
```

(A) False [1, 2, 3, 4] [1, 2, 3]
(B) True [1, 2, 3, 4] [1, 2, 3, 4]
(C) True [1, 2, 3, 4] [1, 2, 3]
(D) False [1, 2, 3, 4] [1, 2, 3, 4]

Q.30 MCQ · 1 mark · Stacks and queues

A sequence of stack operations on empty S: push(a), push(b), pop, push(c), pop, pop. The sequence of popped values is

(A) b, c, a
(B) a, b, c
(C) b, a, c
(D) c, b, a

Q.31 NAT · 1 mark · Trees

Keys 8, 3, 10, 1, 6, 14, 4 are inserted in that order into an empty BST. The key in the left child of 6 is

NAT · enter the numerical value

Q.32 NAT · 2 marks · Hash tables

Table size 11, h(x)=x mod 11, quadratic probing i-th probe (h(x)+i^2) mod 11, i=0,1,2,... Insert 10,22,31,4,15. Index of 15 is

NAT · enter the numerical value

Q.33 MSQ · 2 marks · Searching

Linear search on an unordered array of n distinct keys. Which statements are correct?

(A) Worst-case successful search does n key comparisons
(B) Best-case successful search does 1 comparison
(C) Average successful search (uniform) does (n+1)/2 comparisons
(D) Unsuccessful search can be done in O(1) expected time without hashing

Q.34 NAT · 1 mark · Sorting

Selection sort (swap the current index with the min of the suffix, skip if already in place) on QuizMarks = [4, 3, 2, 1, 5]. Number of swaps is

NAT · enter the numerical value

Q.35 MCQ · 2 marks · Graphs

Undirected unweighted graph Harbor has edges Pier-Dock, Dock-Yard, Yard-Gate, Pier-Shed, Shed-Gate, Dock-Shed. Recursive DFS from Pier with neighbours tried in alphabetical order. The discovery order is

(A) Pier Dock Shed Gate Yard
(B) Pier Dock Shed Yard Gate
(C) Yard Gate Shed Dock Pier
(D) Yard Shed Pier Gate Dock

Q.36 MCQ · 1 mark · Python

```
The Python fragment
row = [2, 4, 6, 8, 10]
row[1:4] = [0]
print(row)
prints
```

(A) [2, 0, 10]
(B) [2, 0, 8, 10]
(C) [2, 0, 6, 8, 10]
(D) [0, 0, 0, 0, 10]

Q.37 MSQ · 2 marks · Stacks and queues

Which data structures can implement a FIFO queue with amortized O(1) enqueue and dequeue?

(A) A singly linked list with both head and tail pointers
(B) A singly linked list with only a head pointer (insert/delete at head)
(C) Two stacks (input stack and output stack)
(D) A singly linked list with only a tail pointer (insert at tail)

Q.38 MCQ · 1 mark · Trees

Keys 8, 3, 10, 1, 6, 14, 4 are inserted in that order into an empty BST. The inorder traversal is

(A) 1, 3, 4, 6, 8, 10, 14
(B) 8, 3, 1, 6, 4, 10, 14
(C) 1, 4, 6, 3, 14, 10, 8
(D) 1, 3, 6, 4, 8, 10, 14

Q.39 MCQ · 1 mark · Hash tables

Chaining is used in a table of size 5 with h(x)=x mod 5. Keys 12, 22, 7, 17, 27 are inserted (each chain is a list, new key at the head). The chain at index 2, from head to tail, is

(A) 27 - > 17 - > 12
(B) 12 - > 17 - > 27
(C) 22 - > 7
(D) 27 - > 22 - > 7

Q.40 NAT · 1 mark · Searching

Unordered array ExamMarks = [41, 17, 63, 8, 55, 29]. Linear search from the left for 55 performs how many comparisons?

NAT · enter the numerical value

Q.41 NAT · 1 mark · Sorting

Selection sort (swap the current index with the min of the suffix, skip if already in place) on LabQueue = [5, 1, 4, 2, 8]. Number of swaps is

NAT · enter the numerical value

Q.42 NAT · 1 mark · Graphs

In graph Harbor (edges Pier-Dock, Dock-Yard, Yard-Gate, Pier-Shed, Shed-Gate, Dock-Shed), BFS from Pier with alphabetical neighbour order visits how many vertices?

NAT · enter the numerical value

Q.43 MCQ · 1 mark · Python

```
After
buf = [7, 8, 9]
view = buf[0:2]
view[0] = 1
print(buf)
buf is
```

(A) [1, 8, 9]
(B) [7, 8, 9]
(C) [1, 8]
(D) [7, 1, 9]

Q.44 MCQ · 2 marks · Stacks and queues

To evaluate the postfix expression 5 1 2 + 4 * + 3 - with a stack, the value remaining on the stack at the end is

(A) 14
(B) 10
(C) 9
(D) 18

Q.45 NAT · 1 mark · Trees

Keys 50, 30, 70, 20, 40, 60, 80 are inserted into an empty BST. The height of the tree (root height 0) is

NAT · enter the numerical value

Q.46 NAT · 1 mark · Hash tables

A hash table of 13 slots uses chaining. 10 distinct keys are stored. The minimum possible length of the longest chain is

NAT · enter the numerical value

Q.47 NAT · 1 mark · Searching

Unordered array PacketLen = [20, 11, 9, 40, 7]. Linear search for 6 (absent) performs how many comparisons?

NAT · enter the numerical value

Q.48 MCQ · 2 marks · Sorting

Array CabinBags = [4, 3, 2, 1, 5] is sorted ascending. After exactly two passes of selection sort (each pass places the next minimum), the array is

(A) [1, 2, 3, 4, 5]
(B) [2, 1, 3, 4, 5]
(C) [2, 3, 4, 1, 5]
(D) [1, 3, 2, 4, 5]

Q.49 MCQ · 2 marks · Graphs

Undirected unweighted graph Metro has edges North-Hub, Hub-East, Hub-West, East-South, West-South. BFS from Hub scans neighbours in alphabetical order. The dequeue order is

(A) Hub East North West South
(B) Hub East South West North
(C) South West North East Hub
(D) East Hub North South West

Q.50 MCQ · 2 marks · Python

```
The fragment
grid = [[0] * 2] * 3
grid[0][0] = 5
print(grid)
prints
```

(A) [[5, 0], [0, 0], [0, 0]]
(B) [[5, 0], [5, 0], [5, 0]]
(C) [[5, 5], [5, 5], [5, 5]]
(D) [[5, 0], [0, 0], [5, 0]]

Q.51 NAT · 1 mark · Stacks and queues

The postfix 3 4 + 2 * 7 / is evaluated with integer floor division on a stack. The final integer is

NAT · enter the numerical value

Q.52 MCQ · 1 mark · Trees

A full binary tree has 9 internal nodes (every node has 0 or 2 children). The number of leaves is

(A) 10
(B) 9
(C) 8
(D) 18

Q.53 NAT · 1 mark · Hash tables

A hash table of 13 slots uses chaining. 10 distinct keys are stored. The maximum possible length of the longest chain is

NAT · enter the numerical value

Q.54 MCQ · 1 mark · Searching

Sentinel linear search puts the key at the end of an n-element array (now n+1 cells). For an unsuccessful search the number of key comparisons is

(A) n+1
(B) n
(C) 1
(D) 2n

Q.55 MCQ · 2 marks · Sorting

Array CabinBags = [4, 3, 2, 1, 5]. After exactly two passes of bubble sort (adjacent, each pass sinks the next maximum to the right), the array is

(A) [2, 1, 3, 4, 5]
(B) [1, 2, 3, 4, 5]
(C) [3, 2, 1, 4, 5]
(D) [4, 3, 2, 1, 5]

Q.56 MCQ · 2 marks · Graphs

Undirected unweighted graph Metro has edges North-Hub, Hub-East, Hub-West, East-South, West-South. Recursive DFS from Hub with neighbours tried in alphabetical order. The discovery order is

(A) Hub East South West North
(B) Hub East North West South
(C) North West South East Hub
(D) West South North Hub East

Q.57 MCQ · 1 mark · Python

```
Executing
try:
s = { (1, 2), (3, 4), [5, 6] }
print('ok', len(s))
except TypeError:
print('err')
prints
```

(A) ok 3
(B) ok 2
(C) err
(D) ok 1

Q.58 MCQ · 2 marks · Stacks and queues

Which of the following sequences of push(1), push(2), push(3) and three pops cannot occur as the output of a stack (output is the pop sequence)?

(A) 1 2 3
(B) 3 2 1
(C) 2 3 1
(D) 3 1 2

Q.59 NAT · 1 mark · Trees

A complete binary tree with 12 nodes is stored in a 1-indexed array. The parent of the node at index 12 is at index

NAT · enter the numerical value

Q.60 MCQ · 1 mark · Hash tables

Load factor alpha of an open-address table of size m holding n keys is n/m. For uniform probing, the expected probes in an unsuccessful search is about 1/(1-alpha). If m=1000 and n=800, this estimate is

(A) 5
(B) 1.25
(C) 0.8
(D) 4

Q.61 NAT · 1 mark · Searching

Binary search on a sorted array of size 1 for a present key uses how many mid comparisons?

NAT · enter the numerical value

Q.62 MCQ · 2 marks · Sorting

Insertion sort on CabinBags = [4, 3, 2, 1, 5] after processing the first three keys (i.e. the prefix of length 3 is sorted, the rest untouched) yields

(A) [2, 3, 4, 1, 5]
(B) [1, 2, 3, 4, 5]
(C) [2, 1, 3, 4, 5]
(D) [3, 4, 2, 1, 5]

Q.63 NAT · 1 mark · Graphs

In graph Metro (edges North-Hub, Hub-East, Hub-West, East-South, West-South), BFS from Hub with alphabetical neighbour order visits how many vertices?

NAT · enter the numerical value

Q.64 NAT · 1 mark · Python

```
The fragment
s = set()
s.add((1, (2, 3)))
s.add((1, (2, 3)))
print(len(s))
prints which integer?
```

NAT · enter the numerical value

Q.65 NAT · 1 mark · Stacks and queues

A circular queue is implemented in an array of size 6 (indices 0..5) with front=2 and rear=5, holding 3 elements in the usual convention that the rear index is the last occupied slot. Using count = (rear - front + N) mod N with front pointing at the first occupied slot, the number of stored elements is

NAT · enter the numerical value

Q.66 NAT · 1 mark · Trees

A complete binary tree with 12 nodes (1-indexed heap layout) has how many leaves?

NAT · enter the numerical value

Q.67 MSQ · 2 marks · Hash tables

Which statements about hashing are correct?

(A) Linear probing suffers from primary clustering
(B) Quadratic probing eliminates secondary clustering completely
(C) Double hashing uses a second hash to choose the step
(D) Chaining remains correct even when n exceeds m

Q.68 MCQ · 1 mark · Searching

Ternary search on a unimodal discrete array splits into three thirds. Compared with binary search on a strictly increasing array of n keys, ternary search's comparison count is

(A) still Theta(log n) but with a larger constant if each step does two comparisons
(B) O(1)
(C) Theta(n)
(D) Theta(n log n)

Q.69 NAT · 2 marks · Sorting

In-place Quicksort with Lomuto partition and last-element pivot on PivotTrace = [5, 2, 8, 1, 9, 3]. A swap is counted whenever two distinct indices exchange values (including the final pivot swap in a partition). The number of swaps is

NAT · enter the numerical value

Q.70 MCQ · 2 marks · Graphs

Undirected unweighted graph Forest has edges Oak-Pine, Pine-Birch, Birch-Maple, Oak-Elm, Elm-Maple, Pine-Elm. BFS from Oak scans neighbours in alphabetical order. The dequeue order is

(A) Oak Elm Pine Maple Birch
(B) Oak Elm Maple Birch Pine
(C) Birch Maple Pine Elm Oak
(D) Birch Elm Maple Oak Pine

Q.71 NAT · 2 marks · Python

```
In the Python function
def fib(n, c):
c[0] += 1
if n < 2:
return n
return fib(n-1, c) + fib(n-2, c)
c = [0]
fib(5, c)
print(c[0])
the printed call count (including the initial call) is
```

NAT · enter the numerical value

Q.72 MCQ · 2 marks · Stacks and queues

A queue is implemented using two stacks In and Out. Enqueue always pushes onto In. Dequeue pops from Out, but if Out is empty it first pops every element of In onto Out. Starting empty, perform: enq 1, enq 2, deq, enq 3, deq, deq. The last dequeued value is

(A) 1
(B) 2
(C) 3
(D) the queue is empty too early

Q.73 MCQ · 1 mark · Trees

Which pair of traversals uniquely determines a binary tree (no duplicate keys)?

(A) inorder and preorder
(B) preorder and postorder
(C) preorder and level-order
(D) postorder and level-order

Q.74 MCQ · 2 marks · Hash tables

Double hashing: h1(k)=k mod 11, h2(k)=1+(k mod 10), table size 11. Probe sequence is (h1 + i*h2) mod 11. The first three probes for k=14 are

(A) 3, 8, 2
(B) 3, 4, 5
(C) 3, 7, 0
(D) 14, 15, 16

Q.75 NAT · 2 marks · Sorting

Lomuto-quicksort, last-element pivot, count swaps of distinct indices, on LastPivot = [10, 7, 8, 9, 1, 5]. Swap count is

NAT · enter the numerical value

Q.76 MCQ · 2 marks · Graphs

Undirected unweighted graph Forest has edges Oak-Pine, Pine-Birch, Birch-Maple, Oak-Elm, Elm-Maple, Pine-Elm. Recursive DFS from Oak with neighbours tried in alphabetical order. The discovery order is

(A) Oak Elm Maple Birch Pine
(B) Oak Elm Pine Maple Birch
(C) Pine Birch Maple Elm Oak
(D) Pine Oak Maple Elm Birch

Q.77 NAT · 2 marks · Python

```
Let boom(n) return boom(n-1)+boom(n-1) with boom(0)=1.
Including the first call, how many calls occur while
evaluating boom(4)? Equivalent program:
def count(n):
c = [0]
def b(n):
c[0] += 1
if n == 0:
return 1
return b(n-1) + b(n-1)
b(n)
return c[0]
print(count(4))
```

NAT · enter the numerical value

Q.78 NAT · 1 mark · Stacks and queues

Minimum number of stacks needed to implement a queue (with the two-stack method) is

NAT · enter the numerical value

Q.79 MSQ · 2 marks · Trees

Let T be a nonempty binary tree. Which statements are always true?

(A) The inorder of a BST is a sorted sequence of its keys
(B) The preorder of a BST is always sorted
(C) A heap-ordered complete tree is always a BST
(D) The height of a complete binary tree on n nodes is floor(log2 n)

Q.80 NAT · 1 mark · Hash tables

In a table of size 8 with h(x)=x mod 8 and linear probing, insert 16, 24, 32. The index of 32 is

NAT · enter the numerical value

Q.81 NAT · 2 marks · Sorting

Lomuto-quicksort, last-element pivot, distinct-index swaps, on ReverseFour = [4, 3, 2, 1]. Swap count is

NAT · enter the numerical value

Q.82 NAT · 1 mark · Graphs

In graph Forest (edges Oak-Pine, Pine-Birch, Birch-Maple, Oak-Elm, Elm-Maple, Pine-Elm), BFS from Oak with alphabetical neighbour order visits how many vertices?

NAT · enter the numerical value

Q.83 MCQ · 1 mark · Python

```
After
x = [1, 2, 3]
y = x[:]
y.append(4)
print(len(x), len(y), x is y)
the printed triple is
```

(A) 3 4 False
(B) 4 4 True
(C) 3 4 True
(D) 4 4 False

Q.84 MCQ · 2 marks · Stacks and queues

While converting the infix (A+B)*C - D/E to postfix with an operator stack (precedence * / over + -, left-associative, parentheses), the postfix is

(A) AB+C*DE/-
(B) AB+C*D-E/
(C) ABC*+DE/-
(D) AB+CD-E/*

Q.85 NAT · 1 mark · Trees

Number of distinct BSTs that can be formed with keys {W, X, Y} (three distinct keys) is

NAT · enter the numerical value

Q.86 MCQ · 2 marks · Hash tables

Deleting a key from a linear-probing table without a tombstone can break later searches. After inserting 5, 12 into size-7 table h(x)=x mod 7 (both hash to 5: 5 at 5, 12 at 6) and physically emptying slot 5, a search for 12

(A) may stop at the empty slot 5 and miss 12
(B) always finds 12 at slot 6
(C) rehashes 12 to a new slot
(D) is guaranteed to scan the whole table

Q.87 MCQ · 2 marks · Sorting

Quicksort on n distinct keys with the first element of the current subarray as pivot, keys in random order, partition in linear time. Expected time T(n) satisfies

(A) T(n) = (1/n) * sum_{k=0..n-1} (T(k)+T(n-1-k)) + Theta(n)
(B) T(n) = T(n-1) + T(0) + Theta(n)
(C) T(n) = 2 T(n/2) + Theta(1)
(D) T(n) = T(n/4) + T(3n/4) + Theta(1)

Q.88 MCQ · 2 marks · Graphs

Directed graph Airline has edges Delhi- > Mumbai, Delhi- > Pune, Mumbai- > Goa, Pune- > Goa, Goa- > Kochi. BFS from Delhi scans neighbours in alphabetical order. The dequeue order is

(A) Delhi Mumbai Pune Goa Kochi
(B) Delhi Mumbai Goa Kochi Pune
(C) Kochi Goa Pune Mumbai Delhi
(D) Delhi Goa Kochi Mumbai Pune

Q.89 MCQ · 2 marks · Python

```
After the shallow copy
a = [[1], [2]]
b = a[:]
b[0].append(9)
print(a)
the value of a is
```

(A) [[1], [2]]
(B) [[1, 9], [2]]
(C) [[1, 9], [2, 9]]
(D) [[9], [2]]

Q.90 NAT · 1 mark · Stacks and queues

Using an operator stack to convert A*(B+C) to postfix, the number of times an operator (* or +) is popped onto the output is

NAT · enter the numerical value

Q.91 MCQ · 2 marks · Trees

A max-heap is stored as the array [90, 40, 70, 20, 30, 10, 50]. After one extract-max (standard sink), the new root is

(A) 70
(B) 40
(C) 50
(D) 90

Q.92 NAT · 1 mark · Hash tables

A chain-hash table of size 4 stores keys 1,5,9,13,17 (all hash to 1). The number of key comparisons to search for 17 if each chain is a list from head=most-recently-inserted is

NAT · enter the numerical value

Q.93 MCQ · 1 mark · Sorting

Quicksort with a pivot that is always the smallest key of the subarray (adversarial) has recurrence

(A) T(n) = T(n-1) + Theta(n)
(B) T(n) = 2 T(n/2) + Theta(n)
(C) T(n) = T(n/2) + Theta(1)
(D) T(n) = 2 T(n-1) + Theta(n)

Q.94 MCQ · 2 marks · Graphs

Directed graph Airline has edges Delhi- > Mumbai, Delhi- > Pune, Mumbai- > Goa, Pune- > Goa, Goa- > Kochi. Recursive DFS from Delhi with neighbours tried in alphabetical order. The discovery order is

(A) Delhi Mumbai Goa Kochi Pune
(B) Delhi Mumbai Pune Goa Kochi
(C) Pune Kochi Goa Mumbai Delhi
(D) Pune Mumbai Kochi Goa Delhi

Q.95 MCQ · 1 mark · Python

```
The fragment
vals = [3, 1, 2]
print(sorted(vals), vals)
prints
```

(A) [1, 2, 3] [1, 2, 3]
(B) [1, 2, 3] [3, 1, 2]
(C) [3, 1, 2] [1, 2, 3]
(D) [3, 1, 2] [3, 1, 2]

Q.96 MCQ · 2 marks · Stacks and queues

A linked list holds 8 - > 3 - > 5 - > 1 (head 8). The following pointer surgery is done, where p points at the node 8 and q at the node 3: p.next = q.next; q.next = p. The sequence of keys from the new head q is

(A) 3 - > 8 - > 5 - > 1
(B) 8 - > 5 - > 1
(C) 3 - > 5 - > 1
(D) 3 - > 8 - > 3 - > ... (cycle)

Q.97 NAT · 1 mark · Trees

In a min-heap array [2, 4, 3, 9, 7, 8], the number of heap-order violations is

NAT · enter the numerical value

Q.98 NAT · 1 mark · Hash tables

Same chain as above (head 17 - > 13 - > 9 - > 5 - > 1). Unsuccessful search for 3, which hashes to 1, performs how many key comparisons?

NAT · enter the numerical value

Q.99 NAT · 1 mark · Sorting

Mergesort's merge counts split inversions. For MergeTrace = [2, 4, 1, 3, 5] the number of inversions (pairs i < j, a[i] > a[j]) is

NAT · enter the numerical value

Q.100 NAT · 1 mark · Graphs

In graph Airline (edges Delhi- > Mumbai, Delhi- > Pune, Mumbai- > Goa, Pune- > Goa, Goa- > Kochi), BFS from Delhi with alphabetical neighbour order visits how many vertices?

NAT · enter the numerical value

Q.101 MCQ · 1 mark · Python

```
After
vals = [3, 1, 2]
r = vals.sort()
print(r, vals)
the printed pair is
```

(A) [1, 2, 3] [1, 2, 3]
(B) None [1, 2, 3]
(C) None [3, 1, 2]
(D) [3, 1, 2] [1, 2, 3]

Q.102 NAT · 1 mark · Stacks and queues

A singly linked list of 9 nodes is traversed by two pointers starting at the head: slow moves one node per step, fast moves two. When fast first reaches the last node, slow is at node number (1-indexed from the head)

NAT · enter the numerical value

Q.103 MCQ · 1 mark · Trees

The array [2, 8, 3, 9, 7, 4] (1-indexed heap layout) is tested as a min-heap. How many parent-child pairs violate the min-heap property?

(A) 0
(B) 1
(C) 2
(D) 3

Q.104 MCQ · 1 mark · Hash tables

Universal hashing is used so that for any two distinct keys the collision probability under a random hash from the family is

(A) at most 1/m for table size m
(B) exactly 1/2
(C) 0
(D) n/m

Q.105 NAT · 1 mark · Sorting

Number of inversions in ReverseFive = [5, 4, 3, 2, 1] is

NAT · enter the numerical value

Q.106 MCQ · 2 marks · Graphs

Directed graph Pipeline has edges Src- > A, Src- > B, A- > C, B- > C, C- > Sink, A- > Sink. BFS from Src scans neighbours in alphabetical order. The dequeue order is

(A) Src A B C Sink
(B) Src A C Sink B
(C) Sink C B A Src
(D) A B C Sink Src

Q.107 MCQ · 2 marks · Python

```
The program
def wrap(xs=[]):
xs.append(len(xs))
return xs
wrap()
wrap()
print(wrap())
prints
```

(A) [0]
(B) [0, 1, 2]
(C) [2]
(D) [0, 1]

Q.108 MCQ · 1 mark · Stacks and queues

To reverse a singly linked list in one pass with three pointers prev, cur, nxt, the number of next-pointer assignments per node (the rotation nxt=cur.next; cur.next=prev; prev=cur; cur=nxt) that write into a node's next field is

(A) 1
(B) 2
(C) 3
(D) 0

Q.109 NAT · 1 mark · Trees

A binary tree has preorder F B A D C E G I H and inorder A B C D E F G H I. How many nodes are in the left subtree of F?

NAT · enter the numerical value

Q.110 NAT · 1 mark · Hash tables

Simple uniform hashing, chaining, n=2m keys in m slots. Expected chain length (load factor) is

NAT · enter the numerical value

Q.111 MCQ · 1 mark · Sorting

Mergesort on n keys uses the standard two-way merge. The extra array space (besides the input) is

(A) Theta(n)
(B) Theta(1)
(C) Theta(log n)
(D) Theta(n^2)

Q.112 MCQ · 2 marks · Graphs

Directed graph Pipeline has edges Src- > A, Src- > B, A- > C, B- > C, C- > Sink, A- > Sink. Recursive DFS from Src with neighbours tried in alphabetical order. The discovery order is

(A) Src A C Sink B
(B) Src A B C Sink
(C) B Sink C A Src
(D) Src Sink C B A

Q.113 MCQ · 1 mark · Python

```
Executing
t = (1, 2, [3])
t[2].append(4)
print(t)
prints
```

(A) (1, 2, [3])
(B) (1, 2, [3, 4])
(C) TypeError
(D) (1, 2, 3, 4)

Q.114 NAT · 1 mark · Stacks and queues

A circular singly linked list has 10 nodes. Starting at the head, the minimum number of next-pointer follows needed to return to the head is

NAT · enter the numerical value

Q.115 MCQ · 2 marks · Trees

Tree with preorder F B A D C E G I H and inorder A B C D E F G H I. The right child of G is

(A) I
(B) H
(C) E
(D) none (G is a leaf)

Q.116 MSQ · 2 marks · Sorting

Which algorithms are stable in their usual textbook form?

(A) insertion sort
(B) mergesort (left half preferred on ties)
(C) heapsort
(D) selection sort (swap min into place)

Q.117 NAT · 1 mark · Graphs

In graph Pipeline (edges Src- > A, Src- > B, A- > C, B- > C, C- > Sink, A- > Sink), BFS from Src with alphabetical neighbour order visits how many vertices?

NAT · enter the numerical value

Q.118 MCQ · 1 mark · Python

```
The program
k = 0
def inc(n):
global k
k += n
return k
print(inc(2), inc(3), k)
prints
```

(A) 2 5 5
(B) 2 3 5
(C) 2 5 0
(D) 0 0 5

Q.119 MCQ · 1 mark · Stacks and queues

A dummy-head singly linked list is used to delete the first key equal to 7 in 7 - > 2 - > 7. After a correct dummy-head deletion of the first 7, the list of real keys is

(A) 2 - > 7
(B) 7 - > 2
(C) 2
(D) 7 - > 2 - > 7

Q.120 NAT · 1 mark · Trees

In a rooted binary tree every node has 0 or 2 children (full). If there are 17 leaves, the number of nodes with two children is

NAT · enter the numerical value

Q.121 MSQ · 2 marks · Sorting

Which algorithms sort in Theta(n^2) comparisons in the worst case on an array?

(A) insertion sort
(B) selection sort
(C) heapsort
(D) bubble sort (no early abort needed for the bound)

Q.122 MCQ · 2 marks · Graphs

Undirected unweighted graph Gridlock has edges A1-A2, A2-A3, A1-B1, A2-B2, A3-B3, B1-B2, B2-B3. BFS from A1 scans neighbours in alphabetical order. The dequeue order is

(A) A1 A2 B1 A3 B2 B3
(B) A1 A2 A3 B3 B2 B1
(C) B3 B2 A3 B1 A2 A1
(D) A1 A2 A3 B1 B2 B3

Q.123 NAT · 1 mark · Python

```
For the Python function
def f(n):
if n < = 1:
return n
return f(n-1) + 2*f(n-2)
print(f(4))
the printed value is
```

NAT · enter the numerical value

Q.124 NAT · 2 marks · Stacks and queues

Two sorted lists 1- > 4- > 7 and 2- > 3- > 8 are merged by the standard two-pointer merge (stable: take from the first list on ties). The number of next-pointer comparisons of keys until one list is exhausted is

NAT · enter the numerical value

Q.125 MCQ · 2 marks · Trees

Level-order of a binary tree is 10, 6, 15, 4, 8, 12, 20 and inorder is 4, 6, 8, 10, 12, 15, 20. This tree is

(A) a BST that is also a complete binary tree
(B) a BST that is not complete
(C) complete but not a BST
(D) neither a BST nor complete

Q.126 MCQ · 2 marks · Sorting

Among insertion, selection and bubble (adjacent) sorts on the array [4,3,2,1,5], which one has the prefix of length 2 equal to [1,2] after exactly two passes?

(A) selection sort only
(B) bubble sort only
(C) insertion sort only
(D) all three

Q.127 MCQ · 2 marks · Graphs

Undirected unweighted graph Gridlock has edges A1-A2, A2-A3, A1-B1, A2-B2, A3-B3, B1-B2, B2-B3. Recursive DFS from A1 with neighbours tried in alphabetical order. The discovery order is

(A) A1 A2 A3 B3 B2 B1
(B) A1 A2 B1 A3 B2 B3
(C) B1 B2 B3 A3 A2 A1
(D) B3 B2 B1 A3 A2 A1

Q.128 MCQ · 2 marks · Python

```
After
a = [1, 2, 3, 4]
a[::-1] = a
print(a)
the list a is
```

(A) [4, 3, 2, 1]
(B) [1, 2, 3, 4]
(C) [1, 4, 3, 2]
(D) an error is raised

Q.129 MCQ · 2 marks · Stacks and queues

Floyd's tortoise and hare start together at the head of 1- > 2- > 3- > 4- > 2 (the next of 4 is 2). They move (slow +1, fast +2) until they meet. The key at the meeting node is

(A) 2
(B) 3
(C) 4
(D) 1

Q.130 NAT · 1 mark · Trees

A perfect binary tree of height 3 (root height 0) has how many nodes?

NAT · enter the numerical value

Q.131 NAT · 1 mark · Sorting

Standard mergesort splits in half. For n=8 the number of merge-pass levels (times the whole array is merged from pairs, 4s, 8s) is

NAT · enter the numerical value

Q.132 NAT · 1 mark · Graphs

In graph Gridlock (edges A1-A2, A2-A3, A1-B1, A2-B2, A3-B3, B1-B2, B2-B3), BFS from A1 with alphabetical neighbour order visits how many vertices?

NAT · enter the numerical value

Q.133 MCQ · 2 marks · Python

```
After
a = [1, 2, 3, 4]
a[::-1] = [10, 20, 30, 40]
print(a)
a equals
```

(A) [10, 20, 30, 40]
(B) [40, 30, 20, 10]
(C) [1, 2, 3, 4]
(D) [40, 20, 30, 10]

Q.134 MSQ · 2 marks · Stacks and queues

A dequeue (double-ended queue) supports insert/delete at both ends in O(1). Which statements are correct?

(A) A deque can simulate a stack using only one end
(B) A deque can simulate a FIFO queue using opposite ends
(C) A singly linked list with only a head pointer is already a deque
(D) BFS of a graph is naturally implemented with a deque used as a FIFO queue

Q.135 MCQ · 1 mark · Trees

Inserting 25 into the BST whose inorder is 10,20,30,40 and whose root is 20 (20 has right child 30, 30 has right 40, 20 has left 10) places 25 as

(A) the left child of 30
(B) the right child of 20
(C) the left child of 40
(D) the right child of 10

Q.136 NAT · 1 mark · Sorting

A single merge of two sorted runs of lengths 3 and 5 uses how many key comparisons in the worst case?

NAT · enter the numerical value

Q.137 MCQ · 2 marks · Graphs

Undirected unweighted graph Club has edges Ram-Sam, Sam-Tom, Tom-Uma, Ram-Uma, Sam-Uma. BFS from Ram scans neighbours in alphabetical order. The dequeue order is

(A) Ram Sam Uma Tom
(B) Ram Sam Tom Uma
(C) Tom Uma Sam Ram
(D) Uma Tom Sam Ram

Q.138 MCQ · 1 mark · Python

```
After
s = 'gate'
t = s
s += 'da'
print(s, t, s is t)
the printed line is
```

(A) gateda gateda True
(B) gateda gate False
(C) gate gate True
(D) gateda gateda False

Q.139 NAT · 1 mark · Stacks and queues

An empty stack of integers processes: push 10, push 20, pop, push 30, push 40, pop, pop. The sum of the remaining elements is

NAT · enter the numerical value

Q.140 NAT · 1 mark · Trees

The minimum possible height (root height 0) of a binary tree with 100 nodes is

NAT · enter the numerical value

Q.141 MCQ · 1 mark · Sorting

Quicksort vs mergesort on arrays. Which statement is correct?

(A) mergesort's worst-case comparisons are Theta(n log n); quicksort's are Theta(n^2)
(B) both are Theta(n) worst case
(C) quicksort is stable in the usual in-place Lomuto form
(D) mergesort is in-place with Theta(1) extra memory in the usual form

Q.142 MCQ · 2 marks · Graphs

Undirected unweighted graph Club has edges Ram-Sam, Sam-Tom, Tom-Uma, Ram-Uma, Sam-Uma. Recursive DFS from Ram with neighbours tried in alphabetical order. The discovery order is

(A) Ram Sam Tom Uma
(B) Ram Sam Uma Tom
(C) Uma Tom Sam Ram
(D) Sam Tom Uma Ram

Q.143 MCQ · 2 marks · Python

```
The program
def g(x, y=[]):
y.append(x)
return id(y)
u = g(1)
v = g(2)
w = g(3, [])
print(u == v, v == w)
prints
```

(A) True False
(B) False False
(C) True True
(D) False True

Q.144 NAT · 1 mark · Stacks and queues

An empty FIFO queue processes: enq 5, enq 8, enq 6, deq, enq 9, deq, deq. The remaining front element is

NAT · enter the numerical value

Q.145 NAT · 1 mark · Trees

The maximum possible height (root height 0) of a binary tree with 100 nodes is

NAT · enter the numerical value

Q.146 NAT · 1 mark · Sorting

Selection sort on n=6 keys always performs how many key comparisons (independent of the input order)?

NAT · enter the numerical value

Q.147 NAT · 1 mark · Graphs

In graph Club (edges Ram-Sam, Sam-Tom, Tom-Uma, Ram-Uma, Sam-Uma), BFS from Ram with alphabetical neighbour order visits how many vertices?

NAT · enter the numerical value

Q.148 MCQ · 2 marks · Python

```
After
xs = [1, 2, 3]
def add(n, acc=xs):
acc.append(n)
return acc
xs = []
print(add(4))
the printed list is
```

(A) [4]
(B) [1, 2, 3, 4]
(C) []
(D) [1, 2, 3]

Q.149 MCQ · 2 marks · Stacks and queues

Next-greater-element to the right is computed with a stack for the array [2, 1, 4, 3]. The next-greater list (use -1 if none) is

(A) 4, 4, -1, -1
(B) 4, 4, 3, -1
(C) 1, 4, 3, -1
(D) 4, 4, -1, 4

Q.150 MSQ · 2 marks · Trees

Which of the following sequences can be the preorder of some BST on keys {1,2,3,4,5}?

(A) 1, 2, 3, 4, 5
(B) 5, 4, 3, 2, 1
(C) 3, 1, 2, 5, 4
(D) 3, 5, 1, 2, 4

Q.151 NAT · 1 mark · Sorting

Bubble sort without early abort on n=6 keys performs how many adjacent comparisons?

NAT · enter the numerical value

Q.152 MCQ · 2 marks · Graphs

Directed graph LabNet has edges S0- > S1, S0- > S2, S1- > S3, S2- > S3, S3- > S4, S1- > S4. BFS from S0 scans neighbours in alphabetical order. The dequeue order is

(A) S0 S1 S2 S3 S4
(B) S0 S1 S3 S4 S2
(C) S4 S3 S2 S1 S0
(D) S1 S2 S3 S4 S0

Q.153 MCQ · 1 mark · Python

```
The fragment
d = { (1, 2): 'ok' }
try:
d[[1, 2]] = 'no'
print('assigned')
except TypeError:
print(list(d.values())[0])
prints
```

(A) assigned
(B) ok
(C) no
(D) nothing (crash without except)

Q.154 NAT · 1 mark · Stacks and queues

Using a stack to check parentheses on (()())(()), scan the given 11-character string. The maximum number of '(' sitting on the stack at any moment is

NAT · enter the numerical value

Q.155 MCQ · 1 mark · Trees

A threaded binary tree (inorder threads) of n nodes stores two extra bits per node. Compared with a recursive inorder walk, an inorder walk on the threaded tree uses

(A) O(1) auxiliary stack space
(B) O(n) auxiliary stack space
(C) O(log n) auxiliary stack space always
(D) O(n^2) time

Q.156 MCQ · 1 mark · Sorting

Insertion sort on a strictly increasing array of n keys performs how many key comparisons?

(A) n-1 (each new key compared once and already in place)
(B) n(n-1)/2
(C) n log n
(D) 0

Q.157 MCQ · 2 marks · Graphs

Directed graph LabNet has edges S0- > S1, S0- > S2, S1- > S3, S2- > S3, S3- > S4, S1- > S4. Recursive DFS from S0 with neighbours tried in alphabetical order. The discovery order is

(A) S0 S1 S3 S4 S2
(B) S0 S1 S2 S3 S4
(C) S2 S4 S3 S1 S0
(D) S4 S3 S2 S1 S0

Q.158 NAT · 1 mark · Python

```
The Python function
def f(n, m):
if n == 0:
return m
return f(n-1, m+n)
print(f(4, 0))
prints
```

NAT · enter the numerical value

Q.159 MCQ · 1 mark · Stacks and queues

A priority queue (min-heap) is built on keys 7, 2, 9, 4, 5 by successive insertions into an initially empty binary heap (array layout, swim on insert). The root key after all insertions is

(A) 2
(B) 7
(C) 4
(D) 9

Q.160 MCQ · 1 mark · Sorting

A 3-way partition quicksort (Dutch-national-flag style) on keys with many duplicates is attractive because

(A) equal-to-pivot keys are excluded from both recursive calls
(B) it uses no extra memory versus Lomuto
(C) it makes the algorithm stable
(D) the worst case becomes Theta(n log n) for all inputs

Q.161 NAT · 1 mark · Graphs

In graph LabNet (edges S0- > S1, S0- > S2, S1- > S3, S2- > S3, S3- > S4, S1- > S4), BFS from S0 with alphabetical neighbour order visits how many vertices?

NAT · enter the numerical value

Q.162 NAT · 2 marks · Python

```
How many calls (including the first) does a naive
Fibonacci implementation make on input 6? Equivalent
code:
def calls(n):
c = [0]
def g(n):
c[0] += 1
return n if n < = 1 else g(n-1) + g(n-2)
g(n)
return c[0]
print(calls(6))
```

NAT · enter the numerical value

Q.163 NAT · 2 marks · Stacks and queues

Insert 7,2,9,4,5 into an empty binary min-heap (1-indexed array, swim). After all inserts, the key at index 2 (left child of the root) is

NAT · enter the numerical value

Q.164 NAT · 1 mark · Sorting

Heap-sort first build-heap on n=7 keys using Floyd's method. The number of sink operations during build-heap is equal to the number of non-leaves, which is

NAT · enter the numerical value

Q.165 MCQ · 2 marks · Graphs

Undirected unweighted graph River has edges North-Island, Island-South, North-West, West-South. BFS from North scans neighbours in alphabetical order. The dequeue order is

(A) North Island West South
(B) North Island South West
(C) South West Island North
(D) Island North South West

Q.166 MCQ · 1 mark · Python

```
After
a, *b, c = [9, 8, 7, 6, 5]
print(a, b, c)
the printed line is
```

(A) 9 [8, 7, 6] 5
(B) 9 8 [7, 6, 5]
(C) [9, 8, 7, 6] 5 None
(D) 9 [8, 7, 6, 5] error

Q.167 MCQ · 1 mark · Stacks and queues

A queue of size 3 (capacity 3) is used as a sliding window. Stream 4,1,7,2 is enqueued, and a dequeue happens whenever the queue is full before an enqueue. After processing the whole stream this way, the queue contents front-to-rear are

(A) 1, 7, 2
(B) 4, 1, 7
(C) 7, 2
(D) 4, 1, 7, 2

Q.168 MCQ · 1 mark · Sorting

After the first extract-max pass of heapsort on a max-heap [9, 6, 8, 2, 5, 7, 3] (n=7), the last array position holds

(A) 9
(B) 8
(C) 3
(D) 2

Q.169 MCQ · 2 marks · Graphs

Undirected unweighted graph River has edges North-Island, Island-South, North-West, West-South. Recursive DFS from North with neighbours tried in alphabetical order. The discovery order is

(A) North Island South West
(B) North Island West South
(C) West South Island North
(D) West South North Island

Q.170 MCQ · 1 mark · Python

```
The expression
print(bool([]), bool([0]), bool(''), bool('0'))
prints
```

(A) False False False True
(B) False True False True
(C) False True True True
(D) True True False True

Q.171 MSQ · 2 marks · Stacks and queues

Which of the following problems are naturally solved with a stack?

(A) Matching HTML tags / parentheses
(B) BFS layer-order traversal of a tree
(C) Evaluating postfix arithmetic
(D) Implementing recursive DFS with an explicit data structure

Q.172 MSQ · 2 marks · Sorting

Array PairSwap = [2, 1, 4, 3, 5]. Which claims hold?

(A) the array has exactly two inversions
(B) adjacent bubble sort performs exactly two swaps
(C) adjacent insertion sort performs exactly two swaps
(D) mergesort extra memory is Theta(1)

Q.173 NAT · 1 mark · Graphs

In graph River (edges North-Island, Island-South, North-West, West-South), BFS from North with alphabetical neighbour order visits how many vertices?

NAT · enter the numerical value

Q.174 MCQ · 1 mark · Python

```
The program
def f(x):
return x or 7
print(f(0), f(2), f([]), f([0]))
prints
```

(A) 7 2 7 [0]
(B) 0 2 [] [0]
(C) 7 2 [] [0]
(D) 7 2 7 7

Q.175 NAT · 1 mark · Stacks and queues

A singly linked list has nodes A,B,C,D in that order. Pointer p points to A. The assignments p = p.next; p = p.next; q = p; p = p.next are executed. Node q now holds letter (A=1,B=2,C=3,D=4)

NAT · enter the numerical value

Q.176 NAT · 2 marks · Sorting

Using T(n)=T(floor n/2)+T(ceil n/2)+(n-1), T(1)=0 (each merge uses n-1 comparisons), T(16) equals

NAT · enter the numerical value

Q.177 MCQ · 1 mark · Graphs

Undirected graph Campus has edges Admin-Lab, Lab-Canteen, Canteen-Hostel, Admin-Library, Library-Hostel. How many connected components does it have?

(A) 1
(B) 2
(C) 3
(D) 5

Q.178 MCQ · 2 marks · Python

```
After
n = 0
def side():
global n
n += 1
return n
print(side() and side() and side())
print(n)
the two printed lines are
```

(A) 1 then 1
(B) 3 then 3
(C) 1 then 3
(D) True then 3

Q.179 MCQ · 1 mark · Stacks and queues

Deleting the successor of node p in a singly linked list (p is not the tail) is done by p.next = p.next.next. If the list is 6- > 8- > 1- > 4 and p points at 8, the resulting sequence is

(A) 6 - > 8 - > 4
(B) 6 - > 1 - > 4
(C) 6 - > 8 - > 1
(D) 8 - > 1 - > 4

Q.180 MCQ · 1 mark · Sorting

A lower bound on comparisons to sort n distinct keys in the worst case in the comparison model is

(A) ceil(log2(n!))
(B) n-1
(C) n
(D) 2^n

Q.181 NAT · 1 mark · Graphs

Undirected graph Forest has 6 vertices Oak,Pine,Birch,Maple,Elm,Ash and edges Oak-Pine, Pine-Birch, Birch-Maple, Oak-Elm, Elm-Maple, Pine-Elm. Ash is isolated. The number of connected components is

NAT · enter the numerical value

Q.182 MCQ · 1 mark · Python

```
After
n = 0
def side():
global n
n += 1
return n
print(0 and side())
print(n)
the printed pair is
```

(A) 0 then 0
(B) 0 then 1
(C) 1 then 1
(D) False then 0

Q.183 NAT · 1 mark · Stacks and queues

A doubly linked list of 5 nodes is reversed by swapping prev/next at every node and moving the head. The number of nodes whose both pointer fields are rewritten is

NAT · enter the numerical value

Q.184 NAT · 1 mark · Graphs

A tree on 12 vertices has how many edges?

NAT · enter the numerical value

Q.185 MCQ · 1 mark · Python

```
The expression
print(5 / 2, 5 // 2, -5 // 2)
prints
```

(A) 2.5 2 -2
(B) 2.5 2 -3
(C) 2 2 -2
(D) 2.5 2.0 -2.0

Q.186 MCQ · 2 marks · Stacks and queues

An output-restricted deque allows deletions only at the front, but insertions at both ends. Which ADT can it still implement in O(1) per operation?

(A) a stack, by inserting and deleting at the front
(B) a stack, only if we insert at the rear
(C) a FIFO queue is impossible
(D) neither stack nor queue

Q.187 MSQ · 2 marks · Graphs

Let G be an undirected simple graph on n > =2 vertices. Which statements are true?

(A) If G is a tree then G is connected and has n-1 edges
(B) If G is connected and has n-1 edges then G is a tree
(C) If G has n-1 edges then G is a tree
(D) Every tree is bipartite

Q.188 MCQ · 1 mark · Python

```
The expression
print(2 ** 3 ** 2, (2 ** 3) ** 2)
prints
```

(A) 64 64
(B) 512 64
(C) 64 512
(D) 256 64

Q.189 NAT · 1 mark · Stacks and queues

Push the integers 1,2,3,4,5 in that order onto a stack. You may pop at any time. One possible output sequence is 2,3,1,5,4. How many pop operations occur in total to produce that full output?

NAT · enter the numerical value

Q.190 MCQ · 1 mark · Graphs

Degree sequence of undirected graph Club with edges Ram-Sam, Sam-Tom, Tom-Uma, Ram-Uma, Sam-Uma is (vertices Ram,Sam,Tom,Uma in that name order)

(A) 2,3,2,3
(B) 3,3,3,3
(C) 1,2,2,3
(D) 2,2,2,2

Q.191 MCQ · 2 marks · Python

```
The program
fns = []
for i in range(3):
fns.append(lambda: i)
print([f() for f in fns])
prints
```

(A) [0, 1, 2]
(B) [2, 2, 2]
(C) [3, 3, 3]
(D) [0, 0, 0]

Q.192 MCQ · 2 marks · Stacks and queues

Which input sequence of {push, pop} on an initially empty stack leaves the stack empty and never pops an empty stack? Here + means push one new item, - means pop.

(A) + - + - + + - -
(B) + + - - - + + -
(C) + - - + + - + -
(D) - + + - + - + -

Q.193 NAT · 1 mark · Graphs

Handshaking: undirected Harbor has edges Pier-Dock, Dock-Yard, Yard-Gate, Pier-Shed, Shed-Gate, Dock-Shed. The sum of degrees is

NAT · enter the numerical value

Q.194 MCQ · 2 marks · Python

```
The program
fns = []
for i in range(3):
fns.append(lambda i=i: i)
print([f() for f in fns])
prints
```

(A) [0, 1, 2]
(B) [2, 2, 2]
(C) [0, 0, 0]
(D) [1, 1, 1]

Q.195 MCQ · 2 marks · Graphs

Which of the following degree sequences is graphic for a simple undirected graph on 4 vertices?

(A) 3,1,1,1
(B) 3,3,3,1
(C) 4,2,2,2
(D) 3,3,1,1

Q.196 MCQ · 2 marks · Python

```
After
d = dict.fromkeys(['a', 'b'], [])
d['a'].append(1)
print(d)
d is
```

(A) {'a': [1], 'b': []}
(B) {'a': [1], 'b': [1]}
(C) {'a': [1], 'b': None}
(D) {'a': 1, 'b': []}

Q.197 NAT · 2 marks · Graphs

Directed weighted graph RoadNet with edges S- > A(4), S- > B(1), B- > A(2), A- > C(1), B- > C(5), C- > T(3), B- > T(8). Dijkstra from S (non-negative weights). Shortest-path distance to T is

NAT · enter the numerical value

Q.198 MCQ · 1 mark · Python

```
The function
def f():
try:
return 1
finally:
return 2
print(f())
prints
```

(A) 1
(B) 2
(C) None
(D) an error is raised

Q.199 NAT · 2 marks · Graphs

Directed weighted graph PacketNet with edges U- > V(2), U- > W(5), V- > W(1), V- > X(4), W- > X(1). Dijkstra from U (non-negative weights). Shortest-path distance to X is

NAT · enter the numerical value

Q.200 MCQ · 1 mark · Python

```
The function
def f():
try:
return 1
except:
return 3
else:
return 4
print(f())
prints
```

(A) 1
(B) 4
(C) 3
(D) None

Q.201 NAT · 2 marks · Graphs

Directed weighted graph HillWalk with edges Base- > Ridge(6), Base- > Lake(3), Lake- > Ridge(2), Ridge- > Peak(4), Lake- > Peak(9). Dijkstra from Base (non-negative weights). Shortest-path distance to Peak is

NAT · enter the numerical value

Q.202 MCQ · 1 mark · Python

```
After slice assignment
xs = [1, 2, 3]
ys = xs
xs[0:3] = [9, 9, 9]
print(ys)
ys is
```

(A) [1, 2, 3]
(B) [9, 9, 9]
(C) [9, 2, 3]
(D) [1, 2, 3, 9, 9, 9]

Q.203 NAT · 2 marks · Graphs

Directed weighted graph CityCab with edges Home- > X(7), Home- > Y(2), Y- > X(3), X- > Office(1), Y- > Office(8). Dijkstra from Home (non-negative weights). Shortest-path distance to Office is

NAT · enter the numerical value

Q.204 MCQ · 1 mark · Python

```
The list comprehension
print([i for i in range(5) if i % 2 == 0])
evaluates to
```

(A) [0, 2, 4]
(B) [1, 3]
(C) [0, 1, 2, 3, 4]
(D) [2, 4]

Q.205 NAT · 2 marks · Graphs

Directed weighted graph RailLink with edges Pune- > Mumbai(10), Pune- > Lonavala(4), Lonavala- > Mumbai(5), Mumbai- > Goa(8), Lonavala- > Goa(20). Dijkstra from Pune (non-negative weights). Shortest-path distance to Goa is

NAT · enter the numerical value

Q.206 MCQ · 2 marks · Python

```
The program
acc = []
print([acc.append(i) or i for i in range(3)])
print(acc)
prints
```

(A) [0, 1, 2] then [0, 1, 2]
(B) [None, None, None] then [0, 1, 2]
(C) [0, 1, 2] then []
(D) [None, None, None] then []

Q.207 MCQ · 1 mark · Graphs

BFS on an unweighted graph computes

(A) a shortest path in number of edges from the source to every reachable vertex
(B) a minimum spanning tree
(C) shortest paths for negative weights
(D) a topological order of every directed graph

Q.208 MCQ · 1 mark · Python

```
After
z = zip([1, 2, 3], [10, 20])
print(list(z), list(z))
the printed pair is
```

(A) [(1, 10), (2, 20), (3, None)] []
(B) [(1, 10), (2, 20)] [(1, 10), (2, 20)]
(C) [(1, 10), (2, 20)] []
(D) [(1, 10), (2, 20), (3, 0)] []

Q.209 MSQ · 2 marks · Graphs

Dijkstra's algorithm (as taught with a min-priority queue, non-negative weights) . Which are correct?

(A) It may fail if some edge weight is negative
(B) It computes shortest paths from one source
(C) It requires the graph to be a DAG
(D) On an unweighted graph it is no better asymptotically than BFS if implemented with a binary heap on dense graphs, and BFS is the natural choice

Q.210 NAT · 1 mark · Python

```
The expression
print((lambda x: x*x)(4))
prints
```

NAT · enter the numerical value

Q.211 MCQ · 1 mark · Graphs

A directed graph has edges A- > B, B- > C, A- > C, C- > D. How many topological orders exist?

(A) 2
(B) 1
(C) 3
(D) 6

Q.212 NAT · 1 mark · Python

```
The decorated function
def deco(f):
def g(x):
return f(x) + 1
return g
@deco
def h(x):
return x * 2
print(h(3))
prints
```

NAT · enter the numerical value

Q.213 MCQ · 1 mark · Graphs

Directed edges P- > Q, P- > R, Q- > S, R- > S. The number of topological orders is

(A) 2
(B) 1
(C) 3
(D) 4

Q.214 NAT · 1 mark · Python

```
The program
class C:
n = 0
def __init__(self):
C.n += 1
C(); C(); print(C.n)
prints
```

NAT · enter the numerical value

Q.215 NAT · 1 mark · Graphs

Undirected cycle of 8 vertices (C8). BFS from one vertex visits how many vertices at distance exactly 2?

NAT · enter the numerical value

Q.216 MCQ · 1 mark · Python

```
The expression
print(list(map(lambda p: p[0]+p[1],
zip([1,2,3],[4,5,6]))))
evaluates to
```

(A) [5, 7, 9]
(B) [1, 2, 3, 4, 5, 6]
(C) [(1, 4), (2, 5), (3, 6)]
(D) [14]

Q.217 NAT · 1 mark · Graphs

Complete graph K5 has how many edges?

NAT · enter the numerical value

Q.218 MCQ · 1 mark · Python

```
After
s = {1, 2, 3}
t = {3, 4}
print(sorted(s & t), sorted(s | t), sorted(s - t))
the printed line is
```

(A) [3] [1, 2, 3, 4] [1, 2]
(B) [3] [1, 2, 3, 4] [4]
(C) [1, 2] [1, 2, 3, 4] [3]
(D) [] [1, 2, 3, 4] [1, 2, 3]

Q.219 NAT · 1 mark · Graphs

Complete bipartite K(3,4) has how many edges?

NAT · enter the numerical value

Q.220 MCQ · 1 mark · Python

```
After rebinding
a = [1, 2, 3]
b = a
a = [0]
print(b)
b is
```

(A) [0]
(B) [1, 2, 3]
(C) [1, 2, 3, 0]
(D) None

Q.221 MCQ · 1 mark · Graphs

Which graph is not bipartite?

(A) an undirected triangle C3
(B) any tree
(C) C6
(D) K(2,5)

Q.222 NAT · 1 mark · Python

```
The function
def f(n):
if n < = 0:
return 0
return 1 + f(n-2)
print(f(7))
prints
```

NAT · enter the numerical value

Q.223 MCQ · 1 mark · Graphs

DFS on an undirected connected graph produces a DFS tree. A back edge in that classification indicates

(A) a cycle
(B) a bridge
(C) that the graph is bipartite
(D) that the graph is a tree

Q.224 MCQ · 1 mark · Python

```
The fragment
x = [1, 2, 3, 4, 5]
print(x[-1], x[-3:], x[:-2])
prints
```

(A) 5 [3, 4, 5] [1, 2, 3]
(B) 5 [4, 5] [1, 2, 3]
(C) 1 [1, 2, 3] [4, 5]
(D) 5 [3, 4, 5] [1, 2]

Q.225 NAT · 1 mark · Graphs

A connected undirected graph has 10 vertices and 12 edges. The minimum number of edges that must be removed to make it a tree is

NAT · enter the numerical value

Q.226 MCQ · 2 marks · Python

```
After
def mut(t):
t[0].append('x')
t = t + ([9],)
return t
u = ([1], [2])
v = mut(u)
print(u, v)
the pair (u, v) is
```

(A) (([1, 'x'], [2]), ([1, 'x'], [2], [9]))
(B) (([1], [2]), ([1, 'x'], [2], [9]))
(C) (([1, 'x'], [2], [9]), ([1, 'x'], [2], [9]))
(D) (([1], [2]), ([1], [2], [9]))

Q.227 MCQ · 1 mark · Graphs

Bellman-Ford vs Dijkstra on a directed graph with a negative edge but no negative cycle. Which is correct?

(A) Bellman-Ford still computes correct single-source distances; Dijkstra may not
(B) Dijkstra is always correct here because there is no negative cycle
(C) Both require all weights positive
(D) BFS computes the same distances as both

Q.228 MCQ · 2 marks · Python

```
The call in
def f(a, b, c=1, *rest, d=2, **kw):
return (a, b, c, rest, d, kw)
print(f(9, 8, 7, 6, d=5, z=4))
returns
```

(A) (9, 8, 7, (6,), 5, {'z': 4})
(B) (9, 8, 1, (7, 6), 5, {'z': 4})
(C) (9, 8, 7, (6,), 2, {'d': 5, 'z': 4})
(D) (9, 8, 7, (6, 5), 2, {'z': 4})

Q.229 NAT · 1 mark · Graphs

BFS from vertex Src in directed Pipeline with edges Src- > A, Src- > B, A- > C, B- > C, C- > Sink, A- > Sink, neighbours in alpha order. The hop-distance to Sink is

NAT · enter the numerical value

Q.230 NAT · 1 mark · Python

```
The generator expression
print(sum(i*i for i in range(1, 5) if i % 2 == 1))
prints
```

NAT · enter the numerical value

Q.231 MCQ · 1 mark · Graphs

An adjacency-matrix representation of a simple undirected graph on n vertices uses Theta(n^2) space. BFS from one vertex then runs in

(A) Theta(n^2) time in the worst case
(B) Theta(n+m) time always, independent of the matrix
(C) Theta(n) time
(D) Theta(n^3) time

Q.232 MCQ · 1 mark · Python

```
After
A = [1, 2, 3]
B = [4, 5, 6]
A.extend(B)
print(A)
A is
```

(A) [1, 2, 3, [4, 5, 6]]
(B) [1, 2, 3, 4, 5, 6]
(C) [1, 2, 3]
(D) [4, 5, 6]

Q.233 MCQ · 1 mark · Graphs

Adjacency lists of a directed graph with n vertices and m edges support BFS from one vertex in

(A) Theta(n+m) time
(B) Theta(n^2) time always
(C) Theta(m log n) time
(D) Theta(n m) time

Q.234 MCQ · 1 mark · Python

```
After
A = [1, 2, 3]
B = [4, 5, 6]
A.append(B)
print(len(A), A[-1])
the printed pair is
```

(A) 6 [4, 5, 6]
(B) 4 [4, 5, 6]
(C) 4 6
(D) 6 6

Q.235 MSQ · 2 marks · Graphs

Undirected simple graph G. Which are equivalent to G being a tree?

(A) G is connected and has no cycle
(B) G is connected and has n-1 edges
(C) there is a unique path between every pair of vertices
(D) G is bipartite and has n-1 edges

Q.236 MCQ · 1 mark · Python

```
The call in
def rec(n, acc):
acc.append(n)
if n == 0:
return acc
return rec(n-1, acc)
print(rec(3, []))
prints
```

(A) [3, 2, 1, 0]
(B) [0, 1, 2, 3]
(C) [3]
(D) [0]

Q.237 NAT · 1 mark · Graphs

A DAG on vertices W,X,Y,Z has edges W- > X, W- > Y, X- > Z, Y- > Z. The number of distinct topological sorts is

NAT · enter the numerical value

Q.238 MCQ · 1 mark · Python

```
The call in
def rec(n):
if n == 0:
return [0]
return rec(n-1) + [n]
print(rec(3))
prints
```

(A) [3, 2, 1, 0]
(B) [0, 1, 2, 3]
(C) [3]
(D) [0, 3]

Q.239 MCQ · 1 mark · Graphs

In an undirected unweighted graph, if BFS from s first reaches t via parent p, then

(A) dist(s,t)=dist(s,p)+1 and this dist is minimum hop count
(B) the path s..p,t is a unique shortest path
(C) DFS would find the same parent
(D) t is a neighbour of s

Q.240 NAT · 1 mark · Python

```
The loop
i = 1
while i < 10:
i *= 2
print(i)
prints
```

NAT · enter the numerical value

Q.241 NAT · 1 mark · Graphs

Directed graph Airline: Delhi- > Mumbai, Delhi- > Pune, Mumbai- > Goa, Pune- > Goa, Goa- > Kochi. Number of simple paths from Delhi to Kochi is

NAT · enter the numerical value

Q.242 NAT · 1 mark · Python

```
The nested loops
cnt = 0
for i in range(1, 5):
for j in range(i):
cnt += 1
print(cnt)
print
```

NAT · enter the numerical value

Q.243 MCQ · 1 mark · Graphs

A spanning tree of a connected undirected graph with 20 vertices has how many edges?

(A) 19
(B) 20
(C) 21
(D) 10

Q.244 NAT · 1 mark · Python

```
The function
def f(n):
r = 0
while n:
r += n & 1
n > > = 1
return r
print(f(13))
prints
```

NAT · enter the numerical value

Q.245 NAT · 1 mark · Graphs

How many spanning trees does K3 (a triangle) have?

NAT · enter the numerical value

Q.246 NAT · 1 mark · Python

```
The expression
print(int('1011', 2) + int('12', 8))
prints
```

NAT · enter the numerical value

Q.247 MCQ · 1 mark · Graphs

If Dijkstra is run on RoadNet with non-negative weights from S, the set of finalized vertices grows by extracting the min tentative distance. The first vertex finalized after S, given edges S- > A(4), S- > B(1), B- > A(2), A- > C(1), B- > C(5), C- > T(3), B- > T(8), is

(A) B
(B) A
(C) C
(D) T

Q.248 MCQ · 1 mark · Python

```
The fragment
s = 'abca'
print(s.find('a'), s.rfind('a'), s.find('z'))
prints
```

(A) 0 3 -1
(B) 0 0 -1
(C) 1 4 0
(D) 0 3 4

Q.249 NAT · 1 mark · Graphs

On that same RoadNet, after Dijkstra from S, the shortest-path distance of A is

NAT · enter the numerical value

Q.250 MCQ · 1 mark · Python

```
The expression
print(''.join(reversed('gate')))
prints
```

(A) etag
(B) gate
(C) e t a g
(D) ['e','t','a','g']

Q.251 MSQ · 2 marks · Graphs

Which claims about BFS and DFS on a finite directed graph G (no self-loops) are always true?

(A) From a vertex v, BFS and DFS visit exactly the same set of vertices (the descendants of v in the reachable sense)
(B) The order of first visits in BFS is always the reverse of DFS
(C) If G^R is G with all edges reversed, u reachable from v in G implies v reachable from u in G^R
(D) BFS requires a queue and DFS a stack (or recursion stack)

Q.252 MCQ · 1 mark · Python

```
After
def f(x, y):
x.append(y)
y = y + 1
a = [1]
b = 2
f(a, b)
print(a, b)
the printed pair is
```

(A) [1, 2] 2
(B) [1, 2] 3
(C) [1] 2
(D) [1] 3

Q.253 MCQ · 1 mark · Graphs

An undirected graph is Eulerian (has a closed trail using every edge) if and only if

(A) it is connected (ignoring isolated vertices) and every degree is even
(B) it is a tree
(C) it is complete
(D) it has exactly two odd-degree vertices

Q.254 MCQ · 2 marks · Python

```
The expression
print([n for n in range(8) if n % 3][::2])
evaluates to
```

(A) [1, 4, 7]
(B) [1, 2, 4, 5, 7]
(C) [1, 5]
(D) [0, 3, 6]

Q.255 NAT · 1 mark · Graphs

Hypercube Q3 (the cube graph) has how many vertices and how many edges? Report the number of edges.

NAT · enter the numerical value

Q.256 MCQ · 2 marks · Python

```
After calling memo_fib(6) in
def memo_fib(n, m={}):
if n < 2:
return n
if n not in m:
m[n] = memo_fib(n-1) + memo_fib(n-2)
return m[n]
memo_fib(6)
print(sorted(memo_fib.__defaults__[0].keys()))
the printed key list is
```

(A) [2, 3, 4, 5, 6]
(B) [0, 1, 2, 3, 4, 5, 6]
(C) [6]
(D) []

Q.257 MCQ · 1 mark · Graphs

A cut-vertex (articulation point) of a connected graph is a vertex whose removal

(A) increases the number of connected components
(B) decreases the number of edges by 1 always
(C) always leaves the graph a tree
(D) cannot lie on a cycle

Q.258 MCQ · 2 marks · Python

```
The generator
def f(n):
yield n
yield from f(n-1) if n else ()
print(list(f(3)))
yields the list
```

(A) [3, 2, 1, 0]
(B) [0, 1, 2, 3]
(C) [3]
(D) [3, 2, 1]

Q.259 NAT · 1 mark · Graphs

Grid graph of a 3x3 lattice of vertices with edges between 4-neighbours. Number of vertices is 9; number of edges is

NAT · enter the numerical value

Q.260 MSQ · 2 marks · Python

Which of the following Python objects are hashable and may be placed in a set?

(A) (1, [2])
(B) (1, (2, 3))
(C) frozenset({1, 2})
(D) {1: 2}

Q.261 MCQ · 1 mark · Graphs

If every edge of an undirected connected graph has weight 1, Dijkstra's distances from s coincide with

(A) BFS hop distances from s
(B) DFS discovery times
(C) the number of spanning trees
(D) the maximum degree

Q.262 MSQ · 2 marks · Python

Let L be a list and T a tuple of the same elements. Which statements are always true in Python 3?

(A) L.append(0) mutates L in place
(B) T.append(0) mutates T in place
(C) L += [0] reuses the same list object
(D) T += (0,) reuses the same tuple object

Q.263 MSQ · 2 marks · Python

```
Regarding default arguments in a Python def statement,
which claims hold?
```

(A) A default list object is evaluated once, at function definition time
(B) A default list object is evaluated fresh on every call that omits the argument
(C) Using None as a sentinel and allocating a list inside the body avoids cross-call sharing
(D) Integer defaults such as n=0 share mutations across calls the same way lists do

Q.264 MCQ · 1 mark · Python

```
The program
def f(x):
x = x + [1]
return x
a = [0]
print(f(a), a)
prints
```

(A) [0, 1] [0, 1]
(B) [0, 1] [0]
(C) [0] [0]
(D) [1] [0]

Q.265 MCQ · 1 mark · Python

```
The program
def f(x):
x += [1]
return x
a = [0]
print(f(a), a)
prints
```

(A) [0, 1] [0]
(B) [0, 1] [0, 1]
(C) [0] [0]
(D) [1] [1]

Q.266 MCQ · 1 mark · Python

```
The expression
print(all([]), any([]), all([1, 0]), any([0, 0, 2]))
prints
```

(A) True False False True
(B) False False False True
(C) True True True True
(D) True False True True

Q.267 MCQ · 1 mark · Python

```
The expression
print(sorted('cba'), list('cba'))
prints
```

(A) ['a', 'b', 'c'] ['c', 'b', 'a']
(B) abc ['c', 'b', 'a']
(C) ['c', 'b', 'a'] ['c', 'b', 'a']
(D) ['a', 'b', 'c'] cba

Q.268 MCQ · 1 mark · Python

```
After
m = {'x': 1, 'y': 2}
print(list(m.keys())[0], m.get('z', 9))
a possible printed line (insertion order is preserved)
is
```

(A) x 9
(B) z 9
(C) x None
(D) y None

Q.269 NAT · 1 mark · Python

```
The factorial function
def f(n):
if n < = 1:
return 1
return n * f(n-1)
print(f(5))
prints
```

NAT · enter the numerical value

Q.270 NAT · 1 mark · Python

```
The function
def depth(n):
return 1 if n == 0 else 1 + depth(n-1)
print(depth(6))
prints the number of stack frames created by the chain
(including the first). The printed value is
```

NAT · enter the numerical value

Q.271 MCQ · 1 mark · Python

```
After
xs = [0, 1, 2, 3, 4, 5]
del xs[1:4]
print(xs)
xs is
```

(A) [0, 4, 5]
(B) [0, 1, 5]
(C) [4, 5]
(D) [0, 1, 2, 3]

Q.272 MCQ · 1 mark · Python

```
After inserting into an empty slice
xs = [0, 1, 2, 3]
xs[1:1] = [8, 9]
print(xs)
xs is
```

(A) [0, 8, 9, 1, 2, 3]
(B) [8, 9, 0, 1, 2, 3]
(C) [0, 1, 8, 9, 2, 3]
(D) [0, 8, 9, 2, 3]

Q.273 MCQ · 1 mark · Python

```
The expression
print(2 < 3 < 2, 2 < 3 and 3 < 2)
prints
```

(A) True False
(B) False False
(C) False True
(D) True True

Q.274 MCQ · 1 mark · Python

```
The fragment
a = [1, 2, 3]
print(a * 2, [a] * 2)
prints
```

(A) [1, 2, 3, 1, 2, 3] [[1, 2, 3], [1, 2, 3]]
(B) [2, 4, 6] [[1, 2, 3], [1, 2, 3]]
(C) [1, 2, 3, 1, 2, 3] [1, 2, 3, 1, 2, 3]
(D) [1, 2, 3, 2] [[1, 2, 3], [1, 2, 3]]

Q.275 MCQ · 1 mark · Python

```
The expression
print({1, 2, 3} == {3, 2, 1}, [1, 2, 3] == [3, 2, 1])
prints
```

(A) True False
(B) True True
(C) False False
(D) False True

Q.276 MCQ · 2 marks · Python

```
The program
def f(n, s=[]):
s.append(n)
if n == 0:
return list(s)
return f(n-1)
print(f(2))
print(f(1))
prints
```

(A) [2, 1, 0] then [1, 0]
(B) [2, 1, 0] then [2, 1, 0, 1, 0]
(C) [0] then [0]
(D) [2, 1, 0] then [2, 1, 0]

Q.277 MCQ · 1 mark · Python

```
The comprehension
print([i // 2 for i in range(-3, 3)])
evaluates to
```

(A) [-2, -1, -1, 0, 0, 1]
(B) [-1, -1, 0, 0, 1, 1]
(C) [-2, -2, -1, -1, 0, 1]
(D) [-1, -1, -1, 0, 0, 1]

Q.278 MCQ · 1 mark · Python

```
The expression
print(list(enumerate(['p', 'q'], start=2)))
evaluates to
```

(A) [(2, 'p'), (3, 'q')]
(B) [(0, 'p'), (1, 'q')]
(C) [(1, 'p'), (2, 'q')]
(D) [(2, 'p'), (2, 'q')]

Q.279 MCQ · 1 mark · Python

```
In Python 3, bankers rounding is used. The expression
print(round(2.5), round(3.5), round(2.51))
prints
```

(A) 2 4 3
(B) 3 4 3
(C) 2 3 3
(D) 3 3 3

Q.280 MCQ · 1 mark · Python

```
The function
def f(n):
return [n] + f(n-1) if n else []
print(f(3))
prints
```

(A) [3, 2, 1]
(B) [1, 2, 3]
(C) [3, 2, 1, 0]
(D) []

Q.281 MCQ · 1 mark · Python

```
After in-place set union
a = {1, 2, 3}
b = a
a |= {4}
print(sorted(b))
sorted(b) is
```

(A) [1, 2, 3]
(B) [1, 2, 3, 4]
(C) [4]
(D) [1, 2, 3, 4, 4]

Q.282 MCQ · 1 mark · Python

```
After
a = {1, 2, 3}
b = a
a = a | {4}
print(sorted(b), sorted(a))
the printed pair is
```

(A) [1, 2, 3] [1, 2, 3, 4]
(B) [1, 2, 3, 4] [1, 2, 3, 4]
(C) [1, 2, 3] [4]
(D) [1, 2, 3, 4] [1, 2, 3]

Q.283 MCQ · 1 mark · Python

```
The expression
print(min(['10','2','3'], key=int), min(['10','2','3']))
prints
```

(A) 2 10
(B) 10 10
(C) 2 2
(D) 10 2

Q.284 MCQ · 1 mark · Python

```
The fragment
xs = [1, 2, 3, 2, 1]
print(xs.index(2), xs.count(1))
prints
```

(A) 1 2
(B) 2 2
(C) 1 1
(D) 3 2

Q.285 MCQ · 1 mark · Python

```
The nested function
def f(n):
k = 0
def g():
nonlocal k
k += 1
return k
return [g() for _ in range(n)]
print(f(3))
prints
```

(A) [1, 2, 3]
(B) [0, 0, 0]
(C) [1, 1, 1]
(D) [3, 3, 3]

Q.286 MCQ · 2 marks · Python

```
The double comprehension
print([x*y for x in range(3) for y in range(3) if x < =
y])
evaluates to
```

(A) [0, 0, 0, 1, 2, 4]
(B) [0, 0, 0, 1, 2, 2, 4]
(C) [0, 1, 2, 0, 2, 4]
(D) [0, 1, 4]

UNIT 4 SOLUTIONS

Q.1 [MCQ] B — The default list is created once at def-time, so the first two calls share it. The third call passes a fresh list.

Q.2 [NAT] 3 — After the sequence the stack holds only 3.

Q.3 [MCQ] A — R is root; left subtree inorder CAD / pre ACD so A with left C right D; right subtree inorder EB / pre BE so B with left E. Postorder C D A E B R.

Q.4 [MCQ] A — h(1)=3, h(4)=2, h(5)=5, h(6)=8, h(14)=2 (collision with 4, probe 3 occupied by 1, then 4), h(15)=5 (collision, probe 6). Simulate carefully in sol: compute positions {1: 3, 4: 2, 5: 5, 6: 8, 14: 4, 15: 6}.

Q.5 [NAT] 3 — Standard binary search on this concrete array performs 3 comparisons; found=True.

Q.6 [NAT] 6 — Each inversion is swapped exactly once in this bubble-sort variant; the inversion count is 6.

Q.7 [MCQ] A — BFS with alpha neighbour order dequeues Admin Lab Library Canteen Hostel.

Q.8 [MCQ] B — box=None is immutable; a new list is allocated on every call that omits the second argument.

Q.9 [MCQ] A — Dequeue removes 4 then 9, leaving 2 then 6.

Q.10 [NAT] 2 — Longest root-to-leaf path has length 2 (R-A-C or R-A-D or R-B-E).

Q.11 [NAT] 9 — h=1,0,10,9? 12 mod 11=1, 22 mod 11=0, 32 mod 11=10, 42 mod 11=9. No collisions actually if those slots empty. 1,0,10,9 are distinct so 42 at 9. {12: 1, 22: 0, 32: 10, 42: 9}

Q.12 [NAT] 1 — Standard binary search on this concrete array performs 1 comparisons; found=True.

Q.13 [NAT] 4 — Each inversion is swapped exactly once in this bubble-sort variant; the inversion count is 4.

Q.14 [MCQ] A — DFS discovery order is Admin Lab Canteen Hostel Library.

Q.15 [MCQ] B — rack and shelf alias the same list object, so append mutates both names.

Q.16 [MCQ] A — After insertRear 1: [1]; insertFront 2: [2,1]; insertRear 3: [2,1,3]; deleteFront removes 2: [1,3]; insertFront 4: [4,1,3]; deleteRear removes 3: [4,1].

Q.17 [NAT] 3 — Leaves are C, D and E.

Q.18 [NAT] 2 — Positions {10: 10, 22: 0, 31: 9, 4: 4, 15: 5, 28: 6, 17: 7} and probes {10: 1, 22: 1, 31: 1, 4: 1, 15: 2, 28: 1, 17: 2}. 17 mod 11=6; collisions force extra probes.

Q.19 [NAT] 9 — Simulate unsuccessful search on 0..999 for -1 (always go left) or a value larger than all. Count=9.

Q.20 [NAT] 6 — Adjacent insertion sort also swaps once per inversion, so the count equals the inversion number 6.

Q.21 [NAT] 5 — The reachable set has size 5.

Q.22 [MCQ] B — p + [4] builds a new list and rebinds p; q still names the original list.

Q.23 [NAT] 4 — E is pushed last and popped first; its code is 4.

Q.24 [MSQ] A;B;D — Left subtree of P is Q with left S and right T; right child of P is leaf R. Postorder S T Q R P.

Q.25 [MCQ] D — 50 goes to 1; 15 collides and takes 2; 36 takes 3; 8 takes 4.

Q.26 [MCQ] B — Need O(1) index access and sorted order. Linked lists make mid access O(n).

Q.27 [NAT] 4 — Adjacent insertion sort also swaps once per inversion, so the count equals the inversion number 4.

Q.28 [MCQ] A — BFS with alpha neighbour order dequeues Pier Dock Shed Yard Gate.

Q.29 [MCQ] B — For lists, += is extend in-place, so p and q remain the same object.

Q.30 [MCQ] A — Stack is a, then a b; pop b; push c; pop c; pop a.

Q.31 [NAT] 4 — 6 is the right child of 3; 4 is between 3 and 6 so it becomes 6's left child.

Q.32 [NAT] 5 — Positions {10: 10, 22: 0, 31: 9, 4: 4, 15: 5}.

Q.33 [MSQ] A;B;C — Without extra structure an unsuccessful linear scan is Theta(n).

Q.34 [NAT] 2 — Selection sort does at most n-1 swaps; here it does 2.

Q.35 [MCQ] A — DFS discovery order is Pier Dock Shed Gate Yard.

Q.36 [MCQ] A — Slice assignment replaces the whole slice [4,6,8] by the single value 0, so the list shrinks.

Q.37 [MSQ] A;C — Head+tail list gives O(1) both ends. Two stacks give amortized O(1). Head-only list makes the rear expensive; tail-only cannot delete the head in O(1).

Q.38 [MCQ] A — BST inorder is sorted order of the keys.

Q.39 [MCQ] A — 12,17,27 all hash to 2; inserting at head reverses insertion order: 27,17,12.

Q.40 [NAT] 5 — 55 is the 5th element (1-based), so 5 comparisons.

Q.41 [NAT] 2 — Selection sort does at most n-1 swaps; here it does 2.

Q.42 [NAT] 5 — The reachable set has size 5.

Q.43 [MCQ] B — A slice of a list copies; mutating view does not touch buf.

Q.44 [MCQ] A — 5, then 1 2 + - > 3, 3 4 * - > 12, 5+12=17, 17-3=14.

Q.45 [NAT] 2 — This is a perfectly balanced tree of height 2.

Q.46 [NAT] 1 — 10 < 13 so the keys can occupy 10 distinct slots; longest chain can be 1.

Q.47 [NAT] 5 — Every element is compared; then the search fails.

Q.48 [MCQ] A — Pass1 swaps 4 with 1 - > [1,3,2,4,5]; pass2 swaps 3 with 2 - > [1,2,3,4,5]. Selection after 2 passes has the two smallest in place.

Q.49 [MCQ] A — BFS with alpha neighbour order dequeues Hub East North West South.

Q.50 [MCQ] B — [[0]*2]*3 reuses one inner list three times, so writing grid[0][0] is visible in every row.

Q.51 [NAT] 2 — (3+4)*2 = 14, then 14/7 = 2.

Q.52 [MCQ] A — In a full binary tree L = I + 1, so 10 leaves.

Q.53 [NAT] 10 — All keys may collide in one slot.

Q.54 [MCQ] A — The sentinel is found after n failures plus 1 match against the sentinel.

Q.55 [MCQ] A — After pass1 the largest 5 is already at the end and 4 has bubbled; after pass2 the suffix of length 2 is sorted.

Q.56 [MCQ] A — DFS discovery order is Hub East South West North.

Q.57 [MCQ] C — A list is unhashable, so inserting [5,6] into a set raises TypeError.

Q.58 [MCQ] D — 3 1 2 would require 1 to be popped before 2 while 3 is already gone, which violates stack discipline: after popping 3 the top is 2, so 1 cannot come next.

Q.59 [NAT] 6 — Parent of i is floor(i/2)=6.

Q.60 [MCQ] A — 1/(1-0.8)=5.

Q.61 [NAT] 1 — lo=hi=0, one comparison at mid, success.

Q.62 [MCQ] A — After two insertions the prefix of length 3 is [2,3,4] and the tail is [1,5].

Q.63 [NAT] 5 — The reachable set has size 5.

Q.64 [NAT] 1 — Nested tuples of immutables are hashable; the second add is a duplicate.

Q.65 [NAT] 3 — (5-2+6) mod 6 = 3.

Q.66 [NAT] 6 — Leaves start after the last internal node floor(12/2)=6, so indices 7..12: six leaves.

Q.67 [MSQ] A;C;D — Quadratic probing still has some clustering; it does not fully remove secondary clustering. Double hashing and chaining claims are standard.

Q.68 [MCQ] A — log3 n steps, each up to 2 comparisons, still logarithmic.

Q.69 [NAT] 5 — Simulated Lomuto quicksort performs 5 distinct-index swaps and yields [1, 2, 3, 5, 8, 9].

Q.70 [MCQ] A — BFS with alpha neighbour order dequeues Oak Elm Pine Maple Birch.

Q.71 [NAT] 15 — The recursion tree of fib(5) has 15 nodes: T(n)=T(n-1)+T(n-2)+1 with T(0)=T(1)=1 gives 15.

Q.72 [MCQ] C — First deq yields 1 after transferring 2,1 onto Out (pop order 1 then 2). Then enq 3 onto In. Next deq pops remaining Out which is 2. Last deq transfers 3 and yields 3.

Q.73 [MCQ] A — Inorder plus one of preorder/postorder/level-order determines the tree. Preorder+postorder does not, in general.

Q.74 [MCQ] A — h1=3, h2=1+4=5. Probes 3, 8, 13 mod 11=2.

Q.75 [NAT] 5 — Simulation gives 5 swaps.

Q.76 [MCQ] A — DFS discovery order is Oak Elm Maple Birch Pine.

Q.77 [NAT] 31 — A perfect binary recursion of depth 4 makes 2^{5}-1 = 31 activations.

Q.78 [NAT] 2 — The standard simulation uses an input stack and an output stack.

Q.79 [MSQ] A;D — BST inorder is sorted; preorder is not. Heaps are not BSTs. Complete-tree height is floor(log2 n) with root height 0.

Q.80 [NAT] 2 — All hash to 0; they occupy 0,1,2 so 32 is at 2.

Q.81 [NAT] 2 — Worst-case split; simulation gives 2.

Q.82 [NAT] 5 — The reachable set has size 5.

Q.83 [MCQ] A — x[:] shallow-copies the top-level list, so append on y does not change x.

Q.84 [MCQ] A — (A+B)*C becomes AB+C*, then D/E becomes DE/, subtraction last: AB+C*DE/-.

Q.85 [NAT] 5 — The 3rd Catalan number C3=5.

Q.86 [MCQ] A — Search follows probes until an empty slot; a hole at the home address hides later clustered keys. Tombstones are required.

Q.87 [MCQ] A — A random (or random-input first) pivot is equally likely to have rank 0..n-1, giving the averaging recurrence.

Q.88 [MCQ] A — BFS with alpha neighbour order dequeues Delhi Mumbai Pune Goa Kochi.

Q.89 [MCQ] B — The outer list is copied but inner lists are shared, so b[0].append mutates a[0].

Q.90 [NAT] 2 — On seeing ')' we pop '+'; at the end we pop '*'. Parentheses themselves are not output.

Q.91 [MCQ] A — 90 is removed, last key 50 is placed at root and sinks: 50 vs max(40,70) swaps with 70. Root becomes 70.

Q.92 [NAT] 1 — Head is 17, so the successful search compares once.

Q.93 [MCQ] A — One side is empty, the other has n-1, plus linear partition: Theta(n^2).

Q.94 [MCQ] A — DFS discovery order is Delhi Mumbai Goa Kochi Pune.

Q.95 [MCQ] B — sorted returns a new list and leaves vals unchanged.

Q.96 [MCQ] A — p.next skips 3 and points at 5; q.next points at 8. From q: 3,8,5,1.

Q.97 [NAT] 0 — Every parent is < = its children.

Q.98 [NAT] 5 — The whole chain of 5 keys is scanned.

Q.99 [NAT] 3 — Inversions: (2,1),(4,1),(4,3) so 3. Simulation=3.

Q.100 [NAT] 5 — The reachable set has size 5.

Q.101 [MCQ] B — list.sort sorts in place and returns None.

Q.102 [NAT] 5 — After 4 fast steps fast is at node 9; slow moved 4 steps to node 5 (the middle).

Q.103 [MCQ] B — The only violation is parent 8 (index 2) versus child 7 (index 5).

Q.104 [MCQ] A — That is the definition of a universal family.

Q.105 [NAT] 10 — C(5,2)=10.

Q.106 [MCQ] A — BFS with alpha neighbour order dequeues Src A B C Sink.

Q.107 [MCQ] B — Three calls share one default list: append 0, then 1, then 2.

Q.108 [MCQ] A — Only cur.next = prev writes a next field; the others only move local pointers.

Q.109 [NAT] 5 — Inorder left of F is A B C D E, five nodes.

Q.110 [NAT] 2 — alpha = n/m = 2.

Q.111 [MCQ] A — The merge buffer is linear; the recursion stack is only log n.

Q.112 [MCQ] A — DFS discovery order is Src A C Sink B.

Q.113 [MCQ] B — The tuple is immutable but its third component is a mutable list, which can be mutated in place.

Q.114 [NAT] 10 — Ten steps around the cycle.

Q.115 [MCQ] A — Right of F is G I H with inorder G H I, so G has right subtree H I with pre I H, hence I with left H. Right child of G is I.

Q.116 [MSQ] A;B — Heapsort and typical selection sort swap non-adjacent equal keys and are not stable.

Q.117 [NAT] 5 — The reachable set has size 5.

Q.118 [MCQ] A — global k is updated by both calls: 0+2=2 then 2+3=5.

Q.119 [MCQ] A — Dummy lets us unlink the first node without a special case; the later 7 remains.

Q.120 [NAT] 16 — I = L - 1 = 16.

Q.121 [MSQ] A;B;D — Heapsort is Theta(n log n) worst case.

Q.122 [MCQ] A — BFS with alpha neighbour order dequeues A1 A2 B1 A3 B2 B3.

Q.123 [NAT] 5 — f(0)=0, f(1)=1, f(2)=1, f(3)=3, f(4)=3+2*1=5.

Q.124 [NAT] 5 — Compare (1,2),(4,2),(4,3),(4,8),(7,8) - five key comparisons; then the rest is appended.

Q.125 [MCQ] A — Inorder is sorted so it is a BST; level-order fills levels left-to-right with 7 nodes, so it is complete (even perfect).

Q.126 [MCQ] A — Selection places the global min then the next min at the front. Bubble places maxima at the back. Insertion only sorts a growing prefix from the left, which after two insertions is [2,3,4] plus tail.

Q.127 [MCQ] A — DFS discovery order is A1 A2 A3 B3 B2 B1.

Q.128 [MCQ] B — The RHS is evaluated first (the original list), then written into the reversed slice, restoring [1,2,3,4].

Q.129 [MCQ] C — Positions after moves: (2,3), (3,2), (4,4). They meet at 4.

Q.130 [NAT] 15 — 2^{4}-1 = 15.

Q.131 [NAT] 3 — log2 8 = 3 levels.

Q.132 [NAT] 6 — The reachable set has size 6.

Q.133 [MCQ] B — Assigning to the reversed slice writes 10 into a[3], 20 into a[2], 30 into a[1], 40 into a[0].

Q.134 [MSQ] A;B;D — A head-only singly linked list cannot delete at the tail in O(1), so it is not a deque.

Q.135 [MCQ] A — 25 goes right of 20 then left of 30.

Q.136 [NAT] 7 — Worst case leaves one key in a run: 3+5-1=7 comparisons.

Q.137 [MCQ] A — BFS with alpha neighbour order dequeues Ram Sam Uma Tom.

Q.138 [MCQ] B — Strings are immutable; += rebinds s to a new object while t still names 'gate'.

Q.139 [NAT] 10 — Pops remove 20, then 40, then 30; 10 remains.

Q.140 [NAT] 6 — A complete tree has height floor(log2 100)=6.

Q.141 [MCQ] A — Standard facts: in-place QS is not stable; usual MS needs a buffer.

Q.142 [MCQ] A — DFS discovery order is Ram Sam Tom Uma.

Q.143 [MCQ] A — The first two calls share the default list (same id); the third uses a freshly passed list.

Q.144 [NAT] 9 — Dequeues remove 5,8,6 leaving 9.

Q.145 [NAT] 99 — A skew chain of 100 nodes has height 99.

Q.146 [NAT] 15 — 5+4+3+2+1=15.

Q.147 [NAT] 4 — The reachable set has size 4.

Q.148 [MCQ] B — Default acc is bound to the original list object at def-time; later rebinding of the name xs is irrelevant.

Q.149 [MCQ] A — 2 and 1 both see 4; 4 and 3 have nothing larger on their right.

Q.150 [MSQ] A;B;C — A preorder of a BST cannot present a key outside the current subtree window. 3,5,1 is invalid because after going right from 3 to 5, 1 cannot appear.

Q.151 [NAT] 15 — Same triangular number 15.

Q.152 [MCQ] A — BFS with alpha neighbour order dequeues S0 S1 S2 S3 S4.

Q.153 [MCQ] B — A list cannot be a dict key; the except path prints the existing value 'ok'.

Q.154 [NAT] 3 — Trace depths: 1,2,1,2,1,0,1,2,3,2,1. Peak is 3.

Q.155 [MCQ] A — Threads replace the recursion/stack; time remains O(n).

Q.156 [MCQ] A — Best case: each of n-1 insertions stops after one comparison.

Q.157 [MCQ] A — DFS discovery order is S0 S1 S3 S4 S2.

Q.158 [NAT] 10 — It accumulates 4+3+2+1 = 10.

Q.159 [MCQ] A — A min-heap always stores the minimum at the root.

Q.160 [MCQ] A — The middle band of equals is finished; worst case on distinct keys is still quadratic if pivots are bad.

Q.161 [NAT] 5 — The reachable set has size 5.

Q.162 [NAT] 25 — Let C(n)=C(n-1)+C(n-2)+1, C(0)=C(1)=1. Then C(6)=25.

Q.163 [NAT] 4 — Simulate: [7]; 2 swims - > [2,7]; 9 - > [2,7,9]; 4 swims with 7 - > [2,4,9,7]; 5 - > [2,4,9,7,5]. Index 2 is 4.

Q.164 [NAT] 3 — Non-leaves in a complete 7-node tree: floor(7/2)=3.

Q.165 [MCQ] A — BFS with alpha neighbour order dequeues North Island West South.

Q.166 [MCQ] A — Starred assignment captures the middle remainder as a list.

Q.167 [MCQ] A — Enq 4,1,7 (full). Next 2: deq 4 then enq 2 - > 1,7,2.

Q.168 [MCQ] A — Extract-max swaps the root 9 with the last key and shrinks the heap; 9 sits at the end in its final sorted position.

Q.169 [MCQ] A — DFS discovery order is North Island South West.

Q.170 [MCQ] B — Empty list and empty string are falsey; a non-empty list (even [0]) and '0' are truthy.

Q.171 [MSQ] A;C;D — BFS needs a queue. Parentheses, postfix and DFS are stack-shaped.

Q.172 [MSQ] A;B;C — Inversions (2,1),(4,3); adjacent bubble/insertion swap once per inversion. Usual mergesort needs a linear buffer.

Q.173 [NAT] 4 — The reachable set has size 4.

Q.174 [MCQ] A — or returns the first truthy operand, else the last: 0 and [] are falsey so 7 is returned.

Q.175 [NAT] 3 — Two nexts from A land on C; q aliases C.

Q.176 [NAT] 49 — For powers of two T(n)=n lg n - n + 1 = 64-16+1=49.

Q.177 [MCQ] A — Every vertex is reachable from Admin; the graph is connected.

Q.178 [MCQ] B — Each side() returns a positive int (truthy), so all three calls run; and yields the last value 3.

Q.179 [MCQ] A — The node 1 is unlinked; 8 now points at 4.

Q.180 [MCQ] A — There are n! permutations; each comparison has 2 outcomes.

Q.181 [NAT] 2 — Five trees form one connected piece; Ash is a second component.

Q.182 [MCQ] A — 0 is falsey so and short-circuits and side() is never called.

Q.183 [NAT] 5 — Every node swaps its two pointers.

Q.184 [NAT] 11 — A tree on n vertices has n-1 edges.

Q.185 [MCQ] B — True division gives 2.5; floor division rounds toward -inf, so -5//2 = -3.

Q.186 [MCQ] A — Front insert + front delete is a stack. Front-delete plus rear-insert is also a queue, so C is false.

Q.187 [MSQ] A;B;D — n-1 edges alone do not imply a tree (a triangle plus an isolated vertex has n=4, e=3). Trees have no odd cycles, hence are bipartite.

Q.188 [MCQ] B — ** is right-associative: 2**(3**2)=2**9=512, while (2**3)**2=64.

Q.189 [NAT] 5 — Every pushed item is popped once; there are 5 pops.

Q.190 [MCQ] A — Ram: Sam,Uma (2); Sam: Ram,Tom,Uma (3); Tom: Sam,Uma (2); Uma: Tom,Ram,Sam (3).

Q.191 [MCQ] B — Late binding: each lambda reads the same loop variable i, which is 2 after the loop.

Q.192 [MCQ] A — The running height must stay non-negative and end at 0. A is 1,0,1,0,1,2,1,0. B goes negative. C goes negative. D starts with a pop.

Q.193 [NAT] 12 — 6 edges contribute 12 to the degree sum.

Q.194 [MCQ] A — Default i=i captures the current integer at each iteration, so the lambdas close over 0,1,2.

Q.195 [MCQ] A — 3,1,1,1 is a star (K1,3). 3,3,3,1 is not graphic (Havel-Hakimi fails). 4 exceeds n-1. 3,3,1,1 forces two vertices of degree 3 on 4 vertices, hence the other two have degree at least 2, contradiction with two 1s.

Q.196 [MCQ] B — fromkeys reuses the same list object as the default value for every key.

Q.197 [NAT] 7 — Dijkstra distance {'A': 3, 'S': 0, 'T': 7, 'C': 4, 'B': 1}.

Q.198 [MCQ] B — A return in finally overrides the return in try.

Q.199 [NAT] 4 — Dijkstra distance {'U': 0, 'V': 2, 'W': 3, 'X': 4}.

Q.200 [MCQ] A — else runs only if try did not return/raise; here try already returned 1.

Q.201 [NAT] 9 — Dijkstra distance {'Peak': 9, 'Ridge': 5, 'Base': 0, 'Lake': 3}.

Q.202 [MCQ] B — Slice assignment mutates the list object in place; ys aliases xs.

Q.203 [NAT] 6 — Dijkstra distance {'Office': 6, 'Home': 0, 'Y': 2, 'X': 5}.

Q.204 [MCQ] A — Even i in 0..4 are 0,2,4.

Q.205 [NAT] 17 — Dijkstra distance {'Goa': 17, 'Pune': 0, 'Lonavala': 4, 'Mumbai': 9}.

Q.206 [MCQ] A — append returns None (falsey) so 'or i' yields i; the side-effect fills acc.

Q.207 [MCQ] A — Each BFS layer is the next hop distance.

Q.208 [MCQ] C — zip stops at the shorter iterable and the iterator is exhausted after the first list().

Q.209 [MSQ] A;B;D — Dijkstra is single-source and needs non-negative weights; it does not need a DAG (that's for one DP method).

Q.210 [NAT] 16 — The lambda computes 4*4.

Q.211 [MCQ] B — A must be first, then B before C, C before D, and B before C. A,B,C,D is the only order (A then B then C then D); A,C,... is illegal because B must precede C. Only

ABCD.

Q.212 [NAT] 7 — h(3) computes 3*2 then the wrapper adds 1, giving 7.

Q.213 [MCQ] A — P first, S last; Q and R may be swapped: P Q R S and P R Q S.

Q.214 [NAT] 2 — Each constructor increments the class variable n.

Q.215 [NAT] 2 — The two vertices two hops away, one each direction, plus not the opposite which is distance 4.

Q.216 [MCQ] A — Pairwise sums 1+4, 2+5, 3+6.

Q.217 [NAT] 10 — C(5,2)=10.

Q.218 [MCQ] A — Intersection {3}, union {1,2,3,4}, difference {1,2}.

Q.219 [NAT] 12 — Every of 3 vertices meets 4 vertices: 12 edges.

Q.220 [MCQ] B — Rebinding the name a does not mutate the list still named by b.

Q.221 [MCQ] A — Odd cycles are not bipartite.

Q.222 [NAT] 4 — It counts how many times 2 can be subtracted from 7 until < =0, i.e. 4.

Q.223 [MCQ] A — A back edge to an ancestor closes a cycle.

Q.224 [MCQ] A — x[-1] is last; x[-3:] is the last three; x[:-2] drops the last two.

Q.225 [NAT] 3 — A tree would have 9 edges; remove 12-9=3 (and keep connected, which is possible by dropping extra edges of cycles).

Q.226 [MCQ] A — t[0].append mutates the shared inner list of u; t = t+... rebinds only the local name.

Q.227 [MCQ] A — Negative edges (without negative cycles) are the setting Bellman-Ford handles and Dijkstra does not.

Q.228 [MCQ] A — c takes 7, 6 goes to *rest, keyword d=5, leftover z in **kw.

Q.229 [NAT] 2 — Src-A-Sink is a 2-edge path; BFS first reaches Sink at layer 2.

Q.230 [NAT] 10 — Odd i in 1..4 are 1 and 3; 1+9=10.

Q.231 [MCQ] A — Scanning a row to find neighbours costs Theta(n) per vertex, total Theta(n^2).

Q.232 [MCQ] B — extend concatenates elements of B into A in place.

Q.233 [MCQ] A — Each vertex and each edge is scanned O(1) times.

Q.234 [MCQ] B — append adds B as a single nested element, so A has length 4.

Q.235 [MSQ] A;B;C — A bipartite graph with n-1 edges need not be connected (or can have a cycle if not connected properly); D fails. A,B,C are standard tree characterizations.

Q.236 [MCQ] A — n is appended before the recursive call, so the list is 3,2,1,0.

Q.237 [NAT] 2 — WXYZ and WYXZ.

Q.238 [MCQ] B — The recursive result is prepended: [0]+[1]+[2]+[3].

Q.239 [MCQ] A — BFS parents encode some shortest path, not necessarily the unique one.

Q.240 [NAT] 16 — i takes 1,2,4,8 then 16 which exits the while i < 10 test.

Q.241 [NAT] 2 — Delhi-Mumbai-Goa-Kochi and Delhi-Pune-Goa-Kochi.

Q.242 [NAT] 10 — Inner iterations: 1+2+3+4=10.

Q.243 [MCQ] A — Any tree on 20 vertices has 19 edges.

Q.244 [NAT] 3 — 13 is 1101b, which has three 1-bits.

Q.245 [NAT] 3 — Any two of the three edges form a spanning tree.

Q.246 [NAT] 21 — 0b1011=11 and 0o12=10; sum 21.

Q.247 [MCQ] A — Tentative: A=4, B=1. Extract B next.

Q.248 [MCQ] A — find leftmost index 0, rfind rightmost 3, missing char yields -1.

Q.249 [NAT] 3 — S-B-A = 1+2 = 3, better than the direct 4.

Q.250 [MCQ] A — reversed on a string yields characters in reverse, joined with empty separator.

Q.251 [MSQ] A;C;D — Reachable sets coincide for BFS and DFS; visit orders need not be reverses. Reachability reverses with the edges.

Q.252 [MCQ] A — The list is mutated via the shared object; rebinding local y does not change b.

Q.253 [MCQ] A — Euler's theorem. Exactly two odd degrees give an open Euler trail, not a circuit.

Q.254 [MCQ] A — n%3 truthy keeps 1,2,4,5,7; then [::2] takes 1,4,7.

Q.255 [NAT] 12 — Q3 has 8 vertices, each degree 3, so 12 edges.

Q.256 [MCQ] A — Bases n < 2 return without writing the cache; keys 2..6 are stored in the default dict.

Q.257 [MCQ] A — That is the definition. Cut-vertices often do lie on cycles (e.g. two cycles sharing a vertex).

Q.258 [MCQ] A — It yields n then recurses until n=0, which still yields 0 then stops.

Q.259 [NAT] 12 — Horizontal: 3 rows * 2 = 6; vertical: 3 cols * 2 = 6; total 12.

Q.260 [MSQ] B;C — A tuple is hashable only if all components are. frozenset is hashable; list and dict are not.

Q.261 [MCQ] A — Unit weights make Dijkstra degenerate to BFS (with a queue).

Q.262 [MSQ] A;C — Tuples have no append; T += (0,) rebinds T to a new tuple because tuples are immutable.

Q.263 [MSQ] A;C — Mutable defaults are classic sharing bugs. Integers are immutable so mutation across calls does not apply.

Q.264 [MCQ] B — x = x+[1] builds a new list; a is unchanged.

Q.265 [MCQ] B — For lists x += [1] is in-place extend, so a is mutated.

Q.266 [MCQ] A — all on empty is True (vacuous); any on empty is False.

Q.267 [MCQ] A — sorted always returns a list of characters; list() preserves original order.

Q.268 [MCQ] A — Insertion order puts 'x' first; get with default 9 is used for missing 'z'.

Q.269 [NAT] 120 — 5!=120.

Q.270 [NAT] 7 — A linear chain of 6 recursive steps plus the base call is 7 activations.

Q.271 [MCQ] A — del xs[1:4] removes indices 1,2,3 i.e. values 1,2,3.

Q.272 [MCQ] A — xs[1:1] is the empty slice at index 1, so 8,9 are inserted there.

Q.273 [MCQ] B — Chained 2 < 3 < 2 is (2 < 3) and (3 < 2), both False overall.

Q.274 [MCQ] A — List * k repeats elements; [a]*2 is a list of two aliases of a.

Q.275 [MCQ] A — Sets are unordered; lists compare order-sensitively.

Q.276 [MCQ] B — The default list survives between top-level calls, so the second call appends onto [2,1,0].

Q.277 [MCQ] A — range(-3,3) is -3,-2,-1,0,1,2; floor-div by 2 gives -2,-1,-1,0,0,1.

Q.278 [MCQ] A — enumerate with start=2 numbers from 2.

Q.279 [MCQ] A — round ties-to-even: 2.5- > 2, 3.5- > 4; 2.51 is nearer 3.

Q.280 [MCQ] A — Prepend n until n is 0, which contributes [].

Q.281 [MCQ] B — |= mutates the set object aliased by a and b.

Q.282 [MCQ] A — a | {4} builds a new set and rebinds a; b still names {1,2,3}.

Q.283 [MCQ] A — key=int compares numeric values so min is '2'; lexicographic min of those strings is '10'.

Q.284 [MCQ] A — index returns the first occurrence (position 1); 1 appears twice.

Q.285 [MCQ] A — nonlocal k is incremented on each g() call.

Q.286 [MCQ] A — Pairs (x,y) with x < =y: (0,0),(0,1),(0,2),(1,1),(1,2),(2,2) products 0,0,0,1,2,4.

UNIT 5 Database Management and Warehousing · 263 questions

MCQ negative -1/3 or -2/3. MSQ / NAT no negative. Work first, then the solutions section.

Q.1 MCQ · 1 mark · ER model

An ER diagram for Clinic has entity Patient (key pid) and entity Doctor (key did) linked by Treats with attributes date and fee. Treats is M:N. The minimum number of tables needed in a standard relational mapping (no extra attributes on entities besides keys/names) is

(A) 3
(B) 2
(C) 1
(D) 4

Q.2 NAT · 2 marks · Relational model and FDs

Let schema Voyage(A,B,C,D,E) with FDs A- > BC, CD- > E, E- > A. The number of candidate keys is

NAT · enter the numerical value

Q.3 MCQ · 1 mark · Relational algebra and tuple calculus

Relations Team(name) and Player(name, team). RA expression that returns team names that have at least one player is

(A) pi_name(Team) ∩ pi_team(Player) [with rename so attributes match]
(B) Team - Player
(C) Player ÷ Team
(D) Team × Player

Q.4 NAT · 2 marks · SQL

```
Relation Raider
id name team points
1 Ajay Patna 12
2 Naveen Patna 18
3 Pawan Jaipur 12
4 Pardeep Jaipur 20
5 Maninder Delhi 18
6 Nitin Delhi 9
Query: SELECT team FROM Raider WHERE points > (SELECT
AVG(points) FROM Raider);
The number of rows returned is
```

NAT · enter the numerical value

Q.5 MSQ · 2 marks · Integrity constraints

Relations r(A,B) and s(B,C) with FK r.B - > s.B (every r.B must appear in s). Which operations NEVER violate this FK, regardless of the current tuples (assuming they currently satisfy the FK)?

(A) inserting a tuple into s
(B) deleting a tuple from r
(C) inserting a tuple into r
(D) deleting a tuple from s

Q.6 NAT · 2 marks · Files and indexing

A B+ tree is built on relation Voyage. Node size = 4096 bytes, node-pointer size = 10 bytes, search-key size = 11 bytes, data-record pointer = 12 bytes. A non-leaf stores n node pointers and n-1 keys, packed with no extra header. The maximum n (fanout) is

NAT · enter the numerical value

Q.7 NAT · 2 marks · Data types and transformation

Attribute AgeYears has values [10, 20, 30, 40, 50]. Min-max normalisation into [0,1] maps the last value to (round to two decimals if needed; if it is a terminating decimal, give that). Report the exact value as a simplified decimal or integer.

NAT · enter the numerical value

Q.8 MCQ · 1 mark · Warehousing and OLAP

A star schema for RetailSales has a fact table SalesFact(date_id, store_id, sku_id, qty, amount) and dimension tables DateDim, StoreDim, SkuDim. This is a star rather than a snowflake because

(A) dimensions are denormalised single tables, not split into sub-dimensions
(B) there is no fact table
(C) every dimension is in BCNF with many hops
(D) measures are stored in every dimension

Q.9 MCQ · 1 mark · ER model

In an ER model of Library, Copy is a weak entity identified by (isbn, copyno) relative to Book. The identifying relationship HasCopy is

(A) 1:N from Book to Copy, with total participation of Copy
(B) M:N between Book and Copy
(C) 1:1 with partial participation of Copy
(D) an ordinary strong relationship that can drop copyno

Q.10 MCQ · 2 marks · Relational model and FDs

Let schema Voyage(A,B,C,D,E) with FDs A- > BC, CD- > E, E- > A. Assuming the schema is already in 1NF, the highest normal form it is guaranteed to be in is

(A) 1NF
(B) 2NF
(C) 3NF
(D) BCNF

Q.11 MCQ · 2 marks · Relational algebra and tuple calculus

Relations Defender(name) and Forward(name) and Team(name). Which RA expression is empty exactly when every Team.name appears in Defender or Forward?

(A) pi_name(Team) - (pi_name(Defender) ∪ pi_name(Forward))
(B) Defender ∩ Forward ∩ Team
(C) Team × Defender
(D) (Defender ∪ Forward) - Team

Q.12 NAT · 1 mark · SQL

```
On the same Raider table, SELECT team, COUNT(*) FROM
Raider GROUP BY team HAVING COUNT(*) > =2 returns how
many groups?
```

NAT · enter the numerical value

Q.13 MCQ · 1 mark · Integrity constraints

ON DELETE CASCADE on FK Child.pid - > Parent.pid means that deleting a Parent row

(A) deletes matching Child rows
(B) is rejected if any Child references it
(C) sets Child.pid to NULL
(D) moves Child rows to an archive table automatically

Q.14 NAT · 2 marks · Files and indexing

For the same B+ tree on Voyage (block 4096, key 11, record-ptr 12, sibling/node-ptr 10), a leaf holds n keys and n record pointers plus one sibling pointer. Maximum n is

NAT · enter the numerical value

Q.15 NAT · 2 marks · Data types and transformation

On the same AgeYears sample [10, 20, 30, 40, 50], using population standard deviation, the z-score of the last value rounded to two decimals is

NAT · enter the numerical value

Q.16 MCQ · 1 mark · Warehousing and OLAP

Snowflake schema for the same warehouse splits StoreDim into Store, City, State. Compared with star,

(A) snowflake reduces dimension redundancy and adds more joins at query time
(B) snowflake always speeds every query
(C) star cannot represent hierarchies
(D) fact tables disappear

Q.17 MSQ · 2 marks · ER model

Which statements about ER constructs are correct?

(A) A weak entity has no complete key of its own
(B) Total participation of A in R means every entity of A appears in R
(C) An M:N relationship can always be mapped without creating a new relation
(D) A composite attribute is typically flattened into component columns

Q.18 NAT · 2 marks · Relational model and FDs

Let schema Roster(A,B,C,D) with FDs AB- > C, C- > D, D- > A. The number of candidate keys is

NAT · enter the numerical value

Q.19 NAT · 1 mark · Relational algebra and tuple calculus

R has 5 tuples, S has 4 tuples, R and S are union-compatible and share 2 tuples. |R ∪ S| is

NAT · enter the numerical value

Q.20 NAT · 1 mark · SQL

```
On Raider, SELECT DISTINCT points FROM Raider returns how many rows?
```

NAT · enter the numerical value

Q.21 MCQ · 1 mark · Integrity constraints

ON DELETE SET NULL requires that the FK column is

(A) nullable
(B) part of the child's primary key
(C) a candidate key of the parent
(D) composite

Q.22 NAT · 2 marks · Files and indexing

A B+ tree is built on relation Invoice. Node size = 2048 bytes, node-pointer size = 8 bytes, search-key size = 12 bytes, data-record pointer = 8 bytes. A non-leaf stores n node pointers and n-1 keys, packed with no extra header. The maximum n (fanout) is

NAT · enter the numerical value

Q.23 NAT · 2 marks · Data types and transformation

Attribute RainMm has values [0, 2, 4, 6, 18]. Min-max normalisation into [0,1] maps the last value to (round to two decimals if needed; if it is a terminating decimal, give that). Report the exact value as a simplified decimal or integer.

NAT · enter the numerical value

Q.24 MCQ · 1 mark · Warehousing and OLAP

Cube SalesQty is visualised by ProductType × Month × Country. To see ProductType × Month × State (State is a finer geographic level under Country) the OLAP operation is

(A) drill-down
(B) roll-up
(C) slice
(D) dice

Q.25 MCQ · 1 mark · ER model

University ER: entity sets Student, Course, Department, Instructor. Relationships: Enrolls(Student,Course) M:N with grade; WorksIn(Instructor,Department) N:1. Standard mapping (entity table each; M:N table; 1:N as FK) produces how many relations?

(A) 4
(B) 5
(C) 6
(D) 3

Q.26 MCQ · 2 marks · Relational model and FDs

Let schema Roster(A,B,C,D) with FDs AB- > C, C- > D, D- > A. Assuming the schema is already in 1NF, the highest normal form it is guaranteed to be in is

(A) 3NF
(B) 1NF
(C) 2NF
(D) BCNF

Q.27 NAT · 1 mark · Relational algebra and tuple calculus

R has 5 tuples, S has 4, they share 2. |R ∩ S| is

NAT · enter the numerical value

Q.28 NAT · 2 marks · SQL

```
Relation Loan
lno branch amount
L11 Banjara 90000
L14 Kondapur 50000
L15 SRNagar 40000
L22 SRNagar 25000
L23 Balanagar 80000
L25 Kondapur 70000
L19 SRNagar 65000
SELECT L1.lno FROM Loan L1 WHERE L1.amount > (SELECT
MAX(L2.amount) FROM Loan L2 WHERE L2.branch='SRNagar');
Number of rows returned is
```

NAT · enter the numerical value

Q.29 MSQ · 2 marks · Integrity constraints

Emp(ssn PRIMARY KEY, dno REFERENCES Dept(dno), salary INT CHECK (salary > 0)). Dept currently contains only dno=5. Which new Emp rows are rejected?

(A) ssn that already exists in Emp
(B) dno=9 (absent from Dept)
(C) salary=0 (fails CHECK)
(D) dno=5, salary=10, fresh ssn

Q.30 NAT · 2 marks · Files and indexing

For the same B+ tree on Invoice (block 2048, key 12, record-ptr 8, sibling/node-ptr 8), a leaf holds n keys and n record pointers plus one sibling pointer. Maximum n is

NAT · enter the numerical value

Q.31 NAT · 2 marks · Data types and transformation

On the same RainMm sample [0, 2, 4, 6, 18], using population standard deviation, the z-score of the last value rounded to two decimals is

NAT · enter the numerical value

Q.32 MCQ · 1 mark · Warehousing and OLAP

From the same SalesQty cube, aggregating Month up to Quarter is

(A) roll-up
(B) drill-down
(C) slice
(D) pivot

Q.33 MCQ · 2 marks · ER model

In the ER diagram of Harbour, Berth participates totally in Occupies with Ship, cardinality 1:1. After mapping, the key of Ship may be placed as a FK in Berth. This FK is

(A) NOT NULL and UNIQUE
(B) NULL-able and non-unique
(C) a set-valued attribute
(D) unnecessary because 1:1 needs no FK

Q.34 NAT · 2 marks · Relational model and FDs

Let schema ClinicVisit(A,B,C,D,E) with FDs A- > B, B- > C, CD- > E, E- > A. The number of candidate keys is

NAT · enter the numerical value

Q.35 NAT · 1 mark · Relational algebra and tuple calculus

R has 5 tuples, S has 4, overlap 2. |R - S| is

NAT · enter the numerical value

Q.36 NAT · 2 marks · SQL

```
Car(serial,model,year,color): (10,Swift,2019,red),
(11,Swift,2020,blue), (12,City,2018,red),
(13,Nexon,2021,red).
Make(maker,model): (Maruti,Swift), (Honda,City),
(Tata,Nexon).
Own(owner,serial): (Ravi,10), (Ravi,12), (Asha,11),
(Asha,13), (Isha,13).
SELECT DISTINCT owner FROM Own JOIN Car JOIN Make WHERE
color='red' AND maker='Tata'.
Number of rows is
```

NAT · enter the numerical value

Q.37 MCQ · 1 mark · Integrity constraints

A UNIQUE constraint on column email of User allows how many NULLs in SQL (standard, UNIQUE is satisfied by NULL != NULL)?

(A) any number of NULLs (in most SQL systems UNIQUE allows multiple NULLs)
(B) exactly one NULL
(C) zero NULLs always
(D) NULL is equal to NULL under UNIQUE so at most one

Q.38 NAT · 2 marks · Files and indexing

A B+ tree is built on relation Roster. Node size = 8192 bytes, node-pointer size = 16 bytes, search-key size = 20 bytes, data-record pointer = 16 bytes. A non-leaf stores n node pointers and n-1 keys, packed with no extra header. The maximum n (fanout) is

NAT · enter the numerical value

Q.39 NAT · 2 marks · Data types and transformation

Attribute TempC has values [0, 10, 10, 20, 40]. Min-max normalisation into [0,1] maps the last value to (round to two decimals if needed; if it is a terminating decimal, give that). Report the exact value as a simplified decimal or integer.

NAT · enter the numerical value

Q.40 MCQ · 1 mark · Warehousing and OLAP

Fixing Country = 'India' and viewing ProductType × Month is

(A) a slice
(B) a drill-down
(C) a snowflake
(D) a bitmap join

Q.41 NAT · 1 mark · ER model

A ternary relationship Supplies(Supplier, Part, Project) is M:N:P. In the standard mapping, the relationship table has how many foreign keys (one per entity key, assuming simple keys)?

NAT · enter the numerical value

Q.42 MCQ · 2 marks · Relational model and FDs

Let schema ClinicVisit(A,B,C,D,E) with FDs A- > B, B- > C, CD- > E, E- > A. Assuming the schema is already in 1NF, the highest normal form it is guaranteed to be in is

(A) 3NF
(B) 1NF
(C) 2NF
(D) BCNF

Q.43 NAT · 1 mark · Relational algebra and tuple calculus

R(A,B) has 6 tuples, S(B,C) has 5 tuples. A foreign-key-like join: each R.B matches exactly one S.B and every S.B occurs at least once. |R ⋈ S| equals

NAT · enter the numerical value

Q.44 NAT · 1 mark · SQL

```
Emp(id,dept,bonus): (1,HR,100), (2,HR,NULL), (3,IT,200),
(4,IT,NULL), (5,NULL,50).
SELECT id FROM Emp WHERE bonus > 100 returns how many
rows?
```

NAT · enter the numerical value

Q.45 MCQ · 1 mark · Integrity constraints

PRIMARY KEY versus UNIQUE NOT NULL. In SQL they are similar, but a table may have

(A) at most one PRIMARY KEY and any number of UNIQUE constraints
(B) any number of PRIMARY KEYs
(C) no UNIQUE if it has a PRIMARY KEY
(D) a PRIMARY KEY that contains NULLs

Q.46 NAT · 2 marks · Files and indexing

For the same B+ tree on Roster (block 8192, key 20, record-ptr 16, sibling/node-ptr 16), a leaf holds n keys and n record pointers plus one sibling pointer. Maximum n is

NAT · enter the numerical value

Q.47 NAT · 2 marks · Data types and transformation

On the same TempC sample [0, 10, 10, 20, 40], using population standard deviation, the z-score of the last value rounded to two decimals is

NAT · enter the numerical value

Q.48 MCQ · 1 mark · Warehousing and OLAP

Selecting Country in {India, Nepal} and Month in {Jan, Feb} on the cube is

(A) dice
(B) slice
(C) roll-up
(D) star join

Q.49 MCQ · 1 mark · ER model

`ISA hierarchy: Employee is the superclass; Engineer and Manager are disjoint subclasses, total specialisation. A mapping that uses one table per subclass plus one for Employee (with shared ssn) is the`

(A) 8A / superclass-and-subclass method
(B) single-table-with-nulls method only
(C) relationship-table method for M:N
(D) unnormalised blob method

Q.50 NAT · 2 marks · Relational model and FDs

Let schema Invoice(A,B,C,D) with FDs A- > BC, C- > D. The number of candidate keys is

NAT · enter the numerical value

Q.51 MCQ · 1 mark · Relational algebra and tuple calculus

Division: Completed(student, course) and Required(course) with Required = {DB,ML}. student who completed at least those courses is expressed by

(A) Completed ÷ Required
(B) Completed - Required
(C) Completed × Required
(D) pi_student(Completed)

Q.52 NAT · 1 mark · SQL

`On the same Emp table, SELECT id FROM Emp WHERE bonus > 100 OR bonus IS NULL returns how many rows?`

NAT · enter the numerical value

Q.53 NAT · 1 mark · Integrity constraints

Relation Takes(sid, cid, grade) with PRIMARY KEY (sid,cid) and FK sid- > Student, cid- > Course. Inserting (1,10,'A') succeeds once. A second insert of (1,10,'B') is rejected. The number of tuples in Takes after both attempts is

NAT · enter the numerical value

Q.54 NAT · 2 marks · Files and indexing

A B+ tree is built on relation Clinic. Node size = 4096 bytes, node-pointer size = 6 bytes, search-key size = 10 bytes, data-record pointer = 8 bytes. A non-leaf stores n node pointers and n-1 keys, packed with no extra header. The maximum n (fanout) is

NAT · enter the numerical value

Q.55 NAT · 2 marks · Data types and transformation

Attribute DepthM has values [3, 6, 9, 12, 15]. Min-max normalisation into [0,1] maps the last value to (round to two decimals if needed; if it is a terminating decimal, give that). Report the exact value as a simplified decimal or integer.

NAT · enter the numerical value

Q.56 MCQ · 1 mark · Warehousing and OLAP

Cube ClinicVisits is shown at dimensions Hospital, Ward, Day. Moving the last geographic/organisational/time level to the finer grain (Ward to Bed) is

(A) drill-down
(B) roll-up
(C) slice
(D) dice

Q.57 MSQ · 2 marks · ER model

An ER diagram for Festival has entity Performer and entity Stage linked by PlaysOn (M:N) and a recursive relationship Mentors on Performer (1:N). Which mappings are required?

(A) a table Performer
(B) a table Stage
(C) a table PlaysOn with two FKs
(D) Mentors absorbed as a nullable mentor_id FK on Performer

Q.58 MCQ · 2 marks · Relational model and FDs

Let schema Invoice(A,B,C,D) with FDs A- > BC, C- > D. Assuming the schema is already in 1NF, the highest normal form it is guaranteed to be in is

(A) 2NF
(B) 1NF
(C) 3NF
(D) BCNF

Q.59 MCQ · 1 mark · Relational algebra and tuple calculus

Tuple relational calculus: {t | t in Player ∧ t.team = 'Goa'}. This is equivalent to

(A) sigma_{team='Goa'}(Player)
(B) pi_team(Player)
(C) Player ÷ {Goa}
(D) Player × Team

Q.60 MCQ · 2 marks · SQL

On Emp as above, (COUNT(*), COUNT(bonus), COUNT(dept)) is

(A) 5, 3, 4
(B) 5, 5, 5
(C) 5, 3, 5
(D) 4, 3, 4

Q.61 MCQ · 1 mark · Integrity constraints

Referential integrity from R.a to S.b is violated by which one of these, if S currently has b-values {1,2} and R has a-values {1}?

(A) DELETE FROM S WHERE b=1
(B) INSERT INTO S VALUES (3)
(C) DELETE FROM R WHERE a=1
(D) UPDATE S SET b=2 WHERE b=2

Q.62 NAT · 2 marks · Files and indexing

For the same B+ tree on Clinic (block 4096, key 10, record-ptr 8, sibling/node-ptr 6), a leaf holds n keys and n record pointers plus one sibling pointer. Maximum n is

NAT · enter the numerical value

Q.63 NAT · 2 marks · Data types and transformation

On the same DepthM sample [3, 6, 9, 12, 15], using population standard deviation, the z-score of the last value rounded to two decimals is

NAT · enter the numerical value

Q.64 MCQ · 1 mark · Warehousing and OLAP

On cube ClinicVisits (Hospital, Ward, Day), aggregating away the middle dimension entirely (summing the measure over it) is

(A) roll-up / projection of a dimension
(B) drill-down
(C) a foreign-key insert
(D) 2NF decomposition

Q.65 MCQ · 1 mark · ER model

Attribute age derived from dob on entity Citizen should, in a normalised relational schema, usually be

(A) not stored, or stored only as a view/computed column
(B) part of the primary key
(C) a foreign key to Date
(D) a multivalued attribute table

Q.66 NAT · 2 marks · Relational model and FDs

Let schema FlightLeg(A,B,C,D,E) with FDs AB- > C, C- > D, D- > E, E- > A. The number of candidate keys is

NAT · enter the numerical value

Q.67 MSQ · 2 marks · Relational algebra and tuple calculus

Which RA expressions are equivalent to sigma_{A=1}(R □ S) when A is an attribute of R only?

(A) (sigma_{A=1}(R)) □ S
(B) R □ (sigma_{A=1}(S))
(C) sigma_{A=1}(R) × S (if the join is a natural join on other attributes, not in general)
(D) the selection can be pushed to the R side

Q.68 NAT · 1 mark · SQL

```
On Emp, SELECT dept FROM Emp WHERE dept = NULL returns how many rows?
```

NAT · enter the numerical value

Q.69 MSQ · 2 marks · Integrity constraints

Deferred constraint checking (end of transaction) is useful because

(A) a cycle of FKs can be inserted in one transaction
(B) a tuple can temporarily violate a CHECK that a later statement of the same transaction repairs
(C) it makes PRIMARY KEY optional
(D) it disables FOREIGN KEY forever

Q.70 NAT · 2 marks · Files and indexing

A B+ tree is built on relation Harbour. Node size = 512 bytes, node-pointer size = 4 bytes, search-key size = 8 bytes, data-record pointer = 4 bytes. A non-leaf stores n node pointers and n-1 keys, packed with no extra header. The maximum n (fanout) is

NAT · enter the numerical value

Q.71 MCQ · 1 mark · Data types and transformation

Equal-width discretisation of AgeYears on [10,50] into 4 bins uses bin width

(A) 10
(B) 4
(C) 40
(D) 5

Q.72 MCQ · 1 mark · Warehousing and OLAP

Cube HarbourTonnage is shown at dimensions Port, ShipType, Month. Moving the last geographic/organisational/time level to the finer grain (Port to Berth) is

(A) drill-down
(B) roll-up
(C) slice
(D) dice

Q.73 NAT · 1 mark · ER model

Entity Product has a multivalued attribute color (a product may have 3 colours) and a composite attribute size(length,width). Standard mapping: Product table plus a table ProductColor. How many relations result from this fragment?

NAT · enter the numerical value

Q.74 MCQ · 2 marks · Relational model and FDs

Let schema FlightLeg(A,B,C,D,E) with FDs AB- > C, C- > D, D- > E, E- > A. Assuming the schema is already in 1NF, the highest normal form it is guaranteed to be in is

(A) 3NF
(B) 1NF
(C) 2NF
(D) BCNF

Q.75 MCQ · 1 mark · Relational algebra and tuple calculus

A tuple-calculus query {t | ∃p ∈ Player (t.name = p.name ∧ p.team = 'Pune')} is

(A) a projection of a selection of Player
(B) unsafe, because t is not bound to a relation
(C) the Cartesian product Player × Player
(D) a division

Q.76 NAT · 2 marks · SQL

```
On Emp, SELECT id FROM Emp WHERE NOT (bonus > 100) returns how many rows?
```

NAT · enter the numerical value

Q.77 MCQ · 2 marks · Integrity constraints

A domain constraint / CHECK (status IN ('open','closed')) rejects

(A) status='pending'
(B) status='open'
(C) a NULL status if CHECK treats NULL as pass (UNKNOWN is not FALSE)
(D) both pending and NULL in all engines uniformly

Q.78 NAT · 2 marks · Files and indexing

For the same B+ tree on Harbour (block 512, key 8, record-ptr 4, sibling/node-ptr 4), a leaf holds n keys and n record pointers plus one sibling pointer. Maximum n is

NAT · enter the numerical value

Q.79 MCQ · 1 mark · Data types and transformation

Equal-frequency (quantile) discretisation of a sample of 12 distinct values into 4 bins puts how many values in each bin?

(A) 3
(B) 4
(C) 12
(D) 2

Q.80 MCQ · 1 mark · Warehousing and OLAP

On cube HarbourTonnage (Port, ShipType, Month), aggregating away the middle dimension entirely (summing the measure over it) is

(A) roll-up / projection of a dimension
(B) drill-down
(C) a foreign-key insert
(D) 2NF decomposition

Q.81 MCQ · 2 marks · ER model

Partial participation of Employee in Manages (1:1 with Department on the other side, Department total) means

(A) some employees are not managers; every department has a manager
(B) every employee manages a department
(C) some departments have no manager
(D) Manages is M:N

Q.82 NAT · 2 marks · Relational model and FDs

Let schema LabSample(A,B,C,D) with FDs A- > B, B- > C, C- > A, AD- > C. The number of candidate keys is

NAT · enter the numerical value

Q.83 MCQ · 1 mark · Relational algebra and tuple calculus

Unsafe TRC query example: {t | t.A = t.A} ranges over

(A) the infinite universe of all tuples, hence unsafe
(B) one empty tuple
(C) the active domain automatically
(D) a single attribute A of a stored relation

Q.84 NAT · 2 marks · SQL

```
Team: (1,Goa), (2,Pune), (3,Kochi). Player: (10,Amit,1),
(11,Bala,1), (12,Chet,2).
SELECT tname FROM Team LEFT JOIN Player ON
Team.tid=Player.tid WHERE Player.pid IS NULL;
Number of rows is
```

NAT · enter the numerical value

Q.85 NAT · 1 mark · Integrity constraints

```
Parent P(id PK) has rows 1,2. Child C(id PK, pid FK- > P)
has rows (10,1),(11,1). ON DELETE CASCADE. DELETE FROM P
WHERE id=1. How many rows remain in C?
```

NAT · enter the numerical value

Q.86 NAT · 2 marks · Files and indexing

A B+ tree is built on relation SensorLog. Node size = 4096 bytes, node-pointer size = 12 bytes, search-key size = 16 bytes, data-record pointer = 12 bytes. A non-leaf stores n node pointers and n-1 keys, packed with no extra header. The maximum n (fanout) is

NAT · enter the numerical value

Q.87 MSQ · 2 marks · Data types and transformation

Which are lossless compression schemes (in the information-lossless sense)?

(A) Huffman coding of a discrete alphabet
(B) run-length encoding of a bit-vector
(C) JPEG at quality 30 on a photo
(D) dictionary (LZ) compression of a text file

Q.88 MCQ · 1 mark · Warehousing and OLAP

Cube GridLoad is shown at dimensions Region, Fuel, Hour. Moving the last geographic/organisational/time level to the finer grain (Hour to Minute) is

(A) drill-down
(B) roll-up
(C) slice
(D) dice

Q.89 MCQ · 1 mark · ER model

A identifying relationship for weak entity InvoiceLine of Invoice is drawn with a double diamond / double rectangle in many texts. The primary key of InvoiceLine is

(A) invoice_id plus a discriminator line_no
(B) line_no alone
(C) invoice_id alone
(D) a surrogate that ignores Invoice

Q.90 MCQ · 2 marks · Relational model and FDs

Let schema LabSample(A,B,C,D) with FDs A- > B, B- > C, C- > A, AD- > C. Assuming the schema is already in 1NF, the highest normal form it is guaranteed to be in is

(A) 3NF
(B) 1NF
(C) 2NF
(D) BCNF

Q.91 NAT · 1 mark · Relational algebra and tuple calculus

R has attributes (A,B), 4 tuples; S has (B,C), 3 tuples; natural join on B is a full Cartesian because all R.B equal all S.B (single shared B-value). |R ⋈ S| is

NAT · enter the numerical value

Q.92 NAT · 1 mark · SQL

```
On the same Team/Player, SELECT Team.tid FROM Team LEFT
JOIN Player ON Team.tid=Player.tid returns how many
rows?
```

NAT · enter the numerical value

Q.93 NAT · 1 mark · Integrity constraints

```
Same P/C but ON DELETE RESTRICT. DELETE FROM P WHERE id=1
is attempted. Number of rows remaining in P is
```

NAT · enter the numerical value

Q.94 NAT · 2 marks · Files and indexing

For the same B+ tree on SensorLog (block 4096, key 16, record-ptr 12, sibling/node-ptr 12), a leaf holds n keys and n record pointers plus one sibling pointer. Maximum n is

NAT · enter the numerical value

Q.95 MCQ · 1 mark · Data types and transformation
Sampling 20 rows out of 200 without replacement. The probability a given row is included is
(A) 0.1
(B) 0.2
(C) 1/20
(D) 20

Q.96 MCQ · 1 mark · Warehousing and OLAP
On cube GridLoad (Region, Fuel, Hour), aggregating away the middle dimension entirely (summing the measure over it) is
(A) roll-up / projection of a dimension
(B) drill-down
(C) a foreign-key insert
(D) 2NF decomposition

Q.97 MSQ · 2 marks · ER model
Which ER features require an extra relation in the standard mapping (beyond one table per strong entity)?
(A) M:N relationship
(B) multivalued attribute
(C) 1:N relationship (FK can be placed on the N side)
(D) composite attribute (components become columns)

Q.98 NAT · 2 marks · Relational model and FDs
Let schema HarbourBerth(A,B,C,D,E) with FD A- > BCDE. The number of candidate keys is
NAT · enter the numerical value

Q.99 MCQ · 1 mark · Relational algebra and tuple calculus
pi_{A}(R) ∪ pi_{A}(S) versus pi_{A}(R ∪ S) when R and S are union-compatible. They are
(A) equal
(B) unequal in general
(C) equal only if A is a key
(D) equal only if R=S

Q.100 NAT · 2 marks · SQL
Course: (1,DB,CS), (2,ML,CS), (3,Opt,MA). Enroll(sid,cid): (8,1),(8,2),(9,1),(9,3),(7,3).
SQL: sids from Enroll for which NOT EXISTS a CS course they did not take.
Number of rows returned (Enroll-level, not DISTINCT) is
NAT · enter the numerical value

Q.101 MCQ · 1 mark · Integrity constraints
An assertion 'every team name in Team appears in Defender or Forward' is the SQL counterpart of which RA test being empty?
(A) Team_names - (Defender_names ∪ Forward_names)
(B) Team × Defender
(C) Defender ∩ Forward
(D) pi_name(Team)

Q.102 NAT · 2 marks · Files and indexing
A B+ tree is built on relation MatchCard. Node size = 1024 bytes, node-pointer size = 6 bytes, search-key size = 10 bytes, data-record pointer = 8 bytes. A non-leaf stores n node pointers and n-1 keys, packed with no extra header. The maximum n (fanout) is
NAT · enter the numerical value

Q.103 MCQ · 1 mark · Data types and transformation
Sampling with replacement of 20 draws from 200. Expected number of times a given row appears is
(A) 0.1
(B) 1
(C) 20
(D) 0

Q.104 MCQ · 1 mark · Warehousing and OLAP
Cube CourseEnrol is shown at dimensions Dept, Programme, Sem. Moving the last geographic/organisational/time level to the finer grain (Dept to Faculty) is
(A) drill-down
(B) roll-up
(C) slice
(D) dice

Q.105 NAT · 1 mark · ER model
Garage ER: Car, Owner, Mechanic, Job. Relationships Owns(Owner,Car) 1:N, WorksOn(Mechanic,Job) M:N, ForCar(Job,Car) N:1. Number of relations after standard mapping (entity tables + M:N table, 1:N as FKs) is
NAT · enter the numerical value

Q.106 MCQ · 2 marks · Relational model and FDs
Let schema HarbourBerth(A,B,C,D,E) with FD A- > BCDE. Assuming the schema is already in 1NF, the highest normal form it is guaranteed to be in is
(A) BCNF
(B) 1NF
(C) 2NF
(D) 3NF

Q.107 MCQ · 2 marks · Relational algebra and tuple calculus
pi_{A}(R ∩ S) versus pi_{A}(R) ∩ pi_{A}(S). In general they are
(A) the left is a subset of the right; they may differ
(B) always equal
(C) the right is a subset of the left
(D) incomparable

Q.108 NAT · 1 mark · SQL
`T(x,y): (1,10),(1,20),(NULL,30),(NULL,40),(2,50). SELECT x, COUNT(*) FROM T GROUP BY x returns how many groups?`
NAT · enter the numerical value

Q.109 MSQ · 2 marks · Integrity constraints
Which are integrity constraints in the relational model?
(A) entity integrity (no NULL in primary key)
(B) referential integrity of foreign keys
(C) user-defined CHECK / assertions
(D) the physical block size of a file

Q.110 NAT · 2 marks · Files and indexing
For the same B+ tree on MatchCard (block 1024, key 10, record-ptr 8, sibling/node-ptr 6), a leaf holds n keys and n record pointers plus one sibling pointer. Maximum n is
NAT · enter the numerical value

Q.111 MCQ · 1 mark · Data types and transformation

Min-max normalisation to [0,1] is sensitive to

(A) outliers that stretch min or max
(B) the mean only
(C) the mode only
(D) the order of columns in the table

Q.112 MCQ · 1 mark · Warehousing and OLAP

On cube CourseEnrol (Dept, Programme, Sem), aggregating away the middle dimension entirely (summing the measure over it) is

(A) roll-up / projection of a dimension
(B) drill-down
(C) a foreign-key insert
(D) 2NF decomposition

Q.113 MCQ · 1 mark · ER model

Aggregation in ER (a relationship treated as an entity so it can participate in another relationship) is used when, for example,

(A) a WorksOn(Employee,Project) relationship is itself related to a Supplier via Sponsors
(B) two entities have a 1:1 relationship
(C) an attribute is derived
(D) a key is composite

Q.114 NAT · 2 marks · Relational model and FDs

Let schema MatchCard(A,B,C,D) with FDs AB- > CD, C- > A. The number of candidate keys is

NAT · enter the numerical value

Q.115 MSQ · 2 marks · Relational algebra and tuple calculus

Natural join R □ S. Which are always true?

(A) if R and S have no common attributes, □ is Cartesian product
(B) |R □ S| < = |R| * |S|
(C) |R □ S| > = |R| + |S| always
(D) natural join is commutative up to column order

Q.116 NAT · 2 marks · SQL

```
Match(mid,home,away,goals): (1,Goa,Pune,3),(2,Goa,Kochi,1),(3,Pune,Kochi,2),(4,Pune,Goa,0),(5,Kochi,Goa,4).
SELECT home FROM Match GROUP BY home HAVING SUM(goals) > 3 returns how many rows?
```

NAT · enter the numerical value

Q.117 MSQ · 2 marks · Integrity constraints

Foreign key Child.pid REFERENCES Parent.pid. Currently satisfying. Which statements never introduce a violation by themselves?

(A) INSERT INTO Parent
(B) DELETE FROM Child
(C) UPDATE Child SET pid to a value that exists in Parent
(D) DELETE FROM Parent of a referenced pid without CASCADE/SET NULL (NO ACTION)

Q.118 NAT · 2 marks · Files and indexing

A B+ tree is built on relation Payroll. Node size = 4096 bytes, node-pointer size = 8 bytes, search-key size = 14 bytes, data-record pointer = 8 bytes. A non-leaf stores n node pointers and n-1 keys, packed with no extra header. The maximum n (fanout) is

NAT · enter the numerical value

Q.119 MCQ · 1 mark · Data types and transformation

z-score normalisation uses

(A) (x-mean)/std
(B) (x-min)/(max-min)
(C) log(1+x)
(D) rank(x)/n

Q.120 MCQ · 1 mark · Warehousing and OLAP

Cube MatchGoals is shown at dimensions Team, Venue, Week. Moving the last geographic/organisational/time level to the finer grain (Week to MatchDay) is

(A) drill-down
(B) roll-up
(C) slice
(D) dice

Q.121 MSQ · 2 marks · ER model

Mapping a 1:N relationship WorksIn(Instructor - > Department) . Which are correct?

(A) Department's key becomes a FK in Instructor
(B) a separate WorksIn table is mandatory
(C) Instructor's key as FK in Department would force each dept to have at most one instructor if UNIQUE
(D) total participation of Instructor makes the FK NOT NULL

Q.122 MCQ · 2 marks · Relational model and FDs

Let schema MatchCard(A,B,C,D) with FDs AB- > CD, C- > A. Assuming the schema is already in 1NF, the highest normal form it is guaranteed to be in is

(A) 3NF
(B) 1NF
(C) 2NF
(D) BCNF

Q.123 MCQ · 1 mark · Relational algebra and tuple calculus

The RA rename operator ρ_{S(A,B)}(R) is needed so that

(A) union/intersection/difference can be applied after aligning attribute names
(B) selection can test constants
(C) projection can drop columns
(D) division can be rewritten as join

Q.124 NAT · 1 mark · SQL

```
On Match, SELECT COUNT(*) FROM Match M WHERE goals > = ALL (SELECT goals FROM Match) returns
```

NAT · enter the numerical value

Q.125 MCQ · 1 mark · Integrity constraints

Entity integrity says

(A) no primary-key attribute may be NULL
(B) every FK is NULL
(C) every table has two keys
(D) CHECK constraints are forbidden

Q.126 NAT · 2 marks · Files and indexing

For the same B+ tree on Payroll (block 4096, key 14, record-ptr 8, sibling/node-ptr 8), a leaf holds n keys and n record pointers plus one sibling pointer. Maximum n is

NAT · enter the numerical value

Q.127 MSQ · 2 marks · Data types and transformation
Data-type roles in warehousing / mining. Which assignments are reasonable?
(A) Gender: categorical (nominal)
(B) Temperature in C: numeric (interval/ratio depending on scale use)
(C) T-shirt size S < M < L: ordinal
(D) Employee-id as a 6-digit code: numeric quantity on which means are meaningful

Q.128 MCQ · 1 mark · Warehousing and OLAP
On cube MatchGoals (Team, Venue, Week), aggregating away the middle dimension entirely (summing the measure over it) is
(A) roll-up / projection of a dimension
(B) drill-down
(C) a foreign-key insert
(D) 2NF decomposition

Q.129 MCQ · 1 mark · ER model
A recursive relationship MarriedTo on Person that is 1:1 and symmetric is mapped by
(A) a nullable unique FK spouse_id on Person (or a two-column table with a symmetric constraint)
(B) two Person tables
(C) an M:N table with no keys
(D) a CHECK on dates only

Q.130 NAT · 2 marks · Relational model and FDs
Let schema Payroll(A,B,C,D,E) with FDs AB- > C, C- > DE, E- > A. The number of candidate keys is
NAT · enter the numerical value

Q.131 NAT · 1 mark · Relational algebra and tuple calculus
Relation Match(home, away, goals) has 8 rows. pi_home(Match) has 5 distinct homes. The number of rows returned by pi_home(Match) is
NAT · enter the numerical value

Q.132 NAT · 1 mark · SQL
```
Emp2(id,name,mgr):
(1,Amy,NULL),(2,Ben,1),(3,Cal,1),(4,Dee,2),(5,Eve,2).
SELECT E.name FROM Emp2 E JOIN Emp2 M ON E.mgr=M.id WHERE
M.name='Ben' returns how many rows?
```
NAT · enter the numerical value

Q.133 NAT · 1 mark · Integrity constraints
Table R(a INT PRIMARY KEY). Inserts: 1, 2, 1, 3. How many rows does R contain after the four statements (the duplicate is rejected)?
NAT · enter the numerical value

Q.134 MCQ · 1 mark · Files and indexing
A dense secondary index on attribute A of HeapFile Staff has one entry per
(A) record of Staff
(B) block of Staff
(C) distinct A-value only, even if A is not unique, with no posting list
(D) cylinder of the disk

Q.135 NAT · 1 mark · Data types and transformation
Decimal 13 in 4-bit two's complement does not fit. The largest non-negative integer representable in 4-bit two's complement is
NAT · enter the numerical value

Q.136 MCQ · 1 mark · Warehousing and OLAP
Cube SensorAvg is shown at dimensions Station, Metric, Day. Moving the last geographic/organisational/time level to the finer grain (Station to SensorId) is
(A) drill-down
(B) roll-up
(C) slice
(D) dice

Q.137 MCQ · 2 marks · ER model
Total participation of both sides of a 1:1 relationship Married(Person,Person) with two entity sets Man and Woman can be mapped as
(A) either key as a NOT NULL UNIQUE FK in the other table, or a single merged table
(B) an M:N table without uniqueness
(C) two independent tables with no FK
(D) a weak entity with no owner

Q.138 MCQ · 2 marks · Relational model and FDs
Let schema Payroll(A,B,C,D,E) with FDs AB- > C, C- > DE, E- > A. Assuming the schema is already in 1NF, the highest normal form it is guaranteed to be in is
(A) 1NF
(B) 2NF
(C) 3NF
(D) BCNF

Q.139 MCQ · 2 marks · Relational algebra and tuple calculus
TRC query of 'students who take every course in Dept=CS': {s | s ∈ Student ∧ ∀c (c ∈ Course ∧ c.dept='CS' = > ∃e ∈ Enroll (e.sid=s.sid ∧ e.cid=c.cid))}. This is the calculus form of
(A) division of Enroll by the CS courses (after suitable projection)
(B) a simple selection on Student
(C) Student - Course
(D) an unsafe query with no active domain

Q.140 MCQ · 1 mark · SQL
A.x = {1,2,2,3}, B.x={2,3,4}. (|UNION|, |UNION ALL|) is
(A) 4, 7
(B) 4, 4
(C) 3, 7
(D) 4, 6

Q.141 MCQ · 1 mark · Integrity constraints
A composite foreign key (sid,cid) REFERENCES Enroll(sid,cid) requires that
(A) the pair matches some Enroll pair; matching sid and cid from different Enroll rows is not enough
(B) each component separately exists, even if the pair does not
(C) only sid exists
(D) the child table has no PK

Q.142 MSQ · 2 marks · Files and indexing

Which index types support range predicates (A BETWEEN 10 AND 20) efficiently?

(A) B+ tree on A
(B) sorted dense index on A
(C) hash index on A (extendible / static hash)
(D) clustered B+ tree on A

Q.143 MCQ · 1 mark · Data types and transformation

A CHAR(10) column versus VARCHAR(10) for strings of typical length 3. VARCHAR is preferred because

(A) it stores only the actual length plus a length prefix, saving space
(B) it is always faster to index
(C) it forbids trailing spaces by standard
(D) it is a numeric type

Q.144 MCQ · 1 mark · Warehousing and OLAP

On cube SensorAvg (Station, Metric, Day), aggregating away the middle dimension entirely (summing the measure over it) is

(A) roll-up / projection of a dimension
(B) drill-down
(C) a foreign-key insert
(D) 2NF decomposition

Q.145 MCQ · 1 mark · ER model

An attribute of a relationship (e.g. Enrolls.grade) is placed, after mapping an M:N relationship, in

(A) the relationship table
(B) both entity tables
(C) a CHECK constraint only
(D) the log

Q.146 NAT · 2 marks · Relational model and FDs

Let schema StoreSku(A,B,C,D) with FDs AB- > C, AB- > D, C- > D. The number of candidate keys is

NAT · enter the numerical value

Q.147 MCQ · 1 mark · Relational algebra and tuple calculus

Left outer join R ⟕ S, |R|=4, |S|=3, inner join has 2 matching R-tuples (each matches one S). How many rows in the left outer join?

(A) 4
(B) 2
(C) 7
(D) 3

Q.148 MSQ · 2 marks · SQL

SQL three-valued logic. Which evaluations are UNKNOWN (not TRUE)?

(A) NULL = 5
(B) NULL IS NULL
(C) NOT (NULL > 0)
(D) 5 = 5

Q.149 MCQ · 1 mark · Files and indexing

A clustered (primary) index on A usually requires the data file to be

(A) ordered on A
(B) a heap in arbitrary order
(C) hashed on a different key
(D) a bitmap

Q.150 MCQ · 1 mark · Data types and transformation

Discretising a numeric attribute before naive Bayes is done so that

(A) a multinomial / categorical model can be used on bins
(B) the mean of the attribute becomes 0
(C) compression becomes lossy
(D) the attribute becomes a key

Q.151 MCQ · 1 mark · Warehousing and OLAP

Cube PayrollCost is shown at dimensions Org, Grade, Month. Moving the last geographic/organisational/time level to the finer grain (Org to Team) is

(A) drill-down
(B) roll-up
(C) slice
(D) dice

Q.152 MCQ · 2 marks · ER model

`A disjoint total specialisation of Account into Savings and Checking, using subclass tables only (no Account table), is awkward because`

(A) a query over all accounts must UNION the subclasses, and shared attributes are duplicated
(B) FKs cannot reference accounts
(C) 1NF fails
(D) OLAP drill-down fails

Q.153 MCQ · 2 marks · Relational model and FDs

Let schema StoreSku(A,B,C,D) with FDs AB- > C, AB- > D, C- > D. Assuming the schema is already in 1NF, the highest normal form it is guaranteed to be in is

(A) 2NF
(B) 1NF
(C) 3NF
(D) BCNF

Q.154 NAT · 1 mark · Relational algebra and tuple calculus

Full outer join of those R and S (inner matches 2 pairs; 2 unmatched R; 1 unmatched S) has how many rows?

NAT · enter the numerical value

Q.155 MSQ · 2 marks · SQL

Concerning COUNT and NULLs in SQL:

(A) COUNT(*) counts rows including those with NULLs
(B) COUNT(A) ignores rows where A is NULL
(C) SUM(A) ignores NULLs
(D) AVG(A) treats NULL as 0

Q.156 NAT · 1 mark · Files and indexing

A heap file of 10000 records, blocking factor 10 records/block. Number of blocks is

NAT · enter the numerical value

Q.157 NAT · 1 mark · Data types and transformation

`Stratified sampling: a table is 30% class A and 70% class B. A stratified sample of 100 rows (proportional) contains how many class-A rows?`

NAT · enter the numerical value

Q.158 MCQ · 1 mark · Warehousing and OLAP
On cube PayrollCost (Org, Grade, Month), aggregating away the middle dimension entirely (summing the measure over it) is
(A) roll-up / projection of a dimension
(B) drill-down
(C) a foreign-key insert
(D) 2NF decomposition

Q.159 NAT · 2 marks · Relational model and FDs
Let schema Enroll(A,B,C) with FDs AB- > C, C- > A. The number of candidate keys is
NAT · enter the numerical value

Q.160 MSQ · 2 marks · Relational algebra and tuple calculus
Let R and S be union-compatible. Which identities hold?
(A) R ∪ S = S ∪ R
(B) R ∩ S = R - (R - S)
(C) R × S = S × R (up to attribute order)
(D) R - S = S - R always

Q.161 NAT · 1 mark · SQL
`Voyage(vid,ship,day,cargo): (1,Nila,1,40),(2,Nila,2,10),(3,Orca,1,25),(4,Orca,3,25),(5,Pinta,2,60). SELECT ship FROM Voyage GROUP BY ship HAVING MIN(cargo) > =25 returns how many rows?`
NAT · enter the numerical value

Q.162 NAT · 1 mark · Files and indexing
Unsorted heap, 1000 blocks, linear search for a missing equality key (unindexed). Worst-case block reads are
NAT · enter the numerical value

Q.163 MCQ · 1 mark · Data types and transformation
Run-length encoding of the bit-vector 0000011111000 (13 bits) can store it as counts
(A) 5 zeros, 5 ones, 3 zeros
(B) 13 ones
(C) 8 zeros
(D) 2 runs only

Q.164 MCQ · 1 mark · Warehousing and OLAP
Cube LibraryLoans is shown at dimensions Branch, Genre, Month. Moving the last geographic/organisational/time level to the finer grain (Genre to Shelf) is
(A) drill-down
(B) roll-up
(C) slice
(D) dice

Q.165 MCQ · 2 marks · Relational model and FDs
Let schema Enroll(A,B,C) with FDs AB- > C, C- > A. Assuming the schema is already in 1NF, the highest normal form it is guaranteed to be in is
(A) 3NF
(B) 1NF
(C) 2NF
(D) BCNF

Q.166 NAT · 1 mark · Relational algebra and tuple calculus
R has 10 tuples, sigma_P(R) keeps 3. pi_A(sigma_P(R)) has 2 distinct A values. Number of rows in the projection is
NAT · enter the numerical value

Q.167 NAT · 2 marks · SQL
Clinic(pid,ward,bill): (1,ICU,900),(2,ICU,400),(3,GEN,400),(4,GEN,100),(5,DAY,50).
Correlated: pid whose bill > average bill of their own ward. Number of rows is
NAT · enter the numerical value

Q.168 NAT · 1 mark · Files and indexing
Sorted file, 1000 blocks, binary search on the sort key, one record per matching key. Worst-case block reads (approx ceil(log2 1000)) are
NAT · enter the numerical value

Q.169 MCQ · 1 mark · Data types and transformation
Normalising a numeric column to zero mean and unit variance is
(A) z-score standardisation
(B) min-max to [0,1]
(C) equal-width binning
(D) lossy JPEG

Q.170 MCQ · 1 mark · Warehousing and OLAP
On cube LibraryLoans (Branch, Genre, Month), aggregating away the middle dimension entirely (summing the measure over it) is
(A) roll-up / projection of a dimension
(B) drill-down
(C) a foreign-key insert
(D) 2NF decomposition

Q.171 NAT · 2 marks · Relational model and FDs
Let schema Sensor(A,B,C,D,E) with FDs A- > B, BC- > D, D- > E, E- > C. The number of candidate keys is
NAT · enter the numerical value

Q.172 MCQ · 1 mark · Relational algebra and tuple calculus
The five primitive RA operators of Codd's algebra are usually listed as
(A) select, project, union, difference, Cartesian product (rename sometimes added)
(B) join, division, outer join only
(C) GROUP BY and HAVING
(D) star and snowflake

Q.173 NAT · 1 mark · SQL
`Book(isbn,title,copies): (1,Ada,3),(2,C,0),(3,ML,5),(4,DB,1). SELECT title FROM Book WHERE copies BETWEEN 1 AND 4 returns how many rows?`
NAT · enter the numerical value

Q.174 MCQ · 1 mark · Files and indexing
Blocking factor of a file of 200-byte records in 4096-byte blocks with unspanned records is
(A) 20
(B) 21
(C) 19
(D) 4096

Q.175 NAT · 1 mark · Data types and transformation
Attribute DelayMin = [1,1,2,3,13]. After min-max to [0,1], the first value maps to

NAT · enter the numerical value

Q.176 MSQ · 2 marks · Warehousing and OLAP
Measures of a warehouse fact. Which classifications are correct?

(A) quantity sold is additive across all dimensions
(B) account balance is semi-additive (cannot sum over time)
(C) ratio of defective items (a percentage) is typically non-additive
(D) date_id in the fact table is an additive measure

Q.177 MCQ · 2 marks · Relational model and FDs
Let schema Sensor(A,B,C,D,E) with FDs A- > B, BC- > D, D- > E, E- > C. Assuming the schema is already in 1NF, the highest normal form it is guaranteed to be in is

(A) 1NF
(B) 2NF
(C) 3NF
(D) BCNF

Q.178 MSQ · 2 marks · Relational algebra and tuple calculus
Pushing selections. Which are valid for improving an RA plan?

(A) sigma_P(R □ S) = sigma_P(R) □ S if P mentions only R-attributes
(B) sigma_{P and Q}(R) = sigma_P(sigma_Q(R))
(C) sigma_P(R ∪ S) = sigma_P(R) ∪ sigma_P(S)
(D) sigma_P(R × S) cannot ever be pushed

Q.179 NAT · 1 mark · SQL
```
Marks(sid,course,m):
(1,DB,80),(1,ML,70),(2,DB,90),(2,ML,60),(3,DB,90).
SELECT course, AVG(m) FROM Marks GROUP BY course returns
how many rows?
```

NAT · enter the numerical value

Q.180 MCQ · 1 mark · Files and indexing
Extendible hashing doubles the directory when a bucket overflows. The directory is an array of

(A) bucket pointers of size 2^{global depth}
(B) records
(C) B+ leaves
(D) cylinders

Q.181 MCQ · 1 mark · Data types and transformation
Sampling a data cube's facts at 10% (random) mainly affects

(A) approximate aggregates, with variance depending on the sample size
(B) the schema of dimensions
(C) the BCNF status of the fact table
(D) foreign-key directions

Q.182 MCQ · 1 mark · Warehousing and OLAP
Grain of SalesFact is 'one row per sku per store per day'. A query that needs 'per week' should

(A) roll-up the Date dimension along day < week
(B) insert extra fact rows
(C) change the FK direction
(D) dice on sku only

Q.183 MCQ · 1 mark · Relational model and FDs
Relation R(A,B,C) with FDs A- > B and B- > C. Which is a candidate key?

(A) A
(B) B
(C) C
(D) BC

Q.184 NAT · 2 marks · SQL
```
On Marks, students who have a row for every distinct
course: SELECT sid FROM Marks GROUP BY sid HAVING
COUNT(*) = (SELECT COUNT(DISTINCT course) FROM Marks).
Number of rows is
```

NAT · enter the numerical value

Q.185 MSQ · 2 marks · Files and indexing
B+ tree leaves are linked. Which follow?

(A) an equality search is O(height)
(B) a range scan after locating the start is sequential along leaves
(C) non-leaf nodes store record pointers for every key in the usual internal-node design
(D) all record pointers live in the leaves in a typical B+ tree

Q.186 MSQ · 2 marks · Data types and transformation
Given AgeYears = [10,20,30,40,50], which are true?

(A) min-max of 10 to [0,1] is 0
(B) min-max of 50 to [0,1] is 1
(C) the sample mean is 30
(D) equal-width 4 bins on [10,50] have width 20

Q.187 MCQ · 1 mark · Warehousing and OLAP
A concept hierarchy for Location is Store < City < State < Country. The coarsest level is

(A) Country
(B) Store
(C) City
(D) the fact table

Q.188 MSQ · 2 marks · Relational model and FDs
Relation Shop(A,B,C,D) with FDs A- > B, B- > C, C- > D. Which statements are true?

(A) A is the unique candidate key
(B) the schema is in BCNF
(C) B- > C is a 3NF violation (B is not a superkey and C is nonprime)
(D) a lossless BCNF decomposition can produce (A,B), (B,C), (C,D)

Q.189 MCQ · 2 marks · SQL
```
SQL query SELECT * FROM R WHERE A IN (SELECT A FROM S).
If S.A contains NULL and 5, and R.A=5, the row with 5
```

(A) is returned (5 IN (5,NULL) is TRUE)
(B) is rejected because of the NULL
(C) causes a runtime error
(D) is returned only if R has no NULLs

Q.190 NAT · 2 marks · Files and indexing
A B+ tree of order (max keys per node) 4 and height 2 (root height 0, leaves at level 2) can store at most how many keys if every node is full? Internal fanout 5 children, 4 keys/node. Leaves also 4 keys. Max keys = 5^2 * 4 =

NAT · enter the numerical value

Q.191 MCQ · 1 mark · Data types and transformation
Decimal(5,2) can store which of the following?
(A) 123.45
(B) 1234.5 (needs 5 digits to the left? 1234.5 is 6 significant with scale 2: 1234.50 needs precision 6)
(C) 0.0001 stored exactly as four decimals of scale 2
(D) an arbitrary-length string

Q.192 MSQ · 2 marks · Warehousing and OLAP
Star schema facts. Which statements hold?
(A) the fact table is typically in (near) BCNF with FKs to dimensions plus measures
(B) dimension tables are often denormalised and not in BCNF
(C) a snowflake normalises dimension hierarchies into separate tables
(D) a star schema cannot have more than two dimensions

Q.193 NAT · 1 mark · Relational model and FDs
Number of superkeys of R(A,B,C) with only FD A- > BC (so A is the unique candidate key) is
NAT · enter the numerical value

Q.194 MCQ · 2 marks · SQL
`SELECT * FROM R WHERE A NOT IN (SELECT A FROM S) when S.A is {5, NULL} and R.A=7`
(A) the row is filtered out because NOT IN with a NULL in the set is UNKNOWN
(B) the row is returned because 7 != 5
(C) the query errors
(D) NULL is ignored in NOT IN always

Q.195 MCQ · 1 mark · Files and indexing
Static hashing with n buckets. Expected time for equality search under uniform hashing and no overflow is
(A) O(1)
(B) O(log n)
(C) O(n)
(D) O(n log n)

Q.196 MCQ · 2 marks · Data types and transformation
Huffman coding a 4-symbol alphabet with probabilities 1/2, 1/4, 1/8, 1/8 has average code length (bits)
(A) 1.75
(B) 2
(C) 4
(D) 1

Q.197 MCQ · 1 mark · Warehousing and OLAP
Pivot (rotate) on a cube Product × Month showing Sales is
(A) swapping the axes so Month × Product is displayed
(B) deleting the Product dimension
(C) a B+ tree split
(D) an FK cascade

Q.198 MCQ · 1 mark · Relational model and FDs
Armstrong axioms. From X- > Y and Y- > Z we may infer X- > Z. This is
(A) transitivity
(B) augmentation
(C) reflexivity
(D) union

Q.199 NAT · 2 marks · SQL
```
Sensor(id,loc,t,val): (1,Dock,1,3.2),(2,Dock,2,3.2),(3,Yard,1,4.0),(4,Yard,2,1.0),(5,Gate,1,4.0).
SELECT loc FROM Sensor GROUP BY loc HAVING COUNT(DISTINCT val)=1 returns how many rows?
```
NAT · enter the numerical value

Q.200 MCQ · 1 mark · Files and indexing
A bitmap index on a low-cardinality attribute Gender of N rows uses (ignoring compression) about
(A) 2 bit-vectors of N bits
(B) N B+ tree leaves
(C) one float per row
(D) a hash table of size N

Q.201 NAT · 1 mark · Data types and transformation
Equal-width bins on [0,100] of width 20. The number of bins is
NAT · enter the numerical value

Q.202 MCQ · 1 mark · Warehousing and OLAP
A warehouse ETL pipeline's 'T' (transform) step typically includes
(A) cleaning, conforming dimensions, and computing measures
(B) dropping all FKs
(C) replacing SQL by Python loops only
(D) forcing every table into 6NF

Q.203 MCQ · 1 mark · Relational model and FDs
A decomposition of R into R1 and R2 is lossless-join iff
(A) R1 ∩ R2 - > R1 or R1 ∩ R2 - > R2 (the intersection is a key of one piece)
(B) R1 ∩ R2 is empty
(C) R1 ∪ R2 has more attributes than R
(D) both R1 and R2 are in BCNF

Q.204 NAT · 1 mark · SQL
```
Exam(sid,subj,score):
(1,DA,80),(2,DA,80),(3,DA,60),(1,GA,70),(2,GA,90).
SELECT sid FROM Exam WHERE subj='DA' AND score > =80 returns how many rows?
```
NAT · enter the numerical value

Q.205 NAT · 1 mark · Files and indexing
Data file 8000 records, sparse clustered index with one index entry per data block, 40 records/block. Number of index entries is
NAT · enter the numerical value

Q.206 MCQ · 1 mark · Data types and transformation
A DATE versus TIMESTAMP column: TIMESTAMP additionally stores
(A) a time-of-day (and possibly a time zone, depending on the type)
(B) only a year
(C) a B+ tree
(D) a cube measure

Q.207 MCQ · 1 mark · Warehousing and OLAP

A conformed dimension DateDim shared by SalesFact and InventoryFact means

(A) both facts join to the same Date dimension table / meaning of dates
(B) the facts are the same table
(C) dates are stored as TEXT without a dimension
(D) the warehouse has no time hierarchy

Q.208 MSQ · 2 marks · Relational model and FDs

Relation R(A,B,C) with FDs A- > B and C- > B is decomposed into R1(A,B) and R2(C,B). Which statements hold?

(A) the decomposition is lossy (R1 ∩ R2 = {B} is not a key of either)
(B) A- > B is preserved in R1
(C) C- > B is preserved in R2
(D) the decomposition is in BCNF and lossless

Q.209 NAT · 2 marks · SQL

On Exam, sids whose DA score is strictly less than their own GA score. Number of rows is

NAT · enter the numerical value

Q.210 MCQ · 1 mark · Files and indexing

ISAM is a static multilevel index. Inserts that overflow a leaf go to

(A) overflow chains
(B) a freshly split leaf that updates the root always
(C) a hash bucket
(D) the WAL only

Q.211 NAT · 1 mark · Warehousing and OLAP

A 3-dimensional cube Product(10) × Store(5) × Day(30) stored densely has how many cells?

NAT · enter the numerical value

Q.212 NAT · 2 marks · Relational model and FDs

How many FDs of the form X- > Y with X,Y nonempty subsets of {A,B} hold in a relation whose only candidate key is AB (no nontrivial FD)? Count all X- > Y that are implied by Armstrong reflexivity / trivial FDs only, with X,Y subsets of {A,B}, Y nonempty, X nonempty, Ysubseteq X. That number is

NAT · enter the numerical value

Q.213 MSQ · 2 marks · SQL

On Emp(bonus) with some NULLs, which queries count only non-NULL bonuses?

(A) SELECT COUNT(bonus) FROM Emp
(B) SELECT COUNT(*) FROM Emp WHERE bonus IS NOT NULL
(C) SELECT COUNT(*) FROM Emp
(D) SELECT SUM(1) FROM Emp WHERE bonus IS NOT NULL

Q.214 MSQ · 2 marks · Files and indexing

File organisations. Which pairs are matched correctly?

(A) heap file: insert is O(1) (append), equality search is linear
(B) sorted file: range search is efficient, insert is expensive
(C) hashed file: equality search expected O(1), range search poor
(D) sorted file: insert is O(1) append at end regardless of key

Q.215 NAT · 1 mark · Warehousing and OLAP

Slicing that cube on Store='Pune' (one of 5 stores) leaves how many cells in the remaining Product × Day plane (dense)?

NAT · enter the numerical value

Q.216 MCQ · 2 marks · Relational model and FDs

Relation Coach(A,B,C) with FDs AB- > C and C- > A. Candidate keys are

(A) AB and BC
(B) AB and C
(C) A and BC
(D) C only

Q.217 NAT · 1 mark · SQL

```
SELECT 1 FROM dual WHERE 1=1 AND NULL=NULL returns how many rows in standard SQL 3VL?
```

NAT · enter the numerical value

Q.218 MSQ · 2 marks · Files and indexing

A B+ tree index is chosen over a hash index when

(A) queries include range predicates on the key
(B) the key is used in ORDER BY
(C) only point lookups occur and ranges never do (hash would be competitive)
(D) the attribute has two values (bitmap may be even better) but ranges still matter

Q.219 MCQ · 1 mark · Warehousing and OLAP

MOLAP versus ROLAP: MOLAP stores the cube in

(A) a specialised multidimensional array structure
(B) only normalised 3NF tables with no aggregates
(C) a B+ tree of XML
(D) main-memory Python lists exclusively

Q.220 MCQ · 2 marks · Relational model and FDs

The schema Coach(A,B,C) with AB- > C and C- > A is

(A) in 3NF but not in BCNF
(B) in BCNF
(C) not in 3NF
(D) not even in 1NF

Q.221 NAT · 1 mark · SQL

```
SELECT 1 WHERE 1=1 OR NULL=NULL returns how many rows?
```

NAT · enter the numerical value

Q.222 NAT · 1 mark · Files and indexing

Unspanned records of 300 bytes in 2048-byte blocks. Blocking factor is

NAT · enter the numerical value

Q.223 MSQ · 2 marks · Warehousing and OLAP
Which are OLAP operations on cube HarbourTonnage(Port, ShipType, Month)?
(A) roll-up Month to Year
(B) drill-down Port to Berth
(C) slice Month='Jan'
(D) BCNF decomposition of the cube

Q.224 NAT · 1 mark · Relational model and FDs
Attribute closure of CD under FDs A- > B, CD- > E, E- > A on ABCDE. | (CD)+ | (number of attributes in the closure) is
NAT · enter the numerical value

Q.225 MCQ · 1 mark · SQL
`SELECT AVG(x) FROM (VALUES (1),(2),(NULL)) t(x) in standard SQL is`
(A) 1.5
(B) 1
(C) NULL
(D) 0.75

Q.226 NAT · 1 mark · Files and indexing
8000 such records (300 B, 2048 B blocks, unspanned, bf=6). Number of blocks needed is
NAT · enter the numerical value

Q.227 MCQ · 1 mark · Warehousing and OLAP
A degenerate dimension in a fact table is
(A) a dimension key with no attributes (e.g. a bill_no stored in the fact)
(B) a dimension in 6NF
(C) a measure that is additive
(D) an ETL error

Q.228 MSQ · 2 marks · Relational model and FDs
Which are candidate keys of R(A,B,C,D,E) with F={A- > BC, CD- > E, E- > A}?
(A) AD
(B) ED
(C) CD
(D) BD

Q.229 NAT · 1 mark · SQL
`Table Q(a) has rows 1,1,2,NULL. SELECT COUNT(DISTINCT a) FROM Q returns`
NAT · enter the numerical value

Q.230 MCQ · 1 mark · Files and indexing
`A covering index for query SELECT A,B FROM R WHERE A=5 is a B+ tree on`
(A) (A,B) so the query can be answered from the index alone
(B) C only
(C) the heap RID list of an unrelated key
(D) a hash on B only, with A unindexed

Q.231 MCQ · 1 mark · Warehousing and OLAP
Slowly changing dimension Type 2 on CustomerDim handles a city change by
(A) inserting a new dimension row with new surrogate key and version dates
(B) overwriting the city in place with no history (Type 1)
(C) dropping the customer
(D) storing the city in the fact table only

Q.232 MCQ · 1 mark · Relational model and FDs
A relation is in 2NF if it is in 1NF and
(A) no nonprime attribute is partially dependent on a candidate key
(B) every determinant is a candidate key
(C) it has no transitive dependency of a nonprime on a key
(D) it has a single-attribute key

Q.233 NAT · 1 mark · SQL
`Same Q(a)={1,1,2,NULL}. SELECT COUNT(a) FROM Q returns`
NAT · enter the numerical value

Q.234 MCQ · 1 mark · Files and indexing
Primary index versus secondary: a primary index is built on
(A) the ordering key of a sequentially ordered file
(B) any non-ordering attribute, densely
(C) a bitmap of gender only
(D) the log file

Q.235 NAT · 1 mark · Warehousing and OLAP
Factless fact table Attendance(date_id, student_id, course_id) stores event rows with no numeric measure. For a course that appears in 17 fact rows, COUNT(*) for that course is
NAT · enter the numerical value

Q.236 MCQ · 1 mark · Relational model and FDs
A relation is in BCNF if
(A) for every nontrivial FD X- > Y, X is a superkey
(B) for every FD X- > A, X is a superkey or A is prime
(C) it is in 3NF and has no overlapping candidate keys
(D) it is in 1NF and has no partial dependency

Q.237 MCQ · 1 mark · SQL
`SELECT * FROM R NATURAL JOIN S. If R(A,B) and S(B,C) then the result schema has attributes`
(A) A,B,C
(B) A,B,B,C
(C) A,C only
(D) B only

Q.238 MSQ · 2 marks · Files and indexing
Height of a B+ tree (root at level 0) of n keys and minimum leaf fill of d keys, internal min fanout d, is
(A) O(log_d n)
(B) O(n) in the worst balanced case still, because B+ trees are balanced
(C) Theta(log_d n) because the tree is kept balanced
(D) Theta(n) because leaves are linked

Q.239 MCQ · 2 marks · Warehousing and OLAP
Additive measure amount can be summed along Store and along Date. Semi-additive inventory_qty can be summed along Store but not along Date. The right way to get a monthly inventory picture is
(A) take an average or end-of-month snapshot along Date, sum along Store
(B) sum inventory_qty over all days and all stores
(C) ignore Date
(D) use COUNT(DISTINCT) on qty

Q.240 NAT · 1 mark · Relational model and FDs

R(A,B,C,D) with FDs A- > B, B- > C, C- > D. Size of A+ is

NAT · enter the numerical value

Q.241 MSQ · 2 marks · SQL

GROUP BY vs DISTINCT. On Marks(sid,course) with duplicate (sid,course) forbidden by PK, which are true?

(A) SELECT sid FROM Marks GROUP BY sid returns each sid once
(B) SELECT DISTINCT sid FROM Marks returns each sid once
(C) SELECT sid, COUNT(*) FROM Marks GROUP BY sid is legal
(D) SELECT sid, COUNT(*) FROM Marks DISTINCT is the usual legal form

Q.242 NAT · 1 mark · Files and indexing

A dense index on a file of 5000 records has 5000 entries, 50 entries per index block. Number of first-level index blocks is

NAT · enter the numerical value

Q.243 MSQ · 2 marks · Warehousing and OLAP

Concept hierarchy Time: Day < Week < Month < Quarter < Year. Which OLAP moves are correctly named for cube Sales(Product, Day)?

(A) Day to Month: roll-up
(B) Month to Day: drill-down
(C) Day to Year: roll-up
(D) Year to Month: roll-up

Q.244 MCQ · 1 mark · Relational model and FDs

Dependency preservation: a decomposition is dependency-preserving if

(A) the union of local FDs (projected) implies the original cover
(B) the join is lossless
(C) each piece is in BCNF
(D) there are no FDs

Q.245 MCQ · 1 mark · SQL

HAVING versus WHERE: HAVING

(A) filters groups after GROUP BY (and may use aggregates)
(B) always filters rows before grouping and cannot use COUNT(*)
(C) is a synonym of WHERE
(D) is only for ORDER BY

Q.246 MCQ · 1 mark · Warehousing and OLAP

Bitmap indexes are attractive in warehouses because

(A) dimension attributes often have low cardinality and queries are range/aggregate heavy
(B) every fact table is updated thousands of times per second in OLTP style
(C) they replace the need for a Date dimension
(D) they implement BCNF

Q.247 MCQ · 1 mark · Relational model and FDs

If a decomposition is not dependency-preserving, which operator is needed more often to enforce FDs that span pieces?

(A) join
(B) selection
(C) projection
(D) union

Q.248 NAT · 1 mark · SQL

```
SELECT SUM(x) FROM (VALUES (1),(NULL),(3)) t(x) returns
```

NAT · enter the numerical value

Q.249 MCQ · 1 mark · Warehousing and OLAP

A data mart versus an enterprise warehouse: a mart is

(A) a subject-area subset (e.g. sales only), often a star
(B) always a normalised 3NF EDW of the whole company
(C) a B+ tree
(D) an OLTP orders table

Q.250 MSQ · 2 marks · Relational model and FDs

R(A,B,C,D) with FDs AB- > C, C- > D, D- > A. Which are candidate keys?

(A) AB
(B) BC
(C) BD
(D) CD

Q.251 MCQ · 1 mark · SQL

```
Correlated subquery: SELECT * FROM R WHERE R.a > (SELECT
AVG(S.b) FROM S WHERE S.k=R.k). The inner query is
executed
```

(A) once per candidate R-row (conceptually)
(B) exactly once, always
(C) never
(D) only at COMMIT

Q.252 NAT · 1 mark · Warehousing and OLAP

Cube GridLoad(Region 8 × Fuel 4 × Hour 24). Drill-down Hour to 4 fifteen-minute slots (4× finer) multiplies the dense cell count by

NAT · enter the numerical value

Q.253 MCQ · 1 mark · Relational model and FDs

Trivial FD X- > Y means

(A) Y is a subset of X
(B) X is a candidate key
(C) Y is empty only
(D) X and Y are disjoint

Q.254 MCQ · 1 mark · Warehousing and OLAP

HOLAP (hybrid OLAP) typically

(A) keeps detailed facts in relational tables and some aggregates in MOLAP storage
(B) forbids SQL
(C) stores only bitmaps
(D) is identical to 1NF

Q.255 NAT · 1 mark · Relational model and FDs

R(A,B,C) with FDs A- > B, B- > C. Number of candidate keys is

NAT · enter the numerical value

Q.256 MSQ · 2 marks · Warehousing and OLAP

Cube MatchGoals(Team, Venue, Week). Which operations are correctly named?

(A) Week to MatchDay: drill-down
(B) Venue='Pune' only, keep Team × Week: slice
(C) Team in {Goa,Pune} and Week in {1,2}: dice
(D) Week to MatchDay: roll-up

Q.257 MCQ · 2 marks · Relational model and FDs

That same R(A,B,C) with A- > B, B- > C is in

(A) 2NF (single-attribute key, so no partial) but not 3NF (transitive C on A via B)
(B) BCNF
(C) 3NF but not 2NF
(D) not 1NF

Q.258 MCQ · 1 mark · Warehousing and OLAP

A junk dimension packs several low-cardinality flags (pay_type, gift_wrap, ...) into

(A) one dimension table whose key is stored in the fact
(B) the date dimension
(C) a MOLAP cell comment
(D) an ER weak entity of the OLTP system only

Q.259 NAT · 1 mark · Relational model and FDs

A cover F has 4 FDs. A canonical cover of a chain A- > B, B- > C, C- > D, A- > D may drop A- > D as redundant. The size of a minimal cover (number of FDs) is

NAT · enter the numerical value

Q.260 MSQ · 2 marks · Warehousing and OLAP

Kimball-style dimensional modelling. Which are true?

(A) facts are additive (or semi/non) numeric measurements at a declared grain
(B) dimensions provide the labels and hierarchies for slicing
(C) a star join is fact-to-dimension on surrogate keys
(D) every dimension must be a weak entity of the fact in the ER sense of double rectangles

Q.261 NAT · 1 mark · Warehousing and OLAP

A concept hierarchy of 4 levels Store < City < State < Country. Number of roll-up steps from Store to Country is

NAT · enter the numerical value

Q.262 MCQ · 2 marks · Warehousing and OLAP

A snapshot fact InventoryFact(date_id, sku_id, on_hand) at grain 'sku at end of day'. Summing on_hand over a month of dates for one sku is

(A) usually meaningless as a total (semi-additive); use last-day or average
(B) the true number of items manufactured that month
(C) a drill-down
(D) a lossless join

Q.263 MCQ · 1 mark · Warehousing and OLAP

An aggregate navigation / summary table (Product, Month, SUM(qty)) is a

(A) materialised roll-up of the base fact
(B) snowflake of StoreDim
(C) B+ internal node
(D) CHECK constraint

UNIT 5 SOLUTIONS

Q.1 [MCQ] A — M:N relationship becomes its own table with the two keys plus relationship attributes.

Q.2 [NAT] 3 — Candidate keys are AD, CD, DE.

Q.3 [MCQ] A — Intersect the team names with the projected Player.team column.

Q.4 [NAT] 3 — AVG is (12+18+12+20+18+9)/6=14.833..., so 18,20,18: three rows.

Q.5 [MSQ] A;B — Inserting into the referenced table s, or deleting from the referencing table r, cannot create a dangling FK. Inserts into r and deletes from s can.

Q.6 [NAT] 195 — n*nodeptr + (n-1)*key < = block = > n < = floor((block+key)/(nodeptr+key)) = 195.

Q.7 [NAT] 1 — (x-min)/(max-min) = (50-10)/(50-10) = 1.0.

Q.8 [MCQ] A — Star: one denormalised table per dimension. Snowflake: dimensions normalised into subtables.

Q.9 [MCQ] A — A weak entity is identified by a 1:N identifying relationship and depends totally on the owner.

Q.10 [MCQ] A — Candidate keys AD, CD, DE. Highest NF is 1NF.

Q.11 [MCQ] A — The difference is the set of team names missing from both positions; it is empty iff coverage is complete.

Q.12 [NAT] 3 — Each of Patna, Jaipur, Delhi has 2 raiders; three groups.

Q.13 [MCQ] A — CASCADE propagates the delete. RESTRICT/NO ACTION reject; SET NULL is the third option.

Q.14 [NAT] 177 — n*(key+recptr) + nodeptr < = block = > n = floor((block-nodeptr)/(key+recptr)) = 177.

Q.15 [NAT] 1.41 — z=(x-mean)/sigma. mean=30, sigma=14.142135623730951, z=1.414213562373095.

Q.16 [MCQ] A — Normalised dimensions trade join cost for less redundancy.

Q.17 [MSQ] A;B;D — M:N mapping needs a relationship table.

Q.18 [NAT] 3 — Candidate keys are AB, BC, BD.

Q.19 [NAT] 7 — 5+4-2=7.

Q.20 [NAT] 4 — Distinct points: 12,18,20,9 - > 4.

Q.21 [MCQ] A — You cannot set NULL on a NOT NULL FK (or on a PK component that is not nullable).

Q.22 [NAT] 103 — n*nodeptr + (n-1)*key < = block = > n < = floor((block+key)/(nodeptr+key)) = 103.

Q.23 [NAT] 1 — (x-min)/(max-min) = (18-0)/(18-0) = 1.0.

Q.24 [MCQ] A — Drill-down walks down a concept hierarchy (Country to State).

Q.25 [MCQ] B — Four entity tables plus Enrolls. WorksIn is absorbed as a department FK on Instructor.

Q.26 [MCQ] A — Candidate keys AB, BC, BD. Highest NF is 3NF.

Q.27 [NAT] 2 — Intersection size is the overlap.

Q.28 [NAT] 3 — MAX of SRNagar is 65000; amounts > 65000 are 90000,80000,70000: three rows.

Q.29 [MSQ] A;B;C — The last row satisfies PK, FK and CHECK.

Q.30 [NAT] 102 — n*(key+recptr) + nodeptr < = block = > n = floor((block-nodeptr)/(key+recptr)) = 102.

Q.31 [NAT] 1.9 — z=(x-mean)/sigma. mean=6, sigma=6.324555320336759, z=1.8973665961010275.

Q.32 [MCQ] A — Roll-up climbs a hierarchy or drops a dimension by aggregation.

Q.33 [MCQ] A — Total 1:1 participation of Berth forces a unique non-null FK to Ship.

Q.34 [NAT] 4 — Candidate keys are AD, BD, CD, DE.

Q.35 [NAT] 3 — 5-2=3.

Q.36 [NAT] 2 — Only serial 13 is a red Tata; owners Asha and Isha: 2 rows.

Q.37 [MCQ] A — UNIQUE treats two NULLs as not equal, so multiple NULLs are allowed in the usual interpretation. PRIMARY KEY still forbids NULL.

Q.38 [NAT] 228 — n*nodeptr + (n-1)*key < = block = > n < = floor((block+key)/(nodeptr+key)) = 228.

Q.39 [NAT] 1 — (x-min)/(max-min) = (40-0)/(40-0) = 1.0.

Q.40 [MCQ] A — Slice selects a single value of one dimension, reducing dimensionality by one.

Q.41 [NAT] 3 — Each participating entity contributes its key as an FK, and the three together form the relationship key.

Q.42 [MCQ] A — Candidate keys AD, BD, CD, DE. Highest NF is 3NF.

Q.43 [NAT] 6 — Many-to-one from R to S; join cardinality equals |R|.

Q.44 [NAT] 1 — NULL > 100 is UNKNOWN, not TRUE. Only id 3 qualifies.

Q.45 [MCQ] A — One PK; PK columns are NOT NULL; several UNIQUE are fine.

Q.46 [NAT] 227 — n*(key+recptr) + nodeptr < = block = > n = floor((block-nodeptr)/(key+recptr)) = 227.

Q.47 [NAT] 1.77 — z=(x-mean)/sigma. mean=16, sigma=13.564659966250536, z=1.7693034738587656.

Q.48 [MCQ] A — Dice selects ranges/sets on multiple dimensions.

Q.49 [MCQ] A — The standard options are one-table, two-table (subclasses only), or superclass plus subclass tables.

Q.50 [NAT] 1 — Candidate keys are A.

Q.51 [MCQ] A — Division is the RA operator for 'for all' / 'every required course'.

Q.52 [NAT] 3 — id 3 (200) plus ids 2 and 4 (NULL): 3 rows.

Q.53 [NAT] 1 — The PK rejects the duplicate (sid,cid).

Q.54 [NAT] 256 — n*nodeptr + (n-1)*key < = block = > n < = floor((block+key)/(nodeptr+key)) = 256.

Q.55 [NAT] 1 — (x-min)/(max-min) = (15-3)/(15-3) = 1.0.

Q.56 [MCQ] A — Finer grain along a hierarchy is drill-down on cube ClinicVisits.

Q.57 [MSQ] A;B;C;D — Recursive 1:N becomes a self-FK; M:N becomes a table.

Q.58 [MCQ] A — Candidate keys A. Highest NF is 2NF.

Q.59 [MCQ] A — The calculus formula is a selection.

Q.60 [MCQ] A — COUNT(*) counts rows (5). COUNT(col) skips NULLs: bonus has 2 NULLs so 3; dept has 1 NULL so 4.

Q.61 [MCQ] A — Deleting the referenced 1 leaves R.a=1 dangling.

Q.62 [NAT] 227 — n*(key+recptr) + nodeptr < = block = > n = floor((block-nodeptr)/(key+recptr)) = 227.

Q.63 [NAT] 1.41 — z=(x-mean)/sigma. mean=9, sigma=4.242640687119285, z=1.4142135623730951.

Q.64 [MCQ] A — Dropping a dimension by aggregation is a roll-up.

Q.65 [MCQ] A — Derived attributes are computed, not independently stored, to avoid update anomalies.

Q.66 [NAT] 4 — Candidate keys are AB, BC, BD, BE.

Q.67 [MSQ] A;D — Selection on an R-only attribute pushes to R. It cannot be applied to S.

Q.68 [NAT] 0 — In SQL, = NULL is UNKNOWN for every row; use IS NULL. Zero rows.

Q.69 [MSQ] A;B — DEFERRABLE constraints are checked at COMMIT, not after each statement.

Q.70 [NAT] 43 — n*nodeptr + (n-1)*key < = block = > n < = floor((block+key)/(nodeptr+key)) = 43.

Q.71 [MCQ] A — (50-10)/4=10.

Q.72 [MCQ] A — Finer grain along a hierarchy is drill-down on cube HarbourTonnage.

Q.73 [NAT] 2 — Composite size flattens into columns of Product; colors need a separate table.

Q.74 [MCQ] A — Candidate keys AB, BC, BD, BE. Highest NF is 3NF.

Q.75 [MCQ] A — t is determined by p.name; it is the names of Pune players. (If t is a Player tuple it is a selection.)

Q.76 [NAT] 2 — NOT UNKNOWN is UNKNOWN, so NULL bonuses drop. bonus > 100 is true only for 3, false for 1 and 5. NOT false keeps 1 and 5: 2 rows.

Q.77 [MCQ] A — CHECK fails only when the condition is FALSE. UNKNOWN (NULL) typically passes. 'pending' is FALSE.

Q.78 [NAT] 42 — n*(key+recptr) + nodeptr < = block = > n = floor((block-nodeptr)/(key+recptr)) = 42.

Q.79 [MCQ] A — 12/4=3 observations per bin.

Q.80 [MCQ] A — Dropping a dimension by aggregation is a roll-up.

Q.81 [MCQ] A — Partial on Employee, total on Department: manager_id is a unique NOT NULL FK on Department.

Q.82 [NAT] 3 — Candidate keys are AD, BD, CD.

Q.83 [MCQ] A — Safety requires every value of a result tuple to come from the active domain of the query.

Q.84 [NAT] 1 — Kochi has no player: one row. This is the anti-join.

Q.85 [NAT] 0 — Both child rows referenced pid=1 and are deleted. Wait, only those two exist so 0 remain. Yes.

Q.86 [NAT] 146 — n*nodeptr + (n-1)*key < = block = > n < = floor((block+key)/(nodeptr+key)) = 146.

Q.87 [MSQ] A;B;D — JPEG is lossy. Huffman, RLE and LZ are lossless.

Q.88 [MCQ] A — Finer grain along a hierarchy is drill-down on cube GridLoad.

Q.89 [MCQ] A — Weak-entity keys are owner key plus partial key.

Q.90 [MCQ] A — Candidate keys AD, BD, CD. Highest NF is 3NF.

Q.91 [NAT] 12 — 4*3=12 when the join key is constant.

Q.92 [NAT] 4 — Goa matches 2 players, Pune 1, Kochi 1 padded: 4 rows.

Q.93 [NAT] 2 — The delete is rejected; both parent rows stay.

Q.94 [NAT] 145 — n*(key+recptr) + nodeptr < = block = > n = floor((block-nodeptr)/(key+recptr)) = 145.

Q.95 [MCQ] A — Simple random sample: 20/200=0.1.

Q.96 [MCQ] A — Dropping a dimension by aggregation is a roll-up.

Q.97 [MSQ] A;B — 1:N and composite attributes do not by themselves add a table.

Q.98 [NAT] 1 — Candidate keys are A.

Q.99 [MCQ] A — Projection and union commute: both yield the set of A-values appearing in either.

Q.100 [NAT] 2 — Only sid 8 takes both CS courses; sid 8 appears twice in Enroll, so 2 rows without DISTINCT.

Q.101 [MCQ] A — The assertion is that the difference is empty.

Q.102 [NAT] 64 — n*nodeptr + (n-1)*key < = block = > n < = floor((block+key)/(nodeptr+key)) = 64.

Q.103 [MCQ] A — Each draw includes it with p=1/200; 20 draws = > 20/200=0.1.

Q.104 [MCQ] A — Finer grain along a hierarchy is drill-down on cube CourseEnrol.

Q.105 [NAT] 5 — Four entity tables plus WorksOn. Owns and ForCar become FKs.

Q.106 [MCQ] A — Candidate keys A. Highest NF is BCNF.

Q.107 [MCQ] A — An A-value may occur in R and in S with different B's, so it survives the projected intersection but not the intersection of the original tuples.

Q.108 [NAT] 3 — NULLs form a single group. Groups: 1, NULL, 2 - > 3.

Q.109 [MSQ] A;B;C — Block size is physical design, not an integrity constraint.

Q.110 [NAT] 56 — n*(key+recptr) + nodeptr < = block = > n = floor((block-nodeptr)/(key+recptr)) = 56.

Q.111 [MCQ] A — A single extreme value changes min or max and rescales everyone.

Q.112 [MCQ] A — Dropping a dimension by aggregation is a roll-up.

Q.113 [MCQ] A — Aggregation is the ER device for relationships on relationships.

Q.114 [NAT] 2 — Candidate keys are AB, BC.

Q.115 [MSQ] A;B;D — Join size can be 0, which is not at least |R|+|S|.

Q.116 [NAT] 2 — Goa 3+1=4, Pune 2+0=2, Kochi 4. Two homes (Goa, Kochi).

Q.117 [MSQ] A;B;C — D is the classic violation. C is OK if the new pid exists.

Q.118 [NAT] 186 — n*nodeptr + (n-1)*key < = block = > n < = floor((block+key)/(nodeptr+key)) = 186.

Q.119 [MCQ] A — Standard score. Min-max is the other common linear rescaling.

Q.120 [MCQ] A — Finer grain along a hierarchy is drill-down on cube MatchGoals.

Q.121 [MSQ] A;C;D — 1:N is absorbed as a FK on the N-side. A relationship table is optional, not mandatory.

Q.122 [MCQ] A — Candidate keys AB, BC. Highest NF is 3NF.

Q.123 [MCQ] A — Set operators require identical schemas, hence rename.

Q.124 [NAT] 1 — > = ALL is the maximum; only 4 (Kochi vs Goa).

Q.125 [MCQ] A — Entity integrity is the no-NULL-in-PK rule.

Q.126 [NAT] 185 — n*(key+recptr) + nodeptr < = block = > n = floor((block-nodeptr)/(key+recptr)) = 185.

Q.127 [MSQ] A;B;C — IDs are nominal identifiers, not magnitudes.

Q.128 [MCQ] A — Dropping a dimension by aggregation is a roll-up.

Q.129 [MCQ] A — A self-relationship becomes a self-FK (plus uniqueness for 1:1).

Q.130 [NAT] 3 — Candidate keys are AB, BC, BE.

Q.131 [NAT] 5 — Projection is a set of distinct homes.

Q.132 [NAT] 2 — Ben's reports are Dee and Eve: 2.

Q.133 [NAT] 3 — The second 1 fails; 1,2,3 remain.

Q.134 [MCQ] A — Dense means every record is referenced. A sparse index points at blocks.

Q.135 [NAT] 7 — Range is -8..7.

Q.136 [MCQ] A — Finer grain along a hierarchy is drill-down on cube SensorAvg.

Q.137 [MCQ] A — Total 1:1 can even merge the entities; otherwise a mandatory unique FK.

Q.138 [MCQ] A — Candidate keys AB, BC, BE. Highest NF is 1NF.

Q.139 [MCQ] A — Universal quantification over CS courses is division.

Q.140 [MCQ] A — UNION distinct: 1,2,3,4 (4). UNION ALL concatenates 4+3=7.

Q.141 [MCQ] A — FK matching is tuple-wise on the whole key.

Q.142 [MSQ] A;B;D — Hash indexes are for equality, not ranges.

Q.143 [MCQ] A — CHAR pads to the declared width; VARCHAR is variable-length.

Q.144 [MCQ] A — Dropping a dimension by aggregation is a roll-up.

Q.145 [MCQ] A — Relationship attributes become columns of the relationship relation.

Q.146 [NAT] 1 — Candidate keys are AB.

Q.147 [MCQ] A — All 4 R-tuples appear: 2 matched plus 2 padded with nulls.

Q.148 [MSQ] A;C — IS NULL is TRUE for NULL. Comparisons with NULL are UNKNOWN; NOT UNKNOWN is UNKNOWN.

Q.149 [MCQ] A — Clustered means data records are stored in the search-key order.

Q.150 [MCQ] A — Binning turns a continuous feature into a categorical one.

Q.151 [MCQ] A — Finer grain along a hierarchy is drill-down on cube PayrollCost.

Q.152 [MCQ] A — The 'subclasses only' mapping duplicates the superclass attributes and complicates unified queries.

Q.153 [MCQ] A — Candidate keys AB. Highest NF is 2NF.

Q.154 [NAT] 5 — 2 matched + 2 left-only + 1 right-only = 5.

Q.155 [MSQ] A;B;C — AVG ignores NULLs too; it does not treat them as 0.

Q.156 [NAT] 1000 — 10000/10=1000.

Q.157 [NAT] 30 — 0.3*100=30.

Q.158 [MCQ] A — Dropping a dimension by aggregation is a roll-up.

Q.159 [NAT] 2 — Candidate keys are AB, BC.

Q.160 [MSQ] A;B;C — Difference is not commutative.

Q.161 [NAT] 2 — Nila min 10 fails; Orca 25 and Pinta 60 pass: 2.

Q.162 [NAT] 1000 — The whole file is scanned.

Q.163 [MCQ] A — Three runs: 5,5,3.

Q.164 [MCQ] A — Finer grain along a hierarchy is drill-down on cube LibraryLoans.

Q.165 [MCQ] A — Candidate keys AB, BC. Highest NF is 3NF.

Q.166 [NAT] 2 — Projection is a set of size 2.

Q.167 [NAT] 2 — ICU avg 650 so pid 1; GEN avg 250 so pid 3; DAY avg 50 none. 2 rows.

Q.168 [NAT] 10 — ceil(log2 1000)=10.

Q.169 [MCQ] A — That is the definition of z-score / standard-score scaling.

Q.170 [MCQ] A — Dropping a dimension by aggregation is a roll-up.

Q.171 [NAT] 3 — Candidate keys are AC, AD, AE.

Q.172 [MCQ] A — Join and division are derived. Outer join is a later extension.

Q.173 [NAT] 2 — Ada (3) and DB (1); C=0 and ML=5 out: 2.

Q.174 [MCQ] A — floor(4096/200)=20. Unspanned wastes the leftover 96 bytes.

Q.175 [NAT] 0 — min is 1, so (1-1)/(13-1)=0.

Q.176 [MSQ] A;B;C — date_id is a dimension foreign key, not a measure.

Q.177 [MCQ] A — Candidate keys AC, AD, AE. Highest NF is 1NF.

Q.178 [MSQ] A;B;C — Selections do push into products: sigma on R-only attributes of a product goes to R.

Q.179 [NAT] 2 — Two courses DB and ML.

Q.180 [MCQ] A — Global depth d = > 2^d directory entries.

Q.181 [MCQ] A — Sampling is an approximation technique for warehouse queries.

Q.182 [MCQ] A — Time hierarchy day < week < month < year supports roll-up.

Q.183 [MCQ] A — A+ = ABC, so A is a key. B+ = BC misses A; C+ = C.

Q.184 [NAT] 2 — Two distinct courses; sids 1 and 2 have 2 rows, sid 3 has 1. Answer 2.

Q.185 [MSQ] A;B;D — Internal nodes store only keys and child pointers; data pointers are at the leaves.

Q.186 [MSQ] A;B;C — Width is (50-10)/4=10, not 20.

Q.187 [MCQ] A — The top of the hierarchy is the coarsest aggregation.

Q.188 [MSQ] A;C;D — Nontrivial FDs B- > C and C- > D have left sides that are not superkeys, so not BCNF/3NF. The chain decomposition is lossless.

Q.189 [MCQ] A — IN is TRUE if any equality is TRUE; UNKNOWN others do not cancel a TRUE match.

Q.190 [NAT] 100 — 5^2=25 leaves * 4 keys = 100.

Q.191 [MCQ] A — Precision 5 scale 2: up to 999.99 (3+2). 123.45 fits. 1234.5 does not. 0.0001 would round at 2 decimals.

Q.192 [MSQ] A;B;C — Stars routinely have many dimensions (date, store, product, promo, ...).

Q.193 [NAT] 4 — Superkeys are all supersets of {A}: A, AB, AC, ABC. Four of them.

Q.194 [MCQ] A — NOT IN is NOT(OR of equalities). A NULL equality is UNKNOWN, so the OR is UNKNOWN if no TRUE, and NOT UNKNOWN is UNKNOWN.

Q.195 [MCQ] A — One hash computation plus one bucket access.

Q.196 [MCQ] A — Optimal lengths 1,2,3,3: 0.5*1+0.25*2+0.125*3+0.125*3=1.75.

Q.197 [MCQ] A — Pivot/rotate changes which dimensions are on which axis.

Q.198 [MCQ] A — Transitivity is the third Armstrong axiom.

Q.199 [NAT] 2 — Dock all 3.2; Yard two values; Gate one value. Two locs.

Q.200 [MCQ] A — One bitmap per distinct value.

Q.201 [NAT] 5 — 100/20=5.

Q.202 [MCQ] A — Transform is the cleansing / conforming / surrogate-key stage.

Q.203 [MCQ] A — The standard test: common attributes form a superkey of at least one of the two schemas.

Q.204 [NAT] 2 — Sids 1 and 2: two rows.

Q.205 [NAT] 200 — 8000/40=200 data blocks, one entry each.

Q.206 [MCQ] A — DATE is calendar day; TIMESTAMP is date+time.

Q.207 [MCQ] A — Conformed dimensions are reusable across facts (Kimball).

Q.208 [MSQ] A;B;C — {B} determines neither AB nor CB. Each FD is local, but the join is lossy.

Q.209 [NAT] 1 — sid1: DA 80 vs GA 70 no; sid2: 80 vs 90 yes; sid3 has no GA. 1 row.

Q.210 [MCQ] A — Classic ISAM does not rebalance; overflow chains degrade it.

Q.211 [NAT] 1500 — 10*5*30=1500.

Q.212 [NAT] 5 — Nonempty Ysubseteq X: (A- > A), (B- > B), (AB- > A), (AB- > B), (AB- > AB). Five trivial FDs.

Q.213 [MSQ] A;B;D — COUNT(*) over the whole table includes NULL-bonus rows.

Q.214 [MSQ] A;B;C — Inserting into a sorted file while keeping order is not an append (unless the key is maximal).

Q.215 [NAT] 300 — 10*30=300.

Q.216 [MCQ] A — AB+ = ABC. C+ = CA, needs B so BC+ = BCA. AB and BC are keys. C is not.

Q.217 [NAT] 0 — NULL=NULL is UNKNOWN; AND with TRUE is UNKNOWN; WHERE keeps only TRUE.

Q.218 [MSQ] A;B — Hash wins pure equality; B+ wins ranges and ordered scans. The question asks when B+ is chosen over hash: A and B. C argues the opposite.

Q.219 [MCQ] A — MOLAP: multidimensional arrays; ROLAP: relational star/snowflake plus SQL.

Q.220 [MCQ] A — C- > A has left side not a superkey (C is not a key) but A is prime, so 3NF not BCNF.

Q.221 [NAT] 1 — TRUE OR UNKNOWN is TRUE.

Q.222 [NAT] 6 — floor(2048/300)=6.

Q.223 [MSQ] A;B;C — Normal-form decomposition is a relational design step, not an OLAP navigation operator.

Q.224 [NAT] 5 — CD - > E - > A - > B, so CDEAB, all five.

Q.225 [MCQ] A — AVG ignores NULL: (1+2)/2=1.5.

Q.226 [NAT] 1334 — ceil(8000/6)=1334.

Q.227 [MCQ] A — Degenerate dimensions are identifiers without a dimension table.

Q.228 [MSQ] A;B;C — AD+ = AD+BC+E = ABCDE; ED+ similarly; CD+ = CDE+A+BC. BD+ = BD only.

Q.229 [NAT] 2 — DISTINCT a ignores NULL and collapses the two 1s: {1,2}.

Q.230 [MCQ] A — Covering / index-only scan: all needed columns sit in the index.

Q.231 [MCQ] A — Type 2 versions rows. Type 1 overwrites.

Q.232 [MCQ] A — 2NF bans partial dependence of nonprimes. BCNF is 'every determinant is a superkey'. Transitive is 3NF.

Q.233 [NAT] 3 — COUNT(a) skips NULL but not duplicate 1s: three non-null rows.

Q.234 [MCQ] A — Primary/clustered indexes follow the file's sort order.

Q.235 [NAT] 17 — Each fact row is one attendance event.

Q.236 [MCQ] A — BCNF is the superkey-determinant condition. 3NF allows prime-right-hand sides.

Q.237 [MCQ] A — Natural join equalises and keeps one copy of B.

Q.238 [MSQ] A;C — Balance makes height logarithmic. Leaf links do not make height linear. B is self-contradictory.

Q.239 [MCQ] A — Semi-additive measures need a time snapshot policy.

Q.240 [NAT] 4 — A determines the whole chain ABCD.

Q.241 [MSQ] A;B;C — DISTINCT applies to the select list, not as SELECT ... DISTINCT COUNT with a leftover sid without GROUP BY.

Q.242 [NAT] 100 — 5000/50=100.

Q.243 [MSQ] A;B;C — Year to Month is finer, hence drill-down not roll-up.

Q.244 [MCQ] A — Lossless and dependency-preserving are independent goals; BCNF cannot always do both.

Q.245 [MCQ] A — WHERE is pre-aggregation; HAVING is post-aggregation.

Q.246 [MCQ] A — Low-cardinality bits + AND/OR of bitmaps fit OLAP predicates. They are costly to maintain under heavy OLTP updates.

Q.247 [MCQ] A — Checking a cross-piece FD requires joining the pieces.

Q.248 [NAT] 4 — SUM skips NULL: 1+3=4.

Q.249 [MCQ] A — Marts are departmental; the EDW is enterprise-wide.

Q.250 [MSQ] A;B;C — AB+ = ABCD; BC+ : C- > D- > A so BCAD; BD+ : D- > A so ABD then AB- > C. CD+ = CDA misses B.

Q.251 [MCQ] A — Correlation binds S.k to the outer R.k.

Q.252 [NAT] 4 — One dimension becomes 4 times longer; 8*4*96 / (8*4*24) = 4.

Q.253 [MCQ] A — Reflexivity: if Y ⊆ X then X- > Y always holds.

Q.254 [MCQ] A — Hybrid: relational detail + multidimensional aggregates.

Q.255 [NAT] 1 — Only A is a key.

Q.256 [MSQ] A;B;C — Finer time is drill-down, not roll-up.

Q.257 [MCQ] A — Classic 2NF-not-3NF chain with a simple key.

Q.258 [MCQ] A — Junk dimensions collect miscellaneous flags.

Q.259 [NAT] 3 — A- > D follows by transitivity from the chain.

Q.260 [MSQ] A;B;C — Dimensions are not weak entities of the fact; they are conformed master data.

Q.261 [NAT] 3 — Three climbs: Store to City to State to Country.

Q.262 [MCQ] A — on_hand is a level, not a flow; summing over time double-counts stock.

Q.263 [MCQ] A — Warehouse performance uses pre-aggregated fact tables.

UNIT 6 Machine Learning · 275 questions

MCQ negative -1/3 or -2/3. MSQ / NAT no negative. Work first, then the solutions section.

Q.1 NAT · 2 marks · SLR
Chai stall on FC Road records cups x versus revenue in 10 Rs y as (1,2), (2,3), (3,5), (4,6). Fit simple OLS with intercept. The fitted slope equals ____.
NAT · enter the numerical value

Q.2 MCQ · 1 mark · MLR
Design matrix X is n x (p+1) including a column of ones. Unique OLS exists iff
(A) n > p+1 only
(B) X has full column rank p+1
(C) y is orthogonal to col(X)
(D) features are standard-normal

Q.3 MCQ · 1 mark · Ridge
Ridge (intercept unpenalised, features centered) solves
(A) min ||y-Xb||_2^2 + lambda ||b||_2^2 for lambda > 0
(B) min ||y-Xb||_1 + lambda ||b||_2
(C) min ||y-Xb||_2^2 + lambda ||b||_1
(D) min ||b||_2 s.t. ||y-Xb||=0

Q.4 MCQ · 1 mark · Logistic
Logistic regression models P(Y=1|x) as
(A) w^T x (unbounded)
(B) sigmoid(w^T x+b)=1/(1+exp(-(w^T x+b)))
(C) the N(w^T x,1) density
(D) a hard 0-1 threshold of w^T x

Q.5 MCQ · 2 marks · kNN
Train: A(0,0) class0, B(0,2) class0, C(2,0) class1, D(2,2) class1. Query q=(0,1). Euclidean 1-NN predicts
(A) class 0, because A and B are both at distance 1 while C,D are farther
(B) class 1
(C) class 0 only if Manhattan is used
(D) a four-way tie

Q.6 MCQ · 1 mark · NaiveBayes
Bernoulli NB on K classes and one binary feature, counting K priors and K Bernoulli weights, has parameter count
(A) K
(B) 2K
(C) 2K+1
(D) K^2

Q.7 MCQ · 1 mark · LDA
`Fisher's two-class LDA seeks a direction w maximising`
(A) w^T S_W w / w^T S_B w
(B) w^T S_B w / w^T S_W w
(C) ||w||_1
(D) the Gini of the projected labels

Q.8 MCQ · 1 mark · SVM
Hard-margin linear SVM on linearly separable data minimises
(A) (1/2)||w||^2 s.t. y_i(w?x_i+b) > =1
(B) hinge+L1 with C=0
(C) ||w||_1 s.t. y_i(w?x_i+b) > =0
(D) the Gini of slack

Q.9 NAT · 1 mark · DecisionTree
A binary node has 2 positives and 2 negatives. Gini 1-sum p_c^2 equals ____.
NAT · enter the numerical value

Q.10 MCQ · 1 mark · BiasVariance
For k-NN classification, decreasing k from 51 to 1 typically
(A) raises bias and lowers variance
(B) lowers bias and raises variance
(C) does not affect the decomposition
(D) makes the classifier linear

Q.11 MCQ · 1 mark · LOOCV
LOOCV on n distinct training points fits how many models?
(A) 1
(B) k
(C) n
(D) n-1

Q.12 NAT · 1 mark · KFold
n=200, 10-fold CV. Each training fold has how many points (equal folds)?
NAT · enter the numerical value

Q.13 MCQ · 2 marks · MLP
A fully connected net input-4, hidden-5, output-3, biases on hidden and output, no extra layers. Parameter count is
(A) 4*5+5*3
(B) 4*5+5 + 5*3+3
(C) 5*3
(D) 4+5+3

Q.14 MCQ · 1 mark · KMeans
k-means with Euclidean distance updates each centroid to
(A) the coordinate-wise mean of its assigned points
(B) a randomly chosen training label
(C) the median of all features globally
(D) the first principal component

Q.15 MCQ · 1 mark · KMedoid
A medoid of a cluster is
(A) any mean vector, possibly off the data
(B) an actual data point that minimises total distance to other members
(C) the first principal component
(D) a support vector of an SVM

Q.16 MCQ · 1 mark · HAC
Agglomerative (bottom-up) clustering starts with
(A) one cluster containing every point
(B) each point in its own cluster, then merges
(C) a random SVM
(D) PCA scores of labels

Q.17 MCQ · 1 mark · PCA
Principal components of a centered data matrix are
(A) orthogonal directions of successive maximal variance (eigenvectors of the covariance)
(B) the class-mean differences of LDA
(C) support vectors
(D) Gini-maximising splits

Q.18 MCQ · 1 mark · MLMatch

Task: separate two linearly separable classes with a maximum-margin hyperplane. Algorithm:

(A) hard-margin SVM
(B) k-means with k=2
(C) PCA to 1-D then threshold at random
(D) single-linkage HAC

Q.19 NAT · 2 marks · SLR

Using the OLS line fitted on Chai stall on FC Road pairs (1,2), (2,3), (3,5), (4,6) (cups vs revenue in 10 Rs), the prediction at x=5 is ____.

NAT · enter the numerical value

Q.20 MSQ · 2 marks · MLR

In MLR y=X beta+e with full-column-rank OLS, which are ALWAYS true?

(A) The residual vector is orthogonal to every column of X
(B) H=X(X^T X)^{-1}X^T is the orthogonal projector onto col(X)
(C) Unadjusted R^2 cannot decrease when a linearly independent column is added
(D) Every coefficient stays unbiased even if a relevant variable is omitted

Q.21 MSQ · 2 marks · Ridge

`Relative to OLS, increasing ridge lambda typically`

(A) raises bias of the coefficient vector
(B) lowers variance of the coefficient vector
(C) drives coefficients toward 0
(D) makes X^T X + lambda I more ill-conditioned

Q.22 NAT · 1 mark · Logistic

Fitted sigmoid outputs f(x)=0.8. The derivative f(x)(1-f(x)) equals ____.

NAT · enter the numerical value

Q.23 NAT · 2 marks · kNN

Points P(0,0) y=1, Q(4,0) y=2, R(0,3) y=2. Query (1,1). Euclidean 1-NN label is ____.

NAT · enter the numerical value

Q.24 NAT · 1 mark · NaiveBayes

Gaussian NB, K classes, one scalar feature, class-specific means, ONE shared variance, and K priors. Parameter count is ____.

NAT · enter the numerical value

Q.25 MCQ · 2 marks · LDA

`Two-class means m1,m2 and invertible pooled S_W. An optimal Fisher direction is`

(A) S_W (m1+m2)
(B) S_W^{-1} (m1-m2)
(C) S_B^{-1} (m1+m2)
(D) PC1 of the pooled unlabelled data

Q.26 NAT · 1 mark · SVM

Hyperplane w=(3,4), b=0. The Euclidean ||w|| is ____.

NAT · enter the numerical value

Q.27 NAT · 1 mark · DecisionTree

A pure node with 5 positives and 0 negatives has Gini ____.

NAT · enter the numerical value

Q.28 MCQ · 1 mark · BiasVariance

`Raising the ridge penalty lambda typically`

(A) lowers bias, raises variance
(B) raises bias, lowers variance
(C) raises both
(D) has no bias effect

Q.29 NAT · 2 marks · LOOCV

1-NN LOOCV on (0,class0),(1,class0),(10,class1),(11,class1). Number of misclassified left-out points is ____.

NAT · enter the numerical value

Q.30 NAT · 1 mark · KFold

n=30, 5-fold. Number of models fitted is ____.

NAT · enter the numerical value

Q.31 NAT · 2 marks · MLP

Input-2, hidden-2, output-1, with biases on hidden and output. Number of scalar parameters is ____.

NAT · enter the numerical value

Q.32 NAT · 1 mark · KMeans

Points 1,2,3,10 on a line, k=2, initial centroids 1 and 10. After the first assignment, the left cluster size is ____.

NAT · enter the numerical value

Q.33 MCQ · 1 mark · KMedoid

k-medoid (PAM) is often preferred to k-means when

(A) one needs robustness to outliers and a prototype that is a real observation
(B) one needs a differentiable mean for SGD
(C) labels are continuous
(D) PCA must be avoided

Q.34 MCQ · 1 mark · HAC

Divisive (top-down) clustering starts with

(A) n singletons
(B) one cluster of all points, then recursively splits
(C) k random medoids only
(D) a logistic boundary

Q.35 MSQ · 2 marks · PCA

Which are TRUE of PCA?

(A) The principal axes are pairwise orthogonal
(B) The first PC maximises projected variance among unit vectors
(C) PCA uses class labels
(D) The scores on distinct PCs are uncorrelated (for the empirical covariance)

Q.36 MCQ · 1 mark · MLMatch

Task: compress unlabeled 50-D sensor readings to 3 orthogonal directions of maximal variance.

(A) PCA
(B) logistic regression
(C) LDA with 3 classes only
(D) decision tree Gini

Q.37 NAT · 2 marks · SLR

Wheat plots in Nanded records kg urea x versus quintals y as (0,2), (1,3), (2,5), (3,6). Fit simple OLS with intercept. The fitted slope equals ____.

NAT · enter the numerical value

Q.38 NAT · 1 mark · MLR
A house-price model uses intercept, area, and a binary near-metro flag. n=40. Columns of X: ____.
NAT · enter the numerical value

Q.39 NAT · 2 marks · Ridge
1-D no intercept: X^T X=4, X^T y=8, lambda=4. Ridge coefficient is ____.
NAT · enter the numerical value

Q.40 NAT · 1 mark · Logistic
Sigmoid output 0.2. Then f(1-f) equals ____.
NAT · enter the numerical value

Q.41 MCQ · 1 mark · kNN
On P(0,0):1, Q(4,0):2, R(0,3):2, query (1,1), 3-NN majority label is
(A) 1 because P wins
(B) 2 because Q and R out-vote P
(C) undefined because k exceeds n
(D) always equal to 1-NN

Q.42 NAT · 2 marks · NaiveBayes
Gaussian NB with 3 classes and 4 features, class-specific means and variances plus 3 priors. Parameter count is ____.
NAT · enter the numerical value

Q.43 MSQ · 2 marks · LDA
Assumptions of classical Gaussian LDA (two classes) include
(A) class-conditional Gaussians
(B) a shared covariance matrix
(C) equal class priors (mandatory)
(D) a linear log-posterior-odds boundary

Q.44 NAT · 1 mark · SVM
Canonical hard-margin SVM with ||w||=4. Margin width 2/||w|| equals ____.
NAT · enter the numerical value

Q.45 NAT · 1 mark · DecisionTree
Node with 3 yes and 1 no. Gini equals ____.
NAT · enter the numerical value

Q.46 MSQ · 2 marks · BiasVariance
Which moves sit on the high-variance side?
(A) Degree-9 polynomial through 10 noisy points
(B) 1-NN instead of 25-NN
(C) Unpruned deep trees
(D) Intercept-only mean predictor on a clearly linear trend

Q.47 NAT · 2 marks · LOOCV
1-NN LOOCV on 1-D pairs (0,0),(1,0),(2,1), ties broken toward smaller x. Number of LOO errors is ____.
NAT · enter the numerical value

Q.48 MCQ · 2 marks · KFold
2-fold CV, y=(0,0,4,8), folds {0,0} vs {4,8}, intercept-only. The four predictions are
(A) 6,6,0,0
(B) 0,0,4,8
(C) 3,3,3,3
(D) 2,2,2,2

Q.49 NAT · 1 mark · MLP
Sigmoid unit outputs 0.8. Local derivative sigma'(z)=sigma(1-sigma) equals ____.
NAT · enter the numerical value

Q.50 NAT · 1 mark · KMeans
After that assignment {1,2,3} and {10}, the new left centroid is ____.
NAT · enter the numerical value

Q.51 NAT · 2 marks · KMedoid
1-D points 0,1,100. The 1-medoid (min total L1 distance) is the median point, i.e. ____.
NAT · enter the numerical value

Q.52 MCQ · 1 mark · HAC
Single linkage uses, as the distance between clusters A and B,
(A) the maximum pairwise distance between a point of A and a point of B
(B) the minimum pairwise distance between a point of A and a point of B
(C) the distance between centroids only, never points
(D) Gini impurity

Q.53 NAT · 1 mark · PCA
A 2-D centered cloud has covariance diag(4,1). Variance explained by PC1 is ____.
NAT · enter the numerical value

Q.54 MCQ · 1 mark · MLMatch
Task: classify email using word presence under a class-conditional independence model.
(A) naive Bayes
(B) ridge OLS
(C) k-medoid
(D) PCA

Q.55 NAT · 1 mark · SLR
For Wheat plots in Nanded data (0,2), (1,3), (2,5), (3,6) of kg urea against quintals, the OLS intercept is ____.
NAT · enter the numerical value

Q.56 MCQ · 2 marks · MLR
Features satisfy x2=2 x1 on every training row. OLS of y on (1,x1,x2)
(A) has singular X^T X so the usual unique solution does not exist
(B) forces the x2 slope to 2
(C) has VIF of x1 equal to 0
(D) makes predictions undefined for every y

Q.57 NAT · 1 mark · Ridge
Same 1-D Gram 4 and X^T y=8 but lambda=0. The coefficient is ____.
NAT · enter the numerical value

Q.58 NAT · 1 mark · Logistic
Logistic score z=2x-2. At x=1 the predicted probability is ____.
NAT · enter the numerical value

Q.59 MSQ · 2 marks · kNN
Which statements about k-NN classification are TRUE?
(A) Training error of 1-NN is 0 if no two equal-x points have different labels
(B) As k- > n the prediction approaches the global majority class
(C) 1-NN typically has lower bias and higher variance than a large k
(D) k-NN needs an explicit parametric density for each class

Q.60 MCQ · 2 marks · NaiveBayes
Spam: 3/5 docs contain 'buy'; ham: 1/5 contain 'buy'; P(spam)=1/2. A mail with 'buy' has P(spam|buy) equal to
(A) 1/2
(B) 3/4
(C) 3/5
(D) 1/4

Q.61 NAT · 1 mark · LDA
Class means 0 and 4 (1-D), shared variance 1, equal priors. LDA threshold is ____.
NAT · enter the numerical value

Q.62 MCQ · 2 marks · SVM
Points (1,1)+,(1,-1)+,(-1,1)-,(-1,-1)-. A hard-margin SVM using the separator x=0 has how many support vectors?
(A) 1
(B) 2
(C) 4
(D) 0

Q.63 MCQ · 2 marks · DecisionTree
Rows (colour,label): (red,+),(red,+),(blue,+),(blue,-). Split on colour. Weighted child Gini is
(A) 0
(B) 0.25
(C) 0.50
(D) 1

Q.64 MCQ · 1 mark · BiasVariance
Expected test MSE of a regression estimator decomposes (squared loss) as
(A) noise + bias^2 + variance
(B) only variance
(C) only bias
(D) hinge + Gini

Q.65 MCQ · 2 marks · LOOCV
OLS PRESS formula e_i/(1-H_ii) shows that LOOCV residuals
(A) can be computed from a single OLS fit and the leverages
(B) cannot be computed without n separate factorisations even in theory
(C) equal training SSE
(D) are undefined if an intercept is present

Q.66 MSQ · 2 marks · KFold
Which are TRUE of k-fold CV?
(A) k=n recovers LOOCV
(B) k=2 has more bias (smaller trains) than k=10 typically
(C) The k test folds partition the n points in standard CV
(D) k-fold requires a separate extra test set of size k

Q.67 NAT · 1 mark · MLP
ReLU(x)=max(0,x). ReLU(-3) equals ____.
NAT · enter the numerical value

Q.68 MCQ · 2 marks · KMeans
Four shops A(0,0), B(0,6), C(8,0), D(8,6). If A and D are forced into the SAME Euclidean 2-means cluster at a fixed point of k-means, then
(A) B and C may freely sit in the other cluster, because k-means cells need not be convex
(B) the cluster is a Voronoi cell of its centroid, hence convex, so the rectangle's interior is constrained; in particular B or C cannot form a separated pair skipping the diagonal without splitting A-D
(C) k-means ignores coordinates
(D) the objective becomes Gini

Q.69 MSQ · 2 marks · KMedoid
Which are TRUE?
(A) k-medoid cluster centres are members of the dataset
(B) k-means centres need not be members of the dataset
(C) A single swap step in PAM evaluates replacing a medoid by a non-medoid
(D) k-medoid requires class labels

Q.70 MCQ · 1 mark · HAC
Complete linkage uses
(A) min pairwise distance
(B) max pairwise distance
(C) mean of labels
(D) hinge loss

Q.71 NAT · 1 mark · PCA
Same covariance diag(4,1). Fraction of variance on PC1 is ____.
NAT · enter the numerical value

Q.72 MCQ · 1 mark · MLMatch
Task: shrink OLS coefficients to handle multicollinearity without forcing zeros.
(A) ridge
(B) 1-NN
(C) single-linkage
(D) hard-margin SVM

Q.73 NAT · 2 marks · SLR
On Pune-Mumbai express buses the OLS residual sum of squares for (4,8), (5,11), (6,13), (7,16) (speed in 10 km/h, stopping metres) equals ____.
NAT · enter the numerical value

Q.74 NAT · 1 mark · MLR
Fitted MLR yhat=3+2 x1 - x2. At (x1,x2)=(4,5) the prediction is ____.
NAT · enter the numerical value

Q.75 MCQ · 1 mark · Ridge

`As lambda - > infinity with fixed centered design, ridge slopes`

(A) diverge
(B) approach the OLS vector
(C) approach the zero vector
(D) approach PCA loadings

Q.76 MCQ · 1 mark · Logistic

log P/(1-P)=3-6x. The P=0.5 boundary occurs at x equal to

(A) 0
(B) 0.5
(C) 2
(D) 3

Q.77 NAT · 1 mark · kNN

1-D train x=0,1,2,3 labels 0,0,1,1. Query x=1.6, k=1 Euclidean. Predicted label is ____.

NAT · enter the numerical value

Q.78 NAT · 2 marks · NaiveBayes

P(C)=0.4, P(X=1|C)=0.5, P(X=1|D)=0.1, P(D)=0.6. Then P(C|X=1) rounded to 2 decimals is ____.

NAT · enter the numerical value

Q.79 NAT · 1 mark · LDA

`1-D two-class means 0 and 2, equal priors and equal variance. LDA threshold is ____.`

NAT · enter the numerical value

Q.80 MSQ · 2 marks · SVM

Which points can be support vectors in a soft-margin linear SVM?

(A) Points on the margin planes
(B) Misclassified points
(C) Points inside the margin strip but correctly signed
(D) A far, correctly classified point with dual alpha exactly 0

Q.81 NAT · 1 mark · DecisionTree

Parent Gini 0.5, weighted children Gini 0.25. Gini gain is ____.

NAT · enter the numerical value

Q.82 MCQ · 1 mark · BiasVariance

A linear fit to a sinusoid is a prototype of

(A) high bias
(B) zero approximation error
(C) infinite variance always
(D) perfect test R^2

Q.83 NAT · 1 mark · LOOCV

n=5 points, 1-NN. Each LOO prediction uses how many neighbours?

NAT · enter the numerical value

Q.84 NAT · 1 mark · KFold

5-fold CV MSE on folds 2,4,3,5,6. Mean CV estimate is ____.

NAT · enter the numerical value

Q.85 NAT · 1 mark · MLP

ReLU(2.5) equals ____.

NAT · enter the numerical value

Q.86 MCQ · 2 marks · KMeans

Points on a line P=1, Q=2, R=3, S=10. If P and S lie in the same k-means cluster (k=2) after convergence, which MUST be true?

(A) Q and R lie in that same cluster, because clusters are intervals (convex) on the line
(B) Q must be a singleton
(C) R must be a medoid
(D) k-means is undefined on a line

Q.87 NAT · 2 marks · KMedoid

Points A,B,C with pairwise distances AB=2, AC=3, BC=4. The 1-medoid (min total distance to the others) is the point with smallest sum, namely A with sum ____.

NAT · enter the numerical value

Q.88 NAT · 2 marks · HAC

HAC on Oak(0,0), Pine(1,0), Elm(0,2), Teak(4,0) with Manhattan distance. The first single-linkage merge height is ____.

NAT · enter the numerical value

Q.89 MCQ · 2 marks · PCA

Failing to center the data before PCA

(A) can make the leading component point toward the mean rather than a direction of spread
(B) is required for orthogonality
(C) turns PCA into SVM
(D) zeros all eigenvalues

Q.90 MCQ · 1 mark · MLMatch

Task: produce a binary tree of axis-aligned splits on a tiny colour/label table.

(A) decision tree
(B) MLP with linear hidden units only
(C) k-means
(D) PCA

Q.91 NAT · 2 marks · SLR

Using the OLS line fitted on DiyaMart click log pairs (1,2), (3,4), (5,7), (7,9) (thousand impressions vs clicks), the prediction at x=8 is ____.

NAT · enter the numerical value

Q.92 MCQ · 2 marks · MLR

Two centered features are sample-orthogonal. Then each MLR slope

(A) equals the SLR slope of y on that feature (with intercept)
(B) must be 0
(C) equals the inverse SLR slope
(D) cannot be estimated

Q.93 MCQ · 2 marks · Ridge

`Why add lambda I to X^T X in ridge?`

(A) To enforce integer coefficients
(B) To guarantee a unique finite solution even when X is rank-deficient, and to shrink
(C) To turn the loss into hinge loss
(D) To make residuals heteroscedastic on purpose

Q.94 MSQ · 2 marks · Logistic

Which are TRUE of binary logistic regression trained by MLE?

(A) Negative log-likelihood is convex in (w,b)
(B) With linearly separable data and no regularisation, ||w|| may diverge
(C) Predicted probabilities are guaranteed to be exactly 0 or 1 on train points
(D) A feature-scale change can be absorbed into w

Q.95 NAT · 2 marks · kNN

Same 1-D train x=0,1,2,3 labels 0,0,1,1. Query 1.6 with k=3. Majority label is ____.

NAT · enter the numerical value

Q.96 MSQ · 2 marks · NaiveBayes

Which modelling assumptions/facts does naive Bayes involve?

(A) Features are conditionally independent given the class
(B) The class prior may be estimated by class frequencies
(C) Features must be jointly Gaussian with a full covariance
(D) A zero training count for a (feature,class) pair can zero a Bernoulli likelihood unless smoothed

Q.97 MCQ · 2 marks · LDA

S_W is singular when, in generic position,

(A) n is huge and features independent
(B) the feature dimension p exceeds within-class rank (about n-K)
(C) classes are well separated
(D) priors are equal

Q.98 MCQ · 1 mark · SVM

The dual of a kernel SVM depends on data only through

(A) inner products (or kernel evaluations) K(x_i,x_j)
(B) explicit high-dimensional phi(x) coordinates
(C) decision-tree Gini
(D) k-means centroids

Q.99 MCQ · 1 mark · DecisionTree

Entropy of a 50-50 binary node (log base 2) is

(A) 0
(B) 0.5
(C) 1 bit
(D) 2 bits

Q.100 NAT · 1 mark · BiasVariance

Noise=1, bias^2=4, variance=3. Expected MSE is ____.

NAT · enter the numerical value

Q.101 MCQ · 1 mark · LOOCV

LOOCV of the intercept-only mean predictor trains each fold on how many responses when the sample has n points?

(A) n-1
(B) n+1
(C) 2n
(D) k

Q.102 MCQ · 1 mark · KFold

Choosing k=n vs k=5 trades

(A) bias and variance of the estimated risk, plus compute
(B) only the kernel width
(C) whether PCA axes are orthogonal
(D) the SVM margin formula

Q.103 MCQ · 1 mark · MLP

XOR of two bits cannot be realised by a linear (no hidden layer) classifier, but a one-hidden-layer MLP can, because

(A) the hidden units can create a nonlinear feature map
(B) XOR is linearly separable
(C) MLPs always output 0
(D) weights must be integers

Q.104 NAT · 2 marks · KMeans

Seeds for k-means on W(1,1), X(1,2), Y(5,1), Z(5,2) are W and Z. After the first assignment, number of points assigned to Z's cluster is ____.

NAT · enter the numerical value

Q.105 MCQ · 1 mark · KMedoid

Manhattan k-medoid on four named cities uses pairwise L1 distances. Unlike k-means,

(A) updates are restricted to swapping among the observed city locations
(B) centroids can sit at the L1 mean off the cities always as medoids
(C) labels are required
(D) PCA axes must be orthogonal first

Q.106 MCQ · 2 marks · HAC

After merging Oak+Pine in that Manhattan single-linkage run, the next merge (into the dendrogram) joins

(A) the Oak-Pine cluster with Elm, at height min(2,3)=2
(B) Teak with Elm at 6
(C) all four at once
(D) Teak with Oak-Pine at 1

Q.107 NAT · 2 marks · PCA

Centered points (-1,0),(1,0),(0,0). Empirical (1/n) covariance has (1,1) entry ____.

NAT · enter the numerical value

Q.108 MCQ · 1 mark · MLMatch

Task: assign shop locations to k groups by iteratively averaging coordinates.

(A) k-means
(B) logistic regression
(C) A* search
(D) resolution theorem proving

Q.109 NAT · 1 mark · SLR

For Clinic BP desk data (1,4), (2,6), (3,7), (4,9) of BMI units over 20 against systolic excess, the OLS intercept is ____.

NAT · enter the numerical value

Q.110 NAT · 1 mark · MLR

Dummy-coded MLR uses 4 city levels plus intercept and no other features. Dummy columns excluding intercept: ____.

NAT · enter the numerical value

Q.111 NAT · 2 marks · Ridge

Orthogonal centered columns with norms squared 9 and 4; X^T y=(18,8); lambda=3. First ridge coeff is ____.

NAT · enter the numerical value

Q.112 NAT · 2 marks · Logistic

One pair x=2, y=1. Model p=sigmoid(w x), current w=0. Grad of log-loss is (p-y)x. One SGD step eta=0.5 updates w to ____.

NAT · enter the numerical value

Q.113 MCQ · 1 mark · kNN

Manhattan 1-NN on A(0,0):0, B(3,0):1, query (1,1). Predicted class is

(A) 0, since |1|+|1|=2 < |2|+|1|=3
(B) 1
(C) tie
(D) 2

Q.114 NAT · 1 mark · NaiveBayes

Laplace add-1 for a binary word given spam: 0 occurrences in 8 spam mails. Smoothed P(word=1|spam) is ____.

NAT · enter the numerical value

Q.115 NAT · 2 marks · LDA

Class A samples 1,2,3 and class B samples 5,6,7. Pooled sample variance (divisor n1+n2-2) is ____.

NAT · enter the numerical value

Q.116 NAT · 1 mark · SVM

Hinge loss of a point with y f(x)=3 is ____.

NAT · enter the numerical value

Q.117 MSQ · 2 marks · DecisionTree

Which splits are useless for a greedy impurity tree?

(A) Reusing a feature on a child that is already pure (gain 0)
(B) A threshold that sends every row to one child
(C) A split that improves gain on an impure node
(D) A multiway split on a 3-level categorical at the root

Q.118 MCQ · 1 mark · BiasVariance

Training error of 1-NN (no label noise, unique x) is 0, yet test error can be large. This gap signals

(A) high variance / overfit
(B) high bias / underfit
(C) a biased intercept in ridge
(D) d-separation

Q.119 NAT · 1 mark · LOOCV

y=1,3,5. Intercept-only model. LOO prediction of the first point is ____.

NAT · enter the numerical value

Q.120 NAT · 1 mark · KFold

n=100, 4-fold. Each test fold size is ____.

NAT · enter the numerical value

Q.121 MSQ · 2 marks · MLP

Which are TRUE of a feed-forward MLP trained by backprop?

(A) Forward pass computes activations layer by layer
(B) Backprop applies the chain rule to obtain dLoss/dW
(C) A sigmoid hidden layer is a linear function of its input
(D) SGD may use a minibatch gradient instead of the full-batch gradient

Q.122 MSQ · 2 marks · KMeans

Which are TRUE of k-means?

(A) The SSE objective is nonincreasing at each assign/update step
(B) The algorithm can converge to a local minimum depending on initialisation
(C) Empty clusters can occur for bad seeds
(D) The global SSE minimum is always found in one iteration

Q.123 NAT · 2 marks · HAC

Same four trees, complete linkage Manhattan. After Oak-Pine merge at 1, the complete-linkage distance of {Oak,Pine} to Elm is max(2,3)= ____.

NAT · enter the numerical value

Q.124 MCQ · 1 mark · PCA

If two features are exactly linearly dependent, the covariance has

(A) a zero eigenvalue (rank-deficient)
(B) all eigenvalues equal
(C) negative variance
(D) no eigenvectors

Q.125 MCQ · 1 mark · MLMatch

Task: two-class labelled Gaussians with a shared full covariance. Natural linear classifier:

(A) LDA (Fisher)
(B) k-NN with k=n
(C) HAC single linkage
(D) unregularised interpolating tree

Q.126 NAT · 2 marks · SLR

On Ice factory ledger the OLS residual sum of squares for (2,1), (4,2), (6,2), (8,3) (C above 0, tonnes sold) equals ____.

NAT · enter the numerical value

Q.127 MCQ · 1 mark · MLR

Adjusted R^2 differs from R^2 because it

(A) uses a kernel smoother
(B) penalises extra parameters via residual df, so it can fall when a useless feature is added
(C) is always larger than R^2
(D) equals 1-SSE/SST with no n,p

Q.128 MCQ · 2 marks · Ridge

Wide matrix p > n with dependent columns:

(A) OLS has a unique minimiser, ridge does not
(B) ridge has a unique minimiser, OLS does not
(C) both fail to have any minimiser
(D) both yield identical coefficients

Q.129 NAT · 1 mark · Logistic
Current w=1, x=0 so p=0.5, y=0, eta=1, grad=(p-y)x=0. After one SGD step w equals ____.
NAT · enter the numerical value

Q.130 MCQ · 1 mark · kNN
A large k in k-NN, relative to k=1, typically
(A) fits more wiggly decision boundaries
(B) smooths the decision boundary and raises bias
(C) is undefined for classification
(D) always overfits more than 1-NN

Q.131 MCQ · 2 marks · NaiveBayes
Gaussian NB with diagonal class-covariances differs from LDA because LDA
(A) does not use class priors
(B) shares one covariance across classes and does not force feature independence unless that covariance is diagonal
(C) cannot produce a linear boundary
(D) is identical to 1-NN

Q.132 MCQ · 1 mark · LDA
`Multi-class LDA with K classes projects onto at most how many useful Fisher directions?`
(A) K
(B) K-1
(C) n
(D) 1 always

Q.133 NAT · 1 mark · SVM
Hinge loss of a point with y f(x)=-1 is ____.
NAT · enter the numerical value

Q.134 NAT · 2 marks · DecisionTree
Right child has counts 1+ and 4-. Its Gini is 0.32. Report 100 times that Gini, i.e. ____.
NAT · enter the numerical value

Q.135 MSQ · 2 marks · BiasVariance
Which statements are TRUE?
(A) Bayes error is a lower bound no model can beat
(B) Variance of an estimator usually falls with n
(C) A more flexible model always has lower test error
(D) Regularisation is one way to move along the bias-variance curve

Q.136 NAT · 1 mark · LOOCV
y=1,3,5 intercept-only. Squared LOO error on the first point is ____.
NAT · enter the numerical value

Q.137 MCQ · 2 marks · KFold
Nested CV is needed when hyperparameters are tuned because otherwise
(A) the risk estimate is optimistically biased by selection on the same loops
(B) the design matrix is square
(C) one uses only LDA
(D) n=2 always

Q.138 NAT · 2 marks · MLP
Linear unit y=w x with w=0.5, x=4, target t=3, loss (1/2)(y-t)^2. dLoss/dw equals (y-t)*x = ____.
NAT · enter the numerical value

Q.139 NAT · 1 mark · KMeans
Cluster { (0,0),(0,2),(2,0),(2,2) }. Euclidean centroid is (a,a) with a= ____.
NAT · enter the numerical value

Q.140 NAT · 2 marks · HAC
Cities Pune(0,0), Nashik(0,4), Kolhapur(6,0), Solapur(6,5). Manhattan single-linkage first merge height is ____.
NAT · enter the numerical value

Q.141 NAT · 1 mark · PCA
Eigenvalues 5, 3, 2, 0. Number of PCs needed to capture ALL variance is ____.
NAT · enter the numerical value

Q.142 MCQ · 1 mark · MLMatch
Task: probability of default from a linear score squeezed to (0,1).
(A) logistic regression
(B) PCA
(C) k-medoid
(D) complete-linkage HAC

Q.143 MCQ · 1 mark · SLR
Four GATE mock pairs (hours, score) are (1,10), (2,20), (3,30), (4,40). After fitting OLS, the residual at hours=3 is
(A) 0
(B) 10
(C) -10
(D) 5

Q.144 MSQ · 2 marks · MLR
Which models stay inside multiple linear regression (linear in parameters)?
(A) y=b0+b1 x+b2 x^2+e
(B) y=b0+b1 log x+e with x > 0
(C) y=b0+b1 x1+b2 x2+b3 x1 x2+e
(D) y=sigmoid(b0+b1 x) with Bernoulli likelihood

Q.145 MCQ · 1 mark · Ridge
Standardising features before ridge is recommended because
(A) lambda then penalises coefficients on a comparable scale
(B) OLS already requires standardising to exist
(C) standardising removes the need for a penalty
(D) it converts ridge into LDA

Q.146 MCQ · 1 mark · Logistic
Binary cross-entropy on a single point with y=1 and p=0.5 equals (natural log)
(A) 0
(B) log 2
(C) 1
(D) 0.5

Q.147 NAT · 1 mark · kNN
`Five neighbour labels 1,1,1,0,0. 5-NN majority class is ____.`
NAT · enter the numerical value

Q.148 NAT · 1 mark · NaiveBayes
`2-class Bernoulli NB on 5 binary features uses 2 priors and 5 coin-weights per class. Parameter count is ____.`
NAT · enter the numerical value

Q.149 MCQ · 2 marks · LDA
LDA vs logistic on well-separated Gaussian classes with shared Sigma:
(A) LDA is the generative counterpart and is statistically efficient when its model is true
(B) logistic always has strictly lower error
(C) LDA cannot output a posterior
(D) they fit identical parameters for any finite sample

Q.150 NAT · 1 mark · SVM
Hinge loss at y f(x)=0.2 is ____.
NAT · enter the numerical value

Q.151 MCQ · 1 mark · DecisionTree
Unpruned trees that split to purity tend to
(A) underfit with high bias
(B) overfit: low training error, high variance
(C) match ridge with large lambda
(D) be identical to LDA

Q.152 MCQ · 1 mark · BiasVariance
k-fold CV is used in this story mainly to
(A) estimate prediction error and help pick a complexity knob
(B) remove the noise term
(C) force bias to 0
(D) compute S_W^{-1} S_B

Q.153 NAT · 2 marks · LOOCV
y=1,3,5 intercept-only LOOCV MSE (average of three squared errors) is ____.
NAT · enter the numerical value

Q.154 NAT · 1 mark · KFold
n=45, 9-fold. Train size per fold is ____.
NAT · enter the numerical value

Q.155 NAT · 2 marks · MLP
One SGD step on that unit: w=0.5, grad=-4, eta=0.1. Updated w is ____.
NAT · enter the numerical value

Q.156 MCQ · 1 mark · KMeans
k-means vs k-medoid: a centroid in k-means
(A) must be an actual data point
(B) need not be a data point (it is a mean)
(C) is always a support vector
(D) minimises Gini

Q.157 MCQ · 2 marks · HAC
On those four cities, Manhattan complete linkage first merge is still Pune-Nashik at 4. Next complete-linkage distance of {Pune,Nashik} to Solapur is max(11,7)= ____ wait as MCQ: the pair merged second under single linkage is
(A) Kolhapur with Solapur at 5
(B) Pune with Solapur at 11
(C) all remaining at 0
(D) Nashik with Kolhapur at 4

Q.158 MCQ · 2 marks · PCA
Projecting onto the first k PCs gives the best rank-k linear reconstruction in
(A) Frobenius / total squared error for centered data
(B) hinge loss
(C) Gini
(D) 0-1 loss

Q.159 MCQ · 2 marks · SLR
OLS of y on x is y=4+2x. The same sample is used to regress x on y. Which identity ALWAYS holds?
(A) The reverse slope must be 1/2
(B) The reverse slope equals 2
(C) The product of the two OLS slopes equals r^2
(D) The reverse intercept is 4

Q.160 NAT · 1 mark · MLR
n=12, p=3 predictors plus intercept. Residual df for OLS SSE is ____.
NAT · enter the numerical value

Q.161 MSQ · 2 marks · Ridge
Which statements about ridge are TRUE?
(A) Effective degrees of freedom fall as lambda grows
(B) Ridge paths are piecewise linear in lambda like lasso
(C) For a fixed design, b_ridge=(X^T X + lambda I)^{-1} X^T y
(D) Ridge zeros coefficients exactly for large finite lambda

Q.162 MCQ · 1 mark · Logistic
Unlike LDA, logistic regression
(A) requires class-conditional Gaussians
(B) models P(Y|x) directly and does not need a Gaussian class-conditional assumption
(C) cannot use a linear decision boundary
(D) cannot output probabilities

Q.163 MCQ · 2 marks · kNN
k-NN regression k=2 on (0,0),(2,4),(10,0). Query x=1. Prediction is
(A) 0
(B) 2
(C) 4
(D) 1

Q.164 MCQ · 2 marks · NaiveBayes
Four spam bags contribute 8 tokens with 'win' appearing twice. Multinomial NB MLE P(win|spam) without smoothing is
(A) 1/2
(B) 2/4
(C) 2/8
(D) 1/8

Q.165 NAT · 1 mark · LDA
Means m1=(0,0), m2=(2,0), S_W=I. Fisher vector S_W^{-1}(m1-m2) has second coordinate ____.
NAT · enter the numerical value

Q.166 MCQ · 1 mark · SVM

Increasing C in min 1/2||w||^2 + C sum xi typically

(A) allows more margin violations
(B) penalises slack more, tending toward a harder margin
(C) changes the kernel degree
(D) has no primal effect

Q.167 MCQ · 1 mark · DecisionTree

Axis-aligned binary trees in 2-D produce regions that are

(A) always balls
(B) rectangles (products of intervals), possibly unbounded
(C) linear halfspaces only
(D) circles centred at means

Q.168 NAT · 2 marks · BiasVariance

Two uncorrelated unbiased estimators have variances 4 and 9. Variance of their average is ____.

NAT · enter the numerical value

Q.169 MCQ · 2 marks · LOOCV

Compared with 5-fold CV, LOOCV on the same data typically

(A) has higher variance of the estimated extra-sample error and is nearly unbiased for training size n-1
(B) always has lower variance
(C) cannot be used for k-NN
(D) requires a validation set besides the n points

Q.170 MCQ · 2 marks · KFold

A student reports LOOCV accuracy by training on all n points then evaluating n times. This is

(A) correct LOOCV
(B) not LOOCV: each evaluation must train without the held-out point
(C) k-fold with k=1
(D) ridge with lambda=n

Q.171 MCQ · 2 marks · MLP

Vanishing gradients in a deep sigmoid MLP occur because

(A) sigmoid derivative < = 0.25, so a long product of derivatives shrinks
(B) ReLU derivative is always 0
(C) SGD cannot use minibatches
(D) the output must be one-hot

Q.172 NAT · 1 mark · KMeans

1-D points 0,0,0,10, k=1. The k-means centroid (the mean) is ____.

NAT · enter the numerical value

Q.173 NAT · 1 mark · HAC

Points Anu(0,0), Bala(2,0), Chetan(0,1), Diya(5,0), Manhattan. Single-linkage first merge height (Anu-Chetan) is ____.

NAT · enter the numerical value

Q.174 NAT · 1 mark · PCA

2-D points (1,1),(1,1),(1,1) already equal. After centering, all PCA eigenvalues are ____.

NAT · enter the numerical value

Q.175 MSQ · 2 marks · SLR

Let e_i be OLS residuals from simple linear regression with intercept. Which identities ALWAYS hold?

(A) sum e_i = 0
(B) sum e_i x_i = 0
(C) sum e_i y_i = 0
(D) sum e_i yhat_i = 0

Q.176 MCQ · 2 marks · MLR

True map y=1+2 x1+0 x2+noise; we still include x2 in OLS. As n grows, the x2 coefficient

(A) converges to 0 in probability (correctly specified zero)
(B) forces R^2 to 0
(C) makes X^T X singular
(D) makes the intercept unestimable

Q.177 NAT · 1 mark · Ridge

Scalar ridge on x=1,1,1 and y=3,5,7 (no intercept). X^T X=3, X^T y=15, lambda=2. Coefficient is ____.

NAT · enter the numerical value

Q.178 NAT · 1 mark · Logistic

Odds P/(1-P)=3. Then P equals ____.

NAT · enter the numerical value

Q.179 MCQ · 1 mark · kNN

Feature A in millimetres, B in kilometres. Euclidean k-NN without scaling will

(A) treat both axes as equally important automatically
(B) be dominated by the large-magnitude axis unless features are scaled
(C) become logistic regression
(D) ignore the training labels

Q.180 NAT · 1 mark · NaiveBayes

Priors 0.7 and 0.3, likelihoods 0.2 and 0.9. MAP class index (1 or 2) is ____.

NAT · enter the numerical value

Q.181 MSQ · 2 marks · LDA

Which are TRUE of S_B and S_W in two-class Fisher LDA?

(A) S_B is rank-1, proportional to (m1-m2)(m1-m2)^T
(B) S_W is a pooled within-class scatter
(C) The Rayleigh quotient leads to the generalised eigenproblem S_B w = lambda S_W w
(D) S_W^{-1} S_B is always symmetric even if S_W is not a multiple of I

Q.182 MCQ · 1 mark · SVM

A linear SVM weight w is a linear combination of training points with nonzero coefficients only for

(A) all points equally
(B) support vectors
(C) the class means as in LDA
(D) the first PC

Q.183 NAT · 1 mark · DecisionTree

A full binary classification tree has 5 leaves. Internal split nodes: ____.

NAT · enter the numerical value

Q.184 MCQ · 1 mark · BiasVariance

As polynomial degree grows to n-1 on n noiseless distinct x, train SSE

(A) stays constant
(B) tends to 0 (interpolation)
(C) must increase
(D) equals SST always

Q.185 NAT · 1 mark · LOOCV

1-NN LOOCV on two points with different labels. Number of errors is ____.

NAT · enter the numerical value

Q.186 NAT · 1 mark · MLP

`Softmax on logits (0,0,0) for 3 classes. Probability of class 1 is ____.`

NAT · enter the numerical value

Q.187 MCQ · 1 mark · KMeans

After assignment, a point x is in cluster j at a Euclidean k-means fixed point only if

(A) ||x-mu_j|| < = ||x-mu_i|| for every i (Voronoi / nearest centroid)
(B) x is a medoid
(C) x has label j in a supervised file
(D) ||x|| is maximal

Q.188 MCQ · 2 marks · HAC

Anu, Bala, Chetan, Diya as above. After merging Anu-Chetan, single-linkage next joins

(A) that cluster with Bala at min(AB,CB)=min(2,3)=2
(B) Diya with Bala at 3 as the first merge
(C) all four at height 1
(D) Chetan with Diya at 1

Q.189 MCQ · 1 mark · PCA

LDA vs PCA: LDA maximises the between/within ratio using labels; PCA maximises

(A) total variance without labels
(B) Gini
(C) the SVM margin
(D) k-means SSE

Q.190 MCQ · 1 mark · SLR

For a sample, corr(x,y)=0.6, s_x=2, s_y=5. The OLS slope of y on x equals

(A) 0.60
(B) 1.50
(C) 2.40
(D) 0.24

Q.191 MCQ · 1 mark · MLR

yhat=10+3 x1-2 x2. Holding x1 fixed, a unit increase in x2 changes mean response by

(A) 3
(B) -2
(C) 1
(D) 10

Q.192 MCQ · 2 marks · Ridge

`A report claims raising lambda increased training R^2 versus OLS on the same design. This is`

(A) expected because ridge always fits better
(B) impossible: OLS minimises training SSE among linear predictors, and ridge is a shrinkage of it
(C) required by Gauss-Markov
(D) proof that lambda was negative

Q.193 MCQ · 1 mark · Logistic

`A 1-D logistic classifier predicts class 1 iff 4x-2 > =0. The threshold on x is`

(A) 0
(B) 0.5
(C) 2
(D) 4

Q.194 NAT · 2 marks · kNN

Train on the line -2,-1,0,8 labels 1,1,1,0. 3-NN at query 7. Predicted label is ____.

NAT · enter the numerical value

Q.195 MCQ · 1 mark · NaiveBayes

Why sum log class-conditionals in NB?

(A) Logs cancel the independence assumption
(B) To avoid underflow of a long product of probabilities
(C) Because MLE is undefined in log space
(D) So that features become dependent

Q.196 MCQ · 1 mark · LDA

`If class covariances are equal but features are correlated, Gaussian NB (diagonal) is misspecified while LDA`

(A) can still model that correlation through the shared Sigma
(B) also forces diagonal Sigma
(C) becomes k-means
(D) cannot use priors

Q.197 NAT · 1 mark · SVM

Two 1-D SVs at x=-2 (label -1) and x=4 (label +1). The midpoint threshold is at x= ____.

NAT · enter the numerical value

Q.198 MCQ · 1 mark · DecisionTree

To predict a numeric house price with piecewise-constant axis splits one uses a

(A) classification tree with Gini
(B) regression tree, typically with squared-error split criterion
(C) hard-margin SVM
(D) k-medoid

Q.199 NAT · 1 mark · BiasVariance

Bias of an estimator is 2, so bias^2 contribution is ____.

NAT · enter the numerical value

Q.200 MCQ · 2 marks · MLP

A 2-2-1 MLP with linear hidden and linear output (no nonlinearities) is

(A) still universal
(B) equivalent to a single linear layer, hence cannot represent XOR
(C) a decision tree
(D) k-means

Q.201 NAT · 1 mark · KMeans
Two clusters with SSE 4 and 5. Total k-means objective is ____.
NAT · enter the numerical value

Q.202 NAT · 1 mark · HAC
Same four people, complete linkage. After Anu-Chetan at 1, complete distance of {Anu,Chetan} to Bala is max(2,3)= ____.
NAT · enter the numerical value

Q.203 NAT · 1 mark · PCA
Covariance [[2,0],[0,2]]. Any unit vector is a PC direction of eigenvalue ____.
NAT · enter the numerical value

Q.204 NAT · 1 mark · SLR
A lab reports SST=80 and SSE=20 for an OLS fit of absorbance on concentration (with intercept). R^2 is ____.
NAT · enter the numerical value

Q.205 NAT · 2 marks · MLR
Centered orthogonal features: sum x1^2=10, sum x2^2=5, sum x1 y=20, sum x2 y=-10. OLS slope of x1 is ____.
NAT · enter the numerical value

Q.206 NAT · 2 marks · Ridge
X^T X=diag(2,2), X^T y=[4,2], lambda=2. Ridge b2 equals ____.
NAT · enter the numerical value

Q.207 NAT · 1 mark · Logistic
Sigmoid derivative at f(x)=0.5 equals ____.
NAT · enter the numerical value

Q.208 NAT · 1 mark · kNN
3-NN, 2-class, neighbours labelled 1,1,0. Predicted class is ____.
NAT · enter the numerical value

Q.209 NAT · 1 mark · NaiveBayes
Add-1 Laplace, word seen 3 times in 14 class-A docs (binary presence). P(word=1|A) is ____.
NAT · enter the numerical value

Q.210 NAT · 1 mark · LDA
Projected LDA scores have class means -3 and +3 (balanced). Midpoint threshold is ____.
NAT · enter the numerical value

Q.211 MSQ · 2 marks · SVM
Which are valid PSD kernels commonly used with SVMs?
(A) Linear K=x^T z
(B) Gaussian RBF K=exp(-||x-z||^2/(2 sigma^2))
(C) Polynomial K=(x^T z+c)^d for c > =0 and d a positive integer
(D) K(x,z)=||x-z||^2 (raw squared Euclidean)

Q.212 NAT · 1 mark · DecisionTree
Node [+, -]=[6,2]. Majority-class misclassification rate is ____.
NAT · enter the numerical value

Q.213 MCQ · 1 mark · BiasVariance
1-NN vs large-k NN: the 1-NN classifier is the one with
(A) higher variance and more jagged boundaries
(B) higher bias and a near-linear boundary always
(C) zero Bayes error always
(D) a unique support vector

Q.214 NAT · 1 mark · MLP
Hidden layer of 3 sigmoids, input dim 4, including hidden biases, no output yet. Hidden-layer parameter count is ____.
NAT · enter the numerical value

Q.215 MCQ · 1 mark · KMeans
Increasing k from 1 to n (one cluster per point) makes the k-means objective
(A) increase
(B) drop to 0
(C) become the Gini of y
(D) undefined

Q.216 MSQ · 2 marks · HAC
Which are TRUE of hierarchical agglomerative clustering?
(A) Single linkage can produce chaining (elongated clusters)
(B) Complete linkage prefers compact clusters
(C) The dendrogram of n points has n-1 merge levels
(D) HAC requires a pre-specified k at the start, unlike k-means

Q.217 MCQ · 1 mark · PCA
Whitening / sphering using PCA rescales along PCs so that
(A) the empirical covariance becomes I (on the span of nonzero eigenvalues)
(B) labels become balanced
(C) k-means SSE is maximised
(D) the intercept of OLS vanishes

Q.218 MCQ · 2 marks · SLR
Simple OLS is fitted through the origin to (x,y) in {(1,1),(2,2),(3,1)}. The slope minimising SSE is
(A) 1
(B) 6/7
(C) 4/7
(D) 5/7

Q.219 MCQ · 1 mark · MLR
Including an interaction x1 x2 in MLR means
(A) the partial effect of x1 is allowed to depend on the value of x2
(B) x1 and x2 must be independent in the population
(C) OLS becomes nonlinear in beta
(D) the intercept must be dropped

Q.220 MCQ · 1 mark · Ridge
Bias-variance reading of moderate ridge vs OLS in an ill-posed problem:
(A) cuts variance more than it adds bias, often lowering expected prediction error
(B) cuts bias and raises variance
(C) is identical to dropping the intercept
(D) is the same as k-means

Q.221 MCQ · 2 marks · Logistic

`L2-regularised logistic adds (lambda/2)||w||^2. As lambda grows, predictions`

(A) must become quadratic
(B) approach an intercept-only (prior) classifier as w shrinks to 0
(C) interpolate every label
(D) become k-means

Q.222 MCQ · 1 mark · kNN

k-NN is a lazy learner meaning

(A) it updates a global w by SGD at train time
(B) it stores the training set and defers computation to query time
(C) it discards the training set after building a tree of depth k
(D) it requires linearly separable data

Q.223 MCQ · 2 marks · NaiveBayes

If two features are identical copies, naive Bayes

(A) automatically drops one copy
(B) double-counts that evidence, over-confidently pushing posteriors
(C) becomes SVM
(D) is still well-specified because copies are independent

Q.224 MCQ · 1 mark · LDA

QDA drops which LDA assumption?

(A) Gaussian class-conditionals
(B) shared covariance across classes
(C) existence of class priors
(D) use of Bayes rule

Q.225 MCQ · 1 mark · SVM

XOR in the plane is not linearly separable. A degree-2 polynomial kernel

(A) cannot help
(B) can make the classes linearly separable in the implicit feature space
(C) reduces XOR to k-means
(D) forces the margin to 0

Q.226 MSQ · 2 marks · DecisionTree

Compared with a linear SVM, a deep axis-aligned tree

(A) can represent XOR without a kernel
(B) has a piecewise-constant decision surface
(C) always maximises the geometric margin
(D) is more prone to high variance when unpruned

Q.227 MCQ · 1 mark · MLP

Cross-entropy + softmax at the output of an MLP is preferred over SSE for classification because

(A) it matches the multinomial likelihood and avoids saturation issues of SSE+sigmoid
(B) it forces linear hidden units
(C) it maximises Gini
(D) it is the k-means objective

Q.228 MCQ · 1 mark · KMeans

k-means is sensitive to feature scaling because

(A) SSE uses Euclidean geometry
(B) it maximises Gini
(C) it is a kernel SVM
(D) labels must be scaled

Q.229 MCQ · 1 mark · HAC

Cutting a dendrogram at a height that yields 3 clusters is a way to

(A) choose k=3 after seeing the hierarchy
(B) train an SVM
(C) compute R^2
(D) estimate S_W

Q.230 NAT · 1 mark · PCA

Scores on PC1 of three centered points sum to ____.

NAT · enter the numerical value

Q.231 MCQ · 1 mark · SLR

In OLS y=b0+b1 x+e, adding 5 to every x-value and leaving y unchanged will

(A) increase b1 by 5
(B) leave b1 unchanged and decrease b0 by 5 b1
(C) leave both b0 and b1 unchanged
(D) multiply b1 by 5

Q.232 NAT · 1 mark · MLR

yhat=5+0*x1+4*x2. At (10, 0.5) predicted y is ____.

NAT · enter the numerical value

Q.233 NAT · 2 marks · Ridge

`OLS 1-D slope (no intercept) is 5. Ridge uses lambda so that (X^T X+lambda)=2(X^T X). Ridge slope is ____.`

NAT · enter the numerical value

Q.234 MCQ · 1 mark · Logistic

A logistic weight on hours_studied is +1.2. Holding others fixed, one extra hour

(A) adds 1.2 to the probability
(B) adds 1.2 to the log-odds
(C) multiplies the probability by 1.2
(D) is illegal in logistic models

Q.235 NAT · 1 mark · kNN

Points (0,0),(0,2),(6,0). Query (1,0). Euclidean 1-NN distance to the winner is ____.

NAT · enter the numerical value

Q.236 NAT · 1 mark · NaiveBayes

`Three classes, uniform priors, likelihoods 0.1, 0.2, 0.7. Posterior of class 3 is ____.`

NAT · enter the numerical value

Q.237 NAT · 1 mark · LDA

n1=n2=2, class1: 0,2 and class2: 8,10. Difference of means m2-m1 is ____.

NAT · enter the numerical value

Q.238 NAT · 1 mark · SVM

Canonical w=(0,2), b=-2. Point x=(0,2), y=+1 has y(w?x+b) equal to ____.

NAT · enter the numerical value

Q.239 NAT · 1 mark · DecisionTree

Parent entropy 1 bit, children weights 1/2 with entropies 0 and 1. Information gain (bits) is ____.

NAT · enter the numerical value

Q.240 NAT · 1 mark · MLP
Forward: z=w x+b with w=2, b=-1, x=1, activation tanh. Argument z equals ____.
NAT · enter the numerical value

Q.241 NAT · 1 mark · HAC
n=6 points, agglomerative clustering. Number of merge operations until one cluster is ____.
NAT · enter the numerical value

Q.242 MSQ · 2 marks · PCA
Which preprocessing steps are the usual prelude to PCA on heterogeneous features?
(A) Center each feature
(B) Optionally scale to unit variance (standardise) so scale does not dominate
(C) Replace features by their labels
(D) Drop orthogonality of PCs on purpose

Q.243 NAT · 1 mark · SLR
Pairs (2,3), (4,7), (6,11) lie on a straight line. OLS slope of y on x is ____.
NAT · enter the numerical value

Q.244 MCQ · 1 mark · MLR
Multicollinearity among predictors primarily inflates
(A) the residual degrees of freedom
(B) the sampling variance of the involved coefficients
(C) the number of columns of y
(D) R^2 toward -1

Q.245 MCQ · 1 mark · Ridge
Ridge closed form replaces the OLS normal equation X^T X b = X^T y by
(A) (X^T X + lambda I) b = X^T y
(B) X^T X b + lambda = y
(C) X b = lambda y
(D) b = lambda X^T y

Q.246 NAT · 1 mark · Logistic
Score z=0. Predicted class-1 probability is ____.
NAT · enter the numerical value

Q.247 MCQ · 1 mark · kNN
If k equals n, unweighted majority-vote k-NN ignores the query location and returns
(A) the 1-NN label
(B) the training majority class
(C) a uniformly random class always
(D) the mean of the features

Q.248 MSQ · 2 marks · NaiveBayes
Which statements are TRUE of naive Bayes?
(A) NB can still classify well even when independence is mildly false
(B) A missing category count can zero a whole posterior path without smoothing
(C) Gaussian NB boundaries are linear if covariances are shared and diagonal (a special LDA)
(D) NB requires linearly separable classes

Q.249 MCQ · 1 mark · LDA
Using PC1 of pooled unlabelled data as the LDA direction is
(A) always identical to Fisher LDA
(B) generally different, because PCA maximises total variance not the between/within ratio
(C) impossible in 2-D
(D) required when S_W is invertible

Q.250 MCQ · 2 marks · SVM
If training data are linearly separable, a hard-margin SVM
(A) has a unique maximum-margin w (in general position)
(B) does not exist
(C) always coincides with the OLS hyperplane
(D) has margin 0

Q.251 MCQ · 2 marks · DecisionTree
A categorical ID with as many levels as rows, if allowed to split fully, will
(A) have zero training error and almost surely terrible generalisation
(B) be ignored by Gini because gain is 0
(C) become a support vector
(D) estimate a ridge penalty

Q.252 MCQ · 1 mark · MLP
Weight decay in MLP training is essentially
(A) an L2 penalty on weights, akin to ridge
(B) k-medoid swapping
(C) PCA whitening of labels
(D) dropout of the loss

Q.253 MCQ · 1 mark · HAC
Average linkage defines cluster distance as
(A) the average of all cross-pair distances
(B) the min cross-pair distance
(C) the max cross-pair distance
(D) Gini of labels

Q.254 MCQ · 1 mark · PCA
The loading vector of PC2 is orthogonal to PC1 because
(A) covariance (or Gram) eigenvectors for distinct eigenvalues can be chosen orthonormal, and PCA enforces that
(B) labels force orthogonality
(C) k-means does
(D) Gini is orthogonal

Q.255 MCQ · 1 mark · SLR
A residual plot of OLS of marks on hours fans out as hours grow. This most directly suggests
(A) the intercept is biased toward 0
(B) heteroscedasticity, so equal-variance Gauss-Markov fails
(C) the slope must be negative
(D) R^2 is necessarily 0

Q.256 MCQ · 1 mark · Logistic
For a linear logistic model in 2-D with w nonzero, {x: P(Y=1|x)=0.5} is
(A) a line (a hyperplane in 2-D)
(B) always a circle
(C) a union of k-means cells
(D) empty

Q.257 MCQ · 1 mark · kNN

Distance-weighted 1-NN (weight 1/d) on two neighbours at distances 1 and 3 with labels 0 and 1. The query is labelled

(A) 0, because the nearer neighbour gets all the 1-NN vote; with k=1 only the nearest is used
(B) 1
(C) the mean 0.5 as a class
(D) undefined

Q.258 NAT · 1 mark · NaiveBayes

Binary feature d=1, classes K=4, Bernoulli NB with K priors. Parameter count is ____.

NAT · enter the numerical value

Q.259 MCQ · 2 marks · SVM

A dual variable pinned at alpha_i=C (box 0 < =alpha < =C) typically corresponds to

(A) a non-support vector
(B) a margin violator (slack or misclassified)
(C) the unique class mean
(D) hinge 0 necessarily

Q.260 NAT · 2 marks · DecisionTree

A node of 10 rows splits into pure children of sizes 7 and 3. Parent majority-class error was ____.

NAT · enter the numerical value

Q.261 NAT · 1 mark · MLP

A minibatch of 8, dataset of 80, one epoch. Number of SGD steps per epoch is ____.

NAT · enter the numerical value

Q.262 MCQ · 1 mark · HAC

Single-linkage HAC on a chain of points equally spaced on a line will

(A) merge neighbours sequentially, producing a chain
(B) always produce balanced complete blobs
(C) equal k-means with k=2 after one merge
(D) require labels

Q.263 NAT · 1 mark · PCA

Variance on PC1 is 9, on PC2 is 1. Cumulative fraction after 1 PC is ____.

NAT · enter the numerical value

Q.264 NAT · 1 mark · SLR

For n=5 observations, OLS with intercept uses how many degrees of freedom for SSE?

NAT · enter the numerical value

Q.265 NAT · 1 mark · Logistic

p=0.9. Then p(1-p) equals ____.

NAT · enter the numerical value

Q.266 MCQ · 2 marks · NaiveBayes

P(Y=1)=0.2, P(X=red|Y=1)=0.9, P(X=red|Y=0)=0.3. Then P(Y=1|X=red) is closest to

(A) 0.20
(B) 0.43
(C) 0.90
(D) 0.06

Q.267 NAT · 2 marks · SVM

1-D labels: -3,-1 are -1 and +2,+5 are +1. Hard-margin SV gap (margin width) is ____.

NAT · enter the numerical value

Q.268 MCQ · 1 mark · DecisionTree

Matching task to algorithm: piecewise-constant 2-way classification on a tiny labelled table of colour vs label is done by a

(A) decision tree using Gini or entropy
(B) ridge with lambda=10^6
(C) PCA projection
(D) k-means with k=10

Q.269 MCQ · 1 mark · MLP

If every hidden unit uses identity activation, stacking layers without residual tricks

(A) does not increase representational power beyond one linear layer
(B) always solves XOR
(C) computes a decision tree
(D) forces Gini splits

Q.270 MCQ · 2 marks · PCA

A student runs PCA on 100 features with n=20 without regularisation. The covariance estimate is

(A) at most rank 19 (centered), so at least 81 eigenvalues are 0
(B) full rank 100 always
(C) equal to S_W of LDA
(D) a kernel of SVM

Q.271 MCQ · 1 mark · SLR

If all x-values are identical, simple OLS with intercept

(A) has a unique slope
(B) cannot estimate the slope because Sxx=0
(C) forces R^2=1
(D) makes every residual equal to x

Q.272 MCQ · 1 mark · SVM

Soft-margin slack xi_i is 0 whenever

(A) y_i f(x_i) > = 1
(B) the point is a support vector
(C) C=0
(D) the kernel is RBF

Q.273 NAT · 1 mark · MLP

Output sigmoid 0.5, binary target 1, log-loss -log p. Using log base e, -log 0.5 = log 2 ~ 0.693. Rounded to 2 decimals: ____.

NAT · enter the numerical value

Q.274 MCQ · 1 mark · SLR

R^2=0 for an OLS line with intercept means

(A) every residual is 0
(B) the fitted slope is 0 (horizontal line at ybar)
(C) SST=0
(D) the intercept is 0

Q.275 MCQ · 1 mark · MLP

Backprop through a ReLU unit with pre-activation $z < 0$ sends

(A) zero gradient to its inputs
(B) gradient 1 always
(C) gradient -1
(D) undefined NaN always

UNIT 6 SOLUTIONS

Q.1 [NAT] 1.4 — b1 = Sxy/Sxx = 1.4.

Q.2 [MCQ] B — Need X^T X invertible, i.e. rank(X)=p+1.

Q.3 [MCQ] A — Squared loss plus L2 penalty on slopes.

Q.4 [MCQ] B — Logistic sigmoid maps a linear score to (0,1).

Q.5 [MCQ] A — d(q,A)=d(q,B)=1, d(q,C)=d(q,D)=sqrt5.

Q.6 [MCQ] B — K priors + K success probabilities = 2K.

Q.7 [MCQ] B — Between-class scatter over within-class scatter.

Q.8 [MCQ] A — Canonical hard-margin primal.

Q.9 [NAT] 0.5 — 1-2*0.25=0.5.

Q.10 [MCQ] B — 1-NN is wiggly (high variance, low bias).

Q.11 [MCQ] C — One fit per left-out point.

Q.12 [NAT] 180 — Test 20, train 180.

Q.13 [MCQ] B — Weights 20+15, biases 5+3, total 43. Option B is 20+5+15+3=43.

Q.14 [MCQ] A — The mean minimises within-cluster SSE.

Q.15 [MCQ] B — Medoid is a prototype from the dataset.

Q.16 [MCQ] B — Bottom-up: n singletons, n-1 merges.

Q.17 [MCQ] A — PCA = eigendecomposition of covariance (or SVD of the centered data).

Q.18 [MCQ] A — That's the SVM primal.

Q.19 [NAT] 7.5 — yhat = 0.5 + 1.4*5 = 7.5.

Q.20 [MSQ] A;B;C — Omitted-variable bias can wreck unbiasedness. A-C are geometry/algebra of OLS.

Q.21 [MSQ] A;B;C — Adding lambda I improves conditioning; it does not worsen it.

Q.22 [NAT] 0.16 — 0.8*0.2=0.16.

Q.23 [NAT] 1 — d(P)~1.41, d(R)~2.24, d(Q)~3.16 so P wins.

Q.24 [NAT] 2K+1 — K priors + K means + 1 variance = 2K+1.

Q.25 [MCQ] B — w ? S_W^{-1}(m1-m2).

Q.26 [NAT] 5 — 3-4-5 triangle.

Q.27 [NAT] 0 — 1-1^2=0.

Q.28 [MCQ] B — Shrinkage trades bias for variance.

Q.29 [NAT] 0 — Each point's NN among the other three is its same-class twin.

Q.30 [NAT] 5 — One per held-out fold.

Q.31 [NAT] 9 — 2*2+2 + 2*1+1 = 4+2+2+1=9.

Q.32 [NAT] 3 — 1,2,3 nearer to 1 than to 10.

Q.33 [MCQ] A — Medoids are robust and interpretable as actual points.

Q.34 [MCQ] B — Top-down is divisive.

Q.35 [MSQ] A;B;D — PCA is unsupervised; it ignores labels.

Q.36 [MCQ] A — PCA is the unlabeled variance maximiser.

Q.37 [NAT] 1.4 — b1 = Sxy/Sxx = 1.4.

Q.38 [NAT] 3 — Intercept + area + dummy = 3.

Q.39 [NAT] 1 — 8/(4+4)=1.

Q.40 [NAT] 0.16 — 0.2*0.8=0.16 by symmetry.

Q.41 [MCQ] B — Labels 1,2,2 majority 2.

Q.42 [NAT] 27 — 3 + 12 means + 12 vars = 27.

Q.43 [MSQ] A;B;D — Priors need not be equal; they only shift the threshold.

Q.44 [NAT] 0.5 — 2/4=0.5.

Q.45 [NAT] 0.375 — 1-9/16-1/16=0.375.

Q.46 [MSQ] A;B;C — The mean predictor is high bias, low variance.

Q.47 [NAT] 1 — Only the point at 2 is classified as 0 when left out.

Q.48 [MCQ] A — Mean 6 predicts the two zeros; mean 0 predicts 4 and 8.

Q.49 [NAT] 0.16 — 0.8*0.2=0.16.

Q.50 [NAT] 2 — Mean of 1,2,3 is 2.

Q.51 [NAT] 1 — Median of 0,1,100 is 1; L1 total 1+0+99=100, better than 0 (101) or 100 (199).

Q.52 [MCQ] B — Single = min, complete = max.

Q.53 [NAT] 4 — Largest eigenvalue 4.

Q.54 [MCQ] A — Bernoulli/multinomial NB.

Q.55 [NAT] 1.9 — b0 = ybar - b1 xbar = 1.9.

Q.56 [MCQ] A — Perfect collinearity: rank(X) < 3.

Q.57 [NAT] 2 — 8/4=2, recovering OLS.

Q.58 [NAT] 0.5 — z=0, sigmoid(0)=1/2.

Q.59 [MSQ] A;B;C — k-NN is a nonparametric local vote; no class-density model is fitted.

Q.60 [MCQ] B — 0.6*0.5=0.3 and 0.2*0.5=0.1, posterior 0.3/0.4=0.75.

Q.61 [NAT] 2 — Midpoint 2.

Q.62 [MCQ] C — All four lie on the margin planes after scaling.

Q.63 [MCQ] B — 0.5*0 + 0.5*0.5=0.25.

Q.64 [MCQ] A — Standard decomposition.

Q.65 [MCQ] A — Classic hat-matrix shortcut.

Q.66 [MSQ] A;B;C — Standard k-fold partitions the training sample.

Q.67 [NAT] 0 — Negative part is zeroed.

Q.68 [MCQ] B — Euclidean k-means cells are Voronoi, hence convex.

Q.69 [MSQ] A;B;C — Both are unsupervised; PAM uses swap.

Q.70 [MCQ] B — Complete = max.

Q.71 [NAT] 0.8 — 4/(4+1)=0.8.

Q.72 [MCQ] A — L2 shrinkage.

Q.73 [NAT] 0.2 — SSE = sum (y-yhat)^2 = 0.2.

Q.74 [NAT] 6 — 3+8-5=6.

Q.75 [MCQ] C — Penalty dominates: b - > 0.

Q.76 [MCQ] B — 3-6x=0 = > x=1/2.

Q.77 [NAT] 1 — Nearest is x=2 (dist 0.4) label 1.

Q.78 [NAT] 0.77 — 0.2 / (0.2+0.06) = 0.2/0.26 ~ 0.77.

Q.79 [NAT] 1 — Midpoint 1.

Q.80 [MSQ] A;B;C — SVs have alpha > 0. Well-inside correct points have alpha=0.

Q.81 [NAT] 0.25 — 0.5-0.25=0.25.

Q.82 [MCQ] A — Systematic misspecification.

Q.83 [NAT] 1 — 1-NN uses one of the remaining n-1.

Q.84 [NAT] 4 — Average 4.

Q.85 [NAT] 2.5 — Positive part unchanged.

Q.86 [MCQ] A — Convex Voronoi cells on a line are intervals, so everything between P and S joins them.

Q.87 [NAT] 5 — A: 2+3=5; B: 2+4=6; C: 3+4=7.

Q.88 [NAT] 1 — Closest pair Oak-Pine at L1=1.

Q.89 [MCQ] A — Uncentered Gram is dominated by the mean vector.

Q.90 [MCQ] A — CART-style tree.

Q.91 [NAT] 10.3 — yhat = 0.7 + 1.2*8 = 10.3.

Q.92 [MCQ] A — X^T X diagonal = > coefficients decouple to simple-regression formulae.

Q.93 [MCQ] B — X^T X + lambda I is SPD for lambda > 0.

Q.94 [MSQ] A;B;D — NLL is convex; separability sends unregularised MLE to infinity; p stays in (0,1).

Q.95 [NAT] 1 — Neighbours x=2,1,3 labels 1,0,1 majority 1.

Q.96 [MSQ] A;B;D — Full joint Gaussian is LDA/QDA, not NB.

Q.97 [MCQ] B — Within scatter lives in an (n-K)-dimensional space.

Q.98 [MCQ] A — Kernel trick.

Q.99 [MCQ] C — H=1 bit.

Q.100 [NAT] 8 — 1+4+3=8.

Q.101 [MCQ] A — Each left-out is predicted by the mean of the other n-1.

Q.102 [MCQ] A — Standard CV design trade-off.

Q.103 [MCQ] A — Hidden layer provides the nonlinear lifting.

Q.104 [NAT] 2 — Y and Z nearer to Z; W,X nearer to W.

Q.105 [MCQ] A — Medoids remain data points (cities).

Q.106 [MCQ] A — d(OP,Elm)=min(Oak-Elm,Pine-Elm)=min(2,3)=2 < d(OP,Teak)=min(4,3)=3.

Q.107 [NAT] 2/3 — ((-1)^2+1+0)/3=2/3.

Q.108 [MCQ] A — k-means centroids.

Q.109 [NAT] 2.5 — b0 = ybar - b1 xbar = 2.5.

Q.110 [NAT] 3 — k levels need k-1 dummies with intercept.

Q.111 [NAT] 1.5 — 18/(9+3)=1.5.

Q.112 [NAT] 0.5 — p=0.5, grad=(0.5-1)*2=-1, w := 0-0.5*(-1)=0.5.

Q.113 [MCQ] A — d1(A)=2, d1(B)=3.

Q.114 [NAT] 0.1 — (0+1)/(8+2)=0.1.

Q.115 [NAT] 1 — Each class has sample var 1; pooled=1.

Q.116 [NAT] 0 — max(0,1-3)=0.

Q.117 [MSQ] A;B — Zero-gain / degenerate splits are useless.

Q.118 [MCQ] A — Zero train error with poor test is overfit.

Q.119 [NAT] 4 — Mean of 3 and 5.

Q.120 [NAT] 25 — 100/4=25.

Q.121 [MSQ] A;B;D — Sigmoid is nonlinear. SGD/minibatch is standard.

Q.122 [MSQ] A;B;C — k-means is locally convergent, not globally optimal.

Q.123 [NAT] 3 — Complete uses max pairwise.

Q.124 [MCQ] A — Rank at most 1 in that plane.

Q.125 [MCQ] A — LDA is the model.

Q.126 [NAT] 0.2 — SSE = sum (y-yhat)^2 = 0.2.

Q.127 [MCQ] B — R_adj^2 = 1-(SSE/(n-p-1))/(SST/(n-1)).

Q.128 [MCQ] B — Ridge is strictly convex; OLS loss is flat along the null space.

Q.129 [NAT] 1 — Gradient 0 because x=0.

Q.130 [MCQ] B — Larger neighbourhoods average more.

Q.131 [MCQ] B — LDA: shared Sigma, possibly correlated features.

Q.132 [MCQ] B — Rank(S_B) < =K-1.

Q.133 [NAT] 2 — max(0,1-(-1))=2.

Q.134 [NAT] 32 — 1-(1/5)^2-(4/5)^2=0.32.

Q.135 [MSQ] A;B;D — Flexibility can overfit; test error is U-shaped.

Q.136 [NAT] 9 — (1-4)^2=9.

Q.137 [MCQ] A — Tuning+evaluating on the same CV leaks.

Q.138 [NAT] -4 — y=2, (2-3)*4=-4.

Q.139 [NAT] 1 — Mean x=1, mean y=1.

Q.140 [NAT] 4 — Pune-Nashik L1=4 is the unique minimum (Kolhapur-Solapur is 5).

Q.141 [NAT] 3 — The zero eigenvalue carries no variance; three PCs suffice.

Q.142 [MCQ] A — Sigmoid score.

Q.143 [MCQ] A — The points lie on y=10x, so every residual is 0.

Q.144 [MSQ] A;B;C — A-C are linear in beta. D is logistic regression.

Q.145 [MCQ] A — L2 penalty is not scale-invariant.

Q.146 [MCQ] B — -log(0.5)=log 2.

Q.147 [NAT] 1 — Votes 3-2.

Q.148 [NAT] 12 — 2+10=12.

Q.149 [MCQ] A — LDA uses p(x,y); logistic uses only p(y|x).

Q.150 [NAT] 0.8 — max(0,1-0.2)=0.8.

Q.151 [MCQ] B — Deep trees interpolate labels.

Q.152 [MCQ] A — CV estimates extra-sample error.

Q.153 [NAT] 6 — Errors 3,0,3 squared 9,0,9 average 6.

Q.154 [NAT] 40 — Test 5, train 40.

Q.155 [NAT] 0.9 — w := 0.5 - 0.1*(-4)=0.9.

Q.156 [MCQ] B — Means generally lie off the data cloud.

Q.157 [MCQ] A — After PN, single-linkage next smallest unused is Kolhapur-Solapur=5.

Q.158 [MCQ] A — Eckart-Young / PCA optimality.

Q.159 [MCQ] C — b_yx * b_xy = r^2 always.

Q.160 [NAT] 8 — 12-4=8.

Q.161 [MSQ] A;C — Ridge shrinks but does not zero; paths are smooth.

Q.162 [MCQ] B — Logistic is a discriminative model for the log-odds.

Q.163 [MCQ] B — Nearest (0,0) and (2,4); mean response 2.

Q.164 [MCQ] C — 2/8=1/4.

Q.165 [NAT] 0 — I^{-1}(-2,0)=(-2,0).

Q.166 [MCQ] B — Large C ~ hard-margin.

Q.167 [MCQ] B — Each split is x_j ? t.

Q.168 [NAT] 3.25 — (4+9)/4=3.25.

Q.169 [MCQ] A — n highly correlated leave-one-out errors.

Q.170 [MCQ] B — Training on all n leaks the test point.

Q.171 [MCQ] A — Chain of factors each at most 1/4.

Q.172 [NAT] 2.5 — (0+0+0+10)/4=2.5.

Q.173 [NAT] 1 — Anu-Chetan L1=1 is smallest.

Q.174 [NAT] 0 — No variance left.

Q.175 [MSQ] A;B;D — Normal equations give 1^T e = 0 and x^T e = 0. yhat lives in col(X), so yhat^T e = 0.

Q.176 [MCQ] A — No omitted-variable bias; OLS is consistent and that entry - > 0.

Q.177 [NAT] 3 — 15/(3+2)=3.

Q.178 [NAT] 0.75 — P=3(1-P) = > P=0.75.

Q.179 [MCQ] B — Distance k-NN is not scale-invariant.

Q.180 [NAT] 2 — 0.14 vs 0.27, class 2.

Q.181 [MSQ] A;B;C — S_W^{-1} S_B need not be symmetric; one solves a generalised eigenproblem.

Q.182 [MCQ] B — w = sum alpha_i y_i x_i, alpha_i=0 for non-SVs.

Q.183 [NAT] 4 — internals=leaves-1.

Q.184 [MCQ] B — Degree n-1 interpolates n points.

Q.185 [NAT] 2 — Each is classified as the other label.

Q.186 [NAT] 1/3 — Equal logits = > uniform 1/3.

Q.187 [MCQ] A — Assignment step is nearest-centroid.

Q.188 [MCQ] A — Next min is Anu-Bala=2.

Q.189 [MCQ] A — Unsupervised vs supervised linear projection.

Q.190 [MCQ] B — b1 = r * s_y / s_x = 0.6*5/2 = 1.5.

Q.191 [MCQ] B — Partial slope of x2 is -2.

Q.192 [MCQ] B — Ridge cannot beat OLS on train SSE unless lambda=0.

Q.193 [MCQ] B — x > =1/2.

Q.194 [NAT] 1 — Neighbours 8,0,-1 labels 0,1,1 majority 1.

Q.195 [MCQ] B — Numerical underflow of a long product.

Q.196 [MCQ] A — LDA's shared Sigma may have off-diagonals.

Q.197 [NAT] 1 — (-2+4)/2=1.

Q.198 [MCQ] B — Regression trees minimise child SSE.

Q.199 [NAT] 4 — 2^2=4.

Q.200 [MCQ] B — Composition of linear maps is linear.

Q.201 [NAT] 9 — Sum of SSEs.

Q.202 [NAT] 3 — Complete = max.

Q.203 [NAT] 2 — Isotropic: every direction variance 2.

Q.204 [NAT] 0.75 — R^2 = 1-SSE/SST = 1-20/80 = 0.75.
Q.205 [NAT] 2 — b1=(x1^T y)/(x1^T x1)=20/10=2.
Q.206 [NAT] 0.5 — 2/(2+2)=0.5.
Q.207 [NAT] 0.25 — 0.5*0.5=0.25.
Q.208 [NAT] 1 — Majority 1.
Q.209 [NAT] 0.25 — (3+1)/(14+2)=4/16=0.25.
Q.210 [NAT] 0 — (-3+3)/2=0.
Q.211 [MSQ] A;B;C — Raw squared Euclidean is not a kernel by itself.
Q.212 [NAT] 0.25 — 2/8=0.25.
Q.213 [MCQ] A — Small k = > flexible, noisy boundary.
Q.214 [NAT] 15 — 4*3 weights + 3 biases = 15.
Q.215 [MCQ] B — Singletons have SSE 0.
Q.216 [MSQ] A;B;C — k is chosen by cutting the dendrogram later; it is not needed to start agglomerative HAC.
Q.217 [MCQ] A — Divide by sqrt(eigenvalue).
Q.218 [MCQ] C — b = (sum xy)/(sum x^2) = 8/14 = 4/7.
Q.219 [MCQ] A — dyhat/dx1 = b1 + b3 x2.
Q.220 [MCQ] A — Classic shrinkage trade-off under multicollinearity.
Q.221 [MCQ] B — Shrinkage drives w to 0.
Q.222 [MCQ] B — No parametric model is fitted at train time.
Q.223 [MCQ] B — Independence is badly false; likelihood is squared.
Q.224 [MCQ] B — Class-specific Sigma yields quadratic boundaries.
Q.225 [MCQ] B — phi includes the product x1 x2.
Q.226 [MSQ] A;B;D — XOR is two splits; no margin objective.
Q.227 [MCQ] A — Standard classification loss.
Q.228 [MCQ] A — Distances change under rescaling.
Q.229 [MCQ] A — Post-hoc cut of HAC.
Q.230 [NAT] 0 — PC scores of centered data have mean 0.
Q.231 [MCQ] B — Slope is shift-invariant in x; intercept adjusts by -b1*5.
Q.232 [NAT] 7 — 5+2=7.
Q.233 [NAT] 2.5 — Half of OLS: 2.5.
Q.234 [MCQ] B — Coefficients are log-odds slopes.
Q.235 [NAT] 1 — Winner (0,0), distance 1.
Q.236 [NAT] 0.7 — Uniform priors: posterior equals normalised likelihoods already summing to 1.
Q.237 [NAT] 8 — Means 1 and 9.
Q.238 [NAT] 2 — 4-2=2.
Q.239 [NAT] 0.5 — 1-0.5=0.5.
Q.240 [NAT] 1 — 2*1-1=1.
Q.241 [NAT] 5 — n-1 merges.
Q.242 [MSQ] A;B — Center always; standardise when units differ.
Q.243 [NAT] 2 — Slope (7-3)/(4-2)=2.
Q.244 [MCQ] B — Near-singular X^T X = > large diagonal of (X^T X)^{-1}.
Q.245 [MCQ] A — The defining linear system of ridge.
Q.246 [NAT] 0.5 — sigmoid(0)=0.5.
Q.247 [MCQ] B — Every training point is a neighbour.
Q.248 [MSQ] A;B;C — Separability is not required. Zero counts are fatal without smoothing.
Q.249 [MCQ] B — PCA is not LDA.
Q.250 [MCQ] A — Max-margin separator is unique in w for generic separable data.
Q.251 [MCQ] A — ID splits isolate each row.
Q.252 [MCQ] A — L2 regularisation on W.
Q.253 [MCQ] A — UPGMA-style average.
Q.254 [MCQ] A — Symmetric eigenproblem.
Q.255 [MCQ] B — A fan shape is non-constant variance.
Q.256 [MCQ] A — P=0.5 iff w^T x+b=0.
Q.257 [MCQ] A — 1-NN uses only the single nearest point.
Q.258 [NAT] 8 — 4+4=8.
Q.259 [MCQ] B — KKT: alpha=C usually means slack.
Q.260 [NAT] 0.3 — 7 vs 3 opposite labels, error 0.3.
Q.261 [NAT] 10 — 80/8=10.
Q.262 [MCQ] A — Classic chaining behaviour.
Q.263 [NAT] 0.9 — 9/10=0.9.
Q.264 [NAT] 3 — Two parameters, so df=5-2=3.
Q.265 [NAT] 0.09 — 0.9*0.1=0.09.
Q.266 [MCQ] B — 0.18/(0.18+0.24)=0.18/0.42=3/7~0.43.
Q.267 [NAT] 3 — Closest opposite pair -1 and +2, gap 3.
Q.268 [MCQ] A — That's the running example of a classification tree.
Q.269 [MCQ] A — Linear o linear = linear.
Q.270 [MCQ] A — Centered n=20 = > rank < =19.
Q.271 [MCQ] B — No x-variation means the slope is unidentified.
Q.272 [MCQ] A — No slack once the point is on the correct side of the margin.
Q.273 [NAT] 0.69 — ln 2 ~ 0.693 ~ 0.69.
Q.274 [MCQ] B — SSR=0 iff the slope is 0 when an intercept is present.
Q.275 [MCQ] A — ReLU'(z)=0 for z < 0.

UNIT 7 Artificial Intelligence · 269 questions

MCQ negative -1/3 or -2/3. MSQ / NAT no negative. Work first, then the solutions section.

Q.1 MCQ · 1 mark · UninformedSearch
Which of the following is an uninformed search strategy (no goal heuristic h(n))?
(A) A* with a consistent heuristic
(B) Greedy best-first using straight-line distance
(C) Breadth-first search
(D) Hill climbing on an evaluation function

Q.2 MCQ · 1 mark · InformedSearch
A* search ranks the frontier by
(A) f(n)=g(n)+h(n)
(B) h(n) only
(C) g(n) only
(D) depth only

Q.3 MCQ · 1 mark · Adversarial
In a two-player zero-sum game of perfect information, minimax at a MAX root returns
(A) the max over legal moves of the min-value of the resulting position
(B) the average of leaf values
(C) a random legal move
(D) A* f-values

Q.4 MCQ · 1 mark · PropLogic
A propositional formula is a tautology iff it is
(A) true under every truth assignment
(B) true under some assignment
(C) false under every assignment
(D) built only from implication

Q.5 MCQ · 1 mark · FOL
The FOL sentence forall x (Cat(x) - > Mammal(x)) together with Cat(Nori) entails
(A) Mammal(Nori)
(B) ~Mammal(Nori)
(C) forall x Cat(x)
(D) exists x ~Mammal(x)

Q.6 MCQ · 1 mark · BayesNet
A Bayesian network on variables V encodes a joint by
(A) the product of each node's CPT given its parents
(B) a single full table of size 2^{|V|} always stored explicitly as the only option
(C) k-means centroids
(D) CNF of the DAG

Q.7 MCQ · 1 mark · VarElim
Variable elimination computes a marginal by
(A) summing out hidden variables one at a time, multiplying and summing factors
(B) drawing Gibbs samples only
(C) running A* on the DAG
(D) converting the BN to CNF and SAT-solving always

Q.8 MCQ · 1 mark · Sampling
Prior (ancestral) sampling from a BN draws each node
(A) in topological order from P(node | already sampled parents)
(B) in reverse topological order always
(C) uniformly ignoring CPTs
(D) by minimax

Q.9 MSQ · 2 marks · UninformedSearch
Which of these search algorithms are uninformed?
(A) Breadth-first search
(B) Depth-first search
(C) Uniform-cost search
(D) A* search

Q.10 MCQ · 1 mark · InformedSearch
Greedy best-first search ranks the frontier by
(A) f=g+h
(B) h(n) only
(C) g(n) only
(D) alpha

Q.11 NAT · 2 marks · Adversarial
MAX root with two moves: Left to a MIN node with leaves 3 and 5; Right to a MIN node with leaves 2 and 9. The minimax value of the root is ____.
NAT · enter the numerical value

Q.12 MCQ · 1 mark · PropLogic
A propositional formula is satisfiable iff
(A) some assignment makes it true
(B) every assignment makes it true
(C) no assignment makes it true
(D) it is necessarily a tautology

Q.13 MCQ · 2 marks · FOL
exists x P(x) does NOT entail
(A) exists y P(y) (alpha-equivalent)
(B) P(A) for a previously unused constant A without existential instantiation care
(C) ~~ exists x P(x)
(D) exists x (P(x) v Q(x))

Q.14 MCQ · 2 marks · BayesNet
Alarm BN: B- > A < -E, A- > J, A- > M. The joint factorises as
(A) P(B)P(E)P(A|B,E)P(J|A)P(M|A)
(B) P(B)P(E)P(A)P(J)P(M)
(C) P(J)P(M)P(A|J,M)P(B,E)
(D) P(B,E,A,J,M) as 32 unconstrained numbers

Q.15 MCQ · 1 mark · VarElim
VE is an exact inference algorithm, whereas Gibbs sampling is
(A) approximate (MCMC)
(B) also exact after one sample
(C) uninformed search
(D) resolution

Q.16 MCQ · 1 mark · Sampling
Rejection sampling for P(Q|e) draws ancestral samples and
(A) throws away any sample inconsistent with evidence e, then histograms Q on the rest
(B) weights every sample by 1 regardless of evidence
(C) never uses the CPTs
(D) solves VE factors

Q.17 MCQ · 1 mark · UninformedSearch
On a tree with unit step costs, BFS is optimal because
(A) it expands by depth, so the first goal found is a shallowest and hence cheapest goal
(B) it expands the highest-h node first
(C) it is identical to DFS
(D) it requires an admissible heuristic

Q.18 MCQ · 1 mark · InformedSearch
A heuristic h is admissible if for every node n
(A) h(n) > = h*(n) (never underestimates)
(B) h(n) < = h*(n) (never overestimates true remaining cost)
(C) h(n)=0 for every n
(D) h is inconsistent

Q.19 MCQ · 1 mark · Adversarial
In that same tree (Left leaves 3,5; Right leaves 2,9), MAX's best first move is
(A) Left
(B) Right
(C) either, values equal
(D) pass

Q.20 MCQ · 1 mark · PropLogic
A formula is unsatisfiable (a contradiction) iff it is
(A) true in every model
(B) false under every truth assignment
(C) a tautology
(D) Horn

Q.21 MSQ · 2 marks · FOL
Which FOL statements are TRUE?
(A) forall x P(x) entails P(A) for any ground term A in the language
(B) P(A) entails exists x P(x)
(C) exists x forall y Likes(x,y) entails forall y exists x Likes(x,y)
(D) forall y exists x Likes(x,y) entails exists x forall y Likes(x,y)

Q.22 MSQ · 2 marks · BayesNet
In the Alarm network B- > A < -E, A- > J, A- > M, which conditional independences hold by d-separation?
(A) J ? M | A
(B) B ? E (unconditionally)
(C) B ? E | A (explaining-away collider)
(D) J ? B | A

Q.23 MCQ · 1 mark · VarElim
The complexity of VE is governed mainly by
(A) the size of the largest intermediate factor (induced width / treewidth related)
(B) only the number of edges, never clique size
(C) the minimax depth
(D) the CNF length

Q.24 MCQ · 1 mark · Sampling
Likelihood weighting, compared with rejection on a tiny BN with rare evidence,
(A) never rejects: it clamps evidence nodes and multiplies a weight P(e_i | parents)
(B) rejects more often than plain rejection
(C) is exact after one sample always
(D) requires a closed-form VE factor

Q.25 MCQ · 2 marks · UninformedSearch
DFS as graph-search with a closed set on a finite graph is
(A) complete on that finite graph but not cost-optimal in general
(B) optimal for any positive costs
(C) incomplete even with a closed set on a finite graph
(D) identical to uniform-cost search

Q.26 MCQ · 1 mark · InformedSearch
A heuristic is consistent (monotone) if for every edge n- > n' of cost c
(A) h(n) < = c + h(n') (triangle inequality)
(B) h(n) > = c + h(n')
(C) h(n)=g(n)
(D) h is zero only at the start

Q.27 NAT · 2 marks · Adversarial
MAX root, three MIN children with leaf-pairs (4,1), (6,6), (2,8). Minimax root value is ____.
NAT · enter the numerical value

Q.28 MSQ · 2 marks · PropLogic
Which of the following propositional formulae are tautologies?
(A) P v ~P
(B) P - > P
(C) ~(P & ~P)
(D) P & ~P

Q.29 MCQ · 1 mark · FOL
Negating forall x P(x) yields
(A) exists x ~P(x)
(B) forall x ~P(x)
(C) exists x P(x)
(D) ~P(x) with free x as a sentence

Q.30 MCQ · 2 marks · BayesNet
A collider chain A - > C < - B is blocked when
(A) C is not observed and no descendant of C is observed
(B) C is observed
(C) A is observed
(D) always never blocked

Q.31 NAT · 1 mark · VarElim
Factors f(A,B) and g(B,C) on binary vars. Their product h(A,B,C) has how many entries?
NAT · enter the numerical value

Q.32 MSQ · 2 marks · Sampling
On the tiny Rain- > Wet BN, evidence Wet=true (a child). Which are TRUE?
(A) Rejection sampling discards worlds with Wet=false
(B) Likelihood weighting sets Wet=true and weights by P(Wet=true | Rain)
(C) Both methods (with infinite samples) can estimate P(Rain | Wet=true)
(D) VE is approximate here while LW is exact in finite samples

Q.33 MCQ · 1 mark · UninformedSearch
Uniform-cost search expands nodes in order of increasing
(A) heuristic h(n)
(B) path cost g(n)
(C) depth only, ignoring action costs
(D) f=g+h with a nonzero h

Q.34 MSQ · 2 marks · InformedSearch
h1 and h2 are both admissible. Which combinations remain admissible?
(A) h(n)=max(h1(n),h2(n))
(B) h(n)=min(h1(n),h2(n))
(C) h(n)=h1(n)+h2(n) in general (no further assumption)
(D) h(n)=0

Q.35 MCQ · 1 mark · Adversarial
MAX root, MIN children (4,1), (6,6), (2,8). The unique best first move is toward the child whose min is
(A) 1
(B) 6
(C) 2
(D) 8

Q.36 MCQ · 1 mark · PropLogic
De Morgan: ~(P & Q) is equivalent to
(A) ~P & ~Q
(B) ~P v ~Q
(C) P v Q
(D) P - > Q

Q.37 MCQ · 1 mark · FOL
Negating exists x P(x) yields
(A) forall x ~P(x)
(B) exists x ~P(x)
(C) forall x P(x)
(D) P(x)

Q.38 MCQ · 1 mark · BayesNet
A chain A - > C - > B is blocked when
(A) C is observed
(B) C is unobserved
(C) A is a leaf
(D) the DAG has no edges

Q.39 NAT · 1 mark · VarElim
Summing h(A,B,C) over C (binary) yields a factor on A,B with how many entries?
NAT · enter the numerical value

Q.40 MCQ · 1 mark · Sampling
Gibbs sampling in a BN is a Markov chain that, at each step, resamples one variable from
(A) its full conditional given the current values of all other variables (or the Markov blanket)
(B) the prior P(node) ignoring the rest
(C) a uniform coin always
(D) A* f-values

Q.41 MCQ · 1 mark · UninformedSearch
Iterative-deepening search (IDS) is designed to combine
(A) the memory of DFS with the completeness of BFS on trees
(B) A* optimality with greedy memory
(C) minimax with UCS
(D) alpha-beta with variable elimination

Q.42 MCQ · 1 mark · InformedSearch
If h1 and h2 are admissible, the dominating choice among them is
(A) min(h1,h2)
(B) max(h1,h2)
(C) h1+h2
(D) h1-h2

Q.43 NAT · 1 mark · Adversarial
A MAX leaf that is also a terminal with payoff 7 contributes minimax value ____.
NAT · enter the numerical value

Q.44 MCQ · 1 mark · PropLogic
Implication P - > Q is equivalent to
(A) ~P v Q
(B) P v Q
(C) ~P & Q
(D) P & Q

Q.45 MCQ · 1 mark · FOL
Unification of P(x,A) and P(B,y) succeeds with mgu
(A) {x/B, y/A}
(B) {x/A, y/B}
(C) fail
(D) {x/y}

Q.46 MCQ · 1 mark · BayesNet
A fork A < - C - > B is blocked when
(A) C is observed
(B) C is unobserved
(C) B is a parent of A
(D) always open

Q.47 MCQ · 2 marks · VarElim
Elimination order can change
(A) intermediate factor sizes (efficiency) but not the exact answer (in exact arithmetic)
(B) the exact numerical marginal always
(C) d-separation
(D) the DAG edges

Q.48 MCQ · 1 mark · Sampling
The Markov blanket of a node in a BN is
(A) parents, children, and co-parents (other parents of the children)
(B) only the parents
(C) all non-descendants
(D) the empty set always

Q.49 NAT · 1 mark · UninformedSearch
A full binary tree, root at depth 0, a unique goal at depth 3. Number of nodes at depths 0 through 3 is ____.
NAT · enter the numerical value

Q.50 NAT · 1 mark · InformedSearch
True remaining cost h*=6. Heuristics 4 and 5, both admissible. max(h1,h2) at this node equals ____.
NAT · enter the numerical value

Q.51 MSQ · 2 marks · Adversarial
Which claims about minimax (deterministic, zero-sum, perfect information) are TRUE?
(A) It assumes the opponent plays optimally against you
(B) It can be computed bottom-up on a finite game tree
(C) It maximises the worst-case payoff
(D) It requires a heuristic h like A* even at terminals with known utility

Q.52 NAT · 1 mark · PropLogic
Number of distinct truth assignments to 3 atoms P,Q,R is ____.
NAT · enter the numerical value

Q.53 MCQ · 1 mark · FOL
Unification of P(x,x) and P(A,B) for distinct constants A,B
(A) succeeds with {x/A}
(B) fails occurs/constant clash
(C) succeeds with {A/B}
(D) succeeds with empty theta

Q.54 MCQ · 2 marks · BayesNet
Explaining away: observing Alarm, then also observing Burglary, typically
(A) lowers P(Earthquake | Alarm, Burglary) relative to P(Earthquake | Alarm)
(B) raises P(Earthquake | Alarm) necessarily to 1
(C) d-connects J and M given A
(D) makes B independent of E given A still

Q.55 MCQ · 1 mark · VarElim
To compute P(Q | e) by VE one typically
(A) reduces factors by evidence, eliminates hidden vars, then normalises the remaining factor on Q
(B) ignores evidence
(C) runs DFS on the moral graph only
(D) uses hinge loss

Q.56 NAT · 2 marks · Sampling
Rejection sampling: 100 ancestral draws, 40 match evidence, of those 10 have Q=true. Estimate of P(Q=true|e) is ____.
NAT · enter the numerical value

Q.57 MCQ · 2 marks · UninformedSearch
Road map: S-A cost 1, S-B cost 4, A-G cost 5, B-G cost 1. Uniform-cost search from S to G returns a path of cost
(A) 6 via A
(B) 5 via B
(C) 1
(D) 9

Q.58 NAT · 1 mark · InformedSearch
True remaining cost h*=6. Heuristics 4 and 5. The sum 4+5 equals 9, which is (1 if admissible at this node, else 0): report 0 because 9 > 6. That indicator is ____.
NAT · enter the numerical value

Q.59 MCQ · 1 mark · Adversarial
Alpha-beta pruning, with optimal move ordering, can reduce time to about
(A) O(b^{d/2}) instead of O(b^d)
(B) O(1)
(C) O(d)
(D) O(b^d) still, never better

Q.60 NAT · 1 mark · PropLogic
Number of models of P v Q on atoms {P,Q} is ____.
NAT · enter the numerical value

Q.61 MCQ · 2 marks · FOL
Occurs check: unify x and f(x)
(A) fails, because x occurs in f(x)
(B) succeeds with {x/f(x)} as a finite unifier
(C) succeeds with empty theta
(D) is the same as resolution

Q.62 NAT · 1 mark · BayesNet
Binary Alarm net B,E,A,J,M as above. Naive full joint table size is 2^5 = ____.
NAT · enter the numerical value

Q.63 NAT · 1 mark · VarElim
A factor on 5 binary variables has size ____.
NAT · enter the numerical value

Q.64 NAT · 1 mark · Sampling
Likelihood weighting: two samples with weights 0.2 and 0.6, Q values true and true. Weighted estimate of P(Q=true) is ____.
NAT · enter the numerical value

Q.65 NAT · 1 mark · UninformedSearch
Same map S-A=1, S-B=4, A-G=5, B-G=1. The cheapest S-to-G cost is ____.
NAT · enter the numerical value

Q.66 MCQ · 1 mark · InformedSearch
Straight-line distance to Bucharest is admissible for road-map path cost (km) because
(A) a straight line is a lower bound on any road distance
(B) a straight line always overestimates road distance
(C) it equals g(n)
(D) it is inconsistent on every map

Q.67 MCQ · 1 mark · Adversarial
Alpha in alpha-beta is
(A) the best (highest) value MAX can currently guarantee
(B) the best (lowest) value MIN can currently guarantee
(C) always 0
(D) the A* heuristic

Q.68 NAT · 1 mark · PropLogic
Number of models of P & Q on atoms {P,Q} is ____.
NAT · enter the numerical value

Q.69 MCQ · 2 marks · FOL
Prenex form pulls quantifiers front. A prenex equivalent of forall x P(x) - > Q (Q sentence, x not free in Q) is
(A) exists x (P(x) - > Q)
(B) forall x (P(x) - > Q)
(C) Q - > forall x P(x)
(D) P(x) - > Q with free x

Q.70 NAT · 2 marks · BayesNet
Same Alarm net: CPT parameters (unconstrained, not using last-prob=1-p) : 1+1+4+2+2 = ____.
NAT · enter the numerical value

Q.71 MSQ · 2 marks · VarElim
Which claims about VE versus sampling are TRUE?
(A) VE is exact (modulo numeric precision)
(B) Gibbs is approximate MCMC
(C) VE can be infeasible when treewidth is large even if n is moderate
(D) Gibbs always returns the exact marginal after 2 samples

Q.72 NAT · 2 marks · Sampling
LW: weights 0.2 (Q=1) and 0.6 (Q=0). Estimate of P(Q=1) is ____.
NAT · enter the numerical value

Q.73 MCQ · 2 marks · UninformedSearch
BFS versus DFS on a complete binary tree of depth 12 looking for a shallow goal at depth 2: BFS will typically
(A) find it after exploring only a small prefix of the tree, while DFS may dive to depth 12 first
(B) use less memory than DFS on that tree
(C) be identical to IDS at every step
(D) require an admissible heuristic

Q.74 MCQ · 2 marks · InformedSearch
Graph: S-A cost 1, S-B cost 2, A-G cost 3, B-G cost 1. Heuristic h(S)=2, h(A)=2, h(B)=1, h(G)=0. Is h admissible on these four nodes?
(A) Yes, because h < = true remaining: h*(S)=min(4,3)=3 > =2, h*(A)=3 > =2, h*(B)=1 > =1, h*(G)=0
(B) No, h(A)=2 underestimates nothing wait it is admissible as listed
(C) No, because h(B)=1 equals h*(B) which is forbidden
(D) No, because h(G) must be 1

Q.75 MCQ · 1 mark · Adversarial
Beta in alpha-beta is
(A) MAX's lower bound
(B) the best (lowest) value MIN can currently guarantee
(C) always +inf
(D) path cost g

Q.76 MCQ · 2 marks · PropLogic
CNF of (P - > Q) & (Q - > R) is
(A) (~P v Q) & (~Q v R)
(B) (P v Q) & (Q v R)
(C) ~P & ~Q & R
(D) P & Q & R

Q.77 MCQ · 2 marks · FOL
Skolemisation of exists y forall x P(x,y) introduces
(A) a Skolem constant c with forall x P(x,c)
(B) a Skolem function f(x) with P(x,f(x))
(C) nothing
(D) a propositional atom only

Q.78 MCQ · 2 marks · BayesNet
Student network: Difficulty - > Grade < - Intelligence, Intelligence - > SAT, Grade - > Letter. Then SAT ? Letter | Intelligence, Grade is
(A) true by d-separation? Path SAT-I-Grade-Letter: observing I blocks the fork at I, but Grade is also observed so the chain Grade- > Letter is not needed... Path SAT-I-Grade-Letter: I observed blocks SAT-I-Grade. No other path. So SAT ? Letter | I,G. The question says | Intelligence, Grade. TRUE.
(B) false because a chain remains
(C) undefined
(D) equivalent to k-means

Q.79 MCQ · 1 mark · VarElim
Pointwise product of factors f(X,Y) and g(Y,Z) is defined on
(A) the union of scopes {X,Y,Z}
(B) the intersection {Y} only
(C) the empty scope
(D) CNF clauses

Q.80 MCQ · 2 marks · Sampling
If evidence is a root node, ancestral rejection versus clamping that root
(A) wastes draws that do not match the root; clamping the root is just sampling from the posterior slice and is exact in distribution for descendants
(B) is identical to minimax
(C) cannot be done
(D) requires CNF

Q.81 MCQ · 1 mark · UninformedSearch
Depth-limited search with limit L is incomplete if
(A) every goal lies at depth greater than L
(B) the branching factor is 2
(C) step costs are 1
(D) the graph is a tree

Q.82 NAT · 1 mark · InformedSearch
In that graph S-A=1, S-B=2, A-G=3, B-G=1, cheapest S-G cost is ____.
NAT · enter the numerical value

Q.83 MCQ · 2 marks · Adversarial
A MIN node can prune a remaining child when
(A) its current value is already < = alpha (MAX would never choose this node)
(B) beta is +inf
(C) the child is a MAX node only
(D) h is admissible

Q.84 MCQ · 2 marks · PropLogic

CNF of ~(P - > Q): first ~(~P v Q) ? P & ~Q. The CNF clauses are

(A) the unit clauses {P} and {~Q}
(B) the single clause {P,~Q}
(C) {~P,Q}
(D) the empty set of clauses

Q.85 MCQ · 1 mark · FOL

Skolemisation of forall x exists y P(x,y) introduces

(A) a Skolem function f with forall x P(x,f(x))
(B) a single constant c and P(x,c) for all x, which is stronger
(C) nothing
(D) CNF of Q

Q.86 MCQ · 2 marks · BayesNet

WetGrass: Cloudy - > Sprinkler, Cloudy - > Rain, Sprinkler - > Wet, Rain - > Wet. Unconditionally Sprinkler ? Rain is

(A) false: they share the fork Cloudy, which is unobserved, so they are dependent
(B) true by a collider at Wet
(C) true because they are roots
(D) true iff Wet is observed

Q.87 MCQ · 1 mark · VarElim

Summing out Y from f(X,Y) produces a factor on

(A) X
(B) Y
(C) X,Y still
(D) nothing always empty

Q.88 MSQ · 2 marks · Sampling

Which are approximate inference methods in the GATE DA syllabus?

(A) rejection sampling
(B) likelihood weighting
(C) Gibbs sampling
(D) variable elimination

Q.89 NAT · 1 mark · UninformedSearch

IDS with successive limits 0,1,2,3 on a binary tree. Number of distinct depth-limit runs is ____.

NAT · enter the numerical value

Q.90 MCQ · 1 mark · InformedSearch

A* with a consistent h on graph-search expands nodes in nondecreasing f-order and is

(A) optimal (returns a cheapest path)
(B) suboptimal always
(C) identical to DFS
(D) uninformed

Q.91 MCQ · 2 marks · Adversarial

A MAX node can prune a remaining child when

(A) its current value is already > = beta (MIN would never choose this node)
(B) alpha is -inf always
(C) the branching factor is 1
(D) UCS is running

Q.92 MCQ · 1 mark · PropLogic

CNF of P < - > Q uses (P- > Q)&(Q- > P), i.e.

(A) (~P v Q) & (~Q v P)
(B) P v Q
(C) ~P v ~Q
(D) P & Q only

Q.93 MSQ · 2 marks · FOL

Regarding entailment, which hold?

(A) forall x (P(x) - > Q(x)), P(A) |= Q(A)
(B) exists x (P(x) & Q(x)) |= (exists x P(x)) & (exists x Q(x))
(C) (exists x P(x)) & (exists x Q(x)) |= exists x (P(x) & Q(x))
(D) forall x (P(x) & Q(x)) |= (forall x P(x)) & (forall x Q(x))

Q.94 MSQ · 2 marks · BayesNet

WetGrass network C- > S, C- > R, S- > W, R- > W. Which hold?

(A) S ? R | C
(B) S not? R (marginally, generally)
(C) S not? R | W (collider W opens explaining away)
(D) W ? C | S,R is false in general? actually W is a child of S,R so given S,R, W is independent of C (no path around). W ? C | S,R holds.

Q.95 NAT · 2 marks · VarElim

Chain A-B-C all binary. Eliminating B from f(A,B)*g(B,C) produces a factor of size ____.

NAT · enter the numerical value

Q.96 MCQ · 1 mark · Sampling

Gibbs sampling needs a burn-in because

(A) the chain starts from an arbitrary state and only approaches the target posterior after mixing
(B) VE factors must warm up
(C) alpha-beta cutoffs burn nodes
(D) CNF conversion is random

Q.97 MSQ · 2 marks · UninformedSearch

Which statements about uniform-cost search are TRUE?

(A) With nonnegative costs, the first time a node is expanded it has an optimal g-value (graph-search)
(B) UCS with all costs equal to 1 behaves like BFS (up to tie-breaking)
(C) UCS is informed because it uses g(n)
(D) Negative action costs can break UCS optimality

Q.98 MCQ · 2 marks · InformedSearch

A* with an admissible but inconsistent h, if one uses tree-search (reopen paths), is still

(A) optimal
(B) incomplete on finite graphs
(C) the same as greedy
(D) forbidden in GATE

Q.99 NAT · 2 marks · Adversarial

MAX root; left MIN leaves 3 then 5; right MIN first leaf 2, then a second leaf x. Alpha-beta (left-to-right) prunes the leaf x no matter what x is. The root value is ____.

NAT · enter the numerical value

Q.100 MSQ · 2 marks · PropLogic
Which formulae are equivalent to P - > (Q - > P)?
(A) a tautology (true under all four assignments of P,Q)
(B) ~P v (~Q v P)
(C) (P & Q) - > P
(D) P & ~P

Q.101 MCQ · 2 marks · FOL
The FOL implication forall x (Student(x) - > Smart(x)) is NOT equivalent to
(A) forall x (~Student(x) v Smart(x))
(B) exists x (Student(x) & ~Smart(x))
(C) ~ exists x (Student(x) & ~Smart(x))
(D) the same sentence with a renamed bound variable y

Q.102 MCQ · 1 mark · BayesNet
Local Markov condition in a BN: a node is independent of its non-descendants given
(A) its parents
(B) its children
(C) the empty set always
(D) all variables

Q.103 MCQ · 2 marks · VarElim
A bad elimination order on a grid-like BN can create
(A) factors whose scope is a long cutset, exponential in the cut width
(B) only size-2 factors always
(C) CNF tautologies
(D) A* nodes

Q.104 NAT · 1 mark · Sampling
Three Gibbs samples of a binary Q after burn-in: 1,0,1. Estimate of P(Q=1) is ____.
NAT · enter the numerical value

Q.105 MCQ · 1 mark · UninformedSearch
A search node is a data structure storing at least
(A) a state, a parent pointer (or path), and a path cost g
(B) only a heuristic, never a state
(C) only the goal test
(D) a full Bayesian network

Q.106 NAT · 1 mark · InformedSearch
Node with g=4, h=3. A* f-value is ____.
NAT · enter the numerical value

Q.107 MCQ · 2 marks · Adversarial
In that left-to-right alpha-beta tree (left leaves 3,5; right first leaf 2 then x), the pruned leaf is
(A) the 3
(B) the 5
(C) the 2
(D) the second right leaf x

Q.108 MCQ · 1 mark · PropLogic
Modus ponens: from P and (P - > Q) one infers
(A) Q
(B) ~Q
(C) P & ~Q
(D) ~P

Q.109 MCQ · 1 mark · FOL
Ground atomic sentence P(A) in a Herbrand interpretation is true iff
(A) the interpretation's set of true ground atoms contains P(A)
(B) A is a variable
(C) P is a function symbol
(D) the sentence is CNF of Q

Q.110 MCQ · 1 mark · BayesNet
d-separation is a graphical test for
(A) conditional independences implied by the DAG (Markov condition)
(B) minimax values
(C) CNF tautology
(D) A* consistency

Q.111 MCQ · 1 mark · VarElim
Evidence V=v is incorporated in VE by
(A) restricting factors whose scope includes V to the slice V=v
(B) adding a random CPT
(C) dropping the DAG
(D) running minimax

Q.112 MCQ · 1 mark · Sampling
A problem with rejection sampling when P(e) is 10^{-6} is
(A) almost every sample is thrown away, so the estimator is high-variance / starved
(B) the method becomes exact too quickly
(C) VE is then approximate
(D) d-separation fails

Q.113 NAT · 1 mark · UninformedSearch
Graph: S-A 2, S-C 1, A-G 1, C-G 4. Cheapest S-G cost is ____.
NAT · enter the numerical value

Q.114 MCQ · 1 mark · InformedSearch
If h=0 everywhere, A* reduces to
(A) DFS
(B) UCS
(C) greedy best-first
(D) alpha-beta

Q.115 NAT · 1 mark · Adversarial
Worst-case alpha-beta (adversarial ordering) on a tree of 8 leaves still evaluates how many leaves?
NAT · enter the numerical value

Q.116 MCQ · 1 mark · PropLogic
Resolution on clauses {P,Q} and {~P,R} yields
(A) {Q,R}
(B) {P,~P}
(C) the empty clause always
(D) {Q} only

Q.117 NAT · 1 mark · FOL
Language with one constant A and one unary function f. Number of ground terms of syntactic depth at most 1 (A and f(A)) is ____.
NAT · enter the numerical value

Q.118 NAT · 2 marks · BayesNet
A chain of 4 binary nodes A- > B- > C- > D, each CPT with 2 free params except A with 1. Free parameter count using 1-p normalisation: 1+2+2+2= ____.
NAT · enter the numerical value

Q.119 NAT · 2 marks · VarElim
Binary Q with unnormalised VE result [2, 6]. P(Q=1) after normalising is ____.
NAT · enter the numerical value

Q.120 MCQ · 2 marks · Sampling
Likelihood weighting still has high variance when
(A) evidence nodes are deep descendants with unlikely values given randomly sampled parents
(B) there is no evidence
(C) the BN has one node
(D) all weights equal 1 and evidence is a root that was clamped consistently

Q.121 MCQ · 1 mark · UninformedSearch
On an infinite tree with no goal, DFS is
(A) complete
(B) incomplete (it can follow an infinite branch)
(C) optimal
(D) equivalent to A*

Q.122 MCQ · 1 mark · InformedSearch
If one ignores g and ranks by h only, the algorithm is
(A) UCS
(B) greedy best-first
(C) BFS
(D) IDS

Q.123 NAT · 1 mark · Adversarial
A complete binary game tree of depth 2 (four leaves). Minimax examines all 4 leaves. Report that count: ____.
NAT · enter the numerical value

Q.124 NAT · 1 mark · PropLogic
Resolution of the unit clauses {P} and {~P} produces a clause with how many literals?
NAT · enter the numerical value

Q.125 MCQ · 1 mark · FOL
Universal instantiation: from forall x P(x) infer
(A) P(t) for a ground term t
(B) exists x ~P(x)
(C) P(x) with x still bound incorrectly as a sentence
(D) minimax(P)

Q.126 MCQ · 1 mark · BayesNet
If a BN's DAG is fully connected (topological complete DAG), the number of free parameters is
(A) the same order as an unconstrained joint (no CI savings)
(B) always 1
(C) always |V|
(D) zero

Q.127 NAT · 1 mark · VarElim
Unnormalised [1,1,2] on a 3-value var. P(first value) is ____.
NAT · enter the numerical value

Q.128 NAT · 1 mark · Sampling
Ancestral sample on A- > B with P(A=1)=0.5, P(B=1|A=1)=1, P(B=1|A=0)=0. Draw A=1 then B is ____.
NAT · enter the numerical value

Q.129 MCQ · 1 mark · UninformedSearch
BFS graph-search memory, branching factor b, goal depth d, is on the order of
(A) O(bd) like DFS
(B) O(b^d) nodes in the frontier
(C) O(1)
(D) O(d) only

Q.130 MSQ · 2 marks · InformedSearch
Which heuristics for 8-puzzle Manhattan / misplaced-tiles are admissible?
(A) Number of misplaced tiles
(B) Sum of Manhattan distances of tiles to goals
(C) Twice the Manhattan sum (generally)
(D) h=0

Q.131 MCQ · 1 mark · Adversarial
Evaluation function at a truncated depth is used because
(A) the full tree may be too large; a heuristic estimates the utility of a nonterminal
(B) minimax is undefined on terminals
(C) alpha is always a heuristic
(D) BFS needs h

Q.132 MCQ · 1 mark · PropLogic
The empty clause in resolution proofs means
(A) a contradiction has been derived (the clause set is unsatisfiable)
(B) a tautology was derived
(C) the set is satisfiable
(D) a Horn query that is true

Q.133 MCQ · 1 mark · FOL
Existential generalisation: from P(A) infer
(A) exists x P(x)
(B) forall x P(x)
(C) ~P(A)
(D) A=A only

Q.134 MCQ · 2 marks · BayesNet
Moralisation of a DAG (for conversion to an undirected Markov network) adds edges between
(A) co-parents (nodes sharing a child) then drops directions
(B) all non-adjacent pairs
(C) leaves only
(D) roots only

Q.135 MCQ · 1 mark · VarElim
If all hidden variables are summed out and Q is binary, the remaining factor has
(A) 2 entries (then normalised to a Bernoulli)
(B) 2^n entries always
(C) 0 entries
(D) n leaves of a game tree

Q.136 MCQ · 2 marks · Sampling
Importance-sampling view: LW is importance sampling whose proposal
(A) samples non-evidence nodes from the prior CPTs with evidence clamped
(B) samples from the exact posterior always
(C) samples uniformly in {0,1}^n ignoring CPTs
(D) is Gibbs' full conditional

Q.137 NAT · 1 mark · UninformedSearch
Unit-cost tree, goal at depth 2, branching 3. Number of nodes at the goal depth is ____.
NAT · enter the numerical value

Q.138 NAT · 1 mark · InformedSearch
8-puzzle: two tiles each 1 step from home, others home. Misplaced-tile h equals ____.
NAT · enter the numerical value

Q.139 NAT · 1 mark · Adversarial
MAX root with one move to a terminal 4. Minimax value is ____.
NAT · enter the numerical value

Q.140 MCQ · 1 mark · PropLogic
A Horn clause has at most one positive literal. Which of these is Horn?
(A) ~P v ~Q v R
(B) P v Q
(C) P v Q v R
(D) ~P v Q v R

Q.141 MCQ · 2 marks · FOL
Conversion of FOL to clausal form includes: implications out, negations in, standardise variables, prenex, Skolemise, drop universals, distribute. Dropping universals is valid after Skolemisation because
(A) remaining variables are implicitly universally quantified in clauses
(B) there are never variables left
(C) Skolem functions remove all variables
(D) CNF cannot have variables

Q.142 NAT · 1 mark · BayesNet
Binary node with 3 binary parents. CPT rows: 2^3 = ____.
NAT · enter the numerical value

Q.143 MSQ · 2 marks · VarElim
Exact inference algorithms in the GATE DA uncertainty syllabus include
(A) variable elimination
(B) enumeration of the joint (brute force)
(C) Gibbs sampling as an exact closed form
(D) likelihood weighting as an exact closed form

Q.144 MCQ · 1 mark · Sampling
To estimate P(Wet=true) with no evidence on WetGrass, the most direct sampler is
(A) ancestral (prior) sampling, no rejection needed
(B) rejection with empty evidence still throwing 50%
(C) alpha-beta
(D) resolution

Q.145 MCQ · 2 marks · UninformedSearch
Bidirectional uninformed search (when a predecessor function exists) meets in the middle to cut
(A) the exponent in b^{d/2} + b^{d/2} versus b^d
(B) the need for a goal test
(C) all path costs to 0
(D) the closed set

Q.146 NAT · 1 mark · InformedSearch
Same two tiles each Manhattan 1, others 0. Manhattan h equals ____.
NAT · enter the numerical value

Q.147 MCQ · 1 mark · Adversarial
Negamax is a reformulation of minimax that
(A) uses a single recursive function with a sign flip, equivalent on zero-sum games
(B) solves Bayesian networks
(C) is uninformed search
(D) requires CNF

Q.148 MCQ · 1 mark · PropLogic
Forward chaining on Horn clauses is complete for
(A) atomic entailments from a Horn KB
(B) all tautologies of arbitrary CNF
(C) first-order validity in general
(D) minimax values

Q.149 MCQ · 1 mark · FOL
Which pair is a unifier of Knows(John,x) and Knows(John,Jane)?
(A) {x/Jane}
(B) {x/John}
(C) fail
(D) {Jane/x, John/Jane}

Q.150 NAT · 1 mark · BayesNet
That CPT with 1-p normalisation has how many free parameters?
NAT · enter the numerical value

Q.151 MCQ · 2 marks · VarElim
VE on a polytree (singly connected BN) with a good order is
(A) polynomial in the CPT sizes (belief propagation style)
(B) always exponential in n regardless of structure
(C) undecidable
(D) the same as alpha-beta

Q.152 NAT · 1 mark · Sampling
Prior sampling 20 draws of Rain, 4 are true. Estimate of P(Rain=true) is ____.
NAT · enter the numerical value

Q.153 MCQ · 1 mark · UninformedSearch
A Grok campus map is an unweighted graph. To find a fewest-hops walk from Lab to Canteen one should run
(A) BFS
(B) DFS (which is hop-optimal)
(C) greedy best-first with a random h
(D) alpha-beta

Q.154 MCQ · 1 mark · InformedSearch
Greedy best-first can be badly suboptimal because
(A) it ignores g(n), so a cheap-looking fringe node may sit on an expensive path
(B) it is the same as UCS
(C) h is always 0
(D) it expands every node

Q.155 MSQ · 2 marks · Adversarial
Which are TRUE of alpha-beta versus plain minimax on the same tree and same left-to-right order?
(A) The returned root value is the same
(B) Some leaves may not be evaluated
(C) Alpha-beta can change the identity of an optimal root move only if there are ties, not the value
(D) Alpha-beta can return a different numeric value than minimax on the same finite tree

Q.156 NAT · 1 mark · PropLogic
Truth table of P - > Q: number of T rows among 4 is ____.
NAT · enter the numerical value

Q.157 MCQ · 2 marks · FOL
Most general unifier of P(x,f(y)) and P(g(z),f(A)) is
(A) {x/g(z), y/A} (or {x/g(z), y/A} up to renaming z)
(B) {x/A, y/g(z)}
(C) fail because f occurs
(D) {z/x}

Q.158 MCQ · 2 marks · BayesNet
A v-structure (collider) A- > C < -B with A not adjacent to B is the pattern that
(A) distinguishes essential graph orientation in many structure-learning arguments
(B) is illegal in BNs
(C) forces A ? B | C
(D) equals a chain

Q.159 NAT · 1 mark · VarElim
Product of two factors each of size 4, scopes {A,B} and {A,B} (same scope, binary). Result size is ____.
NAT · enter the numerical value

Q.160 MSQ · 2 marks · Sampling
Comparing VE, LW, and Gibbs on a 5-node binary Alarm net for P(B | J=true, M=true):
(A) VE can compute the exact rational (or floating) value from the CPTs
(B) LW gives a weighted Monte Carlo estimate
(C) Gibbs gives an MCMC estimate using Markov-blanket updates
(D) all three are guaranteed to match to 10 decimals after 2 samples

Q.161 NAT · 2 marks · UninformedSearch
UCS on S-A=10, S-G=100, A-G=1. Cost of the path UCS returns is ____.
NAT · enter the numerical value

Q.162 MCQ · 2 marks · InformedSearch
Weighted A* with f=g+W h, W > 1, typically
(A) trades optimality for speed (may return a suboptimal path)
(B) is always optimal for any W
(C) becomes DFS
(D) zeros the heuristic

Q.163 NAT · 2 marks · Adversarial
Leaves of a MAX-over-MIN tree in left-to-right order: 1,8 | 6,4. Minimax root is max(min(1,8),min(6,4))=max(1,4)= ____.
NAT · enter the numerical value

Q.164 MCQ · 1 mark · PropLogic
KB entails sentence S (written KB |= S) iff
(A) every model of KB is a model of S
(B) some model of KB is a model of S
(C) KB is unsatisfiable always
(D) S is a Horn query only

Q.165 MSQ · 2 marks · FOL
X is forall x (P(x)- > Q(x)). Which sentences does X entail?
(A) P(A) - > Q(A)
(B) forall y (P(y)- > Q(y))
(C) ~Q(A) - > ~P(A)
(D) exists x P(x) (without further premises)

Q.166 MSQ · 2 marks · BayesNet
Named factorisation P(X)P(Y)P(Z|X,Y). Which independences are implied?
(A) X ? Y (no edge, no common child observed)
(B) X ? Z | Y is not implied (X is a parent of Z)
(C) X not? Z
(D) Z ? Y | X is not implied (Y is a parent of Z)

Q.167 NAT · 1 mark · VarElim
After eliminating all hidden vars, unnormalised [5, 15] for binary Q. P(Q=1) is ____.
NAT · enter the numerical value

Q.168 MCQ · 1 mark · Sampling
A valid Gibbs update for node X given the rest uses only
(A) X's Markov blanket values (parents, children, co-parents)
(B) the leaves of a game tree
(C) CNF clauses of X
(D) the A* open list

Q.169 MCQ · 1 mark · UninformedSearch
Closed-set graph-search is used to
(A) avoid re-expanding a repeated state
(B) compute d-separation
(C) convert a formula to CNF
(D) sample a Bayesian network

Q.170 NAT · 1 mark · InformedSearch
Start f=g+h with g=0, h(S)=5. The first f(S) is ____.
NAT · enter the numerical value

Q.171 MCQ · 2 marks · Adversarial
Leaves 1,8 | 6,4 as above, left-to-right alpha-beta. After left MIN=1, alpha=1. Right MIN sees 6, then 4. Is 4 pruned?
(A) No, because 6 > alpha so MIN must look further; 4 is evaluated and min becomes 4
(B) Yes, 6 already prunes
(C) The 1 is pruned
(D) The 8 is pruned

Q.172 MCQ · 2 marks · PropLogic
If KB is unsatisfiable, then for every S
(A) KB |= S (explosion / ex falso)
(B) KB entails nothing
(C) S must be a tautology independently
(D) resolution cannot run

Q.173 MCQ · 2 marks · FOL
The sentence exists x (P(x) - > Q(x)) is true in a structure if
(A) some element satisfies ~P or Q (quite weak)
(B) every element is P and Q
(C) P is empty necessarily
(D) Q is empty necessarily

Q.174 MCQ · 1 mark · BayesNet
Flu - > Fever, Flu - > Cough. Observing Flu, Fever ? Cough if there is no other path. This is a
(A) fork blocked by observing the common cause
(B) collider opening
(C) chain through Fever to Cough
(D) uninformed search

Q.175 NAT · 1 mark · VarElim
Eliminating a binary var from a factor of scope 3 binaries (size 8) yields size ____.
NAT · enter the numerical value

Q.176 NAT · 1 mark · Sampling
LW weight for evidence child E=true with P(E=true|parents)=0.3 is a factor ____.
NAT · enter the numerical value

Q.177 MSQ · 2 marks · UninformedSearch
Which uninformed methods are complete on finite graphs with nonnegative costs (graph-search)?
(A) BFS
(B) UCS
(C) DFS with a closed set
(D) DLS with a limit smaller than the goal depth

Q.178 MCQ · 2 marks · InformedSearch
A* graph-search with a consistent heuristic never needs to reopen a closed node because
(A) the first time a node is expanded, g is already optimal
(B) h is always 0
(C) the frontier is a stack
(D) cycles have zero cost

Q.179 NAT · 2 marks · Adversarial
MAX-MIN tree leaves 7,3 | 2,9. Root minimax value is ____.
NAT · enter the numerical value

Q.180 MSQ · 2 marks · PropLogic
Which statements are TRUE?
(A) If S is a tautology then KB |= S for every KB
(B) P |= P v Q
(C) P & ~P |= Q
(D) P v Q |= P always

Q.181 MCQ · 2 marks · FOL
FOL validity is
(A) undecidable in general (Church/Turing) though semi-decidable by proof procedures
(B) decidable by a 2-SAT algorithm
(C) the same as 2-SAT
(D) solved by k-means

Q.182 MCQ · 1 mark · BayesNet
Named factorisation P(Cloudy)P(Rain|Cloudy)P(Sprinkler|Cloudy)P(Wet|Rain,Sprinkler). Wet's parents are
(A) Rain and Sprinkler
(B) only Cloudy
(C) the empty set
(D) every variable

Q.183 MCQ · 1 mark · Sampling
If every LW sample has weight 0, the usual cause is
(A) the clamped evidence is impossible given the sampled parents (CPT zeros)
(B) VE failed
(C) the DAG has no edges and evidence is a root with P=1
(D) minimax pruned the BN

Q.184 MCQ · 1 mark · UninformedSearch
The frontier (open list) of BFS is a
(A) FIFO queue
(B) LIFO stack
(C) priority queue on h(n)
(D) minimax game tree

Q.185 MCQ · 1 mark · InformedSearch
An overestimating heuristic (not admissible) can make A*
(A) miss a truly cheaper path, hence suboptimal
(B) always faster and still optimal
(C) identical to BFS always
(D) refuse to run

Q.186 MCQ · 1 mark · Adversarial
MAX-MIN leaves 7,3 | 2,9. Best first move is the left child because
(A) min(7,3)=3 > min(2,9)=2
(B) min(2,9) is larger
(C) 9 is the root value
(D) 7 is already the answer

Q.187 MCQ · 1 mark · PropLogic
Distributivity needed in CNF: P v (Q & R) becomes
(A) (P v Q) & (P v R)
(B) (P & Q) v R
(C) P v Q v R
(D) ~P

Q.188 NAT · 1 mark · FOL
Arity of a binary predicate Likes(x,y) is ____.
NAT · enter the numerical value

Q.189 MCQ · 2 marks · BayesNet
In Alarm, JohnCalls is a descendant of Alarm. J ? E | A holds because the path J-A-E is a
(A) chain/fork through A, blocked by observing A
(B) collider at A that opens
(C) direct edge J-E
(D) cycle

Q.190 MCQ · 1 mark · Sampling
Mixing of Gibbs is slow when variables are
(A) strongly dependent (tight CPTs / near-deterministic relations)
(B) all independent (then Gibbs is unnecessary but mixes immediately)
(C) absent
(D) CNF atoms

Q.191 MCQ · 1 mark · UninformedSearch
The frontier of DFS is a
(A) FIFO queue
(B) LIFO stack
(C) priority queue on g(n)
(D) CNF clause set

Q.192 NAT · 1 mark · InformedSearch
h1=3, h2=5, both admissible at n, h*=8. The dominating admissible combo max= ____.
NAT · enter the numerical value

Q.193 NAT · 2 marks · Adversarial
A MIN root (MIN to move) with MAX children having leaves (2,5) and (4,1). Minimax value at the MIN root is ____.
NAT · enter the numerical value

Q.194 NAT · 1 mark · PropLogic
CNF of P v (Q & R) has how many clauses?
NAT · enter the numerical value

Q.195 MCQ · 1 mark · FOL
A term is built from
(A) variables, constants, and function symbols; it is not a formula by itself
(B) quantifiers only
(C) implication only
(D) clauses of CNF only

Q.196 NAT · 1 mark · Sampling
Rejection: 50 samples, 0 consistent with evidence. The number of usable samples for the histogram is ____.
NAT · enter the numerical value

Q.197 MCQ · 1 mark · UninformedSearch
The frontier of UCS is a
(A) FIFO queue
(B) LIFO stack
(C) priority queue ordered by g(n)
(D) random set

Q.198 MCQ · 1 mark · InformedSearch
Manhattan distance on a 4-connected grid with only unit orthogonal moves is
(A) admissible and consistent
(B) inadmissible because it overestimates
(C) equal to Euclidean always
(D) a minimax value

Q.199 NAT · 1 mark · Adversarial
Tic-style toy: MAX can move to terminals 1, 0, -1. Minimax chooses the move of value ____.
NAT · enter the numerical value

Q.200 MCQ · 1 mark · PropLogic
A literal is
(A) an atom or a negated atom
(B) a conjunction of clauses
(C) a tautology only
(D) an A* node

Q.201 MCQ · 2 marks · FOL
Substitution {x/A} applied to forall x P(x) (bound x)
(A) does not capture the bound x; the sentence is unchanged
(B) becomes forall A P(A) which is ill-formed and intended
(C) entails P(A) by substitution into the binder
(D) fails occurs check

Q.202 MCQ · 2 marks · Sampling
An unbiasedness remark: rejection sampling's histogram on accepted samples is unbiased for P(Q|e) (given at least one accept). LW's weighted average is
(A) also unbiased for P(Q|e) (self-normalised LW is consistent, slightly biased in finite samples)
(B) exact for n=1 always
(C) a minimax value
(D) a tautology

Q.203 NAT · 1 mark · UninformedSearch
A path S-A-B-G with costs 2,2,2. Path cost g(G) is ____.
NAT · enter the numerical value

Q.204 MCQ · 2 marks · InformedSearch
Euclidean distance on that same 4-connected unit grid is
(A) still admissible (a lower bound) but looser than Manhattan
(B) inadmissible
(C) always larger than Manhattan
(D) equal to g(n)

Q.205 MCQ · 1 mark · Adversarial
A chance node in expectiminimax (dice) is evaluated by
(A) the probability-weighted average of children, not min or max
(B) always min
(C) always max
(D) A* f-value

Q.206 MCQ · 1 mark · PropLogic
Unit propagation: a unit clause {L} forces
(A) L true, so ~L is deleted from other clauses and clauses containing L are satisfied
(B) the empty clause immediately in every KB
(C) all atoms true
(D) greedy search

Q.207 MCQ · 1 mark · FOL
From ~ exists x P(x) one may infer
(A) forall x ~P(x)
(B) exists x P(x)
(C) P(A)
(D) forall x P(x)

Q.208 MCQ · 2 marks · Sampling
On a two-node BN A- > E with evidence E=true, likelihood weighting draws A from P(A) then
(A) sets E=true and weights by P(E=true|A)
(B) rejects if A=false always
(C) runs minimax on A
(D) unifies A with E

Q.209 MCQ · 1 mark · UninformedSearch
If every action cost is strictly positive, UCS cannot have
(A) an infinite cheap-loop problem of zero-cost cycles
(B) a finite graph
(C) a priority queue
(D) a start state

Q.210 NAT · 1 mark · InformedSearch
Grid from (0,0) to (3,4). Manhattan h at the start is ____.
NAT · enter the numerical value

Q.211 MCQ · 1 mark · Adversarial
Horizon effect in depth-limited minimax is
(A) a bad tactic pushed just beyond the cutoff looking fine at the leaf evaluator
(B) alpha always equalling beta
(C) UCS failing on unit costs
(D) CNF explosion

Q.212 NAT · 2 marks · PropLogic
Atoms P,Q. Number of inequivalent propositional formulae (up to logical equivalence) is ____.
NAT · enter the numerical value

Q.213 NAT · 1 mark · FOL
Unification of f(x) with f(A) uses how many bindings in the mgu {x/A}?
NAT · enter the numerical value

Q.214 MCQ · 2 marks · Sampling
Metropolis-Hastings is a general MCMC; Gibbs is the special case where the proposal is
(A) the full conditional, so the acceptance probability is 1
(B) uniform over the whole joint always without conditionals
(C) A* with h=0
(D) unit-clause propagation

Q.215 MCQ · 2 marks · UninformedSearch
IDS time complexity on a tree (goal depth d, branching b) is still O(b^d) because
(A) the last iteration dominates the geometric series of earlier iterations
(B) each iteration is O(1)
(C) IDS is exponential in the memory
(D) IDS uses A* internally

Q.216 NAT · 1 mark · InformedSearch
Grid from (0,0) to (3,4). Euclidean h at the start is ____.
NAT · enter the numerical value

Q.217 NAT · 1 mark · Adversarial
Branching factor 3, depth-2 full game tree. Number of leaves is ____.
NAT · enter the numerical value

Q.218 MCQ · 1 mark · PropLogic
P |= Q is FALSE in particular when
(A) P is true and Q is false in some assignment (a countermodel)
(B) P is a contradiction
(C) Q is a tautology
(D) P is P & P

Q.219 MCQ · 1 mark · FOL
Resolution in FOL uses unification to
(A) align complementary literals before resolving, via an mgu
(B) compute A* f-values
(C) d-separate a BN
(D) sample Gibbs

Q.220 MCQ · 2 marks · Sampling
Gibbs versus VE on Alarm P(B|J,M): after 10^7 well-mixed Gibbs samples the histogram
(A) approximates the same number VE would compute exactly from the CPTs
(B) is always 0.5 regardless of CPTs
(C) solves SAT
(D) returns a tautology

Q.221 NAT · 1 mark · UninformedSearch
Depth-limited search with L=0 on a nonempty state space expands (goal-test on the root only) how many nodes?
NAT · enter the numerical value

Q.222 MCQ · 1 mark · InformedSearch
IDA* is iterative-deepening A*, i.e.
(A) depth-first searches with a rising f-bound
(B) BFS with h=0
(C) minimax of heuristics
(D) Gibbs sampling of paths

Q.223 NAT · 1 mark · Adversarial
Branching 2, depth 3. Number of leaves is ____.
NAT · enter the numerical value

Q.224 MCQ · 2 marks · PropLogic
Conversion of (P - > Q) - > R to CNF: first rewrite as ~(~P v Q) v R ? (P & ~Q) v R ? (P v R) & (~Q v R). The CNF has clauses
(A) {P,R} and {~Q,R}
(B) {P,Q,R}
(C) {~P,~Q,~R}
(D) {R} only

Q.225 MSQ · 2 marks · FOL
Which are closed sentences (no free variables)?
(A) forall x P(x)
(B) exists y Q(y)
(C) P(x) with free x
(D) forall x exists y R(x,y)

Q.226 MCQ · 1 mark · UninformedSearch
A robot on a grid can move N,S,E,W at cost 1. Fewest moves from (0,0) to (2,2) is
(A) 2
(B) 4
(C) 8
(D) 0

Q.227 MSQ · 2 marks · InformedSearch
h=0 everywhere. Which algorithms then coincide (up to tie-breaking) with UCS?
(A) A*
(B) greedy best-first
(C) IDA* with f=g
(D) DFS

Q.228 MCQ · 1 mark · Adversarial
Move ordering that tries the eventual best move first in alpha-beta
(A) maximises pruning
(B) prevents all pruning
(C) changes the minimax value
(D) turns the game into UCS

Q.229 NAT · 1 mark · PropLogic
How many clauses in that CNF of (P - > Q) - > R?
NAT · enter the numerical value

Q.230 MCQ · 1 mark · FOL
The formula forall x exists y (y = x+1) over integers uses equality. It is true in (Z, +1) because
(A) every integer has a successor
(B) some integer has no successor
(C) addition is not a function
(D) it is unsatisfiable

Q.231 NAT · 1 mark · UninformedSearch
Grid (0,0) to (3,0) with 4-moves of cost 1. UCS/BFS optimal cost is ____.
NAT · enter the numerical value

Q.232 MCQ · 1 mark · InformedSearch
A* is called informed because it uses
(A) a problem-specific estimate h(n) of remaining cost
(B) only the branching factor
(C) only a stack
(D) a CNF encoding of the map

Q.233 MCQ · 1 mark · Adversarial
If alpha > = beta at a node, alpha-beta
(A) prunes remaining unexamined children of that node
(B) restarts minimax from the root
(C) sets all leaves to 0
(D) switches to uninformed DFS

Q.234 MSQ · 2 marks · PropLogic
Which are valid equivalences?
(A) P - > Q ? ~Q - > ~P (contrapositive)
(B) ~~P ? P
(C) P v P ? P
(D) P & ~P ? P v ~P

Q.235 MCQ · 1 mark · FOL
Sentence X: forall x (Bird(x) - > Flies(x)). Sentence Y: Bird(Tweety) - > Flies(Tweety). Then
(A) X entails Y
(B) Y entails X
(C) neither entails the other
(D) X is unsatisfiable

Q.236 MCQ · 1 mark · UninformedSearch
Tree-search DFS (no closed set) on a graph with cycles is
(A) complete
(B) in danger of infinite loops along a cycle
(C) optimal for unit costs
(D) the same as UCS

Q.237 MCQ · 2 marks · InformedSearch
If two admissible heuristics satisfy h1(n) > = h2(n) for all n, then A* with h1 generally
(A) expands a subset of the nodes that h2 would expand (never more, ignoring ties)
(B) always expands more nodes
(C) becomes inadmissible
(D) ignores g

Q.238 NAT · 1 mark · Adversarial
Leaves under a MIN node: 4, 9, 1. The MIN value is ____.
NAT · enter the numerical value

Q.239 MCQ · 2 marks · PropLogic
SAT is NP-complete. 2-SAT (clauses of size < =2) is
(A) solvable in polynomial time (e.g. implication graph / SCCs)
(B) undecidable
(C) the same as FOL validity
(D) minimax

Q.240 MCQ · 1 mark · FOL
From forall x ~Likes(x,x) one may infer
(A) ~Likes(John,John)
(B) Likes(John,John)
(C) exists x Likes(x,x)
(D) forall x Likes(x,x)

Q.241 MCQ · 1 mark · UninformedSearch
Matching: 'expand shallowest first' names
(A) BFS
(B) DFS
(C) greedy best-first
(D) minimax

Q.242 NAT · 1 mark · InformedSearch
Path S-A-G with costs 2, 5 and h(A)=4, h(G)=0, h(S)=6. f(A) after reaching A via S is ____.
NAT · enter the numerical value

Q.243 NAT · 1 mark · Adversarial
Leaves under a MAX node: 4, 9, 1. The MAX value is ____.
NAT · enter the numerical value

Q.244 MCQ · 1 mark · PropLogic
A knowledge base {P - > Q, Q - > R, P} entails R by
(A) two applications of modus ponens
(B) alpha-beta
(C) k-means
(D) PCA

Q.245 MCQ · 1 mark · UninformedSearch
Matching: 'expand deepest first' names
(A) BFS
(B) DFS
(C) UCS
(D) A*

Q.246 MCQ · 1 mark · InformedSearch
A heuristic with h(G)!=0 at the goal is
(A) not admissible unless the true remaining cost is at least that value; typically we require h(G)=0
(B) always consistent
(C) required by UCS
(D) the definition of greedy

Q.247 MCQ · 1 mark · Adversarial
A two-ply (MAX then MIN) search with an evaluator at depth 2 is
(A) minimax to depth 2 using the evaluator as stand-in utilities
(B) BFS on the game graph
(C) resolution on CNF
(D) Gibbs sampling

Q.248 MCQ · 1 mark · PropLogic
Which search problem is uninformed among: (i) BFS on a formula's implication graph, (ii) A* with a SAT heuristic. The uninformed one is
(A) (i) BFS
(B) (ii) A*
(C) both informed
(D) neither a search

Q.249 MCQ · 1 mark · UninformedSearch
Matching: 'expand cheapest-path-so-far first' names
(A) DFS
(B) UCS
(C) greedy best-first
(D) random walk

Q.250 NAT · 1 mark · InformedSearch
Three heuristics 2, 6, 4 all admissible at a node (h*=10). max of the three is ____.
NAT · enter the numerical value

Q.251 MSQ · 2 marks · Adversarial
Which algorithms belong to adversarial search as in GATE DA?
(A) minimax
(B) alpha-beta pruning
(C) expectiminimax (chance nodes)
(D) k-means clustering of board positions as the primary solver

Q.252 MCQ · 2 marks · PropLogic
Formula (P v Q) & (~P v Q) & (P v ~Q). One satisfying assignment is
(A) P=T, Q=T (check: T,T,T all hold? (T v T)&(~T v T)&(T v ~T)=T&T;&T;)
(B) P=F, Q=F ((F v F) already false)
(C) none, it is unsat
(D) only Q=F, P=T: (T v F)&(~T v F)&(T v T)= T & F & T = F

Q.253 MCQ · 1 mark · UninformedSearch
A librarian wants a cheapest rupee-cost walk on a metro map with unequal fares. The uninformed algorithm that returns an optimal path is
(A) uniform-cost search
(B) DFS
(C) greedy best-first with h=straight-line
(D) depth-limited search with L=1

Q.254 MCQ · 1 mark · InformedSearch
Best-first search is a family; A* is the instance whose evaluation is
(A) f=g+h
(B) f=h
(C) f=-g
(D) f=depth

Q.255 MCQ · 1 mark · Adversarial
If two root moves both attain the minimax value, then
(A) either is minimax-optimal
(B) neither is legal
(C) alpha-beta is invalid
(D) the smaller index is required

Q.256 NAT · 2 marks · PropLogic
That formula (P v Q)&(~P v Q)&(P v ~Q) has how many models on {P,Q}?
NAT · enter the numerical value

Q.257 MCQ · 1 mark · UninformedSearch
Romania-style: Arad-Sibiu 140, Arad-Timisoara 118, Arad-Zerind 75. UCS from Arad expands first the neighbour of cost
(A) 75 (Zerind)
(B) 140 (Sibiu)
(C) 118 (Timisoara)
(D) 0 (Arad twice)

Q.258 MCQ · 2 marks · InformedSearch
Two admissible heuristics for a campus map: h_walk = walking minutes lower bound, h_crow = crow-flies minutes. A* should use
(A) max(h_walk, h_crow), still admissible and dominating
(B) h_walk + h_crow always
(C) min of them only, never max
(D) their product

Q.259 NAT · 1 mark · Adversarial
Root MAX, four terminal children 0, 2, 2, -5. Minimax value is ____.
NAT · enter the numerical value

Q.260 NAT · 1 mark · UninformedSearch
Costs 75, 118, 140 from Arad. The g-value of the first expanded neighbour in UCS is ____.
NAT · enter the numerical value

Q.261 MCQ · 2 marks · InformedSearch
Arad to Bucharest, h(Arad)=366, a step Arad-Sibiu cost 140, h(Sibiu)=253. A* f(Sibiu) via Arad is
(A) 140+253=393
(B) 366
(C) 253
(D) 140

Q.262 MCQ · 1 mark · Adversarial
Quiescence search extends the horizon in tactical lines to reduce
(A) horizon-effect blunders on captures just beyond the depth limit
(B) d-separation work
(C) CNF conversion cost
(D) likelihood-weighting variance

Q.263 NAT · 1 mark · InformedSearch
g=140, h=253. f equals ____.
NAT · enter the numerical value

Q.264 NAT · 2 marks · Adversarial
MAX-MIN leaves 5,0 | 1,1. Root value max(min(5,0),min(1,1))= ____.
NAT · enter the numerical value

Q.265 MCQ · 1 mark · Adversarial
MAX-MIN leaves 5,0 | 1,1. Best first move is
(A) Right, because it guarantees 1 > 0
(B) Left, because 5 is large
(C) pass
(D) the leaf 0

Q.266 NAT · 1 mark · Adversarial
Alpha starts at -inf, beta at +inf. After MAX sees a child value 4, alpha becomes ____.
NAT · enter the numerical value

Q.267 NAT · 1 mark · Adversarial
MAX root children are terminals 8, -2, 5. Best move value is ____.
NAT · enter the numerical value

Q.268 MCQ · 1 mark · Adversarial
In alpha-beta, the pair (alpha,beta) at the root starts at
(A) (-inf, +inf)
(B) (0,0)
(C) (1,0)
(D) (h,g)

Q.269 NAT · 1 mark · Adversarial
Depth-1 MAX with three terminals 2, 7, -4. After examining 2 then 7, alpha is ____.
NAT · enter the numerical value

UNIT 7 SOLUTIONS

Q.1 [MCQ] C — BFS uses only the graph structure; it does not consult h(n).

Q.2 [MCQ] A — Standard A* evaluation.

Q.3 [MCQ] A — Definition of minimax.

Q.4 [MCQ] A — Tautology = valid = true in all models.

Q.5 [MCQ] A — Universal instantiation + MP.

Q.6 [MCQ] A — P(v) = product_i P(v_i | pa_i).

Q.7 [MCQ] A — Exact inference by factor algebra.

Q.8 [MCQ] A — Forward sampling.

Q.9 [MSQ] A;B;C — A* is informed. UCS uses path cost g(n) but not a goal heuristic.

Q.10 [MCQ] B — Greedy looks only at the heuristic.

Q.11 [NAT] 3 — Left min=3, Right min=2, max=3.

Q.12 [MCQ] A — SAT = at least one model.

Q.13 [MCQ] B — You may not replace exists x P(x) by P(A) for an arbitrary old constant A; EI needs a fresh Skolem.

Q.14 [MCQ] A — Parents of A are B,E; parents of J,M are A; B,E are roots.

Q.15 [MCQ] A — Contrast asked in GATE-style.

Q.16 [MCQ] A — Classic rejection.

Q.17 [MCQ] A — Unit costs make shallowest equivalent to cheapest.

Q.18 [MCQ] B — Admissible = optimistic: h < = true cost-to-go.

Q.19 [MCQ] A — Left yields 3 > Right's 2.

Q.20 [MCQ] B — No model exists.

Q.21 [MSQ] A;B;C — Quantifier order: a single liker-of-all implies everyone is liked by someone, not conversely.

Q.22 [MSQ] A;B;D — Conditioning on the collider A opens B-E. J ? M | A and J ? B | A hold.

Q.23 [MCQ] A — Large factors blow up.

Q.24 [MCQ] A — LW is designed for rare evidence.

Q.25 [MCQ] A — Closed-set DFS cannot loop forever on a finite graph; it need not find a cheapest path.

Q.26 [MCQ] A — Consistency implies admissibility.

Q.27 [NAT] 6 — mins 1,6,2; max=6.

Q.28 [MSQ] A;B;C — P & ~P is a contradiction.

Q.29 [MCQ] A — Standard quantifier negation.

Q.30 [MCQ] A — Colliders block unless they (or a descendant) are in the evidence.

Q.31 [NAT] 8 — 2^3=8.

Q.32 [MSQ] A;B;C — VE is the exact one; LW is approximate in finite samples.

Q.33 [MCQ] B — UCS is Dijkstra on the implicit state graph.

Q.34 [MSQ] A;B;D — Max of admissible is admissible and dominating. Sum can double-count and overestimate. Zero is admissible. Min is admissible but weaker.

Q.35 [MCQ] B — Only the (6,6) branch yields 6.

Q.36 [MCQ] B — ~(P&Q;) ? ~P v ~Q.

Q.37 [MCQ] A — ~exists = forall not.

Q.38 [MCQ] A — Serial chain blocked by observing the middle.

Q.39 [NAT] 4 — 2^2=4.

Q.40 [MCQ] A — MCMC Gibbs update.

Q.41 [MCQ] A — IDS reruns DFS with a rising depth bound.

Q.42 [MCQ] B — Larger still-admissible h prunes more.

Q.43 [NAT] 7 — Terminals equal their utility.

Q.44 [MCQ] A — Standard rewrite used en route to CNF.

Q.45 [MCQ] A — Substitute x by B and y by A.

Q.46 [MCQ] A — Common-cause fork blocked by observing C.

Q.47 [MCQ] A — Associativity of sum/product; cost depends on order.

Q.48 [MCQ] A — d-separated from the rest given the blanket.

Q.49 [NAT] 15 — 1+2+4+8=15.

Q.50 [NAT] 5 — max(4,5)=5 still < =6.

Q.51 [MSQ] A;B;C — Terminal utilities are given; a heuristic is used only when the tree is truncated.

Q.52 [NAT] 8 — 2^3=8.

Q.53 [MCQ] B — x cannot be both A and B.

Q.54 [MCQ] A — Collider opening + competing explanations.

Q.55 [MCQ] A — Standard VE query.

Q.56 [NAT] 0.25 — 10/40=0.25.

Q.57 [MCQ] B — S-B-G costs 5, S-A-G costs 6. UCS returns the cheaper path S-B-G.

Q.58 [NAT] 0 — Sum 9 overestimates, so not admissible here.

Q.59 [MCQ] A — Best-case branching ~ sqrt of minimax.

Q.60 [NAT] 3 — Only FF fails.

Q.61 [MCQ] A — No finite unifier; occurs check rejects it.

Q.62 [NAT] 32 — Five binary vars.

Q.63 [NAT] 32 — 2^5=32.

Q.64 [NAT] 1 — Both samples have Q=true, so 1.

Q.65 [NAT] 5 — S-B-G = 5.

Q.66 [MCQ] A — Euclidean distance < = any path length.

Q.67 [MCQ] A — Alpha = MAX's current lower bound.

Q.68 [NAT] 1 — Only TT.

Q.69 [MCQ] A — Implication (forall x P) - > Q ? exists x (P(x)- > Q) when x not free in Q.

Q.70 [NAT] 10 — P(B),P(E) one each; P(A|B,E) four; P(J|A) two; P(M|A) two. (If using 1-p: 1+1+4+2+2 wait same if we count both rows... GATE often counts free params 1+1+4+2+2=10 unconstrained or 1+1+4+1+1=8 with 1-p. We count listed CPT entries as 10.)

Q.71 [MSQ] A;B;C — A few Gibbs samples do not give exactness.

Q.72 [NAT] 0.25 — 0.2/(0.2+0.6)=0.25.

Q.73 [MCQ] A — BFS explores level by level; DFS can go deep down the leftmost branch.

Q.74 [MCQ] A — Admissible allows equality with h*. Cheapest S-G is S-B-G=3.

Q.75 [MCQ] B — Beta = MIN's current upper bound.

Q.76 [MCQ] A — Rewrite each implication as a clause.

Q.77 [MCQ] A — Existential first (not in the scope of a universal) = > constant.

Q.78 [MCQ] A — The only path SAT-I-Grade-Letter is blocked at I (fork).

Q.79 [MCQ] A — Scope union.

Q.80 [MCQ] A — Evidence at roots is easy: just fix them.

Q.81 [MCQ] A — Goals beyond L are invisible to DLS.

Q.82 [NAT] 3 — S-B-G=3 < S-A-G=4.

Q.83 [MCQ] A — Beta cutoff: MIN has already a value < = alpha.

Q.84 [MCQ] A — A conjunction of two unit clauses.

Q.85 [MCQ] A — Existential in the scope of x becomes f(x).

Q.86 [MCQ] A — Common cause Cloudy, unobserved = > dependent. Observing Cloudy would block.

Q.87 [MCQ] A — Marginalisation drops Y.

Q.88 [MSQ] A;B;C — VE is exact.

Q.89 [NAT] 4 — Limits 0 through 3: four runs.

Q.90 [MCQ] A — Consistency = > optimality of graph-search A*.

Q.91 [MCQ] A — Alpha cutoff.

Q.92 [MCQ] A — Biconditional as two clauses.

Q.93 [MSQ] A;B;D — Two possibly different witnesses do not give a common witness. C fails.

Q.94 [MSQ] A;B;C — Option D is actually TRUE (W ? C | S,R) so if the stem lists D as a true CI, the answer would include D.

The option text says 'is false in general' making D a false claim. So A;B;C.

Q.95 [NAT] 4 — Scope {A,C}, size 4.

Q.96 [MCQ] A — MCMC mixing.

Q.97 [MSQ] A;B;D — g(n) is not a heuristic; UCS is uninformed. Negative costs invalidate Dijkstra-style proofs.

Q.98 [MCQ] A — Tree-search A* + admissible h is optimal; graph-search needs consistency (or reopening).

Q.99 [NAT] 3 — Left min=3; right min < =2 already, so MAX keeps 3. x is never needed.

Q.100 [MSQ] A;B;C — P - > (Q - > P) is a tautology; D is a contradiction.

Q.101 [MCQ] B — B is the negation, not the original.

Q.102 [MCQ] A — P(node | parents, non-descendants) = P(node | parents).

Q.103 [MCQ] A — Induced width depends on order.

Q.104 [NAT] 2/3 — Two 1s out of 3.

Q.105 [MCQ] A — Standard node: state, parent, action, g.

Q.106 [NAT] 7 — f=g+h=7.

Q.107 [MCQ] D — After seeing 2, MIN's node is < =2 < =alpha=3, cutoff.

Q.108 [MCQ] A — Inference rule MP.

Q.109 [MCQ] A — Herbrand models are sets of ground atoms.

Q.110 [MCQ] A — Sound (and complete for faithful distributions).

Q.111 [MCQ] A — Factor reduction / observation.

Q.112 [MCQ] A — Rare evidence kills rejection.

Q.113 [NAT] 3 — S-A-G = 3 beats S-C-G = 5.

Q.114 [MCQ] B — f=g, which is UCS.

Q.115 [NAT] 8 — No pruning in the worst order; all leaves are seen.

Q.116 [MCQ] A — Resolve on P.

Q.117 [NAT] 2 — A, f(A). (Depth-0 and applying f once.)

Q.118 [NAT] 7 — Root 1, three 2-row CPTs with 2 free each? Wait each child given 1 parent has 2 params (P(child=1|parent=0) and P(child=1|parent=1)), root 1, total 7.

Q.119 [NAT] 0.75 — 6/(2+6)=0.75.

Q.120 [MCQ] A — Parents sampled from the prior, not the posterior, can make weights tiny.

Q.121 [MCQ] B — An infinite branch traps DFS.

Q.122 [MCQ] B — Greedy best-first.

Q.123 [NAT] 4 — No pruning claimed; 4 terminals.

Q.124 [NAT] 0 — The empty clause.

Q.125 [MCQ] A — UI.

Q.126 [MCQ] A — No missing edges = > no CI.

Q.127 [NAT] 0.25 — 1/4=0.25.

Q.128 [NAT] 1 — Deterministically 1.

Q.129 [MCQ] B — The frontier at level d has Theta(b^d) nodes.

Q.130 [MSQ] A;B;D — Each move fixes at most one Manhattan unit and at most one misplaced tile. Doubling Manhattan can overestimate.

Q.131 [MCQ] A — Standard depth-limited minimax / heuristic play.

Q.132 [MCQ] A — Empty clause = False.

Q.133 [MCQ] A — EG.

Q.134 [MCQ] A — Marry the parents, then undirect.

Q.135 [MCQ] A — Scope {Q}.

Q.136 [MCQ] A — Proposal = prior with evidence fixed.

Q.137 [NAT] 9 — 3^2=9.

Q.138 [NAT] 2 — Two tiles are misplaced.

Q.139 [NAT] 4 — Forced payoff 4.

Q.140 [MCQ] A — ~P v ~Q v R is (P & Q) - > R, definite Horn. D has two positives Q and R.

Q.141 [MCQ] A — Clause variables are universal.

Q.142 [NAT] 8 — One row per parent configuration.

Q.143 [MSQ] A;B — Sampling methods are approximate. Enumeration is exact but exponential.

Q.144 [MCQ] A — Forward sampling estimates any prior marginal.

Q.145 [MCQ] A — Two frontiers of depth d/2.

Q.146 [NAT] 2 — 1+1=2.

Q.147 [MCQ] A — negamax(n) = max_c -negamax(c).

Q.148 [MCQ] A — Horn-SAT / definite-clause FC.

Q.149 [MCQ] A — x binds to Jane.

Q.150 [NAT] 8 — One free prob per row, 8 rows.

Q.151 [MCQ] A — Treewidth 1 / polytree.

Q.152 [NAT] 0.2 — 4/20=0.2.

Q.153 [MCQ] A — Fewest hops = BFS on an unweighted graph.

Q.154 [MCQ] A — Classic greedy failure mode.

Q.155 [MSQ] A;B;C — Pruning is correctness-preserving for the value.

Q.156 [NAT] 3 — Only P=T,Q=F is false.

Q.157 [MCQ] A — x=g(z), f(y)=f(A) so y=A.

Q.158 [MCQ] A — Unshielded collider.

Q.159 [NAT] 4 — Same scope, still 4 entries.

Q.160 [MSQ] A;B;C — Two samples do not certify 10-decimal agreement.

Q.161 [NAT] 11 — S-A-G=11 beats the direct 100.

Q.162 [MCQ] A — Inflation of h can overestimate, losing admissibility.

Q.163 [NAT] 4 — max(1,4)=4.

Q.164 [MCQ] A — Semantic entailment.

Q.165 [MSQ] A;B;C — A universal implication does not by itself assert existence of a P-object.

Q.166 [MSQ] A;C — X and Y are marginally independent; each is dependent with Z. B and D are false claims of CI.

Q.167 [NAT] 0.75 — 15/20=0.75.

Q.168 [MCQ] A — Full conditional factorises on the blanket.

Q.169 [MCQ] A — The explored set blocks duplicates.

Q.170 [NAT] 5 — 0+5=5.

Q.171 [MCQ] A — 6 > alpha=1, no cutoff yet; 4 is seen. (8 is seen too: left MIN looks at both 1 and 8.)

Q.172 [MCQ] A — From False everything follows.

Q.173 [MCQ] A — exists x (~P(x) v Q(x)) is easy to satisfy.

Q.174 [MCQ] A — Common-cause independence given the cause.

Q.175 [NAT] 4 — 2^{2}=4.

Q.176 [NAT] 0.3 — Multiply by that CPT entry.

Q.177 [MSQ] A;B;C — DLS with too small a limit misses the goal.

Q.178 [MCQ] A — Monotonicity of f along paths.

Q.179 [NAT] 3 — mins 3 and 2; max=3.

Q.180 [MSQ] A;B;C — P v Q does not entail P (countermodel P false, Q true).

Q.181 [MCQ] A — Validity is r.e. but not recursive.

Q.182 [MCQ] A — Read off the conditioning.

Q.183 [MCQ] A — Hard zeros in CPTs.

Q.184 [MCQ] A — BFS = queue.

Q.185 [MCQ] A — Overestimates can rank the optimal path behind a worse one that looks cheap.

Q.186 [MCQ] A — Left guarantees 3.

Q.187 [MCQ] A — Or-over-and.

Q.188 [NAT] 2 — Two arguments.

Q.189 [MCQ] A — J < - A < - E is a chain blocked at A; also J < - A - > ... no E. Path J-A-E is A as middle of a chain from E to J? E- > A- > J is a chain, blocked by A.

Q.190 [MCQ] A — Strong coupling = > slow MCMC.

Q.191 [MCQ] B — DFS = stack.

Q.192 [NAT] 5 — max(3,5)=5 < =8.

Q.193 [NAT] 4 — MAX children: max(2,5)=5 and max(4,1)=4; MIN chooses min(5,4)=4.

Q.194 [NAT] 2 — (P v Q) and (P v R).

Q.195 [MCQ] A — Terms vs formulae.

Q.196 [NAT] 0 — All rejected.

Q.197 [MCQ] C — UCS pops least g.

Q.198 [MCQ] A — Each move cuts Manhattan by at most 1.

Q.199 [NAT] 1 — MAX picks the largest terminal 1.

Q.200 [MCQ] A — P or ~P.

Q.201 [MCQ] A — Bound variables are not free for substitution.

Q.202 [MCQ] A — Self-normalised importance sampling is consistent.

Q.203 [NAT] 6 — 2+2+2=6.

Q.204 [MCQ] A — Euclidean < = Manhattan < = true 4-move cost.

Q.205 [MCQ] A — Chance nodes average.

Q.206 [MCQ] A — DPLL unit prop.

Q.207 [MCQ] A — Quantifier negation.

Q.208 [MCQ] A — Canonical one-edge LW.

Q.209 [MCQ] A — Positive costs make cycles strictly more expensive.

Q.210 [NAT] 7 — 3+4=7.

Q.211 [MCQ] A — Search-horizon pathology.

Q.212 [NAT] 16 — 2^(2^2)=16 Boolean functions of 2 vars.

Q.213 [NAT] 1 — One binding.

Q.214 [MCQ] A — Gibbs acceptance = 1.

Q.215 [MCQ] A — b^0+...+b^d = Theta(b^d).

Q.216 [NAT] 5 — 3-4-5 triangle.

Q.217 [NAT] 9 — 3^2=9.

Q.218 [MCQ] A — Entailment fails iff a model of P is not a model of Q.

Q.219 [MCQ] A — Binary resolution + unification.

Q.220 [MCQ] A — MCMC consistency vs exact VE.

Q.221 [NAT] 1 — Only the root, then the limit cuts every child.

Q.222 [MCQ] A — IDA* = DFS-style f-limited iterations.

Q.223 [NAT] 8 — 2^3=8.

Q.224 [MCQ] A — Two binary clauses.

Q.225 [MSQ] A;B;D — P(x) is open.

Q.226 [MCQ] B — Manhattan distance 4; BFS finds it.

Q.227 [MSQ] A;C — Greedy with h=0 ranks all nodes equal (not UCS). DFS is not cost-ordered.

Q.228 [MCQ] A — Best-first ordering is ideal for alpha-beta.

Q.229 [NAT] 2 — Two clauses.

Q.230 [MCQ] A — Successor always exists in Z.

Q.231 [NAT] 3 — Three east moves.

Q.232 [MCQ] A — That is the definition of informed search.

Q.233 [MCQ] A — Cutoff condition.

Q.234 [MSQ] A;B;C — Contradiction is not equivalent to a tautology.

Q.235 [MCQ] A — UI on Tweety.

Q.236 [MCQ] B — Without a closed set, a cycle can be re-entered forever.

Q.237 [MCQ] A — Dominating admissible heuristics prune at least as much.

Q.238 [NAT] 1 — min=1.

Q.239 [MCQ] A — 2-SAT is in P.

Q.240 [MCQ] A — UI + the negated atom.

Q.241 [MCQ] A — Definition of BFS.

Q.242 [NAT] 6 — g(A)=2, f=2+4=6.

Q.243 [NAT] 9 — max=9.

Q.244 [MCQ] A — P, P- > Q = > Q; Q, Q- > R = > R.

Q.245 [MCQ] B — Definition of DFS.

Q.246 [MCQ] A — h*(G)=0 so admissibility forces h(G)=0.

Q.247 [MCQ] A — Bounded-depth game search.

Q.248 [MCQ] A — BFS remains uninformed even if the graph encodes logic.

Q.249 [MCQ] B — UCS orders by g(n).

Q.250 [NAT] 6 — max=6.

Q.251 [MSQ] A;B;C — k-means is unsupervised ML, not adversarial search.

Q.252 [MCQ] A — P=T,Q=T works. P=F,Q=T: (F v T)&(T v T)&(F v F)= T&T;&F;=F. So unique model TT? P=T,Q=F fails as above. Unique TT.

Q.253 [MCQ] A — Unequal nonnegative costs: UCS.

Q.254 [MCQ] A — A* is best-first with f=g+h.

Q.255 [MCQ] A — Ties among optimal actions are all optimal.

Q.256 [NAT] 1 — Only TT, as FF, FT, TF fail.

Q.257 [MCQ] A — Least-cost fringe node Zerind.

Q.258 [MCQ] A — Max of admissible remains admissible.

Q.259 [NAT] 2 — max=2.

Q.260 [NAT] 75 — Zerind.

Q.261 [MCQ] A — f=g+h=393.

Q.262 [MCQ] A — Continue while the position is noisy.

Q.263 [NAT] 393 — 140+253=393.

Q.264 [NAT] 1 — max(0,1)=1.

Q.265 [MCQ] A — Right's min is 1.

Q.266 [NAT] 4 — alpha := max(alpha,4)=4.

Q.267 [NAT] 8 — MAX picks 8.

Q.268 [MCQ] A — No bounds yet.

Q.269 [NAT] 7 — max(2,7)=7.

UNIT 8 General Aptitude · 355 questions

MCQ negative -1/3 or -2/3. MSQ / NAT no negative. Work first, then the solutions section.

Q.1 MCQ · 1 mark · GA/Verbal
Ephemeral : Fleeting :: Perennial : _____
(A) Seasonal
(B) Enduring
(C) Fragile
(D) Annual

Q.2 MCQ · 1 mark · GA/Quant
A two-digit number is 7 times the sum of its digits, and the tens digit is 3 more than the units digit. The number is
(A) 63
(B) 36
(C) 72
(D) 54

Q.3 MCQ · 2 marks · GA/Analytical
Five people P, Q, R, S, T sit in a row facing north. P is second to the left of R. Q sits immediately right of P. T occupies an end. S is not at an end. R sits in the middle. Who is at the right end?
(A) P
(B) Q
(C) S
(D) T

Q.4 MCQ · 2 marks · GA/Spatial
A regular hexagon is inscribed in a circle of radius 6. The ratio of the hexagon's area to the circle's area is
(A) 3*sqrt(3)/(2*pi)
(B) 2/pi
(C) 3/pi
(D) sqrt(3)/pi

Q.5 MCQ · 1 mark · GA/Verbal
Miser : Hoard :: Spendthrift : _____
(A) Invest
(B) Squander
(C) Bargain
(D) Audit

Q.6 MCQ · 1 mark · GA/Quant
The product of the digits of a two-digit number is 18, and the number is 6 times its units digit. The number is
(A) 36
(B) 63
(C) 27
(D) 54

Q.7 MCQ · 2 marks · GA/Analytical
Six colleagues A,B,C,D,E,F sit around a circle. A sits opposite D. B sits immediately clockwise from A. C sits opposite E. Who sits opposite B?
(A) C
(B) E
(C) F
(D) D

Q.8 NAT · 1 mark · GA/Spatial
A 3x3x3 cube is painted on all faces and cut into 27 unit cubes. The number of unit cubes with no paint is _____.
NAT · enter the numerical value

Q.9 MCQ · 1 mark · GA/Verbal
Scalpel : Surgeon :: Gavel : _____
(A) Carpenter
(B) Judge
(C) Blacksmith
(D) Sculptor

Q.10 MCQ · 1 mark · GA/Quant
A two-digit number plus its reverse is 110, and the tens digit exceeds the units digit by 4. The original number is
(A) 73
(B) 37
(C) 64
(D) 46

Q.11 MCQ · 2 marks · GA/Analytical
Pointing to a woman, a man says: 'Her mother's only brother is my father's only son.' The man has no brother. The woman is the man's
(A) daughter
(B) sister
(C) niece
(D) mother

Q.12 NAT · 1 mark · GA/Spatial
A 6x6x6 cube is painted on all faces and diced into unit cubes. The number of unit cubes with paint on exactly three faces is _____.
NAT · enter the numerical value

Q.13 MCQ · 1 mark · GA/Verbal
Oasis : Arid expanse :: Archipelago : _____
(A) Peninsula
(B) Ocean
(C) Glacier
(D) Canyon

Q.14 NAT · 2 marks · GA/Quant
A three-digit number has distinct positive digits in arithmetic progression with common difference 2. The number is 26 times the sum of its digits. The number is _____.
NAT · enter the numerical value

Q.15 MCQ · 2 marks · GA/Analytical
A is B's sister. C is B's mother. D is C's father. E is D's mother. Then A is E's
(A) daughter
(B) granddaughter
(C) great-granddaughter
(D) niece

Q.16 NAT · 1 mark · GA/Spatial
A 6x6x6 cube is painted on all faces and diced into unit cubes. The number of unit cubes with no paint is _____.
NAT · enter the numerical value

Q.17 MCQ · 1 mark · GA/Verbal
Lethargy : Vigour :: Apathy : _____
(A) Fatigue
(B) Concern
(C) Sloth
(D) Inertia

Q.18 MCQ · 1 mark · GA/Quant
The units digit of 7^2025 is
(A) 1
(B) 3
(C) 7
(D) 9

Q.19 MCQ · 1 mark · GA/Analytical
If 1 is opposite 6, 2 opposite 5, and 3 opposite 4 on a cube, and 4 is on top while the four side faces clockwise from the front are 2, 1, 5, 6, then the bottom face is
(A) 1
(B) 3
(C) 5
(D) 6

Q.20 NAT · 2 marks · GA/Spatial
A 6x6x6 cube is painted on all faces and diced into unit cubes. The number of unit cubes with paint on exactly one face is _____.
NAT · enter the numerical value

Q.21 MCQ · 1 mark · GA/Verbal
Alleviate : Aggravate :: Pacify : _____
(A) Soothe
(B) Provoke
(C) Mediate
(D) Console

Q.22 MCQ · 1 mark · GA/Quant
The remainder when 3^25 is divided by 5 is
(A) 1
(B) 2
(C) 3
(D) 4

Q.23 MCQ · 2 marks · GA/Analytical
A cube net is a row of four squares with one square attached to the second square's top edge and one attached to the third square's bottom edge. This net
(A) cannot fold to a cube
(B) folds to a cube
(C) folds only to a cuboid that is not a cube
(D) folds to a tetrahedron

Q.24 MCQ · 1 mark · GA/Spatial
Six unit squares in a single row of length 6 are used as a net. This net folds to a cube?
(A) yes
(B) no
(C) only if one square is removed
(D) only after a twist

Q.25 MCQ · 1 mark · GA/Verbal
Cartographer : Charts of land :: Lexicographer : _____
(A) Statutes
(B) Dictionaries
(C) Portraits
(D) Ledgers

Q.26 NAT · 2 marks · GA/Quant
The number of 3-digit positive integers whose digits sum to 4 is _____.
NAT · enter the numerical value

Q.27 MCQ · 2 marks · GA/Analytical
Four colleagues F, G, H, J each teach exactly two of Algebra, Botany, Coding. Algebra is taught by F, G and H. Botany is taught by F and J only. Who teaches Coding but not Algebra?
(A) F
(B) G
(C) H
(D) J

Q.28 MCQ · 2 marks · GA/Spatial
A plus-shaped net of five squares plus a sixth square attached to an outer edge of one arm
(A) cannot be a cube net
(B) is a cube net for some attachments and not others
(C) always folds to two tetrahedra
(D) always leaves a hole

Q.29 MCQ · 1 mark · GA/Verbal
Opaque : Light :: Dielectric : _____
(A) Heat
(B) Electric charge
(C) Sound
(D) Mass

Q.30 MCQ · 1 mark · GA/Quant
The sequence 2, 6, 12, 20, 30, ... has next term
(A) 36
(B) 40
(C) 42
(D) 48

Q.31 MCQ · 2 marks · GA/Analytical
Four players Ami, Bo, Cy, Di each play exactly two of Go, Hex, Yote. Ami and Bo both play Go. Cy does not play Hex. Di plays Hex and Yote. Who else besides Ami and Bo plays Go?
(A) Cy
(B) Di
(C) nobody
(D) both Cy and Di

Q.32 MCQ · 1 mark · GA/Spatial
A right circular cylinder of radius 3 and height 4 is unrolled. The lateral surface is a rectangle of sides
(A) 3 and 4
(B) 6*pi and 4
(C) 3*pi and 4
(D) 6 and 4

Q.33 MCQ · 1 mark · GA/Verbal
Dearth : Shortage :: Plethora : _____
(A) Balance
(B) Surplus
(C) Rarity
(D) Void

Q.34 MCQ · 1 mark · GA/Quant
Terms of 1, 2, 6, 15, 31, 56 follow adding successive squares. The next term is
(A) 80
(B) 84
(C) 92
(D) 96

Q.35 NAT · 2 marks · GA/Analytical
Five positive integers have unique mode 5 (appearing twice), median 7, mean 8, and all entries other than the two 5s are distinct and different from 5 and 7. The number of such lists, up to order, is _____.
NAT · enter the numerical value

Q.36 MCQ · 1 mark · GA/Spatial
A cone of slant height 5 and base radius 3 is unrolled. The sector's arc length is
(A) 3
(B) 5
(C) 6*pi
(D) 3*pi

Q.37 MCQ · 1 mark · GA/Verbal
Novice : Seasoned craft :: Pauper : _____
(A) Debt
(B) Affluence
(C) Labour
(D) Charity

Q.38 NAT · 1 mark · GA/Quant
A geometric progression has first term 3 and ratio 2. An arithmetic progression has first term 5 and common difference 3. The 4th geometric term minus the 4th arithmetic term equals _____.
NAT · enter the numerical value

Q.39 NAT · 2 marks · GA/Analytical
The unique 4-digit perfect square whose first two digits are equal and whose last two digits are equal is _____.
NAT · enter the numerical value

Q.40 MCQ · 1 mark · GA/Spatial
A square piece of paper is folded once along a diagonal and a small hole is punched through both layers near the midpoint of a short folded edge. Unfolded, the number of holes is
(A) 1
(B) 2
(C) 3
(D) 4

Q.41 MCQ · 1 mark · GA/Verbal
Archaeology : Buried settlements :: Paleontology : _____
(A) Minerals
(B) Fossils
(C) Manuscripts
(D) Constellations

Q.42 MCQ · 1 mark · GA/Quant
The pattern 2, 3, 8, 27, 112, ... multiplies by n and then adds n. The next term is
(A) 545
(B) 560
(C) 565
(D) 580

Q.43 NAT · 1 mark · GA/Analytical
Letters stand for distinct digits and I+I=ME with ME a two-digit number. The value of I+M+E is _____.
NAT · enter the numerical value

Q.44 MCQ · 2 marks · GA/Spatial
A cube has faces painted so that opposite faces share a colour (three colours, each used twice). After a rotation that brings a red face to the bottom, the top face is
(A) red
(B) possibly red
(C) never red
(D) always blue

Q.45 MCQ · 1 mark · GA/Verbal
Cacophony : Discordant :: Euphony : _____
(A) Shrill
(B) Harmonious
(C) Muted
(D) Rhythmic

Q.46 MCQ · 1 mark · GA/Quant
If every term after the second of 4, 10, 28, 82, ... equals 3 times the previous minus 2, the fifth term is
(A) 244
(B) 246
(C) 248
(D) 250

Q.47 NAT · 1 mark · GA/Analytical
A three-digit palindrome with middle digit 6 equals 11 times a two-digit number. The palindrome is _____.
NAT · enter the numerical value

Q.48 MCQ · 1 mark · GA/Spatial
Four unit cubes are joined face-to-face in a straight tetromino (1x4). The surface area of the solid is
(A) 16
(B) 18
(C) 20
(D) 24

Q.49 MCQ · 1 mark · GA/Verbal
Flock : Geese in flight :: Shoal : _____
(A) Wolves
(B) Fish
(C) Crows
(D) Cattle

Q.50 NAT · 1 mark · GA/Quant

The 5th term of 2, 5, 10, 17, ... (adding successive odd integers starting at 3) is _____.

NAT · enter the numerical value

Q.51 NAT · 1 mark · GA/Analytical

In a class of 40, 25 take maths, 20 take physics, and 10 take both. The number who take neither is _____.

NAT · enter the numerical value

Q.52 MCQ · 1 mark · GA/Spatial

A regular tetrahedron has 4 triangular faces. The number of edges is

(A) 4
(B) 6
(C) 8
(D) 12

Q.53 MCQ · 1 mark · GA/Verbal

Telescope : Far-off objects :: Microscope : _____

(A) Loud sounds
(B) Minute specimens
(C) Heavy loads
(D) Deep shafts

Q.54 MCQ · 1 mark · GA/Quant

A mixed series is 5, 10, 20, 35, 55, ... The missing 6th term is

(A) 75
(B) 80
(C) 85
(D) 90

Q.55 NAT · 1 mark · GA/Analytical

Among 50 people, 30 drink tea, 25 drink coffee, and 10 drink both. The number who drink neither is _____.

NAT · enter the numerical value

Q.56 MCQ · 2 marks · GA/Spatial

A plane cuts a sphere through a circle of radius 6. If the sphere's radius is 10, the distance from the centre to the plane is

(A) 4
(B) 6
(C) 8
(D) 10

Q.57 MCQ · 1 mark · GA/Verbal

Traitor : Treason :: Patriot : _____

(A) Exile
(B) Allegiance
(C) Conquest
(D) Tribute

Q.58 MCQ · 1 mark · GA/Quant

In 3, 8, 15, 24, 34, 48 the intended pattern is n^2-1. The term that does not fit is

(A) 15
(B) 24
(C) 34
(D) 48

Q.59 MCQ · 1 mark · GA/Analytical

All poets are dreamers. Some dreamers are critics. Which conclusion must follow?

(A) Some poets are critics
(B) Some dreamers are poets
(C) All critics are poets
(D) No dreamer is a poet

Q.60 MCQ · 2 marks · GA/Spatial

A 2x2x2 cube made of 8 unit cubes is painted outside and disassembled. The number of unit cubes with paint on exactly two faces is

(A) 0
(B) 4
(C) 8
(D) 12

Q.61 MCQ · 1 mark · GA/Verbal

Brittle : Snap under load :: Ductile : _____

(A) Shatter
(B) Draw into wire
(C) Rust
(D) Melt

Q.62 NAT · 1 mark · GA/Quant

The value of log2(3)*log3(4)*log4(8) is _____.

NAT · enter the numerical value

Q.63 MCQ · 1 mark · GA/Analytical

No square is a circle. All circles are round. Which conclusion follows?

(A) No square is round
(B) Some squares are round
(C) No circle is a square
(D) All round things are circles

Q.64 MCQ · 1 mark · GA/Spatial

A unit cube sits on a table. The number of faces not touching the table is

(A) 4
(B) 5
(C) 6
(D) 3

Q.65 MCQ · 1 mark · GA/Verbal

Vaccine : Prevent infection :: Antidote : _____

(A) Diagnose fever
(B) Counter a poison
(C) Sterilise tools
(D) Measure dose

Q.66 MCQ · 1 mark · GA/Quant

If log2(x-1)+log2(x+1)=3 with x > 1, then x equals

(A) 2
(B) 3
(C) 4
(D) 5

Q.67 MCQ · 2 marks · GA/Analytical

All maps are charts. No chart is a sketch. Some sketches are drafts. Which is valid?

(A) Some maps are sketches
(B) No map is a sketch
(C) All drafts are charts
(D) Some charts are drafts

Q.68 MCQ · 2 marks · GA/Spatial
Three unit cubes are stacked as a 1x1x3 tower. The exposed surface area, including the base sitting on a table (base not exposed), is
(A) 10
(B) 13
(C) 14
(D) 18

Q.69 MCQ · 1 mark · GA/Verbal
Anchor : Vessel at rest :: Foundation : _____
(A) Scaffold
(B) Building
(C) Rudder
(D) Mast

Q.70 NAT · 1 mark · GA/Quant
If log3(x)=2 and log_x(y)=3, then y equals _____.
NAT · enter the numerical value

Q.71 MCQ · 1 mark · GA/Analytical
If the first two statements are true, which completion is correct? All mesas are plateaus. Some plateaus are inhabited. Therefore
(A) all mesas are inhabited
(B) some mesas are inhabited
(C) some plateaus are mesas
(D) no mesa is inhabited

Q.72 MCQ · 1 mark · GA/Spatial
A regular octahedron has 8 faces. The number of vertices is
(A) 4
(B) 6
(C) 8
(D) 12

Q.73 MCQ · 1 mark · GA/Verbal
Sculptor : Chisel :: Calligrapher : _____
(A) Anvil
(B) Nib
(C) Trowel
(D) Lathe

Q.74 MCQ · 1 mark · GA/Quant
The number of digits in 2^10 * 5^8, written in decimal, is
(A) 8
(B) 9
(C) 10
(D) 11

Q.75 MCQ · 2 marks · GA/Analytical
A pair jointly sufficient to decide whether a positive integer m is divisible by 6 is
(A) m even; m divisible by 3
(B) m > 6; m odd
(C) m prime; m > 2
(D) m ends with 2; m < 10

Q.76 MCQ · 1 mark · GA/Verbal
Eclipse : Hide a disc :: Illuminate : _____
(A) Orbit
(B) Reveal a surface
(C) Rotate
(D) Reflect

Q.77 MCQ · 1 mark · GA/Quant
If 2^{log2(9)} + 10^{log10(4)} = k, then k equals
(A) 11
(B) 12
(C) 13
(D) 18

Q.78 MCQ · 1 mark · GA/Analytical
Eight friends sit at a square table, two per side, all facing inward. K sits immediately clockwise from L, and K and L are on the same side. M sits on the opposite side from K. Seats on M's side number
(A) 1
(B) 2
(C) 3
(D) 4

Q.79 MCQ · 1 mark · GA/Verbal
If intensity rises as faint to dim to murky, then cool to chilly to _____ is analogous.
(A) tepid
(B) frigid
(C) lukewarm
(D) balmy

Q.80 NAT · 1 mark · GA/Quant
log10(50)+log10(20)-log10(10) equals _____.
NAT · enter the numerical value

Q.81 MCQ · 1 mark · GA/Analytical
In a linear queue of six, X is ahead of Y, Y is not last, and exactly two people stand between X and Y. If X is in position 2 (front is 1), Y is in position
(A) 4
(B) 5
(C) 6
(D) 3

Q.82 MCQ · 1 mark · GA/Verbal
Pedantic : Minute rules :: Ostentatious : _____
(A) Quiet merit
(B) Showy display
(C) Humble craft
(D) Secret hoard

Q.83 MCQ · 1 mark · GA/Quant
If log_b(64)=2, then log_2(b) equals
(A) 2
(B) 3
(C) 4
(D) 8

Q.84 MCQ · 1 mark · GA/Analytical
If 'some machines are noisy' and 'all noisy things are distracting' are true, which must be true?
(A) All machines are distracting
(B) Some machines are distracting
(C) No machine is distracting
(D) All distracting things are machines

Q.85 MCQ · 1 mark · GA/Verbal
Hive : Colony of bees :: Rookery : _____
(A) Herd of deer
(B) Breeding seals
(C) Pack of hounds
(D) Pride of lions

Q.86 MCQ · 2 marks · GA/Quant
A list price is 25% above cost. A 10% discount is then given on the list price. The profit percent on cost is
(A) 10
(B) 12.5
(C) 15
(D) 17.5

Q.87 MCQ · 2 marks · GA/Analytical
A, B, C, D sit in a circle facing the centre. A sits to the immediate left of B. C sits opposite A. Who sits opposite B?
(A) A
(B) C
(C) D
(D) cannot be determined

Q.88 MCQ · 1 mark · GA/Verbal
Malleable : Beaten into sheets :: Viscous : _____
(A) Boiled away
(B) Flows sluggishly
(C) Shatters cleanly
(D) Conducts heat

Q.89 MCQ · 2 marks · GA/Quant
A vessel holds 40 L of milk. 8 L is removed and replaced by water; the same operation is repeated once more. Milk remaining, in litres, is
(A) 24
(B) 25.6
(C) 27.2
(D) 32

Q.90 MCQ · 1 mark · GA/Analytical
If 'P is the brother of Q' and 'R is the father of P', then Q is R's
(A) brother
(B) child
(C) uncle
(D) grandfather

Q.91 MCQ · 1 mark · GA/Verbal
Quisling : Collaborator :: Polyglot : _____
(A) Silent monk
(B) Speaker of many tongues
(C) Lone traveller
(D) Court scribe

Q.92 MCQ · 1 mark · GA/Quant
A population grows 10% in one decade and then falls 10% in the next. The net change over both decades is
(A) no change
(B) 1% decrease
(C) 1% increase
(D) 2% decrease

Q.93 MCQ · 2 marks · GA/Analytical
Four houses in a row are red, blue, green, yellow (not necessarily in that order). The red house is left of blue. Green is at an end. Yellow is immediately right of red. Blue is not at an end. The leftmost house is
(A) red
(B) blue
(C) green
(D) yellow

Q.94 MCQ · 1 mark · GA/Verbal
Tundra : Treeless cold plain :: Atoll : _____
(A) Volcanic peak
(B) Ring-shaped coral reef
(C) River delta
(D) Salt pan

Q.95 MCQ · 2 marks · GA/Quant
In a paper of T marks, a candidate scoring 45% fails by 15 marks, while another scoring 60% passes by 21. Then T equals
(A) 200
(B) 220
(C) 240
(D) 260

Q.96 MCQ · 2 marks · GA/Analytical
Kiran, Leela, Mira, Nisha like four sports: polo, squash, rowing, archery, one each. Kiran does not like polo. Leela likes rowing. Mira does not like squash or polo. Nisha likes polo. Mira likes
(A) polo
(B) squash
(C) rowing
(D) archery

Q.97 MCQ · 1 mark · GA/Verbal
Ink : Fountain pen :: Graphite : _____
(A) Chisel
(B) Pencil
(C) Brush
(D) Stylus of wax

Q.98 MCQ · 2 marks · GA/Quant
A grocer uses a 900 g weight in place of 1 kg and sells at cost price. The profit percent is
(A) 9
(B) 10
(C) 100/9
(D) 11

Q.99 MCQ · 2 marks · GA/Analytical
A clock shows 3:15. The smaller angle between the hands, in degrees, is
(A) 0
(B) 7.5
(C) 15
(D) 22.5

Q.100 MCQ · 1 mark · GA/Verbal
Hypothesis : Tentative claim :: Verdict : _____
(A) Idle rumour
(B) Settled judgment
(C) Open query
(D) Draft note

Q.101 NAT · 1 mark · GA/Quant
Successive discounts of 10% and 20% on a marked price are equivalent to a single discount of _____ percent.
NAT · enter the numerical value

Q.102 MCQ · 1 mark · GA/Analytical
If all Bloops are Razzies and all Razzies are Lazzies, then all Bloops are Lazzies. This argument is
(A) invalid
(B) valid
(C) true only for finite sets
(D) circular

Q.103 MCQ · 1 mark · GA/Verbal
Gesticulate : Hands :: Articulate : _____
(A) Feet
(B) Speech
(C) Eyes
(D) Silence

Q.104 MCQ · 1 mark · GA/Quant
A's salary is 25% less than B's. B's salary is greater than A's by
(A) 20%
(B) 25%
(C) 100/3 %
(D) 40%

Q.105 MCQ · 2 marks · GA/Analytical
Five boxes are stacked. P is above Q. R is immediately above P. S is at the bottom. T is not adjacent to S. Who is immediately above S?
(A) P
(B) Q
(C) R
(D) T

Q.106 MCQ · 1 mark · GA/Verbal
Archive : Stored records :: Reservoir : _____
(A) Mined ore
(B) Stored water
(C) Spun yarn
(D) Fired clay

Q.107 MCQ · 1 mark · GA/Quant
A share price rises 50% on Monday and falls 50% on Tuesday. Relative to Friday's close, the net change is
(A) no change
(B) 25% fall
(C) 25% rise
(D) 50% fall

Q.108 MCQ · 1 mark · GA/Analytical
Among J, K, L, M: J is taller than K. L is shorter than K. M is taller than J. The tallest is
(A) J
(B) K
(C) L
(D) M

Q.109 MCQ · 1 mark · GA/Verbal
By the time the auditor arrived, the clerk _____ the ledgers twice.
(A) has reconciled
(B) had reconciled
(C) is reconciling
(D) reconciles

Q.110 MCQ · 1 mark · GA/Quant
To pass a 250-mark paper, 40% is required. A candidate scores 85. The candidate fails by
(A) 10 marks
(B) 15 marks
(C) 20 marks
(D) 25 marks

Q.111 MCQ · 1 mark · GA/Analytical
A statement 'All birds fly' is falsified by a single flightless bird. The logical name of that counterexample pattern is closest to
(A) affirming the consequent
(B) universal claim defeated by one negative instance
(C) begging the question
(D) false dilemma

Q.112 MCQ · 1 mark · GA/Verbal
Neither the surveys nor the census _____ a decline in literacy.
(A) show
(B) shows
(C) are showing
(D) have shown

Q.113 MCQ · 2 marks · GA/Quant
An article is sold at 20% profit. If both cost and selling price rise by Rs 120, the profit percent becomes 10%. The original cost, in rupees, is
(A) 80
(B) 100
(C) 120
(D) 140

Q.114 MCQ · 1 mark · GA/Analytical
In a coded language TREE is USFF. Then LEAF is coded as
(A) MFBG
(B) MDBE
(C) MFBH
(D) KEZE

Q.115 MCQ · 1 mark · GA/Verbal
If the committee _____ quorum last Monday, the motion would have passed.
(A) has
(B) had had
(C) would have
(D) have had

Q.116 MCQ · 1 mark · GA/Quant
If A:B = 2:3 and B:C = 4:5, then A:B:C equals
(A) 2:3:5
(B) 8:12:15
(C) 2:4:5
(D) 8:12:5

Q.117 MCQ · 1 mark · GA/Analytical
A man walks 3 km north, 4 km east, then 3 km south. Distance from the start is
(A) 3 km
(B) 4 km
(C) 5 km
(D) 10 km

Q.118 MCQ · 1 mark · GA/Verbal
Each of the prototypes _____ tested under salt spray before shipment.
(A) were
(B) was
(C) are
(D) have been

Q.119 MCQ · 1 mark · GA/Quant
If 2A = 3B = 4C, then A:B:C equals
(A) 2:3:4
(B) 6:4:3
(C) 4:3:2
(D) 3:4:6

Q.120 MCQ · 2 marks · GA/Analytical
If 15 August of a year is a Friday, 16 September of the same non-leap year is a
(A) Monday
(B) Tuesday
(C) Wednesday
(D) Thursday

Q.121 MCQ · 1 mark · GA/Verbal
She insisted that the report _____ on her desk by noon.
(A) is
(B) be
(C) was being
(D) has been

Q.122 NAT · 1 mark · GA/Quant
The number 720 divided in the ratio 2:3:4 gives the largest share equal to _____.
NAT · enter the numerical value

Q.123 MCQ · 2 marks · GA/Analytical
Three switches A,B,C control three lamps in another room, one-to-one but unknown. You may toggle in the first room, then visit the lamp room once. A strategy that identifies all three pairings uses the fact that a bulb can be
(A) only on or off
(B) on, off, or warm
(C) broken
(D) coloured

Q.124 MCQ · 1 mark · GA/Verbal
The data _____ inconsistent with the claimed linear trend.
(A) is
(B) are
(C) was
(D) has been

Q.125 NAT · 1 mark · GA/Quant
A 16 L blend of milk and water is in the ratio 3:1. The litres of water that must be added to make the ratio 2:1 is _____.
NAT · enter the numerical value

Q.126 MCQ · 1 mark · GA/Analytical
A Venn diagram of two sets A,B with |A|=12, |B|=8, |A intersect B|=5 has |A union B| equal to
(A) 15
(B) 20
(C) 25
(D) 13

Q.127 MCQ · 1 mark · GA/Verbal
Hardly _____ the gates opened when the crowd surged in.
(A) did
(B) had
(C) have
(D) were

Q.128 NAT · 1 mark · GA/Quant
A 500 g alloy is gold and copper in the ratio 3:2. The mass of gold, in grams, is _____.
NAT · enter the numerical value

Q.129 MCQ · 1 mark · GA/Analytical
From 16, 18, 21, 24 the integer that is not a multiple of 3 is
(A) 16
(B) 18
(C) 21
(D) 24

Q.130 MCQ · 1 mark · GA/Verbal
The panel recommended him _____ the liaison officer.
(A) as
(B) to be
(C) for being
(D) that he is

Q.131 MCQ · 1 mark · GA/Quant
A and B are in the ratio 4:5. After A's quantity is raised by 25%, the new ratio A:B is
(A) 1:1
(B) 5:4
(C) 4:5
(D) 5:6

Q.132 NAT · 1 mark · GA/Analytical
A 3x3x3 cube is painted on all six faces and then diced into unit cubes. The number of unit cubes with paint on exactly two faces is _____.
NAT · enter the numerical value

Q.133 MCQ · 1 mark · GA/Verbal
Between you and _____, the draft still needs a figure caption.
(A) I
(B) me
(C) myself
(D) mine

Q.134 MCQ · 1 mark · GA/Quant
A finishes a job in 10 days and B in 15 days. Working together they finish it in
(A) 5 days
(B) 6 days
(C) 8 days
(D) 12.5 days

Q.135 MCQ · 1 mark · GA/Analytical
If NORTH is coded as OPSUI, then SOUTH is coded as
(A) TPVUI
(B) TPUUI
(C) TPVVI
(D) RNTSG

Q.136 MCQ · 1 mark · GA/Verbal
It is one of those errors that _____ even careful editors.
(A) escapes
(B) escape
(C) has escaped
(D) is escaping

Q.137 NAT · 2 marks · GA/Quant
A and B take 12 days, A and C take 15 days, B and C take 20 days. The days A, B and C together take is _____.
NAT · enter the numerical value

Q.138 MCQ · 2 marks · GA/Analytical
A is looking at B. B is looking at C. A is married, C is unmarried, B's status is unknown. The statement 'a married person is looking at an unmarried person' is
(A) always true
(B) always false
(C) true iff B is married
(D) true iff B is unmarried

Q.139 MCQ · 1 mark · GA/Verbal
No sooner _____ the fuse replaced than the motor hummed again.
(A) was
(B) had been
(C) has been
(D) is

Q.140 NAT · 1 mark · GA/Quant
A is twice as efficient as B, and together they finish a job in 8 days. The days A alone would take is _____.
NAT · enter the numerical value

Q.141 MCQ · 2 marks · GA/Analytical
Three people U, V, W. Exactly one always tells the truth. U says 'V is the truth-teller.' V says 'U is the truth-teller.' W says 'U is not the truth-teller.' The truth-teller is
(A) U
(B) V
(C) W
(D) none

Q.142 MCQ · 1 mark · GA/Verbal
The chemist, along with two technicians, _____ the titration.
(A) perform
(B) performs
(C) are performing
(D) have performed

Q.143 NAT · 1 mark · GA/Quant
Pipe P fills a tank in 6 hours; pipe Q empties it in 8 hours. Hours needed to fill the empty tank with both open is _____.
NAT · enter the numerical value

Q.144 MCQ · 2 marks · GA/Analytical
A calendar shows that a month has five Mondays. The fewest days that month can have is
(A) 28
(B) 29
(C) 30
(D) 31

Q.145 MCQ · 1 mark · GA/Verbal
I would rather you _____ the raw counts tonight.
(A) submit
(B) submitted
(C) have submitted
(D) will submit

Q.146 MCQ · 2 marks · GA/Quant
A, B, C take 6, 8 and 12 days alone. A and B work 3 days, then C finishes the rest. C works for
(A) 1 day
(B) 1.5 days
(C) 2 days
(D) 3 days

Q.147 MCQ · 1 mark · GA/Verbal
Scarcely a page _____ free of a misplaced modifier.
(A) were
(B) was
(C) are
(D) have been

Q.148 NAT · 2 marks · GA/Quant
Three men can finish a job in 10 days, or six women can. Days that one man and one woman together take is _____.
NAT · enter the numerical value

Q.149 MCQ · 1 mark · GA/Verbal
The board had the contract _____ by an outside counsel.
(A) review
(B) reviewed
(C) reviewing
(D) to review

Q.150 MCQ · 1 mark · GA/Quant
Speeds of two walkers are in the ratio 3:4. The ratio of times they take for the same road is
(A) 3:4
(B) 4:3
(C) 9:16
(D) 1:1

Q.151 MCQ · 1 mark · GA/Verbal
Whoever the committee chooses _____ to file a disclosure.
(A) have
(B) has
(C) are
(D) were

Q.152 MCQ · 2 marks · GA/Quant
A can do a job in 18 days, B in 24 days, C in 36 days. They start together; A leaves after 4 days, then B leaves after 2 more days. Days C still needs after that is
(A) 8
(B) 10
(C) 13
(D) 16

Q.153 MCQ · 1 mark · GA/Verbal
Ten kilometres _____ a long walk in this heat.
(A) are
(B) is
(C) were
(D) have been

Q.154 MCQ · 2 marks · GA/Quant
Two pipes fill a cistern in 20 min and 30 min. A drain empties it in 15 min. With all three open, the cistern fills in
(A) 30 min
(B) 60 min
(C) 120 min
(D) it never fills

Q.155 MCQ · 1 mark · GA/Verbal
He spoke as though he _____ the laboratory himself.
(A) designs
(B) had designed
(C) has designed
(D) is designing

Q.156 MCQ · 2 marks · GA/Quant
A tap fills a tank in 8 hours. After 3 hours a leak that empties a full tank in 12 hours opens. Further hours needed to fill from that moment are
(A) 6
(B) 8
(C) 10
(D) 15

Q.157 MCQ · 1 mark · GA/Verbal
Many a researcher _____ overnight in that archive.
(A) have stayed
(B) has stayed
(C) are staying
(D) were staying

Q.158 NAT · 1 mark · GA/Quant
Simple interest on Rs 2000 at 5% per annum for 3 years is _____ rupees.
NAT · enter the numerical value

Q.159 MCQ · 1 mark · GA/Verbal
The number of outliers _____ after the filter was applied.
(A) were reduced
(B) was reduced
(C) have reduced
(D) are reduced

Q.160 NAT · 1 mark · GA/Quant
Compound interest on Rs 1000 at 10% per annum for 2 years, compounded annually, is _____ rupees.
NAT · enter the numerical value

Q.161 MCQ · 1 mark · GA/Verbal
Let the samples _____ at room temperature overnight.
(A) keeps
(B) keep
(C) kept
(D) keeping

Q.162 NAT · 1 mark · GA/Quant
The difference between CI and SI on Rs 5000 for 2 years at 10% per annum is _____ rupees.
NAT · enter the numerical value

Q.163 MCQ · 1 mark · GA/Verbal
Not only the abstract but also the captions _____ rewriting.
(A) needs
(B) need
(C) has needed
(D) is needing

Q.164 MCQ · 2 marks · GA/Quant
A principal becomes 9/4 of itself in 2 years at annual compound interest. The rate percent per annum is
(A) 25
(B) 40
(C) 50
(D) 75

Q.165 MCQ · 1 mark · GA/Verbal
This is the first time she _____ a keynote abroad.
(A) delivered
(B) has delivered
(C) had delivered
(D) delivers

Q.166 MCQ · 1 mark · GA/Quant
The years needed for a sum to double at 8% per annum simple interest is
(A) 8
(B) 10
(C) 12.5
(D) 16

Q.167 MCQ · 1 mark · GA/Verbal
The minutes of the meeting _____ yet to be signed.
(A) is
(B) are
(C) has
(D) was

Q.168 NAT · 1 mark · GA/Quant
The amount on Rs 8000 at 5% per annum simple interest for 4 years is _____ rupees.
NAT · enter the numerical value

Q.169 MCQ · 1 mark · GA/Verbal
The petition was disposed _____ within a fortnight.
(A) off
(B) of
(C) with
(D) away

Q.170 MCQ · 1 mark · GA/Quant
Interest on a sum at 10% SI for 3 years equals interest on the same sum at 15% SI for 2 years. This equality holds
(A) for every principal
(B) only if the principal is 100
(C) only if the principal is 0
(D) never

Q.171 MCQ · 1 mark · GA/Verbal
She is averse _____ taking credit for joint work.
(A) of
(B) to
(C) from
(D) against

Q.172 NAT · 2 marks · GA/Quant
How many 4-digit numbers with distinct digits can be formed from {1,2,3,4,5,6} so that the number is divisible by 3?
NAT · enter the numerical value

Q.173 MCQ · 1 mark · GA/Verbal
The findings are consistent _____ last year's census.
(A) to
(B) with
(C) of
(D) for

Q.174 NAT · 2 marks · GA/Quant
How many 3-digit even numbers with distinct digits can be formed from {0,1,2,3,4}?
NAT · enter the numerical value

Q.175 MCQ · 1 mark · GA/Verbal
He was debarred _____ sitting the make-up paper.
(A) of
(B) from
(C) to
(D) off

Q.176 NAT · 2 marks · GA/Quant
How many 4-digit numbers with distinct digits from 1 to 9 are divisible by 5?
NAT · enter the numerical value

Q.177 MCQ · 1 mark · GA/Verbal
The lecture hinged _____ a single counter-example.
(A) at
(B) on
(C) with
(D) over

Q.178 NAT · 1 mark · GA/Quant
The number of distinct permutations of the letters of GATE that begin with G is _____.
NAT · enter the numerical value

Q.179 MCQ · 1 mark · GA/Verbal
They acquiesced _____ the revised timeline.
(A) with
(B) in
(C) to
(D) for

Q.180 NAT · 1 mark · GA/Quant
Five distinct books are placed in a line. The number of possible orders is _____.
NAT · enter the numerical value

Q.181 MCQ · 1 mark · GA/Verbal
The specimen was deficient _____ potassium.
(A) of
(B) in
(C) with
(D) from

Q.182 NAT · 2 marks · GA/Quant
The number of binary strings of length 4 with no two consecutive 1s is _____.
NAT · enter the numerical value

Q.183 MCQ · 1 mark · GA/Verbal
Nothing can atone _____ that breach of trust.
(A) of
(B) for
(C) with
(D) to

Q.184 NAT · 1 mark · GA/Quant
The number of ways to choose a president and a treasurer from 6 people (distinct offices) is _____.
NAT · enter the numerical value

Q.185 MCQ · 1 mark · GA/Verbal
The clause is binding _____ all signatories.
(A) with
(B) on
(C) to
(D) over

Q.186 NAT · 1 mark · GA/Quant
The number of 3-person committees from 8 people is _____.
NAT · enter the numerical value

Q.187 MCQ · 1 mark · GA/Verbal
She prevailed _____ him to withdraw the remark.
(A) on
(B) with
(C) over
(D) for

Q.188 NAT · 1 mark · GA/Quant
The number of 3-digit palindromes (first digit nonzero) is _____.
NAT · enter the numerical value

Q.189 MCQ · 1 mark · GA/Verbal
To bury the hatchet, in the sentence 'After the audit they buried the hatchet', means
(A) hide evidence
(B) end a quarrel
(C) defer a vote
(D) split a bonus

Q.190 MCQ · 1 mark · GA/Quant
The number of ways to seat 4 people around a circular table, treating rotations as the same, is
(A) 6
(B) 12
(C) 24
(D) 4

Q.191 MCQ · 1 mark · GA/Verbal
If a plan is 'on the anvil', it is
(A) already scrapped
(B) being forged or prepared
(C) publicly advertised
(D) legally frozen

Q.192 MCQ · 2 marks · GA/Quant
A family has two children. Given that the elder child is a girl, the probability that both are girls (assuming each child is boy or girl with equal chance, independently) is
(A) 1/4
(B) 1/3
(C) 1/2
(D) 2/3

Q.193 MCQ · 1 mark · GA/Verbal
'To read between the lines' is closest in meaning to
(A) skip footnotes
(B) infer unstated intent
(C) quote verbatim
(D) index a paragraph

Q.194 MCQ · 1 mark · GA/Quant
Two fair dice are thrown. The probability that the sum is 7 is
(A) 1/12
(B) 1/8
(C) 1/6
(D) 1/4

Q.195 MCQ · 1 mark · GA/Verbal
A report that 'cuts no ice' with the board
(A) is too brief
(B) fails to persuade
(C) lacks figures
(D) arrives late

Q.196 MCQ · 1 mark · GA/Quant
A bag has 4 red and 6 blue balls. Two balls are drawn at random without replacement. The probability both are red is
(A) 2/15
(B) 1/5
(C) 4/25
(D) 1/6

Q.197 MCQ · 1 mark · GA/Verbal
'To take the bull by the horns' means
(A) delegate a risk
(B) face a difficulty directly
(C) delay a decision
(D) blame a rival

Q.198 MCQ · 1 mark · GA/Quant
A fair coin is tossed three times. The probability of at least two heads is
(A) 1/8
(B) 3/8
(C) 1/2
(D) 5/8

Q.199 MCQ · 1 mark · GA/Verbal
If talks are 'at sixes and sevens', they are
(A) nearly finished
(B) in disorder
(C) strictly confidential
(D) evenly split

Q.200 MCQ · 1 mark · GA/Quant
An integer is drawn uniformly from 1 to 20. The probability it is a multiple of 3 or of 5 is
(A) 2/5
(B) 9/20
(C) 1/2
(D) 11/20

Q.201 MCQ · 1 mark · GA/Verbal
'A storm in a teacup' describes
(A) a coastal warning
(B) fuss over a trivial matter
(C) a sudden promotion
(D) a packed agenda

Q.202 MCQ · 1 mark · GA/Quant
Two fair dice are thrown. The probability that the product of the faces is even is
(A) 1/2
(B) 2/3
(C) 3/4
(D) 5/6

Q.203 MCQ · 1 mark · GA/Verbal
To 'throw down the gauntlet' is to
(A) offer a truce
(B) issue a challenge
(C) abandon a claim
(D) seal a pact

Q.204 MCQ · 1 mark · GA/Quant
A card is drawn from a shuffled 52-card deck. The probability it is an ace is
(A) 1/13
(B) 1/12
(C) 1/4
(D) 4/13

Q.205 MCQ · 1 mark · GA/Verbal
'To feather one's nest' means
(A) restore a habitat
(B) enrich oneself opportunistically
(C) host a guest
(D) file a return

Q.206 MCQ · 2 marks · GA/Quant
A regular hexagon is inscribed in a circle of radius 6. The area of the hexagon is
(A) 54
(B) 36*sqrt(3)
(C) 54*sqrt(3)
(D) 108

Q.207 MCQ · 1 mark · GA/Verbal
If someone 'lets the cat out of the bag', they
(A) rescue a pet
(B) reveal a secret
(C) cancel a trip
(D) lose a wager

Q.208 MCQ · 1 mark · GA/Quant
An equilateral triangle has side 6. Its area is
(A) 9*sqrt(3)
(B) 12*sqrt(3)
(C) 18*sqrt(3)
(D) 6*sqrt(3)

Q.209 MCQ · 2 marks · GA/Verbal
A coastal town banned single-use sachets after a survey found them in 80% of beach transects. Local shops switched to refill stations within a month, and a follow-up survey showed a sharp drop in sachet litter but no change in bottle litter. Which statement is best supported?
(A) Bottles were never a litter problem.
(B) The ban and refill shift likely reduced sachet debris.
(C) Tourists caused all of the sachet waste.
(D) Refill stations increased shop profits.

Q.210 MCQ · 1 mark · GA/Quant
A right circular cone has base radius 3 and height 4. Its volume is
(A) 12*pi
(B) 16*pi
(C) 18*pi
(D) 36*pi

Q.211 MCQ · 2 marks · GA/Verbal
In a lab notebook, every titration is initialled by two people. Last Tuesday only one set of initials appears on page 17, yet the recorded titre matches a duplicate run on page 18 that is fully initialled. The most reasonable inference is that
(A) page 17 was forged
(B) the second signatory may have missed page 17
(C) the instrument was uncalibrated
(D) the duplicate run is invalid

Q.212 MCQ · 1 mark · GA/Quant
The space diagonal of a cube of side 6 is
(A) 6*sqrt(2)
(B) 6*sqrt(3)
(C) 9
(D) 12

Q.213 MCQ · 2 marks · GA/Verbal
A language app reports that users who finish ten lessons in a week retain 30% more vocabulary than users who finish the same ten lessons spread over a month. The study did not randomise learners. Which critique is most apt?
(A) Ten lessons are too few to matter.
(B) Motivated learners may both cluster lessons and retain more.
(C) Vocabulary tests are always biased.
(D) A month is not long enough.

Q.214 MCQ · 2 marks · GA/Quant
A square is inscribed in a circle of radius 5. The area of the square is
(A) 25
(B) 50
(C) 75
(D) 100

Q.215 MCQ · 2 marks · GA/Verbal
The museum's night shift recorded three false alarms in May, all on days when humidity exceeded 80%. Alarms in June, a drier month, were zero. The curator concluded that humidity causes false alarms. The weakest point in this reasoning is that
(A) May has 31 days
(B) other seasonal factors may co-vary with humidity
(C) alarms are always electrical
(D) June follows May

Q.216 NAT · 1 mark · GA/Quant
A right triangle has legs 5 and 12. Its area is _____.
NAT · enter the numerical value

Q.217 MCQ · 2 marks · GA/Verbal
A city replaced diesel buses with electric ones on a single corridor and then advertised a city-wide drop in NOx. Roadside monitors exist only on that corridor. Which inference is safest?
(A) City-wide NOx definitely fell.
(B) NOx along the instrumented corridor likely fell.
(C) Electric buses emit no pollutants of any kind.
(D) Diesel buses were the only NOx source.

Q.218 MCQ · 1 mark · GA/Quant
A 60-degree sector of a circle of radius 6 has area
(A) 3*pi
(B) 6*pi
(C) 9*pi
(D) 12*pi

Q.219 MCQ · 2 marks · GA/Verbal
Researchers found that students who annotated PDFs by hand scored higher on a later closed-book quiz than students who highlighted digitally. Both groups had the same reading time. The result most directly suggests that
(A) hand annotation may aid later recall relative to digital highlighting
(B) PDFs are inferior to paper books
(C) quizzes should be open-book
(D) digital tools reduce intelligence

Q.220 MCQ · 1 mark · GA/Quant
An annulus has outer radius 5 and inner radius 3. Its area is
(A) 8*pi
(B) 16*pi
(C) 15*pi
(D) 4*pi

Q.221 MCQ · 2 marks · GA/Verbal
A seed bank stores duplicates at two sites. A heatwave disabled refrigeration at Site A for 36 hours; germination tests later showed no drop relative to Site B. Which conclusion is most cautious?
(A) All seeds are heat-proof.
(B) This batch tolerated that outage under the tested conditions.
(C) Site A should be closed.
(D) Germination tests are unnecessary.

Q.222 MCQ · 1 mark · GA/Quant
The curved surface area of a cylinder of radius 2 and height 7 is
(A) 14*pi
(B) 28*pi
(C) 32*pi
(D) 56*pi

Q.223 MCQ · 2 marks · GA/Verbal
An editor notes that papers with shorter abstracts are cited more, but also that celebrity authors write shorter abstracts. The claim that brevity itself causes citations is weak because
(A) abstracts cannot be counted
(B) author fame may drive both brevity and citations
(C) citations never matter
(D) short abstracts lack keywords

Q.224 NAT · 1 mark · GA/Quant
Using pi=22/7, the area of a circle of radius 7 is _____.
NAT · enter the numerical value

Q.225 MCQ · 2 marks · GA/Verbal
After a hospital posted waiting times on a public board, median waits fell by 12 minutes. Staffing was unchanged. A plausible mechanism consistent with the report is that
(A) patients left when they saw long waits, thinning the queue
(B) the board repaired the HVAC
(C) doctors worked fewer hours
(D) the calendar skipped a week

Q.226 NAT · 1 mark · GA/Quant
A rectangle is 8 by 6. The length of a diagonal is _____.
NAT · enter the numerical value

Q.227 MCQ · 2 marks · GA/Verbal
A wildlife camera on a trail recorded 40 deer in winter and 8 in summer. The biologist did not count off-trail. The fairest reading is that
(A) the deer population collapsed
(B) on-trail detections were lower in summer
(C) cameras fail in heat
(D) poaching explains the gap

Q.228 MCQ · 2 marks · GA/Quant
A plane parallel to the base of a cone cuts a smaller cone similar to the original. If the cut is at half the height, the small cone's volume is what fraction of the original?
(A) 1/2
(B) 1/4
(C) 1/8
(D) 1/6

Q.229 MCQ · 2 marks · GA/Verbal
A cookbook claims a batter 'must rest 30 minutes or the crumb will be dense'. In a test kitchen, rested and unrested batches had the same crumb score. The test most directly
(A) proves rest is harmful
(B) fails to support the necessity of the rest
(C) shows the oven was cold
(D) invalidates all recipes

Q.230 MCQ · 2 marks · GA/Quant
The lateral faces of a regular square pyramid are equilateral triangles of side 6. The slant height from the midpoint of a base edge to the apex is
(A) 3*sqrt(3)
(B) 6
(C) 3
(D) 2*sqrt(3)

Q.231 MCQ · 2 marks · GA/Verbal
Country P taxes sugary drinks. After two years, sales of taxed drinks fell 15% while sales of untaxed sweet snacks rose 10%. Total sugar from packaged foods was almost unchanged. A supported inference is that
(A) the tax had no effect on any purchases
(B) substitution toward untaxed sweets may have offset drink-side sugar cuts
(C) snacks became healthier
(D) P should ban water

Q.232 MCQ · 2 marks · GA/Quant
A cube is cut by a plane through two parallel edges that lie on opposite faces and are not on the same face. The cross-section is a
(A) equilateral triangle
(B) rectangle
(C) regular hexagon
(D) circle

Q.233 MCQ · 2 marks · GA/Verbal
A university offered free bus passes and later found fewer cars in staff lots. Parking fees also rose the same month. The drop in cars cannot be attributed solely to bus passes because
(A) buses are green
(B) two interventions changed together
(C) lots were never full
(D) staff dislike walking

Q.234 MCQ · 1 mark · GA/Quant
Six identical equilateral triangles of side 4 are joined to form a regular hexagon. The perimeter of that hexagon is
(A) 16
(B) 20
(C) 24
(D) 28

Q.235 MCQ · 2 marks · GA/Verbal

Transcripts show a speaker used the word 'unprecedented' six times in a 12-minute briefing about a storm that matched a 1998 event. The most defensible comment is that

(A) the storm was unique in Earth's history
(B) the speaker's wording overstated novelty relative to 1998
(C) briefings cannot mention years
(D) six is a lucky number

Q.236 MCQ · 1 mark · GA/Quant

A firm's yearly outlay of Rs 50 lakh is split as salaries 40%, rent 20%, materials 25%, and the rest miscellaneous. The materials outlay, in lakh rupees, is

(A) 10
(B) 12.5
(C) 15
(D) 20

Q.237 MCQ · 2 marks · GA/Verbal

A trial of a sleep app enrolled only night-shift nurses. Sleep scores rose. Marketing then claimed the app 'works for everyone'. The overreach is that

(A) nurses do not sleep
(B) the sample is not representative of all sleepers
(C) apps cannot be timed
(D) scores cannot rise

Q.238 NAT · 1 mark · GA/Quant

In a class of 80, the grade pie is A 30%, B 40%, C 20%, D 10%. The number of B grades is _____.

NAT · enter the numerical value

Q.239 MCQ · 2 marks · GA/Verbal

Two translators rendered the same poem. Reviewers preferred Translator A's version for rhythm but Translator B's for fidelity to names. A journal can fairly say that

(A) A is objectively better
(B) preference depends on the criterion (cadence vs. onomastic accuracy)
(C) poetry cannot be translated
(D) B copied A

Q.240 NAT · 2 marks · GA/Quant

Quarterly output shares are Q1 20%, Q2 30%, Q3 25%, Q4 25%. If Q2 produced 600 units, Q1 produced _____ units.

NAT · enter the numerical value

Q.241 MCQ · 1 mark · GA/Verbal

Choose the word that best fills the blank: The witness's account was so _____ that the jury needed no further exhibits.

(A) equivocal
(B) lucid
(C) turgid
(D) vacuous

Q.242 MCQ · 2 marks · GA/Quant

A table of three funds shows: Fund P returned 8% on 10 units, Fund Q returned 12% on 15 units, Fund R returned 6% on 25 units. The unit-weighted average return percent is

(A) 8.0
(B) 8.2
(C) 8.6
(D) 9.0

Q.243 MCQ · 1 mark · GA/Verbal

The committee's praise was _____: every compliment hid a demand for unpaid overtime.

(A) fulsome
(B) candid
(C) laconic
(D) plaintive

Q.244 MCQ · 1 mark · GA/Quant

Exports of a port over four years were 40, 50, 45 and 65 (in 1000 tonnes). The year-on-year growth was largest from

(A) year 1 to 2
(B) year 2 to 3
(C) year 3 to 4
(D) cannot be told

Q.245 MCQ · 1 mark · GA/Verbal

A one-word substitute for 'a person who lives by begging' is

(A) mendicant
(B) misanthrope
(C) martinet
(D) maverick

Q.246 MCQ · 1 mark · GA/Quant

A pie of a 360-student poll is: hostel 90 deg, day-scholar 150 deg, PG 120 deg. The number of day-scholars is

(A) 90
(B) 120
(C) 150
(D) 180

Q.247 MCQ · 1 mark · GA/Verbal

A one-word substitute for 'a statement open to more than one interpretation' is

(A) axiom
(B) ambiguity
(C) apothegm
(D) anodyne

Q.248 MCQ · 1 mark · GA/Quant

The angle between the hands of a clock at 3:00 is (in degrees)

(A) 0
(B) 60
(C) 90
(D) 120

Q.249 MCQ · 1 mark · GA/Verbal

The opposite of 'paucity' in 'a paucity of citations' is

(A) dearth
(B) abundance
(C) brevity
(D) candour

Q.250 NAT · 1 mark · GA/Quant
The smaller angle between the hands of a clock at 4:00, in degrees, is _____.
NAT · enter the numerical value

Q.251 MCQ · 1 mark · GA/Verbal
In 'the argument is specious', specious means
(A) loudly stated
(B) apparently sound but false
(C) overly brief
(D) drawn from data

Q.252 MCQ · 1 mark · GA/Quant
Between 12:00 and 1:00 a clock's hands are at right angles
(A) once
(B) twice
(C) three times
(D) not at all

Q.253 MCQ · 1 mark · GA/Verbal
Choose the correct usage: 'The new rule will _____ next Monday.'
(A) effect from
(B) take effect from
(C) affect from
(D) come affect on

Q.254 MCQ · 1 mark · GA/Quant
If a day is Monday, the day 60 days later is
(A) Wednesday
(B) Thursday
(C) Friday
(D) Saturday

Q.255 MCQ · 1 mark · GA/Verbal
Which sentence uses 'comprise' correctly?
(A) The set is comprised of three subsets.
(B) The set comprises three subsets.
(C) Three subsets comprise of the set.
(D) The set comprises of three subsets.

Q.256 MCQ · 2 marks · GA/Quant
1 January is a Friday in a non-leap year. 1 March of that year is a
(A) Friday
(B) Saturday
(C) Sunday
(D) Monday

Q.257 MCQ · 1 mark · GA/Verbal
Match the opening to the most apt reply. P: 'Could you send the raw traces?' Q: 'Is the embargo lifted?' Replies: 1. 'Yes, as of Tuesday.' 2. 'They are in the annex.'
(A) P-1, Q-2
(B) P-2, Q-1
(C) P-1, Q-1
(D) P-2, Q-2

Q.258 NAT · 1 mark · GA/Quant
A 31-day month begins on Sunday. The number of Sundays in that month is _____.
NAT · enter the numerical value

Q.259 MCQ · 1 mark · GA/Verbal
Match greeting to register. P: 'Kindly find enclosed the invoice.' Q: 'Here's the bill.' Registers: 1. formal written 2. informal spoken
(A) P-2, Q-1
(B) P-1, Q-2
(C) P-1, Q-1
(D) P-2, Q-2

Q.260 MCQ · 1 mark · GA/Quant
The year 2000 is a leap year because
(A) it is even
(B) it is divisible by 4 but not by 100
(C) it is divisible by 400
(D) every century year is a leap year

Q.261 MCQ · 1 mark · GA/Verbal
P: 'The motion fails for want of a seconder.' Q: 'Seconded.' Meanings: 1. support is offered 2. support is absent
(A) P-1, Q-2
(B) P-2, Q-1
(C) P-1, Q-1
(D) P-2, Q-2

Q.262 NAT · 1 mark · GA/Quant
The remainder when 7^5 is divided by 4 is _____.
NAT · enter the numerical value

Q.263 MCQ · 1 mark · GA/Verbal
Choose the option that is NOT a synonym of 'abate' as used for a storm.
(A) subside
(B) intensify
(C) ebb
(D) relent

Q.264 MCQ · 2 marks · GA/Quant
If n is an odd integer, n^2 divided by 8 leaves remainder
(A) 0
(B) 1
(C) 3
(D) 5

Q.265 MCQ · 1 mark · GA/Verbal
The word closest to 'intransigent' is
(A) flexible
(B) uncompromising
(C) cheerful
(D) verbose

Q.266 NAT · 1 mark · GA/Quant
The average of five numbers is 20. If 15 is replaced by 25, the new average is _____.
NAT · enter the numerical value

Q.267 MCQ · 1 mark · GA/Verbal
Fill in: The treaty was a _____ peace, likely to unravel at the first border incident.
(A) durable
(B) fragile
(C) perpetual
(D) unanimous

Q.268 NAT · 1 mark · GA/Quant
The average of 10 numbers is 12. The average of the first 6 is 10. The average of the last 4 is _____.
NAT · enter the numerical value

Q.269 MCQ · 1 mark · GA/Verbal
A 'red herring' in an argument is
(A) a colour proof
(B) an irrelevant distraction
(C) a conceded lemma
(D) a numerical error

Q.270 MCQ · 1 mark · GA/Quant
A 120 m train running at 36 km/h crosses a pole in
(A) 10 s
(B) 12 s
(C) 14 s
(D) 18 s

Q.271 MCQ · 1 mark · GA/Verbal
Which pair is an antonym pair?
(A) laconic / terse
(B) profligate / thrifty
(C) adroit / deft
(D) banal / trite

Q.272 NAT · 1 mark · GA/Quant
A 120 m train at 36 km/h crosses a 240 m platform. The time, in seconds, is _____.
NAT · enter the numerical value

Q.273 MCQ · 1 mark · GA/Verbal
Fill: She gave a _____ denial, leaving no room for a follow-up question.
(A) qualified
(B) categorical
(C) tentative
(D) oblique

Q.274 NAT · 1 mark · GA/Quant
Two trains 150 km apart head toward each other at 60 km/h and 90 km/h. They meet after _____ hour(s).
NAT · enter the numerical value

Q.275 MCQ · 1 mark · GA/Verbal
The phrase 'beg the question' in careful usage means
(A) ask a question politely
(B) assume the point at issue
(C) raise a new topic
(D) request more time

Q.276 MCQ · 2 marks · GA/Quant
A person walks 5 km at 5 km/h and then 5 km at 15 km/h. The average speed for the 10 km is
(A) 7.5 km/h
(B) 8 km/h
(C) 10 km/h
(D) 12 km/h

Q.277 MCQ · 1 mark · GA/Verbal
Match: P 'Please RSVP by Friday.' Q 'Regrets only.' 1. reply if you cannot come 2. reply either way
(A) P-1, Q-2
(B) P-2, Q-1
(C) P-1, Q-1
(D) P-2, Q-2

Q.278 MCQ · 1 mark · GA/Quant
A boat's speed in still water is 12 km/h and the current is 3 km/h. Downstream speed is
(A) 9 km/h
(B) 12 km/h
(C) 15 km/h
(D) 18 km/h

Q.279 MCQ · 1 mark · GA/Verbal
A one-word substitute for 'lasting for a very short time' is
(A) perennial
(B) ephemeral
(C) chronic
(D) interminable

Q.280 MCQ · 2 marks · GA/Quant
A father is thrice as old as his son. In 12 years he will be twice as old as the son. The son's present age is
(A) 10
(B) 12
(C) 14
(D) 18

Q.281 MCQ · 1 mark · GA/Verbal
Choose the correctly punctuated sentence.
(A) The samples which were wet were discarded, the rest were kept.
(B) The samples that were wet were discarded; the rest were kept.
(C) The samples that were wet were discarded the rest were kept.
(D) The samples, that were wet, were discarded; the rest were kept.

Q.282 MCQ · 2 marks · GA/Quant
To obtain a 30% solution by mixing a 20% solution with a 50% solution, the ratio of 20% solution to 50% solution is
(A) 1:2
(B) 2:1
(C) 1:1
(D) 3:2

Q.283 MCQ · 1 mark · GA/Verbal
Fill: He is _____ to acknowledge error than his coauthor is.
(A) more readier
(B) readier
(C) most ready
(D) readyer

Q.284 MCQ · 1 mark · GA/Quant
Which of the following is equivalent to the inequality $2x-5 > 7$?
(A) $x > 4$
(B) $x > 6$
(C) $x < 6$
(D) $x > 1$

Q.285 MCQ · 1 mark · GA/Verbal
Which sentence is unambiguous?
(A) Visiting relatives can be tiring.
(B) Relatives who visit can be tiring.
(C) Tiring can be visiting relatives.
(D) Can visiting be relatives tiring.

Q.286 MCQ · 1 mark · GA/Quant
The inequality |x-3| < 2 is equivalent to
(A) 1 < x < 5
(B) x < 1 or x > 5
(C) x > 5
(D) 0 < x < 3

Q.287 MCQ · 1 mark · GA/Verbal
The blank in 'a _____ of lions' is
(A) school
(B) pride
(C) murder
(D) exaltation

Q.288 MCQ · 1 mark · GA/Quant
The solution set of x^2 - 5x + 6 < 0 is
(A) x < 2
(B) x > 3
(C) 2 < x < 3
(D) x < 2 or x > 3

Q.289 MCQ · 1 mark · GA/Verbal
Soporific : Sleep :: Mnemonic : _____
(A) Forgetfulness
(B) Memory
(C) Hunger
(D) Anger

Q.290 MCQ · 2 marks · GA/Quant
The least positive integer n such that n == 3 (mod 5) and n == 2 (mod 4) is
(A) 7
(B) 11
(C) 13
(D) 18

Q.291 MCQ · 1 mark · GA/Verbal
Erode : Soil :: Corrode : _____
(A) Timber
(B) Metal
(C) Fabric
(D) Paper

Q.292 NAT · 1 mark · GA/Quant
The HCF of 84 and 144 is _____.
NAT · enter the numerical value

Q.293 MCQ · 1 mark · GA/Verbal
Garrulous : Talkative :: Taciturn : _____
(A) Noisy
(B) Reserved
(C) Angry
(D) Fluent

Q.294 NAT · 1 mark · GA/Quant
The LCM of 12, 18 and 30 is _____.
NAT · enter the numerical value

Q.295 MCQ · 1 mark · GA/Verbal
The sentence 'Walking to the lab, the rain started' is faulty because of a
(A) split infinitive
(B) dangling modifier
(C) double negative
(D) subject-verb clash

Q.296 MCQ · 2 marks · GA/Quant
If 3/x + 4/y = 5 and xy=6, a possible value of 3y+4x is
(A) 20
(B) 24
(C) 30
(D) 36

Q.297 MCQ · 1 mark · GA/Verbal
Choose the correct sentence.
(A) The pair of scissors are on the bench.
(B) The pair of scissors is on the bench.
(C) The pair of scissors were on the bench.
(D) The pair of scissors have been on the bench.

Q.298 NAT · 2 marks · GA/Quant
The exponent of 5 in 25! is _____.
NAT · enter the numerical value

Q.299 MCQ · 1 mark · GA/Verbal
Fill: The results were _____ than anyone on the team had hoped.
(A) more better
(B) better
(C) best
(D) more best

Q.300 MCQ · 1 mark · GA/Quant
The harmonic mean of 4 and 12 is
(A) 6
(B) 8
(C) 7
(D) 9

Q.301 MCQ · 1 mark · GA/Verbal
Fill: He objected _____ the clause being added without notice.
(A) at
(B) to
(C) against
(D) for

Q.302 MCQ · 1 mark · GA/Quant
The sum of three consecutive odd integers is 57. The middle integer is
(A) 17
(B) 19
(C) 21
(D) 23

Q.303 MCQ · 1 mark · GA/Verbal
Fill: The plan is contingent _____ board approval.
(A) to
(B) on
(C) with
(D) for

Q.304 NAT · 1 mark · GA/Quant
The smallest 3-digit number that leaves remainder 3 when divided by 7 is _____.
NAT · enter the numerical value

Q.305 MCQ · 1 mark · GA/Verbal
'To steal a march on' someone is to
(A) rob a parade
(B) gain an advantage by acting first
(C) join a protest
(D) lose a race

Q.306 MCQ · 1 mark · GA/Quant
If x + 1/x = 3, then x^2 + 1/x^2 equals
(A) 7
(B) 8
(C) 9
(D) 11

Q.307 MCQ · 1 mark · GA/Verbal
The idiom 'a white elephant' refers to
(A) a pale animal
(B) a costly possession of little use
(C) a lucky charm
(D) a legal loophole

Q.308 NAT · 1 mark · GA/Quant
The value of (0.2)^3 / 0.008 is _____.
NAT · enter the numerical value

Q.309 MCQ · 1 mark · GA/Verbal
A one-word substitute for 'a person who hates humankind' is
(A) philanthropist
(B) misanthrope
(C) misogynist
(D) agnostic

Q.310 MCQ · 1 mark · GA/Quant
A shopkeeper mixes 20 kg of rice at Rs 30/kg with 30 kg at Rs 40/kg. The average cost per kg is
(A) 34
(B) 35
(C) 36
(D) 38

Q.311 MCQ · 1 mark · GA/Verbal
A one-word substitute for 'occurring after death' is
(A) antebellum
(B) posthumous
(C) contemporary
(D) prenatal

Q.312 MCQ · 1 mark · GA/Quant
A can complete 40% of a job in 8 days. At that rate, days for the whole job are
(A) 16
(B) 18
(C) 20
(D) 24

Q.313 MCQ · 1 mark · GA/Verbal
The antonym of 'obsequious' is
(A) servile
(B) assertive
(C) humble
(D) polite

Q.314 MCQ · 1 mark · GA/Quant
The simple interest on a sum for 2 years at 5% is Rs 80. The sum is
(A) 600
(B) 700
(C) 800
(D) 900

Q.315 MCQ · 1 mark · GA/Verbal
In 'the evidence is tenuous', tenuous means
(A) robust
(B) thin or weak
(C) illegal
(D) recent

Q.316 MCQ · 1 mark · GA/Quant
The number of diagonals of a convex octagon is
(A) 16
(B) 20
(C) 24
(D) 28

Q.317 MCQ · 1 mark · GA/Verbal
A brief note says: 'The samples arrived late; consequently the run was aborted.' The word 'consequently' signals
(A) contrast
(B) cause-effect
(C) addition
(D) restatement

Q.318 MCQ · 1 mark · GA/Quant
An interior angle of a regular octagon, in degrees, is
(A) 108
(B) 120
(C) 135
(D) 150

Q.319 MCQ · 1 mark · GA/Verbal
Match: P 'Would you mind closing the sash?' Q 'Close the sash.' 1. polite request 2. imperative order
(A) P-2, Q-1
(B) P-1, Q-2
(C) P-1, Q-1
(D) P-2, Q-2

Q.320 MCQ · 1 mark · GA/Quant
A chord of length 8 is at distance 3 from the centre of a circle. The radius is
(A) 4
(B) 5
(C) 6
(D) 8

Q.321 MCQ · 1 mark · GA/Verbal
Which option best completes: 'Not until the backup finished _____ the team leave.'
(A) did
(B) had
(C) have
(D) was

Q.322 MCQ · 1 mark · GA/Quant
The volume of a sphere of radius 3 is
(A) 12*pi
(B) 24*pi
(C) 36*pi
(D) 108*pi

Q.323 MCQ · 1 mark · GA/Verbal
Choose the sentence with parallel structure.
(A) She likes hiking, to swim, and bicycles.
(B) She likes hiking, swimming, and cycling.
(C) She likes to hike, swimming, and cycle.
(D) She likes hike, swim, and cycling.

Q.324 MCQ · 1 mark · GA/Quant
If 5 workers finish a wall in 12 days, 8 workers of the same rate finish it in
(A) 6.5 days
(B) 7.5 days
(C) 8 days
(D) 9 days

Q.325 MCQ · 1 mark · GA/Verbal
The word 'mitigate' is closest to
(A) worsen
(B) lessen
(C) ignore
(D) announce

Q.326 MCQ · 1 mark · GA/Quant
A number is increased by 25% and the result is decreased by 20%. Relative to the start, the net change is
(A) no change
(B) 5% increase
(C) 5% decrease
(D) 10% increase

Q.327 MCQ · 2 marks · GA/Verbal
A passage states that a town banned fireworks after two warehouse fires, then notes that both fires began in electrical panels, not in fireworks. The fairest comment is that
(A) the ban is uniquely justified by those two fires
(B) the cited fires do not, by themselves, show that fireworks caused the damage
(C) electrical panels cannot burn
(D) fireworks are harmless

Q.328 NAT · 1 mark · GA/Quant
The units digit of 3^23 is _____.
NAT · enter the numerical value

Q.329 MCQ · 1 mark · GA/Verbal
Cacophony : Orchestra in chaos :: Drought : _____
(A) Flooded valley
(B) Parched field
(C) Snowy peak
(D) Tidal pool

Q.330 MCQ · 1 mark · GA/Quant
If log10(2)=0.3010 approximately, the number of digits of 2^10 is
(A) 3
(B) 4
(C) 5
(D) 6

Q.331 MCQ · 1 mark · GA/Verbal
Fill: Neither of the two protocols _____ been validated.
(A) have
(B) has
(C) are
(D) were

Q.332 MCQ · 1 mark · GA/Quant
The probability of drawing a king or a heart from 52 cards is
(A) 1/2
(B) 4/13
(C) 1/4
(D) 3/13

Q.333 MCQ · 1 mark · GA/Verbal
The phrase 'in the nick of time' means
(A) after a delay
(B) at the last possible moment
(C) ahead of schedule
(D) during a pause

Q.334 MCQ · 1 mark · GA/Quant
The mean of 3, 7, 8, 10 and x is 8. Then x equals
(A) 10
(B) 12
(C) 14
(D) 16

Q.335 MCQ · 1 mark · GA/Verbal
A one-word substitute for 'a speech to oneself' is
(A) dialogue
(B) soliloquy
(C) epilogue
(D) prologue

Q.336 NAT · 1 mark · GA/Quant
A number when divided by 12 leaves remainder 7. If twice that number is divided by 12, the remainder is _____.
NAT · enter the numerical value

Q.337 MCQ · 1 mark · GA/Quant
The next term of 11, 23, 47, 95, ... (each term 2*previous +1) is
(A) 189
(B) 191
(C) 193
(D) 195

Q.338 MCQ · 2 marks · GA/Quant
If 2^x = 8^{y} and 3^y = 9^z, then x/z equals
(A) 2
(B) 3
(C) 4
(D) 6

Q.339 NAT · 1 mark · GA/Quant
A shop offers 15% off a Rs 2000 item, then a further Rs 100 off. The final price, in rupees, is _____.
NAT · enter the numerical value

Q.340 MCQ · 1 mark · GA/Quant
The ratio of two numbers is 3:5 and their HCF is 8. Their LCM is
(A) 48
(B) 80
(C) 120
(D) 240

Q.341 MCQ · 1 mark · GA/Quant
A and B together earn Rs 420 in 6 days. A alone earns Rs 40 a day. B's daily wage, in rupees, is
(A) 20
(B) 25
(C) 30
(D) 35

Q.342 MCQ · 1 mark · GA/Quant
A rectangular field is 60 m by 40 m. The cost of fencing it at Rs 12 per metre is
(A) Rs 2000
(B) Rs 2400
(C) Rs 2880
(D) Rs 3200

Q.343 NAT · 1 mark · GA/Quant
The area of a trapezium with parallel sides 8 and 12 and height 5 is _____.
NAT · enter the numerical value

Q.344 MCQ · 1 mark · GA/Quant
If 15% of x is 45, then 25% of x is
(A) 60
(B) 75
(C) 90
(D) 100

Q.345 MCQ · 1 mark · GA/Quant
A clock gains 5 minutes in 4 hours. In 24 hours it will gain
(A) 20 min
(B) 25 min
(C) 30 min
(D) 35 min

Q.346 MCQ · 1 mark · GA/Quant
The sum of interior angles of a hexagon, in degrees, is
(A) 540
(B) 720
(C) 900
(D) 1080

Q.347 NAT · 1 mark · GA/Quant
A fair die is rolled once. If P(even face) is written as a/b in lowest terms, then a+b equals _____.
NAT · enter the numerical value

Q.348 MCQ · 1 mark · GA/Quant
The median of 4, 9, 1, 7, 3 is
(A) 3
(B) 4
(C) 7
(D) 9

Q.349 MCQ · 1 mark · GA/Quant
If 40% of a number is 64, the number is
(A) 120
(B) 140
(C) 160
(D) 180

Q.350 MCQ · 1 mark · GA/Quant
A man sells an article at 10% loss. If the selling price is Rs 450, the cost price is
(A) 400
(B) 500
(C) 495
(D) 550

Q.351 NAT · 1 mark · GA/Quant
The compound interest on Rs 2000 at 10% for 1 year, compounded annually, is _____ rupees.
NAT · enter the numerical value

Q.352 MCQ · 1 mark · GA/Quant
The 10th term of the AP 3, 7, 11, ... is
(A) 35
(B) 39
(C) 43
(D) 47

Q.353 NAT · 1 mark · GA/Quant
The 4th term of a GP with first term 2 and ratio 3 is _____.
NAT · enter the numerical value

Q.354 MCQ · 1 mark · GA/Quant
If 7 workers take 8 days, 4 workers take how many days (same rate)?
(A) 12
(B) 14
(C) 16
(D) 18

Q.355 MCQ · 1 mark · GA/Quant
A square has diagonal 10. Its area is
(A) 25
(B) 50
(C) 75
(D) 100

UNIT 8 SOLUTIONS

Q.1 [MCQ] B — Ephemeral and fleeting are synonyms; perennial pairs with enduring.

Q.2 [MCQ] A — Let tens=u+3. Then 10(u+3)+u=7(u+3+u) gives 11u+30=14u+21 so u=3 and the number is 63.

Q.3 [MCQ] D — The unique row is P, Q, R, S, T, so T is at the right end.

Q.4 [MCQ] A — Hexagon area 54*sqrt(3), circle 36*pi, ratio 3*sqrt(3)/(2*pi).

Q.5 [MCQ] B — A miser hoards money; a spendthrift squanders it.

Q.6 [MCQ] A — 10a+b=6b and ab=18 give 10a=5b so b=2a. Then 2a^2=18, a=3, b=6: the number is 36.

Q.7 [MCQ] C — With A at a seat and D opposite, B clockwise from A forces F opposite B.

Q.8 [NAT] 1 — Only the inner cube is unpainted.

Q.9 [MCQ] B — A scalpel is the surgeon's tool; a gavel is the judge's.

Q.10 [MCQ] A — 11(a+b)=110 so a+b=10, and a-b=4, hence a=7, b=3: 73.

Q.11 [MCQ] C — The man is the woman's maternal uncle, so she is his niece.

Q.12 [NAT] 8 — Only the eight corners have three faces painted.

Q.13 [MCQ] B — An oasis is a wet patch in an arid expanse; an archipelago is a cluster of islands in an ocean.

Q.14 [NAT] 468 — Digits 4,6,8: 468=26*(4+6+8).

Q.15 [MCQ] C — D is A's grandfather, so E is A's great-grandmother.

Q.16 [NAT] 64 — The inner 4x4x4 cube is unpainted: 64.

Q.17 [MCQ] B — Lethargy is the lack of vigour; apathy is the lack of concern.

Q.18 [MCQ] C — Units of powers of 7 cycle 7,9,3,1. Since 2025 mod 4=1, the units digit is 7.

Q.19 [MCQ] B — The four sides use 2,1,5,6, so remaining face 3 is opposite the top 4.

Q.20 [NAT] 96 — Each of 6 faces contributes a 4x4 block of centre-face cubes: 96.

Q.21 [MCQ] B — Alleviate is the opposite of aggravate; pacify is the opposite of provoke.

Q.22 [MCQ] C — Powers of 3 mod 5 cycle 3,4,2,1. 25 mod 4=1, so the remainder is 3.

Q.23 [MCQ] B — This hexomino is a standard valid cube net.

Q.24 [MCQ] B — A strip of six squares overlaps when folded; it is not a cube net.

Q.25 [MCQ] B — A cartographer compiles maps; a lexicographer compiles dictionaries.

Q.26 [NAT] 10 — Let hundreds=1+x with x > =0. Then x+y+z=3 in non-negative integers, C(3+2,2)=10, and each part is at most 9 automatically.

Q.27 [MCQ] D — F already has two courses. J must add Coding to Botany. G and H add Coding to Algebra. Only J teaches Coding without Algebra.

Q.28 [MCQ] B — The extra square must not cause overlap; some attachments work.

Q.29 [MCQ] B — An opaque body does not transmit light; a dielectric does not conduct electric charge.

Q.30 [MCQ] C — The terms are n(n+1) for n=1,2,3,... so the sixth is 6*7=42.

Q.31 [MCQ] A — Di skips Go. Cy skips Hex so Cy plays Go and Yote.

Q.32 [MCQ] B — The circumference 6*pi is one side; the height 4 is the other.

Q.33 [MCQ] B — Dearth means shortage; plethora means surplus.

Q.34 [MCQ] C — Differences 1,4,9,16,25 then 36: 56+36=92.

Q.35 [NAT] 4 — Sorted lists are 5,5,7,a,b with a+b=23 and 8 < =a < b, giving four pairs.

Q.36 [MCQ] C — The arc equals the base circumference 6*pi.

Q.37 [MCQ] B — A novice lacks seasoned craft; a pauper lacks affluence.

Q.38 [NAT] 10 — Geometric: 3,6,12,24. Arithmetic: 5,8,11,14. 24-14=10.

Q.39 [NAT] 7744 — 88^2=7744 is the only 4-digit square of form AABB.

Q.40 [MCQ] B — One fold gives two layers, hence two holes after unfolding.

Q.41 [MCQ] B — Archaeology studies buried settlements; paleontology studies fossils.

Q.42 [MCQ] C — 112*5+5=565.

Q.43 [NAT] 6 — 5+5=10, so I=5, M=1, E=0 and the sum is 6.

Q.44 [MCQ] A — Opposite faces stay opposite under rotation, so the new top is the other red face.

Q.45 [MCQ] B — Cacophony is discordant sound; euphony is harmonious sound.

Q.46 [MCQ] A — 3*82-2=244.

Q.47 [NAT] 363 — 11*33=363.

Q.48 [MCQ] B — 4*6 minus 3 joins each hiding 2 faces: 24-6=18.

Q.49 [MCQ] B — Geese gather as a flock; fish gather as a shoal.

Q.50 [NAT] 26 — Differences 3,5,7, then 9: 17+9=26. Equivalently n^2+1 at n=5 is 26.

Q.51 [NAT] 5 — Union is 35, so neither is 5.

Q.52 [MCQ] B — Euler: V=4, F=4, E=V+F-2=6.

Q.53 [MCQ] B — A telescope reveals far-off objects; a microscope reveals minute specimens.

Q.54 [MCQ] B — Differences 5,10,15,20, then 25: 55+25=80.

Q.55 [NAT] 5 — Union is 45, so neither is 5.

Q.56 [MCQ] C — 6^2 + d^2 = 10^2 so d=8.

Q.57 [MCQ] B — A traitor is defined by treason; a patriot is defined by allegiance.

Q.58 [MCQ] C — 2^2-1=3, 3^2-1=8, 4^2-1=15, 5^2-1=24, 6^2-1=35, 7^2-1=48. So 34 is the misfit (should be 35).

Q.59 [MCQ] B — All poets are dreamers already yields that some dreamers are poets.

Q.60 [MCQ] A — Every unit cube is a corner, so each has three painted faces.

Q.61 [MCQ] B — Brittle materials snap; ductile materials can be drawn into wire.

Q.62 [NAT] 3 — The product telescopes to log2(8)=3.

Q.63 [MCQ] C — The converse of 'no square is a circle' is 'no circle is a square'.

Q.64 [MCQ] B — Five faces are not on the table.

Q.65 [MCQ] B — A vaccine prevents infection; an antidote counters a poison.

Q.66 [MCQ] B — log2(x^2-1)=3 so x^2-1=8, x=3.

Q.67 [MCQ] B — Maps sit inside charts, which are disjoint from sketches.

Q.68 [MCQ] B — 3*6 - 4 hidden join faces - 1 base = 18-4-1=13.

Q.69 [MCQ] B — An anchor holds a vessel; a foundation holds a building.

Q.70 [NAT] 729 — x=9 and y=x^3=729.

Q.71 [MCQ] C — Every mesa is a plateau, so some plateaus are mesas.

Q.72 [MCQ] B — Dual of a cube: 6 vertices, 12 edges, 8 faces.

Q.73 [MCQ] B — A sculptor works with a chisel; a calligrapher works with a nib.

Q.74 [MCQ] B — 2^10 * 5^8 = 4 * 10^8, which is 400000000: 9 digits.

Q.75 [MCQ] A — Even and divisible by 3 together mean divisible by 6.

Q.76 [MCQ] B — An eclipse hides a disc; to illuminate is to reveal a surface.

Q.77 [MCQ] C — Each exponential undoes the log: 9+4=13.

Q.78 [MCQ] B — Each side of the square table has two seats.

Q.79 [MCQ] B — The first triple grows darker; cool to chilly to frigid grows colder.

Q.80 [NAT] 2 — log10(50*20/10)=log10(100)=2.

Q.81 [MCQ] B — Two between X and Y with X at 2 puts Y at 5.

Q.82 [MCQ] B — Pedantic fuss is over minute rules; ostentatious fuss is over showy display.

Q.83 [MCQ] B — b^2=64 so b=8 (b > 0, b!=1). Then log2(8)=3.

Q.84 [MCQ] B — The noisy machines are distracting.

Q.85 [MCQ] B — A hive houses a bee colony; a rookery houses breeding seals (or similar colonial nesters).

Q.86 [MCQ] B — Selling price = 1.25*0.90=1.125 times cost, so profit is 12.5%.

Q.87 [MCQ] C — Opposite of A's left-neighbour B is D, the remaining person, who sits opposite B.

Q.88 [MCQ] B — Malleable metal can be beaten into sheets; a viscous fluid flows sluggishly.

Q.89 [MCQ] B — Milk left = 40*(32/40)^2=25.6 L.

Q.90 [MCQ] B — R is parent of P, and P and Q are siblings, so Q is R's child.

Q.91 [MCQ] B — A quisling is a collaborator; a polyglot is a speaker of many tongues.

Q.92 [MCQ] B — Factor 1.1*0.9=0.99, a 1% decrease.

Q.93 [MCQ] C — Green at an end, blue not at an end, red-yellow-blue as a block, red left of blue: the unique row is Green, Red, Yellow, Blue.

Q.94 [MCQ] B — Tundra names a treeless cold plain; atoll names a ring-shaped coral reef.

Q.95 [MCQ] C — 0.45T+15=0.60T-21 gives 36=0.15T so T=240.

Q.96 [MCQ] D — Nisha=polo, Leela=rowing. Mira skips squash and polo, so Mira=archery, Kiran=squash.

Q.97 [MCQ] B — Ink is the writing medium of a fountain pen; graphite is that of a pencil.

Q.98 [MCQ] C — He delivers 900 g for the price of 1000 g, so profit/cost=100/900=100/9 %.

Q.99 [MCQ] B — Minute hand at 90, hour hand at 90+7.5=97.5; difference 7.5.

Q.100 [MCQ] B — A hypothesis is a tentative claim; a verdict is a settled judgment.

Q.101 [NAT] 28 — The factor is 0.9*0.8=0.72, i.e. 28% off.

Q.102 [MCQ] B — It is a valid syllogism (Barbara).

Q.103 [MCQ] B — To gesticulate is to express with hands; to articulate is to express in speech.

Q.104 [MCQ] C — If B=100, A=75, so B exceeds A by 25/75=1/3, i.e. 100/3 %.

Q.105 [MCQ] B — S is bottom. T cannot sit on S, so the consecutive triple R-P-Q occupies the three seats above S with Q on S, and T is on top: T, R, P, Q, S.

Q.106 [MCQ] B — An archive stores records; a reservoir stores water.

Q.107 [MCQ] B — Factor 1.5*0.5=0.75, a 25% fall.

Q.108 [MCQ] D — M > J > K > L, so M is tallest.

Q.109 [MCQ] B — A completed action before a past point takes past perfect: had reconciled.

Q.110 [MCQ] B — Pass mark is 100, so the shortfall is 15.

Q.111 [MCQ] B — A universal is false if one counterexample exists.

Q.112 [MCQ] B — With neither-nor, the verb agrees with the nearer subject census (singular): shows.

Q.113 [MCQ] C — 1.2C+120=1.1(C+120) simplifies to C=120.

Q.114 [MCQ] A — Each letter shifts +1: L- > M, E- > F, A- > B, F- > G.

Q.115 [MCQ] B — A past counterfactual conditional needs had + past participle: had had quorum.

Q.116 [MCQ] B — Scale B to 12: A:B = 8:12 and B:C = 12:15, so A:B:C = 8:12:15.

Q.117 [MCQ] B — Net displacement is 4 km east.

Q.118 [MCQ] B — Each is singular, so the verb is was.

Q.119 [MCQ] B — Set 2A=3B=4C=12k. Then A=6k, B=4k, C=3k.

Q.120 [MCQ] B — From 15 Aug to 16 Sep is 32 days = 4 weeks + 4 days; Friday+4=Tuesday.

Q.121 [MCQ] B — Insist that takes the mandative subjunctive: be.

Q.122 [NAT] 320 — Parts sum to 9, so each part is 80 and the largest share is 4*80=320.

Q.123 [MCQ] B — Leave one on long enough to heat, one on just now, one off; warmth is a third state.

Q.124 [MCQ] B — In scientific usage data is treated as plural: are.

Q.125 [NAT] 2 — Milk is 12 L, so water must become 6 L; 2 L is added.

Q.126 [MCQ] A — 12+8-5=15.

Q.127 [MCQ] B — Hardly/scarcely is followed by had + subject + past participle.

Q.128 [NAT] 300 — Gold is 3/5 of 500 g, i.e. 300 g.

Q.129 [MCQ] A — 16 = 3*5+1.

Q.130 [MCQ] A — Recommend X as Y is the standard collocation here; as is correct.

Q.131 [MCQ] A — A becomes 5 parts against B's 5 parts, so 1:1.

Q.132 [NAT] 12 — Each of the 12 edges contributes one non-corner edge-centre cube with exactly two faces painted.

Q.133 [MCQ] B — The object of a preposition is me, not I.

Q.134 [MCQ] B — Rates 1/10+1/15=1/6 per day.

Q.135 [MCQ] A — Each letter shifts +1: S- > T, O- > P, U- > V, T- > U, H- > I.

Q.136 [MCQ] B — The relative that refers to errors (plural), so escape.

Q.137 [NAT] 10 — Adding the three pair rates gives twice the triple rate equal to 1/5, so the triple rate is 1/10.

Q.138 [MCQ] A — If B is unmarried then A (married) looks at B; if B is married then B looks at C (unmarried).

Q.139 [MCQ] A — No sooner was X than Y is idiomatic with a simple past passive.

Q.140 [NAT] 12 — Rates 2k and k sum to 3k=1/8, so A's rate is 1/12.

Q.141 [MCQ] C — U true would force V to be the truth-teller as well. V true would force U to be. So W is the sole truth-teller, and both U and V are false.

Q.142 [MCQ] B — Along with does not pluralise the subject; chemist takes performs.

Q.143 [NAT] 24 — Net rate 1/6-1/8=1/24 tank per hour.

Q.144 [MCQ] B — Five Mondays need at least 29 days (Monday start in a 29-day month).

Q.145 [MCQ] B — Would rather + subject takes a past-tense form for present preference: submitted.

Q.146 [MCQ] B — A+B do 3*(1/6+1/8)=21/24, leaving 1/8. C's rate is 1/12, so time is 1.5 days.

Q.147 [MCQ] B — Scarcely a page is singular: was.

Q.148 [NAT] 20 — A man does 1/30 per day and a woman 1/60, together 1/20.

Q.149 [MCQ] B — Causative have + object + past participle: had the contract reviewed.

Q.150 [MCQ] B — Time is inversely proportional to speed.

Q.151 [MCQ] B — Whoever ... chooses is a singular clause and takes has.

Q.152 [MCQ] C — Four days of A+B+C finish 1/2. Two days of B+C finish 5/36. Remainder 13/36, so C needs 13 days.

Q.153 [MCQ] B — A measure used as a single distance is singular: is.

Q.154 [MCQ] B — Net rate 1/20+1/30-1/15=1/60 per min.

Q.155 [MCQ] B — As though referring to an unreal past uses had designed.

Q.156 [MCQ] D — 3/8 is already in. Remaining 5/8 at net 1/8-1/12=1/24 per hour takes 15 hours.

Q.157 [MCQ] B — Many a + singular noun takes a singular verb: has stayed.

Q.158 [NAT] 300 — SI = 2000*5*3/100 = 300.

Q.159 [MCQ] B — The number of is singular: was reduced. (A number of would be plural.)

Q.160 [NAT] 210 — Amount = 1000*1.1^2 = 1210, so CI = 210.

Q.161 [MCQ] B — Let + object + base verb: keep.

Q.162 [NAT] 50 — For two years the gap is P*(r/100)^2 = 50.

Q.163 [MCQ] B — Not only ... but also agrees with the nearer subject captions: need.

Q.164 [MCQ] C — (1+r/100)^2 = 9/4 so r = 50.

Q.165 [MCQ] B — This is the first time takes present perfect: has delivered.

Q.166 [MCQ] C — Doubling at 8% SI takes 100/8 = 12.5 years.

Q.167 [MCQ] B — Minutes (record of a meeting) is plural: are.

Q.168 [NAT] 9600 — SI = 1600, so the amount is 9600.

Q.169 [MCQ] B — Dispose of (handle/settle) is the required phrasal verb.

Q.170 [MCQ] A — Both products are 30% of P, so they match for every P.

Q.171 [MCQ] B — Averse to is the standard preposition.

Q.172 [NAT] 120 — Five 4-subsets have digit-sum divisible by 3, and each subset yields 4! = 24 numbers: 120.

Q.173 [MCQ] B — Consistent with is the collocation.

Q.174 [NAT] 30 — Units 0: 4*3=12; units 2 or 4: 3*3 each. Total 30.

Q.175 [MCQ] B — Debarred from doing something.

Q.176 [NAT] 336 — Units digit must be 5; the first three places are P(8,3)=336.

Q.177 [MCQ] B — Hinge on means depend on.

Q.178 [NAT] 6 — The remaining three distinct letters may be arranged in 3! = 6 ways.

Q.179 [MCQ] B — Acquiesce in a decision is the usual formal collocation.

Q.180 [NAT] 120 — 5! = 120.

Q.181 [MCQ] B — Deficient in a nutrient.

Q.182 [NAT] 8 — The strings are 0000, 0001, 0010, 0100, 0101, 1000, 1001, 1010.

Q.183 [MCQ] B — Atone for a wrong.

Q.184 [NAT] 30 — P(6,2)=30.

Q.185 [MCQ] B — Binding on the parties.

Q.186 [NAT] 56 — C(8,3)=56.

Q.187 [MCQ] A — Prevail on someone to do something.

Q.188 [NAT] 90 — The first digit has 9 choices and the middle digit 10; the last digit is then fixed.

Q.189 [MCQ] B — Bury the hatchet means make peace / end a quarrel.

Q.190 [MCQ] A — (4-1)! = 6.

Q.191 [MCQ] B — On the anvil means under preparation.

Q.192 [MCQ] C — The elder-girl condition leaves (G,G) and (G,B) equally likely.

Q.193 [MCQ] B — It means grasp what is implied rather than said.

Q.194 [MCQ] C — Six of the 36 outcomes sum to 7.

Q.195 [MCQ] B — Cut no ice means have no influence or fail to convince.

Q.196 [MCQ] A — C(4,2)/C(10,2)=6/45=2/15.

Q.197 [MCQ] B — The idiom means confront a problem head-on.

Q.198 [MCQ] C — HHH, HHT, HTH, THH are 4 of 8 outcomes.

Q.199 [MCQ] B — At sixes and sevens means in confusion or disorder.

Q.200 [MCQ] B — The favourable set has 9 integers, so 9/20.

Q.201 [MCQ] B — The phrase means an exaggerated fuss about something small.

Q.202 [MCQ] C — The product is odd only for 9 odd-odd pairs, so 27/36=3/4 are even.

Q.203 [MCQ] B — Casting a gauntlet was a medieval challenge.

Q.204 [MCQ] A — There are 4 aces, so 4/52=1/13.

Q.205 [MCQ] B — It means exploit a position to accumulate personal gain.

Q.206 [MCQ] C — Six equilateral triangles of side 6 give area (3*sqrt(3)/2)*36=54*sqrt(3).

Q.207 [MCQ] B — The idiom means disclose something meant to be hidden.

Q.208 [MCQ] A — Area = sqrt(3)/4 * 36 = 9*sqrt(3).

Q.209 [MCQ] B — The temporal link and the sachet-specific drop support B; the other claims go beyond the passage.

Q.210 [MCQ] A — Volume = (1/3)*pi*r^2*h = 12*pi.

Q.211 [MCQ] B — A missing initial is a process gap, not proof of fraud or instrument failure.

Q.212 [MCQ] B — Space diagonal = side*sqrt(3)=6*sqrt(3).

Q.213 [MCQ] B — Without randomisation, a confounder (motivation) can explain both spacing and retention.

Q.214 [MCQ] B — The diagonal of the square is the diameter 10, so area = 50.

Q.215 [MCQ] B — Correlation across two months does not isolate humidity from other seasonal covariates.

Q.216 [NAT] 30 — Area = (1/2)*5*12=30.

Q.217 [MCQ] B — Only the corridor is measured, so only a local drop is supported.

Q.218 [MCQ] B — Area = (60/360)*pi*36=6*pi.

Q.219 [MCQ] A — The controlled comparison supports a relative benefit of hand annotation, not sweeping claims.

Q.220 [MCQ] B — pi(25-9)=16*pi.

Q.221 [MCQ] B — The data speak only to this batch and this outage.

Q.222 [MCQ] B — CSA=2*pi*r*h=28*pi.

Q.223 [MCQ] B — A third variable (fame) can produce the observed association.

Q.224 [NAT] 154 — 22/7 * 49 = 154.

Q.225 [MCQ] A — Public information can change arrival/reneging behaviour even if staffing is fixed.

Q.226 [NAT] 10 — sqrt(64+36)=10.

Q.227 [MCQ] B — Only trail detections are observed; seasonal trail use may change.

Q.228 [MCQ] C — Linear scale 1/2, so volume scale 1/8.

Q.229 [MCQ] B — Equal scores undercut the 'must rest or else' claim without proving the opposite harm.

Q.230 [MCQ] A — That altitude of an equilateral triangle of side 6 is 3*sqrt(3).

Q.231 [MCQ] B — Offsetting movements in two categories can leave total sugar flat.

Q.232 [MCQ] B — The two edges are equal and parallel, so the section is a rectangle.

Q.233 [MCQ] B — Simultaneous fee hikes confound attribution.

Q.234 [MCQ] C — A regular hexagon of side 4 has perimeter 24.

Q.235 [MCQ] B — A documented analogue contradicts 'unprecedented' in a strict sense.

Q.236 [MCQ] B — 25% of 50 is 12.5.

Q.237 [MCQ] B — A specialised sample does not license a universal claim.

Q.238 [NAT] 32 — 40% of 80 is 32.

Q.239 [MCQ] B — The split reviews show criterion-dependence, not a single ranking.

Q.240 [NAT] 400 — Total is 600/0.30=2000, so Q1 is 20% of 2000=400.

Q.241 [MCQ] B — Lucid means clear; the other options mean ambiguous, swollen/pompous, or empty.

Q.242 [MCQ] B — Total units 50 and weighted sum 410, so 410/50=8.2.

Q.243 [MCQ] A — Fulsome here is insincere or excessive praise with an ulterior edge.

Q.244 [MCQ] C — Increments 10, -5, 20; the largest rise is 20, from year 3 to 4.

Q.245 [MCQ] A — Mendicant is a beggar; the others are a people-hater, a strict disciplinarian, and a nonconformist.

Q.246 [MCQ] C — 150/360 * 360 students = 150.

Q.247 [MCQ] B — Ambiguity is multiple meanings; an axiom is taken as given.

Q.248 [MCQ] C — At 3:00 the hands are perpendicular.

Q.249 [MCQ] B — Paucity means scarcity; abundance is the antonym.

Q.250 [NAT] 120 — The hour hand is at 120 degrees from 12; the minute hand is at 0.

Q.251 [MCQ] B — Specious = deceptively plausible.

Q.252 [MCQ] A — In that hour the hands form a right angle exactly once.

Q.253 [MCQ] B — Take effect from a date is standard; affect is the verb 'to influence'.

Q.254 [MCQ] C — 60 = 8*7+4, so four days after Monday is Friday.

Q.255 [MCQ] B — The whole comprises the parts; 'comprised of' and 'comprises of' are nonstandard.

Q.256 [MCQ] D — Jan+Feb contribute 31+28=59 days, and 59 mod 7=3, so Friday+3=Monday.

Q.257 [MCQ] B — Traces sit in the annex (2); the embargo question is answered by the date (1).

Q.258 [NAT] 5 — Sundays fall on the 1st, 8th, 15th, 22nd and 29th.

Q.259 [MCQ] B — Kindly/enclosed is formal written; here's the bill is informal spoken.

Q.260 [MCQ] C — Century years are leap years only when divisible by 400.

Q.261 [MCQ] B — Want of a seconder means no support; 'Seconded' offers support.

Q.262 [NAT] 3 — 7 == -1 (mod 4), so 7^5 == -1 == 3 (mod 4).

Q.263 [MCQ] B — Abate means lessen; intensify is the opposite.

Q.264 [MCQ] B — n=2k+1 gives n^2=4k(k+1)+1 with k(k+1) even, hence 8m+1.

Q.265 [MCQ] B — Intransigent means refusing to compromise.

Q.266 [NAT] 22 — The total rises by 10, so the average rises by 2.

Q.267 [MCQ] B — Likely to unravel implies fragile, not durable or perpetual.

Q.268 [NAT] 15 — Grand total 120; the first six total 60; the last four total 60, average 15.

Q.269 [MCQ] B — A red herring diverts attention from the issue.

Q.270 [MCQ] B — 36 km/h = 10 m/s, so time = 120/10 = 12 s.

Q.271 [MCQ] B — Profligate (wasteful) vs thrifty (sparing); the others are synonyms.

Q.272 [NAT] 36 — Relative length 360 m at 10 m/s gives 36 s.

Q.273 [MCQ] B — Categorical means absolute and unconditional.

Q.274 [NAT] 1 — Relative speed 150 km/h covers 150 km in 1 hour.

Q.275 [MCQ] B — Classically it means circularly assume what must be proved.

Q.276 [MCQ] A — Times 1 h and 1/3 h; 10 km in 4/3 h is 7.5 km/h.

Q.277 [MCQ] B — RSVP asks for a reply either way; regrets only asks for a reply only if absent.

Q.278 [MCQ] C — Downstream speed is 12+3=15 km/h.

Q.279 [MCQ] B — Ephemeral = short-lived.

Q.280 [MCQ] B — 3s+12=2(s+12) yields s=12.

Q.281 [MCQ] B — Restrictive that (no commas) plus a semicolon joining two independent clauses.

Q.282 [MCQ] B — Alligation gives (50-30):(30-20)=2:1.

Q.283 [MCQ] B — Readier is the comparative of ready; more readier is a double comparative.

Q.284 [MCQ] B — 2x > 12 so x > 6.

Q.285 [MCQ] B — B fixes the gerund/modifier ambiguity of A.

Q.286 [MCQ] A — |x-3| < 2 means 1 < x < 5.

Q.287 [MCQ] B — Lions gather as a pride.

Q.288 [MCQ] C — (x-2)(x-3) < 0 between the roots 2 and 3.

Q.289 [MCQ] B — A soporific induces sleep; a mnemonic aids memory.

Q.290 [MCQ] D — n=5k+3 and 5k+3 == 2 (mod 4) force k == 3 (mod 4), hence n=18.

Q.291 [MCQ] B — Erosion wears soil; corrosion wears metal.

Q.292 [NAT] 12 — 84=2^2*3*7 and 144=2^4*3^2, so HCF=2^2*3=12.

Q.293 [MCQ] B — Garrulous matches talkative; taciturn matches reserved.

Q.294 [NAT] 180 — LCM uses 2^2 * 3^2 * 5 = 180.

Q.295 [MCQ] B — Walking... incorrectly modifies the rain.

Q.296 [MCQ] C — Multiply the first equation by xy=6 to get 3y+4x=30.

Q.297 [MCQ] B — Pair is singular, so is.

Q.298 [NAT] 6 — floor(25/5)+floor(5/5)=5+1=6.

Q.299 [MCQ] B — Better is already comparative.

Q.300 [MCQ] A — 2/(1/4+1/12)=2/(1/3)=6.

Q.301 [MCQ] B — Object to is the collocation.

Q.302 [MCQ] B — Three consecutive odds average 19, and 3*19=57.

Q.303 [MCQ] B — Contingent on (or upon) a condition.

Q.304 [NAT] 101 — 98=14*7, so 98+3=101.

Q.305 [MCQ] B — The idiom means get ahead by acting earlier.

Q.306 [MCQ] A — (x+1/x)^2=9=x^2+2+1/x^2, so x^2+1/x^2=7.

Q.307 [MCQ] B — The phrase names a burdensome, unused asset.

Q.308 [NAT] 1 — 0.008/0.008=1.

Q.309 [MCQ] B — Misanthrope is a hater of humankind.

Q.310 [MCQ] C — Total cost 600+1200=1800 for 50 kg, so 36 per kg.

Q.311 [MCQ] B — Posthumous means after death.

Q.312 [MCQ] C — 40% in 8 days means 100% in 20 days.

Q.313 [MCQ] B — Obsequious means fawning; assertive is opposed.

Q.314 [MCQ] C — P*5*2/100=80 so P=800.

Q.315 [MCQ] B — Tenuous means thin, slight, or weak.

Q.316 [MCQ] B — n(n-3)/2=8*5/2=20.

Q.317 [MCQ] B — Consequently marks a result.

Q.318 [MCQ] C — ((8-2)*180)/8=135.

Q.319 [MCQ] B — Would you mind is a polite request; the bare verb is an order.

Q.320 [MCQ] B — Half-chord 4, so r^2=16+9=25.

Q.321 [MCQ] A — Negative inversion: Not until ... did + subject + verb.

Q.322 [MCQ] C — (4/3)*pi*27=36*pi.

Q.323 [MCQ] B — Three gerunds keep the structure parallel.

Q.324 [MCQ] B — Person-days 60, so 60/8=7.5.

Q.325 [MCQ] B — Mitigate means make less severe.

Q.326 [MCQ] A — 1.25*0.80=1, so no net change.

Q.327 [MCQ] B — The stated ignition source undercuts a fireworks-based causal claim from those fires.

Q.328 [NAT] 7 — Units of powers of 3 cycle 3,9,7,1. 23 mod 4 = 3, so 7.

Q.329 [MCQ] B — Cacophony is disordered sound; drought is a parched field's condition.

Q.330 [MCQ] B — 2^10=1024 has 4 digits; also 1+floor(10*0.3010)=4.

Q.331 [MCQ] B — Neither takes a singular verb: has.

Q.332 [MCQ] B — 4 kings + 13 hearts - 1 overlap = 16, and 16/52=4/13.

Q.333 [MCQ] B — It means just in time.

Q.334 [MCQ] B — Sum 3+7+8+10+x=40, so x=12.

Q.335 [MCQ] B — A soliloquy is a speech to oneself.

Q.336 [NAT] 2 — n=12k+7, 2n=24k+14=12(2k+1)+2.

Q.337 [MCQ] B — 2*95+1=191.

Q.338 [MCQ] D — 2^x=(2^3)^y so x=3y. 3^y=(3^2)^z so y=2z. Thus x=6z.

Q.339 [NAT] 1600 — 15% off leaves 1700, then 100 off leaves 1600.

Q.340 [MCQ] C — Numbers 24 and 40; LCM=120.

Q.341 [MCQ] C — Together they earn 70 a day, so B earns 30.

Q.342 [MCQ] B — Perimeter 200 m times 12 is 2400.

Q.343 [NAT] 50 — Area = (1/2)*(8+12)*5=50.

Q.344 [MCQ] B — x=300, so 25% is 75.

Q.345 [MCQ] C — Gain rate 5/4 min per hour times 24 is 30 min.

Q.346 [MCQ] B — (6-2)*180=720.

Q.347 [NAT] 3 — P(even)=3/6=1/2, so a+b=3.

Q.348 [MCQ] B — Sorted 1,3,4,7,9; the middle is 4.

Q.349 [MCQ] C — n=64/0.4=160.

Q.350 [MCQ] B — 0.9 C=450 so C=500.

Q.351 [NAT] 200 — For one year CI equals SI: 200.

Q.352 [MCQ] B — a+(n-1)d=3+9*4=39.

Q.353 [NAT] 54 — 2*3^3=54.

Q.354 [MCQ] B — 56 person-days / 4 = 14.

Q.355 [MCQ] B — Area = (diagonal^2)/2 = 50.

MOCK PAPER A Mock Paper 01 · Guwahati skeleton (full mix)

Duration: 3 hours. Maximum marks: 100. Q.1-5 GA one mark · Q.6-10 GA two marks · Q.11-35 DA one mark · Q.36-65 DA two marks. MCQ: -1/3 / -2/3. MSQ and NAT: no negative.

Q.1 - Q.5 Carry ONE mark each (General Aptitude)

Q.1 NAT · 1 mark · GA/Quant

The smallest 3-digit number that leaves remainder 3 when divided by 7 is _____.

NAT · enter the numerical value

Q.2 MCQ · 1 mark · GA/Verbal

Fill: The plan is contingent _____ board approval.

(A) to
(B) on
(C) with
(D) for

Q.3 NAT · 1 mark · GA/Quant

Two trains 150 km apart head toward each other at 60 km/h and 90 km/h. They meet after _____ hour(s).

NAT · enter the numerical value

Q.4 MCQ · 1 mark · GA/Verbal

I would rather you _____ the raw counts tonight.

(A) submit
(B) submitted
(C) have submitted
(D) will submit

Q.5 NAT · 1 mark · GA/Quant

The average of five numbers is 20. If 15 is replaced by 25, the new average is _____.

NAT · enter the numerical value

Q.6 - Q.10 Carry TWO marks each (General Aptitude)

Q.6 MCQ · 2 marks · GA/Analytical

A is looking at B. B is looking at C. A is married, C is unmarried, B's status is unknown. The statement 'a married person is looking at an unmarried person' is

(A) always true
(B) always false
(C) true iff B is married
(D) true iff B is unmarried

Q.7 NAT · 2 marks · GA/Quant

How many 4-digit numbers with distinct digits from 1 to 9 are divisible by 5?

NAT · enter the numerical value

Q.8 MCQ · 2 marks · GA/Spatial

A cube has faces painted so that opposite faces share a colour (three colours, each used twice). After a rotation that brings a red face to the bottom, the top face is

(A) red
(B) possibly red
(C) never red
(D) always blue

Q.9 NAT · 2 marks · GA/Quant

A three-digit number has distinct positive digits in arithmetic progression with common difference 2. The number is 26 times the sum of its digits. The number is _____.

NAT · enter the numerical value

Q.10 MCQ · 2 marks · GA/Verbal

An editor notes that papers with shorter abstracts are cited more, but also that celebrity authors write shorter abstracts. The claim that brevity itself causes citations is weak because

(A) abstracts cannot be counted
(B) author fame may drive both brevity and citations
(C) citations never matter
(D) short abstracts lack keywords

Q.11 - Q.35 Carry ONE mark each (DA)

Q.11 MCQ · 1 mark · Python

```
In Python 3, bankers rounding is used. The expression
print(round(2.5), round(3.5), round(2.51))
prints
```

(A) 2 4 3
(B) 3 4 3
(C) 2 3 3
(D) 3 3 3

Q.12 NAT · 1 mark · Logistic

Sigmoid output 0.2. Then f(1-f) equals ____.

NAT · enter the numerical value

Q.13 MCQ · 1 mark · Vandermonde

The Vandermonde matrix on distinct nodes is

(A) invertible
(B) always singular
(C) orthogonal
(D) nilpotent

Q.14 NAT · 1 mark · UninformedSearch

Costs 75, 118, 140 from Arad. The g-value of the first expanded neighbour in UCS is ____.

NAT · enter the numerical value

Q.15 MCQ · 1 mark · Python

```
The program
def f(x):
x += [1]
return x
a = [0]
print(f(a), a)
prints
```

(A) [0, 1] [0]
(B) [0, 1] [0, 1]
(C) [0] [0]
(D) [1] [1]

Q.16 NAT · 1 mark · InformedSearch

True remaining cost h*=6. Heuristics 4 and 5, both admissible. max(h1,h2) at this node equals ____.

NAT · enter the numerical value

Q.17 MCQ · 1 mark · P&S; / axioms

For events A,B with P(A)=0.6, P(B)=0.7, the largest possible value of P(A n B) is

(A) 0.6
(B) 0.42
(C) 1.3
(D) 0.7

Q.18 NAT · 1 mark · Python

```
For the Python function
def f(n):
if n < = 1:
return n
return f(n-1) + 2*f(n-2)
print(f(4))
the printed value is
```

NAT · enter the numerical value

Q.19 MCQ · 1 mark · P&S; / t-dist

Student's t with one degree of freedom is equal in law to which named distribution?

(A) Uniform(-1,1)
(B) Exponential(1)
(C) standard Cauchy
(D) chi^2_1

Q.20 NAT · 1 mark · Perm det

det of the permutation matrix for the transposition (12) in R^2, i.e. [[0,1],[1,0]], is _____.

NAT · enter the numerical value

Q.21 MCQ · 1 mark · Log concave

A Gaussian density is log-concave, so 1-var MLE for the mean of N(mu,1) is the sample mean, unique. For one sample x=4, MLE is _____.

(A) 4
(B) 0
(C) 1
(D) infty

Q.22 NAT · 1 mark · Taylor ln

ln(1+x)=x-x^2/2+x^3/3-... for |x| < 1. Truncating after x for x=0.2 gives error about x^2/2=_____. (two decimals)

NAT · enter the numerical value

Q.23 MCQ · 1 mark · Adversarial

Alpha-beta pruning, with optimal move ordering, can reduce time to about

(A) O(b^{d/2}) instead of O(b^d)
(B) O(1)
(C) O(d)
(D) O(b^d) still, never better

Q.24 NAT · 1 mark · Relational model and FDs

Number of superkeys of R(A,B,C) with only FD A- > BC (so A is the unique candidate key) is

NAT · enter the numerical value

Q.25 MCQ · 1 mark · SLR

If all x-values are identical, simple OLS with intercept

(A) has a unique slope
(B) cannot estimate the slope because Sxx=0
(C) forces R^2=1
(D) makes every residual equal to x

Q.26 NAT · 1 mark · Sorting

Selection sort on n=6 keys always performs how many key comparisons (independent of the input order)?

NAT · enter the numerical value

Q.27 MCQ · 1 mark · Graphs

BFS on an unweighted graph computes

(A) a shortest path in number of edges from the source to every reachable vertex
(B) a minimum spanning tree
(C) shortest paths for negative weights
(D) a topological order of every directed graph

Q.28 NAT · 1 mark · GS

Gram-Schmidt on (1,1) then (0,1): the second orthonormal vector's first component is _____ (two decimals, -1/sqrt(2)~-0.71).

NAT · enter the numerical value

Q.29 MCQ · 1 mark · Integrity constraints

Entity integrity says

(A) no primary-key attribute may be NULL
(B) every FK is NULL
(C) every table has two keys
(D) CHECK constraints are forbidden

Q.30 NAT · 1 mark · Add19

The critical point of f(x)=x-ln x (x > 0) solves 1-1/x=0, x=_____.

NAT · enter the numerical value

Q.31 MCQ · 1 mark · Warehousing and OLAP

A concept hierarchy for Location is Store < City < State < Country. The coarsest level is

(A) Country
(B) Store
(C) City
(D) the fact table

Q.32 NAT · 1 mark · HAC

Points Anu(0,0), Bala(2,0), Chetan(0,1), Diya(5,0), Manhattan. Single-linkage first merge height (Anu-Chetan) is ____.

NAT · enter the numerical value

Q.33 MCQ · 1 mark · Block mul

[[A,B],[0,C]] [[X,Y],[0,Z]] equals [[AX, AY+BZ],[0, CZ]] when shapes allow. This is

(A) false in general
(B) the rule for conforming block-upper-triangular products
(C) only for scalars
(D) LU

Q.34 NAT · 1 mark · Gram rank

rank(A^T A) equals rank(A). If A is 4x3 of rank 2, rank(A^T A) is _____.

NAT · enter the numerical value

Q.35 MCQ · 1 mark · FTC2

int_a^b f' = f(b)-f(a) if f' is integrable. For f(x)=x^3, a=0,b=1, the integral of 3x^2 is

(A) 1
(B) 0
(C) 3
(D) 1/4

Q.36 - Q.65 Carry TWO marks each (DA)

Q.36 MSQ · 2 marks · BiasVariance

Which statements are TRUE?

(A) Bayes error is a lower bound no model can beat
(B) Variance of an estimator usually falls with n
(C) A more flexible model always has lower test error
(D) Regularisation is one way to move along the bias-variance curve

Q.37 NAT · 2 marks · SQL

Clinic(pid,ward,bill): (1,ICU,900),(2,ICU,400),(3,GEN,400),(4,GEN,100),(5,DAY,50).
Correlated: pid whose bill > average bill of their own ward. Number of rows is

NAT · enter the numerical value

Q.38 MSQ · 2 marks · Warehousing and OLAP

Concept hierarchy Time: Day < Week < Month < Quarter < Year. Which OLAP moves are correctly named for cube Sales(Product, Day)?

(A) Day to Month: roll-up
(B) Month to Day: drill-down
(C) Day to Year: roll-up
(D) Year to Month: roll-up

Q.39 MSQ · 2 marks · Integrity constraints

Relations r(A,B) and s(B,C) with FK r.B - > s.B (every r.B must appear in s). Which operations NEVER violate this FK, regardless of the current tuples (assuming they currently satisfy the FK)?

(A) inserting a tuple into s
(B) deleting a tuple from r
(C) inserting a tuple into r
(D) deleting a tuple from s

Q.40 MSQ · 2 marks · NaiveBayes

Which statements are TRUE of naive Bayes?

(A) NB can still classify well even when independence is mildly false
(B) A missing category count can zero a whole posterior path without smoothing
(C) Gaussian NB boundaries are linear if covariances are shared and diagonal (a special LDA)
(D) NB requires linearly separable classes

Q.41 NAT · 2 marks · P&S; / joint

If f(x,y)=e^{-x-y} for x > 0,y > 0, then P(X+Y < = 1) equals 1-2/e. Numerically this is _____ (rounded to two decimals).

NAT · enter the numerical value

Q.42 MSQ · 2 marks · SVM

Which points can be support vectors in a soft-margin linear SVM?

(A) Points on the margin planes
(B) Misclassified points
(C) Points inside the margin strip but correctly signed
(D) A far, correctly classified point with dual alpha exactly 0

Q.43 MSQ · 2 marks · PropLogic

Which are valid equivalences?

(A) P - > Q ? ~Q - > ~P (contrapositive)
(B) ~~P ? P
(C) P v P ? P
(D) P & ~P ? P v ~P

Q.44 NAT · 2 marks · Python

```
In the Python function
def fib(n, c):
c[0] += 1
if n < 2:
return n
return fib(n-1, c) + fib(n-2, c)
c = [0]
fib(5, c)
print(c[0])
the printed call count (including the initial call) is
```

NAT · enter the numerical value

Q.45 MSQ · 2 marks · P&S; / normal

Let X~N(0,1). Which are true?

(A) P(X > 0)=0
(B) P(X > a)=P(X < -a) for all a
(C) The density is even: f(x)=f(-x)
(D) E[X^3]=0

Q.46 MSQ · 2 marks · Independence

Which sets in R^3 are linearly independent?

(A) {(1,0,0),(0,1,0)}
(B) {(1,1,1),(2,2,2)}
(C) {(1,0,0),(0,1,0),(0,0,1)}
(D) {(0,0,0),(1,0,0)}

Q.47 MSQ · 2 marks · P&S; / binomial

X~Binomial(n=6, p=1/3). Which identities hold?

(A) P(X=0)=(2/3)^6
(B) X is approximately N(2, 4/3) with no error even for n=6 (exact, not approximate)
(C) Var(X)=6*(1/3)*(2/3)=4/3
(D) E[X]=2

Q.48 MSQ · 2 marks · Adversarial

Which algorithms belong to adversarial search as in GATE DA?

(A) minimax
(B) alpha-beta pruning
(C) expectiminimax (chance nodes)
(D) k-means clustering of board positions as the primary solver

Q.49 NAT · 2 marks · Relational model and FDs

Let schema Sensor(A,B,C,D,E) with FDs A- > B, BC- > D, D- > E, E- > C. The number of candidate keys is

NAT · enter the numerical value

Q.50 MSQ · 2 marks · Python

`Regarding default arguments in a Python def statement, which claims hold?`

(A) A default list object is evaluated once, at function definition time
(B) A default list object is evaluated fresh on every call that omits the argument
(C) Using None as a sentinel and allocating a list inside the body avoids cross-call sharing
(D) Integer defaults such as n=0 share mutations across calls the same way lists do

Q.51 MCQ · 2 marks · P&S; / counting

From a standard 52-card deck, a 5-card hand is dealt. The number of hands that are flushes (all five cards of one suit, including straight flushes) is

(A) 5148
(B) 649740.0
(C) 0
(D) 5108

Q.52 MCQ · 2 marks · MLMatch

`Task: two-class labelled Gaussians with a shared full covariance. Natural linear classifier:`

(A) LDA (Fisher)
(B) k-NN with k=n
(C) HAC single linkage
(D) unregularised interpolating tree

Q.53 NAT · 2 marks · P&S; / cond-exp

N~Poisson(3) and given N=n, X is Binomial(n, 0.5) (X=0 if n=0). Then E[X] equals _____ (one decimal).

NAT · enter the numerical value

Q.54 MCQ · 2 marks · SVM

If training data are linearly separable, a hard-margin SVM

(A) has a unique maximum-margin w (in general position)
(B) does not exist
(C) always coincides with the OLS hyperplane
(D) has margin 0

Q.55 NAT · 2 marks · PCA

Centered points (-1,0),(1,0),(0,0). Empirical (1/n) covariance has (1,1) entry ___.

NAT · enter the numerical value

Q.56 MSQ · 2 marks · Searching

Linear search on an unordered array of n distinct keys. Which statements are correct?

(A) Worst-case successful search does n key comparisons
(B) Best-case successful search does 1 comparison
(C) Average successful search (uniform) does (n+1)/2 comparisons
(D) Unsuccessful search can be done in O(1) expected time without hashing

Q.57 NAT · 2 marks · P&S; / CI

n=9 Normal observations, s=3, t_{8, 0.025}=2.306. Half-width of the 95% t-interval is 2.306*s/sqrt(n) = _____ (two decimals).

NAT · enter the numerical value

Q.58 MCQ · 2 marks · Consistency

The system [[1,2],[2,4]] [x,y]^T = [3,6]^T is

(A) inconsistent
(B) consistent with infinitely many solutions
(C) a unique solution
(D) empty of solutions and of variables

Q.59 MCQ · 2 marks · Sampling

A valid Gibbs update for node X given the rest uses only

(A) X's Markov blanket values (parents, children, co-parents)
(B) the leaves of a game tree
(C) CNF clauses of X
(D) the A* open list

Q.60 MCQ · 2 marks · P&S; / CLT

A de Moivre-Laplace application: n=16 independent Bernoulli(1/2). Approximate P(S=8) using a continuity correction as P(7.5 < S* < 8.5) where S*~N(8,4). This equals Phi((8.5-8)/2)-Phi((7.5-8)/2) = Phi(0.25)-Phi(-0.25). Using Phi(0.25)~0.5987 the approximation rounded to two decimals is

(A) 0.50
(B) 0.08
(C) 0.20
(D) 0.40

Q.61 MSQ · 2 marks · Diff facts

Which are differentiable at 0?

(A) x^2 sin(1/x) (with 0 at 0)
(B) |x|
(C) x^{1/3}
(D) e^x

Q.62 MSQ · 2 marks · P&S; / independence

`Which constructions yield independent events (assuming fair independent tosses of two coins)?`

(A) A = first heads, B = coins show equal faces
(B) A = empty event, B = any event
(C) A = first heads, B = second heads
(D) A = first heads, B = at least one head (these are dependent)

Q.63 MSQ · 2 marks · Subspace

Which of the following are subspaces of R^3?

(A) {x : x1+x2+x3=0}
(B) {x : x1+x2+x3=1}
(C) {0}
(D) {x : x1 x2 = 0}

Q.64 NAT · 2 marks · Char

Characteristic polynomial of [[0,1],[-2,3]] is t^2-3t+2. The roots (eigenvalues) sum to _____.

NAT · enter the numerical value

Q.65 MSQ · 2 marks · Add26

Which functions are convex on R?

(A) e^x
(B) x^2
(C) -ln(1+x^2) is NOT convex everywhere? Wait - skip
(D) cosh x

MOCK A ANSWER KEY AND SHORT SOLUTIONS

Q.1 [NAT] 101 — 98=14*7, so 98+3=101.

Q.2 [MCQ] B — Contingent on (or upon) a condition.

Q.3 [NAT] 1 — Relative speed 150 km/h covers 150 km in 1 hour.

Q.4 [MCQ] B — Would rather + subject takes a past-tense form for present preference: submitted.

Q.5 [NAT] 22 — The total rises by 10, so the average rises by 2.

Q.6 [MCQ] A — If B is unmarried then A (married) looks at B; if B is married then B looks at C (unmarried).

Q.7 [NAT] 336 — Units digit must be 5; the first three places are P(8,3)=336.

Q.8 [MCQ] A — Opposite faces stay opposite under rotation, so the new top is the other red face.

Q.9 [NAT] 468 — Digits 4,6,8: 468=26*(4+6+8).

Q.10 [MCQ] B — A third variable (fame) can produce the observed association.

Q.11 [MCQ] A — round ties-to-even: 2.5- > 2, 3.5- > 4; 2.51 is nearer 3.

Q.12 [NAT] 0.16 — 0.2*0.8=0.16 by symmetry.

Q.13 [MCQ] A — Distinct x_i.

Q.14 [NAT] 75 — Zerind.

Q.15 [MCQ] B — For lists x += [1] is in-place extend, so a is mutated.

Q.16 [NAT] 5 — max(4,5)=5 still < =6.

Q.17 [MCQ] A — P(A n B) < = min(P(A),P(B))=0.6.

Q.18 [NAT] 5 — f(0)=0, f(1)=1, f(2)=1, f(3)=3, f(4)=3+2*1=5.

Q.19 [MCQ] C — Student t with 1 df is Cauchy.

Q.20 [NAT] -1 — Odd permutation.

Q.21 [MCQ] A — 4.

Q.22 [NAT] 0.02 — 0.02.

Q.23 [MCQ] A — Best-case branching ~ sqrt of minimax.

Q.24 [NAT] 4 — Superkeys are all supersets of {A}: A, AB, AC, ABC. Four of them.

Q.25 [MCQ] B — No x-variation means the slope is unidentified.

Q.26 [NAT] 15 — 5+4+3+2+1=15.

Q.27 [MCQ] A — Each BFS layer is the next hop distance.

Q.28 [NAT] -0.71 — v2=(0,1)-proj= (-1/2,1/2), normalize.

Q.29 [MCQ] A — Entity integrity is the no-NULL-in-PK rule.

Q.30 [NAT] 1 — 1.

Q.31 [MCQ] A — The top of the hierarchy is the coarsest aggregation.

Q.32 [NAT] 1 — Anu-Chetan L1=1 is smallest.

Q.33 [MCQ] B — Block multiplication.

Q.34 [NAT] 2 — Same rank.

Q.35 [MCQ] A — 1.

Q.36 [MSQ] A;B;D — Flexibility can overfit; test error is U-shaped.

Q.37 [NAT] 2 — ICU avg 650 so pid 1; GEN avg 250 so pid 3; DAY avg 50 none. 2 rows.

Q.38 [MSQ] A;B;C — Year to Month is finer, hence drill-down not roll-up.

Q.39 [MSQ] A;B — Inserting into the referenced table s, or deleting from the referencing table r, cannot create a dangling FK. Inserts into r and deletes from s can.

Q.40 [MSQ] A;B;C — Separability is not required. Zero counts are fatal without smoothing.

Q.41 [NAT] 0.26 — Integrate e^{-x-y} over x+y < =1, x,y > 0: 1-2e^{-1} ~ 0.264.

Q.42 [MSQ] A;B;C — SVs have alpha > 0. Well-inside correct points have alpha=0.

Q.43 [MSQ] A;B;C — Contradiction is not equivalent to a tautology.

Q.44 [NAT] 15 — The recursion tree of fib(5) has 15 nodes: T(n)=T(n-1)+T(n-2)+1 with T(0)=T(1)=1 gives 15.

Q.45 [MSQ] B;C;D — Standard normal is symmetric about 0, odd moments vanish, P(X > 0)=1/2.

Q.46 [MSQ] A;C — A scalar multiple pair is dependent; the zero vector kills independence.

Q.47 [MSQ] A;C;D — Mean, variance and empty-success probability are exact. CLT is an approximation, not an identity for n=6.

Q.48 [MSQ] A;B;C — k-means is unsupervised ML, not adversarial search.

Q.49 [NAT] 3 — Candidate keys are AC, AD, AE.

Q.50 [MSQ] A;C — Mutable defaults are classic sharing bugs. Integers are immutable so mutation across calls does not apply.

Q.51 [MCQ] A — Choose suit in 4 ways then 5 ranks from 13: 4*C(13,5)=5148. (Straight-flush exclusion is not asked.)

Q.52 [MCQ] A — LDA is the model.

Q.53 [NAT] 1.5 — E[X]=E[E[X|N]]=E[0.5 N]=1.5.

Q.54 [MCQ] A — Max-margin separator is unique in w for generic separable data.

Q.55 [NAT] 2/3 — ((-1)^2+1+0)/3=2/3.

Q.56 [MSQ] A;B;C — Without extra structure an unsuccessful linear scan is Theta(n).

Q.57 [NAT] 2.31 — SE=1, half-width 2.31.

Q.58 [MCQ] B — Second equation is twice the first; one free variable.

Q.59 [MCQ] A — Full conditional factorises on the blanket.

Q.60 [MCQ] C — 2*0.5987-1=0.1974 ~ 0.20.

Q.61 [MSQ] A;D — |x| has a corner; cube root has vertical tangent.

Q.62 [MSQ] A;B;C — First and second heads are independent; first heads and equality: P both = 1/4 = (1/2)(1/2). Empty event is independent of everything. 'At least one head' is dependent on first heads.

Q.63 [MSQ] A;C — Must contain 0 and be closed under addition. The product constraint is not a subspace; the affine plane misses 0.

Q.64 [NAT] 3 — Trace.

Q.65 [MSQ] A;B;D — e^x, x^2, cosh are convex.

MOCK PAPER B Mock Paper 02 · Bayes, 2x2 eigs, Python defaults

Duration: 3 hours. Maximum marks: 100. Q.1-5 GA one mark · Q.6-10 GA two marks · Q.11-35 DA one mark · Q.36-65 DA two marks. MCQ: -1/3 / -2/3. MSQ and NAT: no negative.

Q.1 - Q.5 Carry ONE mark each (General Aptitude)

Q.1 NAT · 1 mark · GA/Quant

The average of 10 numbers is 12. The average of the first 6 is 10. The average of the last 4 is _____.

NAT · enter the numerical value

Q.2 MCQ · 1 mark · GA/Verbal

Hardly _____ the gates opened when the crowd surged in.

(A) did
(B) had
(C) have
(D) were

Q.3 NAT · 1 mark · GA/Quant

The 5th term of 2, 5, 10, 17, ... (adding successive odd integers starting at 3) is _____.

NAT · enter the numerical value

Q.4 MCQ · 1 mark · GA/Verbal

Fill: He objected _____ the clause being added without notice.

(A) at
(B) to
(C) against
(D) for

Q.5 NAT · 1 mark · GA/Quant

The compound interest on Rs 2000 at 10% for 1 year, compounded annually, is _____ rupees.

NAT · enter the numerical value

Q.6 - Q.10 Carry TWO marks each (General Aptitude)

Q.6 NAT · 2 marks · GA/Quant

The number of binary strings of length 4 with no two consecutive 1s is _____.

NAT · enter the numerical value

Q.7 MCQ · 2 marks · GA/Analytical

Kiran, Leela, Mira, Nisha like four sports: polo, squash, rowing, archery, one each. Kiran does not like polo. Leela likes rowing. Mira does not like squash or polo. Nisha likes polo. Mira likes

(A) polo
(B) squash
(C) rowing
(D) archery

Q.8 NAT · 2 marks · GA/Spatial

A 6x6x6 cube is painted on all faces and diced into unit cubes. The number of unit cubes with paint on exactly one face is _____.

NAT · enter the numerical value

Q.9 MCQ · 2 marks · GA/Quant

A firm's yearly outlay of Rs 50 lakh is split as salaries 40%, rent 20%, materials 25%, and the rest miscellaneous. The materials outlay, in lakh rupees, is

(A) 10
(B) 12.5
(C) 15
(D) 20

Q.10 NAT · 2 marks · GA/Quant

How many 3-digit even numbers with distinct digits can be formed from {0,1,2,3,4}?

NAT · enter the numerical value

Q.11 - Q.35 Carry ONE mark each (DA)

Q.11 NAT · 1 mark · MLR

Fitted MLR yhat=3+2 x1 - x2. At (x1,x2)=(4,5) the prediction is ____.

NAT · enter the numerical value

Q.12 MCQ · 1 mark · Direct

The sum U+W is direct if and only if

(A) U intersect W = {0}
(B) U=W
(C) dim U=0
(D) U+W=R^n always

Q.13 NAT · 1 mark · Python

```
The factorial function
def f(n):
if n < = 1:
return 1
return n * f(n-1)
print(f(5))
prints
```

NAT · enter the numerical value

Q.14 MCQ · 1 mark · P&S; / covariance

The correlation matrix of a non-degenerate random vector must be

(A) symmetric positive semidefinite with ones on the diagonal
(B) orthogonal
(C) skew-symmetric
(D) diagonal with possibly negative entries

Q.15 NAT · 1 mark · Graphs

Grid graph of a 3x3 lattice of vertices with edges between 4-neighbours. Number of vertices is 9; number of edges is

NAT · enter the numerical value

Q.16 MCQ · 1 mark · Log log

d/dx ln(ln x) for x > e is 1/(x ln x). At x=e this derivative is

(A) 1/e
(B) 0
(C) undefined / infinite because ln e=1 wait: 1/(e*1)=1/e
(D) e

Q.17 NAT · 1 mark · FOL

Arity of a binary predicate Likes(x,y) is ____.

NAT · enter the numerical value

Q.18 MCQ · 1 mark · Warehousing and OLAP
Slowly changing dimension Type 2 on CustomerDim handles a city change by
(A) inserting a new dimension row with new surrogate key and version dates
(B) overwriting the city in place with no history (Type 1)
(C) dropping the customer
(D) storing the city in the fact table only

Q.19 NAT · 1 mark · Cofactor
Cofactor C_{11} of [[1,2],[3,4]] is det[4]=____.
NAT · enter the numerical value

Q.20 MCQ · 1 mark · UninformedSearch
Matching: 'expand cheapest-path-so-far first' names
(A) DFS
(B) UCS
(C) greedy best-first
(D) random walk

Q.21 NAT · 1 mark · P&S; / t-dist
Var of t_nu for nu=4 equals nu/(nu-2) = _____.
NAT · enter the numerical value

Q.22 MCQ · 1 mark · Python
```
The fragment
s = 'abca'
print(s.find('a'), s.rfind('a'), s.find('z'))
prints
```
(A) 0 3 -1
(B) 0 0 -1
(C) 1 4 0
(D) 0 3 4

Q.23 NAT · 1 mark · DecisionTree
A full binary classification tree has 5 leaves. Internal split nodes: ____.
NAT · enter the numerical value

Q.24 MCQ · 1 mark · kNN
A large k in k-NN, relative to k=1, typically
(A) fits more wiggly decision boundaries
(B) smooths the decision boundary and raises bias
(C) is undefined for classification
(D) always overfits more than 1-NN

Q.25 NAT · 1 mark · Integrity constraints
```
Same P/C but ON DELETE RESTRICT. DELETE FROM P WHERE id=1
is attempted. Number of rows remaining in P is
```
NAT · enter the numerical value

Q.26 MCQ · 1 mark · InformedSearch
IDA* is iterative-deepening A*, i.e.
(A) depth-first searches with a rising f-bound
(B) BFS with h=0
(C) minimax of heuristics
(D) Gibbs sampling of paths

Q.27 NAT · 1 mark · Log deriv
d/dx log_10 x = 1/(x ln 10). At x=1, 1/ln 10 ~ _____ (two decimals).
NAT · enter the numerical value

Q.28 MCQ · 1 mark · Unique sol
Ax=b has a unique solution for every b in R^n iff A is
(A) square and invertible
(B) any n x (n+1)
(C) singular
(D) a projection

Q.29 NAT · 1 mark · Codim
Codimension of a hyperplane in R^7 is _____.
NAT · enter the numerical value

Q.30 MCQ · 1 mark · SLR
R^2=0 for an OLS line with intercept means
(A) every residual is 0
(B) the fitted slope is 0 (horizontal line at ybar)
(C) SST=0
(D) the intercept is 0

Q.31 NAT · 1 mark · Rank I
rank(I_5) is _____.
NAT · enter the numerical value

Q.32 MCQ · 1 mark · P&S; / binomial
X~Binomial(n=4, p=1/2). Then P(X=2)/P(X=1) equals
(A) 3/2
(B) 2
(C) 1
(D) C(4,2)

Q.33 NAT · 1 mark · DecisionTree
A binary node has 2 positives and 2 negatives. Gini 1-sum p_c^2 equals ____.
NAT · enter the numerical value

Q.34 MCQ · 1 mark · Opt 2nd
A critical point with f''=0 is inconclusive (x^3, x^4, -x^4). For x^4, f''(0)=0 but it is a
(A) min
(B) max
(C) inflection
(D) none of these / saddle in 1d doesn't apply; it is a min

Q.35 NAT · 1 mark · Refl
Householder reflection I-2uu^T for unit u is orthogonal. det equals _____.
NAT · enter the numerical value

Q.36 - Q.65 Carry TWO marks each (DA)

Q.36 MSQ · 2 marks · FOL
X is forall x (P(x)- > Q(x)). Which sentences does X entail?
(A) P(A) - > Q(A)
(B) forall y (P(y)- > Q(y))
(C) ~Q(A) - > ~P(A)
(D) exists x P(x) (without further premises)

Q.37 NAT · 2 marks · VarElim
Binary Q with unnormalised VE result [2, 6]. P(Q=1) after normalising is ____.
NAT · enter the numerical value

Q.38 NAT · 2 marks · Stacks and queues

Characters of GATE are pushed on an empty stack in left-to-right order, then all are popped. If A=1, G=2, T=3, E=4, the first popped value (the code of that character) is

NAT · enter the numerical value

Q.39 MSQ · 2 marks · Limit facts

Which limits equal 0?

(A) lim_{x- > infty} (ln x)/x
(B) lim_{x- > infty} x/e^x
(C) lim_{x- > 0+} x ln x
(D) lim_{x- > 0} (sin x)/x

Q.40 MSQ · 2 marks · Logistic

Which are TRUE of binary logistic regression trained by MLE?

(A) Negative log-likelihood is convex in (w,b)
(B) With linearly separable data and no regularisation, ||w|| may diverge
(C) Predicted probabilities are guaranteed to be exactly 0 or 1 on train points
(D) A feature-scale change can be absorbed into w

Q.41 MSQ · 2 marks · Data types and transformation

Which are lossless compression schemes (in the information-lossless sense)?

(A) Huffman coding of a discrete alphabet
(B) run-length encoding of a bit-vector
(C) JPEG at quality 30 on a photo
(D) dictionary (LZ) compression of a text file

Q.42 MSQ · 2 marks · SQL

On Emp(bonus) with some NULLs, which queries count only non-NULL bonuses?

(A) SELECT COUNT(bonus) FROM Emp
(B) SELECT COUNT(*) FROM Emp WHERE bonus IS NOT NULL
(C) SELECT COUNT(*) FROM Emp
(D) SELECT SUM(1) FROM Emp WHERE bonus IS NOT NULL

Q.43 NAT · 2 marks · P&S; / Bayes

At customs, 5% of travellers carry undeclared goods. A scanner alarms with probability 0.92 if undeclared goods are present and 0.08 otherwise. Given an alarm, P(undeclared goods) equals _____ (two decimals).

NAT · enter the numerical value

Q.44 MCQ · 2 marks · Graphs

Undirected graph Campus has edges Admin-Lab, Lab-Canteen, Canteen-Hostel, Admin-Library, Library-Hostel. How many connected components does it have?

(A) 1
(B) 2
(C) 3
(D) 5

Q.45 MSQ · 2 marks · P&S; / cond-exp

X,Y square-integrable. Which identities always hold?

(A) Var(X)=E[Var(X|Y)]+Var(E[X|Y])
(B) E[X|Y]=E[X] almost surely even when X and Y are dependent
(C) E[E[X|Y]]=E[X]
(D) E[X g(Y)]=E[E[X|Y] g(Y)] for bounded g

Q.46 MCQ · 2 marks · Relational model and FDs

Let schema ClinicVisit(A,B,C,D,E) with FDs A- > B, B- > C, CD- > E, E- > A. Assuming the schema is already in 1NF, the highest normal form it is guaranteed to be in is

(A) 3NF
(B) 1NF
(C) 2NF
(D) BCNF

Q.47 MSQ · 2 marks · Sorting

Array PairSwap = [2, 1, 4, 3, 5]. Which claims hold?

(A) the array has exactly two inversions
(B) adjacent bubble sort performs exactly two swaps
(C) adjacent insertion sort performs exactly two swaps
(D) mergesort extra memory is Theta(1)

Q.48 MSQ · 2 marks · ER model

An ER diagram for Festival has entity Performer and entity Stage linked by PlaysOn (M:N) and a recursive relationship Mentors on Performer (1:N). Which mappings are required?

(A) a table Performer
(B) a table Stage
(C) a table PlaysOn with two FKs
(D) Mentors absorbed as a nullable mentor_id FK on Performer

Q.49 MSQ · 2 marks · Theorems

Which apply to f(x)=x^2 on [0,1]?

(A) EVT (attains min/max)
(B) MVT
(C) Rolle if we shift to match endpoints? f(0)!=f(1) so Rolle fails
(D) IVT on the range [0,1]

Q.50 MCQ · 2 marks · P&S; / continuous

The CDF of X is 0 for x < 0, x^2 for 0 < =x < =1, and 1 for x > 1. Then P(0.25 < X < = 0.75) equals

(A) 0.5625
(B) 0.5
(C) 0.75
(D) 0.25

Q.51 NAT · 2 marks · P&S; / Bayes

A box contains two coins: one fair and one two-headed. A coin is picked at random and flipped; it shows heads. P(the coin is two-headed | heads) equals _____ (two decimals).

NAT · enter the numerical value

Q.52 MSQ · 2 marks · UninformedSearch

Which statements about uniform-cost search are TRUE?

(A) With nonnegative costs, the first time a node is expanded it has an optimal g-value (graph-search)
(B) UCS with all costs equal to 1 behaves like BFS (up to tie-breaking)
(C) UCS is informed because it uses g(n)
(D) Negative action costs can break UCS optimality

Q.53 NAT · 2 marks · Relational model and FDs

Let schema Voyage(A,B,C,D,E) with FDs A- > BC, CD- > E, E- > A. The number of candidate keys is

NAT · enter the numerical value

Q.54 MSQ · 2 marks · Stacks and queues

A dequeue (double-ended queue) supports insert/delete at both ends in O(1). Which statements are correct?

(A) A deque can simulate a stack using only one end
(B) A deque can simulate a FIFO queue using opposite ends
(C) A singly linked list with only a head pointer is already a deque
(D) BFS of a graph is naturally implemented with a deque used as a FIFO queue

Q.55 MCQ · 2 marks · Onb

An orthonormal basis of R^2 is

(A) {(1,0),(0,1)}
(B) {(1,1),(1,-1)} without scaling
(C) {(1,0),(1,0)}
(D) {(0,0),(1,0)}

Q.56 MSQ · 2 marks · Python

Let L be a list and T a tuple of the same elements. Which statements are always true in Python 3?

(A) L.append(0) mutates L in place
(B) T.append(0) mutates T in place
(C) L += [0] reuses the same list object
(D) T += (0,) reuses the same tuple object

Q.57 MCQ · 2 marks · Stochastic

A row-stochastic matrix has

(A) each row summing to 1
(B) each column summing to 0
(C) trace 1
(D) det 1

Q.58 MSQ · 2 marks · P&S; / t-dist

Which are true for Student's t_nu as nu - > infinity?

(A) t_nu becomes Uniform(-1,1)
(B) t_nu converges in distribution to N(0,1)
(C) the variance of t_nu (for nu > 2) tends to 1
(D) P(|t_nu| > 2) tends to P(|Z| > 2) for Z~N(0,1)

Q.59 MCQ · 2 marks · PCA

If two features are exactly linearly dependent, the covariance has

(A) a zero eigenvalue (rank-deficient)
(B) all eigenvalues equal
(C) negative variance
(D) no eigenvectors

Q.60 MSQ · 2 marks · P&S; / mixed

Let X~Binomial(n,p) and Y~Poisson(np) with the same mean. Which comparisons are correct for 0 < p < 1, n > =2?

(A) X and Y are equal almost surely
(B) E[X]=E[Y]
(C) P(X=0)=(1-p)^n which is not equal to e^{-np} in general
(D) Var(X)=np(1-p) < np = Var(Y)

Q.61 NAT · 2 marks · More der 2

d/dx (sin x / x) at 0, after filling f(0)=1, equals _____.

NAT · enter the numerical value

Q.62 MSQ · 2 marks · LDA

`Which are TRUE of S_B and S_W in two-class Fisher LDA?`

(A) S_B is rank-1, proportional to (m1-m2)(m1-m2)^T
(B) S_W is a pooled within-class scatter
(C) The Rayleigh quotient leads to the generalised eigenproblem S_B w = lambda S_W w
(D) S_W^{-1} S_B is always symmetric even if S_W is not a multiple of I

Q.63 MCQ · 2 marks · P&S; / covariance

X~Uniform(-1,1) and Y=X^2. Then Corr(X,Y) equals

(A) 0
(B) 1/2
(C) 1
(D) -1

Q.64 MCQ · 2 marks · Volume

|det A| is the volume scaling of the unit cube under A. For a 90-degree rotation this factor is

(A) 1
(B) 0
(C) 2
(D) pi

Q.65 MSQ · 2 marks · PCA

Which are TRUE of PCA?

(A) The principal axes are pairwise orthogonal
(B) The first PC maximises projected variance among unit vectors
(C) PCA uses class labels
(D) The scores on distinct PCs are uncorrelated (for the empirical covariance)

MOCK B ANSWER KEY AND SHORT SOLUTIONS

Q.1 [NAT] 15 — Grand total 120; the first six total 60; the last four total 60, average 15.

Q.2 [MCQ] B — Hardly/scarcely is followed by had + subject + past participle.

Q.3 [NAT] 26 — Differences 3,5,7, then 9: 17+9=26. Equivalently n^2+1 at n=5 is 26.

Q.4 [MCQ] B — Object to is the collocation.

Q.5 [NAT] 200 — For one year CI equals SI: 200.

Q.6 [NAT] 8 — The strings are 0000, 0001, 0010, 0100, 0101, 1000, 1001, 1010.

Q.7 [MCQ] D — Nisha=polo, Leela=rowing. Mira skips squash and polo, so Mira=archery, Kiran=squash.

Q.8 [NAT] 96 — Each of 6 faces contributes a 4x4 block of centre-face cubes: 96.

Q.9 [MCQ] B — 25% of 50 is 12.5.

Q.10 [NAT] 30 — Units 0: 4*3=12; units 2 or 4: 3*3 each. Total 30.

Q.11 [NAT] 6 — 3+8-5=6.

Q.12 [MCQ] A — Definition of internal direct sum.

Q.13 [NAT] 120 — 5!=120.

Q.14 [MCQ] A — Corr = D^{-1/2} Cov D^{-1/2} is a Gram matrix, hence SPD/PSD, with unit diagonal.

Q.15 [NAT] 12 — Horizontal: 3 rows * 2 = 6; vertical: 3 cols * 2 = 6; total 12.

Q.16 [MCQ] A — 1/e.

Q.17 [NAT] 2 — Two arguments.

Q.18 [MCQ] A — Type 2 versions rows. Type 1 overwrites.

Q.19 [NAT] 4 — Minor.

Q.20 [MCQ] B — UCS orders by g(n).

Q.21 [NAT] 2 — 4/(4-2)=2.

Q.22 [MCQ] A — find leftmost index 0, rfind rightmost 3, missing char yields -1.

Q.23 [NAT] 4 — internals=leaves-1.

Q.24 [MCQ] B — Larger neighbourhoods average more.

Q.25 [NAT] 2 — The delete is rejected; both parent rows stay.

Q.26 [MCQ] A — IDA* = DFS-style f-limited iterations.

Q.27 [NAT] 0.43 — 1/2.302~0.43.

Q.28 [MCQ] A — Full rank square.

Q.29 [NAT] 1 — 1.

Q.30 [MCQ] B — SSR=0 iff the slope is 0 when an intercept is present.

Q.31 [NAT] 5 — 5.

Q.32 [MCQ] A — C(4,2)/C(4,1)=6/4=3/2, and the p-powers match.

Q.33 [NAT] 0.5 — 1-2*0.25=0.5.

Q.34 [MCQ] A — Min.

Q.35 [NAT] -1 — One eigenvalue -1.

Q.36 [MSQ] A;B;C — A universal implication does not by itself assert existence of a P-object.

Q.37 [NAT] 0.75 — 6/(2+6)=0.75.

Q.38 [NAT] 4 — E is pushed last and popped first; its code is 4.

Q.39 [MSQ] A;B;C — (sin x)/x - > 1, the others vanish.

Q.40 [MSQ] A;B;D — NLL is convex; separability sends unregularised MLE to infinity; p stays in (0,1).

Q.41 [MSQ] A;B;D — JPEG is lossy. Huffman, RLE and LZ are lossless.

Q.42 [MSQ] A;B;D — COUNT(*) over the whole table includes NULL-bonus rows.

Q.43 [NAT] 0.38 — 0.046/(0.046+0.076)=0.046/0.122 ~ 0.38.

Q.44 [MCQ] A — Every vertex is reachable from Admin; the graph is connected.

Q.45 [MSQ] A;C;D — Tower property, law of total variance, and the defining projection property. E[X|Y]=E[X] is independence-like (mean independence), not automatic.

Q.46 [MCQ] A — Candidate keys AD, BD, CD, DE. Highest NF is 3NF.

Q.47 [MSQ] A;B;C — Inversions (2,1),(4,3); adjacent bubble/insertion swap once per inversion. Usual mergesort needs a linear buffer.

Q.48 [MSQ] A;B;C;D — Recursive 1:N becomes a self-FK; M:N becomes a table.

Q.49 [MSQ] A;B;D — Rolle needs equal endpoint values.

Q.50 [MCQ] B — F(0.75)-F(0.25)=0.5625-0.0625=0.5. Density 2x on (0,1).

Q.51 [NAT] 0.67 — P(H)=0.25+0.5=0.75, P(two-headed|H)=0.5/0.75=2/3 ~ 0.67.

Q.52 [MSQ] A;B;D — g(n) is not a heuristic; UCS is uninformed. Negative costs invalidate Dijkstra-style proofs.

Q.53 [NAT] 3 — Candidate keys are AD, CD, DE.

Q.54 [MSQ] A;B;D — A head-only singly linked list cannot delete at the tail in O(1), so it is not a deque.

Q.55 [MCQ] A — Standard. The second needs 1/sqrt(2).

Q.56 [MSQ] A;C — Tuples have no append; T += (0,) rebinds T to a new tuple because tuples are immutable.

Q.57 [MCQ] A — 1 is always an eigenvalue.

Q.58 [MSQ] B;C;D — Standard limit of t. Variance is nu/(nu-2) - > 1.

Q.59 [MCQ] A — Rank at most 1 in that plane.

Q.60 [MSQ] B;C;D — Poisson has the larger variance np vs np(1-p). They are different laws; Poisson is a limit, not an a.s. identity.

Q.61 [NAT] 0 — Even function, odd derivative.

Q.62 [MSQ] A;B;C — S_W^{-1} S_B need not be symmetric; one solves a generalised eigenproblem.

Q.63 [MCQ] A — X is symmetric about 0 and Y is even, so E[XY]=E[X^3]=0=E[X]E[Y]. Uncorrelated but dependent.

Q.64 [MCQ] A — |det|=1.

Q.65 [MSQ] A;B;D — PCA is unsupervised; it ignores labels.

MOCK PAPER C Mock Paper 03 · SQL NULL, PCA axes, A* vs UCS

Duration: 3 hours. Maximum marks: 100. Q.1-5 GA one mark · Q.6-10 GA two marks · Q.11-35 DA one mark · Q.36-65 DA two marks. MCQ: -1/3 / -2/3. MSQ and NAT: no negative.

Q.1 - Q.5 Carry ONE mark each (General Aptitude)

Q.1 NAT · 1 mark · GA/Quant

A 120 m train at 36 km/h crosses a 240 m platform. The time, in seconds, is _____.

NAT · enter the numerical value

Q.2 MCQ · 1 mark · GA/Spatial

A cone of slant height 5 and base radius 3 is unrolled. The sector's arc length is

(A) 3
(B) 5
(C) 6*pi
(D) 3*pi

Q.3 NAT · 1 mark · GA/Quant

The amount on Rs 8000 at 5% per annum simple interest for 4 years is _____ rupees.

NAT · enter the numerical value

Q.4 MCQ · 1 mark · GA/Verbal

Whoever the committee chooses _____ to file a disclosure.

(A) have
(B) has
(C) are
(D) were

Q.5 NAT · 1 mark · GA/Quant

A number when divided by 12 leaves remainder 7. If twice that number is divided by 12, the remainder is _____.

NAT · enter the numerical value

Q.6 - Q.10 Carry TWO marks each (General Aptitude)

Q.6 NAT · 2 marks · GA/Quant

A geometric progression has first term 3 and ratio 2. An arithmetic progression has first term 5 and common difference 3. The 4th geometric term minus the 4th arithmetic term equals _____.

NAT · enter the numerical value

Q.7 MCQ · 2 marks · GA/Spatial

A plus-shaped net of five squares plus a sixth square attached to an outer edge of one arm

(A) cannot be a cube net
(B) is a cube net for some attachments and not others
(C) always folds to two tetrahedra
(D) always leaves a hole

Q.8 NAT · 2 marks · GA/Analytical

The unique 4-digit perfect square whose first two digits are equal and whose last two digits are equal is _____.

NAT · enter the numerical value

Q.9 MCQ · 2 marks · GA/Quant

A can do a job in 18 days, B in 24 days, C in 36 days. They start together; A leaves after 4 days, then B leaves after 2 more days. Days C still needs after that is

(A) 8
(B) 10
(C) 13
(D) 16

Q.10 NAT · 2 marks · GA/Analytical

Five positive integers have unique mode 5 (appearing twice), median 7, mean 8, and all entries other than the two 5s are distinct and different from 5 and 7. The number of such lists, up to order, is _____.

NAT · enter the numerical value

Q.11 - Q.35 Carry ONE mark each (DA)

Q.11 NAT · 1 mark · Quad eval

q(x,y)=x^2+2xy+y^2 at (1,-1) equals _____.

NAT · enter the numerical value

Q.12 MCQ · 1 mark · Nilp exp

For N^2=0, exp(N)=

(A) I+N
(B) I
(C) 0
(D) N

Q.13 NAT · 1 mark · PropLogic

Number of models of P v Q on atoms {P,Q} is ___.

NAT · enter the numerical value

Q.14 MCQ · 1 mark · Hankel

A Hankel matrix is constant along

(A) anti-diagonals
(B) the main diagonal only
(C) the border
(D) eigenvectors

Q.15 NAT · 1 mark · Inv 2x2

Inverse of [[1,1],[0,1]] is [[1,-1],[0,1]]. The (1,2) entry of the inverse is _____.

NAT · enter the numerical value

Q.16 MCQ · 1 mark · Mean value int

There is c in (a,b) with f(c) equal to the average of f, if f is continuous. For f(x)=x on [0,2], c=_____.

(A) 1
(B) 0
(C) 2
(D) 4

Q.17 NAT · 1 mark · kNN

Points (0,0),(0,2),(6,0). Query (1,0). Euclidean 1-NN distance to the winner is ___.

NAT · enter the numerical value

Q.18 MCQ · 1 mark · BiasVariance

k-fold CV is used in this story mainly to

(A) estimate prediction error and help pick a complexity knob
(B) remove the noise term
(C) force bias to 0
(D) compute S_W^{-1} S_B

Q.19 NAT · 1 mark · Sorting
A single merge of two sorted runs of lengths 3 and 5 uses how many key comparisons in the worst case?
NAT · enter the numerical value

Q.20 MCQ · 1 mark · Inflexion
f(x)=x^3 at 0 has f'(0)=f''(0)=0. It is
(A) a local min
(B) a local max
(C) an inflection (not a local extremum)
(D) discontinuous

Q.21 NAT · 1 mark · Data types and transformation
Decimal 13 in 4-bit two's complement does not fit. The largest non-negative integer representable in 4-bit two's complement is
NAT · enter the numerical value

Q.22 MCQ · 1 mark · P&S; / mixed
Let phi be convex and X integrable so that E[phi(X)] exists. Jensen's inequality states that
(A) phi(E[X]) > = E[phi(X)] always
(B) Var(X) < =0
(C) E[X^2] < = (E[X])^2
(D) phi(E[X]) < = E[phi(X)]

Q.23 NAT · 1 mark · Relational model and FDs
R(A,B,C,D) with FDs A- > B, B- > C, C- > D. Size of A+ is
NAT · enter the numerical value

Q.24 MCQ · 1 mark · Warehousing and OLAP
On cube SensorAvg (Station, Metric, Day), aggregating away the middle dimension entirely (summing the measure over it) is
(A) roll-up / projection of a dimension
(B) drill-down
(C) a foreign-key insert
(D) 2NF decomposition

Q.25 NAT · 1 mark · Det 2I
det(2 I_3) is _____.
NAT · enter the numerical value

Q.26 MCQ · 1 mark · FOL
A term is built from
(A) variables, constants, and function symbols; it is not a formula by itself
(B) quantifiers only
(C) implication only
(D) clauses of CNF only

Q.27 NAT · 1 mark · Subspace
Dimension of {x in R^4 : x1=x2 and x3=0} is _____.
NAT · enter the numerical value

Q.28 MCQ · 1 mark · P&S; / mixed
Consider (i) for X~Poisson(lambda), mean equals variance; (ii) for Z~N(0,4), mean is 0 and variance is 1. Which is correct?
(A) both false
(B) both true
(C) (i) false, (ii) true
(D) (i) true, (ii) false

Q.29 NAT · 1 mark · P&S; / counting
A robot on a grid may move only right or up. The number of shortest paths from (0,0) to (4,3) is _____.
NAT · enter the numerical value

Q.30 MCQ · 1 mark · Python
```
After
A = [1, 2, 3]
B = [4, 5, 6]
A.append(B)
print(len(A), A[-1])
the printed pair is
```
(A) 6 [4, 5, 6]
(B) 4 [4, 5, 6]
(C) 4 6
(D) 6 6

Q.31 NAT · 1 mark · HAC
n=6 points, agglomerative clustering. Number of merge operations until one cluster is ____.
NAT · enter the numerical value

Q.32 MCQ · 1 mark · Diff quot
f'(a)=lim h- > 0 (f(a+h)-f(a))/h. For f(x)=x^2, a=3, the difference quotient at h=0.1 is (9.61-9)/0.1=_____.
(A) 6.1
(B) 6
(C) 9
(D) 0.1

Q.33 NAT · 1 mark · PropLogic
Truth table of P - > Q: number of T rows among 4 is ____.
NAT · enter the numerical value

Q.34 MCQ · 1 mark · Eig vec
A=[[2,0],[0,3]]. A corresponding eigenvector for lambda=3 is
(A) [1,0]^T
(B) [0,1]^T
(C) [1,1]^T
(D) [0,0]^T

Q.35 NAT · 1 mark · LDA
```
Projected LDA scores have class means -3 and +3
(balanced). Midpoint threshold is ____.
```
NAT · enter the numerical value

Q.36 - Q.65 Carry TWO marks each (DA)

Q.36 MSQ · 2 marks · PCA
Which preprocessing steps are the usual prelude to PCA on heterogeneous features?
(A) Center each feature
(B) Optionally scale to unit variance (standardise) so scale does not dominate
(C) Replace features by their labels
(D) Drop orthogonality of PCs on purpose

Q.37 MSQ · 2 marks · P&S; / t-dist

A one-sample t-test is applied to n=9 allegedly i.i.d. Normal observations. Which are true?

(A) Under H0 the t statistic has 8 degrees of freedom
(B) The t critical value is smaller than the z critical value for the same tail probability
(C) If sigma were known one would use a z-test instead
(D) Heavy-tailed true data (not Normal, small n) make the nominal 5% level untrustworthy

Q.38 MSQ · 2 marks · Hash tables

Which statements about hashing are correct?

(A) Linear probing suffers from primary clustering
(B) Quadratic probing eliminates secondary clustering completely
(C) Double hashing uses a second hash to choose the step
(D) Chaining remains correct even when n exceeds m

Q.39 MSQ · 2 marks · Sampling

On the tiny Rain- > Wet BN, evidence Wet=true (a child). Which are TRUE?

(A) Rejection sampling discards worlds with Wet=false
(B) Likelihood weighting sets Wet=true and weights by P(Wet=true | Rain)
(C) Both methods (with infinite samples) can estimate P(Rain | Wet=true)
(D) VE is approximate here while LW is exact in finite samples

Q.40 NAT · 2 marks · P&S; / covariance

X is Uniform on {1,2,3} and Y=X^2. Then Cov(X,Y)=E[X^3]-E[X]E[X^2]. This covariance equals _____ (rounded to two decimals).

NAT · enter the numerical value

Q.41 MSQ · 2 marks · SQL

Concerning COUNT and NULLs in SQL:

(A) COUNT(*) counts rows including those with NULLs
(B) COUNT(A) ignores rows where A is NULL
(C) SUM(A) ignores NULLs
(D) AVG(A) treats NULL as 0

Q.42 MSQ · 2 marks · Warehousing and OLAP

Measures of a warehouse fact. Which classifications are correct?

(A) quantity sold is additive across all dimensions
(B) account balance is semi-additive (cannot sum over time)
(C) ratio of defective items (a percentage) is typically non-additive
(D) date_id in the fact table is an additive measure

Q.43 MSQ · 2 marks · P&S; / tests

In a level-alpha two-sided z-test of H0: mu=mu0 (sigma known), which are true?

(A) Failing to reject H0 proves mu=mu0
(B) Power is a function of the true mu, n, sigma and alpha
(C) Type I error probability is alpha when H0 is true (under the Normal/CLT model)
(D) p-value < alpha is the same decision as rejecting at level alpha

Q.44 MSQ · 2 marks · P&S; / joint

A 2x2 joint PMF on {0,1}x{0,1} has all four cells equal to 1/4. Which are true?

(A) P(X=Y)=1/2
(B) Corr(X,Y)=0
(C) P(X=1,Y=1)=1/2
(D) X and Y are independent Bernoulli(1/2)

Q.45 MSQ · 2 marks · Ridge

`Relative to OLS, increasing ridge lambda typically`

(A) raises bias of the coefficient vector
(B) lowers variance of the coefficient vector
(C) drives coefficients toward 0
(D) makes X^T X + lambda I more ill-conditioned

Q.46 MCQ · 2 marks · Relational model and FDs

A relation is in 2NF if it is in 1NF and

(A) no nonprime attribute is partially dependent on a candidate key
(B) every determinant is a candidate key
(C) it has no transitive dependency of a nonprime on a key
(D) it has a single-attribute key

Q.47 MCQ · 2 marks · InformedSearch

A* graph-search with a consistent heuristic never needs to reopen a closed node because

(A) the first time a node is expanded, g is already optimal
(B) h is always 0
(C) the frontier is a stack
(D) cycles have zero cost

Q.48 MCQ · 2 marks · P&S; / chi-squared

Noise samples Z1,...,Zk are i.i.d. N(0,1). The residual sum of squares Z1^2+...+Zk^2 follows

(A) chi^2 with k degrees of freedom
(B) Uniform(0,k)
(C) N(0,k)
(D) t_k

Q.49 MCQ · 2 marks · Graphs

Directed graph Airline has edges Delhi- > Mumbai, Delhi- > Pune, Mumbai- > Goa, Pune- > Goa, Goa- > Kochi. Recursive DFS from Delhi with neighbours tried in alphabetical order. The discovery order is

(A) Delhi Mumbai Goa Kochi Pune
(B) Delhi Mumbai Pune Goa Kochi
(C) Pune Kochi Goa Mumbai Delhi
(D) Pune Mumbai Kochi Goa Delhi

Q.50 MCQ · 2 marks · PSD

A Gram matrix A^T A is always

(A) positive semidefinite
(B) indefinite
(C) orthogonal
(D) nilpotent

Q.51 MCQ · 2 marks · SLR

OLS of y on x is y=4+2x. The same sample is used to regress x on y. Which identity ALWAYS holds?

(A) The reverse slope must be 1/2
(B) The reverse slope equals 2
(C) The product of the two OLS slopes equals r^2
(D) The reverse intercept is 4

Q.52 MSQ · 2 marks · Trees

Which of the following sequences can be the preorder of some BST on keys {1,2,3,4,5}?

(A) 1, 2, 3, 4, 5
(B) 5, 4, 3, 2, 1
(C) 3, 1, 2, 5, 4
(D) 3, 5, 1, 2, 4

Q.53 MCQ · 2 marks · Sorting

After the first extract-max pass of heapsort on a max-heap [9, 6, 8, 2, 5, 7, 3] (n=7), the last array position holds

(A) 9
(B) 8
(C) 3
(D) 2

Q.54 MSQ · 2 marks · DecisionTree

Which splits are useless for a greedy impurity tree?

(A) Reusing a feature on a child that is already pure (gain 0)
(B) A threshold that sends every row to one child
(C) A split that improves gain on an impure node
(D) A multiway split on a 3-level categorical at the root

Q.55 NAT · 2 marks · Abs crit

d/dx |x-3| does not exist at x=_____.

NAT · enter the numerical value

Q.56 MSQ · 2 marks · Adversarial

Which are TRUE of alpha-beta versus plain minimax on the same tree and same left-to-right order?

(A) The returned root value is the same
(B) Some leaves may not be evaluated
(C) Alpha-beta can change the identity of an optimal root move only if there are ties, not the value
(D) Alpha-beta can return a different numeric value than minimax on the same finite tree

Q.57 NAT · 2 marks · Argmin x+1/x

Attained at x=_____.

NAT · enter the numerical value

Q.58 MCQ · 2 marks · P&S; / t-test

A one-sample t-test (unknown sigma, Normal i.i.d. data) of H0: mu=mu0 uses the statistic

(A) z = (Xbar-mu0)/(sigma/sqrt(n)) with sigma replaced by 0
(B) F = Xbar / S
(C) t = (Xbar-mu0)/(S/sqrt(n)) ~ t_{n-1} under H0
(D) chi^2 = n S^2

Q.59 NAT · 2 marks · Cramer x2

Same system: x equals _____.

NAT · enter the numerical value

Q.60 MCQ · 2 marks · Warehousing and OLAP

Cube SalesQty is visualised by ProductType × Month × Country. To see ProductType × Month × State (State is a finer geographic level under Country) the OLAP operation is

(A) drill-down
(B) roll-up
(C) slice
(D) dice

Q.61 MCQ · 2 marks · MLP

Vanishing gradients in a deep sigmoid MLP occur because

(A) sigmoid derivative < = 0.25, so a long product of derivatives shrinks
(B) ReLU derivative is always 0
(C) SGD cannot use minibatches
(D) the output must be one-hot

Q.62 NAT · 2 marks · P&S; / independence

Three independent alarms each fail with probability 0.2. The probability that at least one works is _____ (three decimals).

NAT · enter the numerical value

Q.63 MSQ · 2 marks · Trees

Let T be a nonempty binary tree. Which statements are always true?

(A) The inorder of a BST is a sorted sequence of its keys
(B) The preorder of a BST is always sorted
(C) A heap-ordered complete tree is always a BST
(D) The height of a complete binary tree on n nodes is floor(log2 n)

Q.64 NAT · 2 marks · Graphs

Directed weighted graph PacketNet with edges U- > V(2), U- > W(5), V- > W(1), V- > X(4), W- > X(1). Dijkstra from U (non-negative weights). Shortest-path distance to X is

NAT · enter the numerical value

Q.65 NAT · 2 marks · Add30

The third derivative of e^{2x} at 0 is 8. Enter _____.

NAT · enter the numerical value

MOCK C ANSWER KEY AND SHORT SOLUTIONS

Q.1 [NAT] 36 — Relative length 360 m at 10 m/s gives 36 s.

Q.2 [MCQ] C — The arc equals the base circumference 6*pi.

Q.3 [NAT] 9600 — SI = 1600, so the amount is 9600.

Q.4 [MCQ] B — Whoever ... chooses is a singular clause and takes has.

Q.5 [NAT] 2 — n=12k+7, 2n=24k+14=12(2k+1)+2.

Q.6 [NAT] 10 — Geometric: 3,6,12,24. Arithmetic: 5,8,11,14. 24-14=10.

Q.7 [MCQ] B — The extra square must not cause overlap; some attachments work.

Q.8 [NAT] 7744 — 88^2=7744 is the only 4-digit square of form AABB.

Q.9 [MCQ] C — Four days of A+B+C finish 1/2. Two days of B+C finish 5/36. Remainder 13/36, so C needs 13 days.

Q.10 [NAT] 4 — Sorted lists are 5,5,7,a,b with a+b=23 and 8 < =a < b, giving four pairs.

Q.11 [NAT] 0 — (x+y)^2.

Q.12 [MCQ] A — Series truncates.

Q.13 [NAT] 3 — Only FF fails.

Q.14 [MCQ] A — H_{ij}=h_{i+j}.

Q.15 [NAT] -1 — Shear inverse.

Q.16 [MCQ] A — Average 1.

Q.17 [NAT] 1 — Winner (0,0), distance 1.

Q.18 [MCQ] A — CV estimates extra-sample error.

Q.19 [NAT] 7 — Worst case leaves one key in a run: 3+5-1=7 comparisons.

Q.20 [MCQ] C — Odd cubic.

Q.21 [NAT] 7 — Range is -8..7.

Q.22 [MCQ] D — Convex: phi of mean < = mean of phi. Variance identity is the square case.

Q.23 [NAT] 4 — A determines the whole chain ABCD.

Q.24 [MCQ] A — Dropping a dimension by aggregation is a roll-up.

Q.25 [NAT] 8 — 2^3.

Q.26 [MCQ] A — Terms vs formulae.

Q.27 [NAT] 2 — Free variables x1 and x4.

Q.28 [MCQ] D — Poisson mean=var. N(0,4) has variance 4, not 1..)

Q.29 [NAT] 35 — Need 4 right and 3 up, in some order: C(7,3)=35.

Q.30 [MCQ] B — append adds B as a single nested element, so A has length 4.

Q.31 [NAT] 5 — n-1 merges.

Q.32 [MCQ] A — 6.1, while f'=6.

Q.33 [NAT] 3 — Only P=T,Q=F is false.

Q.34 [MCQ] B — Ae2=3 e2.

Q.35 [NAT] 0 — (-3+3)/2=0.

Q.36 [MSQ] A;B — Center always; standardise when units differ.

Q.37 [MSQ] A;C;D — t_{n-1} is heavier-tailed than N(0,1), so |t|_{crit} > z_crit. Normality matters for small n.

Q.38 [MSQ] A;C;D — Quadratic probing still has some clustering; it does not fully remove secondary clustering. Double hashing and chaining claims are standard.

Q.39 [MSQ] A;B;C — VE is the exact one; LW is approximate in finite samples.

Q.40 [NAT] 2.67 — E[X]=2, E[X^2]=14/3, E[X^3]=12, Cov=12-2*(14/3)=12-28/3=8/3~2.67.

Q.41 [MSQ] A;B;C — AVG ignores NULLs too; it does not treat them as 0.

Q.42 [MSQ] A;B;C — date_id is a dimension foreign key, not a measure.

Q.43 [MSQ] B;C;D — Standard Neyman-Pearson facts. Non-rejection is not a proof of H0.

Q.44 [MSQ] A;B;D — Product structure 1/4=(1/2)(1/2). P(X=Y)=P(00)+P(11)=1/2. P(11)=1/4 not 1/2.

Q.45 [MSQ] A;B;C — Adding lambda I improves conditioning; it does not worsen it.

Q.46 [MCQ] A — 2NF bans partial dependence of nonprimes. BCNF is 'every determinant is a superkey'. Transitive is 3NF.

Q.47 [MCQ] A — Monotonicity of f along paths.

Q.48 [MCQ] A — Definition of chi^2_k.

Q.49 [MCQ] A — DFS discovery order is Delhi Mumbai Goa Kochi Pune.

Q.50 [MCQ] A — x^T A^T A x = ||Ax||^2 > =0.

Q.51 [MCQ] C — b_yx * b_xy = r^2 always.

Q.52 [MSQ] A;B;C — A preorder of a BST cannot present a key outside the current subtree window. 3,5,1 is invalid because after going right from 3 to 5, 1 cannot appear.

Q.53 [MCQ] A — Extract-max swaps the root 9 with the last key and shrinks the heap; 9 sits at the end in its final sorted position.

Q.54 [MSQ] A;B — Zero-gain / degenerate splits are useless.

Q.55 [NAT] 3 — Corner.

Q.56 [MSQ] A;B;C — Pruning is correctness-preserving for the value.

Q.57 [NAT] 1 — 1.

Q.58 [MCQ] C — Definition of the one-sample t statistic.

Q.59 [NAT] 0 — x+y=1, y=1 = > x=0.

Q.60 [MCQ] A — Drill-down walks down a concept hierarchy (Country to State).

Q.61 [MCQ] A — Chain of factors each at most 1/4.

Q.62 [NAT] 0.992 — Complement of all fail: 1-0.008=0.992.

Q.63 [MSQ] A;D — BST inorder is sorted; preorder is not. Heaps are not BSTs. Complete-tree height is floor(log2 n) with root height 0.

Q.64 [NAT] 4 — Dijkstra distance {'U': 0, 'V': 2, 'W': 3, 'X': 4}.

Q.65 [NAT] 8 — 8 e^{2x}.

MOCK PAPER D Mock Paper 04 · BCNF, ridge, FOL implication

Duration: 3 hours. Maximum marks: 100. Q.1-5 GA one mark · Q.6-10 GA two marks · Q.11-35 DA one mark · Q.36-65 DA two marks. MCQ: -1/3 / -2/3. MSQ and NAT: no negative.

Q.1 - Q.5 Carry ONE mark each (General Aptitude)

Q.1 MCQ · 1 mark · GA/Quant

Which of the following is equivalent to the inequality 2x-5 > 7?

(A) x > 4
(B) x > 6
(C) x < 6
(D) x > 1

Q.2 NAT · 1 mark · GA/Spatial

A 6x6x6 cube is painted on all faces and diced into unit cubes. The number of unit cubes with paint on exactly three faces is _____.

NAT · enter the numerical value

Q.3 MCQ · 1 mark · GA/Spatial

A right circular cylinder of radius 3 and height 4 is unrolled. The lateral surface is a rectangle of sides

(A) 3 and 4
(B) 6*pi and 4
(C) 3*pi and 4
(D) 6 and 4

Q.4 NAT · 1 mark · GA/Quant

The HCF of 84 and 144 is _____.

NAT · enter the numerical value

Q.5 MCQ · 1 mark · GA/Verbal

Novice : Seasoned craft :: Pauper : _____

(A) Debt
(B) Affluence
(C) Labour
(D) Charity

Q.6 - Q.10 Carry TWO marks each (General Aptitude)

Q.6 MCQ · 2 marks · GA/Quant

A tap fills a tank in 8 hours. After 3 hours a leak that empties a full tank in 12 hours opens. Further hours needed to fill from that moment are

(A) 6
(B) 8
(C) 10
(D) 15

Q.7 NAT · 2 marks · GA/Quant

Three men can finish a job in 10 days, or six women can. Days that one man and one woman together take is _____.

NAT · enter the numerical value

Q.8 MCQ · 2 marks · GA/Verbal

After a hospital posted waiting times on a public board, median waits fell by 12 minutes. Staffing was unchanged. A plausible mechanism consistent with the report is that

(A) patients left when they saw long waits, thinning the queue
(B) the board repaired the HVAC
(C) doctors worked fewer hours
(D) the calendar skipped a week

Q.9 NAT · 2 marks · GA/Quant

The number of 3-digit positive integers whose digits sum to 4 is _____.

NAT · enter the numerical value

Q.10 MCQ · 2 marks · GA/Quant

A list price is 25% above cost. A 10% discount is then given on the list price. The profit percent on cost is

(A) 10
(B) 12.5
(C) 15
(D) 17.5

Q.11 - Q.35 Carry ONE mark each (DA)

Q.11 MCQ · 1 mark · III

A large condition number means

(A) A is close to singular in relative perturbation sense
(B) A is orthogonal
(C) det A=1
(D) A=0

Q.12 NAT · 1 mark · One sided

lim_{x- > 0+} |x|/x equals _____.

NAT · enter the numerical value

Q.13 MCQ · 1 mark · P&S; / independence

If A and B are mutually exclusive and P(A) > 0, P(B) > 0, then P(A|B) equals

(A) 0
(B) P(A)/P(B)
(C) 1
(D) P(A)

Q.14 NAT · 1 mark · Graphs

A DAG on vertices W,X,Y,Z has edges W- > X, W- > Y, X- > Z, Y- > Z. The number of distinct topological sorts is

NAT · enter the numerical value

Q.15 MCQ · 1 mark · DecisionTree

Axis-aligned binary trees in 2-D produce regions that are

(A) always balls
(B) rectangles (products of intervals), possibly unbounded
(C) linear halfspaces only
(D) circles centred at means

Q.16 NAT · 1 mark · Sampling

Prior sampling 20 draws of Rain, 4 are true. Estimate of P(Rain=true) is ____.

NAT · enter the numerical value

Q.17 MCQ · 1 mark · P&S; / uniform
X~Uniform(0,1), Y=1-X. Then (X,Y) is
(A) independent with Y~Uniform(0,1) but a different copula
(B) jointly standard normal
(C) Uniform on the segment x+y=1, 0 < x < 1 (hence dependent)
(D) i.i.d. Uniform(0,1)

Q.18 NAT · 1 mark · Hash tables
A hash table of 13 slots uses chaining. 10 distinct keys are stored. The maximum possible length of the longest chain is
NAT · enter the numerical value

Q.19 MCQ · 1 mark · Graph lap
The Laplacian of an undirected graph is PSD with kernel dimension equal to
(A) the number of connected components
(B) the number of edges
(C) n always
(D) 0 always

Q.20 NAT · 1 mark · Limit
lim_{x- > infty} (2x+sin x)/x equals _____.
NAT · enter the numerical value

Q.21 MCQ · 1 mark · Data types and transformation
Run-length encoding of the bit-vector 0000011111000 (13 bits) can store it as counts
(A) 5 zeros, 5 ones, 3 zeros
(B) 13 ones
(C) 8 zeros
(D) 2 runs only

Q.22 NAT · 1 mark · Mid SPD
The average of I and 2I is 1.5 I, which is SPD. Smallest eigenvalue is _____ (one decimal).
NAT · enter the numerical value

Q.23 MCQ · 1 mark · Consistency2
[[1,2],[2,4]] [x,y]^T = [3,5]^T is
(A) consistent
(B) inconsistent
(C) underdetermined with a unique solution
(D) homogeneous

Q.24 NAT · 1 mark · Det2
det([[4,3],[2,1]]) is _____.
NAT · enter the numerical value

Q.25 MCQ · 1 mark · Adversarial
In that same tree (Left leaves 3,5; Right leaves 2,9), MAX's best first move is
(A) Left
(B) Right
(C) either, values equal
(D) pass

Q.26 NAT · 1 mark · KFold
n=100, 4-fold. Each test fold size is ____.
NAT · enter the numerical value

Q.27 MCQ · 1 mark · FOL
Which pair is a unifier of Knows(John,x) and Knows(John,Jane)?
(A) {x/Jane}
(B) {x/John}
(C) fail
(D) {Jane/x, John/Jane}

Q.28 NAT · 1 mark · Hadamard ones
(J odot I) for 2x2 is I. det is _____.
NAT · enter the numerical value

Q.29 MCQ · 1 mark · Sig
The inertia of a real symmetric matrix (numbers of +, -, 0 eigenvalues) is
(A) invariant under congruence
(B) changed by any invertible congruence
(C) the same as the Jordan form always
(D) always (n,0,0)

Q.30 NAT · 1 mark · P&S; / chi-squared
If Z~N(0,1) then P(Z^2 < = 3.84) with 3.84 ~ chi^2_{1, 0.95} quantile is approximately _____ (two decimals).
NAT · enter the numerical value

Q.31 MCQ · 1 mark · C1 not C2
f(x)=x^{4/3} is C^1 but f'' blows at 0. f'(x)=(4/3) x^{1/3}, f'(0)=_____.
(A) 0
(B) infty
(C) 1
(D) 4/3

Q.32 NAT · 1 mark · Add1
lim_{x- > 0} (tan 3x)/x equals _____.
NAT · enter the numerical value

Q.33 MCQ · 1 mark · Warehousing and OLAP
A data mart versus an enterprise warehouse: a mart is
(A) a subject-area subset (e.g. sales only), often a star
(B) always a normalised 3NF EDW of the whole company
(C) a B+ tree
(D) an OLTP orders table

Q.34 NAT · 1 mark · Polar 1d
For A=[-3] in R^{1x1}, the polar 'U' is sign=-1 and P=_____.
NAT · enter the numerical value

Q.35 MCQ · 1 mark · n-1
Centered n points span at most n-1 dimensions. For n=2 that max is _____.
(A) 1
(B) 2
(C) 0
(D) 3

Q.36 - Q.65 Carry TWO marks each (DA)

Q.36 MCQ · 2 marks · ER model

In the ER diagram of Harbour, Berth participates totally in Occupies with Ship, cardinality 1:1. After mapping, the key of Ship may be placed as a FK in Berth. This FK is

(A) NOT NULL and UNIQUE
(B) NULL-able and non-unique
(C) a set-valued attribute
(D) unnecessary because 1:1 needs no FK

Q.37 NAT · 2 marks · kNN

Train on the line -2,-1,0,8 labels 1,1,1,0. 3-NN at query 7. Predicted label is ____.

NAT · enter the numerical value

Q.38 MSQ · 2 marks · P&S; / covariance

X,Y have finite positive variances. Which are true?

(A) Corr(X,Y)=0 iff X and Y are independent
(B) Corr(X,Y)=1 iff Y=a+bX a.s. for some b > 0
(C) Corr(X,Y)=Cov(X,Y)/(sdX sdY)
(D) |Corr(X,Y)| < =1

Q.39 MSQ · 2 marks · KFold

Which are TRUE of k-fold CV?

(A) k=n recovers LOOCV
(B) k=2 has more bias (smaller trains) than k=10 typically
(C) The k test folds partition the n points in standard CV
(D) k-fold requires a separate extra test set of size k

Q.40 NAT · 2 marks · P&S; / binomial

A login succeeds with probability 0.4 independently. In 5 attempts, P(first success occurs on the 5th attempt) equals _____ (rounded to four decimals). (This is a geometric probability, not a binomial count.)

NAT · enter the numerical value

Q.41 MCQ · 2 marks · DecisionTree

A categorical ID with as many levels as rows, if allowed to split fully, will

(A) have zero training error and almost surely terrible generalisation
(B) be ignored by Gini because gain is 0
(C) become a support vector
(D) estimate a ridge penalty

Q.42 MSQ · 2 marks · Files and indexing

B+ tree leaves are linked. Which follow?

(A) an equality search is O(height)
(B) a range scan after locating the start is sequential along leaves
(C) non-leaf nodes store record pointers for every key in the usual internal-node design
(D) all record pointers live in the leaves in a typical B+ tree

Q.43 NAT · 2 marks · LOOCV

y=1,3,5 intercept-only LOOCV MSE (average of three squared errors) is ____.

NAT · enter the numerical value

Q.44 MSQ · 2 marks · MLP

Which are TRUE of a feed-forward MLP trained by backprop?

(A) Forward pass computes activations layer by layer
(B) Backprop applies the chain rule to obtain dLoss/dW
(C) A sigmoid hidden layer is a linear function of its input
(D) SGD may use a minibatch gradient instead of the full-batch gradient

Q.45 MSQ · 2 marks · InformedSearch

h=0 everywhere. Which algorithms then coincide (up to tie-breaking) with UCS?

(A) A*
(B) greedy best-first
(C) IDA* with f=g
(D) DFS

Q.46 MSQ · 2 marks · Graphs

Undirected simple graph G. Which are equivalent to G being a tree?

(A) G is connected and has no cycle
(B) G is connected and has n-1 edges
(C) there is a unique path between every pair of vertices
(D) G is bipartite and has n-1 edges

Q.47 MSQ · 2 marks · P&S; / axioms

Let A,B be events in a probability space. Which of the following are ALWAYS true?

(A) P(A n B) = P(A)P(B)
(B) P(A u B) = P(A)+P(B)-P(A n B)
(C) P(A^c) = 1-P(A)
(D) P(A u B) < = P(A)+P(B)

Q.48 NAT · 2 marks · Relational model and FDs

Let schema Roster(A,B,C,D) with FDs AB- > C, C- > D, D- > A. The number of candidate keys is

NAT · enter the numerical value

Q.49 NAT · 2 marks · ER model

Entity Product has a multivalued attribute color (a product may have 3 colours) and a composite attribute size(length,width). Standard mapping: Product table plus a table ProductColor. How many relations result from this fragment?

NAT · enter the numerical value

Q.50 MSQ · 2 marks · P&S; / covariance

Which claims are correct?

(A) Cov(X,X)=0
(B) |Corr(X,Y)| < = 1 whenever the correlation is defined
(C) Var(X+Y) < = (sd X + sd Y)^2 always (finite variances)
(D) Corr(aX+b, cY+d) = sign(ac) Corr(X,Y) for a,c != 0

Q.51 MCQ · 2 marks · Coercive

f(x)=x^4-x is coercive (f- > +infty as |x|- > infty) so a global min exists. f'(x)=4x^3-1=0 at x=4^{-1/3}. This is the unique crit pt, hence the global min. f''(that point) is

(A) positive
(B) negative
(C) zero
(D) undefined

Q.52 MSQ · 2 marks · Relational algebra and tuple calculus

Let R and S be union-compatible. Which identities hold?

(A) R ∪ S = S ∪ R
(B) R ∩ S = R - (R - S)
(C) R × S = S × R (up to attribute order)
(D) R - S = S - R always

Q.53 MSQ · 2 marks · FOL

Which are closed sentences (no free variables)?

(A) forall x P(x)
(B) exists y Q(y)
(C) P(x) with free x
(D) forall x exists y R(x,y)

Q.54 MCQ · 2 marks · Hash tables

Load factor alpha of an open-address table of size m holding n keys is n/m. For uniform probing, the expected probes in an unsuccessful search is about 1/(1-alpha). If m=1000 and n=800, this estimate is

(A) 5
(B) 1.25
(C) 0.8
(D) 4

Q.55 MCQ · 2 marks · Stacks and queues

Deleting the successor of node p in a singly linked list (p is not the tail) is done by p.next = p.next.next. If the list is 6- > 8- > 1- > 4 and p points at 8, the resulting sequence is

(A) 6 - > 8 - > 4
(B) 6 - > 1 - > 4
(C) 6 - > 8 - > 1
(D) 8 - > 1 - > 4

Q.56 MCQ · 2 marks · Trees

A threaded binary tree (inorder threads) of n nodes stores two extra bits per node. Compared with a recursive inorder walk, an inorder walk on the threaded tree uses

(A) O(1) auxiliary stack space
(B) O(n) auxiliary stack space
(C) O(log n) auxiliary stack space always
(D) O(n^2) time

Q.57 NAT · 2 marks · P&S; / multinomial

Dice: a fair six-sided die is rolled 4 times. The number of outcome tuples with each of four distinct faces (a permutation of four different numbers) is P(6,4)= _____.

NAT · enter the numerical value

Q.58 NAT · 2 marks · BayesNet

Same Alarm net: CPT parameters (unconstrained, not using last-prob=1-p) : 1+1+4+2+2 = ____.

NAT · enter the numerical value

Q.59 MSQ · 2 marks · Stacks and queues

Which of the following problems are naturally solved with a stack?

(A) Matching HTML tags / parentheses
(B) BFS layer-order traversal of a tree
(C) Evaluating postfix arithmetic
(D) Implementing recursive DFS with an explicit data structure

Q.60 MCQ · 2 marks · BiasVariance

1-NN vs large-k NN: the 1-NN classifier is the one with

(A) higher variance and more jagged boundaries
(B) higher bias and a near-linear boundary always
(C) zero Bayes error always
(D) a unique support vector

Q.61 MCQ · 2 marks · Limit

lim_{x- > 0} (tan x - x)/x^3 equals

(A) 0
(B) 1/3
(C) 1/2
(D) infty

Q.62 NAT · 2 marks · P&S; / Bayes

City buses are 80% blue and 20% green. A witness identifies a bus as green; the witness is correct 75% of the time (and otherwise names the other colour). P(bus is green | identified green) equals _____ (two decimals).

NAT · enter the numerical value

Q.63 NAT · 2 marks · Graphs

Complete bipartite K(3,4) has how many edges?

NAT · enter the numerical value

Q.64 MSQ · 2 marks · KMeans

Which are TRUE of k-means?

(A) The SSE objective is nonincreasing at each assign/update step
(B) The algorithm can converge to a local minimum depending on initialisation
(C) Empty clusters can occur for bad seeds
(D) The global SSE minimum is always found in one iteration

Q.65 MCQ · 2 marks · P&S; / mixed

A sample of size 5 has values 1,2,3,4,100. Compared with the mean, the median is

(A) 3, and it is far less affected by the 100
(B) 22, the same as the mean
(C) 1
(D) 100

MOCK D ANSWER KEY AND SHORT SOLUTIONS

Q.1 [MCQ] B — 2x > 12 so x > 6.

Q.2 [NAT] 8 — Only the eight corners have three faces painted.

Q.3 [MCQ] B — The circumference 6*pi is one side; the height 4 is the other.

Q.4 [NAT] 12 — 84=2^2*3*7 and 144=2^4*3^2, so HCF=2^2*3=12.

Q.5 [MCQ] B — A novice lacks seasoned craft; a pauper lacks affluence.

Q.6 [MCQ] D — 3/8 is already in. Remaining 5/8 at net 1/8-1/12=1/24 per hour takes 15 hours.

Q.7 [NAT] 20 — A man does 1/30 per day and a woman 1/60, together 1/20.

Q.8 [MCQ] A — Public information can change arrival/reneging behaviour even if staffing is fixed.

Q.9 [NAT] 10 — Let hundreds=1+x with x > =0. Then x+y+z=3 in non-negative integers, C(3+2,2)=10, and each part is at most 9 automatically.

Q.10 [MCQ] B — Selling price = 1.25*0.90=1.125 times cost, so profit is 12.5%.

Q.11 [MCQ] A — Sensitivity of Ax=b.

Q.12 [NAT] 1 — Positive side.

Q.13 [MCQ] A — A n B is empty so the conditional probability is 0. They are dependent.

Q.14 [NAT] 2 — WXYZ and WYXZ.

Q.15 [MCQ] B — Each split is x_j ? t.

Q.16 [NAT] 0.2 — 4/20=0.2.

Q.17 [MCQ] C — Y is a deterministic function of X, supported on the line x+y=1.

Q.18 [NAT] 10 — All keys may collide in one slot.

Q.19 [MCQ] A — Standard.

Q.20 [NAT] 2 — sin/x- > 0.

Q.21 [MCQ] A — Three runs: 5,5,3.

Q.22 [NAT] 1.5 — 1.5.

Q.23 [MCQ] B — 3*2=6 != 5.

Q.24 [NAT] -2 — 4-6=-2.

Q.25 [MCQ] A — Left yields 3 > Right's 2.

Q.26 [NAT] 25 — 100/4=25.

Q.27 [MCQ] A — x binds to Jane.

Q.28 [NAT] 1 — 1.

Q.29 [MCQ] A — Sylvester's law of inertia.

Q.30 [NAT] 0.95 — chi^2_1 quantile 0.95 is 1.96^2~3.84.

Q.31 [MCQ] A — 0.

Q.32 [NAT] 3 — 3 sec^2(0).

Q.33 [MCQ] A — Marts are departmental; the EDW is enterprise-wide.

Q.34 [NAT] 3 — abs.

Q.35 [MCQ] A — They lie on a line through 0 after centering.

Q.36 [MCQ] A — Total 1:1 participation of Berth forces a unique non-null FK to Ship.

Q.37 [NAT] 1 — Neighbours 8,0,-1 labels 0,1,1 majority 1.

Q.38 [MSQ] B;C;D — Cauchy-Schwarz: |rho|=1 iff linear relation a.s. with the sign of b. Zero correlation is weaker than independence.

Q.39 [MSQ] A;B;C — Standard k-fold partitions the training sample.

Q.40 [NAT] 0.0518 — Four failures then one success: (0.6)^4 * 0.4 = 0.0518.

Q.41 [MCQ] A — ID splits isolate each row.

Q.42 [MSQ] A;B;D — Internal nodes store only keys and child pointers; data pointers are at the leaves.

Q.43 [NAT] 6 — Errors 3,0,3 squared 9,0,9 average 6.

Q.44 [MSQ] A;B;D — Sigmoid is nonlinear. SGD/minibatch is standard.

Q.45 [MSQ] A;C — Greedy with h=0 ranks all nodes equal (not UCS). DFS is not cost-ordered.

Q.46 [MSQ] A;B;C — A bipartite graph with n-1 edges need not be connected (or can have a cycle if not connected properly); D fails. A,B,C are standard tree characterizations.

Q.47 [MSQ] B;C;D — Inclusion-exclusion and complement are axioms/theorems; Boole's inequality always holds. Product formula needs independence.

Q.48 [NAT] 3 — Candidate keys are AB, BC, BD.

Q.49 [NAT] 2 — Composite size flattens into columns of Product; colors need a separate table.

Q.50 [MSQ] B;C;D — Cauchy-Schwarz gives |rho| < =1 and the triangle inequality for sd. Affine transforms pick up the sign of the slopes. Cov(X,X)=Var(X), not 0.

Q.51 [MCQ] A — 12 x^2 > 0.

Q.52 [MSQ] A;B;C — Difference is not commutative.

Q.53 [MSQ] A;B;D — P(x) is open.

Q.54 [MCQ] A — 1/(1-0.8)=5.

Q.55 [MCQ] A — The node 1 is unlinked; 8 now points at 4.

Q.56 [MCQ] A — Threads replace the recursion/stack; time remains O(n).

Q.57 [NAT] 360 — P(6,4)=360.

Q.58 [NAT] 10 — P(B),P(E) one each; P(A|B,E) four; P(J|A) two; P(M|A) two. (If using 1-p: 1+1+4+2+2 wait same if we count both rows... GATE often counts free params 1+1+4+2+2=10 unconstrained or 1+1+4+1+1=8 with 1-p. We count listed CPT entries as 10.)

Q.59 [MSQ] A;C;D — BFS needs a queue. Parentheses, postfix and DFS are stack-shaped.

Q.60 [MCQ] A — Small k = > flexible, noisy boundary.

Q.61 [MCQ] B — Taylor: x+x^3/3 -x over x^3.

Q.62 [NAT] 0.43 — 0.15/(0.15+0.20)=0.15/0.35 ~ 0.43.

Q.63 [NAT] 12 — Every of 3 vertices meets 4 vertices: 12 edges.

Q.64 [MSQ] A;B;C — k-means is locally convergent, not globally optimal.

Q.65 [MCQ] A — Median is 3. Mean is 22. The outlier inflates the mean.

MOCK PAPER E Mock Paper 05 · CLT, SVD, trees / hash

Duration: 3 hours. Maximum marks: 100. Q.1-5 GA one mark · Q.6-10 GA two marks · Q.11-35 DA one mark · Q.36-65 DA two marks. MCQ: -1/3 / -2/3. MSQ and NAT: no negative.

Q.1 - Q.5 Carry ONE mark each (General Aptitude)

Q.1 MCQ · 1 mark · GA/Verbal

The lecture hinged _____ a single counter-example.

(A) at
(B) on
(C) with
(D) over

Q.2 NAT · 1 mark · GA/Quant

Simple interest on Rs 2000 at 5% per annum for 3 years is _____ rupees.

NAT · enter the numerical value

Q.3 MCQ · 1 mark · GA/Quant

The pattern 2, 3, 8, 27, 112, ... multiplies by n and then adds n. The next term is

(A) 545
(B) 560
(C) 565
(D) 580

Q.4 NAT · 1 mark · GA/Quant

The area of a trapezium with parallel sides 8 and 12 and height 5 is _____.

NAT · enter the numerical value

Q.5 MCQ · 1 mark · GA/Quant

The next term of 11, 23, 47, 95, ... (each term 2*previous +1) is

(A) 189
(B) 191
(C) 193
(D) 195

Q.6 - Q.10 Carry TWO marks each (General Aptitude)

Q.6 NAT · 2 marks · GA/Analytical

A 3x3x3 cube is painted on all six faces and then diced into unit cubes. The number of unit cubes with paint on exactly two faces is _____.

NAT · enter the numerical value

Q.7 MCQ · 2 marks · GA/Analytical

Pointing to a woman, a man says: 'Her mother's only brother is my father's only son.' The man has no brother. The woman is the man's

(A) daughter
(B) sister
(C) niece
(D) mother

Q.8 NAT · 2 marks · GA/Quant

A and B take 12 days, A and C take 15 days, B and C take 20 days. The days A, B and C together take is _____.

NAT · enter the numerical value

Q.9 MCQ · 2 marks · GA/Verbal

Transcripts show a speaker used the word 'unprecedented' six times in a 12-minute briefing about a storm that matched a 1998 event. The most defensible comment is that

(A) the storm was unique in Earth's history
(B) the speaker's wording overstated novelty relative to 1998
(C) briefings cannot mention years
(D) six is a lucky number

Q.10 NAT · 2 marks · GA/Quant

The exponent of 5 in 25! is _____.

NAT · enter the numerical value

Q.11 - Q.35 Carry ONE mark each (DA)

Q.11 MCQ · 1 mark · Direct sum R

R^2 = span{e1} oplus span{e2} is a

(A) direct sum (internal)
(B) not a sum
(C) tensor product only
(D) quotient only

Q.12 NAT · 1 mark · Python

```
The nested loops
cnt = 0
for i in range(1, 5):
for j in range(i):
cnt += 1
print(cnt)
print
```

NAT · enter the numerical value

Q.13 MCQ · 1 mark · P&S; / Poisson

```
Rare independent events in 50 trials with success
probability 0.02 are approximated by Poisson(lambda)
with lambda equal to
```

(A) 50
(B) 0.02
(C) 1
(D) 2.5

Q.14 NAT · 1 mark · UninformedSearch

Depth-limited search with L=0 on a nonempty state space expands (goal-test on the root only) how many nodes?

NAT · enter the numerical value

Q.15 MCQ · 1 mark · KKT 1d

Min f on [a,b]: candidates are interior crit pts and endpoints. This is the 1-var KKT. For |x| on [-1,2] the min is at

(A) 0
(B) -1
(C) 2
(D) 1

Q.16 NAT · 1 mark · Dual n

dim (R^6)* is _____.

NAT · enter the numerical value

Q.17 MCQ · 1 mark · Second enough

On R, f' > 0 everywhere means f is strictly convex, so a critical point is a

(A) unique global min
(B) max
(C) saddle
(D) inflection

Q.18 NAT · 1 mark · Relational algebra and tuple calculus

R has 5 tuples, S has 4, overlap 2. |R - S| is

NAT · enter the numerical value

Q.19 MCQ · 1 mark · P&S; / cond-pdf

If f_{X|Y}(x|y)=f_X(x) for (almost) all x,y in the support, then

(A) the joint density cannot exist
(B) X and Y are independent
(C) Y is Normal
(D) X is constant

Q.20 NAT · 1 mark · Python

```
The generator expression
print(sum(i*i for i in range(1, 5) if i % 2 == 1))
prints
```

NAT · enter the numerical value

Q.21 MCQ · 1 mark · Hadamard prod

The entrywise (Hadamard) product of two PSD matrices is PSD (Schur product theorem). I odot I equals

(A) I
(B) 0
(C) all-ones
(D) 2I

Q.22 NAT · 1 mark · e approx

e ~ _____ (two decimals).

NAT · enter the numerical value

Q.23 MCQ · 1 mark · KMeans

k-means with Euclidean distance updates each centroid to

(A) the coordinate-wise mean of its assigned points
(B) a randomly chosen training label
(C) the median of all features globally
(D) the first principal component

Q.24 NAT · 1 mark · KMeans

After that assignment {1,2,3} and {10}, the new left centroid is ____.

NAT · enter the numerical value

Q.25 MCQ · 1 mark · MLR

Design matrix X is n x (p+1) including a column of ones. Unique OLS exists iff

(A) n > p+1 only
(B) X has full column rank p+1
(C) y is orthogonal to col(X)
(D) features are standard-normal

Q.26 NAT · 1 mark · Poly eig

If Av=2v and p(t)=t^2-t, then p(A)v equals _____ v, i.e. the scalar is _____.

NAT · enter the numerical value

Q.27 MCQ · 1 mark · Cont 2

f(x)=1 for x rational, 0 for x irrational is continuous at

(A) every x
(B) no x
(C) 0 only
(D) integers only

Q.28 NAT · 1 mark · LDA

Class means 0 and 4 (1-D), shared variance 1, equal priors. LDA threshold is ____.

NAT · enter the numerical value

Q.29 MCQ · 1 mark · P&S; / z-test

For the z=1.5 two-sided test of H0: mu=50, using z_{0.025}=1.96, the decision at 5% is

(A) reject H0
(B) do not reject H0
(C) reject only the lower tail
(D) the test is undefined

Q.30 NAT · 1 mark · Inflect count

f=x^4 has f''=12 x^2 > =0, inflection points: _____. (number of them)

NAT · enter the numerical value

Q.31 MCQ · 1 mark · Ridge

Standardising features before ridge is recommended because

(A) lambda then penalises coefficients on a comparable scale
(B) OLS already requires standardising to exist
(C) standardising removes the need for a penalty
(D) it converts ridge into LDA

Q.32 NAT · 1 mark · Stiefel

The Stiefel manifold of 3x2 orthonormal-column matrices: each point has rank _____.

NAT · enter the numerical value

Q.33 MCQ · 1 mark · Adversarial

A chance node in expectiminimax (dice) is evaluated by

(A) the probability-weighted average of children, not min or max
(B) always min
(C) always max
(D) A* f-value

Q.34 NAT · 1 mark · Proj 1d

The orthogonal projection of (3,4) onto span{(1,0)} is the vector (a,0). a equals _____.

NAT · enter the numerical value

Q.35 MCQ · 1 mark · Warehousing and OLAP

HOLAP (hybrid OLAP) typically

(A) keeps detailed facts in relational tables and some aggregates in MOLAP storage
(B) forbids SQL
(C) stores only bitmaps
(D) is identical to 1NF

Q.36 - Q.65 Carry TWO marks each (DA)

Q.36 NAT · 2 marks · P&S; / joint

Joint density f(x,y)=8xy on the triangle 0 < y < x < 1. Then P(Y < 1/2) equals _____ (four decimals).

NAT · enter the numerical value

Q.37 NAT · 2 marks · Bisection

Bisection on [1,2] for that cubic: midpoint 1.5, f(1.5)=3.375-1.5-1=0.875 > 0 so the root is in [1, 1.5]. New interval length is _____ (one decimal).

NAT · enter the numerical value

Q.38 NAT · 2 marks · Stacks and queues

A circular queue is implemented in an array of size 6 (indices 0..5) with front=2 and rear=5, holding 3 elements in the usual convention that the rear index is the last occupied slot. Using count = (rear - front + N) mod N with front pointing at the first occupied slot, the number of stored elements is

NAT · enter the numerical value

Q.39 MSQ · 2 marks · Ridge

Which statements about ridge are TRUE?

(A) Effective degrees of freedom fall as lambda grows
(B) Ridge paths are piecewise linear in lambda like lasso
(C) For a fixed design, b_ridge=(X^T X + lambda I)^{-1} X^T y
(D) Ridge zeros coefficients exactly for large finite lambda

Q.40 MSQ · 2 marks · Integrity constraints

Foreign key Child.pid REFERENCES Parent.pid. Currently satisfying. Which statements never introduce a violation by themselves?

(A) INSERT INTO Parent
(B) DELETE FROM Child
(C) UPDATE Child SET pid to a value that exists in Parent
(D) DELETE FROM Parent of a referenced pid without CASCADE/SET NULL (NO ACTION)

Q.41 NAT · 2 marks · P&S; / cond-exp

A fair die is rolled. If the face is even, a fair coin is tossed once and X=number of heads; if odd, the coin is tossed twice and X=number of heads. E[X] equals _____ (two decimals).

NAT · enter the numerical value

Q.42 NAT · 2 marks · BayesNet

A chain of 4 binary nodes A- > B- > C- > D, each CPT with 2 free params except A with 1. Free parameter count using 1-p normalisation: 1+2+2+2= ____.

NAT · enter the numerical value

Q.43 MSQ · 2 marks · Graphs

Dijkstra's algorithm (as taught with a min-priority queue, non-negative weights) . Which are correct?

(A) It may fail if some edge weight is negative
(B) It computes shortest paths from one source
(C) It requires the graph to be a DAG
(D) On an unweighted graph it is no better asymptotically than BFS if implemented with a binary heap on dense graphs, and BFS is the natural choice

Q.44 MSQ · 2 marks · HAC

Which are TRUE of hierarchical agglomerative clustering?

(A) Single linkage can produce chaining (elongated clusters)
(B) Complete linkage prefers compact clusters
(C) The dendrogram of n points has n-1 merge levels
(D) HAC requires a pre-specified k at the start, unlike k-means

Q.45 NAT · 2 marks · P&S; / binomial

A batter gets a hit with probability 0.3 independently in 8 at-bats. P(at least one hit)=1-(0.7)^8. Rounded to two decimals this is _____.

NAT · enter the numerical value

Q.46 MSQ · 2 marks · Files and indexing

File organisations. Which pairs are matched correctly?

(A) heap file: insert is O(1) (append), equality search is linear
(B) sorted file: range search is efficient, insert is expensive
(C) hashed file: equality search expected O(1), range search poor
(D) sorted file: insert is O(1) append at end regardless of key

Q.47 NAT · 2 marks · Relational model and FDs

A cover F has 4 FDs. A canonical cover of a chain A- > B, B- > C, C- > D, A- > D may drop A- > D as redundant. The size of a minimal cover (number of FDs) is

NAT · enter the numerical value

Q.48 MSQ · 2 marks · Sampling

Which are approximate inference methods in the GATE DA syllabus?

(A) rejection sampling
(B) likelihood weighting
(C) Gibbs sampling
(D) variable elimination

Q.49 MSQ · 2 marks · P&S; / chi2-test

A Pearson chi-squared test of independence is run on an r x c table with large n. Which are true?

(A) The test requires the two classification variables to be Normal
(B) Expected cell counts use the product of fitted marginals
(C) Degrees of freedom equal (r-1)(c-1)
(D) A large statistic (with small p) is evidence against independence

Q.50 NAT · 2 marks · MLP

One SGD step on that unit: w=0.5, grad=-4, eta=0.1. Updated w is ____.

NAT · enter the numerical value

Q.51 MCQ · 2 marks · Isometry tall

A tall matrix with orthonormal columns satisfies A^T A=I. Then ||Ax||=||x||. For A 3x2 such, rank is _____.

(A) 2
(B) 3
(C) 1
(D) 0

Q.52 MSQ · 2 marks · BiasVariance

Which moves sit on the high-variance side?

(A) Degree-9 polynomial through 10 noisy points
(B) 1-NN instead of 25-NN
(C) Unpruned deep trees
(D) Intercept-only mean predictor on a clearly linear trend

Q.53 MCQ · 2 marks · Graphs

If every edge of an undirected connected graph has weight 1, Dijkstra's distances from s coincide with

(A) BFS hop distances from s
(B) DFS discovery times
(C) the number of spanning trees
(D) the maximum degree

Q.54 MSQ · 2 marks · P&S; / continuous

Let f be a PDF on R. Which properties MUST hold?

(A) int_{-inf}^{inf} f(x) dx = 1
(B) f is continuous
(C) f(x) > = 0 for (almost) all x
(D) f(x) < = 1 for all x

Q.55 MCQ · 2 marks · BayesNet

WetGrass: Cloudy - > Sprinkler, Cloudy - > Rain, Sprinkler - > Wet, Rain - > Wet. Unconditionally Sprinkler ? Rain is

(A) false: they share the fork Cloudy, which is unobserved, so they are dependent
(B) true by a collider at Wet
(C) true because they are roots
(D) true iff Wet is observed

Q.56 MSQ · 2 marks · ER model

Which ER features require an extra relation in the standard mapping (beyond one table per strong entity)?

(A) M:N relationship
(B) multivalued attribute
(C) 1:N relationship (FK can be placed on the N side)
(D) composite attribute (components become columns)

Q.57 MSQ · 2 marks · Python

Which of the following Python objects are hashable and may be placed in a set?

(A) (1, [2])
(B) (1, (2, 3))
(C) frozenset({1, 2})
(D) {1: 2}

Q.58 MCQ · 2 marks · Whitening

If Sigma = U D U^T with D > 0, the whitening map D^{-1/2} U^T sends the Gaussian to

(A) isotropic (covariance I)
(B) a Poisson
(C) a Bernoulli
(D) k-means

Q.59 MSQ · 2 marks · BayesNet

In the Alarm network B- > A < -E, A- > J, A- > M, which conditional independences hold by d-separation?

(A) J ? M | A
(B) B ? E (unconditionally)
(C) B ? E | A (explaining-away collider)
(D) J ? B | A

Q.60 MSQ · 2 marks · P&S; / uniform

X~Uniform(0,2). Which are true?

(A) P(X=1)=1/2
(B) E[X]=1
(C) f_X(x)=1/2 for x in (0,2)
(D) Var(X)=(2-0)^2/12=1/3

Q.61 MCQ · 2 marks · Graphs

Directed graph Pipeline has edges Src- > A, Src- > B, A- > C, B- > C, C- > Sink, A- > Sink. Recursive DFS from Src with neighbours tried in alphabetical order. The discovery order is

(A) Src A C Sink B
(B) Src A B C Sink
(C) B Sink C A Src
(D) Src Sink C B A

Q.62 MCQ · 2 marks · Relational model and FDs

Let schema Invoice(A,B,C,D) with FDs A- > BC, C- > D. Assuming the schema is already in 1NF, the highest normal form it is guaranteed to be in is

(A) 2NF
(B) 1NF
(C) 3NF
(D) BCNF

Q.63 MSQ · 2 marks · Graphs

Which claims about BFS and DFS on a finite directed graph G (no self-loops) are always true?

(A) From a vertex v, BFS and DFS visit exactly the same set of vertices (the descendants of v in the reachable sense)
(B) The order of first visits in BFS is always the reverse of DFS
(C) If G^R is G with all edges reversed, u reachable from v in G implies v reachable from u in G^R
(D) BFS requires a queue and DFS a stack (or recursion stack)

Q.64 NAT · 2 marks · Trace eig

Sum of eigenvalues of [[4,1],[2,3]] (counted with multiplicity) is _____.

NAT · enter the numerical value

Q.65 MCQ · 2 marks · P&S; / uniform

The spacings of n i.i.d. Uniform(0,1) order statistics are distributed as

(A) i.i.d. Uniform(0,1)
(B) i.i.d. Normal
(C) degenerate at 1/n
(D) normalised i.i.d. exponentials (Dirichlet(1,..,1) on the simplex)

MOCK E ANSWER KEY AND SHORT SOLUTIONS

Q.1 [MCQ] B — Hinge on means depend on.

Q.2 [NAT] 300 — SI = 2000*5*3/100 = 300.

Q.3 [MCQ] C — 112*5+5=565.

Q.4 [NAT] 50 — Area = (1/2)*(8+12)*5=50.

Q.5 [MCQ] B — 2*95+1=191.

Q.6 [NAT] 12 — Each of the 12 edges contributes one non-corner edge-centre cube with exactly two faces painted.

Q.7 [MCQ] C — The man is the woman's maternal uncle, so she is his niece.

Q.8 [NAT] 10 — Adding the three pair rates gives twice the triple rate equal to 1/5, so the triple rate is 1/10.

Q.9 [MCQ] B — A documented analogue contradicts 'unprecedented' in a strict sense.

Q.10 [NAT] 6 — floor(25/5)+floor(5/5)=5+1=6.

Q.11 [MCQ] A — Intersection 0, sum whole space.

Q.12 [NAT] 10 — Inner iterations: 1+2+3+4=10.

Q.13 [MCQ] C — np=1, the Poisson limit mean.

Q.14 [NAT] 1 — Only the root, then the limit cuts every child.

Q.15 [MCQ] A — 0.

Q.16 [NAT] 6 — 6.

Q.17 [MCQ] A — Convex.

Q.18 [NAT] 3 — 5-2=3.

Q.19 [MCQ] B — Conditional equals marginal iff independence.

Q.20 [NAT] 10 — Odd i in 1..4 are 1 and 3; 1+9=10.

Q.21 [MCQ] A — Entrywise 1 on the diagonal.

Q.22 [NAT] 2.72 — 2.718.

Q.23 [MCQ] A — The mean minimises within-cluster SSE.

Q.24 [NAT] 2 — Mean of 1,2,3 is 2.

Q.25 [MCQ] B — Need X^T X invertible, i.e. rank(X)=p+1.

Q.26 [NAT] 2 — p(2)=4-2=2.

Q.27 [MCQ] B — Thomae's opposite: Dirichlet function.

Q.28 [NAT] 2 — Midpoint 2.

Q.29 [MCQ] B — |1.5| < 1.96 so insufficient evidence against H0 at 5%.

Q.30 [NAT] 0 — f'' vanishes at 0 but does not change sign.

Q.31 [MCQ] A — L2 penalty is not scale-invariant.

Q.32 [NAT] 2 — 2.

Q.33 [MCQ] A — Chance nodes average.

Q.34 [NAT] 3 — First coordinate.

Q.35 [MCQ] A — Hybrid: relational detail + multidimensional aggregates.

Q.36 [NAT] 0.4375 — Integrate 8xy over y < 1/2 inside the triangle: x in (0,1/2] with y < x plus x in (1/2,1] with y < 1/2, giving 1/16+3/8=0.4375.

Q.37 [NAT] 0.5 — 0.5.

Q.38 [NAT] 3 — (5-2+6) mod 6 = 3.

Q.39 [MSQ] A;C — Ridge shrinks but does not zero; paths are smooth.

Q.40 [MSQ] A;B;C — D is the classic violation. C is OK if the new pid exists.

Q.41 [NAT] 0.75 — P(even)=1/2, E[X|even]=1/2; E[X|odd]=1. Total 0.75.

Q.42 [NAT] 7 — Root 1, three 2-row CPTs with 2 free each? Wait each child given 1 parent has 2 params (P(child=1|parent=0) and P(child=1|parent=1)), root 1, total 7.

Q.43 [MSQ] A;B;D — Dijkstra is single-source and needs non-negative weights; it does not need a DAG (that's for one DP method).

Q.44 [MSQ] A;B;C — k is chosen by cutting the dendrogram later; it is not needed to start agglomerative HAC.

Q.45 [NAT] 0.94 — 0.7^8 ~ 0.0576, so 0.94.

Q.46 [MSQ] A;B;C — Inserting into a sorted file while keeping order is not an append (unless the key is maximal).

Q.47 [NAT] 3 — A- > D follows by transitivity from the chain.

Q.48 [MSQ] A;B;C — VE is exact.

Q.49 [MSQ] B;C;D — It is a categorical test; Normality of the labels is not assumed. Expected counts are n * p_row * p_col under independence.

Q.50 [NAT] 0.9 — w := 0.5 - 0.1*(-4)=0.9.

Q.51 [MCQ] A — Full column rank.

Q.52 [MSQ] A;B;C — The mean predictor is high bias, low variance.

Q.53 [MCQ] A — Unit weights make Dijkstra degenerate to BFS (with a queue).

Q.54 [MSQ] A;C — Non-negativity and unit integral. Densities may exceed 1 (e.g. Uniform(0,0.1)) and need not be continuous.

Q.55 [MCQ] A — Common cause Cloudy, unobserved = > dependent. Observing Cloudy would block.

Q.56 [MSQ] A;B — 1:N and composite attributes do not by themselves add a table.

Q.57 [MSQ] B;C — A tuple is hashable only if all components are. frozenset is hashable; list and dict are not.

Q.58 [MCQ] A — Affine.

Q.59 [MSQ] A;B;D — Conditioning on the collider A opens B-E. J ? M | A and J ? B | A hold.

Q.60 [MSQ] B;C;D — Continuous uniform has P(X=1)=0. Mean is the midpoint, variance the interval-length formula.

Q.61 [MCQ] A — DFS discovery order is Src A C Sink B.

Q.62 [MCQ] A — Candidate keys A. Highest NF is 2NF.

Q.63 [MSQ] A;C;D — Reachable sets coincide for BFS and DFS; visit orders need not be reverses. Reachability reverses with the edges.

Q.64 [NAT] 7 — Equals trace.

Q.65 [MCQ] D — Classic fact: n+1 spacings including ends are like n+1 i.i.d. exp / sum.

MOCK PAPER F Mock Paper 06 · Exponential, B+ fanout, minimax

Duration: 3 hours. Maximum marks: 100. Q.1-5 GA one mark · Q.6-10 GA two marks · Q.11-35 DA one mark · Q.36-65 DA two marks. MCQ: -1/3 / -2/3. MSQ and NAT: no negative.

Q.1 - Q.5 Carry ONE mark each (General Aptitude)

Q.1 MCQ · 1 mark · GA/Quant

Between 12:00 and 1:00 a clock's hands are at right angles

(A) once
(B) twice
(C) three times
(D) not at all

Q.2 NAT · 1 mark · GA/Quant

A 31-day month begins on Sunday. The number of Sundays in that month is _____.

NAT · enter the numerical value

Q.3 MCQ · 1 mark · GA/Verbal

If talks are 'at sixes and sevens', they are

(A) nearly finished
(B) in disorder
(C) strictly confidential
(D) evenly split

Q.4 NAT · 1 mark · GA/Quant

A fair die is rolled once. If P(even face) is written as a/b in lowest terms, then a+b equals _____.

NAT · enter the numerical value

Q.5 MCQ · 1 mark · GA/Verbal

A one-word substitute for 'a person who lives by begging' is

(A) mendicant
(B) misanthrope
(C) martinet
(D) maverick

Q.6 - Q.10 Carry TWO marks each (General Aptitude)

Q.6 MCQ · 2 marks · GA/Verbal

Country P taxes sugary drinks. After two years, sales of taxed drinks fell 15% while sales of untaxed sweet snacks rose 10%. Total sugar from packaged foods was almost unchanged. A supported inference is that

(A) the tax had no effect on any purchases
(B) substitution toward untaxed sweets may have offset drink-side sugar cuts
(C) snacks became healthier
(D) P should ban water

Q.7 NAT · 2 marks · GA/Quant

How many 4-digit numbers with distinct digits can be formed from {1,2,3,4,5,6} so that the number is divisible by 3?

NAT · enter the numerical value

Q.8 MCQ · 2 marks · GA/Quant

A square is inscribed in a circle of radius 5. The area of the square is

(A) 25
(B) 50
(C) 75
(D) 100

Q.9 NAT · 2 marks · GA/Quant

Quarterly output shares are Q1 20%, Q2 30%, Q3 25%, Q4 25%. If Q2 produced 600 units, Q1 produced _____ units.

NAT · enter the numerical value

Q.10 MCQ · 2 marks · GA/Analytical

Four houses in a row are red, blue, green, yellow (not necessarily in that order). The red house is left of blue. Green is at an end. Yellow is immediately right of red. Blue is not at an end. The leftmost house is

(A) red
(B) blue
(C) green
(D) yellow

Q.11 - Q.35 Carry ONE mark each (DA)

Q.11 MCQ · 1 mark · UninformedSearch

The frontier (open list) of BFS is a

(A) FIFO queue
(B) LIFO stack
(C) priority queue on h(n)
(D) minimax game tree

Q.12 NAT · 1 mark · InformedSearch

In that graph S-A=1, S-B=2, A-G=3, B-G=1, cheapest S-G cost is ____.

NAT · enter the numerical value

Q.13 MCQ · 1 mark · Span

span{(1,0),(0,1),(1,1)} in R^2 equals

(A) a line
(B) R^2
(C) {0}
(D) R^3

Q.14 NAT · 1 mark · Geom mult

For a Jordan block of size 3 with lambda=2, geometric multiplicity of 2 is _____.

NAT · enter the numerical value

Q.15 MCQ · 1 mark · Files and indexing

A dense secondary index on attribute A of HeapFile Staff has one entry per

(A) record of Staff
(B) block of Staff
(C) distinct A-value only, even if A is not unique, with no posting list
(D) cylinder of the disk

Q.16 NAT · 1 mark · MLE bern

Bernoulli likelihood p^3 (1-p)^1. log L' = 3/p - 1/(1-p)=0 = > p=_____. (two decimals)

NAT · enter the numerical value

Q.17 MCQ · 1 mark · Graphs

An adjacency-matrix representation of a simple undirected graph on n vertices uses Theta(n^2) space. BFS from one vertex then runs in

(A) Theta(n^2) time in the worst case
(B) Theta(n+m) time always, independent of the matrix
(C) Theta(n) time
(D) Theta(n^3) time

Q.18 NAT · 1 mark · Det scale

det(3I_2) is _____.

NAT · enter the numerical value

Q.19 MCQ · 1 mark · Python

```
The function
def f(n):
return [n] + f(n-1) if n else []
print(f(3))
prints
```

(A) [3, 2, 1]
(B) [1, 2, 3]
(C) [3, 2, 1, 0]
(D) []

Q.20 NAT · 1 mark · Files and indexing

Unsorted heap, 1000 blocks, linear search for a missing equality key (unindexed). Worst-case block reads are

NAT · enter the numerical value

Q.21 MCQ · 1 mark · PropLogic

A propositional formula is a tautology iff it is

(A) true under every truth assignment
(B) true under some assignment
(C) false under every assignment
(D) built only from implication

Q.22 NAT · 1 mark · Trace I

tr(I_5) is _____.

NAT · enter the numerical value

Q.23 MCQ · 1 mark · Polar

The polar factorisation A=U P with P=SPD and U orthogonal (for invertible A). P equals

(A) sqrt(A^T A)
(B) A
(C) A^{-1}
(D) 0

Q.24 NAT · 1 mark · Chain 2

d/dx ln(1+x^2) at x=1 equals _____.

NAT · enter the numerical value

Q.25 MCQ · 1 mark · Attained

A continuous function on a compact set

(A) attains its inf and sup
(B) need not attain inf
(C) is unbounded
(D) is polynomial

Q.26 NAT · 1 mark · Neg

That local max value is _____.

NAT · enter the numerical value

Q.27 MCQ · 1 mark · Error term

The remainder after linear Taylor of sqrt(1+x) is negative because the second derivative of sqrt is negative (concave). So the linear approx

(A) overestimates
(B) underestimates
(C) is exact
(D) is random

Q.28 NAT · 1 mark · UninformedSearch

A full binary tree, root at depth 0, a unique goal at depth 3. Number of nodes at depths 0 through 3 is ____.

NAT · enter the numerical value

Q.29 MCQ · 1 mark · P&S; / cond-exp

For a square-integrable response X and a discrete grouping variable Y, the law of total variance expands Var(X) as

(A) E[Var(X|Y)] only
(B) Var(E[X|Y]) only
(C) E[Var(X|Y)] + Var(E[X|Y])
(D) E[X^2] - E[X]

Q.30 NAT · 1 mark · DecisionTree

Parent entropy 1 bit, children weights 1/2 with entropies 0 and 1. Information gain (bits) is ____.

NAT · enter the numerical value

Q.31 MCQ · 1 mark · Asymp line

f(x)=x+1/x for x > 0 has min at x=1, f=2. As x- > infty, f(x)-x tends to

(A) 0
(B) 1
(C) infty
(D) -infty

Q.32 NAT · 1 mark · Stacks and queues

A circular singly linked list has 10 nodes. Starting at the head, the minimum number of next-pointer follows needed to return to the head is

NAT · enter the numerical value

Q.33 MCQ · 1 mark · PA=LU

Partial pivoting writes PA=LU. P is a

(A) permutation matrix
(B) projection
(C) diagonal of eigenvalues
(D) SVD factor

Q.34 NAT · 1 mark · Files and indexing

Sorted file, 1000 blocks, binary search on the sort key, one record per matching key. Worst-case block reads (approx ceil(log2 1000)) are

NAT · enter the numerical value

Q.35 MCQ · 1 mark · HAC

Cutting a dendrogram at a height that yields 3 clusters is a way to

(A) choose k=3 after seeing the hierarchy
(B) train an SVM
(C) compute R^2
(D) estimate S_W

Q.36 - Q.65 Carry TWO marks each (DA)

Q.36 MCQ · 2 marks · P&S; / mixed

Negative-binomial flavour: independent Bernoulli(p=1/2) trials. The number of trials T until (and including) the 3rd success has E[T] equal to

(A) 1.5
(B) 6
(C) 3
(D) 8

Q.37 NAT · 2 marks · Python

```
The function
def depth(n):
return 1 if n == 0 else 1 + depth(n-1)
print(depth(6))
prints the number of stack frames created by the chain
(including the first). The printed value is
```

NAT · enter the numerical value

Q.38 MSQ · 2 marks · Relational algebra and tuple calculus

Natural join R □ S. Which are always true?

(A) if R and S have no common attributes, □ is Cartesian product
(B) |R □ S| < = |R| * |S|
(C) |R □ S| > = |R| + |S| always
(D) natural join is commutative up to column order

Q.39 MSQ · 2 marks · P&S; / axioms

Let A,B be events with 0 < P(A) < 1 and 0 < P(B) < 1. Which statements MUST be true?

(A) P(A n B) < = min(P(A), P(B))
(B) P(A|B) = P(B|A)
(C) P(A|B) + P(A^c|B) = 1
(D) P(A u B) > = max(P(A), P(B))

Q.40 NAT · 2 marks · P&S; / mixed

Three fair coins. Let T = at least two heads and W = at least one tail. P(T n W) equals _____ (three decimals).

NAT · enter the numerical value

Q.41 MSQ · 2 marks · P&S; / mixed

A data set of 7 distinct numbers is summarised. Which are true?

(A) The sample mean is sensitive to that largest value
(B) Changing the largest value does not change the median
(C) The unique mode of 7 distinct numbers is the largest
(D) The sample median is the 4th order statistic

Q.42 MSQ · 2 marks · DecisionTree

Compared with a linear SVM, a deep axis-aligned tree

(A) can represent XOR without a kernel
(B) has a piecewise-constant decision surface
(C) always maximises the geometric margin
(D) is more prone to high variance when unpruned

Q.43 MSQ · 2 marks · UninformedSearch

Which uninformed methods are complete on finite graphs with nonnegative costs (graph-search)?

(A) BFS
(B) UCS
(C) DFS with a closed set
(D) DLS with a limit smaller than the goal depth

Q.44 MCQ · 2 marks · P&S; / axioms

A die with faces {1,2,3,4,5,6} is loaded so that P(face k) is proportional to k. Then P(even face) equals

(A) 9/21
(B) 1/2
(C) 6/21
(D) 12/21

Q.45 MSQ · 2 marks · Graphs

Let G be an undirected simple graph on n > =2 vertices. Which statements are true?

(A) If G is a tree then G is connected and has n-1 edges
(B) If G is connected and has n-1 edges then G is a tree
(C) If G has n-1 edges then G is a tree
(D) Every tree is bipartite

Q.46 NAT · 2 marks · Stacks and queues

Two sorted lists 1- > 4- > 7 and 2- > 3- > 8 are merged by the standard two-pointer merge (stable: take from the first list on ties). The number of next-pointer comparisons of keys until one list is exhausted is

NAT · enter the numerical value

Q.47 MSQ · 2 marks · Sorting

Which algorithms sort in Theta(n^2) comparisons in the worst case on an array?

(A) insertion sort
(B) selection sort
(C) heapsort
(D) bubble sort (no early abort needed for the bound)

Q.48 MCQ · 2 marks · Stacks and queues

To evaluate the postfix expression 5 1 2 + 4 * + 3 - with a stack, the value remaining on the stack at the end is

(A) 14
(B) 10
(C) 9
(D) 18

Q.49 MSQ · 2 marks · P&S; / Bayes

```
A binary classifier for a rare class with prevalence
0.01 has sensitivity 0.99 and specificity 0.99. Which
claims are correct?
```

(A) Sensitivity equals P(predicted 1 | class=1)
(B) P(predicted 1 | class=0) = 0.01
(C) Specificity equals P(predicted 1 | class=0)
(D) P(class=1 | predicted 1) is far below 0.99 because of the low prevalence

Q.50 MCQ · 2 marks · MLP

A 2-2-1 MLP with linear hidden and linear output (no nonlinearities) is

(A) still universal
(B) equivalent to a single linear layer, hence cannot represent XOR
(C) a decision tree
(D) k-means

Q.51 NAT · 2 marks · Weyl

Eigenvalues of A and A+E differ by at most ||E|| (in a suitable norm). If ||E||=0.1 and gap is 5, a crude sin Theta bound 0.1/5 equals _____ (two decimals).

NAT · enter the numerical value

Q.52 MSQ · 2 marks · SVM

Which are valid PSD kernels commonly used with SVMs?

(A) Linear K=x^T z
(B) Gaussian RBF K=exp(-||x-z||^2/(2 sigma^2))
(C) Polynomial K=(x^T z+c)^d for c > =0 and d a positive integer
(D) K(x,z)=||x-z||^2 (raw squared Euclidean)

Q.53 MSQ · 2 marks · P&S; / CLT

Let X1,...,Xn be i.i.d. with mean mu and finite variance sigma^2 > 0. Which are true for large n?

(A) Xbar is approximately N(mu, sigma^2/n)
(B) sqrt(n)(Xbar-mu)/sigma is approximately N(0,1)
(C) Xbar converges in probability to mu (WLLN)
(D) Xbar is exactly t_{n-1} for every n, even if the Xi are Uniform

Q.54 MSQ · 2 marks · FOL

Which FOL statements are TRUE?

(A) forall x P(x) entails P(A) for any ground term A in the language
(B) P(A) entails exists x P(x)
(C) exists x forall y Likes(x,y) entails forall y exists x Likes(x,y)
(D) forall y exists x Likes(x,y) entails exists x forall y Likes(x,y)

Q.55 NAT · 2 marks · P&S; / covariance

A 2D sample of 4 points: (1,2), (2,4), (3,6), (4,8). The (Pearson) sample correlation is _____.

NAT · enter the numerical value

Q.56 NAT · 2 marks · SQL

```
On Emp, SELECT dept FROM Emp WHERE dept = NULL returns how many rows?
```

NAT · enter the numerical value

Q.57 MCQ · 2 marks · MLR

True map y=1+2 x1+0 x2+noise; we still include x2 in OLS. As n grows, the x2 coefficient

(A) converges to 0 in probability (correctly specified zero)
(B) forces R^2 to 0
(C) makes X^T X singular
(D) makes the intercept unestimable

Q.58 MSQ · 2 marks · Data types and transformation

Data-type roles in warehousing / mining. Which assignments are reasonable?

(A) Gender: categorical (nominal)
(B) Temperature in C: numeric (interval/ratio depending on scale use)
(C) T-shirt size S < M < L: ordinal
(D) Employee-id as a 6-digit code: numeric quantity on which means are meaningful

Q.59 MCQ · 2 marks · Warehousing and OLAP

Cube SensorAvg is shown at dimensions Station, Metric, Day. Moving the last geographic/organisational/time level to the finer grain (Station to SensorId) is

(A) drill-down
(B) roll-up
(C) slice
(D) dice

Q.60 MCQ · 2 marks · SVD

For any real m x n matrix, SVD writes A as

(A) U Sigma V^T with U,V orthogonal and Sigma rectangular diagonal
(B) QR only
(C) P^2=P
(D) LDL with no signs

Q.61 NAT · 2 marks · P&S; / descriptive

Five exam scores have mean 12. After a sixth score 18 is included, the new mean is _____ (one decimal).

NAT · enter the numerical value

Q.62 MCQ · 2 marks · GL

GL_n(R) is the set of

(A) invertible n x n real matrices
(B) all matrices
(C) projections
(D) symmetric matrices

Q.63 NAT · 2 marks · BiasVariance

Noise=1, bias^2=4, variance=3. Expected MSE is ____.

NAT · enter the numerical value

Q.64 NAT · 2 marks · NaiveBayes

```
Three classes, uniform priors, likelihoods 0.1, 0.2, 0.7. Posterior of class 3 is ____.
```

NAT · enter the numerical value

Q.65 MSQ · 2 marks · NaiveBayes

Which modelling assumptions/facts does naive Bayes involve?

(A) Features are conditionally independent given the class
(B) The class prior may be estimated by class frequencies
(C) Features must be jointly Gaussian with a full covariance
(D) A zero training count for a (feature,class) pair can zero a Bernoulli likelihood unless smoothed

MOCK F ANSWER KEY AND SHORT SOLUTIONS

Q.1 [MCQ] A — In that hour the hands form a right angle exactly once.

Q.2 [NAT] 5 — Sundays fall on the 1st, 8th, 15th, 22nd and 29th.

Q.3 [MCQ] B — At sixes and sevens means in confusion or disorder.

Q.4 [NAT] 3 — P(even)=3/6=1/2, so a+b=3.

Q.5 [MCQ] A — Mendicant is a beggar; the others are a people-hater, a strict disciplinarian, and a nonconformist.

Q.6 [MCQ] B — Offsetting movements in two categories can leave total sugar flat.

Q.7 [NAT] 120 — Five 4-subsets have digit-sum divisible by 3, and each subset yields 4! = 24 numbers: 120.

Q.8 [MCQ] B — The diagonal of the square is the diameter 10, so area = 50.

Q.9 [NAT] 400 — Total is 600/0.30=2000, so Q1 is 20% of 2000=400.

Q.10 [MCQ] C — Green at an end, blue not at an end, red-yellow-blue as a block, red left of blue: the unique row is Green, Red, Yellow, Blue.

Q.11 [MCQ] A — BFS = queue.

Q.12 [NAT] 3 — S-B-G=3 < S-A-G=4.

Q.13 [MCQ] B — The first two vectors already form a basis.

Q.14 [NAT] 1 — dim ker(A-2I)=1.

Q.15 [MCQ] A — Dense means every record is referenced. A sparse index points at blocks.

Q.16 [NAT] 0.75 — 3/4.

Q.17 [MCQ] A — Scanning a row to find neighbours costs Theta(n) per vertex, total Theta(n^2).

Q.18 [NAT] 9 — 3^2.

Q.19 [MCQ] A — Prepend n until n is 0, which contributes [].

Q.20 [NAT] 1000 — The whole file is scanned.

Q.21 [MCQ] A — Tautology = valid = true in all models.

Q.22 [NAT] 5 — 5.

Q.23 [MCQ] A — Analog of z=e^{i theta} |z|.

Q.24 [NAT] 1 — 2x/(1+x^2)=2/2=1.

Q.25 [MCQ] A — Weierstrass.

Q.26 [NAT] -2 — -2.

Q.27 [MCQ] A — Concave down.

Q.28 [NAT] 15 — 1+2+4+8=15.

Q.29 [MCQ] C — Var(X)=E[Var(X|Y)]+Var(E[X|Y]).

Q.30 [NAT] 0.5 — 1-0.5=0.5.

Q.31 [MCQ] A — f(x)-x=1/x - > 0.

Q.32 [NAT] 10 — Ten steps around the cycle.

Q.33 [MCQ] A — Row swaps.

Q.34 [NAT] 10 — ceil(log2 1000)=10.

Q.35 [MCQ] A — Post-hoc cut of HAC.

Q.36 [MCQ] B — Sum of 3 geometric(1/2) waiting times: 3*(1/p)=6.

Q.37 [NAT] 7 — A linear chain of 6 recursive steps plus the base call is 7 activations.

Q.38 [MSQ] A;B;D — Join size can be 0, which is not at least |R|+|S|.

Q.39 [MSQ] A;C;D — Monotonicity gives the two inequalities. Conditioning on B, A and A^c partition the space. Bayes does not force P(A|B)=P(B|A).

Q.40 [NAT] 0.375 — T n W is exactly two heads (three heads has no tail). Probability 3/8=0.375.

Q.41 [MSQ] A;B;D — For odd n the median is the middle order statistic, robust to extremes. With all values distinct there are 7 modes (or none unique), not 'the largest'.

Q.42 [MSQ] A;B;D — XOR is two splits; no margin objective.

Q.43 [MSQ] A;B;C — DLS with too small a limit misses the goal.

Q.44 [MCQ] D — Weights 1+...+6=21; even faces 2+4+6=12, so 12/21.

Q.45 [MSQ] A;B;D — n-1 edges alone do not imply a tree (a triangle plus an isolated vertex has n=4, e=3). Trees have no odd cycles, hence are bipartite.

Q.46 [NAT] 5 — Compare (1,2),(4,2),(4,3),(4,8),(7,8) - five key comparisons; then the rest is appended.

Q.47 [MSQ] A;B;D — Heapsort is Theta(n log n) worst case.

Q.48 [MCQ] A — 5, then 1 2 + - > 3, 3 4 * - > 12, 5+12=17, 17-3=14.

Q.49 [MSQ] A;B;D — PPV is pulled down by the 99:1 prior odds. FPR=1-specificity=0.01. Specificity is P(predicted 0 | class=0), not the FPR.

Q.50 [MCQ] B — Composition of linear maps is linear.

Q.51 [NAT] 0.02 — Davis-Kahan flavour.

Q.52 [MSQ] A;B;C — Raw squared Euclidean is not a kernel by itself.

Q.53 [MSQ] A;B;C — CLT and WLLN apply. The exact t law needs Gaussian data and an independent sample sd in the denominator, not the raw mean.

Q.54 [MSQ] A;B;C — Quantifier order: a single liker-of-all implies everyone is liked by someone, not conversely.

Q.55 [NAT] 1 — Y=2X exactly on the sample, so r=1.

Q.56 [NAT] 0 — In SQL, = NULL is UNKNOWN for every row; use IS NULL. Zero rows.

Q.57 [MCQ] A — No omitted-variable bias; OLS is consistent and that entry - > 0.

Q.58 [MSQ] A;B;C — IDs are nominal identifiers, not magnitudes.

Q.59 [MCQ] A — Finer grain along a hierarchy is drill-down on cube SensorAvg.

Q.60 [MCQ] A — Existence of SVD.

Q.61 [NAT] 13 — Total 60+18=78, mean 13.0.

Q.62 [MCQ] A — Group under multiplication.

Q.63 [NAT] 8 — 1+4+3=8.

Q.64 [NAT] 0.7 — Uniform priors: posterior equals normalised likelihoods already summing to 1.

Q.65 [MSQ] A;B;D — Full joint Gaussian is LDA/QDA, not NB.

MOCK PAPER G Mock Paper 07 · Joint tables, HAC, d-separation

Duration: 3 hours. Maximum marks: 100. Q.1-5 GA one mark · Q.6-10 GA two marks · Q.11-35 DA one mark · Q.36-65 DA two marks. MCQ: -1/3 / -2/3. MSQ and NAT: no negative.

Q.1 - Q.5 Carry ONE mark each (General Aptitude)

Q.1 MCQ · 1 mark · GA/Quant

The angle between the hands of a clock at 3:00 is (in degrees)

(A) 0
(B) 60
(C) 90
(D) 120

Q.2 NAT · 1 mark · GA/Quant

Pipe P fills a tank in 6 hours; pipe Q empties it in 8 hours. Hours needed to fill the empty tank with both open is _____.

NAT · enter the numerical value

Q.3 MCQ · 1 mark · GA/Verbal

The committee's praise was _____: every compliment hid a demand for unpaid overtime.

(A) fulsome
(B) candid
(C) laconic
(D) plaintive

Q.4 NAT · 1 mark · GA/Analytical

Among 50 people, 30 drink tea, 25 drink coffee, and 10 drink both. The number who drink neither is _____.

NAT · enter the numerical value

Q.5 MCQ · 1 mark · GA/Quant

A can complete 40% of a job in 8 days. At that rate, days for the whole job are

(A) 16
(B) 18
(C) 20
(D) 24

Q.6 - Q.10 Carry TWO marks each (General Aptitude)

Q.6 MCQ · 2 marks · GA/Quant

If n is an odd integer, n^2 divided by 8 leaves remainder

(A) 0
(B) 1
(C) 3
(D) 5

Q.7 NAT · 2 marks · GA/Quant

Compound interest on Rs 1000 at 10% per annum for 2 years, compounded annually, is _____ rupees.

NAT · enter the numerical value

Q.8 MCQ · 2 marks · GA/Analytical

A calendar shows that a month has five Mondays. The fewest days that month can have is

(A) 28
(B) 29
(C) 30
(D) 31

Q.9 NAT · 2 marks · GA/Quant

If log3(x)=2 and log_x(y)=3, then y equals _____.

NAT · enter the numerical value

Q.10 MCQ · 2 marks · GA/Analytical

Three switches A,B,C control three lamps in another room, one-to-one but unknown. You may toggle in the first room, then visit the lamp room once. A strategy that identifies all three pairings uses the fact that a bulb can be

(A) only on or off
(B) on, off, or warm
(C) broken
(D) coloured

Q.11 - Q.35 Carry ONE mark each (DA)

Q.11 MCQ · 1 mark · P&S; / normal

A z-test pivot is claimed to be standard normal. Which option correctly states the mean and variance of N(0,1)?

(A) mean 0 and variance 0
(B) mean 0 and variance 1
(C) mean 1 and variance 0
(D) mean 1 and variance 1

Q.12 NAT · 1 mark · Closed min

Same. Global minimum on [-2,2] is _____.

NAT · enter the numerical value

Q.13 MCQ · 1 mark · Warehousing and OLAP

MOLAP versus ROLAP: MOLAP stores the cube in

(A) a specialised multidimensional array structure
(B) only normalised 3NF tables with no aggregates
(C) a B+ tree of XML
(D) main-memory Python lists exclusively

Q.14 NAT · 1 mark · Hash tables

Same chain as above (head 17 - > 13 - > 9 - > 5 - > 1). Unsuccessful search for 3, which hashes to 1, performs how many key comparisons?

NAT · enter the numerical value

Q.15 MCQ · 1 mark · Adversarial

Horizon effect in depth-limited minimax is

(A) a bad tactic pushed just beyond the cutoff looking fine at the leaf evaluator
(B) alpha always equalling beta
(C) UCS failing on unit costs
(D) CNF explosion

Q.16 NAT · 1 mark · InformedSearch

8-puzzle: two tiles each 1 step from home, others home. Misplaced-tile h equals ____.

NAT · enter the numerical value

Q.17 MCQ · 1 mark · MLP
Backprop through a ReLU unit with pre-activation z < 0 sends
(A) zero gradient to its inputs
(B) gradient 1 always
(C) gradient -1
(D) undefined NaN always

Q.18 NAT · 1 mark · LDA
Means m1=(0,0), m2=(2,0), S_W=I. Fisher vector S_W^{-1}(m1-m2) has second coordinate ____.
NAT · enter the numerical value

Q.19 MCQ · 1 mark · Data types and transformation
Discretising a numeric attribute before naive Bayes is done so that
(A) a multinomial / categorical model can be used on bins
(B) the mean of the attribute becomes 0
(C) compression becomes lossy
(D) the attribute becomes a key

Q.20 NAT · 1 mark · LS 1d
Best least-squares slope through origin for points (1,2),(2,2): xhat = (A^T b)/(A^T A) with A=(1,2)^T, b=(2,2)^T equals _____ (one decimal).
NAT · enter the numerical value

Q.21 MCQ · 1 mark · Projection
An orthogonal projection matrix P satisfies
(A) P^T=P and P^2=P
(B) P^T=-P
(C) P^2=I always
(D) det P=1 always

Q.22 NAT · 1 mark · Sampling
Likelihood weighting: two samples with weights 0.2 and 0.6, Q values true and true. Weighted estimate of P(Q=true) is ____.
NAT · enter the numerical value

Q.23 MCQ · 1 mark · Exp
exp(tA) for a square A is the fundamental matrix of u'=Au. exp(0)=
(A) I
(B) 0
(C) A
(D) tI

Q.24 NAT · 1 mark · Cong I
P^T I P = P^T P, which is SPD if P is invertible. For P=2I_2, P^T P = 4 I, det _____.
NAT · enter the numerical value

Q.25 MCQ · 1 mark · Annihilator
dim W^0 = dim V - dim W. For a plane in R^4, dim annihilator is _____.
(A) 2
(B) 4
(C) 1
(D) 0

Q.26 NAT · 1 mark · PCA
Variance on PC1 is 9, on PC2 is 1. Cumulative fraction after 1 PC is ____.
NAT · enter the numerical value

Q.27 MCQ · 1 mark · Graphs
Adjacency lists of a directed graph with n vertices and m edges support BFS from one vertex in
(A) Theta(n+m) time
(B) Theta(n^2) time always
(C) Theta(m log n) time
(D) Theta(n m) time

Q.28 NAT · 1 mark · Trees
A perfect binary tree of height 3 (root height 0) has how many nodes?
NAT · enter the numerical value

Q.29 MCQ · 1 mark · Constrained 1d
Min x^2 on [1,2] is at the boundary x=1, value 1. The unconstrained min 0 is infeasible. So constraints matter. The constrained min value is
(A) 1
(B) 0
(C) 4
(D) 2

Q.30 NAT · 1 mark · Taylor sin
sin x = x - x^3/6 + O(x^5). The cubic approximation of sin(0.3) uses 0.3 - 0.027/6. 0.027/6=0.0045 so the value is _____ (three decimals).
NAT · enter the numerical value

Q.31 MCQ · 1 mark · Add14
If f'' changes from + to - at c with f'(c)=0, then c is a
(A) local max
(B) local min
(C) neither
(D) discontinuity

Q.32 NAT · 1 mark · MLP
A minibatch of 8, dataset of 80, one epoch. Number of SGD steps per epoch is ____.
NAT · enter the numerical value

Q.33 MCQ · 1 mark · Files and indexing
ISAM is a static multilevel index. Inserts that overflow a leaf go to
(A) overflow chains
(B) a freshly split leaf that updates the root always
(C) a hash bucket
(D) the WAL only

Q.34 NAT · 1 mark · Free
RREF of a 3x4 matrix has two pivot columns. Number of free variables is _____.
NAT · enter the numerical value

Q.35 MCQ · 1 mark · PD
A symmetric matrix is positive definite iff
(A) all eigenvalues are positive
(B) det > 0 only
(C) trace > 0 only
(D) it is orthogonal

Q.36 - Q.65 Carry TWO marks each (DA)

Q.36 MSQ · 2 marks · P&S; / CI

n i.i.d. N(mu, sigma^2) with sigma known. A 95% z-interval I_n = Xbar +/- 1.96 sigma/sqrt(n). Which are true?

(A) P_mu(I_n contains mu)=0.95 for every n > =1
(B) Replacing 1.96 by 2.58 produces a 99% interval of the same centre
(C) The interval is a posterior 95% credible set under every prior
(D) The length of I_n tends to 0 as n tends to inf

Q.37 MCQ · 2 marks · P&S; / counting

A reading list is formed by choosing 3 books from 5 distinct ML texts, 4 distinct systems texts, and 3 distinct theory texts, with at least one ML text. The number of lists (order does not matter) is

(A) 185
(B) 105
(C) 220
(D) 10

Q.38 MSQ · 2 marks · Relational algebra and tuple calculus

Which RA expressions are equivalent to sigma_{A=1}(R ⋈ S) when A is an attribute of R only?

(A) (sigma_{A=1}(R)) ⋈ S
(B) R ⋈ (sigma_{A=1}(S))
(C) sigma_{A=1}(R) × S (if the join is a natural join on other attributes, not in general)
(D) the selection can be pushed to the R side

Q.39 MCQ · 2 marks · P&S; / exponential

X~Exponential with mean 1/ln 2. Let Y=floor(X). Then P(Y=ell) = (1-q) q^ell for ell=0,1,2,... The value of q is

(A) 2
(B) 1/2
(C) 1/4
(D) ln 2

Q.40 MSQ · 2 marks · P&S; / Poisson

Let N~Poisson(lambda) with lambda > 0. Which are true?

(A) E[N(N-1)]=lambda^2
(B) The MGF is exp(lambda(e^t - 1))
(C) N takes values in (0, inf) with a density
(D) P(N=0)=e^{-lambda}

Q.41 NAT · 2 marks · P&S; / counting

The number of onto functions from a 5-element domain to a 3-element codomain is _____.

NAT · enter the numerical value

Q.42 NAT · 2 marks · Graphs

In graph Harbor (edges Pier-Dock, Dock-Yard, Yard-Gate, Pier-Shed, Shed-Gate, Dock-Shed), BFS from Pier with alphabetical neighbour order visits how many vertices?

NAT · enter the numerical value

Q.43 MSQ · 2 marks · MLR

Which models stay inside multiple linear regression (linear in parameters)?

(A) y=b0+b1 x+b2 x^2+e
(B) y=b0+b1 log x+e with x > 0
(C) y=b0+b1 x1+b2 x2+b3 x1 x2+e
(D) y=sigmoid(b0+b1 x) with Bernoulli likelihood

Q.44 MSQ · 2 marks · P&S; / discrete

Let X be integer-valued with CDF F. Which statements are correct?

(A) P(X=k)=F(k)-F(k-1) for integer k
(B) F may have jump discontinuities
(C) F is left-continuous
(D) F is non-decreasing

Q.45 MSQ · 2 marks · P&S; / mixed

Let F be the CDF of Uniform(0,1). Which are true?

(A) F is differentiable at 0.3 with derivative 1
(B) F has a jump at 0
(C) F(x)=0 for x < 0, F(x)=x for x in [0,1], F(x)=1 for x > 1
(D) F is continuous

Q.46 MSQ · 2 marks · LDA

Assumptions of classical Gaussian LDA (two classes) include

(A) class-conditional Gaussians
(B) a shared covariance matrix
(C) equal class priors (mandatory)
(D) a linear log-posterior-odds boundary

Q.47 MCQ · 2 marks · NaiveBayes

Gaussian NB with diagonal class-covariances differs from LDA because LDA

(A) does not use class priors
(B) shares one covariance across classes and does not force feature independence unless that covariance is diagonal
(C) cannot produce a linear boundary
(D) is identical to 1-NN

Q.48 NAT · 2 marks · P&S; / mixed

Skewness of N(mu, sigma^2) is 0. For Exp(rate=1) the skewness E[(X-1)^3]/1^3 equals 2. The difference (Exp skewness minus Normal skewness) is _____.

NAT · enter the numerical value

Q.49 MCQ · 2 marks · UninformedSearch

Bidirectional uninformed search (when a predecessor function exists) meets in the middle to cut

(A) the exponent in b^{d/2} + b^{d/2} versus b^d
(B) the need for a goal test
(C) all path costs to 0
(D) the closed set

Q.50 NAT · 2 marks · SQL

```
On Marks, students who have a row for every distinct course: SELECT sid FROM Marks GROUP BY sid HAVING COUNT(*) = (SELECT COUNT(DISTINCT course) FROM Marks). Number of rows is
```

NAT · enter the numerical value

Q.51 MSQ · 2 marks · PropLogic

Which formulae are equivalent to P - > (Q - > P)?

(A) a tautology (true under all four assignments of P,Q)
(B) ~P v (~Q v P)
(C) (P & Q) - > P
(D) P & ~P

Q.52 MCQ · 2 marks · Python

```
The program
def g(x, y=[]):
y.append(x)
return id(y)
u = g(1)
v = g(2)
w = g(3, [])
print(u == v, v == w)
prints
```

(A) True False
(B) False False
(C) True True
(D) False True

Q.53 MSQ · 2 marks · MLR

In MLR y=X beta+e with full-column-rank OLS, which are ALWAYS true?

(A) The residual vector is orthogonal to every column of X
(B) H=X(X^T X)^{-1}X^T is the orthogonal projector onto col(X)
(C) Unadjusted R^2 cannot decrease when a linearly independent column is added
(D) Every coefficient stays unbiased even if a relevant variable is omitted

Q.54 NAT · 2 marks · P&S; / binomial

A shopper buys a warranty on a visit with probability 0.3, independently across 7 visits. The variance of the number of warranties sold is _____ (two decimals).

NAT · enter the numerical value

Q.55 MSQ · 2 marks · Trees

Binary tree U has preorder P Q S T R and inorder S Q T P R. Which statements are true?

(A) R is the right child of P and is a leaf
(B) Q is the left child of P
(C) S is the right child of Q
(D) The postorder is S T Q R P

Q.56 MCQ · 2 marks · Opt word

A farmer maximises area. This is 1-var calculus once the constraint is used to eliminate a variable. The constraint 2x+y=P is used to write A=x(P-2x). This is

(A) substitution
(B) Lagrange on 2 vars (off syllabus as a method, though equivalent)
(C) linear programming in many vars
(D) k-means

Q.57 MSQ · 2 marks · Sorting

Which algorithms are stable in their usual textbook form?

(A) insertion sort
(B) mergesort (left half preferred on ties)
(C) heapsort
(D) selection sort (swap min into place)

Q.58 NAT · 2 marks · SQL

Course: (1,DB,CS), (2,ML,CS), (3,Opt,MA). Enroll(sid,cid): (8,1),(8,2),(9,1),(9,3),(7,3).
SQL: sids from Enroll for which NOT EXISTS a CS course they did not take.
Number of rows returned (Enroll-level, not DISTINCT) is

NAT · enter the numerical value

Q.59 NAT · 2 marks · Cycle even

A 4-cycle is bipartite so -2 is an adjacency eigenvalue of C4 (regular of degree 2). The largest adjacency eigenvalue is _____.

NAT · enter the numerical value

Q.60 NAT · 2 marks · GL det

The matrix [[1,0],[0,-1]] is in GL_2(R) because det equals _____.

NAT · enter the numerical value

Q.61 MSQ · 2 marks · Adversarial

Which claims about minimax (deterministic, zero-sum, perfect information) are TRUE?

(A) It assumes the opponent plays optimally against you
(B) It can be computed bottom-up on a finite game tree
(C) It maximises the worst-case payoff
(D) It requires a heuristic h like A* even at terminals with known utility

Q.62 MSQ · 2 marks · Stacks and queues

Which data structures can implement a FIFO queue with amortized O(1) enqueue and dequeue?

(A) A singly linked list with both head and tail pointers
(B) A singly linked list with only a head pointer (insert/delete at head)
(C) Two stacks (input stack and output stack)
(D) A singly linked list with only a tail pointer (insert at tail)

Q.63 MCQ · 2 marks · SLR

A residual plot of OLS of marks on hours fans out as hours grow. This most directly suggests

(A) the intercept is biased toward 0
(B) heteroscedasticity, so equal-variance Gauss-Markov fails
(C) the slope must be negative
(D) R^2 is necessarily 0

Q.64 MSQ · 2 marks · Data types and transformation

Given AgeYears = [10,20,30,40,50], which are true?

(A) min-max of 10 to [0,1] is 0
(B) min-max of 50 to [0,1] is 1
(C) the sample mean is 30
(D) equal-width 4 bins on [10,50] have width 20

Q.65 NAT · 2 marks · Add7

The tangent to y=x^2 at x=1 has slope 2 and equation y=2x-1. y-intercept is _____.

NAT · enter the numerical value

MOCK G ANSWER KEY AND SHORT SOLUTIONS

Q.1 [MCQ] C — At 3:00 the hands are perpendicular.

Q.2 [NAT] 24 — Net rate 1/6-1/8=1/24 tank per hour.

Q.3 [MCQ] A — Fulsome here is insincere or excessive praise with an ulterior edge.

Q.4 [NAT] 5 — Union is 45, so neither is 5.

Q.5 [MCQ] C — 40% in 8 days means 100% in 20 days.

Q.6 [MCQ] B — n=2k+1 gives n^2=4k(k+1)+1 with k(k+1) even, hence 8m+1.

Q.7 [NAT] 210 — Amount = 1000*1.1^2 = 1210, so CI = 210.

Q.8 [MCQ] B — Five Mondays need at least 29 days (Monday start in a 29-day month).

Q.9 [NAT] 729 — x=9 and y=x^3=729.

Q.10 [MCQ] B — Leave one on long enough to heat, one on just now, one off; warmth is a third state.

Q.11 [MCQ] B — Definition of N(0,1).

Q.12 [NAT] -2 — -2.

Q.13 [MCQ] A — MOLAP: multidimensional arrays; ROLAP: relational star/snowflake plus SQL.

Q.14 [NAT] 5 — The whole chain of 5 keys is scanned.

Q.15 [MCQ] A — Search-horizon pathology.

Q.16 [NAT] 2 — Two tiles are misplaced.

Q.17 [MCQ] A — ReLU'(z)=0 for z < 0.

Q.18 [NAT] 0 — I^{-1}(-2,0)=(-2,0).

Q.19 [MCQ] A — Binning turns a continuous feature into a categorical one.

Q.20 [NAT] 1.2 — (2+4)/(1+4)=6/5=1.2.

Q.21 [MCQ] A — Symmetric idempotent.

Q.22 [NAT] 1 — Both samples have Q=true, so 1.

Q.23 [MCQ] A — Series.

Q.24 [NAT] 16 — 16.

Q.25 [MCQ] A — 2.

Q.26 [NAT] 0.9 — 9/10=0.9.

Q.27 [MCQ] A — Each vertex and each edge is scanned O(1) times.

Q.28 [NAT] 15 — 2^{4}-1 = 15.

Q.29 [MCQ] A — 1.

Q.30 [NAT] 0.296 — 0.3-0.0045=0.2955~0.296.

Q.31 [MCQ] A — First-derivative test / sign of f''.

Q.32 [NAT] 10 — 80/8=10.

Q.33 [MCQ] A — Classic ISAM does not rebalance; overflow chains degrade it.

Q.34 [NAT] 2 — 4-2.

Q.35 [MCQ] A — Sylvester / spectral.

Q.36 [MSQ] A;B;D — Exact Gaussian z-interval has exact coverage. Length shrinks. Larger z widens a still-centred interval.

Q.37 [MCQ] A — Total C(12,3) minus lists with no ML book C(7,3) = 220-35=185.

Q.38 [MSQ] A;D — Selection on an R-only attribute pushes to R. It cannot be applied to S.

Q.39 [MCQ] B — P(Y > =ell)=P(X > =ell)=e^{-ell ln 2}=2^{-ell}, a geometric ratio q=1/2.

Q.40 [MSQ] A;B;D — Poisson is a discrete law on {0,1,2,...}; it has no density on the line. The factorial-moment and MGF formulae hold.

Q.41 [NAT] 150 — 3^5 - C(3,1)2^5 + C(3,2)1^5 = 243-96+3 = 150. Also 3! S(5,3)=6*25=150.

Q.42 [NAT] 5 — The reachable set has size 5.

Q.43 [MSQ] A;B;C — A-C are linear in beta. D is logistic regression.

Q.44 [MSQ] A;B;D — Standard discrete CDF facts. F is right-continuous, not left-continuous.

Q.45 [MSQ] A;C;D — Continuous uniform CDF is continuous (no jumps) and has density 1 on (0,1).

Q.46 [MSQ] A;B;D — Priors need not be equal; they only shift the threshold.

Q.47 [MCQ] B — LDA: shared Sigma, possibly correlated features.

Q.48 [NAT] 2 — Exponential(1) has skewness 2, Gaussian has 0.

Q.49 [MCQ] A — Two frontiers of depth d/2.

Q.50 [NAT] 2 — Two distinct courses; sids 1 and 2 have 2 rows, sid 3 has 1. Answer 2.

Q.51 [MSQ] A;B;C — P - > (Q - > P) is a tautology; D is a contradiction.

Q.52 [MCQ] A — The first two calls share the default list (same id); the third uses a freshly passed list.

Q.53 [MSQ] A;B;C — Omitted-variable bias can wreck unbiasedness. A-C are geometry/algebra of OLS.

Q.54 [NAT] 1.47 — np(1-p)=7*0.21=1.47.

Q.55 [MSQ] A;B;D — Left subtree of P is Q with left S and right T; right child of P is leaf R. Postorder S T Q R P.

Q.56 [MCQ] A — Eliminate.

Q.57 [MSQ] A;B — Heapsort and typical selection sort swap non-adjacent equal keys and are not stable.

Q.58 [NAT] 2 — Only sid 8 takes both CS courses; sid 8 appears twice in Enroll, so 2 rows without DISTINCT.

Q.59 [NAT] 2 — Degree of regular graph.

Q.60 [NAT] -1 — nonzero.

Q.61 [MSQ] A;B;C — Terminal utilities are given; a heuristic is used only when the tree is truncated.

Q.62 [MSQ] A;C — Head+tail list gives O(1) both ends. Two stacks give amortized O(1). Head-only list makes the rear expensive; tail-only cannot delete the head in O(1).

Q.63 [MCQ] B — A fan shape is non-constant variance.

Q.64 [MSQ] A;B;C — Width is (50-10)/4=10, not 20.

Q.65 [NAT] -1 — -1.

MOCK PAPER H Mock Paper 08 · 3VL SQL, LOOCV, variable elimination

Duration: 3 hours. Maximum marks: 100. Q.1-5 GA one mark · Q.6-10 GA two marks · Q.11-35 DA one mark · Q.36-65 DA two marks. MCQ: -1/3 / -2/3. MSQ and NAT: no negative.

Q.1 - Q.5 Carry ONE mark each (General Aptitude)

Q.1 NAT · 1 mark · GA/Quant

The LCM of 12, 18 and 30 is _____.

NAT · enter the numerical value

Q.2 MCQ · 1 mark · GA/Spatial

Six unit squares in a single row of length 6 are used as a net. This net folds to a cube?

(A) yes
(B) no
(C) only if one square is removed
(D) only after a twist

Q.3 NAT · 1 mark · GA/Quant

In a class of 80, the grade pie is A 30%, B 40%, C 20%, D 10%. The number of B grades is _____.

NAT · enter the numerical value

Q.4 MCQ · 1 mark · GA/Analytical

If NORTH is coded as OPSUI, then SOUTH is coded as

(A) TPVUI
(B) TPUUI
(C) TPVVI
(D) RNTSG

Q.5 NAT · 1 mark · GA/Quant

The number of 3-digit palindromes (first digit nonzero) is _____.

NAT · enter the numerical value

Q.6 - Q.10 Carry TWO marks each (General Aptitude)

Q.6 MCQ · 2 marks · GA/Analytical

Six colleagues A,B,C,D,E,F sit around a circle. A sits opposite D. B sits immediately clockwise from A. C sits opposite E. Who sits opposite B?

(A) C
(B) E
(C) F
(D) D

Q.7 NAT · 2 marks · GA/Quant

The difference between CI and SI on Rs 5000 for 2 years at 10% per annum is _____ rupees.

NAT · enter the numerical value

Q.8 MCQ · 2 marks · GA/Verbal

A trial of a sleep app enrolled only night-shift nurses. Sleep scores rose. Marketing then claimed the app 'works for everyone'. The overreach is that

(A) nurses do not sleep
(B) the sample is not representative of all sleepers
(C) apps cannot be timed
(D) scores cannot rise

Q.9 NAT · 2 marks · GA/Analytical

In a class of 40, 25 take maths, 20 take physics, and 10 take both. The number who take neither is _____.

NAT · enter the numerical value

Q.10 MCQ · 2 marks · GA/Verbal

Two translators rendered the same poem. Reviewers preferred Translator A's version for rhythm but Translator B's for fidelity to names. A journal can fairly say that

(A) A is objectively better
(B) preference depends on the criterion (cadence vs. onomastic accuracy)
(C) poetry cannot be translated
(D) B copied A

Q.11 - Q.35 Carry ONE mark each (DA)

Q.11 MCQ · 1 mark · Data types and transformation

Sampling with replacement of 20 draws from 200. Expected number of times a given row appears is

(A) 0.1
(B) 1
(C) 20
(D) 0

Q.12 NAT · 1 mark · MLR

A house-price model uses intercept, area, and a binary near-metro flag. n=40. Columns of X: ___.

NAT · enter the numerical value

Q.13 MCQ · 1 mark · Graphs

An undirected graph is Eulerian (has a closed trail using every edge) if and only if

(A) it is connected (ignoring isolated vertices) and every degree is even
(B) it is a tree
(C) it is complete
(D) it has exactly two odd-degree vertices

Q.14 NAT · 1 mark · Min x+1/x

Min of x+1/x for x > 0 is _____.

NAT · enter the numerical value

Q.15 MCQ · 1 mark · PropLogic

A Horn clause has at most one positive literal. Which of these is Horn?

(A) ~P v ~Q v R
(B) P v Q
(C) P v Q v R
(D) ~P v Q v R

Q.16 NAT · 1 mark · Jensen

For convex f, (f(a)+f(b))/2 > = f((a+b)/2). For f(x)=x^2, a=0,b=2: left 2, right f(1)=_____.

NAT · enter the numerical value

Q.17 MCQ · 1 mark · Implicit

x^2+y^2=1, dy/dx at (3/5,4/5) is

(A) -3/4
(B) 3/4
(C) -4/3
(D) 0

Q.18 NAT · 1 mark · Graphs
On that same RoadNet, after Dijkstra from S, the shortest-path distance of A is
NAT · enter the numerical value

Q.19 MCQ · 1 mark · P&S; / covariance
Which statement is TRUE for finite-variance random variables?
(A) Independent = > Cov=0, but Cov=0 need not imply independent
(B) Corr is defined even if Var(X)=0
(C) Cov=0 = > independent
(D) Var(X+Y)=Var(X)+Var(Y) always

Q.20 NAT · 1 mark · MLR
Dummy-coded MLR uses 4 city levels plus intercept and no other features. Dummy columns excluding intercept: ____.
NAT · enter the numerical value

Q.21 MCQ · 1 mark · Commute
If AB=BA and A is diagonalizable with distinct eigenvalues, then B
(A) is a polynomial in A (in particular shares the eigenbasis)
(B) must be 0
(C) must be I
(D) cannot exist

Q.22 NAT · 1 mark · P&S; / CLT
Sum S_n of n i.i.d. with mean 0 variance 1. Approximate P(S_n / sqrt(n) < = 0) for large n is _____ (one decimal).
NAT · enter the numerical value

Q.23 MCQ · 1 mark · FTC
d/dx int_0^x t^2 dt equals
(A) x^2
(B) x^3/3
(C) 2x
(D) 0

Q.24 NAT · 1 mark · NII
NLL of N(mu,1) at x=4 is (1/2)(4-mu)^2 + const. Min at mu=_____.
NAT · enter the numerical value

Q.25 MCQ · 1 mark · Partition inv
Block inversion of [[A,B],[C,D]] uses the Schur complement D-C A^{-1} B when A is invertible. If B=0,C=0 the inverse is
(A) diag(A^{-1}, D^{-1})
(B) diag(A,D)
(C) 0
(D) I

Q.26 NAT · 1 mark · Need 260 pad unique
Number of leading principal minors of an n x n matrix is n. For n=4 that is _____.
NAT · enter the numerical value

Q.27 MCQ · 1 mark · Trees
Which pair of traversals uniquely determines a binary tree (no duplicate keys)?
(A) inorder and preorder
(B) preorder and postorder
(C) preorder and level-order
(D) postorder and level-order

Q.28 NAT · 1 mark · Nullity
An invertible 4x4 matrix has nullity _____.
NAT · enter the numerical value

Q.29 MCQ · 1 mark · UninformedSearch
Tree-search DFS (no closed set) on a graph with cycles is
(A) complete
(B) in danger of infinite loops along a cycle
(C) optimal for unit costs
(D) the same as UCS

Q.30 NAT · 1 mark · P&S; / discrete
X takes values 1,2,3,4 each with probability 1/4. Then P(X < = 2.5) equals _____ (two decimals).
NAT · enter the numerical value

Q.31 MCQ · 1 mark · kNN
```
Manhattan 1-NN on A(0,0):0, B(3,0):1, query (1,1).
Predicted class is
```
(A) 0, since |1|+|1|=2 < |2|+|1|=3
(B) 1
(C) tie
(D) 2

Q.32 NAT · 1 mark · Sigma min
For an invertible A, ||A^{-1}||_2 = 1/sigma_min. If sigma = 4 and 0.5, ||A^{-1}||_2 is _____.
NAT · enter the numerical value

Q.33 MCQ · 1 mark · Skew
A real skew-symmetric n x n matrix with n odd
(A) is invertible
(B) has det 0
(C) has all eigenvalues real and positive
(D) equals I

Q.34 NAT · 1 mark · Trees
A binary tree has preorder F B A D C E G I H and inorder A B C D E F G H I. How many nodes are in the left subtree of F?
NAT · enter the numerical value

Q.35 MCQ · 1 mark · kNN
Distance-weighted 1-NN (weight 1/d) on two neighbours at distances 1 and 3 with labels 0 and 1. The query is labelled
(A) 0, because the nearer neighbour gets all the 1-NN vote; with k=1 only the nearest is used
(B) 1
(C) the mean 0.5 as a class
(D) undefined

Q.36 - Q.65 Carry TWO marks each (DA)

Q.36 NAT · 2 marks · PropLogic

Atoms P,Q. Number of inequivalent propositional formulae (up to logical equivalence) is ____.

NAT · enter the numerical value

Q.37 NAT · 2 marks · Graphs

Directed weighted graph RailLink with edges Pune- > Mumbai(10), Pune- > Lonavala(4), Lonavala- > Mumbai(5), Mumbai- > Goa(8), Lonavala- > Goa(20). Dijkstra from Pune (non-negative weights). Shortest-path distance to Goa is

NAT · enter the numerical value

Q.38 NAT · 2 marks · P&S; / CI

To have a 95% z-interval half-width at most 0.5 when sigma=2 and z=1.96, the smallest integer n satisfying n > = (1.96*2/0.5)^2 is _____.

NAT · enter the numerical value

Q.39 MCQ · 2 marks · Python

```
The nested function
def f(n):
k = 0
def g():
nonlocal k
k += 1
return k
return [g() for _ in range(n)]
print(f(3))
prints
```

(A) [1, 2, 3]
(B) [0, 0, 0]
(C) [1, 1, 1]
(D) [3, 3, 3]

Q.40 NAT · 2 marks · Searching

Let A be a sorted array of 1000 distinct integers. Recursive binary search for a missing key y uses a three-way comparison at mid and loops while lo < =hi. An upper bound on the number of comparisons is the number of times n=1000 can be halved, about ceil(log2(1001)). The exact worst-case comparison count of this implementation (floor mid, lo=mid+1 or hi=mid-1) on n=1000 is

NAT · enter the numerical value

Q.41 MCQ · 2 marks · P&S; / mixed

Five i.i.d. N(0,1) r.v.s. Let S = X1^2+...+X5^2 and T = sqrt(5) Xbar. Which is TRUE?

(A) S ~ N(0,5)
(B) T ~ chi^2_5
(C) S and T are equal
(D) S ~ chi^2_5 and T^2 ~ chi^2_1, and S-T^2 ~ chi^2_4 independent of T

Q.42 MSQ · 2 marks · P&S; / counting

Let S be a finite n-set. Which counting formulae are correct?

(A) The number of 2-element sequences with repetition from S is C(n,2)
(B) The number of k-subsets is C(n,k)
(C) The number of permutations of S is n!
(D) The number of injections from a k-set to S is P(n,k)=n!/(n-k)! for k < =n

Q.43 NAT · 2 marks · SQL

```
On Emp, SELECT id FROM Emp WHERE NOT (bonus > 100)
returns how many rows?
```

NAT · enter the numerical value

Q.44 MCQ · 2 marks · Python

```
After
d = dict.fromkeys(['a', 'b'], [])
d['a'].append(1)
print(d)
d is
```

(A) {'a': [1], 'b': []}
(B) {'a': [1], 'b': [1]}
(C) {'a': [1], 'b': None}
(D) {'a': 1, 'b': []}

Q.45 MSQ · 2 marks · kNN

Which statements about k-NN classification are TRUE?

(A) Training error of 1-NN is 0 if no two equal-x points have different labels
(B) As k- > n the prediction approaches the global majority class
(C) 1-NN typically has lower bias and higher variance than a large k
(D) k-NN needs an explicit parametric density for each class

Q.46 MSQ · 2 marks · VarElim

Which claims about VE versus sampling are TRUE?

(A) VE is exact (modulo numeric precision)
(B) Gibbs is approximate MCMC
(C) VE can be infeasible when treewidth is large even if n is moderate
(D) Gibbs always returns the exact marginal after 2 samples

Q.47 MCQ · 2 marks · Warehousing and OLAP

Cube LibraryLoans is shown at dimensions Branch, Genre, Month. Moving the last geographic/organisational/time level to the finer grain (Genre to Shelf) is

(A) drill-down
(B) roll-up
(C) slice
(D) dice

Q.48 MSQ · 2 marks · KMedoid

Which are TRUE?

(A) k-medoid cluster centres are members of the dataset
(B) k-means centres need not be members of the dataset
(C) A single swap step in PAM evaluates replacing a medoid by a non-medoid
(D) k-medoid requires class labels

Q.49 MSQ · 2 marks · SQL

SQL three-valued logic. Which evaluations are UNKNOWN (not TRUE)?

(A) NULL = 5
(B) NULL IS NULL
(C) NOT (NULL > 0)
(D) 5 = 5

Q.50 NAT · 2 marks · Jordan

Number of Jordan blocks for lambda equals geo multiplicity. For a single 2x2 block, geo multiplicity is _____.

NAT · enter the numerical value

Q.51 NAT · 2 marks · PropLogic

Number of distinct truth assignments to 3 atoms P,Q,R is ____.

NAT · enter the numerical value

Q.52 MCQ · 2 marks · KFold

2-fold CV, y=(0,0,4,8), folds {0,0} vs {4,8}, intercept-only. The four predictions are

(A) 6,6,0,0
(B) 0,0,4,8
(C) 3,3,3,3
(D) 2,2,2,2

Q.53 MSQ · 2 marks · FOL

Regarding entailment, which hold?

(A) forall x (P(x) - > Q(x)), P(A) |= Q(A)
(B) exists x (P(x) & Q(x)) |= (exists x P(x)) & (exists x Q(x))
(C) (exists x P(x)) & (exists x Q(x)) |= exists x (P(x) & Q(x))
(D) forall x (P(x) & Q(x)) |= (forall x P(x)) & (forall x Q(x))

Q.54 MSQ · 2 marks · P&S; / tests

Which statements are correct about GATE-style hypothesis tests?

(A) z-tests require a known sigma (or a large-sample plug-in)
(B) A chi^2 GoF statistic is always exactly chi^2 even for n=5 sparse cells
(C) Failing to reject H0 does not prove H0 is true
(D) A small p-value is evidence against H0 (under the modelling assumptions)

Q.55 MSQ · 2 marks · Files and indexing

A B+ tree index is chosen over a hash index when

(A) queries include range predicates on the key
(B) the key is used in ORDER BY
(C) only point lookups occur and ranges never do (hash would be competitive)
(D) the attribute has two values (bitmap may be even better) but ranges still matter

Q.56 NAT · 2 marks · Files and indexing

A B+ tree is built on relation Voyage. Node size = 4096 bytes, node-pointer size = 10 bytes, search-key size = 11 bytes, data-record pointer = 12 bytes. A non-leaf stores n node pointers and n-1 keys, packed with no extra header. The maximum n (fanout) is

NAT · enter the numerical value

Q.57 NAT · 2 marks · Files and indexing

A B+ tree is built on relation Clinic. Node size = 4096 bytes, node-pointer size = 6 bytes, search-key size = 10 bytes, data-record pointer = 8 bytes. A non-leaf stores n node pointers and n-1 keys, packed with no extra header. The maximum n (fanout) is

NAT · enter the numerical value

Q.58 MSQ · 2 marks · SLR

Let e_i be OLS residuals from simple linear regression with intercept. Which identities ALWAYS hold?

(A) sum e_i = 0
(B) sum e_i x_i = 0
(C) sum e_i y_i = 0
(D) sum e_i yhat_i = 0

Q.59 MCQ · 2 marks · P&S; / counting

Let S be a 4-element set. The number of functions f: S - > S that are bijections AND satisfy f(f(x))=x for every x in S (involutions that are permutations) is

(A) 24
(B) 16
(C) 10
(D) 4

Q.60 MSQ · 2 marks · P&S; / independence

A fair coin is tossed twice independently. Let H1 = first toss heads, Eq = both tosses equal, Xor = tosses differ. Which of the following pairs are independent?

(A) H1 and (H1 n Eq)
(B) Eq and Xor
(C) H1 and Xor
(D) H1 and Eq

Q.61 NAT · 2 marks · KMeans

1-D points 0,0,0,10, k=1. The k-means centroid (the mean) is ____.

NAT · enter the numerical value

Q.62 MCQ · 2 marks · Basis

Any two bases of a finite-dimensional vector space

(A) can have different cardinalities
(B) have equally many vectors
(C) must be orthogonal
(D) must contain e1

Q.63 MSQ · 2 marks · P&S; / discrete

X takes values in {0,1,2} with p=(0.2,0.5,0.3). Which are true?

(A) P(X=1.7)=0.7
(B) F(1.7)=P(X < =1.7)=0.7
(C) E[X]=1.1
(D) The mode is 1

Q.64 MCQ · 2 marks · Quotient

dim(V/W) = dim V - dim W. For V=R^4, W a line, this is _____.

(A) 3
(B) 4
(C) 1
(D) 0

Q.65 MCQ · 2 marks · Taylor 1d

The 2nd-order Taylor of f at a is f(a)+f'(a)h+(1/2)f''(a)h^2. For f=e^x, a=0, h=1 this is 1+1+1/2=_____.

(A) 2.5
(B) e
(C) 1
(D) 3

MOCK H ANSWER KEY AND SHORT SOLUTIONS

Q.1 [NAT] 180 — LCM uses 2^2 * 3^2 * 5 = 180.

Q.2 [MCQ] B — A strip of six squares overlaps when folded; it is not a cube net.

Q.3 [NAT] 32 — 40% of 80 is 32.

Q.4 [MCQ] A — Each letter shifts +1: S- > T, O- > P, U- > V, T- > U, H- > I.

Q.5 [NAT] 90 — The first digit has 9 choices and the middle digit 10; the last digit is then fixed.

Q.6 [MCQ] C — With A at a seat and D opposite, B clockwise from A forces F opposite B.

Q.7 [NAT] 50 — For two years the gap is P*(r/100)^2 = 50.

Q.8 [MCQ] B — A specialised sample does not license a universal claim.

Q.9 [NAT] 5 — Union is 35, so neither is 5.

Q.10 [MCQ] B — The split reviews show criterion-dependence, not a single ranking.

Q.11 [MCQ] A — Each draw includes it with p=1/200; 20 draws = > 20/200=0.1.

Q.12 [NAT] 3 — Intercept + area + dummy = 3.

Q.13 [MCQ] A — Euler's theorem. Exactly two odd degrees give an open Euler trail, not a circuit.

Q.14 [NAT] 2 — AM-GM.

Q.15 [MCQ] A — ~P v ~Q v R is (P & Q) - > R, definite Horn. D has two positives Q and R.

Q.16 [NAT] 1 — 1 < =2.

Q.17 [MCQ] A — 2x+2y y'=0 = > y'=-x/y.

Q.18 [NAT] 3 — S-B-A = 1+2 = 3, better than the direct 4.

Q.19 [MCQ] A — Independence implies uncorrelatedness; the converse fails. Additivity of variance needs uncorrelatedness. Correlation needs positive variances.

Q.20 [NAT] 3 — k levels need k-1 dummies with intercept.

Q.21 [MCQ] A — Classical fact for distinct eigs.

Q.22 [NAT] 0.5 — The standardised sum is approx N(0,1).

Q.23 [MCQ] A — FTC.

Q.24 [NAT] 4 — 4.

Q.25 [MCQ] A — Block diagonal.

Q.26 [NAT] 4 — 1x1 up to 4x4.

Q.27 [MCQ] A — Inorder plus one of preorder/postorder/level-order determines the tree. Preorder+postorder does not, in general.

Q.28 [NAT] 0 — Trivial kernel.

Q.29 [MCQ] B — Without a closed set, a cycle can be re-entered forever.

Q.30 [NAT] 0.5 — X=1 or 2, probability 1/2.

Q.31 [MCQ] A — d1(A)=2, d1(B)=3.

Q.32 [NAT] 2 — 1/0.5.

Q.33 [MCQ] B — det A = det(A^T)=det(-A)=(-1)^n det A, n odd = > det=0.

Q.34 [NAT] 5 — Inorder left of F is A B C D E, five nodes.

Q.35 [MCQ] A — 1-NN uses only the single nearest point.

Q.36 [NAT] 16 — 2^(2^2)=16 Boolean functions of 2 vars.

Q.37 [NAT] 17 — Dijkstra distance {'Goa': 17, 'Pune': 0, 'Lonavala': 4, 'Mumbai': 9}.

Q.38 [NAT] 62 — (3.92/0.5)^2=7.84^2=61.4656, so n=62.

Q.39 [MCQ] A — nonlocal k is incremented on each g() call.

Q.40 [NAT] 9 — Simulate unsuccessful search on 0..999 for -1 (always go left) or a value larger than all. Count=9.

Q.41 [MCQ] D — Cochran: sample ss splits into n(Xbar)^2 ~ chi^2_1 and residual chi^2_{n-1}, independent.

Q.42 [MSQ] B;C;D — Sequences with repetition: n^2, not C(n,2). C(n,2) counts unordered pairs.

Q.43 [NAT] 2 — NOT UNKNOWN is UNKNOWN, so NULL bonuses drop. bonus > 100 is true only for 3, false for 1 and 5. NOT false keeps 1 and 5: 2 rows.

Q.44 [MCQ] B — fromkeys reuses the same list object as the default value for every key.

Q.45 [MSQ] A;B;C — k-NN is a nonparametric local vote; no class-density model is fitted.

Q.46 [MSQ] A;B;C — A few Gibbs samples do not give exactness.

Q.47 [MCQ] A — Finer grain along a hierarchy is drill-down on cube LibraryLoans.

Q.48 [MSQ] A;B;C — Both are unsupervised; PAM uses swap.

Q.49 [MSQ] A;C — IS NULL is TRUE for NULL. Comparisons with NULL are UNKNOWN; NOT UNKNOWN is UNKNOWN.

Q.50 [NAT] 1 — 1.

Q.51 [NAT] 8 — 2^3=8.

Q.52 [MCQ] A — Mean 6 predicts the two zeros; mean 0 predicts 4 and 8.

Q.53 [MSQ] A;B;D — Two possibly different witnesses do not give a common witness. C fails.

Q.54 [MSQ] A;C;D — p-values are not proof of H0. Chi^2 is an approximation requiring adequate expected counts.

Q.55 [MSQ] A;B — Hash wins pure equality; B+ wins ranges and ordered scans. The question asks when B+ is chosen over hash: A and B. C argues the opposite.

Q.56 [NAT] 195 — n*nodeptr + (n-1)*key < = block = > n < = floor((block+key)/(nodeptr+key)) = 195.

Q.57 [NAT] 256 — n*nodeptr + (n-1)*key < = block = > n < = floor((block+key)/(nodeptr+key)) = 256.

Q.58 [MSQ] A;B;D — Normal equations give 1^T e = 0 and x^T e = 0. yhat lives in col(X), so yhat^T e = 0.

Q.59 [MCQ] C — Count involutions in S_4: the identity, 6 transpositions, and 3 products of two disjoint transpositions, total 10.

Q.60 [MSQ] C;D — P(H1 n Eq)=P(HH)=1/4=P(H1)P(Eq). Same for Xor. But Eq and Xor are complementary, hence disjoint with positive probability, so dependent.

Q.61 [NAT] 2.5 — (0+0+0+10)/4=2.5.

Q.62 [MCQ] B — Dimension is well-defined.

Q.63 [MSQ] B;C;D — E[X]=0.5+0.6=1.1. CDF at 1.7 includes 0 and 1. A discrete law has P(X=1.7)=0.

Q.64 [MCQ] A — 3.

Q.65 [MCQ] A — 2.5 vs e~2.718.

MOCK PAPER I Mock Paper 09 · Rotation matrices, logistic, sampling

Duration: 3 hours. Maximum marks: 100. Q.1-5 GA one mark · Q.6-10 GA two marks · Q.11-35 DA one mark · Q.36-65 DA two marks. MCQ: -1/3 / -2/3. MSQ and NAT: no negative.

Q.1 - Q.5 Carry ONE mark each (General Aptitude)

Q.1 MCQ · 1 mark · GA/Verbal

Vaccine : Prevent infection :: Antidote : _____

(A) Diagnose fever
(B) Counter a poison
(C) Sterilise tools
(D) Measure dose

Q.2 NAT · 1 mark · GA/Quant

A shop offers 15% off a Rs 2000 item, then a further Rs 100 off. The final price, in rupees, is _____.

NAT · enter the numerical value

Q.3 MCQ · 1 mark · GA/Verbal

The antonym of 'obsequious' is

(A) servile
(B) assertive
(C) humble
(D) polite

Q.4 NAT · 1 mark · GA/Quant

The remainder when 7^5 is divided by 4 is _____.

NAT · enter the numerical value

Q.5 MCQ · 1 mark · GA/Verbal

Pedantic : Minute rules :: Ostentatious : _____

(A) Quiet merit
(B) Showy display
(C) Humble craft
(D) Secret hoard

Q.6 - Q.10 Carry TWO marks each (General Aptitude)

Q.6 NAT · 2 marks · GA/Quant

Using pi=22/7, the area of a circle of radius 7 is _____.

NAT · enter the numerical value

Q.7 MCQ · 2 marks · GA/Verbal

A wildlife camera on a trail recorded 40 deer in winter and 8 in summer. The biologist did not count off-trail. The fairest reading is that

(A) the deer population collapsed
(B) on-trail detections were lower in summer
(C) cameras fail in heat
(D) poaching explains the gap

Q.8 NAT · 2 marks · GA/Spatial

A 6x6x6 cube is painted on all faces and diced into unit cubes. The number of unit cubes with no paint is _____.

NAT · enter the numerical value

Q.9 MCQ · 2 marks · GA/Spatial

Three unit cubes are stacked as a 1x1x3 tower. The exposed surface area, including the base sitting on a table (base not exposed), is

(A) 10
(B) 13
(C) 14
(D) 18

Q.10 NAT · 2 marks · GA/Analytical

Letters stand for distinct digits and I+I=ME with ME a two-digit number. The value of I+M+E is _____.

NAT · enter the numerical value

Q.11 - Q.35 Carry ONE mark each (DA)

Q.11 NAT · 1 mark · Power

d/dx x^{1/2} at x=4 equals _____ (two decimals).

NAT · enter the numerical value

Q.12 MCQ · 1 mark · Graphs

A cut-vertex (articulation point) of a connected graph is a vertex whose removal

(A) increases the number of connected components
(B) decreases the number of edges by 1 always
(C) always leaves the graph a tree
(D) cannot lie on a cycle

Q.13 NAT · 1 mark · Hash tables

In a table of size 8 with h(x)=x mod 8 and linear probing, insert 16, 24, 32. The index of 32 is

NAT · enter the numerical value

Q.14 MCQ · 1 mark · Nowhere deriv

Weierstrass function is continuous everywhere, differentiable nowhere. It shows continuity

(A) does not imply differentiability
(B) implies differentiability
(C) equals analyticity
(D) is impossible

Q.15 NAT · 1 mark · Poly crit count

A degree-4 polynomial has at most how many critical points? _____.

NAT · enter the numerical value

Q.16 MCQ · 1 mark · Four fund

ker(A) and row(A) in R^n are

(A) the same space
(B) orthogonal complements
(C) both {0}
(D) equal to Col(A)

Q.17 NAT · 1 mark · PCA

2-D points (1,1),(1,1),(1,1) already equal. After centering, all PCA eigenvalues are ____.

NAT · enter the numerical value

Q.18 MCQ · 1 mark · Python

```
After
a, *b, c = [9, 8, 7, 6, 5]
print(a, b, c)
the printed line is
```

(A) 9 [8, 7, 6] 5
(B) 9 8 [7, 6, 5]
(C) [9, 8, 7, 6] 5 None
(D) 9 [8, 7, 6, 5] error

Q.19 NAT · 1 mark · Files and indexing

8000 such records (300 B, 2048 B blocks, unspanned, bf=6). Number of blocks needed is

NAT · enter the numerical value

Q.20 MCQ · 1 mark · Graphs

Directed edges P- > Q, P- > R, Q- > S, R- > S. The number of topological orders is

(A) 2
(B) 1
(C) 3
(D) 4

Q.21 NAT · 1 mark · Graphs

In graph Club (edges Ram-Sam, Sam-Tom, Tom-Uma, Ram-Uma, Sam-Uma), BFS from Ram with alphabetical neighbour order visits how many vertices?

NAT · enter the numerical value

Q.22 MCQ · 1 mark · Strict diag dom

A strictly diagonally dominant matrix is

(A) nonsingular
(B) always orthogonal
(C) a projection
(D) nilpotent

Q.23 NAT · 1 mark · Skew 2

det of [[0,2],[-2,0]] is _____.

NAT · enter the numerical value

Q.24 MCQ · 1 mark · Maclaurin

Maclaurin series is Taylor at

(A) 0
(B) 1
(C) infty
(D) a generic a

Q.25 NAT · 1 mark · MLP

Hidden layer of 3 sigmoids, input dim 4, including hidden biases, no output yet. Hidden-layer parameter count is ____.

NAT · enter the numerical value

Q.26 MCQ · 1 mark · P&S; / joint

X,Y have joint density f(x,y)=1 on the unit square (0,1)^2. Then X and Y are

(A) uniform but dependent
(B) neither uniform nor independent
(C) independent but not uniform
(D) independent Uniform(0,1)

Q.27 NAT · 1 mark · Trees

A complete binary tree with 12 nodes is stored in a 1-indexed array. The parent of the node at index 12 is at index

NAT · enter the numerical value

Q.28 MCQ · 1 mark · Unbdd below

-x^4 has

(A) a global max at 0 and no global min
(B) a global min
(C) no extremum
(D) only inflections

Q.29 NAT · 1 mark · Comp min

Min of e^{-x} on [0,1] is e^{-1} ~ _____ (two decimals).

NAT · enter the numerical value

Q.30 MCQ · 1 mark · BiasVariance

As polynomial degree grows to n-1 on n noiseless distinct x, train SSE

(A) stays constant
(B) tends to 0 (interpolation)
(C) must increase
(D) equals SST always

Q.31 NAT · 1 mark · Sum dim 2

dim(span{e1}+span{e2}) in R^2 is _____.

NAT · enter the numerical value

Q.32 MCQ · 1 mark · ER model

An attribute of a relationship (e.g. Enrolls.grade) is placed, after mapping an M:N relationship, in

(A) the relationship table
(B) both entity tables
(C) a CHECK constraint only
(D) the log

Q.33 NAT · 1 mark · Hash tables

A hash table of 13 slots uses chaining. 10 distinct keys are stored. The minimum possible length of the longest chain is

NAT · enter the numerical value

Q.34 MCQ · 1 mark · P&S; / chi2-test

A 2x2 contingency table independence test (no cell fusion) uses chi^2 with degrees of freedom

(A) 2
(B) 3
(C) 1
(D) 4

Q.35 NAT · 1 mark · Perp

In R^3, the orthogonal complement of span{(1,0,0),(0,1,0)} has dimension _____.

NAT · enter the numerical value

Q.36 - Q.65 Carry TWO marks each (DA)

Q.36 MSQ · 2 marks · SQL

GROUP BY vs DISTINCT. On Marks(sid,course) with duplicate (sid,course) forbidden by PK, which are true?

(A) SELECT sid FROM Marks GROUP BY sid returns each sid once
(B) SELECT DISTINCT sid FROM Marks returns each sid once
(C) SELECT sid, COUNT(*) FROM Marks GROUP BY sid is legal
(D) SELECT sid, COUNT(*) FROM Marks DISTINCT is the usual legal form

Q.37 MSQ · 2 marks · P&S; / exponential

Waiting times W1, W2 are i.i.d. Exponential(rate=1). Which are true?

(A) max(W1,W2) ~ Exponential(rate=1)
(B) W1+W2 ~ Gamma(shape=2, rate=1) (Erlang-2)
(C) min(W1,W2) ~ Exponential(rate=2)
(D) P(W1 < W2)=1/2

Q.38 MSQ · 2 marks · P&S; / z-test

n=100, sigma=10, Xbar=103, H0: mu=100 vs H1: mu != 100, alpha=0.05, z_crit=1.96. Which are true?

(A) An approximate p-value is 2(1-Phi(3)) ~ 0.0027
(B) The 95% z-interval for mu is centred at 100
(C) z = (103-100)/(10/10) = 3
(D) The two-sided test rejects H0

Q.39 MCQ · 2 marks · MLP

A fully connected net input-4, hidden-5, output-3, biases on hidden and output, no extra layers. Parameter count is

(A) 4*5+5*3
(B) 4*5+5 + 5*3+3
(C) 5*3
(D) 4+5+3

Q.40 MSQ · 2 marks · InformedSearch

Which heuristics for 8-puzzle Manhattan / misplaced-tiles are admissible?

(A) Number of misplaced tiles
(B) Sum of Manhattan distances of tiles to goals
(C) Twice the Manhattan sum (generally)
(D) h=0

Q.41 NAT · 2 marks · P&S; / mixed

A hypergeometric flavour without the name: an urn has 5 red and 7 blue tokens. Three tokens are drawn without replacement. P(all red) = C(5,3)/C(12,3) equals _____ (rounded to three decimals).

NAT · enter the numerical value

Q.42 NAT · 2 marks · P&S; / Bayes

In a mailbox, 30% of messages are spam. A filter tags spam with probability 0.90 and tags ham as spam with probability 0.05. Given a tagged message, P(spam) equals _____ (two decimals).

NAT · enter the numerical value

Q.43 MCQ · 2 marks · SVM

A dual variable pinned at alpha_i=C (box 0 < =alpha < =C) typically corresponds to

(A) a non-support vector
(B) a margin violator (slack or misclassified)
(C) the unique class mean
(D) hinge 0 necessarily

Q.44 MSQ · 2 marks · Warehousing and OLAP

Which are OLAP operations on cube HarbourTonnage(Port, ShipType, Month)?

(A) roll-up Month to Year
(B) drill-down Port to Berth
(C) slice Month='Jan'
(D) BCNF decomposition of the cube

Q.45 NAT · 2 marks · Adversarial

MAX root with two moves: Left to a MIN node with leaves 3 and 5; Right to a MIN node with leaves 2 and 9. The minimax value of the root is ____.

NAT · enter the numerical value

Q.46 MSQ · 2 marks · Integrity constraints

Emp(ssn PRIMARY KEY, dno REFERENCES Dept(dno), salary INT CHECK (salary > 0)). Dept currently contains only dno=5. Which new Emp rows are rejected?

(A) ssn that already exists in Emp
(B) dno=9 (absent from Dept)
(C) salary=0 (fails CHECK)
(D) dno=5, salary=10, fresh ssn

Q.47 MCQ · 2 marks · Sampling

If evidence is a root node, ancestral rejection versus clamping that root

(A) wastes draws that do not match the root; clamping the root is just sampling from the posterior slice and is exact in distribution for descendants
(B) is identical to minimax
(C) cannot be done
(D) requires CNF

Q.48 MCQ · 2 marks · NaiveBayes

Four spam bags contribute 8 tokens with 'win' appearing twice. Multinomial NB MLE P(win|spam) without smoothing is

(A) 1/2
(B) 2/4
(C) 2/8
(D) 1/8

Q.49 NAT · 2 marks · Relational model and FDs

Let schema FlightLeg(A,B,C,D,E) with FDs AB- > C, C- > D, D- > E, E- > A. The number of candidate keys is

NAT · enter the numerical value

Q.50 MCQ · 2 marks · Warehousing and OLAP

Cube PayrollCost is shown at dimensions Org, Grade, Month. Moving the last geographic/organisational/time level to the finer grain (Org to Team) is

(A) drill-down
(B) roll-up
(C) slice
(D) dice

Q.51 NAT · 2 marks · P&S; / joint

Using the same joint table P(X=1,Y=1)=0.10, P(X=1,Y=2)=0.20, P(X=2,Y=1)=0.30, P(X=2,Y=2)=0.40, the value of P(Y=1|X=1) is _____ (rounded to two decimals).

NAT · enter the numerical value

Q.52 MCQ · 2 marks · LDA

Two-class means m1,m2 and invertible pooled S_W. An optimal Fisher direction is

(A) S_W (m1+m2)
(B) S_W^{-1} (m1-m2)
(C) S_B^{-1} (m1+m2)
(D) PC1 of the pooled unlabelled data

Q.53 NAT · 2 marks · UninformedSearch

Grid (0,0) to (3,0) with 4-moves of cost 1. UCS/BFS optimal cost is ____.

NAT · enter the numerical value

Q.54 MSQ · 2 marks · BayesNet

WetGrass network C- > S, C- > R, S- > W, R- > W. Which hold?

(A) S ? R | C
(B) S not? R (marginally, generally)
(C) S not? R | W (collider W opens explaining away)
(D) W ? C | S,R is false in general? actually W is a child of S,R so given S,R, W is independent of C (no path around). W ? C | S,R holds.

Q.55 NAT · 2 marks · SLR

On Pune-Mumbai express buses the OLS residual sum of squares for (4,8), (5,11), (6,13), (7,16) (speed in 10 km/h, stopping metres) equals ____.

NAT · enter the numerical value

Q.56 NAT · 2 marks · PCA frac

Eigenvalues of covariance 4 and 1. Fraction of variance on PC1 is _____ (one decimal).

NAT · enter the numerical value

Q.57 MCQ · 2 marks · Real schur

A real matrix with a complex conjugate eigenvalue pair has a real Schur form containing

(A) a 2x2 block
(B) only 1x1 blocks always
(C) a zero row
(D) P^2=P

Q.58 MCQ · 2 marks · Sampling

On a two-node BN A- > E with evidence E=true, likelihood weighting draws A from P(A) then

(A) sets E=true and weights by P(E=true|A)
(B) rejects if A=false always
(C) runs minimax on A
(D) unifies A with E

Q.59 MCQ · 2 marks · Eig 2x2b

Let A=[[0,-1],[1,0]]. The eigenvalues are

(A) 1 and -1
(B) i and -i
(C) 0 and 0
(D) 2 and 0

Q.60 MCQ · 2 marks · Hash tables

Open-address table of size 7 with h(x)=x mod 7 and linear probing. Keys 50, 15, 36, 8 are inserted (empty start). Key 8 occupies index

(A) 1
(B) 2
(C) 3
(D) 4

Q.61 MCQ · 2 marks · P&S; / joint

The joint PMF of (X,Y) on {0,1}x{0,1} is p(0,0)=0.1, p(0,1)=0.2, p(1,0)=0.3, p(1,1)=0.4. Then P(X=1|Y=1) equals

(A) 0.4/0.7
(B) 0.7
(C) 0.4
(D) 0.4/0.6

Q.62 MSQ · 2 marks · P&S; / chi-squared

Let X1,...,Xn (n > =2) be i.i.d. N(0,1) and Xbar the sample mean. Which statements are correct?

(A) sum_{i=1}^n X_i^2 ~ chi^2_n
(B) X1^2 + X2^2 ~ Exponential with mean 2
(C) (sqrt(n) Xbar)^2 ~ chi^2_2
(D) sum_{i=1}^n (X_i - Xbar)^2 ~ chi^2_{n-1}

Q.63 NAT · 2 marks · kNN

Five neighbour labels 1,1,1,0,0. 5-NN majority class is ____.

NAT · enter the numerical value

Q.64 MCQ · 2 marks · Affine

The set {x: Ax=b} for consistent Ax=b is

(A) a subspace if b!=0
(B) an affine translate of ker A
(C) always {0}
(D) a sphere

Q.65 MCQ · 2 marks · P&S; / mixed

Assume the MGF M(t)=E[e^{tX}] exists in a neighbourhood of 0. Then M'(0) equals

(A) Var(X)
(B) E[X]
(C) 1
(D) E[X^2]

MOCK I ANSWER KEY AND SHORT SOLUTIONS

Q.1 [MCQ] B — A vaccine prevents infection; an antidote counters a poison.

Q.2 [NAT] 1600 — 15% off leaves 1700, then 100 off leaves 1600.

Q.3 [MCQ] B — Obsequious means fawning; assertive is opposed.

Q.4 [NAT] 3 — 7 == -1 (mod 4), so 7^5 == -1 == 3 (mod 4).

Q.5 [MCQ] B — Pedantic fuss is over minute rules; ostentatious fuss is over showy display.

Q.6 [NAT] 154 — 22/7 * 49 = 154.

Q.7 [MCQ] B — Only trail detections are observed; seasonal trail use may change.

Q.8 [NAT] 64 — The inner 4x4x4 cube is unpainted: 64.

Q.9 [MCQ] B — 3*6 - 4 hidden join faces - 1 base = 18-4-1=13.

Q.10 [NAT] 6 — 5+5=10, so I=5, M=1, E=0 and the sum is 6.

Q.11 [NAT] 0.25 — 1/(2sqrt x)=1/4.

Q.12 [MCQ] A — That is the definition. Cut-vertices often do lie on cycles (e.g. two cycles sharing a vertex).

Q.13 [NAT] 2 — All hash to 0; they occupy 0,1,2 so 32 is at 2.

Q.14 [MCQ] A — Classic.

Q.15 [NAT] 3 — Degree of derivative is 3.

Q.16 [MCQ] B — Fundamental theorem.

Q.17 [NAT] 0 — No variance left.

Q.18 [MCQ] A — Starred assignment captures the middle remainder as a list.

Q.19 [NAT] 1334 — ceil(8000/6)=1334.

Q.20 [MCQ] A — P first, S last; Q and R may be swapped: P Q R S and P R Q S.

Q.21 [NAT] 4 — The reachable set has size 4.

Q.22 [MCQ] A — Levy-Desplanques.

Q.23 [NAT] 4 — 4.

Q.24 [MCQ] A — a=0.

Q.25 [NAT] 15 — 4*3 weights + 3 biases = 15.

Q.26 [MCQ] D — f factors as 1*1 and each marginal is Uniform(0,1).

Q.27 [NAT] 6 — Parent of i is floor(i/2)=6.

Q.28 [MCQ] A — Goes to -infty.

Q.29 [NAT] 0.37 — 0.368.

Q.30 [MCQ] B — Degree n-1 interpolates n points.

Q.31 [NAT] 2 — 2.

Q.32 [MCQ] A — Relationship attributes become columns of the relationship relation.

Q.33 [NAT] 1 — 10 < 13 so the keys can occupy 10 distinct slots; longest chain can be 1.

Q.34 [MCQ] C — (2-1)(2-1)=1.

Q.35 [NAT] 1 — The z-axis.

Q.36 [MSQ] A;B;C — DISTINCT applies to the select list, not as SELECT ... DISTINCT COUNT with a leftover sid without GROUP BY.

Q.37 [MSQ] B;C;D — Min of independent exponentials adds rates. Sum is Erlang. Symmetry gives P(W1 < W2)=1/2.

Q.38 [MSQ] A;C;D — SE=1, z=3 > 1.96 so reject. The CI is centred at Xbar=103, not at the H0 value.

Q.39 [MCQ] B — Weights 20+15, biases 5+3, total 43. Option B is 20+5+15+3=43.

Q.40 [MSQ] A;B;D — Each move fixes at most one Manhattan unit and at most one misplaced tile. Doubling Manhattan can overestimate.

Q.41 [NAT] 0.045 — 10/220=1/22~0.045.

Q.42 [NAT] 0.89 — 0.27/(0.27+0.035)=0.27/0.305 ~ 0.89.

Q.43 [MCQ] B — KKT: alpha=C usually means slack.

Q.44 [MSQ] A;B;C — Normal-form decomposition is a relational design step, not an OLAP navigation operator.

Q.45 [NAT] 3 — Left min=3, Right min=2, max=3.

Q.46 [MSQ] A;B;C — The last row satisfies PK, FK and CHECK.

Q.47 [MCQ] A — Evidence at roots is easy: just fix them.

Q.48 [MCQ] C — 2/8=1/4.

Q.49 [NAT] 4 — Candidate keys are AB, BC, BD, BE.

Q.50 [MCQ] A — Finer grain along a hierarchy is drill-down on cube PayrollCost.

Q.51 [NAT] 0.33 — P(X=1)=0.30 so 0.10/0.30=1/3 ~ 0.33.

Q.52 [MCQ] B — w ? S_W^{-1}(m1-m2).

Q.53 [NAT] 3 — Three east moves.

Q.54 [MSQ] A;B;C — Option D is actually TRUE (W ? C | S,R) so if the stem lists D as a true CI, the answer would include D. The option text says 'is false in general' making D a false claim. So A;B;C.

Q.55 [NAT] 0.2 — SSE = sum (y-yhat)^2 = 0.2.

Q.56 [NAT] 0.8 — 4/5.

Q.57 [MCQ] A — 2x2 rotation-scale blocks.

Q.58 [MCQ] A — Canonical one-edge LW.

Q.59 [MCQ] B — Rotation by 90 degrees: characteristic t^2+1=0.

Q.60 [MCQ] D — 50 goes to 1; 15 collides and takes 2; 36 takes 3; 8 takes 4.

Q.61 [MCQ] D — P(Y=1)=0.2+0.4=0.6, so P(X=1|Y=1)=0.4/0.6=2/3.

Q.62 [MSQ] A;B;D — The last claim would require 2 degrees of freedom but sqrt(n) Xbar is N(0,1), so its square is chi^2_1. chi^2_2 is Exp with mean 2.

Q.63 [NAT] 1 — Votes 3-2.

Q.64 [MCQ] B — Particular + homogeneous.

Q.65 [MCQ] B — The first derivative of the MGF at 0 is the mean.

MOCK PAPER J Mock Paper 10 · Full syllabus mix

Duration: 3 hours. Maximum marks: 100. Q.1-5 GA one mark · Q.6-10 GA two marks · Q.11-35 DA one mark · Q.36-65 DA two marks. MCQ: -1/3 / -2/3. MSQ and NAT: no negative.

Q.1 - Q.5 Carry ONE mark each (General Aptitude)

Q.1 MCQ · 1 mark · GA/Quant

If x + 1/x = 3, then x^2 + 1/x^2 equals

(A) 7
(B) 8
(C) 9
(D) 11

Q.2 NAT · 1 mark · GA/Quant

The 4th term of a GP with first term 2 and ratio 3 is _____.

NAT · enter the numerical value

Q.3 MCQ · 1 mark · GA/Verbal

A report that 'cuts no ice' with the board

(A) is too brief
(B) fails to persuade
(C) lacks figures
(D) arrives late

Q.4 NAT · 1 mark · GA/Quant

A rectangle is 8 by 6. The length of a diagonal is _____.

NAT · enter the numerical value

Q.5 MCQ · 1 mark · GA/Quant

If log_b(64)=2, then log_2(b) equals

(A) 2
(B) 3
(C) 4
(D) 8

Q.6 - Q.10 Carry TWO marks each (General Aptitude)

Q.6 NAT · 2 marks · GA/Quant

A right triangle has legs 5 and 12. Its area is _____.

NAT · enter the numerical value

Q.7 MCQ · 2 marks · GA/Quant

A family has two children. Given that the elder child is a girl, the probability that both are girls (assuming each child is boy or girl with equal chance, independently) is

(A) 1/4
(B) 1/3
(C) 1/2
(D) 2/3

Q.8 NAT · 2 marks · GA/Quant

The number of ways to choose a president and a treasurer from 6 people (distinct offices) is _____.

NAT · enter the numerical value

Q.9 MCQ · 2 marks · GA/Quant

A cube is cut by a plane through two parallel edges that lie on opposite faces and are not on the same face. The cross-section is a

(A) equilateral triangle
(B) rectangle
(C) regular hexagon
(D) circle

Q.10 NAT · 2 marks · GA/Quant

A 500 g alloy is gold and copper in the ratio 3:2. The mass of gold, in grams, is _____.

NAT · enter the numerical value

Q.11 - Q.35 Carry ONE mark each (DA)

Q.11 MCQ · 1 mark · P&S; / tests

Power of a level-alpha test against a particular alternative is

(A) always equal to alpha
(B) the p-value
(C) P(Type I error)
(D) 1 - P(Type II error | that alternative)

Q.12 NAT · 1 mark · Onb size

An orthonormal set in R^n has size at most _____.

NAT · enter the numerical value

Q.13 MCQ · 1 mark · Partial off

Partial derivatives of F(x,y) are

(A) in the DA 2027 calculus syllabus
(B) NOT in the official DA calculus syllabus (single variable only)
(C) the same as d/dx
(D) Lagrange multipliers

Q.14 NAT · 1 mark · NaiveBayes

Add-1 Laplace, word seen 3 times in 14 class-A docs (binary presence). P(word=1|A) is ____.

NAT · enter the numerical value

Q.15 MCQ · 1 mark · P&S; / t-test

For t=2 on 8 df, using the critical value t_{8, 0.025}~2.306, a two-sided 5% test

(A) rejects H0
(B) has p-value 0
(C) requires known sigma
(D) does not reject H0

Q.16 NAT · 1 mark · Units

Number of matrix units in M_{3,2} is _____.

NAT · enter the numerical value

Q.17 MCQ · 1 mark · Trees

The array [2, 8, 3, 9, 7, 4] (1-indexed heap layout) is tested as a min-heap. How many parent-child pairs violate the min-heap property?

(A) 0
(B) 1
(C) 2
(D) 3

Q.18 NAT · 1 mark · KFold

5-fold CV MSE on folds 2,4,3,5,6. Mean CV estimate is ____.

NAT · enter the numerical value

Q.19 MCQ · 1 mark · Data types and transformation

Decimal(5,2) can store which of the following?

(A) 123.45
(B) 1234.5 (needs 5 digits to the left? 1234.5 is 6 significant with scale 2: 1234.50 needs precision 6)
(C) 0.0001 stored exactly as four decimals of scale 2
(D) an arbitrary-length string

Q.20 NAT · 1 mark · P&S; / cond-pdf

Let Y ~ N(0,1) and, given Y=y, let X ~ N(y, 1). Then Var(X) equals _____.

NAT · enter the numerical value

Q.21 MCQ · 1 mark · 2x2 inv formula

[[a,b],[c,d]]^{-1} = (1/(ad-bc)) [[d,-b],[-c,a]]. For ad-bc=1, the (1,1) entry is

(A) d
(B) a
(C) 0
(D) 1 always

Q.22 NAT · 1 mark · VarElim

Product of two factors each of size 4, scopes {A,B} and {A,B} (same scope, binary). Result size is ____.

NAT · enter the numerical value

Q.23 MCQ · 1 mark · Warehousing and OLAP

On cube LibraryLoans (Branch, Genre, Month), aggregating away the middle dimension entirely (summing the measure over it) is

(A) roll-up / projection of a dimension
(B) drill-down
(C) a foreign-key insert
(D) 2NF decomposition

Q.24 NAT · 1 mark · SPD check

Leading minors of [[2,1],[1,2]] are 2 and 3, both positive, so the matrix is SPD. det equals _____.

NAT · enter the numerical value

Q.25 MCQ · 1 mark · Cont not diff

f(x)=|x| at 0 is

(A) differentiable
(B) continuous but not differentiable
(C) discontinuous
(D) twice differentiable

Q.26 NAT · 1 mark · Trees

In a rooted binary tree every node has 0 or 2 children (full). If there are 17 leaves, the number of nodes with two children is

NAT · enter the numerical value

Q.27 MCQ · 1 mark · Not needed

GATE DA calculus is single-variable: no partials, no Lagrange. The number of variables in Section 3 optimisation is

(A) 1
(B) 2
(C) n
(D) unspecified

Q.28 NAT · 1 mark · Fourier 2

The 2-point DFT matrix F/sqrt(2) with F=[[1,1],[1,-1]] is unitary up to the scale already. F^2 equals 2I. det F is _____.

NAT · enter the numerical value

Q.29 MCQ · 1 mark · P&S; / joint

For a joint density f_{X,Y}(x,y), the conditional density f_{X|Y}(x|y) equals (when f_Y(y) > 0)

(A) f_{X,Y}(x,y)/f_Y(y)
(B) F_X(x)
(C) f_X(x)
(D) f_Y(y)

Q.30 NAT · 1 mark · SQL

```
SELECT SUM(x) FROM (VALUES (1),(NULL),(3)) t(x) returns
```

NAT · enter the numerical value

Q.31 MCQ · 1 mark · Scalar

The matrix aI_n has eigenvalues

(A) all equal to a
(B) 0 and a
(C) 1 repeated
(D) a^n

Q.32 NAT · 1 mark · Skew 90

[[0,-1],[1,0]] represents rotation by 90 degrees. Its squared Frobenius norm is _____.

NAT · enter the numerical value

Q.33 MCQ · 1 mark · UninformedSearch

A robot on a grid can move N,S,E,W at cost 1. Fewest moves from (0,0) to (2,2) is

(A) 2
(B) 4
(C) 8
(D) 0

Q.34 NAT · 1 mark · Flat min

x^4 min value _____.

NAT · enter the numerical value

Q.35 MCQ · 1 mark · Growth

The growth factor in GE measures

(A) how large entries become relative to ||A||
(B) the rank
(C) the trace
(D) k-means k

Q.36 - Q.65 Carry TWO marks each (DA)

Q.36 MCQ · 2 marks · LDA

S_W is singular when, in generic position,

(A) n is huge and features independent
(B) the feature dimension p exceeds within-class rank (about n-K)
(C) classes are well separated
(D) priors are equal

Q.37 NAT · 2 marks · Hash tables

A chain-hash table of size 4 stores keys 1,5,9,13,17 (all hash to 1). The number of key comparisons to search for 17 if each chain is a list from head=most-recently-inserted is

NAT · enter the numerical value

Q.38 MCQ · 2 marks · Ridge

Wide matrix p > n with dependent columns:

(A) OLS has a unique minimiser, ridge does not
(B) ridge has a unique minimiser, OLS does not
(C) both fail to have any minimiser
(D) both yield identical coefficients

Q.39 MCQ · 2 marks · LOOCV

OLS PRESS formula e_i/(1-H_ii) shows that LOOCV residuals

(A) can be computed from a single OLS fit and the leverages
(B) cannot be computed without n separate factorisations even in theory
(C) equal training SSE
(D) are undefined if an intercept is present

Q.40 MSQ · 2 marks · P&S; / exponential

Let X~Exponential(rate=lambda). Which are true?

(A) The MGF is lambda/(lambda-t) for t < lambda
(B) P(X=0)=lambda
(C) E[X^2]=2/lambda^2
(D) P(X > s+t | X > s)=P(X > t) for s,t > 0

Q.41 MCQ · 2 marks · Graphs

Undirected unweighted graph Harbor has edges Pier-Dock, Dock-Yard, Yard-Gate, Pier-Shed, Shed-Gate, Dock-Shed. BFS from Pier scans neighbours in alphabetical order. The dequeue order is

(A) Pier Dock Shed Yard Gate
(B) Pier Dock Shed Gate Yard
(C) Gate Yard Shed Dock Pier
(D) Dock Gate Pier Shed Yard

Q.42 NAT · 2 marks · SQL

```
Exam(sid,subj,score):
(1,DA,80),(2,DA,80),(3,DA,60),(1,GA,70),(2,GA,90).
SELECT sid FROM Exam WHERE subj='DA' AND score > =80
returns how many rows?
```

NAT · enter the numerical value

Q.43 MSQ · 2 marks · Warehousing and OLAP

Kimball-style dimensional modelling. Which are true?

(A) facts are additive (or semi/non) numeric measurements at a declared grain
(B) dimensions provide the labels and hierarchies for slicing
(C) a star join is fact-to-dimension on surrogate keys
(D) every dimension must be a weak entity of the fact in the ER sense of double rectangles

Q.44 MSQ · 2 marks · P&S; / normal

X~N(2, 16) and Z=(X-2)/4. Which are true?

(A) Z ~ N(0,1)
(B) Var(X+Z)=16+1 because X and Z are independent
(C) P(|X-2| < 4)=P(|Z| < 1)
(D) P(X > 2)=1/2

Q.45 MSQ · 2 marks · Sampling

Comparing VE, LW, and Gibbs on a 5-node binary Alarm net for P(B | J=true, M=true):

(A) VE can compute the exact rational (or floating) value from the CPTs
(B) LW gives a weighted Monte Carlo estimate
(C) Gibbs gives an MCMC estimate using Markov-blanket updates
(D) all three are guaranteed to match to 10 decimals after 2 samples

Q.46 NAT · 2 marks · Graphs

In graph Metro (edges North-Hub, Hub-East, Hub-West, East-South, West-South), BFS from Hub with alphabetical neighbour order visits how many vertices?

NAT · enter the numerical value

Q.47 NAT · 2 marks · P&S; / Bayes

A malware scanner flags 98% of infected files and 4% of clean files. Prevalence of infection is 1%. Given a flag, P(infected) equals _____ (two decimals).

NAT · enter the numerical value

Q.48 NAT · 2 marks · Sampling

Ancestral sample on A- > B with P(A=1)=0.5, P(B=1|A=1)=1, P(B=1|A=0)=0. Draw A=1 then B is ____.

NAT · enter the numerical value

Q.49 NAT · 2 marks · BiasVariance

Bias of an estimator is 2, so bias^2 contribution is ____.

NAT · enter the numerical value

Q.50 NAT · 2 marks · LDA

```
Class A samples 1,2,3 and class B samples 5,6,7. Pooled
sample variance (divisor n1+n2-2) is ____.
```

NAT · enter the numerical value

Q.51 NAT · 2 marks · P&S; / counting

A 6-character OTP is formed from digits 0-9 with all characters distinct. The number of possible OTPs is _____.

NAT · enter the numerical value

Q.52 NAT · 2 marks · Data types and transformation

On the same RainMm sample [0, 2, 4, 6, 18], using population standard deviation, the z-score of the last value rounded to two decimals is

NAT · enter the numerical value

Q.53 MSQ · 2 marks · BayesNet

Named factorisation P(X)P(Y)P(Z|X,Y). Which independences are implied?

(A) X ? Y (no edge, no common child observed)
(B) X ? Z | Y is not implied (X is a parent of Z)
(C) X not? Z
(D) Z ? Y | X is not implied (Y is a parent of Z)

Q.54 NAT · 2 marks · Add9

Linearisation of sqrt(x) at x=4: 2+(x-4)/4. At x=4.4 this is 2.1. Enter _____ (one decimal).

NAT · enter the numerical value

Q.55 NAT · 2 marks · Trees

Keys 8, 3, 10, 1, 6, 14, 4 are inserted in that order into an empty BST. The key in the left child of 6 is

NAT · enter the numerical value

Q.56 MSQ · 2 marks · P&S; / independence

A and B are independent with 0 < P(A),P(B) < 1. Which statements MUST be true?

(A) P(A n B)=P(A)P(B)
(B) P(A u B)=P(A)+P(B)-P(A)P(B)
(C) P(A|B)=P(A)
(D) A and B are disjoint

Q.57 MCQ · 2 marks · Python

```
The expression
print(list(enumerate(['p', 'q'], start=2)))
evaluates to
```

(A) [(2, 'p'), (3, 'q')]
(B) [(0, 'p'), (1, 'q')]
(C) [(1, 'p'), (2, 'q')]
(D) [(2, 'p'), (2, 'q')]

Q.58 MCQ · 2 marks · P&S; / joint

X and Y are independent. X is uniform on {1,2} and Y is uniform on {0,1}. Then P(X+Y=2) equals

(A) 1/4
(B) 1/8
(C) 3/8
(D) 1/2

Q.59 MCQ · 2 marks · VarElim

VE on a polytree (singly connected BN) with a good order is

(A) polynomial in the CPT sizes (belief propagation style)
(B) always exponential in n regardless of structure
(C) undecidable
(D) the same as alpha-beta

Q.60 MSQ · 2 marks · Warehousing and OLAP

Star schema facts. Which statements hold?

(A) the fact table is typically in (near) BCNF with FKs to dimensions plus measures
(B) dimension tables are often denormalised and not in BCNF
(C) a snowflake normalises dimension hierarchies into separate tables
(D) a star schema cannot have more than two dimensions

Q.61 NAT · 2 marks · Inv sin

d/dx arcsin x at x=0 equals _____.

NAT · enter the numerical value

Q.62 MCQ · 2 marks · Logistic

Binary cross-entropy on a single point with y=1 and p=0.5 equals (natural log)

(A) 0
(B) log 2
(C) 1
(D) 0.5

Q.63 MCQ · 2 marks · Python

```
The call in
def f(a, b, c=1, *rest, d=2, **kw):
return (a, b, c, rest, d, kw)
print(f(9, 8, 7, 6, d=5, z=4))
returns
```

(A) (9, 8, 7, (6,), 5, {'z': 4})
(B) (9, 8, 1, (7, 6), 5, {'z': 4})
(C) (9, 8, 7, (6,), 2, {'d': 5, 'z': 4})
(D) (9, 8, 7, (6, 5), 2, {'z': 4})

Q.64 MCQ · 2 marks · Area para

Area of the parallelogram spanned by (1,0) and (1,1) is |det| = _____.

(A) 1
(B) 0
(C) 2
(D) sqrt(2)

Q.65 MCQ · 2 marks · KFold

A student reports LOOCV accuracy by training on all n points then evaluating n times. This is

(A) correct LOOCV
(B) not LOOCV: each evaluation must train without the held-out point
(C) k-fold with k=1
(D) ridge with lambda=n

MOCK J ANSWER KEY AND SHORT SOLUTIONS

Q.1 [MCQ] A — (x+1/x)^2=9=x^2+2+1/x^2, so x^2+1/x^2=7.

Q.2 [NAT] 54 — 2*3^3=54.

Q.3 [MCQ] B — Cut no ice means have no influence or fail to convince.

Q.4 [NAT] 10 — sqrt(64+36)=10.

Q.5 [MCQ] B — b^2=64 so b=8 (b > 0, b!=1). Then log2(8)=3.

Q.6 [NAT] 30 — Area = (1/2)*5*12=30.

Q.7 [MCQ] C — The elder-girl condition leaves (G,G) and (G,B) equally likely.

Q.8 [NAT] 30 — P(6,2)=30.

Q.9 [MCQ] B — The two edges are equal and parallel, so the section is a rectangle.

Q.10 [NAT] 300 — Gold is 3/5 of 500 g, i.e. 300 g.

Q.11 [MCQ] D — Power = 1 - beta(alternative).

Q.12 [NAT] n — Independence.

Q.13 [MCQ] B — Stay 1D.

Q.14 [NAT] 0.25 — (3+1)/(14+2)=4/16=0.25.

Q.15 [MCQ] D — |2| < 2.306, so do not reject at 5%.

Q.16 [NAT] 6 — 6.

Q.17 [MCQ] B — The only violation is parent 8 (index 2) versus child 7 (index 5).

Q.18 [NAT] 4 — Average 4.

Q.19 [MCQ] A — Precision 5 scale 2: up to 999.99 (3+2). 123.45 fits. 1234.5 does not. 0.0001 would round at 2 decimals.

Q.20 [NAT] 2 — 1 + Var(Y)=2.

Q.21 [MCQ] A — d.

Q.22 [NAT] 4 — Same scope, still 4 entries.

Q.23 [MCQ] A — Dropping a dimension by aggregation is a roll-up.

Q.24 [NAT] 3 — 4-1=3.

Q.25 [MCQ] B — Corner.

Q.26 [NAT] 16 — I = L - 1 = 16.

Q.27 [MCQ] A — Syllabus.

Q.28 [NAT] -2 — Same as Hadamard.

Q.29 [MCQ] A — Definition of conditional density: joint over marginal.

Q.30 [NAT] 4 — SUM skips NULL: 1+3=4.

Q.31 [MCQ] A — a times I.

Q.32 [NAT] 2 — 1+1.

Q.33 [MCQ] B — Manhattan distance 4; BFS finds it.

Q.34 [NAT] 0 — 0.

Q.35 [MCQ] A — Stability.

Q.36 [MCQ] B — Within scatter lives in an (n-K)-dimensional space.

Q.37 [NAT] 1 — Head is 17, so the successful search compares once.

Q.38 [MCQ] B — Ridge is strictly convex; OLS loss is flat along the null space.

Q.39 [MCQ] A — Classic hat-matrix shortcut.

Q.40 [MSQ] A;C;D — Memoryless, second moment 2/l^2, MGF formula. A continuous law has P(X=0)=0.

Q.41 [MCQ] A — BFS with alpha neighbour order dequeues Pier Dock Shed Yard Gate.

Q.42 [NAT] 2 — Sids 1 and 2: two rows.

Q.43 [MSQ] A;B;C — Dimensions are not weak entities of the fact; they are conformed master data.

Q.44 [MSQ] A;C;D — Z is an affine function of X, hence perfectly dependent with X; Var(X+Z)=Var(1.25 X - 0.5) is not 17.

Q.45 [MSQ] A;B;C — Two samples do not certify 10-decimal agreement.

Q.46 [NAT] 5 — The reachable set has size 5.

Q.47 [NAT] 0.2 — 0.0098/(0.0098+0.0396)=0.0098/0.0494 ~ 0.20.

Q.48 [NAT] 1 — Deterministically 1.

Q.49 [NAT] 4 — 2^2=4.

Q.50 [NAT] 1 — Each class has sample var 1; pooled=1.

Q.51 [NAT] 151200 — P(10,6) = 10!/4! = 151200.

Q.52 [NAT] 1.9 — z=(x-mean)/sigma. mean=6, sigma=6.324555320336759, z=1.8973665961010275.

Q.53 [MSQ] A;C — X and Y are marginally independent; each is dependent with Z. B and D are false claims of CI.

Q.54 [NAT] 2.1 — 2.1.

Q.55 [NAT] 4 — 6 is the right child of 3; 4 is between 3 and 6 so it becomes 6's left child.

Q.56 [MSQ] A;B;C — Independence is the product formula, which implies P(A|B)=P(A) and the inclusion-exclusion form. Disjointness would force a zero product, a contradiction.

Q.57 [MCQ] A — enumerate with start=2 numbers from 2.

Q.58 [MCQ] D — Favourable independent pairs: (1,1) and (2,0), each with probability 1/4, total 1/2.

Q.59 [MCQ] A — Treewidth 1 / polytree.

Q.60 [MSQ] A;B;C — Stars routinely have many dimensions (date, store, product, promo, ...).

Q.61 [NAT] 1 — 1/sqrt(1-x^2).

Q.62 [MCQ] B — -log(0.5)=log 2.

Q.63 [MCQ] A — c takes 7, 6 goes to *rest, keyword d=5, leftover z in **kw.

Q.64 [MCQ] A — 1.

Q.65 [MCQ] B — Training on all n leaks the test point.